



ISLS Annual Meeting 2026
June 15-19, 2026

Partnering with Purpose

6th Annual Meeting of the International
Society of the Learning Sciences (ISLS)

Proceedings

Edited by: Kylie Peppler, Maggie Dahn



June 15-19, 2026
Irvine, California

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University of California, Irvine (UCI)

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Preface

We are delighted to welcome you to the 2026 Annual Meeting of the International Society of the Learning Sciences (ISLS), taking place in beautiful **Irvine, California, USA**, from **June 15–19, 2026**. This year, we come together around the unifying theme **“Partnering with Purpose,”** emphasizing the power of collaboration and connection across sectors, contexts, and communities to amplify the impact of learning sciences research and practice.

In ISLS 2026, we focus on building effective **cross-sector partnerships** that bridge boundary spaces between scholars, practitioners, learners, and institutions — inviting purposeful engagement and knowledge co-creation as essential elements for meaningful and sustained change. This theme reflects our community’s enduring commitment not only to advancing knowledge about learning, but also to designing and enacting equitable, responsive, and consequential learning environments.

The Annual Meeting continues the tradition of weaving together the **International Conference of the Learning Sciences (ICLS)** and the **Computer Supported Collaborative Learning (CSCL)** conference into a shared program that highlights both established and emerging directions in the field. The dual call for proposals has invited a diverse set of contributions that explore collaborative research, design, and practice across formal and informal learning settings, with a range of methodological, theoretical, and empirical approaches.

This year’s program features a rich array of **keynotes, symposia, workshops, panels, visual and media arts installations, live performances, paper sessions, and posters**, with expanded offerings designed to support vibrant exchange and inclusive participation. We are particularly proud of the workshop and tutorial tracks that precede the main conference, including the Doctoral Consortium and Early and MidCareer Workshops, the new Alt-Academia Career Workshop, and Interactive Tools & Demos, which provide deep engagement with current methods, tools, and pressing questions in the learning sciences.

This year’s meeting features an inspiring and diverse keynote lineup that speaks powerfully to our theme of “Partnering with Purpose.” On June 16, Marisa Morán Jahn (designer, artist; instructor and researcher at MIT, Director of Integrated Design at Parsons/The New School) opens the series with a keynote grounded in creative collaborations and community-centered partnership, demonstrating how joyful and purposeful design can amplify collective learning and impact. On June 17, Alyssa Friend Wise (Professor of Technology and Education and Director of the LIVE Learning Innovation Incubator at Vanderbilt University) brings her deep expertise in technology-mediated learning and innovation ecosystems, exploring how collaboration across disciplinary and institutional boundaries can shape equitable and transformative learning opportunities. On June 18, Ananda Marin (associate professor in the Department of Education and faculty in American Indian Studies at the University of California, Los Angeles (Tongva territory)) — whose work foregrounds culture, equity, and narrative in learning contexts — offers a keynote that highlights the role of intentional partnerships in advancing culturally sustaining pedagogies and justice-oriented approaches in the learning sciences. Finally, on June 19, Kareem Edouard (Assistant Professor of Learning Technologies at Drexel University’s School of Education and Co-Director of the Informal Learning Linking Engineering, Science and Technology (ILLEST) Lab) closes the series by illuminating how purposeful collaboration across creative, academic, and community worlds can foster transformative inquiry and collective empowerment.

ISLS 2026 also continues to expand opportunities for connection and inclusion through hybrid and virtual engagement options, ensuring that members of our global community can participate even when travel is not feasible. These efforts reflect our collective value of inclusivity, and our aspiration to build spaces where diverse voices, experiences, and perspectives are welcomed and amplified.

We gratefully acknowledge the hard work of the **ISLS Annual Meeting Committee, Communications Committee, Hybrid Engagement Sub-Committee, Membership Committee**, and the many volunteers across program and support roles. We also thank the School of Education at the University of California, Irvine, our funders, and the broader institutional partners who have supported this year’s event.

Finally, we express our profound gratitude to everyone, including authors, reviewers, presenters, attendees, and volunteers, whose dedication continues to shape and strengthen the ISLS community. Together, we advance not only scientific inquiry, but also collaborative, equitable, and purposeful engagement in learning research and practice. May ISLS 2026 inspire new partnerships, deeper connections, and continued innovation in the learning sciences.

Kylie Peppler and Maggie Dahn
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Keynotes

Imagination, Impact, and Epistemic Justice

Marisa Morán Jahn, MIT, Parsons/The New School, United States, mjahn@mit.edu

Abstract: In this talk, artist and educator Marisa Morán Jahn presents her work, highlighting how a joyful, playful, and imaginative approach serves as a powerful tool to achieving broad-scale social impact — and ask fundamental questions about society. Jahn’s talk draws on two decades of teaching at MIT, Columbia University, and Parsons/The New School as well as K-12 youth and creating civic media toolkits including audionovelas for domestic workers, pocket-sized comics for migrant farmworkers, illustrated maps for New York City’s 13,000 English Language Learners, and more. In 2008, she collaborated with a community in rural Honduras to create “Bibliobandido” (in English, ‘story eater’), a literacy movement that centers around a bandit who EATS stories and whose fame rivals Santa Claus. As a living legend still active today, Bibliobandido has transformed the lives of 25,000 young believers in Central and North America. For another civic-scale artwork called “CareForce” (2010-ongoing), Jahn collaborated with the National Domestic Workers Alliance (NDWA) to create a social practice artwork featuring two mobile studios (NannyVan, CareForce One) and a PBS- and Sundance-supported film that amplifies care workers’ voices. In 2020, “CareForce” informed the creation of “Carehaus,” the U.S.’s first intergenerational, care-based co-housing building which Jahn has presented to members of the U.S. Congress to help shape housing policy. Across these projects, Jahn explores how collaboration, storytelling, and design operate as powerful sites of learning—revealing how people make meaning, build empathy, and co-create knowledge through participation. Jahn notes, “I’m fascinated by differences in how we learn and how epistemologies form the bedrock of the worlds we create. Creating art and civic media invites me to stay the most attuned to human subtleties and my environment — and connecting with others.” Her practice offers a model of creating with purpose.

Bio: Marisa Morán Jahn (she/her) is an artist of Ecuadorian/Chinese descent whose public art, installations, films, and civic media tools explore “civic spaces and the radical art of play” (Chicago Tribune). Jahn’s work directly engages new immigrant families and low-wage workers — and millions more via The United Nations, Obama’s White House, Venice Biennale, Guggenheim Museum, Tribeca Film Festival, The National Public Housing Museum, and international media (The New York Times, Wall Street Journal, Univision, Forbes, BBC, CNN, Fader). Jahn’s highly interdisciplinary work achieves scale and impact through partnerships with organizations such as NBA’s Chicago Bulls and WNBA’s Chicago Sky basketball teams, National Domestic Workers Alliance, the U.S.’ Mine Safety Health Administration and through research teams across Johns Hopkins, MIT (her alma mater and where she is now a Senior Researcher), and Parsons/The New School where is the Director of Integrated Design. With architect and MIT professor Rafi Segal, she co-authored “Design and Solidarity” (Columbia University Press); co-founded Carehaus, the U.S.’s first care-based co-housing project; co-created the HOOPcycle (think MesoAmerican basketball meets tricycle), and collaborates on various architectural/urban-scale projects. Jahn has received awards from Sundance, The Holcim Foundation, Creative Capital, Anonymous Was a Woman, and Rockefeller Foundation and is represented by Sapar Contemporary.

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Productive Entanglements: Patterns of Partnership for Learning Technology Innovation

Alyssa Friend Wise, Vanderbilt University, United States, alyssa.wise@vanderbilt.edu

Abstract: Impactful learning technology innovation doesn't happen in isolation; it grows through relationships, among researchers, educators, students, technologists, designers, and institutional leaders. While some collaborations are carefully structured, others take root more organically, evolving to meet shifting needs and opportunities. In this keynote, Dr. Alyssa Wise explores how purposeful partnerships drive innovation across the technological, social, conceptual, and practical dimensions of learning, tracing three kinds of partnerships that have unfolded over the past decade in her work. Flexible Coupling articulates how institutional partnerships can leverage complementary strengths without becoming over-structured. In a longstanding collaboration with a university learning analytics service unit, in vivo studies of instructors' analytics use informed both theoretical models of sense-making and applied tool design. Constructive Responsiveness describes how partnerships with educators can evolve around a shared object of inquiry when goals are not yet fully defined. In work with dental educators implementing an online reflection tool, the collaboration advanced from examinations of reflection quality to generate domain-specific insights into failure attribution and professional identity formation. Finally, Generative Pluralism characterizes interdisciplinary partnerships among researchers or co-designers where the goal is not convergence but conceptual creation. These collaborations embraced multiplicity, surfacing productive tensions, methodological bridges, and new vocabularies for imagining learning technologies. Together, these patterns illustrate partnerships as adaptable forms of collaboration, each suited to different problems, people, and purposes. This talk highlights how designing, studying, and sustaining learning with the support of technology can be supported by purposeful partnerships that bridge disciplines, organizations, and stakeholder perspectives.

Bio: Dr. Alyssa Wise is Professor of Technology and Education at Vanderbilt University's Peabody College, with a courtesy appointment in Computer Science in the College of Connected Computing. She also serves as Director of the LIVE Learning Innovation Incubator, Vanderbilt's interdisciplinary hub for impactful learning technology innovation. Her research productively integrates tools of computational data science, theories of learning sciences and collaborative learning, and practices of human-centered design to build and study analytic and artificial intelligence systems that promote powerful learning for individuals situated in social context. She also conducts mixed-method investigations into how educational practices are being reshaped by new sources of data and the growing availability of new generative artificial intelligence capabilities. Dr. Wise holds a Ph.D. in Learning Sciences and M.S. in Instructional Systems Technology from Indiana University, as well as a B.S. in Chemistry from Yale University, was previously a Professor of Learning Sciences and Educational Technology and Director of the Learning Analytics Research Network (LEARN) at New York University and has worked in industry, with non-profits, and in classrooms across the Americas.

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“I hear your vibes”: Partnering in the Moment to Moment of Activity

Ananda Marin, University of California, Los Angeles, United States, amarin1@ucla.edu

Abstract: In this talk, Dr. Ananda Marin traces her own history with the methodologies of community-based design research and co-design, linking what she has learned to key organizing constructs in this domain — matters of purpose, place, scale, impact, power, and equity. She shares axioms that were surfaced and re-membered while partnering alongside Indigenous families and most recently with improvisational jazz artists to design generative contexts that move toward collective thriving. From here, Dr. Marin reflects on how partnering can extend the field’s conceptualizations of learning in consequential ways. Sociocultural theories in the learning sciences have conceptualized learning as both the acquisition of practices as well as shifts in the use of cultural and cognitive resources for the purposes of meaning making. She suggests that these theories, which have played a pivotal role in her scholarly trajectory, can be expanded when we theorize alongside scholarship in Indigenous Studies and the work of Black and Indigenous learning scientists who have foregrounded the importance of kinship, relationality, respect, responsibility, reciprocity, and reverence. Turning to questions of human maturation, the talk will invite the learning sciences community to consider the field-wide consequences of theorizing from human centrism. Building with the work of scholars who have reminded us that sociality is organized through the interconnectedness of talk and interaction, as well as four years of conversations and art making with creative musicians, Dr. Marin proposes that wondering about and with sound making can expand our understandings of human learning in ways that are necessary for meeting the challenges we face as a society.

Bio: Ananda Marin is associate professor in the Department of Education and faculty in American Indian Studies at the University of California, Los Angeles (Tongva territory). As a learning scientist she aims to bring research on the socio-ecological and cultural nature of learning and development alongside efforts to co-design more just teaching/learning environments. Broadly speaking, her research explores the diversity of ways that multigenerational groups of people organize attention and observation to participate in joint activity, collaborate, and improvise in everyday and professional contexts. She draws on video-based methodologies to examine how people in science-related and arts-based teaching/learning contexts make meaning in the moment-to-moment unfolding of interaction while accounting for the role of relationality, embodied movement, and place. Her studies in these areas inform her collaborations with educators and practitioners. She has widespread experience partnering with Indigenous communities, community-based organizations, and small collectives to design educational environments that cultivate community well-being. A primary goal of her work is to broaden research methodologies and theories of teaching/learning in ways that are consequential to the communities she partners with and the field of education.

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Prolepsis 'Till Infinity : Designing STEM Futures Through Play, Media, and Imagination

Kareem Edouard, Drexel University, United States, ke388@drexel.edu

Abstract: Prolepsis 'Till Infinity calls for an education that recognizes imagination as infrastructure. Dr. Kareem Edouard invites us to rethink how we build STEM spaces that honor culture, creativity, and care as the foundation of learning. His research at the ILLEST Lab challenges systems that reward compliance over curiosity, asking what becomes possible when play, storytelling, and design are treated as serious forms of knowledge, particularly for those most marginalized in STEAM, Black boys. Drawing from his design practices across learning, children's media, and informal STEAM environments, Dr. Edouard examines how learners develop identity, agency, and understanding through participatory, culturally responsive experiences, highlighting key insights from the learning sciences. For example, in co-design projects with educators and youth, students engage in creative problem-solving and media production, positioning them as knowledge-makers whose ideas actively shape the learning environment. These collaborative partnerships demonstrate how purposeful engagement between educators, researchers, and communities fosters equitable and imaginative learning outcomes. Dr. Edouard envisions an educational future where play, imagination, and community are essential for liberation. This keynote invites us to build toward proleptic, possible, and connected futures already in motion, where creativity and meaning-making shape how we learn, teach, and thrive together. By treating play, imagination, and community as foundational, Dr. Edouard shows how learning and partnership are inseparable in designing educational spaces that cultivate curiosity, empowerment, and cultural relevance.

Bio: Dr. Kareem Edouard is an Assistant Professor of Learning Technologies at Drexel University's School of Education and Co-Director of the Informal Learning Linking Engineering, Science and Technology (ILLEST) Lab. His work explores how design, as a cultural, creative, and communal practice, intersects informal STEM education and children's media to promote meaningful knowledge transfer across learning spaces. He examines how storytelling and co-creation cultivate belonging, purpose, and imagination in environments shaped by inequity.

Through initiatives like the Black Male Animation Lab, Dr. Edouard redefines STEM as a space for cultural creation and collective care, empowering Black boys as designers, storytellers, and innovators. He focuses on Black boys because their experiences sharpen the stakes of equity work in STEM: to support those most marginalized by systemic exclusion is to test whether design can truly be liberatory. His liberatory co-design framework examines how collaborative creation transforms educational systems and expands equity and representation. In the Sneaker Lab, design, material science, and cultural expression are explored for their impact on identity and innovation. The GRIND initiative further investigates how sports, data, and technology can foster community-driven STEM engagement.

Dr. Edouard integrates his co-liberty design framing vision into children's STEM media. This approach combines his experience as a former music video and commercial director with his academic research. As Executive Producer of MayNERD's Wild World of Science (YouTube Kids), Built From Scratch, and Creative Producer of Work It Out Wombats! (PBS KIDS), he uses media as a conduit to share STEM learning and engagement, which remains at the heart of his work with children's media.

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Special Session

Humanizing Institutional Landscapes of Ability Differences and Learning

Cathery Yeh, Erin McCloskey, Alfredo J. Artiles, Mimi Ito

How can schools, universities, and community spaces construct, reinforce, or challenge how ability differences and disabilities are understood and organized in relation to race, language, and culture? This session brings together researchers working in schools, higher education, and out-of-school settings to study, reconceptualize, and redesign institutional, programmatic, and pedagogical practices. Through an engaging, dialogic format, the panelists will share stories and artifacts from four participatory design projects and explore common themes and challenges. The goal is to grapple with institutional landscapes that unevenly distribute learning opportunities, and to consider what equitable, culturally sustaining approaches to ability differences and disabilities in learning might look like.

Cathery Yeh uses participatory mapping with youth, teachers, and families to capture lived experiences, decision points, and institutional boundaries shaping access, labels, and opportunities across bilingual education, special education, and mathematics, tracing patterns of exclusion and possibility to inform asset-based intervention systems. *Erin McCloskey* is investigating multi-generational college classrooms involving formerly incarcerated individuals, examining how storytelling about home, school, and community illuminates systemic patterns of disability, racial, linguistic, and economic inequality. *Alfredo J. Artiles* will describe the critical cultural-historical framework he is bringing to a research-practice partnership to involve practitioners and families in co-developing systems of support for African American students. The project is conducted in a school district that has been chronically cited for racial disparities in special education affecting African American learners. *Mimi Ito* is working with educators and families developing "Neurodiversity Oases"—informal learning environments that offer neurodivergent young people identity-affirming, inclusive, social spaces centered on their technology, media, and creative interests.

Interactive Tools and Demos

Designing an AI-Powered Reflective Partner: Translating Self-Regulated Learning Theory into Generative Chatbot Scaffolds

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Abstract: This interactive tool paper presents the design and development of an AI-powered chatbot that scaffolds college students' self-regulated learning (SRL). Grounded in Pintrich's SRL framework and informed by meta-analytic evidence on effective SRL interventions, the chatbot is designed as a reflective learning partner that supports metacognitive monitoring, strategic deployment, and affective regulation. Using a design-based research (DBR) approach, this study explores how SRL theory can guide the development of generative AI-mediated interactions. We describe the theoretical rationale underlying the tool, articulate key design principles, present the prototype's architecture, and discuss implications for improving evidence-based, equitable AI scaffolds to enhance undergraduate learning outcomes.

Introduction

Self-regulated learning (SRL) is the process by which learners actively manage their cognition, motivation, and behavior to achieve academic goals (Zimmerman, 1986; Pintrich, 2000). While SRL is a strong predictor of academic success in higher education (Broadbent & Poon, 2015), many students continue to underuse metacognitive and motivational strategies despite explicit instruction (Cao & Nietfeld, 2007). Existing SRL interventions, such as semester-long SRL courses (Bail et al., 2008), reflective diaries (Dörrenbächer & Perels, 2016), and LMS-integrated training modules (Bernacki et al., 2020), have demonstrated small to moderate effects (Theobald, 2021). These efforts consistently face limitations related to scalability, personalization, and sustained learner engagement.

Generative artificial intelligence (AI) introduces new opportunities to address these pedagogical constraints by delivering adaptive, context-sensitive scaffolding at scale. Chatbots powered by large language models (LLMs) can engage students in human-like, reflective dialogue and provide timely, personalized feedback. Early studies suggest such systems can enhance motivation, self-efficacy, and strategic learning behaviors (Ng et al., 2024; Pan et al., 2024). This project therefore bridges AI's technological affordances with the pedagogical rigor of learning sciences theory by designing and testing a SRL chatbot grounded in empirical evidence and models of self-regulation to foster reflection, strategy use, and metacognitive growth.

Purpose of the study

The overarching goal is to design, implement, and evaluate an AI-powered SRL chatbot that functions as a reflective partner rather than an answer-giving tutor. Specifically, this study aims to:

1. Translate SRL theory into conversational scaffolds that foster self-reflection, metacognitive monitoring, and strategic regulation.
2. Establish a design-based research (DBR) process that iteratively refines the chatbot through user feedback and pedagogical validation.
3. Examine how learners across subjects engage with and respond to AI-empowered SRL support

Theoretical foundations

The chatbot's architecture draws upon Pintrich's (2000) four-phase SRL model: forethought/planning, monitoring, control, and reflection. This model is highly suitable for educational technology design as it delineates specific, targetable processes while integrating critical motivational and affective dimensions often underrepresented in simpler frameworks. The AI's dialogue is constructed to guide the learner through each phase: setting goals (forethought), self-monitoring, adapting strategies (control), and assessing outcomes (reflection).

Complementing this framework, this design incorporates meta-analytic findings (Jansen et al., 2019; Theobald, 2021) that identify key features of effective SRL interventions:

- Emphasis on metacognitive strategy training ($g = 0.40$) and resource management ($g = 0.39$).
- Integration of Feedback and reflection protocols (learning diaries, goal sheets).
- Contextual tailoring and ongoing, phased support to promote persistence and transfer.

Design principles and AI affordances

Drawing on SRL theory, empirical findings, and the affordances of large language model, the chatbot was designed around seven guiding principles:

1. **Personalized SRL Support:** The chatbot begins by assessing the learner's prior knowledge, current motivation, and strategy use to provide truly tailored prompts and adaptive feedback,
2. **Metacognitive Reflection Prompts:** Frequent, open-ended questions, such as "How do you usually monitor your learning progress?", are used to encourage awareness of cognitive and motivational processes (Winne & Azevedo, 2022).
3. **Guidance through Learning Protocols:** The chatbot supports structured planning and reflection tools (e.g., progress trackers, reflection logs) to foster consistency and intentionality, particularly when integrated with LMS environments.
4. **Ongoing Multiphase Support:** The bot remains available throughout all SRL phases (planning, monitoring, reflection) to provide continuity and temporal scaffolding and feedback.
5. **Domain-Specific Strategy Feedback:** By using retrieval-augmented generation (RAG), the chatbot allows learners to upload syllabi or course readings. This will dramatically reduce hallucinations and enable context-specific strategy suggestions.
6. **Affective Support and Emotion Regulation:** The chatbot is designed to detect cues of frustration or disengagement and responds with affirmations or coping suggestions, supporting persistence
7. **Feedback Usefulness Check-Ins:** Learners periodically evaluate the usefulness and fairness of chatbot feedback. These meta-dialogues enhance metacognitive calibration and help identify potential biases in AI-generated responses.

Together, these principles illustrate how AI affordances (adaptivity, personalization, and natural language generation) can be harnessed within a learning-sciences framework (metacognition, motivation, scaffolding) to promote theoretically grounded learning support.

Prototype interface and functionality

The SRL Learning Assistant prototype (Figures 1 to 6) was developed using Botpress, a platform for building custom AI agents and chatbots, to translate the self-regulated learning model into an interactive conversational environment. Grounded in Pintrich's (2000) four phases of SRL, it uses a structured dialogue flow to guide learners through planning, monitoring, control, and reflections. In line with the study's goal of designing the chatbot as a reflective partner, this role is realized through prompt level rules that keep the interaction focused on learner's SRL process rather than task completion. For example, when students ask for full answers or solutions, the bot is instructed not to provide them directly. Instead, it briefly acknowledges the request, offers a minimal starter example only when needed, suggest a strategy or next step, and poses a follow up question that prompts further thinking.

During onboarding (Figure 1), learners provide a preferred name or nickname to establish social presence and rapport. They are also invited to share relevant course materials, such as a syllabus or assignment brief, so that the chatbot can contextualize its prompts and feedback based on the task.

In the Planning phase (Figure 2), the chatbot invites students to articulate specific, measurable learning goals (e.g., "What's your main academic goal this week?"), and follows up with structured prompts that help break tasks into manageable steps, anticipate potential challenges, and plan for self-monitoring.

Figure 3 illustrates the three remaining phases of the chatbot interface. The Metacognition phase (a) supports the monitoring and evaluation of cognitive processes through reflective questioning, such as "How do you usually track your progress?". The Strategy Control phase (b) engages learners in assessing the effectiveness of their learning strategies and fostering adaptive regulation (e.g., "What strategies have worked well or not as expected?"). Finally, the Reflection phase (c) focuses on emotional regulation and learning transfer, prompting learners to consider their affective experiences and future goals (e.g., "How do you feel about your learning this week? What insights could guide you moving forward?").

Figure 1
Onboarding Interface



SRL Learning Assistant

Tell me a bit about yourself

Your Name

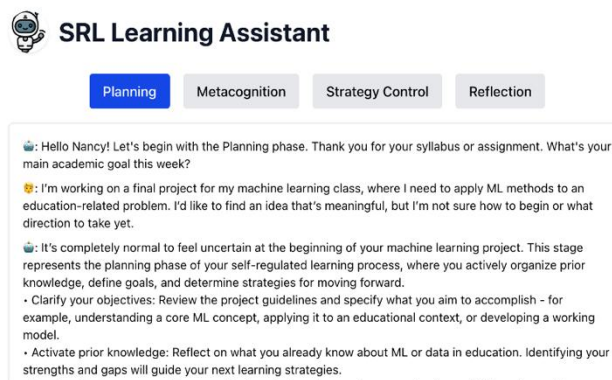
Your Course or Subject

Upload Syllabus or Assignment

Choose File No file chosen

Start Chat

Figure 2
Planning Phase



SRL Learning Assistant

Planning Metacognition Strategy Control Reflection

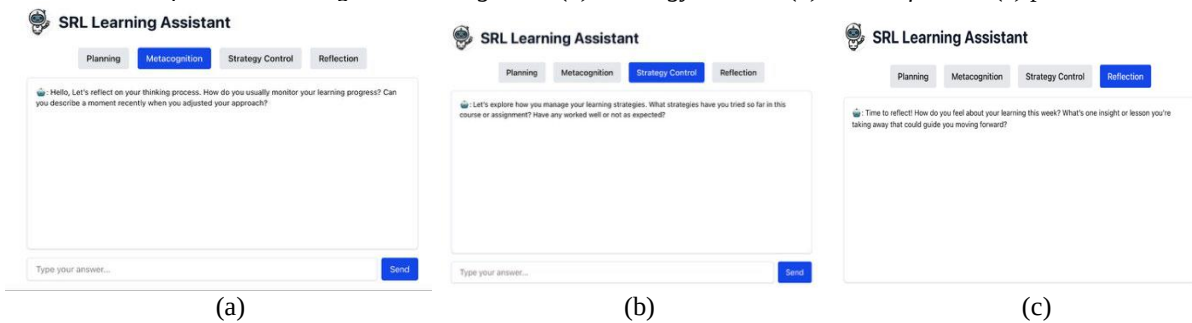
Hello Nancy! Let's begin with the Planning phase. Thank you for your syllabus or assignment. What's your main academic goal this week?

I'm working on a final project for my machine learning class, where I need to apply ML methods to an education-related problem. I'd like to find an idea that's meaningful, but I'm not sure how to begin or what direction to take yet.

It's completely normal to feel uncertain at the beginning of your machine learning project. This stage represents the planning phase of your self-regulated learning process, where you actively organize prior knowledge, define goals, and determine strategies for moving forward.

- Clarify your objectives: Review the project guidelines and specify what you aim to accomplish - for example, understanding a core ML concept, applying it to an educational context, or developing a working model.
- Activate prior knowledge: Reflect on what you already know about ML or data in education. Identifying your strengths and gaps will guide your next learning strategies.

Figure 3
Chatbot interfaces illustrating the Metacognition (a), Strategy Control (b), and Reflection (c) phases.



SRL Learning Assistant

Planning Metacognition Strategy Control Reflection

Hello, Let's reflect on your thinking process. How do you usually monitor your learning progress? Can you describe a moment recently when you adjusted your approach?

Type your answer... Send

(a)

SRL Learning Assistant

Planning Metacognition Strategy Control Reflection

Let's explore how you manage your learning strategies. What strategies have you tried so far in this course or assignment? Have any worked well or not as expected?

Type your answer... Send

(b)

SRL Learning Assistant

Planning Metacognition Strategy Control Reflection

Time to reflect! How do you feel about your learning this week? What's one insight or lesson you're taking away that could guide you moving forward?

Type your answer... Send

(c)

Design-based research approach

The chatbot's development follows a Design-Based Research methodology emphasizing iterative co-design, theory refinement, and empirical grounding. The project is currently in an early DBR cycle focused on translating SRL theory into a functional conversational bot prototype and examining initial feasibility, usability, and alignment between intended scaffolds and actual interaction flow.

- Cycle 1 (Exploratory Design): Defined pedagogical and functional requirements based on SRL theory and empirical work, with particular attention to how planning, monitoring, control, and reflection could be translated into conversational scaffolds.
- Cycle 2 (Prototype Development and Expert Review): Implemented the prototype in Botpress and developed theory informed prompts, workflow logic, and cross session logging functions. Initial expert review provided formative evidence that the chatbot's phase structure, reflective stance, and supportive tone were generally clear. On a 5-point scale, reviewers rated it 4.75 for prompt clarity, 4.5 for avoiding over directing or answer giving, 4.3 for emotional support, and 4.0 for phase structure clarity. Its overall rating as an effective SRL support tool was 3.5. Reviewers also identified several priorities for

refinement, including strengthening task specific prompting, merging the monitoring and control phases, adding attribution support within the reflection phase, and providing phase entry guidance for learners unfamiliar with SRL.

- Cycle 3 (Small Scale Pilot Testing): Will deploy the refined prototype with students in short term, individual task-based testing to further examine usability, engagement, and the quality of students' planning, monitoring, strategy adjustment, and reflection. This cycle will also test the chatbot's cross session information retrieval function to determine whether it can maintain continuity across interactions, provide consistent feedback, and generate weekly or semester level progress reports based on students' activity across multiple sessions.
- Cycle 4 (Refinement and Extended Embedding): Will use the small-scale evidence from Cycle 3 to make final refinements before wider course-based implementation. In this cycle, the study will examine longer term questions about how students use the chatbot across a semester, whether repeated interaction supports improvement in self-regulated learning processes, and how the quality of learner engagement changes over time. Data will be collected longitudinally through chatbot interaction logs, phase completion and transition patterns, the depth and quality of learner responses, indicators of strategy monitoring or adjustment across turns, usability and engagement feedback, and follow up user comments or interviews. These data will be used to refine prompt phrasing, pacing, memory functions, and adaptive logic, while also examining the chatbot's potential influence on students' self-regulated learning and learning outcomes in an authentic course context.

Through this iterative process, the chatbot functions both as a learning tool and as a research instrument for studying AI-mediated self-regulation.

Implications and future directions

This work demonstrates how design principles derived from SRL theory and prior empirical evidence can be translated into an AI supported conversational tools. It highlights the potential for AI to function as a reflective co-partner that fosters learner agency and metacognitive growth, rather than replacing human instruction. At this stage, the contribution lies in showing how SRL principles can be operationalized through prompt design, workflow structure, and task grounded interaction. What remains to be tested more directly is how students engage with the chatbot over time and whether sustained use supports measurable changes in self-regulated learning processes and outcomes.

Future research will focus on three interrelated directions:

- Exploring how SRL prompt and feedback can be applied across different subject areas.
- Evaluating the chatbot's impact on students' metacognitive growth and learning outcomes.
- Examining how this AI-powered SRL chatbot might support underrepresented groups in their transition to college learning.

This prototype contributes to the ISLS community by demonstrating how theory-informed design and AI affordances can converge to enhance SRL at scale. By articulating its design logic and empirical grounding, this work positions the chatbot as both a pedagogical innovation and a research vehicle for advancing understanding of reflective learning with AI in an inclusive way.

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Creaticode: An AI Playground for Every Middle School Student

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Abstract: Artificial intelligence (AI) increasingly influences how we communicate, create, and solve problems, making early AI education essential. Middle school is an ideal time for students to build AI literacy, yet teachers face barriers such as limited tools, steep learning curves, and lack of classroom-ready materials. Creaticode addresses these challenges by extending the familiar Scratch platform with AI tools and AI-enhanced coding blocks that introduce generative and predictive AI concepts in an engaging, visual way. Supporting all four dimensions of AI literacy: understanding, using, evaluating, and creating, Creaticode helps students move from AI consumers to creators. The platform includes free teacher support such as tutorials, lesson plans, and slide decks to simplify classroom integration. This paper introduces the Creaticode AI toolset, describes partnerships with teachers to refine usability and curriculum alignment, and showcases examples of Creaticode in action that highlight its potential as an accessible AI playground for every middle school student.

Introduction

Artificial intelligence (AI) is increasingly impacting all aspects of our daily lives, from how we communicate to how we learn, create, and solve problems. As AI becomes more pervasive, it is essential that all students begin learning about it early, particularly in middle school, when curiosity and computational thinking skills are rapidly developing (Touretzky et al., 2019; Long & Magerko, 2020). Students need to understand what AI is, why it matters, and how it can be used responsibly and creatively.

However, teaching AI at the middle school level presents significant challenges. While there are many AI-related tools available (Druga, 2018; Carney et al., 2020; Ali et al., 2021), most are designed for students to passively observe instead of actively create or experiment. Even when effective tools exist, they often demand substantial preparation, technical expertise, and time from teachers, barriers that can make classroom implementation difficult (Chiu & Chai, 2020; Kim & Kwon, 2025).

Creaticode addresses these challenges directly. Built on the familiar Scratch platform, Creaticode extends block-based programming with custom AI blocks that introduce core AI concepts in an engaging and accessible way. In addition, Creaticode integrates AI-powered assistants and image generators, enabling students to develop their prompting skills while creating their own AI-driven projects. The platform supports all four dimensions of AI literacy (OECD & European Commission, 2025): understanding, using, evaluating, and creating AI, while maintaining the playful, exploratory spirit of Scratch.

To further reduce barriers for teachers, Creaticode includes comprehensive educator resources such as tutorial videos, ready-to-teach lesson plans, and classroom slide decks. Ongoing partnerships with teachers inform continuous improvements to both the tool and curriculum. It is available to everyone: the free version of Creaticode provides all the functionality necessary for effective classroom use. Creaticode is a playground for middle school students to engage with AI.

The Creaticode AI toolset

The Creaticode platform extends MIT Scratch (Resnick et al., 2009) with a wide variety of free coding and learning tools, enabling students to become both consumers and producers of AI. Students not only learn how to select the right tool for the task, but also gain the insight and creativity to harness their full potential.

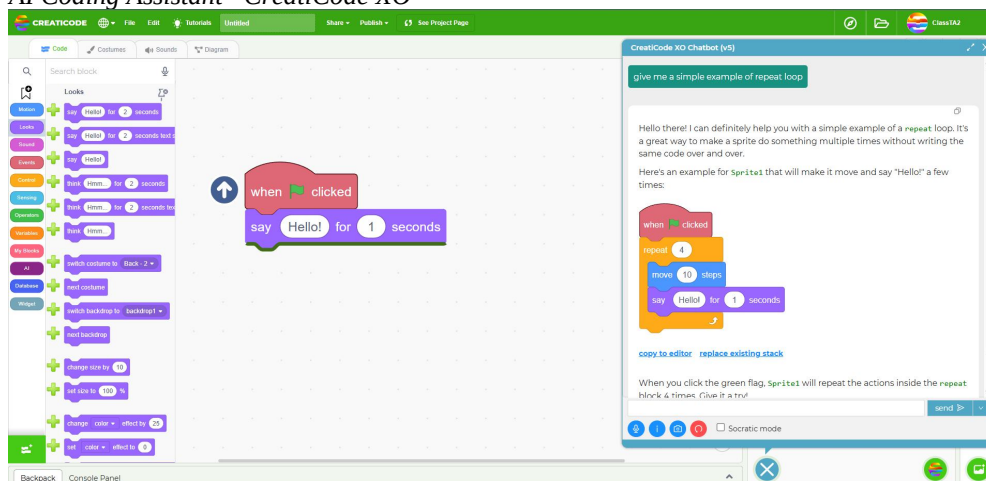
AI coding assistant

The AI assistant “Creaticode XO” is a ChatGPT-like chatbot in the coding playground. It is designed to accommodate the learning needs of middle students in the following ways.

- 1) While most mainstream chatbots can only read or write pseudocode for blocks and tend to make syntactical errors, XO can read students’ block code in the editor directly and can write block code with high accuracy.
- 2) For students who are new to block-based coding, XO can help them catch up quickly by illustrating the coding concepts with intuitive example code.

- 3) XO can chat in the same language as each student, removing a major hurdle for non-English speaking students.
- 4) XO offers a “socratic mode” that withholds direct answers, guiding students with questions and hints instead. Teachers can also enforce it class-wide to ensure students actively work through problems.
- 5) XO can also work in an “agent mode” - it discusses the design with the student and then edits the code directly. This prepares students to work with advanced agentic tools like Claude Code in the future.
- 6) XO maintains a continually updated “learning profile” for each student based on their chat history. It tracks what concepts and skills they’ve mastered or they’re struggling so XO can personalize future responses.

Figure 1
AI Coding Assistant “CreatiCode XO”



AI image library and generator

MIT Scratch offers only a few hundred images for sprites and backdrops, significantly limiting personalization. CreatiCode provides millions of moderated, AI-generated images, and students can create new ones with prompts.

Figure 2
Generate a New Image and then Generate a Variation of It



The image search/generation tool is designed to assist students to engage and create with AI.

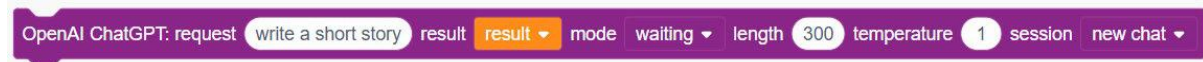
- 1) Students can select an existing image as reference and generate a variation of it. This allows them to improve an image iteratively with better prompts, or generate multiple images with consistent characters.
- 2) Students can view the original prompt used to generate each image to learn how to generate similar images.
- 3) Images generated by any student become part of the public library automatically, so other students can use it as well. This model is similar to how students create and share code, encouraging creativity and collaboration.

Generative AI blocks

While students can learn to write effective prompts by using the AI coding assistant and image generator, a higher level of AI fluency can be achieved by building projects powered by AI.

- Students have the opportunity to design programs with integrated AI blocks and are encouraged to consider the strengths and limitations of AI;
 - Students are challenged to craft highly resilient prompts capable of handling unpredictable user inputs, as these prompts become fixed and uneditable once the projects are released to the public
- To that end, the CreatiCode platform offers generative AI blocks, such as the ChatGPT request block:

Figure 3
The ChatGPT Block



The ChatGPT block sends requests to OpenAI's ChatGPT service via CreatiCode's server, which will reject any inappropriate requests. It also retains all previous messages in the chat session, so the student code can be much simpler. This block is central to many projects in the open-source CreatiCode curriculum. Some projects include: a chatbot for students to interact with a historical figure, a mystery game where the player interrogates characters to find the thief, and an app that generates quiz problems for any subject and skill level (see Figure 4).

Figure 4
AI Projects: <Chat with Einstein>, <Who's the spy?>, <AI Quiz Writer>



Predictive AI blocks

The CreatiCode platform provides predictive AI blocks that enable fast and accurate forecasting and decision-making. Some features that depend on predictive blocks include text-to-speech and speech-to-text, vision models for hand tracking and body tracking, classification models like K-Nearest-Neighbors, and Tensorflow blocks for training Neural Network models. Students apply the predictive AI blocks in the projects where they develop an app that listens to user speech in one language and speaks it in another, a training game that asks players to make various poses and checks their pose in realtime, and a neural-network model that classifies random 2D dots based on the training data (see Figure 5).

Figure 5
AI Projects: <AI Voice Translator>, <Fitness Game>, <Random dots classifier>



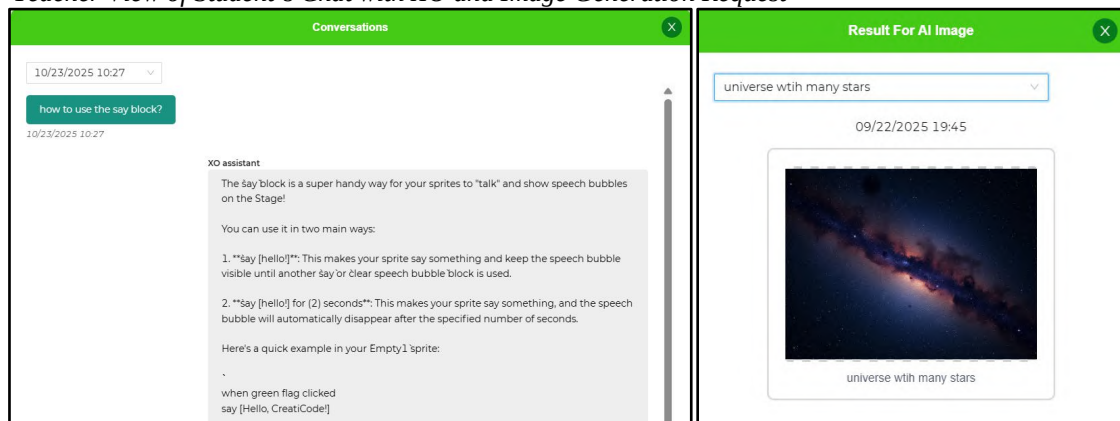
Partnering with teachers to get better perspectives

When it comes to AI-literacy education, a key design decision in CreatiCode is to always include teachers in the loop. Many of CreatiCode's key features are a result of feedback received from teachers who were teaching with CreatiCode in their classrooms.

Monitoring students' interactions with AI tools

Since each student can interact with AI tools individually, a top request from teachers is to have access to transcripts between XO and each individual student. Teachers can also view each image generation question and the image that was generated (see Figure 6.)

Figure 6
Teacher View of Student's Chat with XO and Image Generation Request



Teacher control panel

An equally important feature for teachers is to enable/disable coding tools based on their teaching goals. For example, they can force students to use the “socratic mode” or disable the AI image generator.

Figure 7
Teacher's Control Panel

Student Settings						
	Enable ChatGPT	Enable XO assistant	Enable AI Image	Enable Explore	Force Socratic	XO Code Copy
Entire class:	☒	☒	☐	☒	☒	☒
 castudent01@...	☒	☒	☐	☒	☒	☒
 castudent02@...	☒	☒	☐	☒	☒	☒

Integrated AI curriculum

A year-long AI curriculum with over 50 well-structured lessons is available for free for teachers to make full use of the AI tools in CreatiCode systematically (reference to curriculum paper). All lessons come with lesson plans, slide decks, assessment questions and student workbooks.

The lessons start with using AI tools for coding help and image generation, which are crucial skills that students can practice in later lessons. These later lessons are all project-based activities, where students first build a baseline project following given steps, and then work on a “creative adaptation” of the project to demonstrate the skills/concepts they have learned in a new project.

Demo setup

This 25-minute interactive demo follows a show-then-try cadence for each key feature. For each segment, the presenter gives a 2-3 minute live walkthrough on the projector, then invites a 3-4 minute hands-on tryout plus brief Q&A before moving on. Participants work in pairs on their own laptops; QR codes provide one-click access to starter projects. Materials and starters are distributed via a single short link. For more information, visit <https://www.curriculum.elementarycomputingforall.org/act4>

Conclusion

CreatiCode shows how AI can be approachable, creative, and meaningful for middle school students. By extending the familiar Scratch environment with AI-powered assistants, generative and predictive blocks, and a structured curriculum, it bridges the gap between AI use and creation. Built with teachers, CreatiCode stays accessible and educationally sound. Students learn to use, question, and design with AI, developing literacy in understanding, application, and creation. As AI reshapes learning and life, platforms like CreatiCode model equitable, engaging, and hands-on education that empowers young learners to shape an AI-driven future.

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How AI Can Improve Student Learning Motivation and Engagement While Reducing Faculty and Admin Burden

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Abstract: My Learning Coach (MLC) is an AI-powered software application designed to enhance student engagement by providing real-time, data-driven guidance. MLC addresses persistent student disengagement and the need for durable human skills (critical thinking, resilience, agency) in the age of AI. Grounded in learning science, including Expectancy-Value-Cost (EVC) Theory and Bloom's Taxonomy, MLC detects learning motivation via biweekly "State of Mind" (SOM) check-ins and correlates this affective data with behavioral metrics such as Learning Management System (LMS) data. This demo session showcases the MLC workflow: from a student completing a check-in to receiving personalized guidance, and how faculty utilize the same tool to make in situ pedagogical adjustments. Early randomized controlled trials at UC Irvine demonstrated an increase of 2.1 study hours per week and 11.2% more LMS pageviews among MLC users compared to their peers. MLC directly aligns with the National Academies' recommendations for transforming undergraduate STEM education.

The need for an AI-augmented learning intervention

The landscape of undergraduate education faces significant challenges, notably student disengagement and high dropout rates—three out of five students may not graduate from college (National Center for Education Statistics, 2023), and 41% of employers agree that recent graduates are ready for entry-level roles (American Association of Colleges and Universities, 2023). In the context of computer-supported collaborative learning (CSCL), impactful learning must be culturally responsive, contextually grounded, and collaboratively shaped, requiring solutions that move beyond traditional content delivery. My Learning Coach (MLC) is proposed as a solution: an AI-powered learning coach software that acts as a co-pilot, augmenting in-person assistance during a student's educational journey.

MLC is specifically engineered to foster durable skills—a growth mindset through critical thinking, resilience forged by perseverance, and agency grounded in self-determination. It achieves this by identifying each student's unique learning context, including academic and non-academic constraints, motivational misalignment, and behavioral gaps.

This demonstration aligns with the ISLS 2026 theme, "Partnering with Purpose," by showcasing a technological/design contribution that uses data science and learning theories to create a scalable platform for personalized coaching. The demo will illustrate how MLC provides the deep, real-time data necessary for instructors and institutions to partner more effectively with students to achieve equitable and effective learning.

Theoretical framework for data collection

MLC's design is grounded in established learning science frameworks to ensure diagnostic precision beyond simple engagement metrics.

The Expectancy-Value-Cost (EVC) Theory (Eccles, 1983; Barron & Hulleman, 2014) is central to MLC's assessment of student motivation. The biweekly State of Mind (SOM) check-ins utilize 15 Likert-scale questions (-3 strongly disagree to +3 strongly agree, omitting a neutral midpoint) to measure three domains:

1. Expectancy (I can master content): Measures self-efficacy and belief in the ability to grasp subject matter, aligning with Principle #3 (Social Dimensions).
2. Value (I see value in mastering content): Measures perceived relevance to goals and intrinsic motivation, linking to career aspirations and collaborative learning value.
3. Cost (What is holding me back?): Explicitly addresses non-academic stressors, excessive demands, and fairness concerns that act as barriers, aligning with Principle #6 (Flexibility).

MLC integrates motivation measurement with Bloom's Taxonomy (Build, Practice, Reflect, Transfer Knowledge) (Bloom, Engelhart, Furst, Hill, & Krathwohl, 1956) to track cognitive learning progression. For instance, Expectancy questions map directly to Bloom's levels, such as connecting confidence in mastering material to the *Build Knowledge (Understand)* stage.

Multi-Modal Data Strategy: MLC combines quantitative SOM data with qualitative data from the three-question Retrospective (RETRO) check-in, in which students provide free-form text feedback on "What went well?", "What didn't go well?" and "What can be improved?" This correlates with behavioral data from the LMS,

such as pageviews and assignment submissions. This robust approach allows MLC to deliver diagnostic insights—revealing *why* a student might be struggling (e.g., motivational barrier or cognitive misalignment) rather than just *what* their grades or clicks are.

MLC demonstration components

The demo session will walk through the MLC workflow and the data-rich artifacts it generates, focusing on how the tool supports both the student and the instructor in real time.

1. The Student Workflow: Reflection and Personalized Action. The demo will begin by simulating a student completing their biweekly SOM check-in. The focus then shifts to the resulting Personalized Student Action Report. This report provides the student with insights into their longitudinal progress.

- **Wellness Analysis:** The report visualizes shifts in emotional states (e.g., "Content," "Stressed," "Confused") over the term. For one sample student, wellness shifted from "Content" to primarily "Stressed" due to growing workload pressure, demonstrating "clear comprehension amid growing workload" (resilience).
- **Data-Driven Recommendations:** Based on the analyzed motivation and behavior patterns, the student receives targeted, actionable recommendations. Examples include suggestions to "Schedule short mindfulness breaks" to reduce stress, "Engage in weekly peer study groups" to build accountability (fostering relatedness and agency), or "Review analytics each Friday" (fostering a growth mindset).

2. The Instructor Workflow: In-Situ Adjustments via Faculty Action Reports. The MLC demo will showcase the Faculty Action Report (e.g., the End-of-Term Report). This report aggregates student data on learning motivation, Bloom's progression, sentiment, and wellness.

- **Holistic Evidence for Evaluation:** The report provides "new forms of evidence" beyond traditional student surveys, supporting a holistic evaluation of teaching. One faculty member noted this data provided "additional evidence for my upcoming professional advancement review".
- **Pedagogical Insights:** Faculty can identify class-wide trends for in situ adjustments. For instance, analysis revealed that "Build, Practice, Reflect, and Transfer Knowledge all trended upward," indicating steady student confidence and mastery. Conversely, data might highlight "Stressed" states peaking in Week 4, reflecting early workload pressures, allowing the instructor to incorporate stress management strategies or adjust assignment timing.

3. Future Capabilities: The Agentic AI Chatbot. The demonstration will preview the planned Agentic AI Chatbot. This chatbot will be trained on anonymized historical class data (including student freeform text feedback) to serve as a specialized Teaching Assistant (TA).

- **Proactive Guidance:** The AI can answer student questions (e.g., "How much time have A students spent in this class in the past?") by retrieving relevant historical data.
- **Targeted Assistance for Instructors:** Instructors can query the AI for class issues (e.g., "What is the most common issue that the students scoring 50% overall are struggling with?"), receiving real-time insights based on the collected SOM and RETRO data.

Pilot results and alignment with equity principles

The 2024–2025 MLC pilot at UC Irvine involved 3,195 undergraduates across 26 courses. The data collected support the efficacy of the reflection tool, even before the implementation of automated coaching reports and the AI chatbot.

Proven Impact: In a randomized control trial in the ENGR190W course, MLC users reported an additional 2.1 hours of weekly study time and 11.2% more LMS pageviews than their peers without access. Student feedback confirms MLC's motivational impact: 61% report a 10% increase in class engagement, and 54% note a 12% improvement in keeping up with coursework. Furthermore, 94% of respondents felt comfortable using MLC, and 83% believed it could inform AI-driven or in-person support interventions.

Additionally, the MLC aligns with the National Academies' Principles for Equitable and Effective Teaching (National Academies, 2025). The alignment is summarized in Table 1.

Table 1

My Learning Coach mapping with Bloom's Taxonomy and Expectancy-Value-Cost Learning Theory, and alignment with the National Academies' 7 Principles for Equitable and Effective Teaching

MLC Feature/Mechanism	National Academies Principle	How MLC Meets the Principle
Bloom's Taxonomy Tracking & LMS Data	Principle #1: Active Engagement	Measures and motivates engagement (e.g., +11.2% LMS pageviews) and tracks cognitive progression (Build to Transfer Knowledge), foundational to active learning.
EVC/Cost Questions & Wellness Checks	Principle #6: Flexibility & Responsiveness	Provides real-time data on non-academic "Cost" barriers, allowing faculty and support staff to make "in situ adjustments" based on students' evolving, individualized needs.
Faculty & Student Action Reports	Principle #5: Multiple Forms of Data	Collects quantitative (EVC, LMS activity) and qualitative (wellness, freeform text sentiment) data to provide holistic, actionable insights for continuous improvement at all levels.
Dashboard Function & Proactive Guidance	Principle #7: Intentionality & Transparency	Makes the learning process and overall class state visible, revealing "hidden challenges" like time management struggles and using anonymized historical data to set expectations for future cohorts.
Training Data/Inclusive Design	Principle #4: Identity & Sense of Belonging	Designed using primary research from diverse institutions (including MSIs) and employs culturally responsive frameworks (EVC) to foster agency, resilience, and an identity-safe environment.

MLC is a direct example of the infrastructure called for by Recommendation 11—collecting, monitoring, and providing transparent access to diverse data about student outcomes, experiences, and affective measures for systemic improvement. Furthermore, by grounding its design in EVC, Mindset Theory, and Self-Determination Theory, MLC directly addresses Recommendation 3 by using the Principles from the initial conceptual stage.

CSSL implications

The My Learning Coach demo illustrates how AI and learning science can be combined to operationalize key principles of equitable and effective teaching. The platform provides faculty with timely, nuanced insights necessary for making data-informed pedagogical decisions, while simultaneously empowering students with self-awareness (metacognition) and the skills (growth mindset, agency) needed for long-term academic adaptability.

A crucial area for future development that directly engages the CSSL community is the planned *Social Comparison Assessment*. This feature measures a student's emotional and motivational response to peers who perform better or worse (upward/downward comparisons). The resulting data guides highly targeted coaching strategies, such as promoting mastery goals for envious students or fostering empathy for "elitist" students, demonstrating how the system will facilitate personalized, adaptive, and human-skills-focused collaborative support.

MLC demonstrates that technology can serve as a robust, theory-driven partner—a key objective of the CSSL 2026 conference theme—for managing the vast amounts of data required for individualized, equitable, and effective learning at scale.

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Leveraging Online Videos for Elementary Science (LOVE Science): An AI Tool for Identifying Developmentally Appropriate, NGSS- Aligned Science Videos

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Abstract: Multimodal presentation of science content can support science learning; however, elementary teachers lack the time and support to find age-appropriate, freely accessible, and Next Generation Science Standards (NGSS)-aligned science videos. This interactive demo will showcase our Artificial intelligence (AI)-powered Video Analyzer, a feature of the Leveraging Online Videos for Elementary Science (LOVE Science) platform, which analyzes YouTube videos to determine alignment to NGSS performance expectations, conceptual complexity, vocabulary level, and relevance across NGSS domains. The LOVE Science Video Analyzer helps teachers make more effective use of the open educational resources already available to them—particularly on YouTube—by providing an AI-powered plug-in that streamlines the process of finding, reviewing, and selecting high-quality science videos to enhance instruction.

Introduction: Using videos to enhance elementary science learning.

Scores from the most recent National Assessment of Educational Progress indicate that less than 30% of U.S. 8th graders are proficient in science and that students show post-pandemic losses in science achievement and interest (Jacobson, 2025). Supporting learners through science instruction in the early elementary grades can set them up for later success in science (Curran & Kitchin, 2019; Kaderavek et al., 2020; Morgan et al., 2016). However, at this grade level, teachers generally spend less time teaching science compared to other subjects (Bassok et al., 2016) and often lack access to high-quality instructional materials for teaching science, relying instead on fee-based websites or creating lessons themselves (Plumley, 2019). Elementary teachers need support finding engaging instructional resources suitable for young learners who are still developing language and literacy skills (Kaderavek et al., 2020).

The multimodal nature of video—combining images, narration, and sound effects—supports understanding of science concepts by presenting information through multiple channels, consistent with Mayer’s (2009) Cognitive Theory of Multimedia Learning. Young learners can engage in scientific meaning-making when using multimodal sources (Tang et al., 2014), especially when their reading ability is limited. Videos allow children to observe phenomena directly—such as the movement of the sun, the survival features of animals, and cause-and-effect processes—without needing advanced reading skills to access the content. Videos can be used in science instruction to introduce new concepts in engaging, accessible ways for young learners, or as part of teachers’ formative assessment practices. Video contents can be aligned with NGSS performance expectations (e.g., modeling how plant and animal parts solve human problems) and scaffolded with teacher prompts or formative assessment questions to help teachers surface student thinking and misconceptions while keeping assessments accessible to linguistically and developmentally diverse learners. Text-heavy assessment formats can inadvertently mask students’ conceptual understanding, leading teachers to underestimate their actual grasp of scientific concepts (Billman et al., 2021). Using videos as part of elementary science assessments can be especially helpful for students who are still developing language and literacy, such as English language learners, because the visual and auditory supports reduce barriers that text-heavy assessments often create (Schnotz & Lowe, 2003; Zimmerle et al., 2024). When assessments reveal gaps in student understanding, videos can also be used to reinforce learning and clarify conceptual challenges. Integrating high-quality science videos into assessment and instruction has the potential to enhance student learning, increase engagement, and spark early interest in STEM.

LOVE Science: Using AI to help elementary teachers find appropriate, high-quality science videos.

To harness the benefits of integrating videos into science instruction, teachers need access to high-quality video content. Elementary teachers face a persistent challenge: finding and vetting developmentally appropriate, freely

accessible, and NGSS-aligned science videos can be daunting and time-intensive. We have developed a teacher-facing Artificial Intelligence (AI)-powered YouTube Video Curator and Video Analyzer called Leveraging Online Videos for Elementary Science (LOVE Science) to support elementary teachers in finding multimedia content aligned to science standards. LOVE Science helps teachers find high-quality, NGSS-aligned science videos for instruction and formative assessment. Teachers can enter search terms into the Video Curator to find online science videos. The URLs from the Video Curator can then be entered into the Video Analyzer, which provides a detailed summary of the video that teachers can use to determine the appropriateness of specific video content, evaluations of conceptual complexity, vocabulary level, and a list of prerequisite content (see Figure 1). LOVE Science enables teachers to search for “states of matter, grade 2” or “animal adaptations” and instantly receive vetted, instruction-ready video options, so they do not have to spend valuable preparation time watching several videos to decide what to show students.

In LOVE Science, APIs serve as the connective tissue that brings together Gemini’s large language models (LLMs), YouTube’s vast educational video library, and an intuitive user interface built on Gradio (Figure 1). By integrating these existing technologies through thoughtfully designed API calls, the system can automatically search, filter, and summarize videos using Gemini’s LLMs to analyze video transcripts, visuals, and descriptions to determine alignment with NGSS standards, while YouTube’s API provides access to millions of open educational videos (see Figure 2). Because the LOVE Science platform relies on multiple API connections between Gemini’s LLMs, YouTube, and Gradio, strict and careful prompt engineering is essential to ensure that the system extracts accurate and relevant information for teachers and students. Each API call must be precisely structured to retrieve science-based, age-appropriate, and NGSS-aligned content while filtering out unrelated or low-quality videos. To maintain this level of precision and reliability, we implement verification agents at key points in the data flow; serving as quality checkpoints that review, validate, and refine the AI’s outputs before presenting them to users (Figure 2). The Gradio interface consolidates the process into one simple, web-based environment. Teachers can input a few keywords and receive standards-aligned results without needing technical expertise in AI or programming. This layered design minimizes errors, strengthens trustworthiness, and ensures that every recommended video meets high educational and technical standards. The model achieved a 92.7% match with an expert annotator for a set of 247 videos across the three NGSS content domains (Earth and Space Science, Life Science, Physical Science). Grade-level misclassifications accounted for most of the disagreements, and there were very few content relevance errors. Using APIs to connect existing resources, allows LOVE Science to innovate without reinventing the wheel. LOVE Science can support teachers without requiring them to learn multiple platforms or tools, allowing them to focus their energy where it matters most—helping students engage deeply with science. We are currently working on improving the technical feasibility of LOVE Science and will be piloting with teachers in authentic settings soon.

Session description: Activities and timing

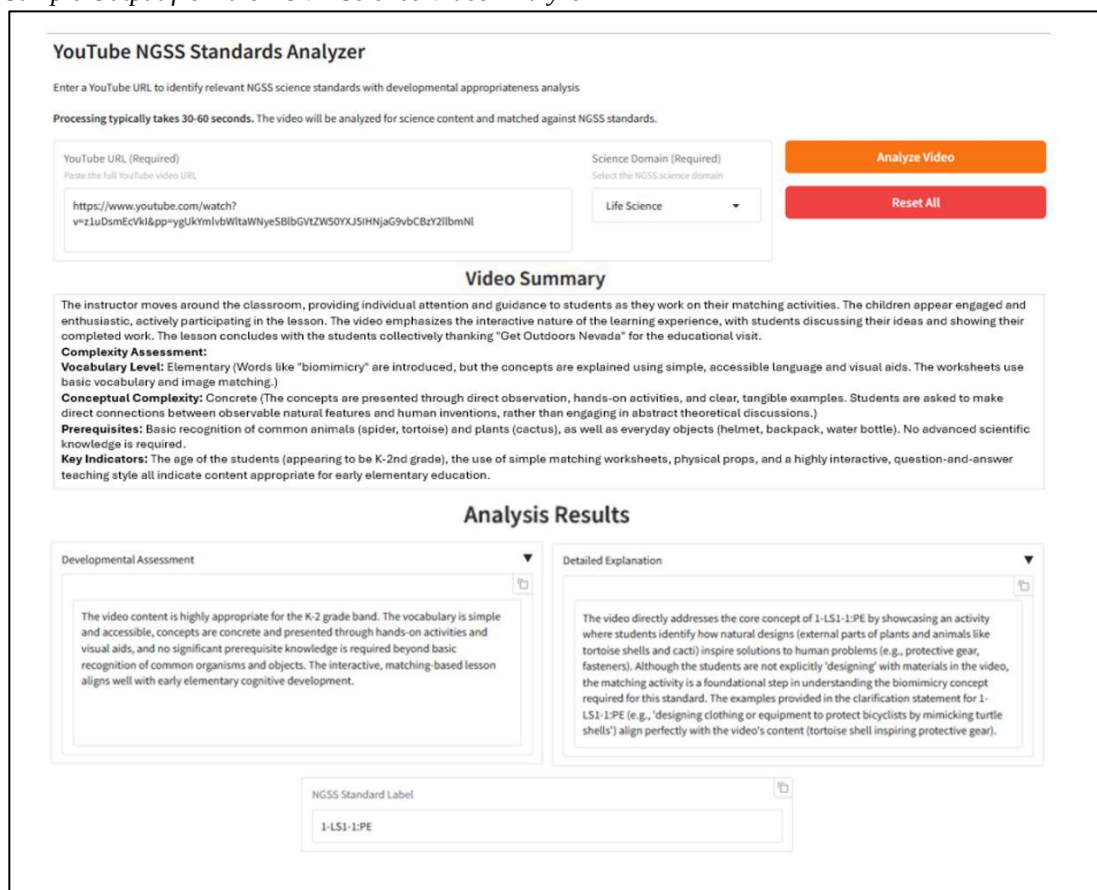
We plan to start the session with a brief presentation and demonstration of the LOVE Science Video Analyzer in the first 10 minutes (in which we will require access to a projector). The presentation will introduce the context for the technology and demonstrate how it can be used to support teachers in vetting YouTube videos. Then we will allow 10-20 minutes for session attendees to interact with the engineering design interface and respond to a few questions raised during our initial presentation. Attendees can interact with the LOVE Science Video Analyzer using their own laptops (either in person or virtually) but will need internet access. We will provide two additional laptops that attendees can use. Attendees will be given example assessment tasks and scenarios that elementary teachers use and experience when integrating videos into instruction or assessment, which will guide the exploration of the Video Analyzer. The goal of this demo is to showcase the LOVE Science Video Analyzer and explore how the tool supports efficient lesson planning, fosters student engagement, and opens opportunities for research on AI-mediated video learning in early science education. The session will invite feedback on how AI-driven tools like this can enhance teacher efficiency, increase student engagement with science concepts, and open new possibilities for adaptive video-based learning environments. The discussion will also explore collaborative opportunities to extend the system’s capabilities – for example, linking it to formative assessment platforms, studying its effects on teacher decision-making, or adapting it for other grade bands and interdisciplinary contexts.

Contributions of this technology

LOVE Science fills a critical need in supporting early science instruction. It can provide teachers with access to vetted resources which can support integration of video content into instruction to support early science learning. The LOVE Science Video Analyzer can analyze video URLs from YouTube, a resource that is commonly used by teachers, which reduces the need for teachers to learn about another new technology. Future iterations of LOVE Science will function as a plug-in for already existing educational platforms and resources. For students, LOVE Science offers engaging, developmentally appropriate science content that reinforces understanding and builds early enthusiasm for science. This work also has the potential to contribute to research on the utility and effectiveness of video-based instruction in science learning. Future work would need to examine how LOVE Science-mediated video selection shapes classroom enactment, student engagement, and science learning outcomes. Participants in this session will have the opportunity to experience how AI can be leveraged to support elementary science teachers in providing accessible, engaging science instruction and assessment aligned to content standards.

Figure 1

Sample Output from the LOVE Science Video Analyzer



YouTube NGSS Standards Analyzer

Enter a YouTube URL to identify relevant NGSS science standards with developmental appropriateness analysis

Processing typically takes 30-60 seconds. The video will be analyzed for science content and matched against NGSS standards.

YouTube URL (Required)
Paste the full YouTube video URL.

Science Domain (Required)
Select the NGSS science domain.

Analyze Video

Reset All

https://www.youtube.com/watch?v=z1uDsMecvki&pp=ygUkYmlybWltaWNyeSBibGVZWS0yXjU5IHJjaG9vbCBzY2libmNI

Life Science

Video Summary

The instructor moves around the classroom, providing individual attention and guidance to students as they work on their matching activities. The children appear engaged and enthusiastic, actively participating in the lesson. The video emphasizes the interactive nature of the learning experience, with students discussing their ideas and showing their completed work. The lesson concludes with the students collectively thanking "Get Outdoors Nevada" for the educational visit.

Complexity Assessment:
Vocabulary Level: Elementary (Words like "biomimicry" are introduced, but the concepts are explained using simple, accessible language and visual aids. The worksheets use basic vocabulary and image matching.)
Conceptual Complexity: Concrete (The concepts are presented through direct observation, hands-on activities, and clear, tangible examples. Students are asked to make direct connections between observable natural features and human inventions, rather than engaging in abstract theoretical discussions.)
Prerequisites: Basic recognition of common animals (spider, tortoise) and plants (cactus), as well as everyday objects (helmet, backpack, water bottle). No advanced scientific knowledge is required.
Key Indicators: The age of the students (appearing to be K-2nd grade), the use of simple matching worksheets, physical props, and a highly interactive, question-and-answer teaching style all indicate content appropriate for early elementary education.

Analysis Results

Developmental Assessment

The video content is highly appropriate for the K-2 grade band. The vocabulary is simple and accessible, concepts are concrete and presented through hands-on activities and visual aids, and no significant prerequisite knowledge is required beyond basic recognition of common organisms and objects. The interactive, matching-based lesson aligns well with early elementary cognitive development.

Detailed Explanation

The video directly addresses the core concept of 1-LS1-1:PE by showcasing an activity where students identify how natural designs (external parts of plants and animals like tortoise shells and cacti) inspire solutions to human problems (e.g., protective gear, fasteners). Although the students are not explicitly 'designing' with materials in the video, the matching activity is a foundational step in understanding the biomimicry concept required for this standard. The examples provided in the clarification statement for 1-LS1-1:PE (e.g., 'designing clothing or equipment to protect bicyclists by mimicking turtle shells') align perfectly with the video's content (tortoise shell inspiring protective gear).

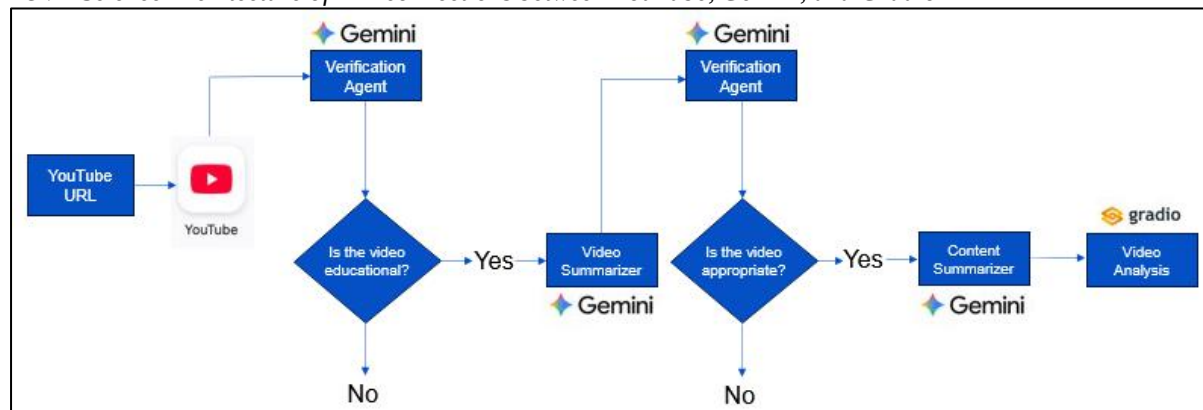
NGSS Standard Label

1-LS1-1:PE

The user enters a YouTube video URL and the Video Analyzer generates a summary of the video including description of the vocabulary, conceptual complexity, and prerequisite knowledge. The Video Analyzer also provides a developmental assessment and provides a label for the corresponding NGSS performance expectation.

Figure 2

LOVE Science Architecture of API connections between YouTube, Gemini, and Gradio



APIs connect Gemini's LLMs, YouTube's video library, and Gradio user interface. Verification agents at key points in the data flow serve as quality checkpoints that review, validate, and refine the AI's outputs before presenting them to users.

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Making Collaborative Creativity Visible: The Viacocrea Dashboard

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Abstract: Addressing complex global challenges requires learners to engage in cocreative processes and regulate their contributions at both individual and group levels. However, supporting and visualizing such multifaceted collaboration is difficult. Building on our previous work and the Viacocrea platform, which guides students through cocreation, we have developed the Viacocrea Dashboard—an interactive tool that visualizes group cocreation dynamics and computes creativity indicators using Natural Language Processing and learning analytics. Key metrics are developed that include fluency (idea quantity), flexibility (topic diversity), elaboration (idea richness), and originality (novelty). The dashboard helps students reflect on and regulate their work, draw inspiration from peers, and enhance cocreation strategies, while providing teachers with actionable insights for feedback and support. In this demo, we will present the Viacocrea dashboard, invite the audience to engage in interactive demonstrations, and discuss how Viacocrea can advance the orchestration of collaborative creativity in education.

Introduction

Creativity in education is a complex and multifaceted phenomenon that emerges through social interaction and situational contexts. It often involves iterative, flexible, and collaborative thinking, particularly when students work together to address open-ended problems (Sawyer, 2021). The creative process is frequently described as encompassing two main stages: *divergent thinking*, where individuals generate a wide range of ideas, and *convergent thinking*, where ideas are refined, evaluated, and selected for implementation (Heinbucher & Bucher, 2022; Sawyer, 2021).

We ground the concept of collaborative creativity in dialogic theory, and it emphasises the emergence of new perspectives through interaction among group members. Within these dialogic spaces, shared dialogue, intersubjective understanding, and reflective discussion foster the development of creative ideas (Wegerif, 2019). Central to this process is the cultivation of constructive tension, where diverse viewpoints encourage students to explore alternatives and refine their contributions.

Sawyer (2022) emphasizes the central role of artifacts in creative collaboration. Artifacts act as tangible representations of evolving ideas, mediating dialogue and enabling students and teachers to track progress, reflect on emerging solutions, and sustain a continuous exchange around creative work. By visualizing the evolution of ideas, artifacts support metacognition, reflection, and iterative improvement.

Recent advances in interactive technologies have expanded opportunities to nurture collaborative creativity in classrooms. Digital tools now enable students to co-construct physical or virtual artifacts that capture group reasoning, dialogue, and actions (Sun et al., 2022; Richardson, 2020). These tools deepen engagement with idea generation, elaboration, and refinement processes, while providing teachers with insights into group dynamics and participation.

The Viacocrea platform offers an interactive, multi-user environment designed to scaffold structured cocreative processes in educational contexts. It organizes creativity into six orchestrated phases—Start Up, Define, Design, Build Up, Sum Up, and Communicate—guiding students through ideation, discussion, elaboration, and refinement while fostering collaboration and creativity. Students typically work in groups of four to five within a synchronous, shared workspace. Each group tackles a specific challenge, progressing through the six phases using targeted techniques that support cocreation—approximately three per phase (Pifarré, 2019). For instance, nursing students might design an educational campaign to promote public health, while others might develop an educational game to tackle a specific learning need. In a typical classroom, multiple groups work on the same challenge, enabling comparison of approaches and promoting a richer cocreative environment.

Building on this foundation, the Viacocrea Dashboard was developed to make cocreative processes visible and actionable. It visualizes key indicators of group creativity—fluency, flexibility, originality, and elaboration—allowing students to reflect on and regulate their contributions, understand group dynamics, and enhance collaborative strategies. Simultaneously, it provides teachers with insights to monitor progress, guide performance, and deliver targeted feedback and support.

Background for Viacocrea dashboard design

The Viacocrea Dashboard builds on the existing Viacocrea platform where students work collaboratively in groups using a structured, multiuser and synchronic space for solving creatively a challenge. Their contributions are systematically captured in a database and exported to spreadsheets, where each row corresponds to an individual student contribution. The dataset is organized with columns that include Group, Phase of Activity, Creative Technique Used, and Idea Text, providing a structured format for subsequent analysis of participation, idea generation, and collaborative dynamics.

The dashboard processes these contributions to provide real-time visualization of each group's cocreative process, while simultaneously allowing groups to observe and compare their contributions with those of other groups working on the same challenge. This feature encourages reflection on one's own strategies, identification of strengths and gaps, and adaptation of collaborative practices. By making group-level processes visible, the dashboard fosters sustainable, evidence-based reflection, supporting iterative improvement and higher-quality cocreation processes and outcomes.

The Viacocrea Dashboard computes creativity indicators based on students' contributions within the platform's shared workspace, using a combination of Natural Language Processing (NLP) and machine learning techniques, the dashboard provide actionable learning analytics for both students and teachers. These indicators capture core dimensions of creativity identified in research as fundamental to collaborative idea generation—namely fluency, flexibility, elaboration, and originality (Guilford, 1950).

Fluency reflects the quantity of ideas generated by the group across different phases of the creative process. By visualizing idea counts per phase and per creative technique used by the group, the dashboard provides insights into students' cognitive productivity and engagement at each stage, supporting reflection and self-regulation.

Flexibility reflects the diversity of topics or conceptual categories addressed by the group. To capture this dimension, topic modelling is applied to identify thematic variation, and ChatGPT is used to generate topic clusters from students' contributions. As a key indicator of creativity, flexibility highlights learners' ability to shift perspectives, explore alternatives, and combine ideas in novel ways (Guilford, 1950).

Elaboration of ideas captures the depth and richness of students' contributions. NLP techniques analyze the richness and coherence of group ideas, considering factors such as average sentence length, sentence structure, and semantic connections between contributions.

Originality (Novelty score) is assessed using the Universal Sentence Encoder, which converts ideas into embeddings. Semantic similarity between embeddings is calculated to measure how distinct each idea is relative to others in the group, allowing identification of the most innovative contributions (Haq, et al., 2024).

Together, these indicators provide an integrated set of analytics that make the cocreative process visible and interpretable for both students and teachers. Students can monitor their progress and adjust their strategies to improve idea generation and collaboration, while teachers obtain insights into group dynamics and cognitive diversity that support targeted feedback and scaffolding. This integration of creativity metrics and learning analytics strengthens the dashboard's capacity to support and guide cocreative work.

Main features of the Viacocrea dashboard

Idea generation: Fluency and elaboration

The quantitative analysis focuses on measuring student contributions in terms of the number of ideas, the total number of words, and the number of questions posed. Specifically, the number of ideas represents the count of ideas contributed by students (fluency), while the number of words captures the total words (elaboration) used by each student or group across different phases. The number of questions reflects student engagement and curiosity within the collaborative activity. To effectively visualize these metrics, pie charts (See Figure 1) are used to depict overall distributions.

Figure 1

Pie Charts for Number of Ideas (fluency), Words (elaboration) and Questions Distribution.



The comparative analysis examines the distribution of ideas, words, and questions across different student groups, phases, and techniques (see Figure 2). This analysis enables the identification of group progress over time, highlighting how student contributions evolve throughout the collaborative activity. It also reveals the techniques most employed by students and the phases in which they are most active, providing insights into engagement patterns and collaborative dynamics. To effectively convey these trends, multi-bar charts are used, allowing for a clear visualization of variations across groups, phases, and techniques, and facilitating an intuitive understanding of student performance and participation in different phases and techniques.

Figure 2

Visualization of Cocreative Indicators Across Groups



Topic modelling: Flexibility

Topic modelling (flexibility) is performed using the ChatGPT API to extract and cluster the main topics present in students' responses. This analysis allows for the examination of how ideas are distributed across different themes, providing insight into the flexibility and diversity of each group's contributions. By identifying the breadth of topics addressed, students and educators can reflect on cognitive diversity and how ideas are explored in different areas of discussion within collaborative activities. We visualize the group topics using a collapsible tree structure, where each group is displayed at the top level, followed by its main themes, with individual topics nested under each theme. This hierarchical representation allows for an intuitive exploration of how ideas are organized, showing both the breadth and depth of each group's contributions.

Semantic novelty score: Originality

The novelty score quantifies the originality of ideas generated by student groups and provides a multi-level perspective on creative contributions. It is calculated using semantic similarity analysis, each student contribution is transformed into a high-dimensional embedding using the Universal Sentence Encoder (USE) (Authors, 2023, 2024), and the similarity between contributions is measured using cosine similarity. Contributions that are semantically distant from others receive higher novelty scores, reflecting their uniqueness within the group context. Scores are calculated at three hierarchical levels:

1. *Overall group novelty*: the total novelty contributed by each group across the entire project.
2. *Phase-wise novelty*: the novelty contributed by each group within different project phases.
3. *Technique-wise novelty*: the novelty contributed by each group based on the techniques they employed.

To visualize these dimensions, stacked grouped bar charts are employed. The X-axis represents the different groups, while the Y-axis depicts novelty scores. Similarly to Figure 2, for each group, bars are grouped by phase or technique, and stacks within the bars represent contributions from different phases or techniques.

This visualization strategy allows a comprehensive understanding of group creativity, enabling easy comparison between groups, identification of the phases and techniques that contributed most to originality, and assessment of the diversity and uniqueness of contributions across the project.

Implementation and evaluation of the dashboard

The Viacocrea Dashboard has been implemented in real-classroom settings, students work collaboratively in multi-user, synchronous spaces, generating textual contributions that are automatically processed and visualized by the dashboard in real time. Both students and teachers can access group-level and cross-group metrics, supporting reflection, self-regulation, and monitoring of cocreative processes.

To evaluate its usability and pedagogical effectiveness, we adopted a mixed-methods approach, combining qualitative and quantitative data. Qualitative data are gathered through classroom observations, screen recordings, and transcriptions of teacher-student interactions to capture how participants engaged with the dashboard. Quantitative data included click logs, heatmaps, and task-based performance measures to analyze navigation patterns, areas of engagement, and efficiency in completing predefined tasks. Additionally, students and teachers completed standardized evaluation questionnaires, including the Technology Acceptance Model (TAM) and the System Usability Scale (SUS), to assess perceived usefulness, ease of use, and overall user satisfaction.

By triangulating behavioral, performance, and perceptual data, we obtained a comprehensive picture of how both teachers and students interact with and benefit from the Viacocrea Dashboard. The findings from these pilot implementations provide evidence of its potential to enhance collaborative creativity, support self-regulation, and offer actionable insights for teachers to guide cocreative learning processes more effectively.

Conclusions

The Viacocrea dashboard is intended to serve a dual purpose: supporting student self-regulation and reflection, as well as providing teachers with insights to guide and scaffold cocreative activities. By making cocreative processes visible, the dashboard aims to enhance collaborative knowledge building and foster higher-quality cocreation outcomes.

In our conference session, we will present the development of the Viacocrea dashboard, demonstrate how it functions in practice, and share insights from its evaluation by both teachers and students. Participants will also engage directly with the Viacocrea environment, allowing them to experience how the tool supports cocreation through hands-on interaction. Through collaborative exploration, attendees will assess its affordances and limitations and consider its potential to serve as a transformative resource for fostering cocreative practices in authentic classroom contexts.

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LexiLearn: Sensory AI-enhanced VR Integration for Literacy Skills Acquisition

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Abstract: *LexiLearn* is an artificial intelligence (AI) enhanced virtual reality (VR) learning experience for literacy education. The system introduces a novel integration of AI with adaptable haptic interfaces, supporting both wearable gloves and handheld VR controllers. The project investigates how multisensory engagement within an adaptive, generative VR environment can support literacy acquisition, including spelling, reading through interactive and playful movement, phoneme–grapheme recognition and recall, and visual–sensory associations. Led by a cross-disciplinary research team, this project explores innovative alternatives to immersive learning environments. It examines how multisensory and AI-driven technologies can be meaningfully implemented for diverse learners, aiming to inform and transform the educational practice through practice-based design and research methods.

Introduction

Supporting students in the early stages of literacy acquisition remains an urgent challenge in global education. Literacy is a fundamental and critical prerequisite for equitable participation in society; ensuring access to effective literacy education is therefore essential for advancing human rights and enabling transformative life opportunities (UNESCO, 2019). However, current educational environments and digital interventions predominantly still rely heavily on visual and auditory modalities (Oakley et al., 2024). Yet, studies indicate that multisensory and embodied learning can enhance cognitive processing, memory, and engagement (Damsgaard et al., 2022; Fredens, 2018). Haptic feedback has been shown to convey information, support task performance, complement or substitute other sensory inputs, and contribute to a greater sense of immersion and realism in virtual environments (Culbertson et al., 2018). The integration of haptic devices has demonstrated positive effects on learning outcomes within STEM education (Orta Martinez et al., 2020). *LexiLearn* builds on these principles by investigating how a literacy-focused VR learning experience equipped with an adaptive and generative AI system, combined with wearable haptic gloves, potentially can support foundational skills such as letter recognition, spelling and phonics. The project is embedded within research practice partnerships involving schools, teachers, and students and is facilitated by a cross-disciplinary team of researchers from diverse fields, spanning design, psychology, education, and engineering. The remainder of this proposal outlines the project, research purpose, and future directions. With ethical approval secured from the Institutional Review Board (IRB), the study is now entering its co-design and participant user-testing phase. Initial insights and reflections on the design process will be presented at the ISLS 2026 conference.

Immersive virtual learning environments and play-based learning

Several studies and systematic literature reviews found that immersive virtual experiences enhanced learners' sense of presence, engagement, and conceptual understanding compared to traditional computer-based settings, with presence and self-efficacy mediating cognitive and affective learning outcomes (Chen, 2016; Makransky et al., 2020; Queiroz et al., 2023). These findings provide a conceptual basis for *LexiLearn*, investigating whether multisensory and adaptive features in immersive virtual experiences can further enhance literacy skills acquisition. Further studies found positive effects using VR experiences and haptics combined in procedural skill-related tasks that require the memorisation of steps or procedures, ranging from dental anaesthesia to piano lessons (Collaço et al., 2021; Pala & Mihci Turker, 2019). Given that spelling relies on procedural knowledge (Steffler, 2001), specifically the memorisation of phoneme–grapheme correspondences and letter sequences, the results from previous studies indicate promising potential for integrating multisensory and adaptive VR experiences into literacy learning domains.

The project employs elements of play such as narrative framing, embodied interaction, multisensory feedback, and reward cues to sustain motivation without fragmenting attention towards competitive metrics. The act of “harvesting words” in the orchard bridges playful exploration with literacy tasks, transforming the act of reading and spelling into a game-like VR experience. Furthermore, Fredens (2018) argues in his work on embodied learning that cognition emerges from the sensory–motor interplay between the brain, body, and environment. Emphasising that the hands stimulate the brain’s frontal lobes, which are crucial for mental development and

bodily experiences, form the conceptual foundations upon which linguistic development is built (Fredens, 2018). Drawing on these principles, Damsgaard et al. (2022) conducted a randomised controlled trial (the PLAYMORE Project) with children, demonstrating that whole-body movement paired with letter sound learning significantly improved children's letter sound knowledge (Damsgaard, 2022). These perspectives substantiate the exploration of immersive and embodied approaches within literacy learning. The shared experience of play also opens future possibilities for social learning, where future iterations of the project could allow collaborative participation in multiplayer settings, fostering co-learning and collective meaning-making based on principles of computer-supported collaborative learning (CSCL).

Literacy learning framework

The literacy learning framework proposed through *LexiLearn* is developed and designed on four core pillars: lexical development, creative visual association, phonetics and embodied learning and collaborative learning ecology, as described below:

Lexical development: Drawing upon cognitive science, we follow the literature that posits that the lexicon constitutes the primary locus for grammatical generalization across multiple representational tiers. Early, frequency-sensitive phonological pattern acquisition has shown to provide the essential mechanism for bootstrapping general grammatical competence in development (Beckman & Edwards, 2000). As learners engage with the words, visuals, phonemes and letters in *LexiLearn*, they practice decoding symbolic representations into sounds and meanings. The VR experience encourages the recognition of whole-word forms and repetition, supporting fluency and the transition from decoding to automaticity.

Creative Visual Association: The association between the written word and the images is based on the work in cognitive psychology and memory, specifically by demonstrating that linking a verbal stimulus (the word) with a non-verbal stimulus (the image) creates a more robust, multi-path memory trace (Paivio, 1990). *LexiLearn* offers visual cues and imagery to support recognition, memory, and comprehension by pairing each word with a corresponding AI-generated image of what the word represents. With this association, we expect the VR experience to support the development of the crucial orthographic-phonological-semantic mapping required for fluent reading (Warrington & Hulme, 2012).

Phonetics and embodied learning: Grounded in Embodied Cognition theory, body movement engagement has shown to facilitate encoding of lexical items by establishing non-abstract, action-based memory traces (Glenberg & Kaschak, 2002). This mechanism seems to be amplified by multisensory integration, wherein the temporal synchronisation of haptic, auditory, and visual input ensures the statistically optimal and highly reliable formation of complex perceptual-motor associations (Ernst & Banks, 2002). Building on this literature, *LexiLearn* associates visual, verbal and non-verbal stimuli, body movement, and the synchronisation between the vibrotactile feedback from the haptic gloves with phonetic audio on spelling skills.

Collaborative learning ecology: The efficacy of digital learning has been shown to be dependent on placing the educator in the centre of the experience's development, implementation and evaluation (Selwyn, 2021). The literature shows that data is only actionable when teachers can contextualise it to inform their professional judgment (Campen & Molenaar, 2020; Mandinach et al., 2006). Furthermore, there is a need for continuity between the digital and physical experiences, as effective learning requires a coherent curriculum that bridges and aligns technological activities with classroom goals, ensuring that embodied, hands-on learning is integrated rather than disrupted by digital tools (Mhlongo et al., 2023). In this context, we developed a dashboard that translates learners' activity in the VR experience into interpretable data, enabling educators to assess progress, identify individual strengths and challenges, and suggest learning tasks in the classroom accordingly. This aims to foster a reciprocal relationship between student and teacher, grounded in pedagogical reflection by evidence-informed insight.

Together, these pillars create a coherent, multisensory learning framework for exploring literacy acquisition through connecting the VR experience with activities in the classroom throughout the curriculum. The novelty of this research lies in its synthesis of multisensory, embodied, adaptive, collaborative and comprehensive dimensions, seeking to contribute to how multisensory digital experiences may potentially support learning pathways within literacy acquisition based on a model that is responsive to learners' cognitive, emotional, and sensory needs.

Design and development methods

The virtual reality experience and environment

LexiLearn is designed as an educational system composed of an adaptive and generative AI-enhanced VR learning experience coupled with sensory feedback facilitated as an educational game. The experience takes place in a calm, nature-inspired virtual orchard, designed to foster presence, curiosity, and sustained attention by providing

a familiar, low-distraction context. Learners navigate freely through the space, selecting from a variety of fruit trees. When a tree is chosen, a target word appears alongside a generated image of the displayed word, along with fruits listed in front of the player containing letters as visual graphemes with associated phonetic delivery by audio cues linked to the interaction of picking up the “fruit-letters”. The learner engages in this embodied and interactive literacy task, grasping, aiming, and throwing the fruit-letters at a basket placed in front of the tree in the correct sequence to complete the word, where they have more letters to choose from than needed to complete the word (see Figure 1 for visuals of gameplay and link to video demo).

Visual and auditory feedback indicate if the letter selected is correct or not, and with a correctly completed word, a new word will appear with a new set of matching “fruit-letters”, fostering a continuous flow. We deliberately avoided scores and visible levels. This approach aims to cultivate a continuous, process-oriented flow, promoting learning through exploration instead of performance pressure. Aligning with the idea of the value of play lies not in goal attainment but in the process of exploration itself, which can support motivation, creativity, and engagement (Skovbjerg & Bekker, 2018). Before entering the orchard, learners encounter a customisation menu that allows them to tailor aspects of their experience. They can select thematic word categories (for example, animals, vehicles, emotions, etc.) to align with their interests, as well as choose background music, fostering resonant, personal, meaningful atmospheres. These were developed to foster agency, ownership, and intrinsic motivation, aligning with established research that more autonomous forms of motivation are associated with greater engagement, improved performance, and higher-quality learning (Ryan & Deci, 2000).

Figure 1

LexiLearn Visuals: Left, In-Experience View; Right, External View



Link to video demo with sound: <https://youtu.be/VluDrSX1wrA>

Adaptive and generative AI-driven scaffolding

The system generates age-appropriate spelling words at calibrated difficulty levels (1–10), while DALL-E 3 produces corresponding visual imagery designed to reinforce meaning and word–image association of the displayed word (Li & Lan, 2022). Complementing these, a phoneme-level audio pipeline generates individual letter-sound pronunciations to support multi-sensory phonological learning. The system dynamically adapts by analysing each learner’s accuracy and specific misspelling patterns at the character level. The first five words serve as a baseline assessment, prioritising age-appropriateness; thereafter, the system shifts to performance-driven calibration, increasing difficulty upon mastery ($\geq 90\%$ accuracy) and maintaining level when errors persist. By feeding prior misspellings directly into the generation prompt, the AI tailors subsequent word selection to target observed weaknesses — functioning as responsive scaffolding that mirrors a teacher’s diagnostic adjustment (Harris et al., 2017). This adaptive mechanism aims to balance challenge and competence while accommodating individual differences in developing literacy. Through these parameters, the AI system functions as an intelligent mediator of literacy learning, guiding the sequencing of content and modulating task difficulty to align with each learner’s evolving proficiency.

Sensory integration

The haptic feedback is delivered through gloves equipped with linear resonant actuators (LRAs) that provide localised vibrotactile stimulation to the hand and fingertips. Hand and finger movements are captured via inertial measurement unit (IMU) sensors, in conjunction with optical hand tracking from the VR headset (Meta Quest 3). This multimodal integration facilitates context-responsive vibrotactile feedback aligned with literacy-oriented tasks within the VR experience. Rather than replicating physical tactile sensations in a one-to-one correspondence with the phonetic audio, the haptic system is designed to utilise more systematic and differential haptic cues. We found, during the design process, that audio-based haptic pairing with phonetic sounds was too subtle for the users to justifiably notice. Through a systematic integration of vibrotactile feedback, this project seeks to support

grapheme–phoneme associations using clearly differentiable and intuitive sensory stimuli, while investigating the impact of haptic feedback on learning outcomes.

Design and curriculum integration

LexiLearn aims not only to create a single standing VR learning experience but also to explore a holistic, comprehensive learning experience that connects the embodied VR learning experience with in-class curriculum activities supporting literacy skills. In this context, we developed a dashboard combined with the *LexiLearn* VR experience that can be accessed through computers, tablets or mobile phones. The dashboard provides educators with access to data recorded from each student interaction within the VR experience. They can see how long and how often the learners played, the letters practised and the learners' accuracy and mistakes. The AI provides insights into the mistakes' patterns (for example, if the learner frequently selected “b” instead of “d”) and suggests personalised student activities that the teacher can implement in the classroom to support the specific skill level of each learner. This way, the VR learning experience is not isolated from the curriculum but integrated with classroom activities. The dashboard supports a continuum of personalised learning throughout the year and functions as a bridge between learner experience and teaching practice. Insights from immersive sessions are translated into actionable pedagogical reflection, aligning with broader efforts to support evidence-informed teaching. In this context, we aim for the AI system to be positioned as a complementary pedagogical tool rather than a substitute for classroom teaching, extending formative assessment practices and supporting personalisation within diverse and mixed-ability classrooms.

Conclusion and future directions

LexiLearn is a comprehensive system that integrates a novel, adaptive AI-enhanced VR learning experience with vibrotactile haptic feedback and a pedagogical data dashboard. The synthesis of adaptive AI-driven content sequencing, synchronised grapheme-phoneme-tactile cueing, and data translation mechanism provides a robust framework for investigating orthographic-phonological-semantic mapping and comprehensively embedding the digital learning experience in the classroom. Future research must prioritise the empirical validation of this integrated system, investigating the differential effects of the multisensory condition on measures of spelling proficiency, phonological awareness, and lexical development. Specifically, longitudinal analyses should quantify the relationship between the dashboard's data analytics—including machine-identified mistake patterns—and the subsequent impact of educator-implemented, evidence-informed instructional modifications on the sustained continuity and alignment across the digital VR learning experience and the curriculum.

Interactive demo session

The researchers will provide attendees with a live, hands-on demonstration of the *LexiLearn* learning system, including the teacher's dashboard. Attendees will be able to experience the VR learning experience directly, exploring how vibrotactile feedback provided by the haptic gloves is coupled with visual and auditory cues to support embodied interaction and literacy learning. Additionally, they will be invited to contribute their feedback on the experience and the dashboard to support further potential applications and broader reach across the educational practice. The aim for the demo session is to engage attendees in trialling the novel set-up and spark curiosity while supporting ISLS's knowledge-sharing space within educational research. The session will also allow the research team to share formative findings from early user-testing with teachers and students. Ultimately, this demo aims to foster dialogue around the integration of new immersive technologies in education, encouraging collaboration, reflection, and new directions for future research and educational pathways.

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NLP4Science: A Tool for Middle School Learners to Explore Fundamentals of Natural Language Processing

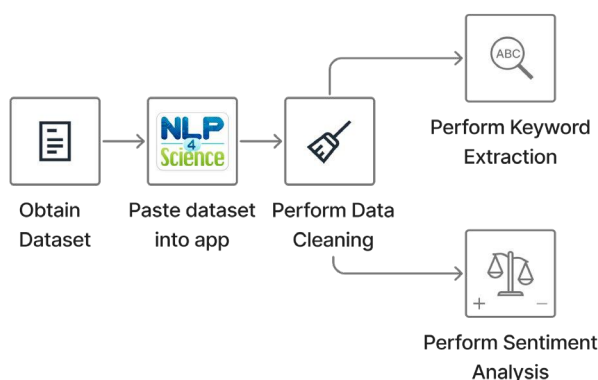
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Abstract: NLP4Science is an interactive tool designed to support AI education by engaging learners with Natural Language Processing (NLP), a subfield of AI focused on analyzing human language. NLP4Science introduces users to three core NLP techniques (data cleaning, keyword extraction, and sentiment analysis). This tool was designed alongside middle school classroom teachers and has applications in personal inquiry, classroom learning, and research. In this demo session, we will showcase use cases of NLP4Science and explore its potential in K-12. The audience will be given the opportunity to engage in hands-on activities using the tool and provide feedback on its educational value and future development opportunities. This session aims to foster discussion around integrating NLP into learning environments and to gather insights for refining the tool and expanding its applications. We invite educators, researchers, and developers interested in AI literacy and NLP to join us in this demonstration.

Introduction

Natural Language Processing (NLP) is an essential component of today's AI. By exploring the fundamentals of NLP, middle school learners can investigate how computers make inferences from human language (Norouzi et al., 2020). NLP4Science is an educational software tool designed to support AI education by engaging middle school learners with fundamentals of NLP. Developed by a multi-institutional team from the University of Florida, Indiana University, and North Carolina State University (Dhama et al., 2023), and alongside numerous middle school teachers from the three different states, NLP4Science is intended to help young learners explore how AI can process and interpret textual human language through personal sensemaking and analysis of consistent, objective results through a scrutable process. The tool focuses on three key NLP techniques (see Figure 1): data cleaning (removing punctuation, numbers, stop words, and standardizing text), keyword extraction (identifying frequently used words and visualizing them through histograms and word clouds), and sentiment analysis (classifying text as positive, negative, or neutral). This tool is applicable for personal inquiry, classroom learning, and research contexts.

Figure 1
NLP4Science Use Case Flow Chart



NLP4Science has been piloted in middle school science and linguistics classrooms in Florida and Indiana (Katuka et al. 2024; Chakraborty et al., 2025) and was built using Python, Django, and ReactJS. It uses the *TweetNLP* (Camacho-Collados et al., 2022) and *VADER* (Hutto & Gilbert, 2014) models for sentiment analysis,

and runs on any modern web browser. In this demo session, we will showcase practical use cases of NLP4Science and explore its potential for K-12 instruction and beyond. Participants will engage in guided, hands-on activities using the tool and share feedback on its usability, learning value, and future development. Participants can access the tool via their own devices or our provided laptops. Sample datasets will be available, although users may also input or paste their own text to explore the tool. This session is designed to spark discussion around how NLP tools can be meaningfully integrated into AI literacy education and to gather insights that will inform the next version of NLP4Science.

Background

Efforts to integrate Artificial Intelligence (AI) Literacy and Natural Language Processing (NLP) into K-12 classrooms have grown significantly in recent years, driven by the need to build foundational AI literacy among young learners (Almatrafi et al., 2024; Lee et al., 2021). In response, researchers and educators have developed a range of tools, which supplement curricula, and frameworks aimed at making AI concepts more accessible and engaging for students. These tools have shown promise in motivating and educating youth about AI. Our work builds on this momentum by addressing the gap of needing a custom NLP tool that aligns with the ethical, legal, and developmental considerations of working with minors in educational research. Even in the context of Generative AI, Foundational AI techniques like NLP (Schmidt et al., 2025) lay the groundwork for more sophisticated AI understandings in the future. NLP4Science is FERPA-compliant, IRB-approved, and designed to ensure student privacy. No data is collected or stored, and login credentials are generated and de-identified by administrators to protect student identities. These safeguards allow researchers to study AI learning in authentic classroom settings while upholding the standard of ethical research for minors.

Main features of NLP4Science

NLP4Science is designed with features that aim to improve AI literacy through Natural Language Processing (NLP) education. The app allows users to perform data cleaning, keyword extraction, and sentiment analysis (Dhama et al., 2023).

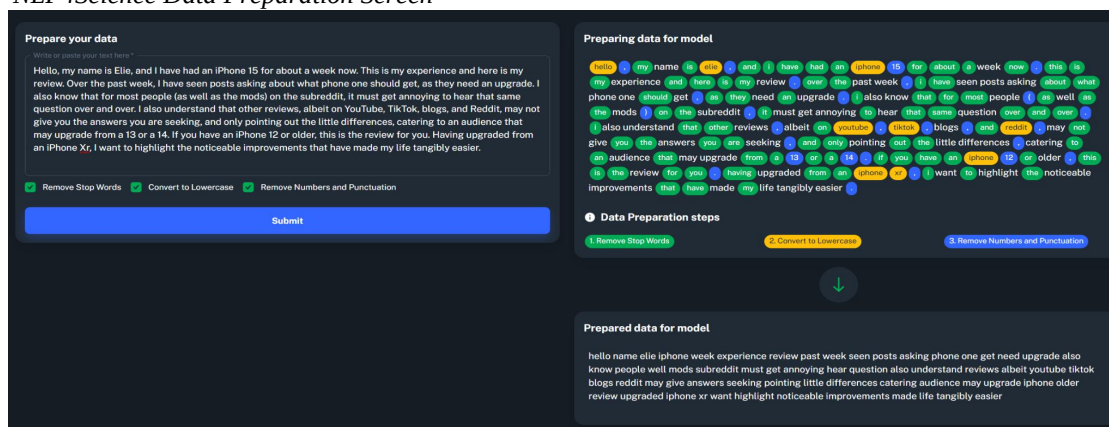
Data preparation

This feature allows users to clean and standardize input text to ensure it is suitable for processing by the AI model. Although this module can be used independently, data preparation is automatically applied when using other features. The process includes (see Figure 2):

- **Stopword Removal:** Eliminates common words with minimal semantic value (e.g., the, is, and). The tool includes a list of predefined stopwords which users have the ability to modify.
- **Case Conversion:** All text is converted to lowercase to avoid case sensitivity of the provided text, allowing the model to treat words like *Science* and *science* as equivalent.
- **Number and Punctuation Removal:** Punctuation and numerical values are removed to allow the model to focus solely on meaningful words. Punctuation is excluded because it primarily aids human readability exclusively, and provides little semantic value to the model. Numbers are also removed, as they typically do not contribute to the model's understanding of the text.

Figure 2

NLP4Science Data Preparation Screen



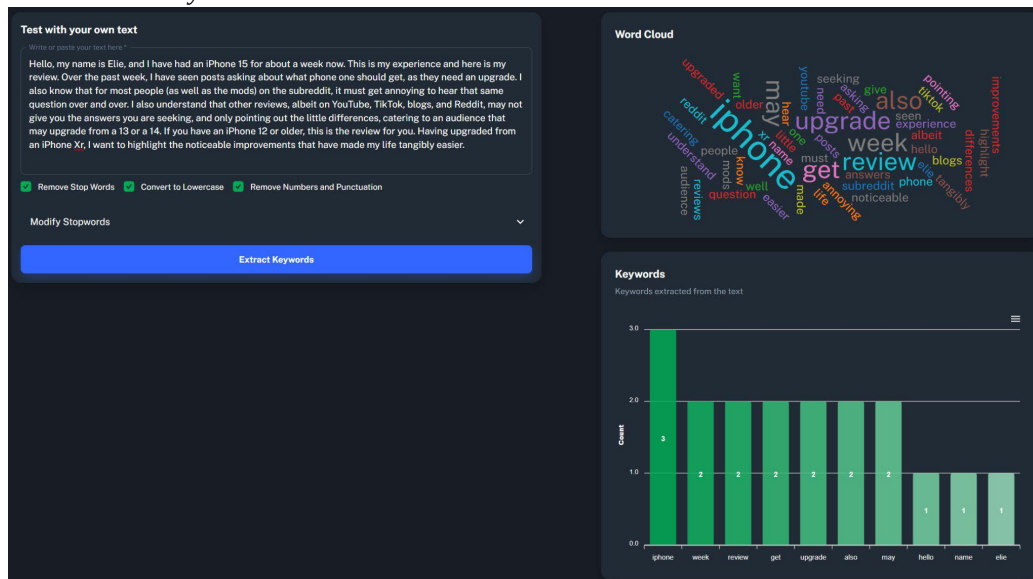
Keyword extraction

This module helps the user identify the most significant words in a text based on frequency, making it useful for summarizing content and identifying key themes. It extracts unique words and categorizes them by frequency. Two visualizations are provided (Figure 3):

- Histogram: Displays the most frequently used words, with bars indicating the number of occurrences of each word.
- Word Cloud: Represents word frequency visually where more frequent words appear larger and less frequent words appear smaller.

Figure 3

NLP4Science Keyword Extraction Screen

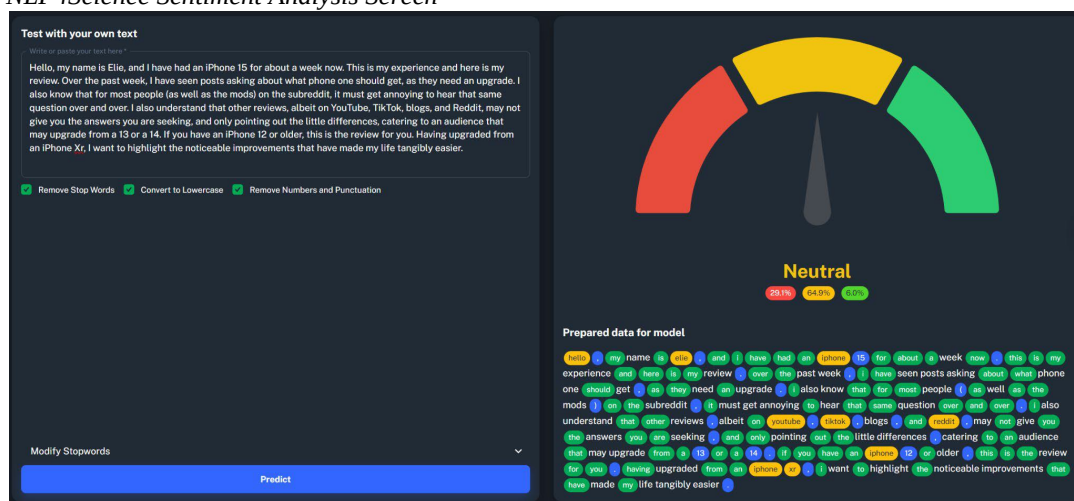


Sentiment analysis

This feature uses an AI model to analyze the emotional tone of a text, classifying it as positive, negative, or neutral. Words associated with positive emotions (e.g., good, joy, awesome) shift the sentiment toward positive, while negative words (e.g., bad, sad, no) influence the sentiment toward negative. A dataset with no sentiment or with an equal mix of positive and negative sentiment shifts it toward neutral (Figure 4).

Figure 4

NLP4Science Sentiment Analysis Screen



We make use of the *TweetNLP* and *VADER* models to calculate the sentiment of a given body of text. These models rely on sentiment lexicons that assign each word a categorical sentiment label and an intensity score, producing an aggregated sentiment classification and overall intensity score.

- Colour-coded Sentiment Labels: Green for positive, red for negative, and yellow for neutral.
- Gauge Chart: Visualizes the overall sentiment with a scale from left (negative) to middle (neutral) to positive (right).
- Confidence Score: Each sentiment category is assigned a percentage score, indicating the proportion of the text that aligns with that sentiment.

Broader applications

NLP4Science supports a range of applications across classroom instruction and educational research. In educational settings, students use the tool to analyze text through data cleaning, keyword extraction, and sentiment analysis. For example, learners can explore the main themes of a scientific article using word clouds and histograms, helping them develop their analytical reading skills and verify their understanding of complex and lengthy texts. Sentiment analysis enables students to interpret tone and emotional context, such as identifying an author's intent or gauging public opinion on topics like climate change. While initially designed for science classrooms, NLP4Science has also been successfully adopted in English Language arts settings, where teachers use its modules (especially data cleaning) to support instruction on English Language syntax and semantics; we expect to publish the findings in the future. The tool allows users to explore frequently used words or overall sentiment in large bodies of text for their personal understanding. NLP4Science not only allows students to explore and interpret real-world text across disciplines, but also provides researchers with a unique lens into how young learners engage with and critique AI-driven tools in both individual and classroom learning contexts. Foundational AI methods like NLP play a distinctive role by offering transparent, scrutable models that produce reproducible results, in contrast to “unpredictable” black boxed AI methods like large language models. NLP4Science enables users to examine the evidence directly, draw their own conclusions, and engage in the scientific inquiry process supported by problem-based learning. It gives users the opportunity to engage in analysis and personal-sense making by counting words, interpreting data visualizations, making comparisons and drawing conclusions based on consistent outputs, a valuable approach at the early childhood developmental stage.

Structure of the session

In this interactive tool and demo session, participants will be provided with an NLP4Science overview outlining the app's modules and features. They will engage in hands-on exploration of the tool, using all three core functionalities: data cleaning, keyword extraction, and sentiment analysis to gain first-hand experience. Following the demo, we will facilitate one on one discussions on how AI tools like NLP4Science can be integrated into teaching and research to support AI literacy, as well as other possible use cases. Participants may use their own devices or provided laptops, and sample datasets will be available, with the option to input custom text. The session is designed to spark dialogue around classroom integration, research use, and future development efforts. This session offers a practical opportunity to explore NLP4Science's classroom-tested design and gather feedback on its role in advancing AI literacy through inquiry-based and multi-disciplinary learning for K-12 education.

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Integrating a Conversational Agent into PBS KIDS Media for Interactive STEM Learning at Home

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Abstract: Interactive media can be effectively enhanced through advancements in artificial intelligence (AI). Through a research–practice partnership with PBS KIDS, we integrated AI into popular PBS KIDS shows to support dialogic interactions that actively engage children in STEM learning. These interactions help ground abstract STEM concepts in socially meaningful experiences. In this work, we present an interactive web app designed to scaffold children’s learning and engagement at home. We discuss design considerations, challenges related to children’s speech recognition, and insights from iterative testing that informed the app’s current stage of development.

Background

Extensive research has shown that interactivity is a major driver of children’s learning and engagement (Ewin et al., 2020; Richert et al., 2011). For example, educational video programs targeted toward young children, such as *Blue’s Clues* and *Dora the Explorer*, primarily use pseudo-interactive conversation, where the main character asks the child a question but cannot respond directly. An alternative approach for providing dialogic interactions is the “Wizard of Oz” approach where researchers themselves will adjust responses to children through a “fake” media character (Calvert et al., 2020). Advances in AI now make it possible for tools to support live, dialogic interaction between children and educational content. Early studies suggest that such tools can lead to significant learning gains (Xu et al., 2024), but much of this research has been conducted in highly controlled lab settings, with few being conducted in schools or homes (He et al., 2025). This limits our understanding of how to design or implement interactive, dialogic tools in real-world contexts. Since a majority of children’s interactions with media take place at home, there is a critical need to investigate how AI-integrated tools can support children’s learning and to determine how these tools can be designed to foster learning in naturalistic home settings.

As part of an NSF-funded project in partnership with PBS KIDS, we developed a web app that allows children to engage with interactive educational media in home settings. Although the primary goal of the project is to evaluate how children learn from interactive media in comparison to non-interactive media, the design choices for this web app and children’s engagement with them could potentially address design challenges in early childhood. One issue that can arise is with automatic speech recognition (ASR) systems embedded into the web app. Research has found that ASR systems lag in accuracy for children in comparison to adults, due to their divergent speech patterns and higher acoustic variability (Bhardwaj et al., 2022). Past studies found that these systems are satisfactory in understanding children’s speech in lab settings (Xu et al., 2024), where the environment is quiet, and the child is typically the only person speaking. However, this performance is more challenging to maintain in home environments, where varying acoustic conditions and background noise may complicate speech recognition. One study examining the performance of ASR systems in classrooms reported a high overall error rate of 84%, with most errors resulting from words being deleted by the system (Southwell et al., 2022). Additionally, the learning content from PBS KIDS is intended for children between the ages of 4 and 8, and designing technology for this age group can be difficult. Children this young are still developing foundational language, cognitive, and self-regulatory skills (Hirsh-Pasek et al., 2015). They may have limited reading proficiency, shorter attention spans, and difficulty following multi-step instructions, which makes it crucial that the design is developmentally appropriate, as well as beneficial for their STEM learning outcomes. Thus, it is important to consider how to design these conversational systems and apps to adapt to the constraints of children’s everyday settings.

Theoretical framework

Learning through conversation is rooted in Vygotsky's sociocultural theory (1978), which emphasizes that children learn through interaction with more knowledgeable others. Within this framework, we draw on three core Vygotskian principles: (1) learning as a socially mediated process, (2) language as a primary tool for thinking and meaning-making, and (3) the role of guidance in advancing children's conceptual understanding beyond their current level. Dialogic interaction supports children in constructing meaning and internalizing concepts. In our design, AI enables the media character to take on the role of a "more knowledgeable other" by engaging the child in guided back-and-forth dialogue. The conversational agent (CA) prompts reasoning, encourages elaboration, and provides gentle corrective feedback, using language as the primary medium through which children engage with and refine their understanding. These interactions are embedded in PBS KIDS media that already hold social and emotional meaning for children, helping them ground abstract STEM concepts in their lived realities. Along with integration into a home watching web app, we build on these principles to supplement the social learning taking place naturally in children's home environments.

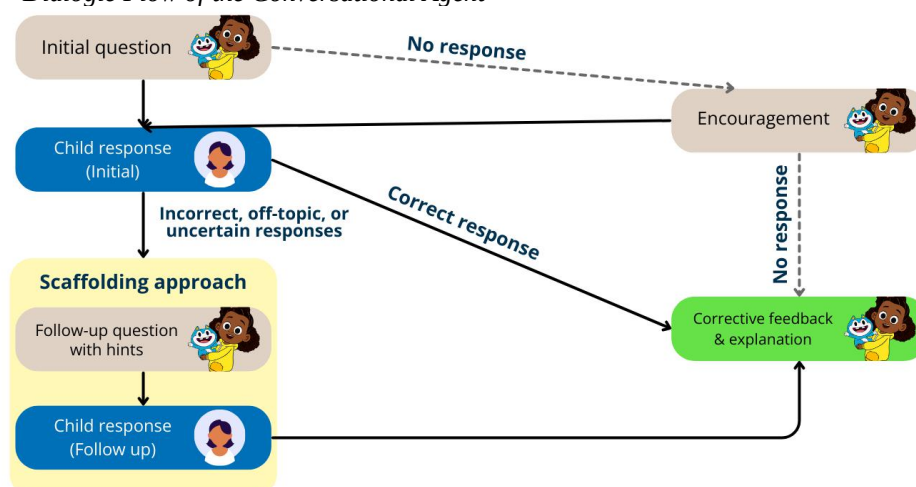
We adopt a pragmatic approach that draws on dialogic interaction as a mechanism for engaging children, while also emphasizing the role of guidance in shaping and extending conceptual understanding. The CA is therefore designed not only to elicit children's ideas, but to direct attention toward key STEM concepts through targeted prompts and feedback. We position Vygotsky's theory as a guiding lens for interaction design rather than a direct operationalization of specific constructs, and note that this reflects an applied interpretation of his work within educational research.

Conversational agent design

Interactive versions of two PBS KIDS shows, *Elinor Wonders Why* and *Lyla in the Loop*, were developed for the web app. Both shows are science-focused and designed for preschool and early elementary children aged 4-8. Each educational episode lasts approximately 11 minutes and covers topics such as molting, aerodynamics, or step-by-step problem solving. For example, one episode of *Lyla in the Loop* teaches sequencing and following steps as Lyla bakes cupcakes for her sisters' birthdays. Children learn to understand and anticipate the correct order of actions, which aligns with core computational thinking skills, including algorithmic thinking and task decomposition.

Using ASR systems, the main character (Elinor or Lyla, depending on the show) can recognize and categorize children's verbal responses and provide feedback tailored to their input. Each episode includes 10 to 11 conversational moments, in which the character asks questions to gauge understanding or prompt reflection on the concepts introduced. Figure 1 illustrates the dialogic flow of these interactions where: the character introduces a question, waits for the child's response (or non-response), offers scaffolding or encouragement as needed, and ultimately delivers appropriate feedback and the correct explanation. All the questions and responses provided by the character were developed based on the Next Generation Science Standards (National Research Council, 2013), along with feedback from PBS KIDS' science learning advisory board.

Figure 1
Dialogic Flow of the Conversational Agent



Interactive media web app design

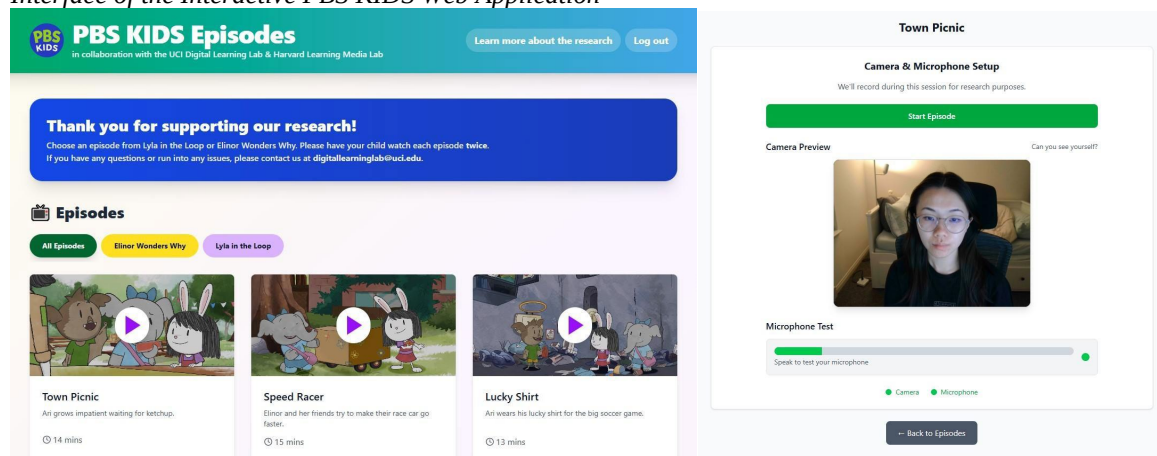
For the initial prototype, the web application was developed using Node.js and informed by feedback and observations from prior studies within the broader project. The application was optimized for consistent performance across phones, tablets, and desktop devices on major operating systems to reduce technological barriers. OpenAI's Whisper model was used for real-time speech-to-text transcription, while intent detection was performed by OpenAI's GPT-4 to classify children's responses into predefined categories.

To address research questions concerning children's interactions and engagement with the CA, video and audio recordings of sessions were collected and securely stored using AWS S3. However, they were used solely for research analysis and were not required for the real-time functioning of the application, as audio input was streamed directly to Whisper for transcription during the interaction. These recordings were crucial for evaluating the accuracy of the ASR system for children, but the recordings could be deleted at any time upon request from participating families.

The prototype underwent several rounds of internal usability testing with children and research team members to refine the interface design, interaction flow, and overall user experience. Designed with families in mind, the interface features a bright and colorful visual scheme, large clickable elements, and simplified navigation. Following deployment to participating families, iterative refinements were made based on user feedback and technical reports. Through these cycles, several key themes have emerged, though we also welcome feedback and collaboration from the ISLS community beyond these:

- **Accessibility and reliability:** Some families expressed frustration with how the app functioned in their home environments (e.g., excessive bandwidth use or lagging on older devices). To address this, we split recordings into smaller clips that are incrementally uploaded to the database, reducing bandwidth demands and improving reliability across different devices.
- **Clarity through visual cues:** Families also noted that it was difficult to track how much of the study their child had completed without closely supervising them. In response to this feedback, we added visual indicators such as progress bars and completion counters so parents could easily monitor their children's progress.
- **Transparency and privacy:** Families expressed concerns about when the app was "listening" and how recordings would be used. To address these concerns, we added a camera and microphone setup screen before each episode to clearly communicate when recordings occur, their purpose, and how families can request data deletion (Figure 2).

Figure 2
Interface of the Interactive PBS KIDS Web Application



Interactive session overview

For an in-person interactive session, we will provide laptops and tablets at the venue and register guest accounts in advance so attendees can test the web app. Attendees can interact with and provide feedback on the CA itself or the web app. Feedback can be provided in two ways: either through an anonymous survey posted via a QR code or by speaking with the first author presenting on-site. The presenter will also be able to give an overview of the PBS KIDS research-practice partnership and facilitate deeper discussion on interactive learning tools. The web app does not require any installation or downloading, so we will follow a similar format in the case of a

virtual session. The link to the web app can be shared virtually, but attendees will need to use their own device to test.

As the ISLS community spans several disciplines and identities, we welcome feedback from anyone who has an interest in or experience with designing learning tools, including other researchers or practitioners. In addition, the insights gathered from this interactive session can inform the ongoing development of our web app and prepare it for future dissemination through the PBS KIDS network. Beyond improving our own tool, we also hope to contribute to knowledge about how to design effective, research-based learning technologies and to advance the field's understanding of how CAs can support children's engagement and learning at home.

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CoReTra: Supporting Learners in Identifying Personally Meaningful Contrasts among Readers' Interpretations

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Abstract: Writing that involves personally reconstructing what one has read is essential for knowledge transformation, although university students often struggle to make meaningful connections between readings and their own ideas. Existing instructional approaches offer procedural guidance but often do not support deep, personally meaningful integration. To address this issue, we developed CoReTra (Collaborative Reflection through conTRast of interpretations), a web-based system designed to support learners in identifying and integrating personally meaningful contrasts. The system includes two views: the Decomposition View, which allows learners to extract content elements from readings and record notes as cards, and the Integration View, which enables learners to compare cards from different perspectives to refine their interpretations. The system uses large language model (LLM)-based semantic mining to adaptively suggest peer-created cards that represent diverse viewpoints. These features support learners in progressively revisiting and reconstructing their interpretations, promoting deeper reflection in CSCL contexts, while keeping contrast identification learner-driven.

Introduction

Writing that involves personally reconstructing what one has read has been conceptualized as a tool for knowledge transformation from a constructivist perspective (Bereiter & Scardamalia, 1987; Spivey, 1997). Such writing is particularly important in higher education, where students are expected to read critically and integrate what they read with their own ideas to produce original texts (MacArthur, 2022). However, university students often struggle to make meaningful connections between what they read and their prior knowledge, and they tend to rely on surface-level strategies (Du & List, 2020). Miyake et al. (2002) compared how university students and professional researchers integrated multiple sources and found that students tended to treat each article as independent, whereas researchers actively decomposed the articles into pieces and then reconstructed them into a new, integrated view. These findings suggest that students have difficulty relating and personally reconstructing information into a coherent, integrated explanation, emphasizing the need for effective support methods to facilitate this process.

Barzilai et al. (2018) conducted a systematic review of empirical studies on text integration support and found that the most frequent instructional practices were collaborative discussions, often combined with strategy instruction (e.g., explicit instruction of integration, modeling of integration), and the use of graphic organizers (e.g., tables, concept maps). These methods provided learners with procedural guidance on how to integrate texts and with frameworks for what to compare, thereby guiding collaborative practices in ways intended by teachers or researchers. In all effective interventions identified, students were taught integration processes, or how to integrate. This finding indicates that explicit instruction can play a major role in successful interventions.

However, Boscolo et al. (2007) found that explicit instruction through modeling of good and poor literature reviews improved students' organization and cohesion, but did not affect content integration. This instruction also reinforced students' view of academic writing as impersonal and did not change their beliefs about the elaborative and transmission roles of writing. Nelson and King (2023) similarly noted that "form- and process-oriented pedagogy" —approaches directed to textual form in written communication (e.g., arguments, literature reviews) while incorporating process-oriented elements such as modeling and collaborative practice—has improved many aspects of students' writing in these genres, but integrative connections remain difficult for many writers, as illustrated by interventions such as Boscolo et al. (2007).

For such integrative reading-and-writing tasks, collaborative learning offers an effective way to address this issue by allowing learners to externalize and compare differences in their ideas. In text comprehension, contrasts emerge in interpretations as a result of learners' differing prior knowledge and beliefs (Kintsch, 1998; Bråten et al., 2011). Miyake (2013) characterizes this kind of socially mediated revision of understanding as conceptual change through collaboration, where collaboration and reflection serve as social mechanisms for

revising understanding, and where role exchange between expressing ideas and monitoring others supports the abstraction of understanding in classroom discussion. However, it is challenging for learners to identify, within classroom discourse, contrasting interpretations that contribute to reconstructing their own understanding.

Large language models (LLMs) have the potential to serve as tools for semantic mining to support learners in exploring personally meaningful contrasts. Previous studies used distributional semantic models to generate semantic relationships based on learners' input, rather than relying on fixed rules set by teachers. One early approach was Latent Semantic Analysis (LSA; Landauer & Dumais, 1997). For example, Wolfe et al. (1998) used LSA to estimate whether a text fell within a student's "zone of learnability" by calculating the semantic distance between the learner's prior knowledge and the text. Unlike LSA, which constructs a static semantic space from word co-occurrence statistics, LLM can generate context-dependent semantic representations dynamically through self-attention mechanisms. As a result, LLM can more precisely emulate contextual semantic differences—though still approximately—than static models. When used appropriately, LLM has the potential to enable a CSCL environment in which students adaptively explore personally meaningful contrasts based on their own writing, without being entirely dependent on externally provided forms.

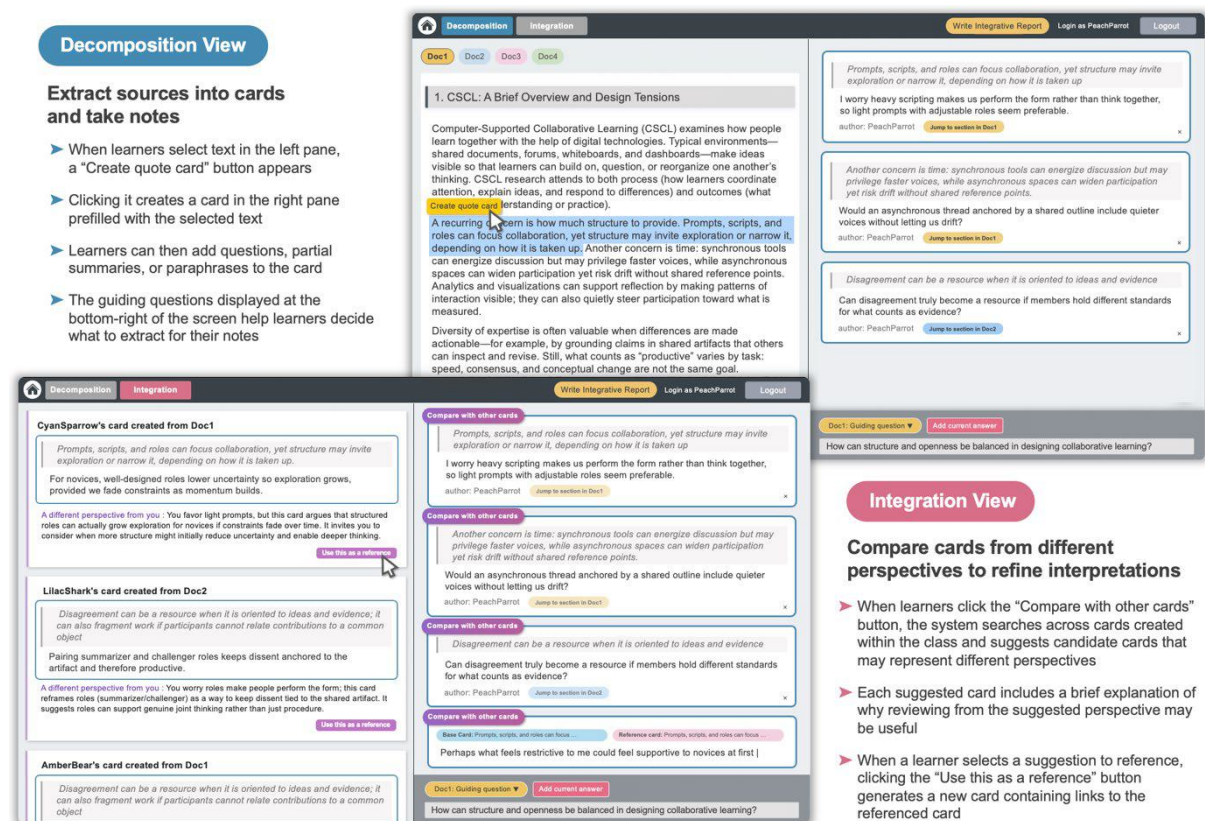
Building on this potential, we designed and developed CoReTra (Collaborative Reflection through conTRast of interpretations), an adaptive integration support system that uses LLM to support learners in identifying meaningful contrasts that they can personally relate to. The system supports learners in progressively revisiting their interpretive notes by contrasting them with those of others, using LLM-based semantic mining. In CoReTra, LLM is not used to generate answers—which could cause hallucinations or overreliance—but serves as a semantic-mining tool that helps learners identify contrasts among interpretations by suggesting potential perspectives. Thus, CoReTra uses LLM as an assistive tool for exploration, not as a decision-maker, ensuring that identifying contrasts remains a human-driven, reflective process.

CoReTra: System design

CoReTra: Collaborative Reflection through Contrast of Interpretations (Figure 1) is a support system that facilitates collaborative activities in which multiple learners share and contrast their interpretations of reading

Figure 1

Overview of CoReTra Interface (On-screen document text and notes are AI-generated examples)



materials, then revisit their own interpretive notes. This system extends Tanaka et al. (2025a), which used LLM

to support identifying student–teacher contrasts, by enabling learners to compare interpretations with peers in a collaborative setting. CoReTra is primarily intended for university students and can be used in both formal and non-formal learning contexts. CoReTra has been piloted in an intensive graduate course on the learning sciences; an initial classroom account is reported in Tanaka et al. (2025b). CoReTra is a web-based system implemented in Next.js with a PostgreSQL backend and deployed on a managed PaaS (currently Railway). The first author developed the system end-to-end.

The system includes two main views: the “Decomposition View,” which breaks down reading materials into elements to create cards, and the “Integration View,” which enables users to compare cards with those of others. Learners can switch between these views using the toggle button at the top of the screen, depending on their reading and writing. In addition, CoReTra provides the “Integrative Report View,” which learners access via the “Write Integrative Report” button to write a provisional, revisable integrative report for the current assignment while revisiting the cards they created. CoReTra organizes activities in a workspace (e.g., a course project) and multiple assignments within it (e.g., class meetings), each with its own reading materials and a shared set of cards.

Decomposition View: Extract sources into cards and take notes

The Decomposition View was designed to support learners in identifying and extracting content elements from reading materials, creating cards as personal units for later comparison. Several reviews (Barzilai et al., 2018; Nelson & King, 2023) have noted that prior studies have found supporting learners in identifying content elements from reading materials and organizing or describing their relationships can effectively promote the creation of new connections. Based on these findings, the Decomposition View allows learners to extract text segments into cards and to write notes or annotations on each segment.

In the Decomposition View, when learners select text in the left pane, a “Create quote card” button appears. Clicking this button generates a card in the right pane, prefilled with the selected text. Learners can add questions, partial summaries, or paraphrases to the card. Each card created from a quotation includes a button that allows learners to jump back to the corresponding location in the original text. The color of this button matches the tab color of the reading material.

In addition, guiding questions displayed at the bottom right of the screen help learners determine what to extract and record as notes. These questions are predefined for each material. By clicking the “Add current answer” button, learners can create a special card that records their provisional responses to the guiding questions at that moment. Because the system supports progressive revision of learners’ descriptions, they can record multiple responses to the same guiding question during the reading process.

These cards can also be revisited in the “Integrative Report View” when learners write an integrative report for the current assignment. As cards are created in the Decomposition View, they become part of the assignment’s shared set of cards, which the Integration View later searches for comparison.

Integration View: Compare cards from different perspectives to refine interpretations

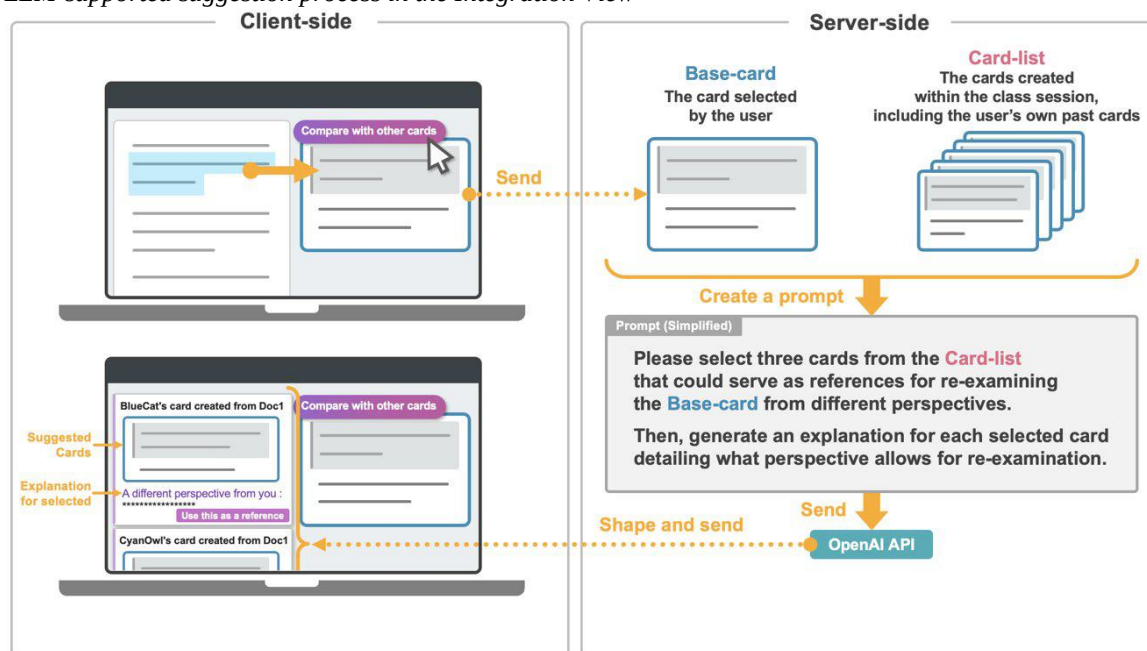
The Integration View was designed to support learners in reconstructing their interpretations through collaborative comparison with others. It includes a function that enables learners to compare the cards they extracted in the Decomposition View with those of others or with their own previous cards, supported by an LLM.

In the Integration View, when learners click the “Compare with other cards” button, the system searches for cards created within the same assignment and suggests three candidate cards that may represent different perspectives. Each suggested card includes a brief explanation of why reviewing from the suggested perspective may be useful. Figure 2 shows the process by which the system generates these suggestions. This process is executed each time the user clicks the “Compare with other cards” button. As a result, even when the same base card is used, different candidate cards and explanations are generated with each execution. Because the Card-list dynamically references the most recent cards created within the same assignment at the time of comparison, the results change as collective reading and writing activities progress. The system currently uses the OpenAI API (LLM) for this process.

When a learner selects a suggestion to reference, clicking the “Use this as a reference” button generates a new card with links to the referenced card. By clicking the inserted link button, learners can access both the original base card and the referenced suggestion. Each suggestion can be reviewed alongside the recommended card and the explanation generated by the LLM, which describes the reason for its selection.

The suggestion function is enabled only after at least three candidate cards are available. Because what constitutes a “meaningful” contrast is inherently subjective, suggestions are intentionally exploratory rather than “best match” recommendations: the system returns up to three candidate contrasts with brief rationales as cues

Figure 2
LLM-supported suggestion process in the Integration View



for reflection, not correctness claims. Learners may ignore unhelpful suggestions and re-run the function to explore alternative contrasts.

For LLM-based suggestion, the server sends only the textual content of the base card and candidate cards (quoted excerpts and learners' notes) along with internal card IDs; it does not include user account identifiers (e.g., names or emails) in the prompt. Candidates are restricted to note-bearing cards created within the same assignment in the workspace (excluding deleted cards and the base card). To bound latency and cost, the candidate list is truncated to fit within a fixed token budget (default: 12k tokens for candidates, measured with tiktoken), removing older cards first until the budget is met.

Interactive demo

In this interactive demo session, participants will use CoReTra to collaboratively read and interpret a short text familiar to the ISLS community. They can write their interpretations in the system and discuss them with other participants through the system interface.

Each participant will receive a temporary account created specifically for the session and may use the system independently on their own computer through a web browser (Chrome, Edge, or Safari recommended). Participants can create cards anonymously, linked to their session accounts, which are identified by randomly generated dummy names consisting of a color and an animal (e.g., BlueCat). All potentially discriminatory terms were excluded from the name set.

For the demo, we will prepare an initial pool of cards so that participants can try the suggestion feature. Participants can also open the "Integrative Report View" via the "Write Integrative Report" button to write an integrative report based on the cards they created.

The reading material used for this demonstration is an open-access article by Pea and Linn (2020), "Personal Perspectives on the Emergence of the Learning Sciences: 1970s–2005." This material is used under the terms of the Creative Commons Attribution (CC BY 4.0) license, with appropriate citation of the original publication. We expect that reading and interpreting this text together will enable ISLS members to identify meaningful contrasts in their interpretations and engage in deeper discussions about issues central to the ISLS community.

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Evidence-Centered Design for AI-Supported Assessment of Scientific Argumentation

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Abstract: This demo showcases an AI-supported, evidence-centered assessment design that scaffolds middle school students' scientific argumentation in the context of ecosystems learning. Grounded in Evidence-Centered Design (ECD), the assessment links constructs of scientific reasoning and argumentation with AI-driven evidence detection and feedback generation. Students investigate ecosystem dynamics through a NetLogo simulation, construct Claim-Evidence-Reasoning (CER) arguments, and receive adaptive feedback from an LLM-based chatbot. Participants will experience the task, explore reasoning analytics, and see how ECD structures both assessment and AI design. The demo highlights how ECD can ensure a valid, interpretable, and instructionally useful AI system that advances students' reasoning and supports classroom instruction.

Introduction

Scientific argumentation is fundamental to how scientists construct knowledge and how students make sense of the world. Engaging learners in argumentation, including formulating claims, supporting them with evidence, and reasoning about underlying mechanisms, promotes deeper conceptual understanding and reflects the epistemic practices of science (Osborne, 2010; Wilson et al., 2024). However, research consistently shows that students struggle to construct well-reasoned arguments: they often state claims without adequate evidence, overlook causal reasoning, and rarely critique or rebut competing ideas (Sampson & Clark, 2009). Teachers also face significant challenges in supporting students' argumentation, including limited time, insufficient training, and lack of resources to provide individualized, timely feedback at scale (Rapanta et al., 2021).

Advances in artificial intelligence (AI), particularly large language models (LLMs), offer promising opportunities to address these challenges. When designed with pedagogical intent, AI systems can analyze students' argument structure and content, identify reasoning patterns, and generate personalized feedback that guides students to strengthen their arguments (Kumar et al., 2024; Watts et al., 2025). Such systems have the potential to extend teachers' capacity to provide timely, diagnostic feedback and create new avenues for investigating how students reason and revise their arguments. However, without principled design, such systems risk offering feedback that is oversimplified or misaligned with learning theories and pedagogical goals (Zawacki-Richter et al., 2019).

Our project applies an Evidence-Centered Design (ECD) framework (Mislevy et al., 2003) to the development of an AI-supported classroom assessment and feedback system that helps students construct evidence-based arguments while making sense of ecosystem phenomena. Grounded in scientific argumentation theory and principles of instructionally supportive feedback, the project integrates the Claim-Evidence-Reasoning (CER) framework (McNeill & Krcjik, 2008) to define high-quality arguments as those that coherently link claims and relevant evidence through scientifically grounded reasoning. The AI feedback system draws from research on scaffolding (Reiser, 2004) and adopts an epistemic agent perspective (Gonzalez-Howard & McNeill, 2020), encouraging students to view themselves as knowledge builders who evaluate the validity of their own reasoning (Stroupe, 2014). Rather than just identifying correct answers or missing components of an argument, the feedback engages students with reflective prompts such as "Do the data you used actually support the claim?," "How does your reasoning explain the why?"

The assessment task to be demonstrated captures and promotes students' scientific reasoning through NGSS-aligned ecosystem scenarios, automated reasoning analysis, and adaptive feedback. By integrating ECD with AI innovation, the project advances both the validity and usefulness of science assessments, offering a model of how principled design can anchor AI systems in learning theory, evidence of construct validity, and pedagogical purpose. This demonstration showcases how an AI-supported assessment enables interactive exploration of

student reasoning in authentic ecosystem contexts, how AI models classify reasoning patterns and generate reflective feedback, and how teachers and researchers can use the system to inform instruction and learning-science research.

Theoretical framing

Research on scientific argumentation demonstrates that students' ability to engage in scientific argumentation significantly influences their understanding of STEM concepts and their success in related courses (Bathgate, et al., 2015; Sampson & Clark, 2009). Research has shown that reasoning is challenging to students (Kuhn, 1991). Furthermore, students rarely rebut potential counterarguments, though some may be able to do so when prompted (Kuhn & Udell, 2007). Students also often lack criteria for distinguishing and evaluating ideas (Sampson & Clark, 2009) or rely on informal criteria (e.g., teacher's authority, fit with personal inferences). Even advanced students struggle to evaluate competing explanations, assess the quality of evidence, or identify underlying assumptions in others' arguments (Klaczynski, 2000).

Like scientists, children begin developing intuitive explanatory frameworks early in life, drawing on everyday experiences to make sense of the natural world. In ecosystem contexts, students tend to interpret changes in the organisms of a food web in linear ways, focusing on simple, one-step relationships (e.g. how changes in prey population influence changes in predator population) rather than complex multivariate causal interactions (Gotwals & Songer, 2010; Leach et al., 1996; Wennersten et al., 2020). These intuitive patterns of reasoning or facets of student thinking reflect deeply held "direct-causal schemas" (Chi et al., 2012) that help students make sense of observations but can also constrain their reasoning about systemic phenomena. Research suggests that building on these intuitive ideas through reflection and guided exploration can foster more sophisticated causal reasoning (Grotzer et al., 2017).

Supporting students' argumentation practice requires teachers to surface and work with students' ideas and engage with them to build scientific understanding. In typical classroom practice, however, students' non-normative ideas are often viewed simply as misconceptions to be corrected, rather than as productive starting points for reasoning. Teachers often lack time, training, or tools to analyze students' reasoning in real time and provide feedback that leverages their ideas to foster the development of scientific reasoning. Innovative, AI-supported assessment design aligned with instructional purposes can bridge this gap by eliciting and analyzing the reasoning patterns students use to make sense of scientific phenomena, enabling feedback that supports conceptual and epistemic growth.

Building on this premise, the demonstrated work applies principles from the learning sciences and argumentation theory to model common reasoning patterns in students' arguments and to design AI-supported assessment tools that both evaluate and scaffold argumentation. Using existing student response data from NGSS-aligned ecosystem assessments, the project identifies recurring reasoning structures, automates their annotation through machine learning, and embeds AI-generated feedback to guide students in strengthening their arguments.

Evidence-centered design (ECD) of an AI-supported argumentation task

The ECD framework provides a principled approach to ensuring that assessment and AI systems embedded within them generate valid, interpretable evidence about student learning. As Mislavy et al. (2003) emphasize, ECD strengthens validity by linking the student model (what we want to infer about learners) with the evidence model (what behaviors and responses constitute evidence for those inferences) and the task model (the situations that elicit that evidence). Applied to AI-supported assessments, ECD extends beyond traditional psychometric design to also guide AI model development to ensure that the design, evaluation, and feedback algorithms remain tightly coupled with the learning constructs and pedagogical intent. In this demonstration, ECD has served as the organizing framework for designing, implementing, and refining the AI-supported argumentation task, which integrates the scientific practice Engaging in Argument from Evidence with the disciplinary core idea of Ecosystem Dynamics, Functioning, and Resilience from the Next Generation Science Standards (NGSS). The development process followed three key phases aligned with ECD principles: target construct identification and domain analysis, domain modeling and AI-integrated evidence model construction, and task design with the embedded AI feedback system.

Target constructs and domain analysis

The construct focus of the classroom assessment is student reasoning within scientific argumentation in the context of ecosystem sensemaking. The domain analysis unpacked relevant NGSS performance expectations to define the knowledge and practices students should demonstrate:

Engaging in Argument from Evidence: Engaging in argument from evidence in Grade 6–8 builds on K–5 experiences and progresses to constructing a convincing argument that supports or refutes claims for either explanations or solutions about the natural and designed world(s).

- Construct an argument supported by empirical evidence and scientific reasoning to support or refute an explanation or a model for a phenomenon or a solution to a problem. (MS-LS2-4)
- Evaluate competing design solutions based on jointly developed and agreed upon design criteria. (MS-LS2-5)

Ecosystem Dynamics, Functioning, and Resilience: Ecosystems are dynamic in nature; their characteristics can vary over time.

- Disruptions to any physical or biological component of an ecosystem can lead to shifts in all its populations.
- Biodiversity describes the variety of species found in Earth’s terrestrial and oceanic ecosystems. The completeness or integrity of an ecosystem’s biodiversity is often used as a measure of its health.

Domain modeling and AI-integrated evidence model construction

The domain model operationalized these constructs into observable evidence features that could inform both human annotation and AI classification. Drawing on established argumentation frameworks (McNeill & Krajcik, 2008; Osborne et al. 2004) and existing assessment data from state assessments and prior research (Liu et al., 2024), the team identified four reasoning patterns: relational, cause-effect, hypothetical, and analogical reasoning. The assessment scenario positioned students as members of a school environmental club investigating a corn rootworm infestation phenomenon. Students interacted with a customized NetLogo predator-prey simulation (Wilensky, 1999), manipulated population variables, collected data, and interpreted outcomes in light of ecological principles. After the simulation activity, students constructed arguments using the claim-evidence-reasoning (CER) structure (McNeill & Krajcik, 2008), explaining whether introducing a predator (harvest spiders) could help restore ecosystem balance. In the evidence model, these student-generated responses were represented as linguistic and conceptual features, serving both as analytic inputs for automated reasoning classification (via GPT-based models) and as interpretive basis for AI-generative formative feedback.

Task design with the embedded AI feedback system

The task model informed the design of the context, iteration flow, and feedback opportunities within the task. Students’ interactions with the simulation and constructed responses were embedded within an AI-supported environment that provides context-sensitive scaffolding through an integrated chatbot interface called SPIN-Bot. The SPIN-Bot, implemented using OpenAI’s GPT-4o was accessible through a “Need Help?” feature and was equipped with conversational memory and item-level awareness in the assessment task. Figure 1 shows how students could interact with the SPIN-Bot in the task.

Figure 1
Task Screenshots Illustrating How to Interact with the Simulation and SPIN-Bot in the Task.



To ensure alignment with ECD principles, the SPIN-Bot system prompt was developed iteratively through human-in-the-loop prompt engineering, explicitly embedding constructs from the student and evidence models (Liu et al., 2024; Watts et al., 2025). Three design iterations progressively refined the prompt structure: Version 0 focused on general tutor role; Version 1 included detailed task context and alignment with dialogic scaffolding principles (Mercer & Howe, 2012); and Version 2 focused on theory-grounded, bulleted prompts emphasizing argumentation scaffolds (e.g., prompting use of evidence, reasoning justification, and self-reflection). A corpus of 162 student responses to synthetic student prompts was evaluated using a four-dimension rubric (adapted from Duschl, 2003; Nicol & MacFarlan-Dick, 2006; Panadero & Lipnevich, 2022) assessing relevance, clarity, engagement, and reflective prompting. The iterative analysis informed refinements to both the system prompt and the AI evaluation criteria embedded in the evidence model. In addition, the project embodies the ECD principle of design-evidence-interpretation cycles through iterative empirical studies spanning cognitive labs, pilot testing, and field studies. Each iteration strengthens the fidelity between what the AI detects, what evidence it interprets, and what pedagogical feedback it provides.

Design principles for ECD-aligned AI systems

Drawing from the above iterative design process, we derived several guiding principles for integrating AI into ECD-based assessment design in support of the development of scientific argumentation:

- *Embed learning-theoretic constructs in the AI system prompt.* Define the student model (e.g., argumentation components and the CER structure, ecosystem core ideas) within the AI system prompt so that generated feedback explicitly targets conceptual reasoning and argumentation goals.
- *Translate evidence rules into algorithmic features.* Operationalize evidence model indicators (e.g., relational reasoning or cause-effect reasoning cues) as input features for AI classifiers or conditions for generating feedback.
- *Design tasks as evidence-eliciting interactions.* Ensure that each student action or response yields interpretable data aligned with the reasoning and argumentation constructs.
- *Scaffold student reasoning and reflection, rather than focusing on correctness.* Prompt reflection and explanation rather than providing direct answers to prioritize students' epistemic agency.
- *Iterate through human-AI calibration cycles.* Use human-in-the-loop evaluation to validate AI responses against expert reasoning and pedagogical quality.

By coupling ECD with AI development, this demonstration will show how validity arguments can extend to algorithmic systems to ensure that both assessment and AI design reflect the same structure of scientific reasoning and feedback purpose.

Demonstration plan

The demonstration will guide participants through the following several interactive segments showcasing how our work integrates simulation-based assessment and AI feedback within the ECD framework. First, it will begin with a live walk-through in which the presenter demonstrates how students explore the NetLogo ecosystem simulation, construct CER arguments, and receive real-time AI feedback. Next, in-person attendees can participate in an interactive exploration through the on-site laptop setups to experience both student and teacher perspectives; and virtual participants can join Zoom breakout rooms to interact with the demonstration. Embedded polls will collect reflections on assessment design, AI feedback usefulness, and improvement suggestions. A behind-the-scenes view then illustrates how the AI pipeline was designed to process student input, classify CER components and reasoning patterns, and generate feedback grounded in the evidence model. Finally, during the post-session engagement, participants will receive author contact information for potential future applications for research or classroom use.

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Learning in the Wild: PAL's New Tools and Methods for Research in Early Childhood Mathematics

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Abstract: PAL (Personal Assistant for Learning) is an adaptive, smart learning system that operationalizes sociocultural theory, Knowledge Space Theory, and learning sciences principles to support and study early mathematics learning in real-world contexts. PAL guides parents and teachers to facilitate hands-on, socially mediated interactions aligned with each child's Zone of Proximal Development while generating frequent, fine-grained evidence about children's mathematical learning. The system integrates a research-grounded knowledge model, an explore-exploit adaptive engine, and a digital assessment system to provide personalized guidance and scalable, ecologically valid measurement. Early pilots show substantial gains in children's learning and strong associations between adult engagement and child learning. Designed as a learning accelerator and a research platform, PAL provides the Learning Sciences community with a tool for examining learning trajectories, scaffolding dynamics, and conceptual and skill development as they naturally unfold across a child's learning ecosystem—while producing evidence needed to connect research insights to practice at scale.

Introduction

Early childhood mathematics is more conceptually complex than it appears on the surface (Geary et al., 2019). Foundational ideas, such as one-to-one correspondence, comparison, subitizing, spatial relations, and patterning, develop along systematic learning trajectories and form the cognitive architecture for later mathematical reasoning (Clements & Sarama, 2020). Yet in practice, these learning trajectories remain largely invisible in curriculum materials, instructional routines, and everyday adult-child interactions, leaving families, teachers, and even researchers without clear markers of how mathematical understanding develops across early concepts and skills (Clements & Sarama, 2024; Schmidt, 2025). Early math is frequently perceived as intuitive or simple, leading to chronic underinvestment in research infrastructure, professional learning, and high-quality assessment tools (Bittner & Bull, 2022; Gallegos, 2025). This misalignment is reinforced by a broader ecosystem-level “expertise gap.” Practitioners have limited professional knowledge for teaching early mathematics (Ban et al., 2024), and the field lacks validated, operationalized learning progressions, granular assessments, and research-grounded instructional tools (Clements & Sarama, 2025). These gaps also appear in emerging educational technologies: most AI systems inherit the same misconceptions and oversimplifications embedded in the materials they were trained on (Betts, 2025), leaving them unable to provide effective, developmentally appropriate recommendations for young children. As a result, the adults who support young children (i.e., teachers, parents, caregivers), often lack access to understandable domain-specific guidance aligned with how mathematical concepts develop for our youngest learners.

Although the Learning Sciences community has developed rich theoretical traditions—such as sociocultural perspectives on adult-child interaction (e.g., “More Knowledgeable Others”; Vygotsky, 1986), conceptual structure theories such as Knowledge Space Theory (Doignon & Falmaigne, 1985), and ecological views of learning environments (e.g., Bloom, 1984)—the field still lacks the infrastructure needed to operationalize and study these perspectives at scale (Motz et al., 2025). There are no widely available, fine-grained knowledge models of early mathematics; few instrumented home-learning environment tools capable of capturing naturalistic learning behaviors; no scalable methods for tracing children's learning sequences and progressions; and no mechanism for linking instructional guidance with concept-level assessments in real-time. These gaps limit the Learning Sciences community's ability to test hypotheses about learning trajectories, conceptual dependencies, scaffolding, and the dynamics of early cognition—not just in math, but in other subject areas as well.

PAL is an adaptive “smart” learning system developed to address this missing infrastructure. Grounded in sociocultural theory, Knowledge Space Theory, and Bloom's Objects of Change, PAL provides personalized, dynamically-adapting support to adults in guiding children's hands-on, real-world math learning while generating research-quality evidence about early learning (Betts, Ryon, & Laski, 2026). Today, PAL integrates a validated,

concept and skill-level knowledge model of early mathematics, interactive assessment items aligned to granular knowledge units, and an adaptive recommendation engine that personalizes activity guidance based on simple adult feedback. Delivered through SMS and a lightweight web interface, PAL embeds developmentally grounded learning moments into daily routines and creates a data-rich environment that captures evidence of growth in children's mathematical knowledge and skills, as well as adult scaffolding practices and engagement across the learning ecosystem (e.g., home, school, community). In doing so, PAL functions both as a learning accelerator that (1) maximizes learning by keeping children within their Zone of Proximal Development and (2) as a research platform that enables the Learning Sciences community to study early mathematical cognition and learning sequences in authentic settings and at scale.

The theoretical grounding of PAL

PAL's design and methodology is grounded in sociocultural learning theory, Knowledge Space Theory (KST), and Bloom's Four Objects of Change. Sociocultural learning theory informs PAL's focus on strengthening adults as effective "more knowledgeable others" who can guide children through meaningful, hands-on activity; the Zone of Proximal Development (ZPD) provides the foundation for PAL's adaptive algorithms, which personalizes recommendations to place each child within an optimal zone for growth (Vygotsky, 1986; Zaretskii, 2009). Knowledge Space Theory (Doignon & Falmagne, 1985) models knowledge as a structured set of possible states defined by prerequisite relations among concepts. For PAL, this framework enables computational inference of which mathematical ideas are reachable, which are dependent on earlier concepts, and which constitute the child's "next learnable" steps. Additionally, Bloom's (1984) Objects of Change emphasize improving the learner, the adult, the materials, and the environment in concert—an approach reflected in PAL's ecosystem design.

Overview of PAL

PAL's design is powered by a comprehensive knowledge model developed through expert synthesis of research-based early math learning trajectories, developmental progressions, and various state, national, and international standards frameworks for early mathematics learning, and continues to evolve based on data and insights from pilot studies. Comprising hundreds of granular early mathematics Knowledge Units (KUs), the model encodes conceptual understanding, prerequisite relationships, and recommended scaffolding strategies derived from decades of developmental and cognitive research and built directly into the core knowledge infrastructure. This enables PAL to generate a probabilistic estimate of a child's current knowledge state, identify concepts they are most ready to learn next, and recommend adult-mediated activities that maintain an optimal level of challenge and provide observable evidence of meaningful learning (Foster & Piacentini, 2025; Mislevy, Almond, & Lukas, 2003).

PAL's adaptive engine uses an explore-exploit strategy to support learning within the child's ZPD in real-world contexts. The system periodically "explores" adjacent KUs to probe emerging learning readiness and receptivity to challenge, while "exploiting" identified strengths by deepening understanding in known areas. Adaptation is informed by two key forms of evidence: (1) lightweight adult feedback (e.g., "did independently," "needed help," "struggled"), which provides immediate and regular insight into children's performance in real-world activities; and (2) performance on periodic, brief formative assessment items generated through PAL's embedded assessment mechanisms. This adaptive cycle reflects learning engineering principles that integrate evidence, inference, and iterative refinement into real-world learning contexts.

To support scalable, developmentally appropriate measurement, items in PAL's KU-aligned digital assessment system approximate hands-on early math tasks and enable concept-level, task-based evaluation that mirrors children's natural math learning activity. These brief, targeted digital assessments (2-3 minutes in duration) offer low-friction, ecologically valid evidence of children's learning and provide a mechanism for validating indirect evidence provided by adults in the ecosystem, verifying conceptual dependencies, identifying misconceptions, and monitoring growth across learning trajectories. These assessments enable concept-level personalization that traditional early childhood assessments (which are typically broad, infrequent, and labor-intensive), are unable to offer at scale (Banerjee & Luckner, 2013).

Beyond its conceptual and adaptive architecture, PAL is intentionally designed for accessibility and equity. Guidance is delivered through simple SMS messages linked to a lightweight web interface. Activities require no special materials and are intended to be integrated into everyday routines, making them feasible for caring adults (parents, caregivers, teachers) to facilitate across diverse settings. This low-bandwidth, adult-supporting design enables use in contexts where device-based or 1:1 instructional technologies may be impractical, developmentally inappropriate, or inconsistent with community preferences regarding screen use.

Taken together, these components position PAL as a learning-acceleration system that integrates concept structure modeling, adaptive guidance, and aligned assessment to support and enhance capacity for high-quality

adult–child interaction, generating structured evidence about children’s learning trajectories in naturalistic settings.

Evidence of PAL’s effectiveness

Early pilots of PAL demonstrate promising effects on children’s early math development, suggesting that a learning-sciences-grounded, adult-facing adaptive system can meaningfully accelerate learning in real-world contexts (Betts, Ryon, & Laski, 2026). In a recent implementation with a cohort of Voluntary Prekindergarten (VPK) classrooms ($N = 19$ children), families in the PAL condition engaged in short, hands-on math activities guided by PAL’s adaptive recommendations over a 3.5-month period. Pre- and post-assessments were conducted using the Test of Early Mathematics Ability, Third Edition (TEMA-3), administered by trained assessors independent of the PAL system. Though the sample was small, the results were promising. Children whose families used PAL showed substantial gains, with 100% improving their TEMA-3 scores compared with 57% in a business-as-usual comparison group. The effect size exceeded 1.0, indicating a significant improvement in early mathematical skills. Importantly, the degree of parent engagement—operationalized as completion of PAL activities and submission of feedback—was positively correlated with children’s learning gains (Spearman $r = .62$), underscoring the role of adult mediation in accelerating children’s conceptual growth.

Building on these early findings, PAL is being implemented in larger, multi-site pilots ($n > 500$) that will provide broader evidence across diverse educational contexts. One pilot is underway in a large, public school district in Glenolden, Pennsylvania, with children in PreK, kindergarten, and first grade. A parallel implementation is running with a small consortium of private schools in Santa Monica, California, serving preschool, PreK, kindergarten, and first-grade students. Across these sites, children, families, and teachers are engaging with PAL-supported math activities, creating opportunities to examine usage patterns, adult engagement, and learning outcomes in both public and independent school settings.

PAL as research infrastructure

PAL currently serves as an extensible research infrastructure for studying early cognitive development in mathematics, learning progressions, and adult–child scaffolding in both naturalistic and classroom settings. The Learning Sciences community has long recognized the importance of early math and informal learning environments; however, it has lacked shared, scalable tools to observe learning as it unfolds outside laboratory or controlled settings (Hornburg et al., 2021; Motz et al., 2025). Rather than examining isolated snapshots of instruction or development, researchers can now investigate early math learning as a dynamic, real-world process, capturing how children’s understanding emerges, shifts, and evolves across everyday interactions.

PAL provides a granular representation of early mathematical understanding through its knowledge model, comprised of hundreds of developmentally sequenced Knowledge Units (KUs). Because each KU encodes conceptual and skill dependencies, misconceptions, and scaffolding strategies, the model supports investigation of learning trajectories at a level of precision often not available in early childhood research. As a directed conceptual graph, the KU structure also enables empirical testing of prerequisite relations, reconstruction of learning sequences, and identification of points where learners accelerate or struggle across contexts. PAL’s explore–exploit adaptive logic further supports research on ZPD-aligned learning mechanisms, continuously probing readiness on adjacent KUs (“explore”) while reinforcing emerging competencies (“exploit”). Behavioral evidence from adult feedback provides insight into how support is applied and how ZPD positioning evolves in real-world contexts.

Complementing these conceptual and behavioral layers, PAL also instruments the early learning ecosystem, capturing lightweight telemetry that spans Bloom’s Objects of Change—learner, adult, materials, and environment—and the dynamic relations among them. The system logs how adults engage with nudges and recommendations; how they select, adapt, or sequence activities; how multiple caregivers or teachers coordinate around a child’s learning; and how contextual rhythms shape opportunities for engagement. Simultaneously, PAL captures children’s performance across hands-on tasks and digital assessments, enabling researchers to link moment-to-moment behavior with longer-term conceptual development. Such multi-actor, multi-context evidence is rarely accessible in early childhood research, where observations are episodic, resource-intensive, and typically limited to laboratory or classroom settings. By making these cross-level interactions visible at scale, PAL opens new possibilities for examining how instruction, environment, and adult beliefs jointly influence young children’s mathematical development in authentic settings. Moreover, PAL’s integrated, data-rich design enables researchers to envision new lines of inquiry, while offering a shared platform that invites collaborative extension and refinement across the Learning Sciences community.

Demonstration format

The demonstration will provide a guided walkthrough of PAL, highlighting how its theoretical foundations, adaptive engine, and assessment architecture operate in practice. The session is structured to provide attendees with direct visibility into how PAL works and its application as a research tool.

1. Knowledge model overview

The demonstration will begin with a brief visualization of PAL's research-grounded Knowledge Model, showing how concepts and prerequisite relations are represented as a structured conceptual graph. This segment illustrates how PAL operationalizes foundational Learning Sciences frameworks to model the systematic nature of early mathematical understanding.

2. Adaptive recommendation logic

The presenter will demonstrate how PAL uses both direct and indirect data to assess a child's likely knowledge state and generate personalized activity recommendations positioned within that child's predicted Zone of Proximal Development. This live walkthrough will demonstrate how the system maps learning trajectories across content, domains, and activities using algorithmic methods and rationales.

3. Digital assessment items

The presenter will demonstrate examples of PAL's KU-aligned digital assessment items, highlighting how each item maps onto specific, individual knowledge units on the knowledge model and provides direct, fine-grained, ecologically valid evidence of conceptual understanding. This segment highlights the potential of the assessment system to support scalable research on early mathematics learning in diverse settings.

4. Real-world activity design

Several sample PAL activities will be presented to illustrate how complex early math concepts are translated into concise, developmentally appropriate, hands-on interactions that naturally fit within home and classroom routines. This segment clarifies how PAL supports adults as effectively more knowledgeable others.

5. Researcher view and data opportunities

The final segment will highlight PAL's value as research infrastructure. Examples of anonymized learner progression data, patterns of adult scaffolding, readiness signals, and trajectory evidence will be displayed to show how PAL can support studies of learning progressions, learning in informal contexts, and the dynamics of ZPD support at scale.

Throughout the demonstration, the presenter will show how PAL integrates developmental theory, knowledge modeling, adaptive inference, and embedded assessment to produce both high-quality learning interactions and research-grade data. Attendees will leave with a clear understanding of how PAL operates and its applications in researching early cognition and learning trajectories in naturalistic settings.

Conclusion

PAL illustrates how developmental research, Knowledge Space Theory, and learning sciences principles can be operationalized into an adaptive system that accelerates early mathematics learning while generating forms of evidence that are largely inaccessible at scale for both research and ongoing system refinement. By strengthening adults as effective partners in children's education, PAL embeds dynamically adapting, ZPD-aligned support into everyday activity and creates an instrumented home-learning environment capable of capturing learning as it naturally unfolds. As both an intelligent learning system and a research platform, PAL enables investigations of learning trajectories, scaffolding dynamics, and conceptual and skill development in authentic, real-world contexts.

Importantly, PAL makes visible the broader learning ecosystem itself—providing scalable ways to examine Bloom's Objects of Change as a coordinated system rather than isolated components. This cross-level visibility enables researchers to link conceptual development to variations in adult practice, environmental rhythms, and material use, providing a more comprehensive understanding of how early learning emerges and evolves. In doing so, PAL offers the Learning Sciences community new research infrastructure for connecting theory and practice, deepening foundational understanding of early learning, and empowering families, teachers, and young children. We envision PAL as shared infrastructure for the field—designed to be used, interrogated, and extended by researchers seeking to advance the next generation of early learning science.

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Trend Trackers: Building Data Literacy Through Problem-Based Learning

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Abstract: In this demo, we discuss the co-design of an interactive, problem-based science supplement called Trend Trackers. Trend Trackers aims to engage middle school students with socioscientific issues that impact cities and communities all over the world. Students take up the role of a data scientist to address public health problems within the fictional city of Springfield. In this demo, we will explore a challenge in which students are asked to design a vaccination plan to reduce the rates of flu in Springfield. Students engage in interactive and immersive activities to collect wastewater samples, measure rates of flu in the community, and generate evidence through a series of wrangling, visualization, and analysis activities. Students then share their findings and collaborate with their team to co-construct a vaccination plan for Springfield.

Introduction

In K-12 education, students are introduced to data science practices such as wrangling, visualizing, and analyzing data by exploring data about themselves or their local communities. Students build on prior knowledge and personal experiences to ask questions about the context in which data were produced and to reason about the social and environmental implications of data (Lee et al. 2021; Wilkerson and Laina 2018). However, data scientists often work with data that are collected from unfamiliar, inaccessible, or dangerous sources (Wilkerson and Polman 2020; Wise 2020). Exploring these types of data sources offers students opportunities to build an understanding of data as situated in complex social and environmental contexts and produced through a series of decisions. However, students need support to think critically about the unfamiliar context in which these data are collected. Building on prior research that highlights modeling (Wolff et al. 2019), filtering, and embodiment as promising strategies to engage students in reasoning about secondary data sets, we co-designed an interactive, immersive science supplement. *Trend Trackers* is an interdisciplinary tool that aims to engage middle school students in a problem-based learning experience, collecting and analyzing data about socioscientific issues through a series of gamified, immersive, and collaborative activities.

In this presentation, we describe and discuss early findings from the co-design of one 45-minute module within *Trend Trackers* in which students are challenged to co-construct a vaccination plan to prevent a flu outbreak in the fictional city of Springfield. Students take up the role of data scientists, working in teams and collaborating with scientific experts, to measure rates of flu across Springfield using wastewater.

During the co-design process, we conducted a total of 12 co-design meetings, seven with a high-school math teacher and a middle-school science teacher and five with an expert virologist. We also conducted cognitive interviews with six individual students in grades 7-10 and with two small groups, before pilot testing *Trend Trackers* with two different schools. We asked: (1) How do youth draw on their virtual experiences sampling and analyzing wastewater to understand data as produced through a process of scientific inquiry? and (2) What data practices do students engage in as they construct and justify evidence for their vaccination plans? This work contributes to literature in data science education by offering a tool that supports students in building data literacies in the context of scientific research by drawing on a problem-based learning model.

Background

In K-12 education, students are introduced to different data science practices in math and science. In science courses, for example, students collect and visualize data through lab activities. In math, students focus on data analysis, exploring sampling strategies, measures of center, and different types of graphs. In addition, students in English language arts and history courses might construct data-based arguments. These practices are data literacies that work together to help students develop “the ability to ask and answer real-world questions from large and small data sets through an inquiry process, with consideration of the ethical use of data” (p. 23, Wolff et al. 2016). Data scientists, however, address real world problems by drawing on these interdisciplinary practices as problem-solving tools, collecting, analyzing, and using data to inform decisions in complex social contexts. To support

students in taking up the role of data scientists to address community-level socioscientific issues, we designed a problem-based learning experience, building on three design principles to help students consider the context in which complex, unfamiliar forms of data are produced and used.

The principle of personal data collection suggests that personally collecting data prompts students to think critically about what is measured in a data set, the context in which the data were produced, and the potential implications or applications of data-based findings (Wolff et al. 2019; Lee et al. 2022). However, when students work with secondary data sets that are unfamiliar, Wolff et al. (2019) suggests integrating simulations or modeling activities to engage students with a data collection experience they can draw on to make sense of data. Interactive and immersive technologies, which build on principles of embodiment (Dunleavy et al. 2009), have potential to support students in generating lived experiences collecting and measuring data. In *Trend Trackers*, students are challenged to analyze rates of flu across the fictional city of Springfield by measuring the amount of flu RNA in wastewater samples. We propose that gamifying the collection of wastewater samples and engaging students in a virtual lab to measure the amount of flu RNA in samples can help students make sense of this novel data source.

The expansion principle suggests that students build confidence in reasoning about a complex dataset by first looking at a snapshot of the full dataset and then expanding their inquiry to the full data set (Wolff et al., 2019). To support this iterative analysis of data, *Trend Trackers* engages students in analyzing data in a three-stage process. To begin, student teams engage in a data collection game, in which each team member has enough time to individually collect about 3 or 4 samples from a region of Springfield. Students first independently wrangle and visualize their sample of data. Then, students have an opportunity to engage in analysis and modeling activities using data collected by the whole group. Finally, student teams share their evidence and collaboratively construct a vaccination plan to minimize the spread of flu within Springfield.

The inquiry principle suggests that students develop data literacies by engaging in a process of asking questions, collecting data, analyzing data, and interpreting results. However, in working with unfamiliar data, Wolff et al. (2019) suggest that students engage in guided inquiry to familiarize themselves with the data and then engage in a process of student-driven inquiry. Building on a problem-based learning framework (Hmelo-Silver 2004), *Trend Trackers* scaffolds student inquiry in the following ways. The first module includes inquiry-based activities that introduce students to the main problem facing Springfield, their role as a data scientist, and the different data sources they will use throughout the four following modules. Within each of the following four modules, students then have opportunities to investigate their own statistical question or choose from a selection of questions. Finally, integrating independent and collaborative activities offers students opportunities to reflect on and compare their process of inquiry with their peers.

In this demo, we discuss findings from an iterative co-design process in which we aimed to support students in understanding wastewater as an innovative source for measuring and monitoring flu rates in a community; build confidence engaging in data science practices; and develop an understanding of data as situated in personal, cultural, and sociopolitical contexts.

Trend Trackers

In *Trend Trackers*, students encounter interdisciplinary challenges as they work to generate evidence-based solutions to socioscientific problems by engaging in a process of scientific inquiry. To design a vaccination plan for Springfield, students measure flu RNA by collecting and processing wastewater samples from districts across Springfield, generate evidence by visualizing and analyzing data, and co-construct a vaccination plan grounded in their team's evidence. Students are also encouraged to consider the role of collaboration in producing and using data throughout the process, first with their peers, then with scientific experts, and finally with community leaders. Next, we discuss how three design principles informed initial activities in *Trend Trackers* and how early pilot testing with students has informed iterations on design features.

Collecting and processing wastewater samples

Building on the principles of personal data collection and the expansion principle, we designed a data collection experience in which students first play a game to collect wastewater samples (see Figure 1) and then process their samples in a virtual lab (see Figure 2). In the initial design, students were given a limited amount of time to collect wastewater samples from as many districts as possible by driving a car around a map of Springfield. The goal was to simulate the real world challenges that scientists face in collecting data and to reinforce the role of collaboration. After collecting samples of wastewater, students enter a virtual lab in which they have an opportunity to get hands-on experiences with the scientific tools used to produce a data set measuring rates of flu from wastewater. To produce their data set, students pasteurize their samples, isolate flu RNA, and measure the amount of RNA.

Preliminary findings based on small group playtesting and early pilot tests in the classroom showed that students enjoyed taking up the role of a scientist, learning about “what real scientists do in a lab,” and “how water

samples are tested.” However, we also found that students needed more support to collaborate in the data collection game. Building on student recommendations, we added a planning stage and an evaluation stage. Prior to the game, student teams now orient themselves to the map of Springfield and the constraints of the game, including where each team member is located in relation to wastewater plants. They are prompted to work together to collect data from different regions of Springfield and enter the game only when the whole team is ready. After students collect their samples, they are shown a visual summary of their team’s data, which highlights districts they personally collected as well as districts collected by the rest of their team.

Figure 1
Initial Designs from the Data Collection Game.

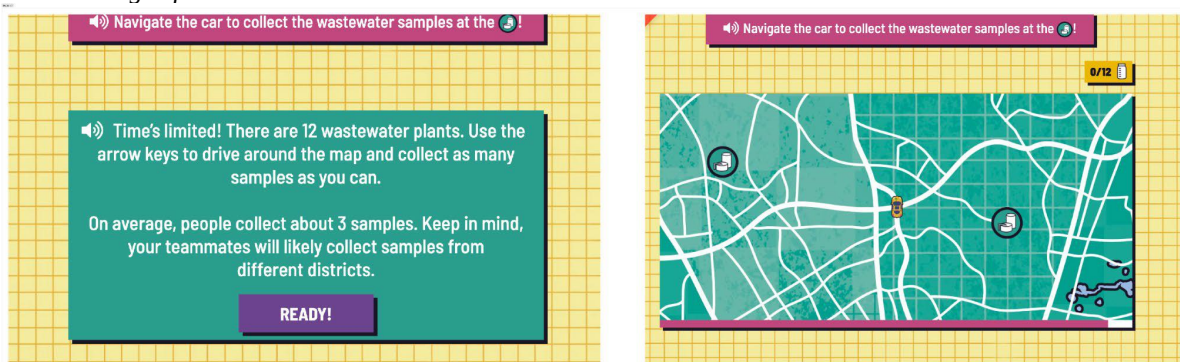
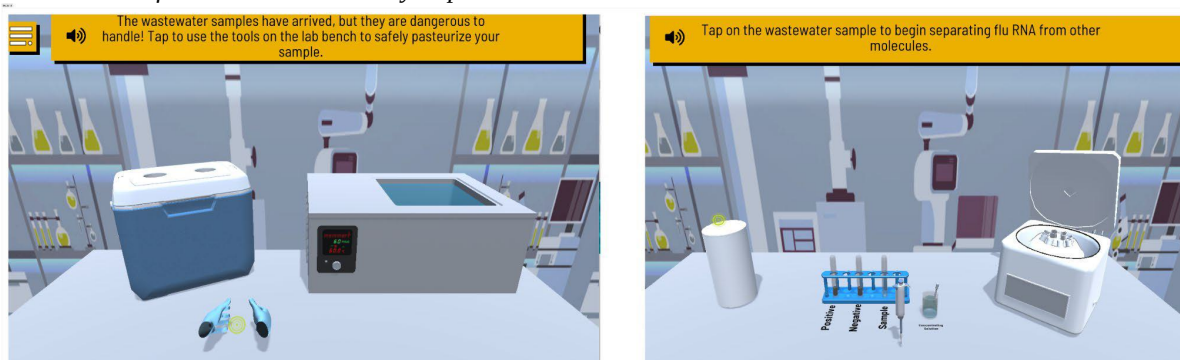


Figure 2
Screenshots of the Virtual Laboratory Experience.

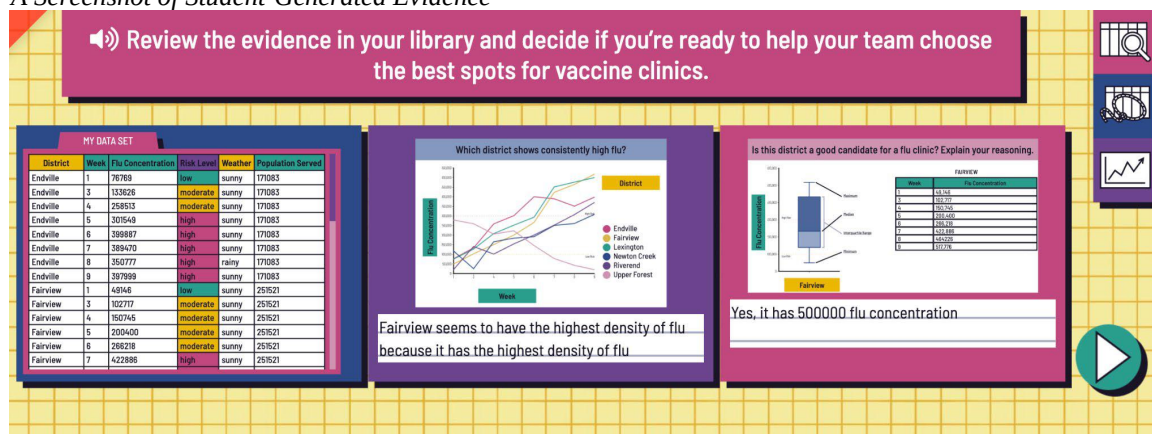


Visualizing and analyzing data

Data exploration in the initial design of *Trend Trackers* aimed to mirror the iterative practices of data scientists, offering students multiple entry points into data inquiry. Data scientists might alternate between wrangling, visualizing, and analyzing as they make observations and ask questions about their data. In *Trend Trackers*, students can choose from a selection of wrangling, visualization, and analysis activities in order to generate evidence. In *Wrangle Data*, students work to improve the quality of their data, addressing issues such as formatting errors and suspicious values. In *Visualize Data*, students construct different types of graphs to answer their investigative questions. In *Analyze Data*, students perform unit transformations, learn about measures of center, and use models to make predictions. In each data science practice, *Trend Trackers*’ activities highlight different data moves (Erickson et al. 2019). For example, in *Wrangle Data* students are encouraged to sort data; in *Visualize Data* students are prompted to group their data, and in *Analyze Data* students engage in calculating and transforming data.

Based on findings from early playtests and interviews with three middle school teachers, we found that students needed more support to wrangle their data. One teacher reflected that “kids trust numbers too much, and they skip that step, and they shouldn’t be able to skip that step,” because it impacts the accuracy of students’ visualizations and analysis. We also found that while students constructed multiple pieces of evidence (see Figure 3), they needed more support to ask and answer different kinds of statistical questions within a socioscientific context.

Figure 3
A Screenshot of Student-Generated Evidence



Co-constructing and communicating data-based arguments

Communicating with and about data includes making data-based claims, negotiating different perspectives through data, and communicating about data with different audiences. In *Trend Trackers*, students are challenged to construct at least two evidence-based claims that they can share with their team. In the initial design, we proposed that the features supporting the expansion principle in data collection might offer students opportunities to share new information about Springfield and co-construct claims as a team. We proposed that opportunities for dialogue and consensus-making would help students select appropriate data to use as evidence, provide sufficient evidence, and document their reasoning (Berland and McNeill 2010).

Early findings suggest that students valued opportunities to compare their findings with those of their peers. For example, one student shared that “it was nice to see what other people concluded from the experiment and to compare answers.” Prior research studies have found that designing for consensus-making in collaborative decision-making encourages students to adapt or refine their original claims based on new information (Felton et al. 2022). While students reflected on collaborative activities as a way to evaluate their own interpretations of data, more research is needed to investigate how *Trend Trackers* can support students in synthesizing data-based findings and negotiating potential solutions.

Interactive demo and discussion

In our demonstration, we will introduce *Trend Trackers*, presenting the design conjectures that have informed our work and initial findings from co-design and play-testing with youth and teachers. We will then engage the audience in a guided exploration of three activities in *Trend Trackers*: the data collection game, the virtual lab, and a data analysis activity. Discussion and feedback will be organized around the following questions: (1) In what ways does *Trend Trackers* offer youth opportunities to consider the context in which data are produced and used? and (2) In what ways might *Trend Trackers* support youth in understanding the role of a data scientist?

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Supporting Classroom Orchestration and Collaborative Programming with Jupyter Analytics

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Abstract: Jupyter notebooks have been widely adopted in STEM courses that have programming or data science integrated. Jupyter Analytics, a learning analytics system, has been built to provide teachers with visibility into student activity. In combination with its existing functions as an awareness and reflection tool, this demo illustrates how the new code-sharing feature can address the real needs of both instructors and learners within Jupyter notebooks. Specifically, we discuss its potential value in two use cases: (1) supporting teachers in orchestrating live-coding lectures, and (2) helping students merge and synchronize work in collaborative programming. This covers two common pedagogical contexts in higher education: teacher-led lectures and student-centered active learning environments with group projects or collaborative sessions.

Introduction and background

The integration of learning analytics (LA) into higher education promises actionable insight for instructors and timely support for learners. According to the Society for Learning Analytics Research (SoLAR, 2025), LA is “the collection, analysis, interpretation and communication of data about learners and their learning that provides theoretically relevant and actionable insights to enhance learning and teaching” (p. 1). The first step of data collection makes the learning process observable in ways that promote reflection and informed action. Taking awareness tools as an example, they can support learners in creating group awareness in collaborative tasks to regulate their work (Janssen et al., 2011); instructors can use them to monitor engagement, spot bottlenecks, and intervene at multiple levels (i.e., individual, group, class) (Holstein, et al., 2018; Alavi et al., 2009), leading to effective classroom orchestration (Dillenbourg, 2013).

In parallel, data science and programming have become more accessible and widely integrated into STEM curricula. Among the platforms and tools in use, computational notebooks like Jupyter notebooks stand out for their web-based versatility and ease of use as interactive textbooks, virtual demonstrations, and more. This convergence has motivated efforts to embed LA directly into Jupyter ecosystems. For example, Jupyter Analytics (Cai et al., 2025) provides an end-to-end toolkit that manages the entire data pipeline—from learner activity to teacher insights—covering data collection, storage, preprocessing, and visualization. JELAI (Valle Torre et al., 2025) extends the idea by coupling fine-grained LA with LLM-based tutoring inside Jupyter, enabling context-aware support and research on student-AI interactions.

Despite these advances, live teaching with notebooks still exposes two gaps. First, live code sharing during a session is inefficient. In practice, instructors push updates via Git or learning-platform links and students pull them—workable for asynchronous distribution but ill-suited to live lecturing where the class needs the latest cell immediately. Second, real-time co-editing in standard Jupyter requires the `jupyter-collaboration` extension and typically grants collaborators broad access to the same server environment; while commercial platforms (e.g., Deepnote, Databricks) offer polished co-authoring, adopting them would require migration effort and costs that many existing courses cannot accommodate.

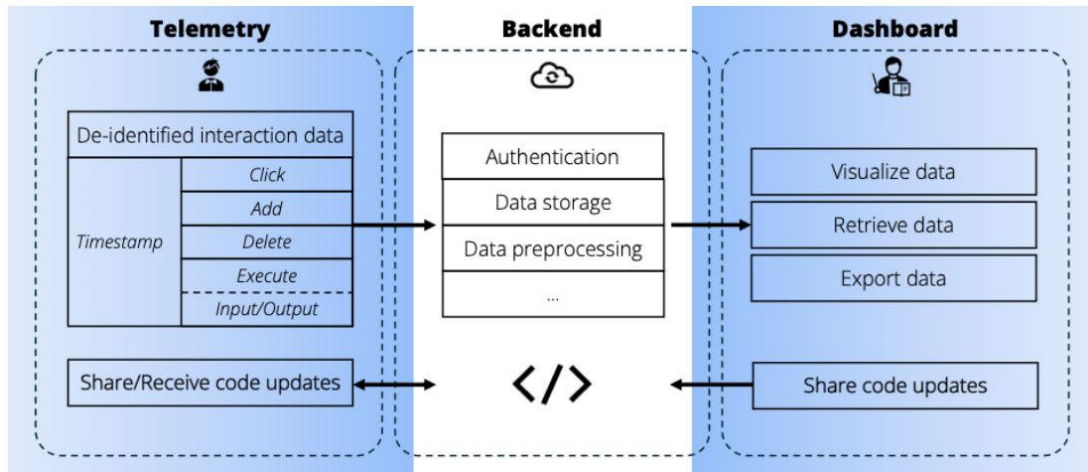
To address these needs, we extend Jupyter Analytics with a lightweight code-sharing channel that leverages its existing architecture. Instructors can broadcast code updates instantly to all online students; students can share updates within their assigned groups—without granting full workspace access or requiring a platform switch. This design preserves current Jupyter-based workflows, keeps orchestration within the instructor’s dashboard, and adds group-level sharing aligned with collaborative learning scenarios.

Tool Overview: Jupyter Analytics

As shown in Figure 1, Jupyter Analytics comprises of a containerized Flask backend and two JupyterLab extensions, Telemetry on the student side, and Dashboard on the teacher side. Two use cases on classroom orchestration and collaborative learning have been reported, justifying its values in ecological learning environments. Previously, its prototype version has also been demonstrated in the 14th Learning Analytics and

Knowledge (LAK) Conference, showcasing the key features that can be used to monitor both synchronous and asynchronous learning activities (Cai et al., 2024).

Figure 1
Overview of Jupyter Analytics



Beyond the public demo features, this release adds seamless code sharing between teachers and students and among students themselves. Figure 2 shows the key interfaces involved in the code-sharing workflow: the teacher (Fig. 2a) is preparing to push a code update in Part 2, while student 1 (Fig. 2b) and student 2 (Fig. 2c) are working on Part 1 and Part 3, respectively. The *Student Location* dashboard integrated into the notebook's table of contents displays each student's current location in real time, allowing the teacher to decide when to push updates on the spot. By right-clicking the code cell in Part 2, the teacher sees the "Push the Selected Cell" option at the bottom of the menu and can use it to send the update to all online students. In addition, this feature can be enhanced by enabling a plugin adapted from RISE—a JupyterLab extension that renders a notebook as a slideshow—allowing the teacher to share updates while in presentation mode as well.

Figure 2
Screenshots of the Code-Sharing Feature: (a) Teacher, (b) Student 1, (c) Student 2

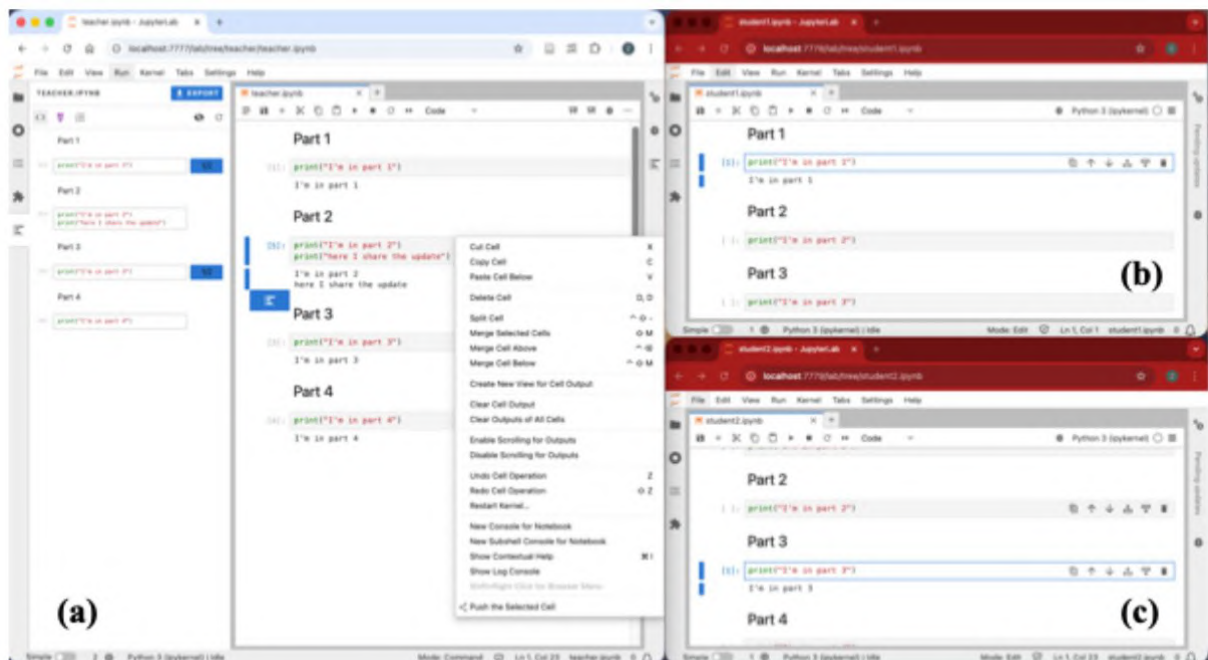
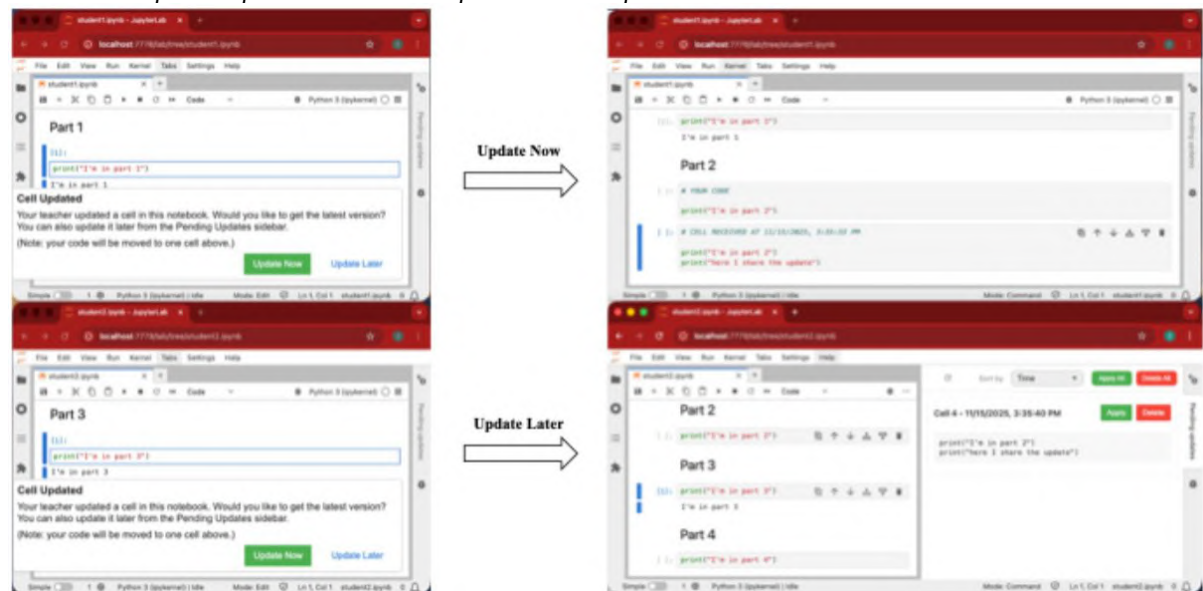


Figure 3 illustrates the process that occurs after the teacher shares the code. Both students receive a notification about the update, with two options: “Update Now” or “Update Later”. If they choose “Update Now”, their original code is moved to a cell above, and the cell replaced by the shared update becomes active. If they choose “Update Later”, they can keep working on their current tasks, and the update is stored in the *Pending Updates* sidebar, where they can apply or discard it later.

Figure 3
Screenshots of the “Update Now” and “Update Later” Options



Moreover, the workflows above also apply in a collaborative programming setting. Once group information is assigned to the class, students can share code updates with their teammates. This offers two key benefits: (1) the code-sharing feature streamlines collaboration by removing the need to switch platforms and manually merge work, and (2) the *Student Location* dashboard strengthens group awareness, supporting better social regulation.

Pedagogical scenarios

In STEM contexts, while students do write code in certain sections, the primary goal is not programming proficiency but rather immediate, tangible feedback on abstract ideas. Our tool is designed accordingly and supports a range of classroom orchestration patterns. Below, we describe how instructors can leverage the code-sharing and awareness features to facilitate different teaching activities with Jupyter notebooks.

Adapting to Pedagogical Goals. Instructors can adapt their code-sharing behavior depending on the activity at hand. When the goal is for students to observe and understand a concept, the instructor pushes complete code cells sequentially; students execute each cell and immediately see the output, allowing them to connect the result to the concept being discussed. When the goal is for students to engage more actively, the instructor pushes incomplete code with guiding hints, inviting students to complete portions of the code, modify parameters, or test hypotheses before running the cell. Although this may involve writing code, the focus remains on understanding system behavior rather than programming syntax.

Transitioning Between Activities. A typical orchestration cycle might proceed as follows. The instructor begins by pushing cells as students observe results that reinforce the lecture material. At a suitable breakpoint, the instructor pauses and prompts student participation, for example, by giving students five minutes to complete a function or vary inputs. During this phase, students experiment with the concepts by manipulating code, adjusting parameters, or both. To reconvene, the instructor pushes a reference solution and leads a discussion on the results and their conceptual implications.

Monitoring, Intervention, and Synchronization. The awareness dashboard allows instructors to monitor student progress without interruption. If the dashboard indicates that students are struggling, the instructor can push a corrective snippet to resolve technical issues quickly. This is particularly valuable in STEM settings, where debugging can distract from the lesson. At the end of each section, instructors can push the complete code to

ensure all students have identical, working notebooks. This synchronization ensures consistent outputs across the class, allowing students to reflect on the material rather than troubleshooting code discrepancies.

Interactive demo and technological setup

As part of this demo, we will present the core features of Jupyter Analytics, and its distinct interaction flows for teachers and students. Moving beyond the functional examples shown in Figures 2 and 3, participants will engage with a pedagogically grounded Jupyter notebook featuring a realistic classroom scenario—one that situates the need for classroom awareness and orchestration.

Hands-on Experience: Attendees will experience the system firsthand by taking on the role of either “student” or “teacher”, working through the notebook and interacting with other participants in real time. To facilitate this, participants will use their own laptops to access our university-hosted JupyterHub via temporary test accounts.

Research & Collaborative Design: We will share preliminary results from real classroom settings to demonstrate the research and practical significance of Jupyter Analytics, particularly regarding its code-sharing capabilities. We will also present outcomes from our participatory design process. Finally, we will invite participants to critique the tool’s design and collaboratively explore new ideas for its application in both research and practice.

Implications and future work

While Jupyter Analytics has been implemented in several university courses for exercise sessions, the new release reported in this paper expands its application to live lectures. Instructors can now synchronize their code with the entire class while demonstrating a workflow. This new capability is complemented by the awareness features (e.g., students’ locations or their error distribution) from the original design, which can help instructors gauge student progress and inform their decision-making (e.g., when to synchronize). Moreover, this combination of features also benefits students’ collaborative learning by enabling code sharing within the same platform. Finally, in both scenarios, instructors can benefit from the data exporting feature, which enables them to reflect on their teaching practices and engage in the teacher inquiry loop.

Future studies will focus on collecting in situ feedback from both teachers and students to iterate on the tool’s design. Additionally, we plan to conduct comparative analyses across multiple sessions for both the lecture and collaborative settings, exploring how the code-sharing feature impacts interaction dynamics within each context.

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Learning from the Experts: Exploring Human–AI Feedback through Expert-AI

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Abstract: This demo introduces Expert-AI, a Retrieval-Augmented Generation (RAG) system designed to emulate aspects of human experts' reasoning and feedback. Drawing on experts' writings, lectures, and interviews, the system provides dialogic, context-sensitive feedback grounded in expert perspectives, allowing experts to review and refine responses over time. We report preliminary findings from a pilot study in a business idea creation course at a private university in Japan, where students used both Expert-AI and general-purpose generative AI systems (e.g., ChatGPT). The findings suggest complementary roles: students used Expert-AI for conceptual framing and strategic evaluation, and general-purpose generative AI for elaboration and implementation planning. These findings suggest learners' interpretations of AI feedback may vary depending on how it is presented. Accordingly, the demo presents the text version of Expert-AI alongside early multimodal prototypes including voice and android embodiments to examine how interaction modality shapes perceived credibility, empathy, trust, and engagement in expert-like AI feedback.

Introduction

Research on expertise has long shown that experts differ from novices not only in how much they know but also in how that knowledge is organized, accessed, and applied (Chi et al., 1981; Ericsson et al., 1993). Hatano and Inagaki (1986) distinguish adaptive experts—individuals who flexibly devise new procedures and restructure their knowledge when confronted with unfamiliar situations—from routine experts, who perform well in familiar contexts but struggle when task demands change. Adaptive experts understand why strategies work, under what conditions they apply, and when new methods must be invented (Schwartz, Bransford, & Sears, 2005). This suggests that expertise involves continual refinement through reflective and flexible thinking rather than mere automatization of skills.

Yet opportunities for learners to observe and internalize such expert thinking processes remain limited. Expert reasoning and tacit decision-making are often invisible in educational contexts. Frameworks such as Cognitive Apprenticeship (Collins et al., 1989) attempt to make expert reasoning visible through modeling and guided reflection, but these depend on sustained expert presence and are difficult to scale.

Recent advances in generative AI raise the possibility of extending access to expert-like reasoning without requiring experts to be physically present. Motivated by this potential, we explore whether Retrieval-Augmented Generation (RAG; Lewis et al., 2020) can surface tacit knowledge, reasoning patterns, and decision processes that characterize adaptive expertise. Building on this idea, we developed Expert-AI, a system designed to reconstruct expert cognition using experts' writings, lectures, and interviews. Rather than serving as a simple information source, Expert-AI simulates aspects of learning from an expert by offering personalized, context-sensitive feedback grounded in authentic expertise. This demo investigates how learners perceive and interpret such feedback across different modalities.

Related work

Tacit knowledge in expert performance

Expert performance draws on structured domain knowledge and context-sensitive reasoning (Chi et al., 1981; Ericsson et al., 1993). Tacit knowledge (Polanyi, 1966) and reflective practice (Schön, 1983) highlight that much of expert cognition is experiential, embodied, and difficult to articulate. Expertise involves adaptive problem solving and refined judgment—skills that are challenging to observe directly. Expert-AI seeks to externalize these implicit cognitive patterns and make them accessible to learners. Because such processes rarely surface in authentic expert–novice interactions, many learners lack opportunities to observe how experts reason through uncertainty, reframe problems, or evaluate competing options.

Building on these perspectives, we conceptualize “knowledge” not only as propositional content but also as situated reasoning practices, tacit judgments, and ways of participating in expert discourse. Accordingly, Expert-AI is not merely a tool for knowledge transmission, but a medium through which learners can engage with and appropriate expert-like ways of thinking. Learning, in this context, is understood as a transformation in how learners frame problems, generate questions, and reflect on their ideas, rather than simply acquiring correct answers. Evidence of such learning may be observed in the emergence of more generative questions, deeper reflective dialogue, and shifts in how learners interpret and extend feedback.

How learners access expert thinking

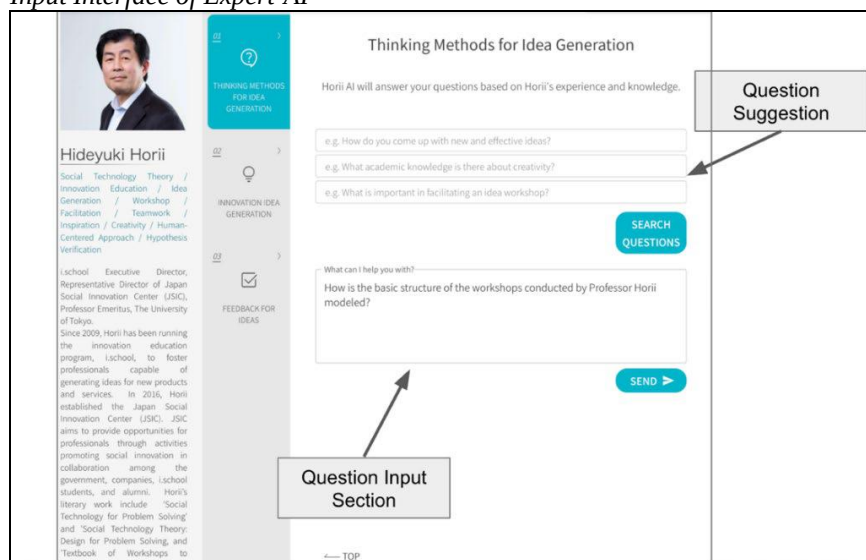
Learning with experts requires participation in expert discourse rather than merely receiving information (Lave & Wenger, 1991). Cognitive Apprenticeship (Collins et al., 1989) aims to make expert reasoning visible through modeling, coaching, and guided reflection. Yet opportunities for direct interaction with experts remain limited due to scarcity of time and availability. Prior technological approaches—such as intelligent tutoring systems and rule-based expert models—have attempted to simulate expert reasoning but often capture only explicit procedures, failing to represent tacit, experience-based aspects of expertise. This gap has motivated interest in systems capable of representing both explicit and tacit expert insights through interactive dialogue.

Feedback plays a central role in learning by helping learners understand the gap between current performance and desired goals, although its effectiveness varies across contexts (Hattie & Timperley, 2007). Its impact depends heavily on how learners interpret feedback and the extent to which they trust its source (Carless & Boud, 2018). Recent studies on generative AI highlight both opportunities and challenges in producing feedback that is perceived as credible and empathetic (Lee & Moore, 2024). Building on these insights, Expert-AI investigates how learners perceive and engage with expert-like AI feedback, particularly across different interaction modalities.

System design for expert-like feedback

Expert-AI is a Retrieval-Augmented Generation (RAG) system (Lewis et al., 2020) designed to emulate the reasoning and communication style of human experts across diverse domains. It is intended for learning contexts with no single correct answer—such as open-ended problem solving and creative decision-making—where learners can benefit from access to diverse expert perspectives and reflective feedback. The system combines retrieval from a curated corpus of expert materials with generative synthesis, producing individualized feedback grounded in authentic expertise. Users interact with the system by selecting a topic or submitting a free-text question or idea through a web-based interface (see Figure 1). Upon submission, the system retrieves semantically relevant passages from a curated corpus of expert materials—including books, unpublished notes, and original interviews—from experts selected based on recognized professional experience. Many of these materials are not publicly available online, and the system synthesizes responses that reflect the expert’s reasoning style.

Figure 1
Input Interface of Expert-AI



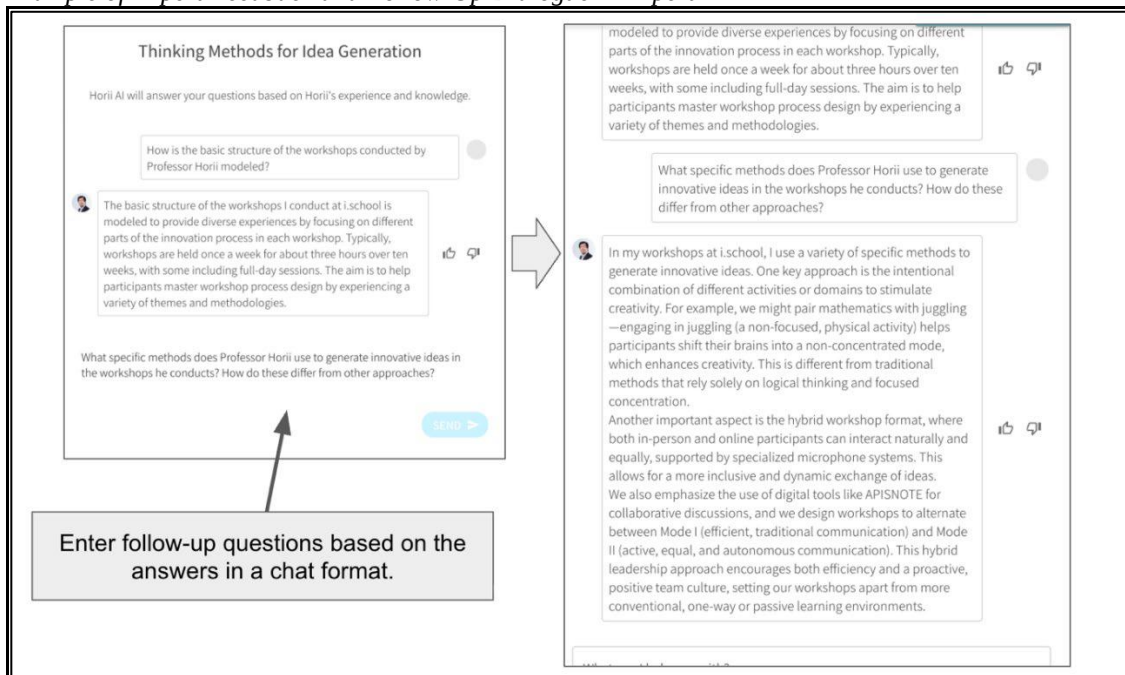
Users can select a topic or submit a free-text question or idea.

Once the query is processed, Expert-AI provides text-based feedback that integrates theoretical reasoning with the expert's practical perspective, enabling learners to reflect on both the logic and the stance underlying the feedback (see Figure 2). Learners can continue the interaction through follow-up questions in a chat format, allowing them to probe deeper into the expert's reasoning process. This dialogic structure supports sustained engagement and scaffolds learners' inquiry.

The system supports both Japanese and English inputs and outputs. While the expert corpus primarily consists of Japanese-language materials, retrieval is conducted across languages using semantic similarity, and responses are generated in the user's input language through LLM-based translation and synthesis. Although cross-lingual interaction is supported, potential differences in retrieval accuracy across languages remain an area for future investigation. A distinctive feature of Expert-AI is its collaborative update mechanism: experts themselves can review and refine the AI's responses or add new reflections, turning the system into a living archive that evolves with their ongoing insights.

Figure 2

Example of Expert Feedback and Follow-Up Dialogue in Expert-AI



Future prototypes will extend this text-based interaction into multimodal feedback—including video and android embodiments—to investigate how embodiment and social presence shape learners' engagement and trust (Abrahamson & Lindgren, 2022).

Pilot study and preliminary findings

This pilot study examined how Expert-AI and general-purpose generative AI (e.g., ChatGPT) function differently within a business idea creation course at a private university in Japan. The course is an undergraduate–graduate integrated lecture, in which a total of 19 students (12 undergraduate and 7 graduate students) received course credit. Participants were provided access to both Expert-AI and ChatGPT and were asked to reflect on their use of these systems in their idea generation and evaluation process.

Analysis of student reports revealed distinct patterns in how the two AI systems were utilized. Across multiple reports, students described using Expert-AI during the early stages of ideation, particularly when identifying core concepts, framing problems, and evaluating the strategic direction of their ideas. In contrast, ChatGPT was more frequently used in later stages to elaborate business models, organize information, and develop implementation plans. Qualitative analysis further suggested differences in how learners perceived the feedback provided by each system. Students tended to perceive Expert-AI as offering more structured, decisive, and insight-driven responses. For instance, one student noted that "ChatGPT provided extensive information, but the conclusion was often buried," whereas "Expert-AI offered clearer judgments and more actionable insights." In particular, Expert-AI responses were valued for introducing perspectives that extended beyond students' initial ideas, such as phased implementation strategies or concrete approaches to differentiation. By contrast, ChatGPT

was recognized for its breadth of information and its ability to support detailed elaboration, although its feedback was sometimes perceived as generic or reiterative of already-identified challenges.

These preliminary findings suggest that Expert-AI and generative AI serve complementary epistemic functions in learning. An effective pedagogical approach is therefore to sequence their use: first leveraging Expert-AI to shape high-level concepts and strategic framing, and then using generative AI to expand, structure, and operationalize those ideas. This staged integration may support both originality and feasibility in student outputs.

For future educational design, these findings highlight the importance of orchestrating multiple AI systems based on their distinct cognitive roles, enabling learners to engage in both conceptual exploration and practical elaboration in a more structured manner.

Implications and future directions for multimodal learning interaction

Building on the pilot findings, an important limitation lies in the current interaction modality. In the pilot study, interactions with Expert-AI and generative AI were conducted exclusively through a text-based web chat interface. While this enabled a clear comparison of their epistemic roles, it does not account for how different modalities may influence learners' perceptions and experiences.

The pilot results also indicated that learners' perceptions of clarity, usefulness, and actionability—factors closely related to perceived credibility and trust—differed between the two systems. This suggests that such perceptions are shaped not only by the content of feedback but also by how that feedback is represented and delivered, highlighting the importance of interaction design in AI-mediated learning.

More broadly, these findings point to the potential of Expert-AI as a form of distributed mentorship, in which expert cognition is translated into a reflective, dialogic format that learners can interact with. By enabling access to representations of authentic expert reasoning, Expert-AI may preserve the depth of expert thinking while enhancing scalability and accessibility. At the same time, this raises important theoretical questions in the learning sciences regarding how trust, credibility, and perceived empathy are constructed in human–AI interaction.

These considerations motivate the need to examine the role of modality more explicitly. Multimodal representations—such as voice, video, or embodied agents (e.g., androids)—may significantly influence how learners interpret feedback, relate to the system, and sustain engagement. To explore these questions, the demo session is designed as an experiential extension of the pilot study, allowing participants to compare different modalities of Expert-AI interaction and reflect on their impact on learning.

Demo session description

This interactive demo session is designed as an experiential extension of the pilot study, enabling participants to explore how different interaction modalities influence their perception of Expert-AI.

The session consists of the following components:

- **Introduction and Concept Overview:** Participants are introduced to the design of Expert-AI as a RAG-based system that reconstructs expert cognition using curated writings and interviews from Japanese experts across domains such as academia, art, and business. The session situates Expert-AI within the broader goal of supporting reflective, expert-like feedback in learning contexts.
- **Interactive Demonstration (Text-Based Interaction):** Participants first engage with the web-based chat interface used in the pilot study. They submit short idea proposals or questions and observe how Expert-AI retrieves relevant materials and generates structured feedback. This serves as a baseline experience aligned with the pilot findings.
- **Multimodal Prototype Experience:** Participants are then introduced to alternative modalities of Expert-AI interaction, including voice-based systems and an embodied android prototype. These experiences allow participants to directly compare how the same underlying expert reasoning is perceived differently when presented through text, voice, or physical embodiment, highlighting the role of modality in shaping interaction.
- **Comparative Reflection and Discussion:** The session concludes with a facilitated discussion on how multimodal Expert-AI systems could support distributed mentorship in educational settings, particularly in relation to trust, credibility, and learner engagement. The discussion also addresses how future research can systematically investigate the role of modality in shaping human–AI learning interaction.

Through this experience, participants will gain insight into how the representation of expert knowledge—whether through text, voice, or embodied agents—affects learners' cognitive and affective responses, and how multimodal AI systems may transform the design of reflective and dialogic learning environments.

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“Where Do I Start?” A Tool for Interactive Literature Discovery in the Learning Sciences

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Abstract: The rapid expansion of Learning Sciences scholarship makes it challenging for newcomers to navigate diverse theories, methods, and contexts. We present the Interactive Literature Review Database Template, an open-source tool that turns any corpus into a dynamic, filterable database. As demonstrated by Journal of the Learning Sciences articles and student-built instances, the tool enables sustainable, transparent, and easily updated literature exploration while lowering technical barriers for researchers.

Introduction

The field of Learning Sciences has grown substantially over the past three decades, producing a rich but increasingly expansive body of scholarship. For newcomers to the learning sciences—many of whom are still developing familiarity with the field’s theoretical traditions, methodological approaches, and evolving areas of inquiry—navigating this landscape can feel overwhelming. Traditional keyword searches often presuppose that readers already “know what they don’t know,” yet, as one of our collaborators noted, “*Sometimes you don’t know what you don’t know. And if I don’t know, how am I supposed to search for it?*” Across our research groups, this sentiment reflected a broader challenge: newcomers to the learning sciences frequently lack accessible, structured tools for exploring disciplinary literature in ways that make underlying patterns, debates, and trajectories visible.

To address this challenge, we developed the Interactive Literature Review Database Template, an open-source, configurable web tool designed to support scholars’ discovery, exploration, and synthesis of research. The tool enables authors to transform any corpus—such as the full set of Journal of the Learning Sciences (JLS) articles used in our demonstration—into an interactive, filterable database organized around meaningful dimensions, including theoretical frameworks, methodologies, study contexts, pedagogical models, and constructs. Unlike traditional systematic reviews that culminate in static narrative summaries or meta-analytic findings, our approach pairs systematic review methods with a dynamic, publicly accessible database that evolves alongside the literature it represents. We envision this tool as a living & crowdsource-able resource to support newcomers to the field and scaffold more transparent and sustainable review practices.

In this demo, we will introduce the design and functionality of the Interactive Literature Review Database Template, showcase the JLS-specific implementation created through a systematic tagging and review process, and invite participants to explore the tool hands-on. We will also highlight how the template can be adapted to other corpora and research areas, offering attendees a practical pathway for integrating interactive literature databases into their own scholarly workflows.

Aims and rationale

Despite the abundance of digital tools available for searching academic literature, two critical groups in the Learning Sciences community remain underserved: (1) scholars-in-training who do not yet know what to search for, and (2) researchers conducting systematic literature reviews whose work quickly becomes outdated, difficult to maintain, and non-interactive. The Interactive Literature Review Database Template was designed explicitly to support these needs.

Supporting LS newcomers’ discovery of literature

For early-career scholars, entering the Learning Sciences means navigating a domain with multiple theoretical traditions, diverse methodological approaches, and highly contextualized phenomena. Traditional keyword searches—whether conducted through library databases, Google Scholar, or increasingly through large language models—assume that users already know the appropriate terminology for their area of inquiry. This epistemic challenge is not merely a matter of inexperience; it reflects the deeply interdisciplinary nature of the field, in which different research communities employ distinct vocabularies to describe similar constructs, contexts, or analytic traditions.

The interactive database provides a scaffold for this early-stage learning process. By enabling users to browse articles through structured filters—frameworks, methods, study contexts, pedagogical models, and constructs—it surfaces the field's conceptual architecture without requiring prior expertise. Users can move fluidly across categories, noticing patterns in theoretical commitments, methodological choices, or the evolution of research across decades. In this way, the tool supports both literature discovery and keyword discovery, helping scholars build the language necessary for deeper engagement.

Addressing the limitations of traditional systematic reviews

The second underserved use case concerns researchers conducting systematic literature reviews. Traditional reviews require considerable time and coordination, often taking months or years to complete, only to become outdated shortly after publication. Moreover, once a systematic review is published, readers must rely on static narrative summaries or tables that flatten the complexity of the underlying research landscape. These formats limit readers' ability—especially for those unfamiliar with the content area—to explore the corpus interactively or identify specific articles aligned with their interests.

The Interactive Literature Review Database Template offers an alternative: a maintainable, updateable infrastructure for systematic reviews that can grow alongside the literature. Because the template relies on lightweight configuration (a single JSON file and a CSV of items), research teams can easily update or extend their databases as new publications emerge. Rather than producing a one-time snapshot, reviewers can create a living systematic review—one that is not only transparent in its tagging categories and inclusion criteria but also navigable by readers with varying degrees of familiarity with the domain.

Considerations of citational justice

As we create tools to enable the discovery of literature, we must acknowledge the historical and current harms that such research has inflicted. As social sciences researchers, our work represents (or misrepresents) the lived experiences of those we study – and excludes those that we don't. Learning and knowledge-production are non-neutral acts. We recognize that this software, the corpus of literature it references, and the unwritten manuscripts it will inform are not objective or politically agnostic. Rather, these products carry the same biases of the humans that designed them and reflect the agendas of government, military, and/or for-profit institutions that funded them.

By including articles in our databases, we are enabling their discovery and dissemination to field newcomers as 'canon' knowledge to be cited. Citational justice scholarship argues that modern academic citation systems operate under the assumption of meritocracy, without attending to how such practices reproduce existing power structures and hierarchies (Kwon, 2022), thereby perpetuating a culture of epistemic injustice in which those with prestige are legitimized as knowledge makers (Gonzales et al., 2024). These scholars call for practices such as corrective citation, citation auditing, including under-cited groups in reference lists, valuing and crediting contributions, and the co-construction of citation criteria with marginalized communities, to name a few (Pereira, 2025; Sauvé, 2025). Indigenous scholars have pointed out that universities play a role in both perpetuating and producing knowledge that supports settler-colonialism (e.g., Grande, 2018). Within this, the call from Palestinian academics for the academic boycott of Israel, modeled after academic boycotts of Apartheid South Africa, highlights that research partnerships and funding models can serve the purposes of normalizing harmful institutions, such as settler-colonial states (PACBI, 2014).

These efforts should push us beyond journals like JLS and possibly traditional academic publishing, towards prioritizing publicly accessible sources (Levendowski, 2025). Other actions can include supporting initiatives to disrupt systems of oppression, of which citational injustices are a part, both inside and outside the university. As such, inclusion in our databases is not meant to be a show of support for a particular work; rather, we hope that this tool increases the discoverability of research for scholars based on topical rather than algorithmically determined relevance. This tool offers an alternative to automated search engines and large-language models that demonstrably privilege Anglophone, cisgendered, racialized white perspectives (Arenas-Castro et al., 2024; Hawkins, 2021), reify existing hierarchies based on citation counts (Kwon, 2022; Sauvé, 2025), and inaccurate/fabricated citation generation (Walters & Wilder, 2023).

Software

At its core, the template is a plug-and-play web repository that requires only two editable files: a single JSON configuration file and a CSV file containing the corpus. All functional logic—HTML, CSS, and JavaScript—is encapsulated in the template, allowing authors to customize, deploy, and update their databases without any programming expertise. This philosophy of lightweight, accessible configuration makes the tool suitable for a wide range of scholars, including those who are new to literature reviews or who wish to provide transparent,

interactive supplements alongside formal publications. By lowering the technical bar, the template supports the creation of living literature reviews that remain relevant and maintainable long after publication.

At the time of submission, the template has undergone three releases (see Table 1):

Table 1
Sample Template Instances

Instance	Version / Access
Medical Specialty Camp Research Database ↳ Gerth, 2025	[alpha] https://jonathangerth.github.io/database.html
JLS Interactive Database ↳ Led by author Vickery & supported by authors Bai, Beyersdorf, Sheng, & Desai	[beta] https://morganavickery.github.io/JLS-interactive-database/
Build Your Own Int. Database Template [v1.0] ↳ Repository & ReadMe with implementation instructions	[v1.0] https://github.com/morganavickery/literature-review-database-template
Graduate student-led mini exploratory reviews ↳ participatory design literature (Bai) ↳ social design in higher education (Beyersdorf) ↳ LLM-supported collaboration (Sheng) ↳ youth interpretation of evidence (Desai)	[forked from v1.0] To be demoed and presented live during ISLS 2026
BYO Interactive Database [v2.0, in progress] ↳ In-progress features include dynamically generated data visualizations & reports exporting ↳ Development led by authors Vickery & Chakraborty	[v2.0] To be demoed and presented live during ISLS 2026

The alpha build of the template has been deployed using an existing corpus aggregated during a systematic review on research conducted by a collaborator on summer camps for youth with disabilities (see Gerth, 2025). This release was used in early usability testing to refine interface features and tagging workflows and demonstrate the cross-disciplinary portability and value for non-LS systematic reviews.

The beta release was used during deployment in parallel with a systematic review of all JLS publications (see Figure 1; publication in progress). Articles were manually tagged across five dimensions—frameworks, methods, study contexts, pedagogical frameworks, and constructs—using a collaboratively developed review procedure (linked from the live JLS Database instance). This implementation sought to demonstrate scalability, stress-test the dynamic filtering, and assess the potential to explore longitudinal trends in LS.

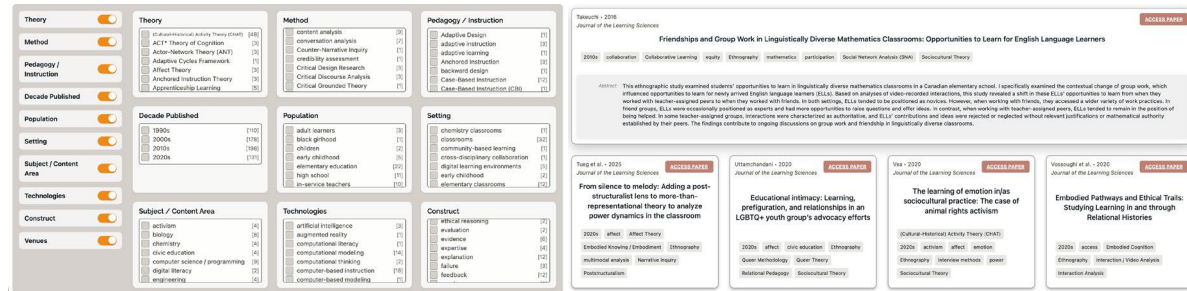
Considering our aims to develop a tool that was context-agnostic, the first author (an early-career learning sciences researcher and the lead software developer) facilitated a workshop for authors 2-5 (all learning sciences graduate students in their first 2 years of study), where each student was tasked with configuring their own instance of the database (using release version 1.0). Here, each student conducted their own exploratory literature review on a niche topic of their interest to populate a smaller (relative to the JLS article corpus) literature database. This workshop functioned to (1) test for usability and accessibility issues, (2) refine the onboarding materials for clarity, and (3) reveal the limitations of the software across varying levels of user expertise, familiarity with their literature, and disciplinary focus.

The template renders each literature item as an interactive tile that displays customizable metadata (e.g., title, authors, year) and provides 1-click access to the article. Users can filter, sort, or combine categories across five tagging dimensions—theoretical frameworks, methodological approaches, study contexts, pedagogical frameworks, and constructs—providing multiple avenues of discovery even for those without prior knowledge of relevant keywords.

Figure 1

JLS Instance Screenshots

Screenshots of (left) Dynamic Filters from the JLS Database Instance & (right) Expanded/Collapsed Article Cards



The tool's architecture includes several features that enhance both usability and transparency:

- **Dynamic, multi-filter navigation:** Users can apply multiple filters simultaneously, generating live cross-tabulations of the corpus that reveal patterns across methods, theories, contexts, and constructs.
- **Browsable tag lists with live frequency counts:** Rather than requiring users to guess keywords, the interface surfaces all available tags and their relative prevalence, allowing scholars to identify co-occurring concepts or underexplored topic areas.
- **Automatic generation of filters based on corpus structure:** The tool infers its filter categories directly from the CSV file, ensuring the interface adapts to the specific needs of any imported corpus.
- **Flexible extensibility:** Although designed with learning sciences corpora in mind, the architecture can be applied to any domain in which literature needs to be sorted, visualized, and explored interactively.

Notably, keyword-search functionality was intentionally deprioritized. Existing scholarly databases already excel at keyword searches, but they do little to scaffold exploration for users who lack preexisting knowledge of the keywords associated with their area of interest. Instead, the template emphasizes structured discoverability—surfacing the conceptual organization of a field in ways that invite browsing, comparison, and curiosity-driven exploration.

Interactive demo session plan

Session Component	Description	Time
Vision & goals	Introduce the purpose of the Interactive Literature Review Database Template and the broader vision of supporting literature discovery, sustainable review practices, and accessible scholarly entry points for in-training LS researchers	5 min
Interactive demo of JLS instance	- Live walkthrough of the JLS implementation, including an overview of the systematic review process, tagging schema, and design decisions that enabled dynamic filtering and cross-category exploration. - Attendees explore the JLS database independently. Presenters circulate (or support virtually) to answer questions, demonstrate combinations of filters, and help participants interpret emergent patterns.	8 min
Instance showcase	Short showcase of how student researchers have used the template in diverse domains	1-2 min each
“Create your own instance” walkthrough	- Provide links to the GitHub repository, documentation, and onboarding materials so participants can clone or adapt the template after the session. - Step-by-step overview of customizing the JSON configuration, editing the CSV, and deploying or updating a new database.	8 min
Development roadmap, feedback, & discussion	- Preview of upcoming enhancements, including tag-based search, expanded crosstabs/data visualizations, and additional filtering capabilities. - Elicit feedback & feature requests in discussion / via a QR-code form	Remaining time

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Virtual Worlds as Ethical Laboratories: Exploring the Risks and Benefits of AI in *Agents of ViTAL: Ethics Missions*

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Abstract: *Agents of ViTAL: Ethics Missions* is a narrative-centered learning environment where middle school students explore the ethical risks and benefits of AI technology in a virtual school. In this demo session, we will give an overview of *Agents of ViTAL*, invite participants to play an activity where they collaboratively rank risks and benefits, and seek feedback on the conversational agent, EthicsBot, embedded in the activity interface.

Introduction and context

It has become commonly accepted that “AI literacy” includes ethics and understanding social impact as a core component (Ng et al., 2021). However, the consensus largely ends there. Although ethical reasoning is considered essential to AI literacy, scholars and practitioners still have significant work to do in identifying what it means to incorporate these dimensions in curriculum and pedagogy (Ma et al., 2025; Chang & Wong, 2025; Wiese et al., 2025). This challenge is further complicated by the fact that ethics instruction itself is often reserved for older students, with formal courses in ethics historically offered only at the tertiary level (Mohr Lone, 2012). Given that there is agreement that (AI) ethics are important and yet that there is a gap in our current understanding of what this means or how instruction should look, especially for K-12 learners, we explore the potential for game-based learning as an opportunity to engage learners in agentic ethical decision-making.

This demo paper presents a narrative-centered learning environment (Lester et al., 2012) intentionally designed to support collaborative ethical reasoning, where students are positioned as influential figures in a school’s AI adoption decisions. As they navigate three dimensions of AI ethics, including privacy, bias, and environmental impact, through authentic scenarios and immediate feedback, students collectively grapple with the messiness of real-world decision-making, ultimately reaching consensus on whether their school should implement classroom monitoring technology. This demonstration exemplifies the ISLS 2026 meeting’s theme “partnering with purpose” by engaging students as co-designers alongside researchers and practitioners. We have been iteratively developing the learning environment, *Agents of ViTAL*, with local middle school students and incorporating their feedback on player characters, EthicsBot features (an in-environment AI-based conversational agent), and ethical dilemmas that felt most relevant, ensuring the experience is grounded in learner identity and voice. Our project’s advisory board, composed of education administrators and teachers, has also provided feedback on interaction design and facilitating classroom implementation feasibility. Additionally, we have sought disciplinary expertise from philosophers throughout the design process to ensure that our treatment of “ethics” is grounded in philosophical practice and that students engage in authentic, disciplinarily-relevant tasks relating to ethical reasoning.

Design principles

Computer-supported collaborative learning (CSCL) enables students to build shared knowledge through meaningful collaboration; this is particularly vital in ethics education, where students have to wrestle with complex dilemmas that demand integrating multiple perspectives that can often be at odds with one another. Extensive research demonstrates that CSCL fosters learning gains, enhances critical thinking, and sustains engagement (e.g., Saleh et al., 2022; Vogel et al., 2017). Our design draws on the synergistic principles of CSCL, emphasizing how technology, pedagogical design, and collaboration work together to support student content learning and ethical reasoning (Hmelo-Silver et al., 2023):

1. **Technology:** *Agents of ViTAL* creates an immersive learning experience by situating students in a virtual school context with real-world relevance (Gee, 2008). Scenes function as dynamic simulations that provide immediate, scenario-based feedback responsive to student choice. For instance, when students decide whether the school's AI feedback tool should store its data locally or on cloud servers, immediate consequences and stakeholder responses are presented. The Ranking Challenge Board serves as a *boundary object* (Star, 2010), visually representing the group's ethical stances and supporting collaborative completion of the joint task. The integrated chat box and the EthicsBot, a conversational agent designed as a thought partner, deliver just-in-time support when students need to negotiate and justify their ranking decisions.
2. **Pedagogical design:** *Agents of ViTAL* is grounded in collaborative inquiry learning, enabling students to collectively build knowledge driven by a shared goal within authentic problem contexts (Slotta et al., 2018). Three design features support this approach: (1) **Authentic problem context:** The overarching narrative--helping the school principal in making a technology adoption decision-- provides relevance grounded in student life. Game narrative events, like a ransomware attack, are based on reports of real-world occurrences and provide a sense of urgency which motivates sustained investigation. (2) **Guided instruction:** Guidance is essential for effective inquiry-based learning (de Jong et al., 2023). Characters (e.g., the principal, HelperBot, and EthicsBot) guide students through navigating the game and building reasoning from provided evidence resources, while maintaining genuine agency in decision-making. The HelperBot also provides adaptive guidance when students encounter questions. (3) **Dynamic feedback:** Timely, multi-source feedback supports both deeper learning and student ownership of reasoning (Jeong & Hmelo-Silver, 2016). Our design embeds feedback at multiple moments: immediate consequences responsive to student choices, peer feedback in their Ranking Challenge discussion, and expert ethics feedback for content support. Students must weigh competing variables and tradeoffs within and across three ethical dimensions (privacy, bias, environmental impact). For instance, when configuring the AI feedback tool for the principal, students must decide whether the interaction data is stored locally or on cloud servers, a decision dynamically linked to the school's limited budget, requiring them to justify ethical tradeoffs grounded in evidence.
3. **Collaboration mode:** *Agents of ViTAL* drives engagement in synchronous interaction through the Ranking Challenge Task and integrated chat box. During the group consensus activity, students engage in shared meaning making by articulating pros and cons, reconciling different perspectives, and collaboratively arriving at a group decision. The chat box supports co-regulation, enabling peer idea negotiation and argument refinement. Additionally, students are encouraged to talk face to face.

Together, these three pillars create conditions where middle school students can engage in authentic reasoning about complex AI ethics issues while maintaining agency and benefiting from timely, adaptive support.

Technological setup

Agents of ViTAL is a browser-based learning environment that uses the Unity game engine and Ink scripts (Inkle, n.d.) to drive the unfolding narrative with branching dialog. Time-stamped log records are created for each action within the learning environment and used for analysis of students' interactions and collaboration with group members.

Learning environment story overview

Students act as agents for the Virtual Technology Assessment Laboratory (ViTAL), tasked with evaluating a new AI feedback tool called PresenterPal in a virtual middle school. On their mission, students gather perspectives from different stakeholders (principals, teachers, students, and IT staff) and decide how to configure the technology within the school's limited budget by weighing each option for its ethical implications, along with its effectiveness and cost. Upon returning to the agency hub space, agents must reach consensus with their group about which aspects of the tool were most beneficial and which were riskiest. Finally, agents prepare a report detailing their ethical analysis to share with the middle school principal.

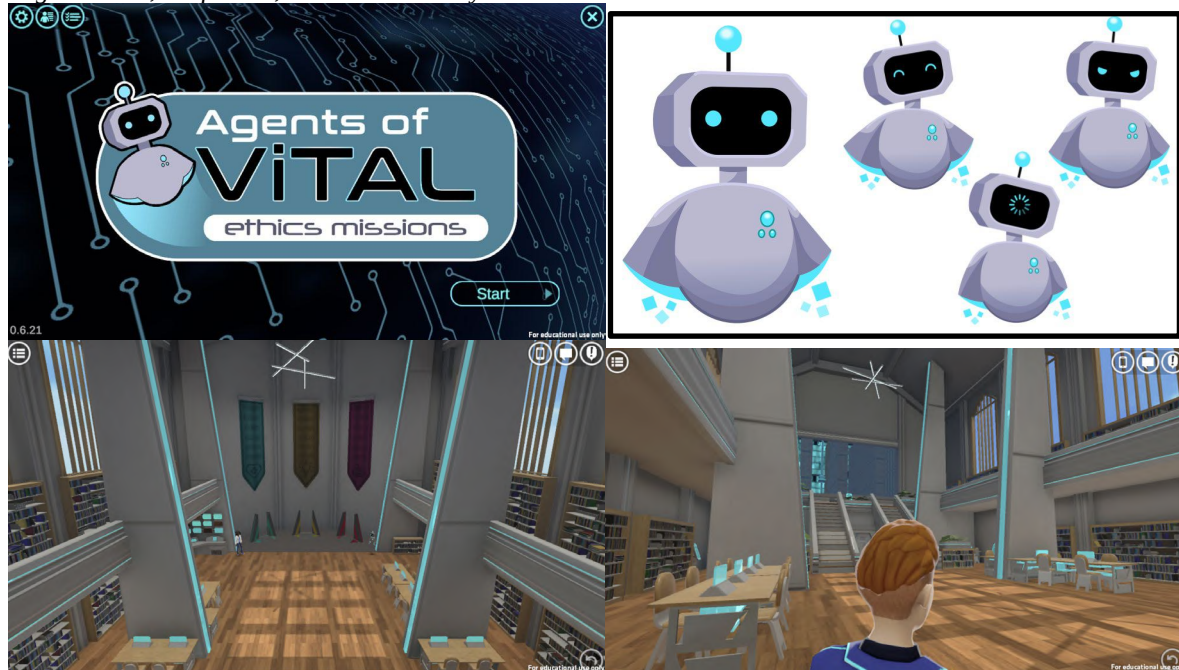
Key elements

Part 1: Agent orientation in the laboratory

Upon logging in to *Agents of ViTAL* and selecting their avatar, students enter the hub space called the "Laboratory" for its combined function as library and virtual laboratory for ethical experimentation. There, they meet the HelperBot (see Figure 1), who orients them to basic game mechanics and directs them to find Professor Mugo, an

ethicist who explains the student's role as an agent of the Virtual Tech Assessment Laboratory (ViTAL) and delivers a mission brief to the student.

Figure 1
Login Screen, HelperBot, and The Libratory



Part 2: Travel to Simulation Middle School, “Home of the Battling Bots!”

After meeting HelperBot and Professor Mugo, players embark on their mission to Simulation Middle School, passing through a portal from the Libratory into the virtual school (see Figure 2). While at the school, students gather artifacts, information, and perspectives--ranging from very enthusiastic to more skeptical of the AI tool--from students, teachers, an IT staff member, and the Principal.

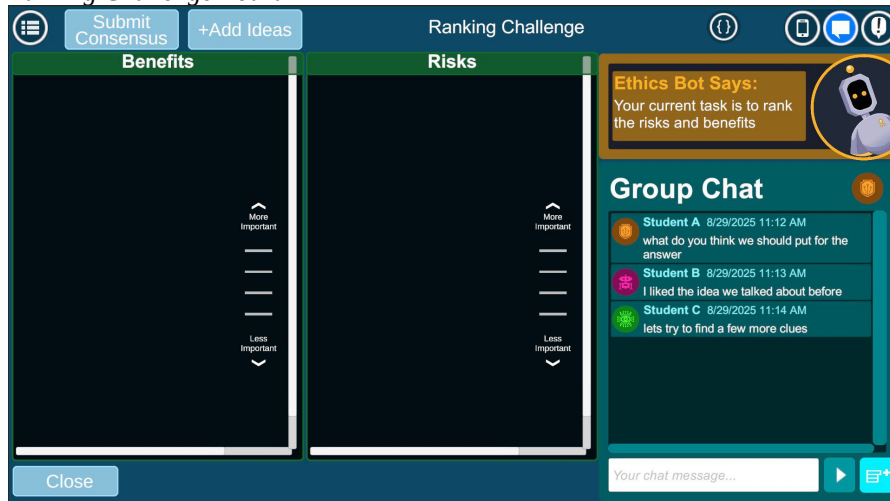
Figure 2
Simulation Middle School, An Artifact Agents Collect, A Student Gives a Presentation Judged by the AI Tool, and Hackers Take Over the School Network



Part 3: Ranking Challenge and Reporting

When players return to the Libratory, they click on a console to open the Ranking Challenge Board (see Figure 3). Players add ethical risks and benefits they have encountered during their visit to the school, then work together to rank them and write a statement sharing what they agree are the major benefits and risks. While they work together to rank the risks and benefits in terms of importance, they can chat with their group members and with the EthicsBot character (see more information on this character below). Visitors to our demo will be able to interact with this user interface in the learning environment to try to come to a consensus with other attendees at the session.

Figure 3
Ranking Challenge Board



Current development and future plans

Current design and development efforts are focused on co-designing and implementing a large language model (LLM)-backed version of the EthicsBot character. The goal of this effort is to deepen ethics learning with a responsive, flexible conversational agent that serves as a thought partner as students evaluate ethical risks and build group consensus. To achieve this, we will conduct co-design sessions with students to define the agent's role and interaction style. The resulting conversational agent (CA) will be designed to support students in reasoning about ethical impact using philosophical lenses (e.g., rules-based and consequence-based frameworks) without requiring the students to have prior training in philosophy. The CA will also scaffold group discussion in supporting constructive disagreement and consensus-building, while providing accurate, reliable, accessible, and safe responses. In addition, accessibility features such as text-to-speech and speech-to-text will broaden participation for learners with varied needs, while the agent's modular architecture will enable adaptation across diverse ethical contexts and curricular domains.

Demo session overview

In this interactive demo session, we will invite attendees to step into the role of a student agent inside *Agents of ViTAL* and grapple with AI ethics dilemmas. This experience will unfold in three parts: (1) We will present a video showcasing the narrative arc and learning environment, orienting attendees to how students navigate three AI ethics dimensions (privacy, bias, and environmental impact). (2) We will invite attendees to participate in an interactive ranking challenge, experiencing firsthand the collaborative reasoning and consensus building process. (3) We will solicit feedback and facilitate a discussion of the effectiveness of the learning environment's narrative scaffolding, HelperBot features, and CA function and integration. This feedback will directly inform ongoing iterations as we prepare *Agents of ViTAL* for full classroom implementation and co-design the EthicsBot agent with students.

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Resonate: An Interactive VR Tool for Experiencing Inclusive Education Through the Student Perspective

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Abstract: Resonate is a virtual reality (VR) design tool developed to enhance educators' understanding and implementation of inclusive teaching practices. By enabling educators to experience classroom environments from the perspective of students with disabilities, Resonate bridges critical gaps in teacher preparation and professional development. Leveraging immersive VR technology, the tool extends beyond traditional disability awareness training by fostering structured observation and authentic, experience-based empathy. Distinct from generic simulations, Resonate supports contextualized instructional reflection by allowing educators to test their own teaching materials within simulated disability scenarios. Grounded in Kolb's experiential learning theory and Universal Design for Learning (UDL) principles, the tool supports customizable parameter configuration, dynamic content integration, and in-situ annotation. This paper describes Resonate's features, applications, and contributions to learning sciences research, aiming to support educators in moving toward more actionable inclusive practices.

Introduction

Resonate addresses a long-standing challenge in teacher preparation: non-disabled educators often design inclusive teaching based on an "imagined deficit," resulting in well-intentioned yet frequently ineffective solutions (Valle & Connor, 2019). While educators may be equipped with inclusive theory, they often lack embodied experience of navigating educational environments with disabilities. To bridge this gap, Resonate harnesses immersive VR to provide embodied perspectives, helping educators identify hidden instructional barriers.

Unlike generic simulation tools, Resonate's core innovation lies in enabling contextualized instructional reflection. The system processes educators' uploaded teaching materials and integrates them into Unity 3D classroom templates. By systematically modulating the original content, such as reducing specific audio frequencies or blurring visuals, the tool simulates the learning experience of students with specific disabilities within real instructional contexts.

Within the field of learning sciences, Resonate contributes to teacher professional vision and inclusive pedagogy. It demonstrates how experiential learning reshapes educators' mental models, transforming abstract empathy into concrete instructional design improvements. Beyond a single VR experience, Resonate is designed as a reusable workflow and research infrastructure that supports iterative, data-informed reflection across lessons and disability-related barriers. This paper will illustrate how Resonate leverages these capabilities to help educators refine their teaching through continuous testing, advancing the practical application of learning sciences in inclusive education.

Theoretical foundations and related work

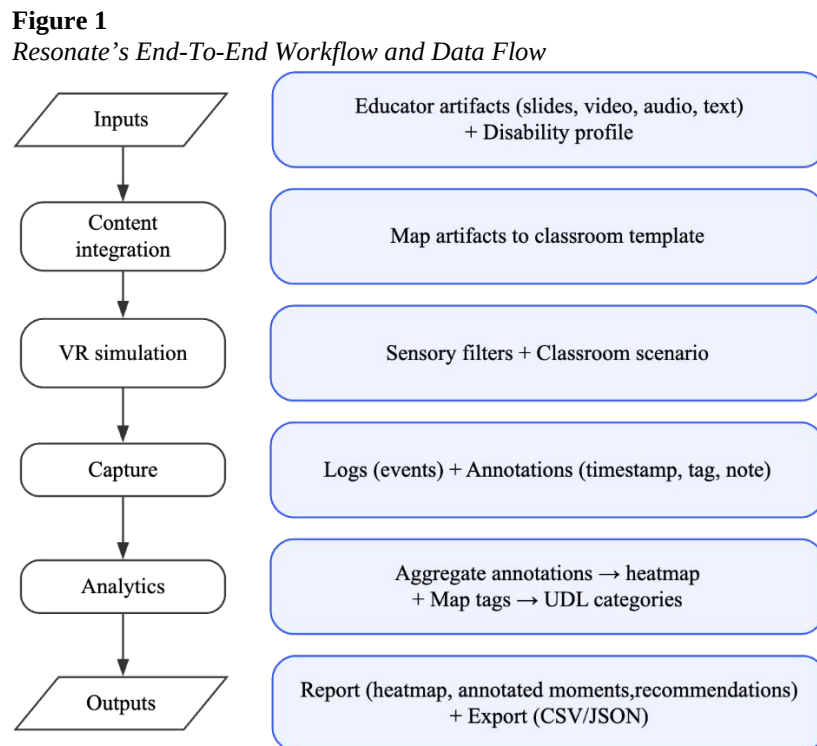
Resonate is grounded in Kolb's Experiential Learning Theory (1984), which posits that transformative learning occurs through a cycle of concrete experience, reflective observation, abstract conceptualization, and active experimentation. This is particularly relevant for developing cross-context empathy, as direct experience reshapes assumptions. The tool also applies UDL principles (CAST, 2024), advocating for multiple means of representation, engagement, and action/expression. By experiencing how disability impacts these dimensions, educators deepen their understanding of the necessity of UDL.

Current research indicates that general education teachers receive limited special education content in pre-service training (Alharbi & Iqtadar, 2024), often leaving them to rely on assumptions to understand student needs (Hwang & Evans, 2011). Meanwhile, VR has been effectively used in disability education (Serafin et al., 2023; Zhang et al., 2022), and studies suggest that teachers who undergo disability simulations via VR are more likely to comprehend the real-life difficulties faced by their students (Stavroulia et al., 2023).

However, despite VR's proven potential for UDL, studies focusing on how educators can employ VR to explicitly improve instructional design remain scarce. Resonate fills this gap by providing a systematic solution that integrates immersive experiences, structured reflection, and UDL-aligned feedback.

Design features

Figure 1 summarizes Resonate's end-to-end workflow and data flow. Educators first provide instructional artifacts (e.g., slides, video, audio, text) and select a parameterized disability simulation profile. These inputs are mapped onto a classroom template and experienced through a Unity-based VR simulation. During the experience, the system captures in-situ annotations with automatic timestamps, which are aggregated into a timeline heatmap and used to support replay and generate UDL-informed recommendations, with exportable outputs.



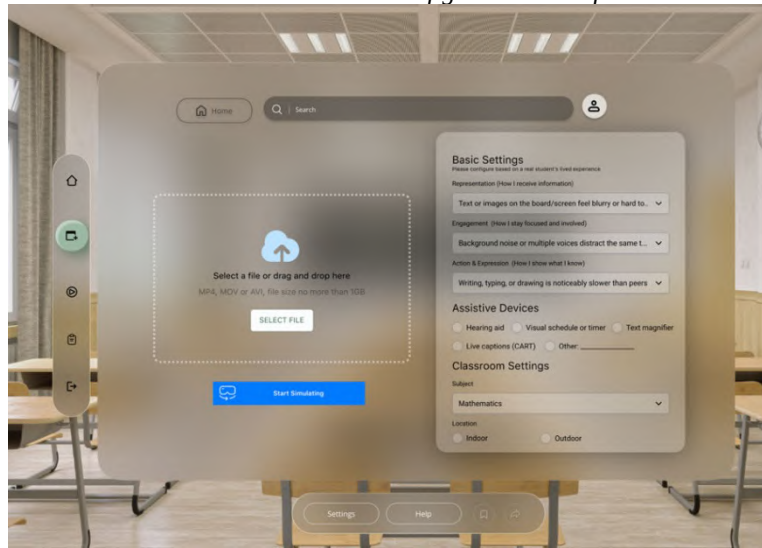
Resonate's design enables educators to customize disability simulations and experience classroom scenarios through VR. Key features include:

UDL-based parameter configuration system

The system deconstructs disability experiences into specific perceptual and interactive parameters, configured based on documented disability-related barriers in educational settings (see Figure 2):

- **Representation:** Configures barriers related to information perception, including visual impairments, hearing loss, and processing delays.
- **Engagement:** Adjusts factors affecting focus and motivation, such as attention regulation difficulties, sensory sensitivities, and specific anxiety triggers.
- **Action & expression:** Simulates challenges in navigation and communication, including motor limitations, speech difficulties, and writing challenges.

Figure 2
Resonate's UDL-Based Parameter Configuration Interface



High-fidelity immersive simulation

Developed using the Unity 3D engine, this module places participants in a high-fidelity virtual classroom template, providing a concrete experience under controlled, repeatable simulation conditions. The system dynamically integrates the educator's uploaded materials while simulating specific perceptual and environmental challenges:

- **Dynamic content filtering:** Applies real-time adjustments to audio and visual clarity based on the input disability profile. The system modulates the user's sensory input, such as blurring visuals or reducing specific audio frequencies, to simulate the learners' perspective.
- **Information delay:** Replicates the real-world Communication Access Real-time Translation (CART) services by introducing a 2–6 second lag between spoken audio and caption display.
- **Environmental distractions:** Simulates the interference caused by real-world classroom dynamics, such as ambient noise and overlapping conversations during group discussions.

Reflection-in-action annotation

During immersion, users can activate a lightweight in-headset annotation panel via simple gestures to instantly annotate recognized issues (e.g., “unclear slide text,” “teacher speaking too fast”) (see Figure 3). All annotations are automatically time-stamped, supporting reflective observation by transforming fleeting subjective perceptions into trackable data points.

Figure 3
First-person VR view of the “Reflection-in-action” annotation panel



Visual analysis and scaffolding feedback

The system converts user annotations into a visual timeline “heatmap.” Annotations are stored with timestamps and tags and aggregated into fixed time bins, allowing the heatmap to highlight periods of high-frequency barriers in the classroom. Beyond providing replay functionality for review, the system generates specific instructional adjustment recommendations based on identified obstacle types and the UDL framework, helping educators translate experiences into concrete teaching design improvements.

Application

Resonate facilitates an inclusive instructional design process, bridging the gap between experiential empathy and evidence-informed revision. By enabling educators to engage with authentic teaching artifacts through immersive simulation conditions, the tool supports evaluation from the perspective of students with disabilities. This process surfaces otherwise “invisible” instructional barriers, allowing educators to pinpoint specific obstacles and translate these insights into actionable classroom strategies.

Resonate flexibly integrates into professional development scenarios:

- Pre-service training: Helps trainees understand and design courses using inclusive design principles.
- In-service workshops: Facilitates team collaboration to redesign courses and discuss barrier identification.
- Special education certification: Serves as an empathy-building tool, allowing trainees to experience diverse student perspectives before entering classrooms with students with disabilities.

Live demonstration plan

During the interactive demonstration, we will provide an overview of Resonate’s design philosophy and workflow. Participants will then use Meta Quest 3 headsets to experience a representative 3-minute classroom scenario from the perspective of a student with disabilities. This scenario will demonstrate how standard instructional materials are transformed by the system, featuring barriers such as caption delays, high-frequency audio loss, and environmental distractions. While one participant is immersed in VR, others will act as “Co-designers” using a tablet interface to view the live simulation feed and discuss potential interventions. Following the demonstration, we will collaboratively review the heatmap and automatically generated enhancement recommendations. Discussions will center on potential barriers and feasible classroom adjustments. We will also open the floor for feedback on the current user experience, as these insights are vital for shaping future iterations. Hardware requirements are minimal, consisting of 2–3 Meta Quest 3 headsets and several tablets for co-designers. A 360° video backup solution is available on-site.

Conclusions and future work

By combining experiential learning theory with practical instructional design support, Resonate transforms how educators understand and respond to student diversity. The integration of immersive simulation, structured reflection, and iterative design refinement creates a comprehensive system for developing both empathy and practical skills. Future development will expand the library of disability profiles and develop longitudinal tracking of implementation outcomes. Through continued iteration based on feedback and research findings, Resonate aims to provide a comprehensive system for educators to translate inclusive aspirations into more concrete, evidence-informed design decisions.

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The Connected Spaces Dashboard: Fostering Peer-to-peer Collaboration Through Ambient Maker Profiles

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Abstract: The Connected Spaces Dashboard is a tool designed to foster maker identity development, peripheral awareness, and the promotion of student help seeking behaviors. Developed iteratively across several years, prototypes of the Dashboard range from lo-fi solutions to in-house developed applications. Beneath these external embodiments, there are a number of frameworks and scaffolds developed to facilitate student interactions and engagement with the Dashboard as a rich identity resource. In this interactive demo, participants will have a chance to utilize these scaffolds alongside a specially tailored version of the Dashboard to develop peripheral awareness with their peers.

Introduction

The Connected Spaces (CxS) Dashboard is a tool that combines aspects of Maker Portfolios with structures that support students' peripheral awareness and participation (Fields et al., 2023, Chan & Holbert, 2020, Lave & Wenger, 2001), as well as the promotion of constructionist learning and the development and reexamination of students' identities (Godwin, 2016). However, finding peers within a distributed community of makers to engage with, particularly when faced with roadblocks in the making process, is a persistent challenge (Aleven et al., 2003). This connection with others and the ability to see their work aligned with work of their peers, is also a strong driver of identity development (Kim et al., 2018). In response, we set out to address the need for peer-to-peer collaboration, as well as ownership of the space as students adapt to a constructivist environment as opposed to direct instruction. Through a system of "affinities" (Kumar & Tissenbaum, 2022) as an alternative to badging, students are able to associate themselves with a number of maker concepts, regardless of whether or not they have gained some arbitrary level of competency sufficient to earn a badge. Through ambient awareness of their peers' affinities (Alavi & Dillenbourg, 2012; Ogata & Yano, 2000), the technology facilitates students' help seeking behavior (Aleven et al., 2003) so that they can locate other students in their community who may be willing to learn alongside them, especially when they encounter a problem. These patterns of behavior support constructionist ideas of knowledge generation through student tutoring (Roscoe & Chi, 2008), as well as acting as a component of a larger system of technology serving to create an extended community of practice across spaces and promoting boundary spanning connections with makers in disparate locations (Smith & Tissenbaum, 2023).

Background

Informal makerspaces are spaces in which people with shared interests and of varying skill levels can gather to share ideas, knowledge and technology. Through the intentional design of their curators, these spaces have the potential to become communities of practice (Lave & Wenger, 1991) by allowing students to collaborate with peers and mentors while also expressing their unique identities through personally relevant making (Bawankule et al., 2019; Keune & Peppler, 2017). Makerspaces can also provide unique opportunities for the use of technology to support student learning alongside deliberate scaffolding from instructors within the space (Pitkänen et al., 2020). One approach that is gaining particular prominence in promoting students' sense of ownership, belonging, and competency in maker environments is the idea of "maker portfolios" (Chan & Holbert, 2020; Lui et al., 2019; Keune & Peppler, 2017), as documentations of students' work and as adaptive representations of their identities. The Dashboard builds on prior work (Kumar & Tissenbaum, 2022) and seeks to combine aspects of maker portfolios with scaffolds for help seeking and peer mentorship. The Dashboard has been developed through a design-based research approach, and has been implemented in a number of settings, including four two-week long summer camps designed to foster computational identity.

Tool overview

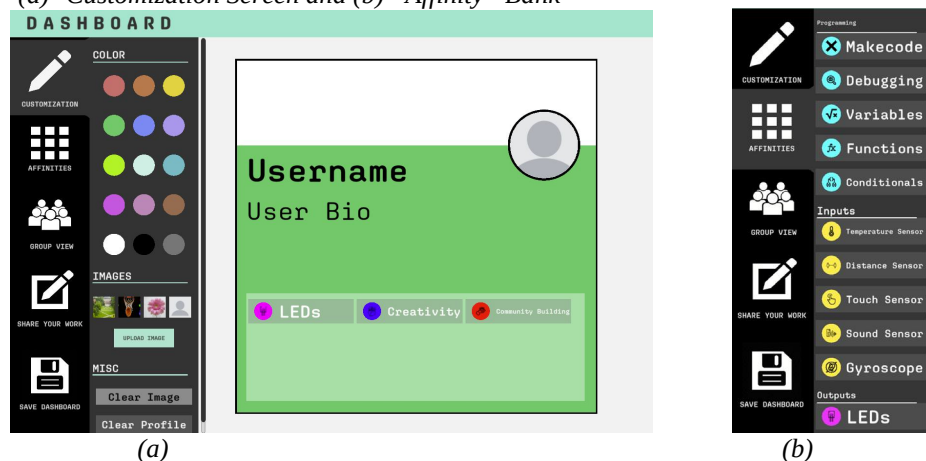
The Dashboard is composed of three main components: an interface by which students can customize and update their personal Dashboard profile, a system of "Affinity" badges that students can tag their profiles with, and a "grid view" screen which is displayed ambiently within the makerspace environment and can also be viewed on a student's personal computer. Through several years of iterations, the Dashboard has been implemented in a

number of different ways, with systems added based on the in-situ needs of students, making the Dashboard a relatively fluid tool, centered around a few core theoretical ideas. Some of the changes that have been previously incorporated have included a virtual “help” signal with alert messages on the ambient display, integration with a “work in progress” photo carousel, reflection prompt “stickers”, and a number of scaffolded activities that provided students an opportunity to utilize the Dashboard as a reflective tool.

Profile personalization/maker affinities

After initial design experiments (Bawankule et al., 2023), we concluded that students really enjoyed and resonated with the ability to personalize their Dashboard profiles, prodding their neighbors to look at what they’d done, collaborating with each other, and growing attached to the representation, even indicating that they didn’t want their profile to be deleted at the end of the camp. When developing an in-house prototype, we slimmed down the customization feature but still left a large amount up to the student to be able to create an authentic representation of self. Alongside the ability to add pictures, colors, and descriptions to their profile, we incorporated a fluid set of affinities – badges that represented both maker concepts and social causes that students could use to add motivation to their projects. These affinities can be quickly modified based on direct student feedback, as well as input from mentors present within the space. Scaffolding included daily reflections on what affinities students chose to include, the act of seeking out other students with similar affinities, and prompts for students to analyze their projects and to add affinities that were related to their components, so that other students could seek help from those who were more experienced (figure 1).

Figure 1
(a) Customization Screen and (b) “Affinity” Bank

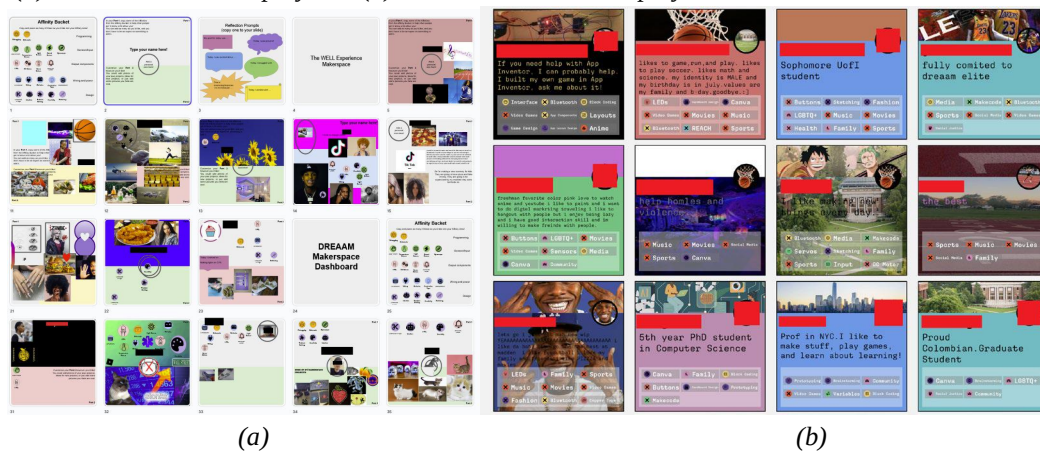


Grid view/ambient display

Through each iteration of the Dashboard, student profiles have been displayed in different ways, either ambiently or on a student’s personal computer. By displaying all profiles simultaneously ambiently in the room, students are now able to passively observe their peers’ particular skills and preferences, removing some of the barriers to collaboration within an informal learning environment (figure 2). Additionally, by being able to filter and explore profiles on their own computers, connected to photos of students’ works in progress through other Connected Spaces tools, students can identify connections between their own needs and other students’ capabilities. As a resource for teacher orchestration, the grid view provides a “quick and dirty” view of students’ interests, and allows a facilitator to point to a shared resource when suggesting student mentors or collaborators. The Dashboard’s grid view acts as a meeting place for making students’ maker identity visible, and is the first step in establishing peer to peer collaboration.

Figure 2

(a) *Lo-Fi Ambient Display* and (b) *In-House Ambient Display*



Dashboard in practice

The Dashboard was implemented in maker education-based summer camps over a period of four years, in which students developed personally meaningful projects after a week of engaging with pre-built physical computing artifacts. Dashboard personalization and scaffolded reflection activities were incorporated into the curriculum, and Dashboard changes were recorded and analyzed, and student feedback was incorporated into the design of the next implementation. From our first iteration to the latest, the Dashboard underwent a number of changes, both within the technology, and in its situated implementation in the environment. We succeeded in incorporating maker portfolios to support students' identity development but recognized the tensions surrounding trying to develop students' computational identities with the Dashboard, if it requires sacrificing their perception of the tool as a help seeking structure. However, based on our observations of the tool in practice, we thought it was possible that students' legitimate representation of identity might make them feel a sense of authenticity and belonging to a community, and that this could potentially be productive when used with scaffolding that allowed for peripheral awareness of their peers, aligning with previous findings about Maker Portfolios (Chan & Holbert, 2020; Lui et al., 2019; Keune & Peppler, 2017).

As we conducted our various camps and programs while using the Dashboard, one question that frequently emerged in our discussions was the role of the instructor in the camp. One of our main goals for the Dashboard was to promote peer-to-peer collaboration and constructivist learning. When instructors short-circuit student questions to avoid momentary chaos, do we deprive students of the opportunity to grow? We considered the possibility of the instructor using the Dashboard as a reference point, a way to redirect student inquiries onto an alternative, more productive path, and to take advantage of learning gains from student tutoring (Roscoe & Chi, 2008). We saw several instances of the Dashboard being used to promote collaboration, but students still demonstrated resistance to our suite of tools, apparently because they were used to a traditional method of engagement with learning – they would come to the instructor before their peers. We attempted to implement a practice of “check the Dashboard before you ask an instructor,” but the scope of our camps made it difficult to fully pull away from traditional instruction, and students didn't have enough time to adapt to the change in classroom culture.

Dashboard fidelity

The affordances of the Dashboard concept varied greatly from iteration to iteration, and our rapid, design-based research approach allowed us to experiment not only with embodiments of design conjectures, but with the design conjectures themselves. When we observed that students weren't responding well to the ambient display, we put it on the backburner and developed additional tools for other aspects of student participation. Because of this, the Dashboard has become a multi-faceted conceptual tool that can be presented in a number of ways. One of our major interests going forward will be what features of the Dashboard make sense to implement within a given environment. For example, our implementation in Google Slides had major benefits due to its easy implementation, flexibility, customization, and in-network connection to many school districts' pre-existing technological networks. On the other hand, the streamlined customization of our in-house implementation prompted students to say that their Dashboards were “less messy” and “easier to understand.” We are considering

whether an even more slimmed down Dashboard, similar to our early conceptualizations (Kumar & Tissenbaum, 2022) may support an environment where students' help seeking behavior is of higher priority than maker identity formation. Additionally, the fidelity of the Dashboard will have an impact on the scalability of the technology. It is important to understand how the underlying infrastructure could be utilized in addition to other technologies, or how it might be utilized in a formal learning environment, for instance. Further research in a wide variety of learning environments, including those where Dashboard interaction and scaffolding could be included and enculturated, in conjunction with differing external makerspace environments, will be a notable point of discussion within this demonstration.

Interactive demo

During the interactive demo session, participants will be able to create their own Dashboard profiles, customized for the ISLS audience, and participate in a peripheral awareness activity based on their own research interests. By creating a virtual space for ISLS participants, the Dashboard will be utilized as a collaborative tool as well as a demo, allowing participants to find common interests, affinities and, potentially, even to foster continuing connections. Participants' Dashboards will be displayed in multiple modalities – they will be welcomed to edit their Dashboard profile or view others' Dashboards on their own web-capable device or at our station, where Dashboards will be displayed ambiently. After the initial demo session, we will keep a version of the Dashboard running for participants who are interested in exploring a representation of ISLS attendees' affinities, and as a low-stakes networking tool.

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From Thought to Action: Designing AI Coaching Support for Time Management with Cognitive Behavioral and Self-Regulated Learning Strategies

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Abstract: This demonstration introduces TimeWise, a conversational agent that integrates self-regulated learning (SRL) and cognitive-behavioral therapy (CBT) to provide personalized, context-aware coaching for time management in online learning. The system leverages learning management system data to visualize study patterns and initiate reflective dialogue grounded in students' actual learning activity. Through guided interaction, TimeWise engages students in examining their thoughts, identifying irrational beliefs, and reconstructing more adaptive perspectives to restore task-oriented behavior. We present the design principles and illustrative interaction cases that demonstrate how the system operationalizes SRL and CBT integration. This work highlights the potential of combining these frameworks to promote sustained academic regulation in online learning contexts.

Background

As online learning modalities gain increasing prominence within higher education, students are given greater control over when, where, and how they engage with academic tasks. While this flexibility promotes access and autonomy, it also requires learners to take a more active role in regulating and sustaining academic effort. Among the various self-regulatory skills needed in online environments, time management has been identified as one of the most critical self-regulatory skills influencing students' learning engagement and academic performance (Wolters et al., 2025). However, many students still struggle to manage their time effectively and delay academic tasks even when they intend to complete them.

Two major approaches have guided efforts to improve procrastination and time management in learning: self-regulated learning (SRL) and cognitive-behavioral therapy (CBT). SRL-based interventions support learners in planning, monitoring, and evaluating their study behaviors through techniques like goal setting and task organization (Zimmerman, 2000). Digital tools such as mobile planners and productivity apps (e.g., Xie et al., 2024) aim to scaffold scheduling and help students break down large tasks. More recently, conversational agents like StudyBuddy (Tian et al., 2021) have applied SRL principles to support pre-task planning through fixed dialogue flows. While these systems improve task visibility, they often assume that upfront planning is sufficient and provide limited support during task execution or post-task reflection. As such, many tools fall short during moments when self-regulation breaks down, underscoring the need for adaptive systems that can intervene during the actual regulation of academic behavior.

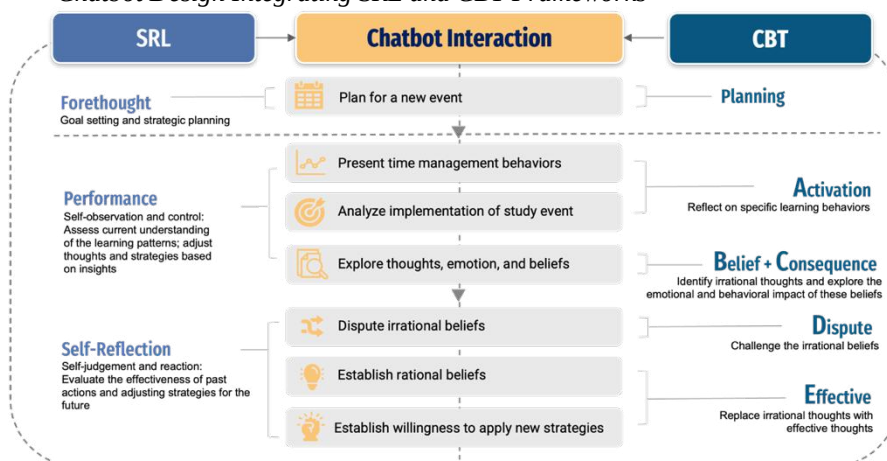
In contrast, CBT-based approaches address the cognitive and emotional barriers underlying procrastination by targeting maladaptive thought patterns (Lukas & Berking, 2018). While therapeutic interventions consistently show strong effects in reducing academic procrastination (van Eerde & Klingsieck, 2018), they are often time- and resource-intensive, limiting their scalability. Digital implementations of CBT offer a more accessible alternative, yet they typically rely on static web-embedded content, reducing their ability to adapt to learners' real-time academic contexts. Although some systems attempt to integrate CBT with task management features, for example, GanttBot (Pereira & Díaz, 2021), these implementations typically concentrate on project-level planning and timeline adjustments rather than addressing the in-the-moment disruptions that prevent learners from engaging with their work.

Despite their distinct origins, SRL and CBT offer highly complementary perspectives for addressing academic procrastination. SRL-based strategies emphasize planning, monitoring, and behavioral regulation, while CBT approaches focus on identifying and restructuring the emotional and cognitive triggers that drive avoidance. Given that procrastination involves both breakdowns in task regulation and maladaptive responses to discomfort or self-doubt, an integrated approach has the potential to support students across both procedural and psychological dimensions of time management. To address these limitations, we designed *TimeWise*, a large language model (LLM)-based conversational agent that integrates principles from both self-regulated learning and cognitive-behavioral therapy to support students in managing their time more effectively in online learning.

Design principles

We grounded our design in theoretical insights from SRL and CBT principles (see Figure 1). TimeWise is designed to provide adaptive support across the full cycle of SRL: forethought, performance, and self-reflection. In the forethought phase, TimeWise assists students with goal setting and planning, aligning with functions seen in earlier SRL-based tools. The system's central contribution lies in its support during the performance phase, where time management often breaks down. TimeWise uses behavioral indicators from LMS data to identify patterns that may signal task avoidance. They are introduced into the conversation to help students raise awareness of the study behaviors, setting the stage for deeper dialogue by anchoring the coaching in concrete behavioral moments. TimeWise then engages the learner at the moment of struggle, initiating the Rational Emotive Behavior Therapy (REBT) process to identify and transform the underlying beliefs related to the procrastination behaviors (Dryden and Neenan, 2004). This process is informed by the CBT ABCDE model, which includes the Activating event, the Beliefs that arise in response to that event, the emotional and behavioral Consequences of those beliefs, the Disputation of irrational beliefs, and the development of more Effective beliefs. In the performance stage, it helps learners identify the activating event that triggered the difficulty, such as encountering a demanding task or feeling overwhelmed, and then guides them to articulate the beliefs shaping their reactions. The system also helps them recognize how those beliefs contribute to emotional responses. This allows the system to intervene precisely when procrastination is unfolding, offering real-time dialogue that helps learners reframe unhelpful thoughts and re-engage with the task. In the reflection phase, TimeWise guides users through disputation of maladaptive beliefs and supports the formulation of more effective, adaptive beliefs that can strengthen future self-regulation. By embedding REBT techniques into each phase of SRL, the system moves beyond surface-level prompting to deliver emotionally responsive coaching that aligns with each learner's moment-to-moment cognitive and affective state.

Figure 1
Chatbot Design Integrating SRL and CBT Frameworks



Illustrative cases: TimeWise chatbot interaction flow

We present illustrative cases that trace the interaction across the design steps. After the learner forms a study plan in the forethought phase, TimeWise tracks learning behavior in the LMS and uses these data to support the next stage of the process. In the performance phase, TimeWise presents LMS-based indicators and visualized activity patterns to help learners monitor how their engagement unfolds in practice. The learner reviews recent study patterns and acknowledges that starting assignments late is not effective for learning, while also noting that some study habits only work in some classes (see Figure 2).

TimeWise extends this phase by exploring thoughts, emotions, and beliefs related to learners' behaviors. It guides the learner translate a general intention into a concrete study event, then identifies likely distractions and emotional reactions, such as feeling overwhelmed and pressed for time, that may disrupt task completion (see Figure 3a). Figures 3b to 3d further illustrate how breakdowns during performance are shaped by students' irrational beliefs. One common thinking pattern was the use of strict "must" or "should" statements, also known as musturbation in REBT. They often interpreted difficulties through inflexible expectations about how they should perform, like "I must do well," "I must finish perfectly," or "I shouldn't fall behind." These demands increased pressure during task engagement and often contributed to avoidance, distress, and harsh self-evaluation.

Figure 2

TimeWise Presents LMS Visuals and Analyzes Implementation of Study Events

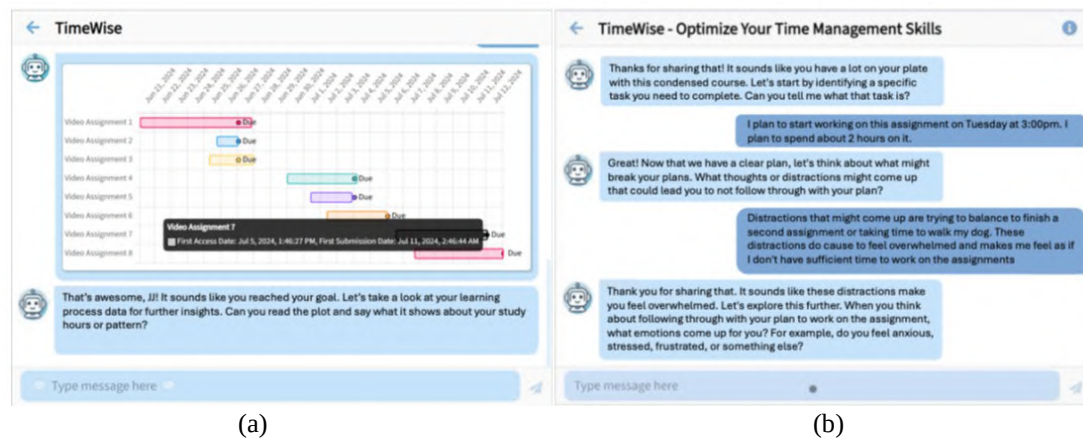
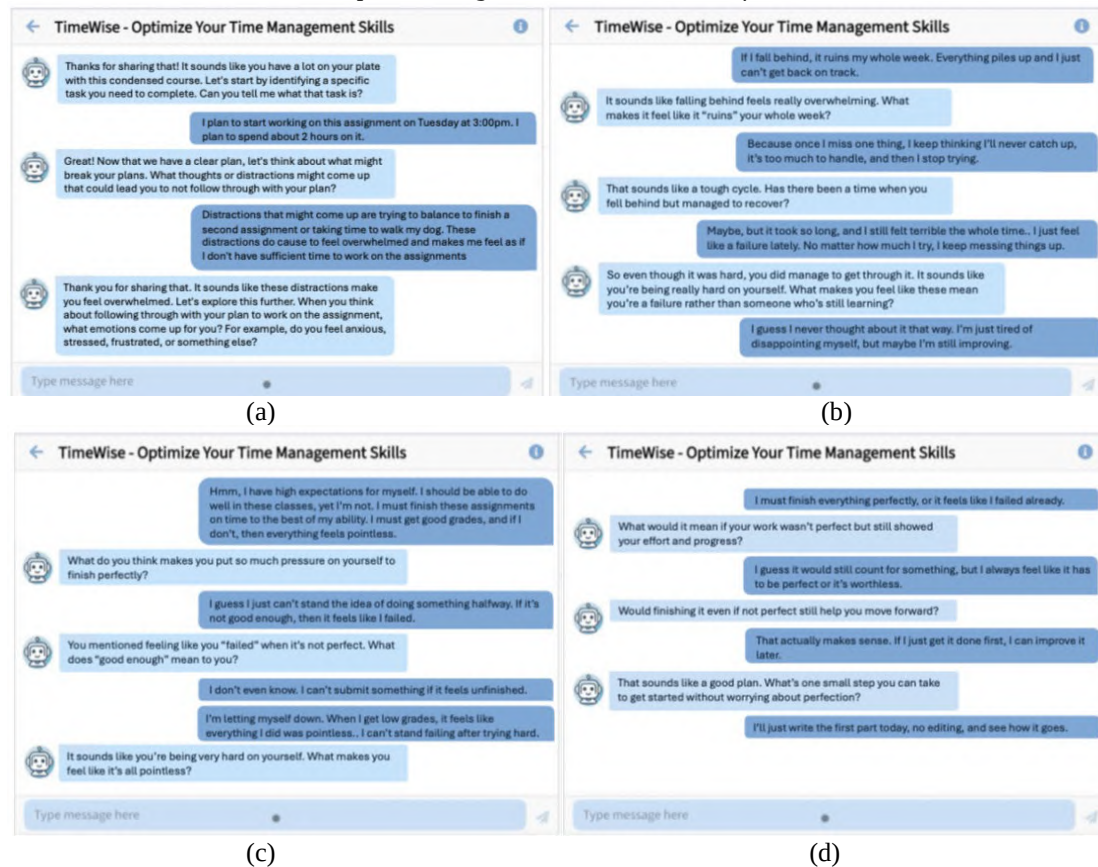


Figure 3

TimeWise Guides Learners to Explore Thoughts, Emotions and Beliefs



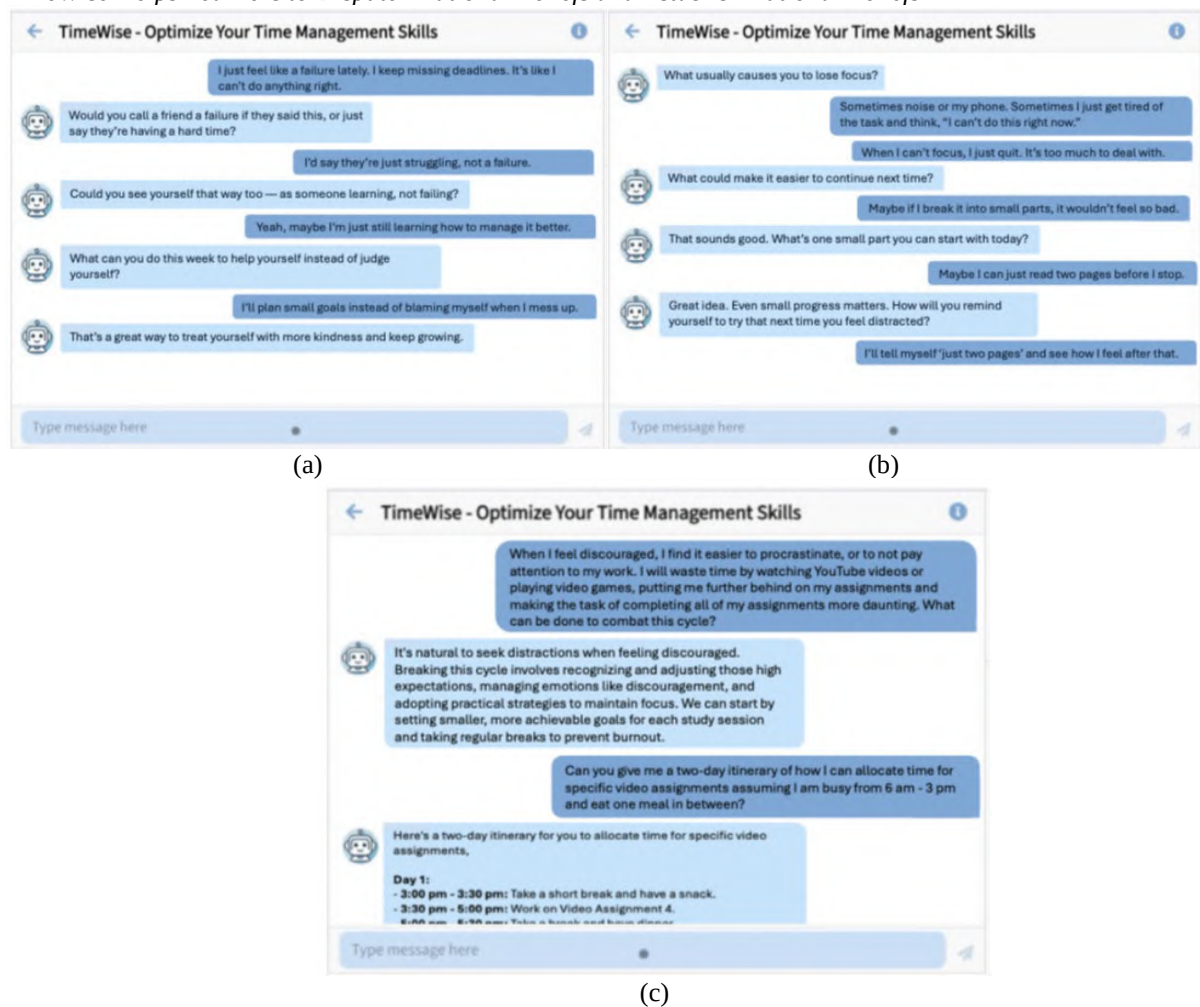
In Figure 3b, must/should thinking appears together with catastrophizing and self-downing. The learner interprets falling behind as something that “ruins” the whole week and then begins to view the setback as evidence of personal failure. In this case, the difficulty is not only the missed progress itself, but also the belief that such setbacks are unacceptable and reflect something negative about the self. In Figure 3c, must/should thinking takes a more clearly perfectionistic form. The learner says, “I must do well” and “I can’t stand the idea of doing something halfway,” which reflects both rigid internal demands and low frustration tolerance. The learner also suggests that if the work is not done properly, then “everything I did was pointless,” showing all-or-nothing thinking in which anything less than perfect feels worthless. The barrier is not only high standards, but also the

belief that imperfect work is intolerable. In Figure 3d, the same perfectionistic pattern remains visible. The learner still starts from a perfectionistic and all-or-nothing position, assuming that work needs to be done properly in order to count. This also reflects low frustration tolerance, because incomplete or unpolished work still feels difficult to accept. But the interaction begins to show the softening of that belief. The learner gradually begins to accept that progress does not have to be perfect to be meaningful.

In the reflection phase, TimeWise helps students recognize, dispute, and reconstruct irrational beliefs about time management. As shown in Figures 4a–4b, when a learner expresses self-downing — interpreting missed deadlines as proof of being "a failure" — TimeWise uses gentle questioning and reflective prompts to challenge rigid self-talk and encourage perspective-taking. Students begin to shift toward more compassionate thinking, saying things like "I'll plan small goals" or "Maybe I'm just still learning." Similarly, when a learner describes distraction as "too much to deal with," TimeWise reframes the task as manageable, helping them move from "I can't do this right now" to tackling it in smaller steps. In Figure 4c, the reflective work extends into action, as the learner connects procrastination to unhelpful habits and works with TimeWise to establish a concrete study plan as a first step toward changing their time management behaviors. During the interactive demonstration, participants will explore selected dialogue flows that illustrate how TimeWise supports students using principles from REBT and SRL. They will have the opportunity to engage with the chatbot directly by entering common student concerns and observing how the system responds. Through this hands-on experience, participants can see how TimeWise identifies irrational beliefs, challenges rigid thinking, and scaffolds the development of adaptive plans, helping students shift from emotional reactivity to proactive time management.

Figure 4

TimeWise Helps Learners to Dispute Irrational Beliefs and Establish Rational Beliefs



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SciStory: A Demo of AI-Supported Socioscientific Learning through Visual Novels

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Abstract: Our tool demo will explore intersections between narrative-centered learning, AI agents as inquiry scaffolds, and the interdisciplinary exploration of socioscientific issues. We will share our current design of SciStory: Pollinators, an interactive visual novel designed to support middle grades students in developing persuasive arguments to solve socioscientific problems in their community. Discussion will explore how we balance ethical issues around generative AI in our agent design, leverage co-design practices to iteratively improve collaborative game features, and draw on principles from problem-based learning to engage learners in developing solutions for complex socioscientific issues.

Introduction

This session will demonstrate SciStory: Pollinators, an interactive visual novel that is designed to teach middle grades students about socioscientific issues, persuasive argumentation, and problem-solving for complex community issues (Humburg et al., in press). Visual novels are a type of interactive story game that use player choice and branching dialogues to engage players as a participant in an unfolding narrative. While there is a dearth of research on the specific impacts of visual novels on learning (Camingue et al., 2020), a handful of educational designers have leveraged the choice-based, exploratory structure of this storytelling medium to engage learners with topics such as endangered species protection (Câmara Olim et al., 2025), empowerment in healthcare settings (Yin et al., 2012), and English conversational practices (Faizal, 2016). More broadly, narrative-centered learning environments that use stories to engage students in disciplinary learning have demonstrated significant benefits for both learning and motivation (Lester et al., 2012; Naul & Liu, 2020). The goal of this design was to leverage the known benefits of digital story games with the emerging capabilities of generative AI to create an adaptive, student-centered narrative learning experience.

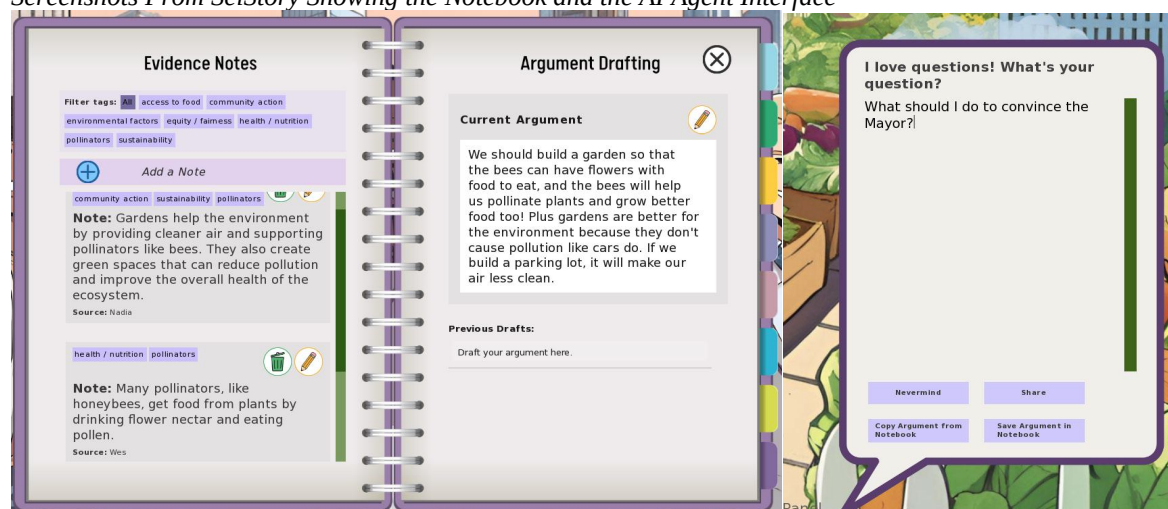
To maximize student engagement with the SciStory narrative, we have designed and embedded AI-driven conversational agents as story characters who can adaptively respond to students' ideas and questions as they play through the visual novel. Our agent design builds on previous studies that have explored the effectiveness of intelligent agents acting as a teachable peer (Biswas et al., 2005), a tutor (Graesser et al., 2018), a knowledge building partner (Gerard et al., 2024), and a conversational facilitator (Dyke et al., 2013). This research has shown that well-designed agents can support students' metacognitive practices during collaborative science learning as well as their scientific inquiry skills (White et al., 1999; Biswas et al., 2005). However, throughout our design process, we have grappled with a variety of ethical concerns in the ways that generative AI can impact student learning and overall community wellness, including the risk of unreliable or inaccurate responses (e.g., "hallucinations"), the negative impact of AI data centers on environmental resources, and concerns about how third-party AI models handle and store student learning data (Kasneci et al., 2023; Selwyn, 2022). As a part of our visual novel demo, we will engage with attendees about our AI agent design process and the ways that our team works to mitigate ethical concerns through cross-disciplinary design conversations as well as teacher and student co-design.

Tool and story overview

The SciStory: Pollinators narrative draws on design principles from problem-based learning (Hmelo-Silver, 2004) by presenting students with an ill-structured, complex community problem that has multiple potential solutions. Students work in pairs in front of a single laptop to negotiate ideas, collect evidence, and determine a solution to present to the town mayor as a team. Players are introduced to local community members throughout the game's narrative and are tasked with helping the community decide what to do with an empty plot

of land. Some characters want to build a community garden, while others want to build a parking lot to attract more businesses to the area. The story is presented to students through a series of 2D backgrounds, much like a digital storybook, and each location the player visits has different selectable characters that, when clicked, trigger a conversation with that character. Students can select responses in the conversation from a list of pre-scripted choices, which lead them down branching conversational pathways. Several tools are built into the story structure to scaffold students' inquiry, including a digital notebook where students can write down ideas, tag notes with different categories, and sort their notes to find key ideas to include in their argument draft (see Fig. 1). Students also have access to a dialogue history to review any conversations they have had. There are also a variety of opportunities for students to type out their own questions and ideas to AI-infused characters in order to receive tailored responses and feedback from AI agents.

Figure 1
Screenshots From SciStory Showing the Notebook and the AI Agent Interface



Agents are designed to fill three key pedagogical roles: 1) Answering content questions (about pollinators, food justice, and plants), 2) Answering help questions (about how to play the game or overcome roadblocks during gameplay), and 3) Providing supportive feedback on students' draft arguments for what to do with the empty lot. Each agent draws on carefully crafted knowledge bases written by learning scientists to cover key scientific and game knowledge as well as examples of effective feedback. The agents also have specific prompts embedded to prevent students from derailing the conversation off-topic or discussing inappropriate or offensive topics. The agent prompts were designed and improved over multiple testing iterations with teachers, students, and interdisciplinary research teams to check for hallucinations, the effectiveness of the safety guardrails, and the overall usefulness of the agent feedback for student learning. We also tried out and compared three different large language models (LLMs) in our agent development and focus group testing: Flan-T5-small (Chung et al., 2022), Llama-3.1-8 billion (Grattafiori et al., 2024), and GPT-4o (OpenAI, 2023). As a part of the demo, we will discuss with attendees how we weighed the pros and cons of different LLMs and tested their effectiveness in student and teacher co-design sessions to iteratively improve the SciStory agent design.

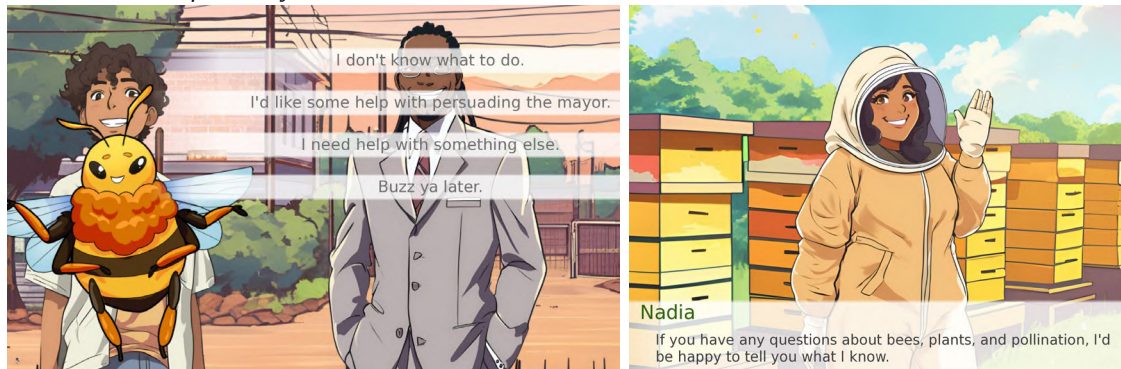
Students can explore the visual novel in any order by visiting different locations, choosing which characters to speak with, and recording digital notes in their in-game notebook. After gathering sufficient evidence, students draft a persuasive argument to convince the town mayor to support their solution for the empty lot. Through this argument drafting, the students engage with concepts around pollinators, ecosystem health, and how human food systems are intertwined with the health of pollinators and our larger environment. Based on whether students successfully convince the mayor to support a garden or a parking lot, the ending of the game changes.

Demo session overview

Our team will present two versions of SciStory: Pollinators for attendees to explore in our demo. The student-facing tool will allow players to move through the story that students see, talking with game characters, gathering notes in their digital notebooks, and drafting a persuasive argument for the Mayor (see Fig. 2). Attendees can also explore the teacher demo version of SciStory (see Fig. 3), which walks the player through the narrative at a high

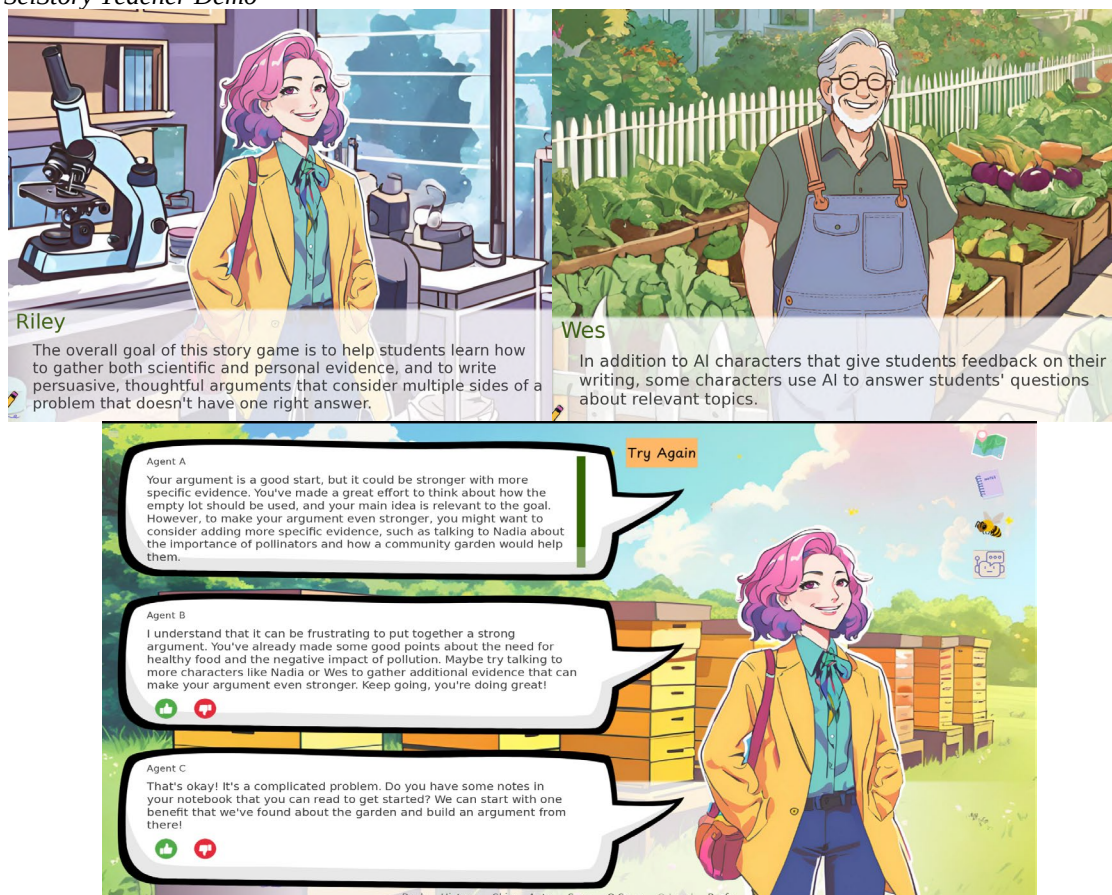
level, explains the core learning goals, and allows the player to iteratively test out how the AI agents behave with different large language models (LLMs) plugged into the game characters. The student version was created for use in classrooms as a student learning activity, while the teacher version was created to gather teacher feedback during focus groups in order to improve the design of the student game.

Figure 2
Student Version of SciStory



Screenshots from the student version of SciStory, demonstrating how students interact with characters via dialogue choices

Figure 3
SciStory Teacher Demo



Screenshots from the SciStory teacher demo, explaining the goals and design of the visual novel (top) and presenting responses from agents using different LLMs for teachers to compare (bottom)

To share these demo tools with attendees, we will bring several laptops running the browser-based visual novels for attendees to test out and ask questions. The tools will require a Wi-Fi internet connection and outlets to plug in laptops. In the case of a virtual or hybrid session, the demo link can be shared with virtual attendees for them to open and load the games on their own computer browsers. In addition to playtesting the tools, we will provide attendees with information about the SciStory narrative development, our AI agent design approach, and our pilot tests with students and teachers. For example, we can discuss how our team shifted from developing for a 3D video game environment to a 2D visual novel design in response to student playtesting and feedback. We can also share details about our approach to improving accessibility for a variety of student reading and writing levels through the integration of automated speech recognition and text-to-speech technologies. Our hope is that demo attendees will find value in learning about several different aspects of the design, including 1) the use of visual novels as a playful way for students to engage in scientific inquiry, 2) the use of AI agents to scaffold open-ended inquiry while promoting student agency and metacognition, and 3) the integration of persuasive writing and scientific inquiry skills to engage students in socioscientific issues relevant to their communities.

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An Interactive Tool for Using Digital Music Note Manipulatives to Compose Rhythms

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Abstract: Previous research utilized paper music note manipulatives to help jazz students brainstorm rhythms to use during improvisation. However, there were issues that can be mitigated by digitizing these music note manipulatives. Therefore, this interactive demo presents a website with these digital music note manipulatives, called Measure, to show math, music, and STEAM educators how to use them with their students.

Problem context

In standard music notation, longer notes often take up less space on the staff than groups of shorter ones, and therefore, notation and audiation are not always logically related, causing students' confusion (Gordon, 2012). Because of this, in previous work, the first author conducted a study at a summer jazz camp which included paper music note manipulatives were designed and students were taught to use them to brainstorm rhythms to use during improvisation (figure 1). While the use of these manipulatives did impact students' performance (DiCioccio, in review), there were some practical and educational obstacles involved that can be mitigated by digitizing these manipulatives. For instance, cutting the music note manipulatives was a time-consuming task that can be eliminated by putting the manipulatives on the computer. Also, the manipulatives were very small and thus would regularly fall on the floor and get lost, which would not be an issue with digital manipulatives. Educationally, there is concern that disconnect from audiation while teaching notation, students will not learn to read the notation fluently and make auditory meaning of it (Gordon, 2012). In my aforementioned study, I taught the students the meaning behind the notation, connecting the symbols to sound, and when students independently brainstormed rhythms, I encouraged students to clap, hum, or play the rhythms on their instruments. However, the paper manipulatives themselves did not enforce the auditory connections like digitized manipulatives can.

Figure 1

Paper Music Note Manipulatives

Existing tools

There are computerized fraction manipulative tools that do not make connections to music, such as the Fraction Strips tool on Toy Theater (Gaspard et al., 2018). Also, there are computerized tools for music composition. For instance, there are tools such as Sibelius (AVID, 2019) that allow the composer to input standard music notation, which do not utilize fraction bars to help users visualize note duration. Also, there exists tools such as GarageBand (Apple, 2025), which allow the user to record musical performance, either on the built-in computer instruments or in real life. Garage Band allows users to clip and organize sections of these recordings but does not allow the user to see the duration of or manipulate individual notes, unless the user records each note individually. Furthermore, there are such tools that are designed for the purpose of music learning. For instance, there is a tool called Learning Music that guides students through lessons on music theory and composition using computer music tools. While this tool utilizes rectangles to show note duration like the tool in this demo, it does not connect these to the numerical values of the note length, nor does it allow for note durations that are smaller than the beat or other, more complicated rhythms, such as swung notes, triplets, or dotted notes. In addition, there is a computerized tool for music composition called Impromptu that, while not built intentionally for interdisciplinary math and music composition, has been studied to be used in this way (Bamberger & DiSessa, 2003). Finally, Toy Theater also has an online tool for composing rhythms using fraction bars. However, Toy Theater's tool does not use standard music symbols for note durations, nor does it provide time signatures or measures to fit the fraction bars into. Also, Toy Theater's tool only provides options for whole, half, quarter, and straight eighth notes. Thus, the current tool fills a gap by linking fraction bars, a visual representation of the fractional length of notes, with their corresponding standard music symbol and putting this in context of the measure and time signature, as well as expanding the note options to include dotted notes and swung eighth notes.

Theoretical framing

As referenced in the problem context, this tool draws upon Gordon's (2012) Music Learning Theory, as this tool aims to make connections between symbols and audiation explicit. Furthermore, this tool draws upon Constructionism (Papert & Harel, 1991), as it encourages "learning-by-making" by guiding students to learn standard music notation by facilitating rhythm composition. Without the tool, students could compose a rhythm by writing various symbols for durations of notes in standard music notation without learning the auditory meaning of the symbols that they wrote. However, the combination of fraction bars, standard music notation, and auditory feedback is designed to help students learn the meaning of the symbols as they compose.

Tool overview

To use Measure (Goehrig & DiCioccio, 2026), the digital music note manipulative tool, users will first pick a time signature and tempo. To input a rhythm, they will click on the button with their desired note duration and then click on the measure where they would like to place the note. This will display a fraction bar with the fractional length of the measure for the user to visualize the length of the note. It will also have the numerical fraction and standard music notation symbol so that the user can connect the numerical value with the music notation. If a user tries to input a note in a measure where there is no room, the tool will split the note into two tied notes, where the first measure is filled and the second measure completes the note duration, teaching the users which fractional values sum to the time signature. Once users have input their rhythm, they can press a button which will play the rhythm so that they can hear how it sounds. We opted to put this tool on the internet so that it is free and available to the public and able to be used on any operating system.

Uses of the tool

This tool has several educational uses. First, it can be used by music educators to teach students how to read rhythms by visually connecting the length of the note to its symbol. To do so, music teachers can have students input rhythms into the tool from pieces they are studying in class and have students practice performing the rhythms along with the tool or after watching and listening to the tool. Also, it can be used by music educators as a tool to teach improvisation and composition, as in my previous work (DiCioccio, in review), where I demonstrated how to use the manipulatives and tasked students to use them to compose their own rhythms. Furthermore, it can be used by math educators to provide an auditory representation of fractions to deepen understanding of the meaning of fractions and fraction operations. To do so, math teachers can represent fractions and fraction operations in the app alongside other representations when teaching, and they can encourage students to model problems in the app when working independently.

Setup for the interactive session

We will first demonstrate the tool and give an overview of its educational purpose. After that, we will ask the attendees if they have any questions. For in-person attendees, we will then have iPads available for conference attendees to test the tool out for themselves. We will also share the link to the tool with attendees on Whova for online participants to use during the session and for all participants to have available for their future use. Furthermore, we will have paper manipulatives available for in-person attendees to test and compare them to the digital tool. We will share a link for all attendees on Whova with a PDF of the paper manipulatives (figure 1) so they can print and use the paper manipulatives in the future, and for the online participants to print and use during the conference to compare to the digital tool.

Discussion

While the use of digital manipulatives has its advantages, there are also limitations to their use in comparison to paper manipulatives. First, digital manipulatives are not as tangible as paper manipulatives, making the manipulatives more abstract and possibly losing some of their educational value. This has been shown in several studies that compared student achievement from learning fractions through concrete and digital manipulatives, many of which found that student achievement was higher for those that learned fractions with concrete manipulatives compared to digital manipulatives (Al Farra et al., 2014; Vessonen et al., 2021). However, others have found that digital manipulatives are at least as effective as concrete manipulatives, and that they are more time efficient (Mendiburo & Hasselbring, 2014). Another concern is that while technology is increasingly available (Gray et al., 2021), not all schools will have access to the necessary technology to use the digital manipulatives, where paper can be more accessible due to its lower cost. Future research can study the ways in which these differences impact accessibility and learning.

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Early Science Hub: Designing Interactive Simulations to Support Preschool Educators' Understanding of Science Core Ideas and Cross Cutting Concepts

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Abstract: To support preschool educators' integration of science in their classrooms, our project is co-designing the Early Science Hub (Hub), a professional learning resource based on promising components identified in prior work, with partner educators and curriculum/media designers. The Hub and its components are being designed to leverage technology and include educative features to make the resource widely accessible across early learning contexts. This paper details the design and iteration of one specific Hub component, the interactive simulations of science core ideas and cross cutting concepts designed to support preschool educators' understanding of these. The session will invite participants to explore the interactive simulations and engage in a discussion to gather their insights and feedback, which can inform potential future iterations.

Introduction

Preschool teachers have strengths that uniquely position them to support science and engineering learning, including their experience orchestrating activities that meaningfully promote learning across disciplines and working closely with families to support children's learning and development. A few recently designed professional development programs have shown evidence of promise and highlighted potential supports or components that may be helpful to preschool teachers as they work to integrate science in their classroom (Gropen et al., 2017; Lange et al., 2021; McWayne et al., 2021). While these findings, together, are encouraging, more research is needed to understand impacts on teachers' instructional science practice and whether and under what conditions improvements in teacher outcomes translate into improvements in child learning. In addition, while these programs have shown preliminary evidence of promise, they require substantial investments in time and resources, which often pose challenges for adoption and broad impact. Our project aims to co-design the Early Science Hub (Hub), a suite of professional development resources that includes some of the key components that have been identified as promising in prior work, uniquely leveraging technology to make professional resources widely available to preschool educators and embedding educative features to make the resources helpful to coaches and communities of practice existing in the ecology of preschool programs.

Our work acknowledges that teachers are learners who develop expertise and build understanding both individually and collectively and that this process unfolds across time and in various contexts (National Academies of Sciences, Engineering, and Medicine [NASEM], 2021). Thus, our work draws on sociocultural theoretical perspectives that emphasize the important role that social and cultural interactions play in learning and development such as sociocultural cognitive theory (Vygotsky, 1978) and cultural historical activity theory (Engeström, 1999). As part of the Hub development, we are currently engaging with partner preschool teachers to co-design the Hub components and resources. For instance, the Hub video library of science practices and simulations of core ideas and cross cutting concepts will be co-designed to support teachers' individual learning and reflection, as well as their collective learning (in communities of practice and through coaching). Drawing on sociocultural perspectives that bridge home and school learning (Gonzalez et al., 2006), the Hub home school connections toolkit will be designed to help teachers understand learners – to build knowledge of who they are, what their families' cultural capital and lived experiences are like, and what role their community plays in their lives – and leverage that knowledge to design meaningful science learning experiences. Finally, our work also draws on ecological theories of development (Bronfenbrenner, 2000) that acknowledge the interconnectedness of early learning. The Hub interactive map of cross-disciplinary connections will be designed to help teachers integrate science meaningfully with other areas of learning, such as mathematics, language and literacy, and engineering.

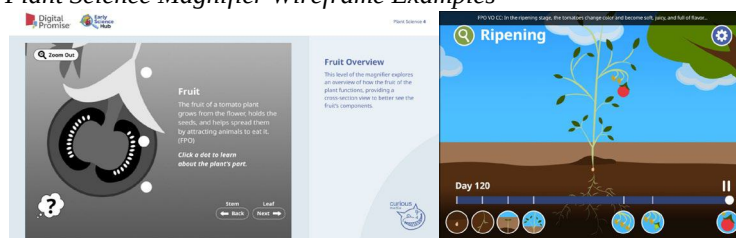
Overview of the interactive simulations

The interactive simulations component of the Hub was designed to support and deepen preschool educators' understanding of developmentally appropriate science core ideas and cross cutting concepts in an engaging and

educative manner. To co-design the simulations, the project team first reviewed findings from our implementation study of science curricula (Dominguez et al., 2023) to identify science learning goals that were most challenging to preschool educators or where they could benefit from professional learning support. Based on recommendations from science educator advisors, our team linked each of these goals to relevant cross cutting concepts and generated seed idea designs across three disciplinary core ideas (Plants, Light and Shadows, Force and Motion). These seed ideas, described below, were meant to be artifacts to spark design discussions with preschool educators during the second co-design workshop.

1. The **Plant Science Magnifier** seed idea emerged from prior findings indicating educators benefitted from support and resources that provide detailed examples of plant parts and their functions. This seed idea aims to allow educators to deepen their understanding of this science learning goal by engaging in a guided walk through with the opportunity to explore additional concepts of interest via a thought bubble feature (see figure 1).

Figure 1
Plant Science Magnifier Wireframe Examples



An initial wireframe depicting a thought bubble feature (left) and a subsequent wireframe with updated color visuals (right).

2. For Shadows, the team conceptualized two simulations: **Shadow Simulator** and **Shadow Science**. The Shadow Simulator seed idea was developed because teaching children how a shadow is created when the light shines on a surface and blocks some of the light was easier for teachers to facilitate when using a man-made light source (lamp or flashlight) rather than a natural light source (sun). Thus, this simulation aims to leverage technology in a way that allows teachers to explore this phenomenon in a controlled manner not possible in real life. The Shadow Science seed idea was conceived to support educators' understanding of how shadows are made while addressing common misconceptions. Similar to the Plant Science Magnifier, this simulation is a guided exploration on how shadows are made yet intentionally incorporates opportunities for educators to isolate and manipulate specific variables that may affect a shadow (see figure 2).

Figure 2
Shadow Science Wireframe Example



An initial wireframe addressing a specific light and shadows learning objective.

3. The **Ramps Physics Simulator** is situated within a miniature golf context where educators engage in a structured environment that allows them to explore how different variables affect the golf ball to learn more about force and motion (see figure 3).

Figure 3
Ramps Physics Simulator Wireframe Example



A second-round wireframe where the user can select different types of objects to test.

Co-design sample and process

Our work included close collaborations with preschool partner sites in CA and VA and a curriculum/media design studio with whom we have established relationships through prior early STEM research projects. Starting in Spring 2025, the project team engaged six preschool educators across the two sites and curriculum/media designers in co-design workshops and will continue to do so for the 2025-2026 school year. The overall aim of these workshops was to build and strengthen relationships with each other so we could collaboratively brainstorm, ideate, iterate and reflect on emerging seed ideas for the Hub components. Depending on educators' availability, the timing of the educator co-design workshops varied from a half-day session to multiple shorter one-hour sessions; however, each workshop included interactive elements and activities designed to address the overarching goal and focused on a specific Hub component. We describe the co-design workshop and related data collection activities below.

Preschool educator survey

Prior to the first co-design workshop, the project team administered an online survey to gather preschool educators' experience implementing science in their classrooms and initial thoughts on the proposed professional learning hub. Survey findings helped the project team better understand and characterize the co-design educator sample and inform the co-design workshop activities. For instance, all educators reported facilitating science learning activities in their classrooms with a majority implementing science "every day" or "a couple of times a week" and only one implementing science "once a week." Despite this variance in the frequency of facilitating science learning activities, all educators reported feeling "very comfortable" with science facilitation and perceive science as "very valuable" for children's learning and development, along with other learning domains. Additionally, educators reported on their comfortability with implementing science practices and core ideas (life science, earth science, physical science). Educators primarily expressed feeling "very comfortable" implementing a majority of the science practices with only one or two educators noting feeling only "somewhat comfortable" with the *questioning* practice and *documenting observations and recording data* practice, respectively. Although these findings are not surprising given educators' prior background participating in early STEM research projects, it also revealed promising areas for further exploration during the co-design phase.

Preschool educator co-design workshop 1

In addition to orienting co-design educators to the project and providing a refresher on early science, the first co-design workshop focused on diving deeper into the science practices. To prepare for the science practices discussion, the project team reviewed early learning frameworks and science standards to create a set of science practices aligned with these multiple sources. As part of this workshop, we explored how a video library could help promote these science practices in early learning classrooms, noting educators' insights on what they liked, what is missing, and any new ideas. For example, educators suggested including different entry points to the videos and broader video library may vary depending on the educators' familiarity with early science, thus highlighting an important design consideration as the Hub development continues. To address this and ensure the developed resource is suitable for newer teachers or those with less experience teaching science in preschool, the project team actively recruited additional preschool partners that fit these criteria.

Preschool educator co-design workshop 2

For the second educator co-design workshop, the primary objective was to discuss and gather educators' input on seed ideas of the interactive simulations of science core ideas and cross cutting practices, which is the focus of

the Interactive Tools and Demos session. During the co-design workshop, the project team walked through wireframes of the seed ideas where teachers were encouraged to discuss elements they liked/disliked, potential improvements and/or different tools that could be designed instead or in addition.

Future co-design activities

Following this second workshop, we synthesized educator and advisor feedback to generate a summary memo to be shared with the designers and help guide design decisions. Based on insights gathered during the second co-design workshop, the designers created another round of wireframes and will have prototypes ready for us to demo with educators in a future meeting. By the end of the co-design phase (Summer 2026), the goal is to have more fully developed simulations ready for this session and a subsequent field study (Fall 2026).

Discussion

Given that co-design workshops continued throughout the 2025-2026 school year, the findings reported in this paper are partial and preliminary. Across both sites, educators generally liked the simulations and rated them highly when asked about the likelihood of using each simulation. While these simulations were described as a teacher-facing resource rather than a child-oriented tool, teachers at one site tended to think about how the simulations could also help their learners, discussing potential ways to utilize the simulations during whole group or small group activities. Thus, some of their design feedback considered the child user experience. To address this at the co-design workshop with the other cohort of teachers, we modified the warm up activity to invite teachers to share about the resources they typically use when planning a science activity, which helped emphasize educators as the target user of the simulations being reviewed. Teachers acknowledged not seeing this type of resource support, indicating its promise, and centered their feedback from the teacher user perspective.

Moreover, educators' feedback highlighted specific design features that they found valuable or worth considering as part of future iterations. For instance, educators liked the thought bubble for numerous reasons. At one site, teachers noted this as a unique feature and liked how the information provided aligns with questions that children may be wondering about. At the other site, teachers liked how the thought bubble was presented as an optional choice and can see how it can be especially valuable for newer teachers or those less familiar with teaching science. Another suggested feature related to audio supports and having an option for a voice to read the text aloud, which can be educative for educators including those who are English learners themselves.

As our team analyzes the co-design workshop data, we will look into the potential factors that may be influencing their view of the simulations as an educator resource versus a child tool, including their classroom context and technology experiences as well as the simulation design and features, and how to triangulate this with their design feedback. In turn, this will help the team identify and prioritize future revisions.

Interactive demo session

During this session, we will share about our co-design process and findings and showcase the latest prototypes of the interactive simulations of the science core ideas and cross cutting concepts. Through hands-on demos, participants will have the opportunity to view and explore the latest drafts of interactive simulations on an iPad from the perspective of a teacher user and share their in-the-moment reflections about feasibility and use via an anonymized Mentimeter link. After they explore and share their insights, we will invite participants to engage in discussion and share their feedback on the simulations, including any revisions or new ideas to consider incorporating prior to the Fall 2026 pilot study, and document these using a Miro design board or chart paper and sticky notes. The use of Mentimeter will also allow us to show trends live to enrich the discussion.

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CLUE: A Learning-Goal Oriented AI-Enhanced STEM Inquiry System

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Abstract: Integrating AI into STEM classrooms risks shortcutting the productive struggle that makes inquiry-based learning effective. This session introduces CLUE, a collaborative platform that addresses this risk through rubric-centered AI personalization: rather than deploying generative AI as an open-ended chat interface, CLUE uses curriculum-tuned categorization rubrics to make student problem-solving strategies visible to both learners and teachers. In this way, CLUE's rubric-centered approach operationalizes sensemaking for any particular curriculum. We conjecture that this design supports teacher noticing, promotes metacognitive reflection, and sustains productive struggle by shifting AI feedback away from answer-giving and toward strategy awareness and peer comparison. We will demonstrate several classroom-tested STEM curricula paired with these AI scaffolds, and attendees will participate in a live collaborative experience to see how CLUE's rubric-based approach makes AI-generated enrichment transparent and purposeful. The session will explore the design opportunities and challenges behind this methodology, invite discussion about alternative approaches, and share directions for future empirical evaluation. Attendees will leave with practical strategies for integrating AI into STEM classes and access to CLUE tools for use in their own classrooms.

Keywords: Artificial Intelligence in Education, Collaborative Learning, STEM Inquiry

Tool demonstration description

Intended audience and duration

This half hour tutorial/demonstration is for teachers of grades 6-12 STEM classrooms wishing to incorporate collaborative learning, interactive STEM inscriptional tools (student-controlled graphs, tables, programs, equations, drawings, etc.), and an AI system that fosters productive metacognitive discussion into their classrooms. This demo will invite teachers to reflect on a system based on the premise that AI should be used to ask the student to reflect on their own answer rather than provide one.

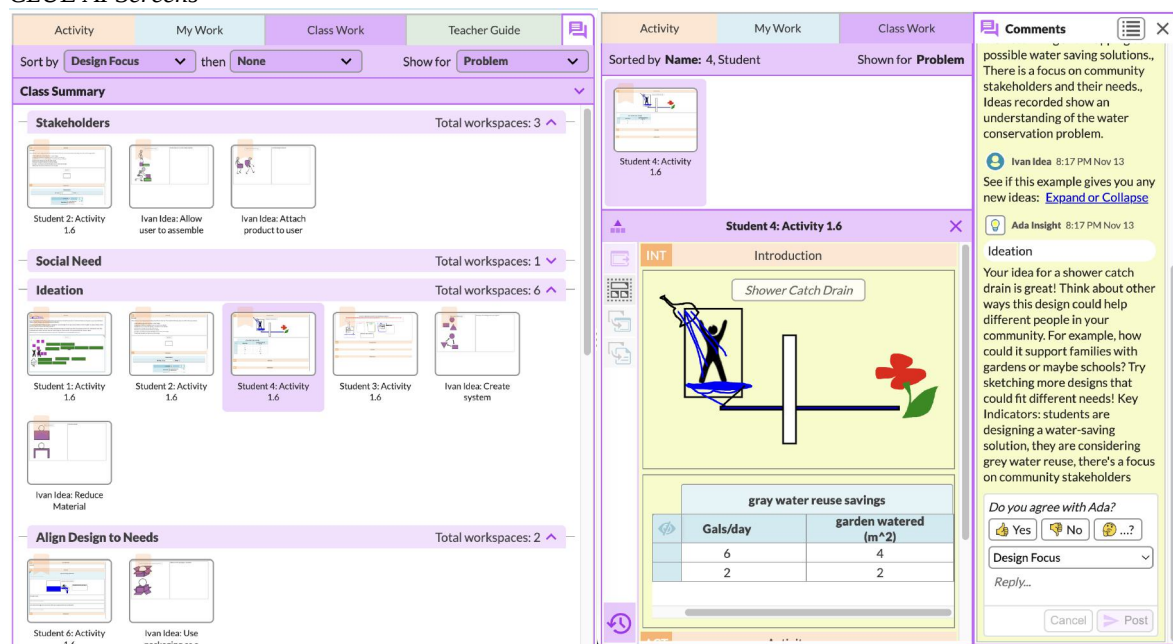
Theme and goals

We will introduce and demonstrate a purpose-built AI-enabled system called the Collaborative Learning User Environment (CLUE) (Bondaryk & Dorsey, 2021). This system allows students to create open-ended digital work with live-linked tools for STEM problem solving including graphs, data tables, programming tools, sensor tools, sketches, and others (Figure 1, right). Students can borrow and expand upon each other's work in real time, and teachers can observe the latest and historical versions of students' documents.

In particular, we will explore CLUE AI features that associate students' work with a categorization rubric that can be tuned to support a variety of curricula (Figure 1, below). We will discuss how centering views of work according to the rubric provides a novel technique for introducing AI assistance into a STEM course. From a theoretical standpoint, this work will be some of the first to use AI to personalize feedback about the students' learning journey rather than focus on problem scoring and sequencing. The ability to identify and track problem-solving strategies across multiple problems and learning goals allows teachers and students to observe how reasoning evolves over time. We will discuss how CLUE AI affordances encourage reflection on key curriculum ideas. These practices are central to students' development as STEM learners, and the CLUE system attempts to provide both automated support to curate volumes of student work and an interface that surfaces topics for discussion.

We will demonstrate and discuss the development and use of sensemaking rubrics for several CLUE curricula as examples both of how to effectively promote productive discussion in STEM environments, and as the guardrails that ensure that AI enhancements in the software do not circumvent the student's experimentation.

Figure 1
CLUE AI Screens



Left view showing the CLUE rubric discussion dashboard. Right view showing individual student AI feedback.

Theoretical background and relevance to field and conference

Collaborative, inquiry-oriented curricula are increasingly used across STEM disciplines, especially in science education (Crippen & Archambault, 2012). These curricula are ideal for generating interest and facility in technical subjects and boosting achievement (Roehrig et al., 2021). In contrast with more traditional math and science teaching, these curricula afford students an opportunity to learn STEM in an environment that more authentically mirrors the practices of practitioners and STEM-literate laypeople (Chinn & Duncan, 2021). Further, collaborative inquiry has been shown to work well for low-performing learners and underrepresented students in STEM disciplines, particularly when combined with peer-to-peer learning that distributes intellectual authority to learners and disrupts status and power hierarchies (Harris et al., 2024). Well-aligned curricula that emphasize a descriptive domain-specific language and progress against outcomes produce more motivation and retention across the subject area (Levine et al., 2008; Winkelmes et al., 2016).

Despite the advantage of this type of curriculum and the productive struggle embodied by inquiry, such courses can be challenging to teach because they require instructors to notice and navigate student thinking with multiple representations and align those noticings with overarching course practices productively in real time (Bolyard et al., 2023; Dreher & Kuntze, 2015). For other teachers, attending to student thinking is a critical part of their planning and reflection (Lampert, 2001; Stein et al., 2008) but they require robust tools to help them summarize efficiently across large numbers of students. Still other teachers face significant challenges when facilitating discussions that build on student thinking (e.g., Leinhard & Steele, 2005).

Supporting these inquiry classrooms full of open-ended descriptive STEM work would seem an obvious application for new large language model (LLM) AI systems. Unfortunately, open-ended AI analysis of student generated content is fraught because of the potential for hallucination, bias, and decline in student critical thinking (Ahmad et al., 2023) exposed by open-ended AI chat interactions. A more structured AI approach is required that enhances teachers' noticing and surfaces moments of productive self-reflection for students. The AI approach taken in CLUE is to try to mimic the best practices of teachers in inquiry classes, supporting both teachers and students to apply the benefits of metacognitive analysis more robustly in their day to day work. For each curriculum developed in the CLUE platform, we work collaboratively with teachers to establish a categorization rubric that specifically allows AI to expose student strategies in problem solving. This AI approach aligns with the needs of the teachers in scaffolding, supports students in self-analysis, and surfaces the larger goals of the curriculum. It also offers a repeatable and influential pathway to codesign, rather than requiring teachers to speculate on 'including AI in their classrooms' which is often too vague or unfamiliar to be productive. Instead

we ask our teacher collaborators to discuss how they evaluate student progress, and what metrics we should look for in student work to identify trajectories of reasoning.

Alongside the strengths of using inquiry to learn, authentic engagement with science involves learning to inquire, individually and as a science community (Chinn & Duncan, 2021; Ford, 2008; Pedaste et al., 2015). Students need to metacognitively develop the practices of the field and understand the nature of the STEM domain(s) they are working with so that they will understand how STEM works and how they can inquire beyond the classroom walls (Veenman, 2012). Accordingly, CLUE's strategy categorization rubrics can be used as a guide to suggest productive directions to students with or without teacher intervention, focusing on expanding strategy use rather than assessing or marking student correctness. CLUE introduces natural options and interfaces for inclusion of the AI categorization that support deep reasoning without revealing answers or over-directing inquiry. The CLUE platform uses a combination of real-time sharing of interactive notational tools (Edson et al., 2023) and AI interventions and summaries that ask students to describe and evaluate what they have learned, what strategies they have employed, and to compare the work of their classmates to observe the variety of approaches that produce comparable results. These results are returned to both students and teachers, allowing them to find better patterns of review and collaboration that support this style of problem-based learning.

We conjecture that ready access to reliable and class-wide rubric-driven categorization will shift classroom discourse toward strategy comparison by making visible the variety of approaches students take, which in turn supports deep thinking about content and the STEM inquiry process, rather than rote memorization of solution techniques. We already have some early evidence that this approach will be productive. For example, in middle grades mathematics classrooms using the Connected Mathematics 4 curriculum (Phillips et al. 2025), we focus on students' proportional reasoning and use a machine learning-supported rubric to analyze their approaches, representations, and strategies rather than correct answers (Edson et al., under review). In this design, AI functions as an analytic layer that identifies and tracks evidence of student reasoning across problems, making patterns of strategy use visible to teachers and students in real time. By structuring feedback around strategy categorization and teacher interpretation instead of open-ended chat responses, this approach offers a principled alternative to answer-oriented AI while directly supporting metacognitive reflection and teacher noticing in inquiry classrooms. The math classrooms engaging in this form of collective summary discussion using CLUE's dashboard views (Figure 1) are able productively use this motivation of class discourse and productive struggle and even reuse and manipulate each other's techniques for greater understanding. Preliminary informal observations from other classroom deployments of CLUE across other research curricula are consistent with this expectation, though we present them here as motivation for future studies rather than as confirmatory evidence.

Outline of planned demonstration

Participants will create logins in the online CLUE system, and will be briefly introduced to the broad features of the system. They will be taken through several lessons in biology, algebra, and engineering design, experiencing first hand the collaborative and AI features of the system. Feedback will be solicited from the audience on the approach employed to incorporate inquiry AI, group work, project-based learning, and interactive tools for exploring dynamic multiple representations.

We will further discuss how AI categorization is technically produced by ChatGPT 4.5 models, including both in the moment and summary information keyed from text and image analysis of student work and logs of sequences of student actions. Participant feedback from the demo will be gathered both on the usability and accessibility of the platform, and ideas about what kinds of rubrics would be effective in different domains and grade levels.

We will provide references to the existing CLUE curricula available through the Concord Consortium's free learning management system, and instruct attendees on how they can create their own custom CLUE curriculum if they desire.

Facilities, equipment, participants

Each participant who wishes to participate in the live collaborative demo will require a laptop with a modern version of the Chrome Browser and internet connectivity. If attendees do not wish to participate in the collaborative demonstration they are still welcome to watch the demo and join the discussion.

Tools demo coordinators

Ms. Bondaryk is the lead designer of the CLUE system, and an investigator on 8 CLUE-related NSF grants. Ms. Bondaryk has introduced new technologies to educational research and school publishing, including the first Web Calculus text, Pearson's Mastering and MyLabs product lines, and Schaum's Interactive Outline Series. She was a defining force behind Mathcad software and the educational version Studyworks. She is widely published on

technology in classrooms, and on collaborative and AI technologies in STEM teaching and learning. The present demonstration brings together a variety of hands-on teacher training experiences Ms. Bondaryk has conducted over the last 6 years.

Dr. Ido Davidesco is an assistant professor and the director of the Lab-to-Classroom Research Group at Boston College's Lynch School of Education. His research is positioned at the intersection of cognitive psychology, neuroscience, and educational technology. He conducts Electroencephalography (EEG) and eye-tracking research in the laboratory and in real-world classrooms to understand fluctuations in attention and engagement over time and their relation to learning. He is also interested in using neuroscience methods to understand how students engage with educational technology.

Dr. Na'ama Y. Av-Shalom is a postdoctoral scholar in the Lab-to-Classroom Research Group at Boston College's Lynch School of Education. Her work centers on students' epistemic thinking and discourse in inquiry-oriented STEM contexts. She is also interested in teachers' learning and implementation of integrated-STEM curricula.

Dr. Alden J. Edson is a research associate professor in the Program for Mathematics Education (PRIME) and the Connected Mathematics Project (CMP) at Michigan State University. AJ is interested in the enactment of problem-based, inquiry-oriented mathematics curriculum by students and their teacher in a digital world and the affordances of innovative mathematics curriculum materials as a context for teacher learning.

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CoDialogue Space: Design of AI-supported Collaborative Learning Platform

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Abstract: This paper introduces CoDialogue Space, a generative AI-supported collaborative learning platform that provides real-time facilitation during group dialogue. While computer-supported collaborative learning holds promise, it often depends on ad hoc analysis, scripted prompts, and static scaffolds. CoDialogue Space addresses these limitations through discursive facilitation strategies, multiuser–AI interaction, adaptive facilitation levels, and instructor-configurable prompts without requiring coding. We present its design features, offering a scalable, theoretically grounded environment for advancing AI-mediated collaborative learning.

Background, goals, and rationales

Learning is social. Computer-supported collaborative learning uses a variety of technological and pedagogical strategies to foster the social nature of learning such as technology-mediated discussion, project-based learning, and online collaborative writing—to support the co-construction of knowledge (Stahl et al., 2006; Ouyang & Zhang, 2024). However, computer-supported collaborative learning may break down when students lack the interpersonal connection and receive limited feedback, which causes greater distraction and a shift toward individual knowledge construction (Cress et al., 2019; Dina & Ciornei, 2013; Strijbos, 2010).

Recent advancements in artificial intelligence (AI), including conversational capabilities and neural networks, may help overcome long-standing limitations of traditional computer-supported collaborative learning (Tan et al., 2022). AI-integrated tools now offer immediate feedback, personalized guidance, adaptive support, and advanced data analytics (Cress & Kimmerle, 2023). Despite the growing trend of integrating AI into collaborative learning, existing research reveals several limitations. Most studies focus on presenting statistical data for ad-hoc analysis, which lacks instructional interventions (e.g., alerts, suggestions, or facilitation) that support learning in the moment (Ouyang & Zhang, 2024). More research is needed to use AI effectively for real-time instruction and richer learning experiences in computer-supported collaborative learning.

Our project addresses this gap by introducing a generative AI-supported collaborative learning platform, CoDialogue Space, in which an AI agent provides real-time facilitation as students engage in goal-oriented tasks assigned by the instructor. Unlike widely used one-to-one AI-user models, our design enables multiuser–AI conversations, supporting collaborative group work. Because instructors vary in their programming expertise, the platform requires no coding and allows instructors to adjust the level of AI facilitation as needed.

This study extends prior research by articulating design principles for generative AI-supported collaborative learning, specifying the features of the AI agent, and examining the learning mechanisms through which the system supports students' progress on goal-oriented tasks. Given the limited literature on the mechanisms and design principles of AI-agent-mediated collaborative learning (Ouyang & Zhang, 2024), this work contributes both a concrete instructional design and empirical insights into how, and under what conditions, such systems enhance learning sciences research.

Conceptual framework: AI-assisted collaborative learning

Our conceptual framework builds on two school of thoughts: 1) Computer-supported collaborative learning, which emphasizes mediated interaction and collaborative knowledge construction (Stahl et al., 2006; Stahl, 2015) and 2) facilitation guides for social learning that highlight discursive strategies such as questioning, revoicing, pressing for reasoning, uptake, and balancing participation (Alexander, 2008; Hmelo-Silver & Brown, 2008; O'Connor & Michaels, 1996). We define AI-assisted collaborative learning as collaborative learning in which multiple users, including students and instructors, interact with an AI facilitator that supports shared inquiry in real time.

AI-assisted collaborative learning extends traditional computer-supported collaborative learning by moving beyond scripted prompts and static representational scaffolds (Kollar et al., 2018; Vogel et al., 2017) toward adaptive, moment-to-moment facilitation (Liu et al., 2024). Recent work on conversational agents and large language models (LLM) demonstrates that AI can monitor participation, detect discourse breakdowns, and trigger targeted facilitation moves such as elaboration prompts, clarification checks, consensus-building cues, and

socio-emotional support (Do et al., 2023; Kim et al., 2021). Yet, existing AI tools rarely operationalize established facilitation moves; most provide generic prompts or participation nudges rather than targeted strategies that shape collaborative reasoning. Many tools rely on fixed rules or predefined cues primarily in highly structured debate or small-scale settings, rather than supporting nuanced interactions that drive deeper peer dialogue. (Do et al., 2023; Sahab et al., 2024). Our system, CoDialogue Space, addresses this gap by integrating real-time discursive facilitation strategies within AI’s adaptivity.

Design features of CoDialogue Space

This section introduces four distinct but inter-related design features of CoDialogue Space for meaningful collaborative learning and instructions.

Design feature 1: Timely, meaningful facilitation by AI for collaborative learning

CoDialogue Space was developed with the belief that dialogue—understood as a structured exchange of ideas in which participants collaboratively explore perspectives to build shared understanding—is a powerful means for learning. CoDialogue uses GPT-3.5 Turbo as the foundational model for the facilitator in our collaborative learning.

Our LLM-powered agent functions as a facilitator rather than an answer-giver. Drawing on established dialogue-facilitation strategies, the agent follows system-level prompt created by the development team. The strategies involve: clarifying ambiguities, connecting earlier ideas, resolving emerging disagreements, encouraging synthesis across contributions, and redirecting off-task drift. Additionally, the system includes participation-support features. For example, if participants remain inactive for a defined period (e.g., two minutes), the agent can send a private, encouraging prompt to re-engage them in the discussion.

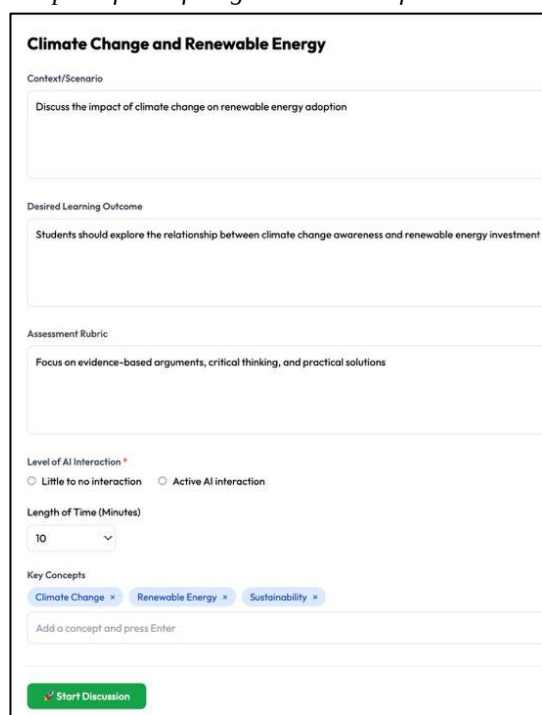
As discursive moves are critical for productive collaborative meaning making and knowledge co-construction (Ouyang & Zhang, 2024), CoDialogue Space provides a not only pedagogically grounded but also scalable model for AI-assisted CL.

Design feature 2: Human-centered AI interface that requires no instructor coding

CoDialogue Space provides an interface for instructors to create personas for AI facilitators. This input will be converted into system prompts once the individual parts are filled out or selected. This design feature enables users to explore various perspectives in dialogue and are contributing towards the learning goals. A sample template for defining the persona of the AI facilitator is shown below (See Figure. 1)

Figure 1

Template for Defining the Persona of the AI Facilitator



Description of each function in Figure 1

- Context/scenario: Overall goal of the discussion
- Desired learning outcome: Primary learning goals that students should achieve in the discussion
- Assessment rubric: Criteria that students will be assessed on based on the quality of their responses in the discussion
- Level of AI interaction: Selection to choose whether or not AI should be a facilitator during the discussion. If Active AI Interaction is selected for the Level of AI Interaction, an additional field (i.e., number of messages) needs to be filled out below. Number of messages indicates how often the AI facilitator should respond in the discussion based on the number of student messages that have been sent.
- Length of time: how long the discussion should last
- Key concepts: the concepts that the discussion should focus on— this will help the AI guide the discussion to focus on these topics

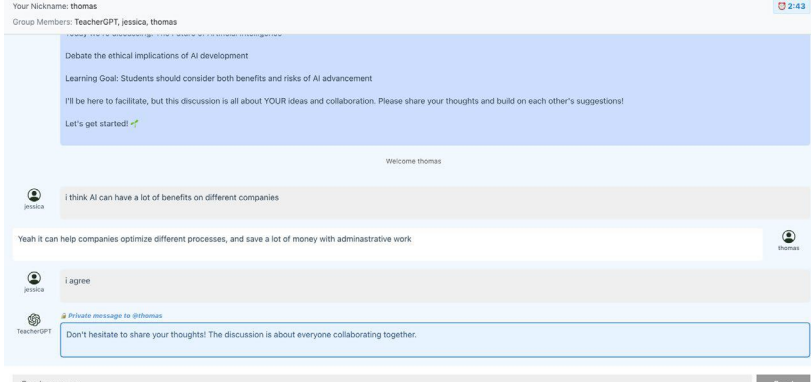
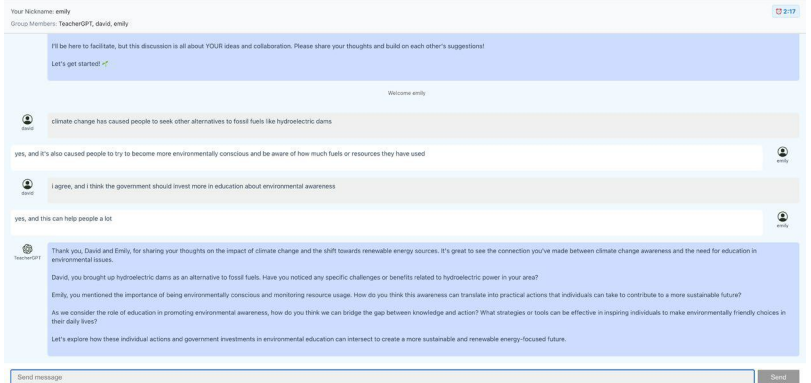
The context/scenario, desired learning outcome, assessment rubric, and key concepts are appended to the existing prompt used for the AI facilitator.

Design feature 3: Multi-user dialogues with an adjustable level of facilitation

Table 1 presents two modes of AI-agent facilitation. The first illustrates “little to no interaction” between the AI agent and students, while the second illustrates active AI facilitation. Descriptions for each mode are provided in the table.

Table 1

An Example of the Level of AI Interaction Provided by CoDialogue Space

Level of AI Interaction	A screen capture of each level
Little to no interaction: Figure on the right shows how an AI agent merely facilitates discussions. Its interaction limits to identifying students who have not contributed to the discussion in the past two minutes. The AI agent sends messages to those students to encourage them to participate.	
Active interaction: This example shows an AI agent’s active facilitation of student discussion. The AI agent detects 4 messages sent by other students. Then, the agent responds by considering the current goals of the discussion and student discussion content. This approach guides students to critically analyze the topics and explore other perspectives.	

Design feature 4: Seamless student workflow that spans learning and evaluation

Students access the CoDialogue Space through a shareable web link provided by the instructor and log in using Single Sign-On or a social identity provider such as Google. Before the dialogue begins, they receive brief onboarding instructions (e.g., “You may type your ideas” and “Take turns when responding”).

Once the dialogue starts, the LLM-powered AI agent initiates the conversation using instructor-specified prompts aligned with the goal-oriented task. The dialogue is designed to be interactive; depending on the level of facilitation set by the instructor (as shown in Table 1). At the conclusion of the activity, when students demonstrate sufficient understanding of the target concepts (as determined by rubric settings provided by the instructor), the AI agent delivers a closing message and suggests next steps for continued learning. It can also direct students to subsequent pages—such as a user experience survey—if enabled by the instructor.

Architecture and security of CoDialogue Space

Our team developed CoDialogue Space using Next.js (Vercel, 2024). MongoDB (MongoDB, 2024) is used as the database to store messages, student names, and other information from chat logs. We use Vercel to host the website for the AI Collaborative Discussion Platform and the frontend page for the discussion interface as well. For the backend, we use Railway and provide resources to enable students to join and leave discussions and the AI to facilitate the discussions (Railway, 2024) We have also created a dashboard for teachers to see student learning engagement data. The platform supports multi-layered analytics combining discourse analysis, temporal learning analytics, network analysis, and process mining. Researchers can trace how meaning-making unfolds over time, analyze shifts in cognitive presence, and identify markers of productive or unproductive collaboration.

The number of users that the platform can support depends on the amount of resources provided by the various pricing plans on Railway and Vercel.

Plan for the presentation

The annual International Society of Learning Sciences (ISLS) conference hosts the Interactive Tools and Demos session. We plan to provide a 20–30 minute presentation designed to introduce our new interactive learning platform for teaching and learning (i.e., CoDialogue Space), showcase activity designs, and connect these tools to ongoing research and practice.

During the session, we will set up a laptop to demonstrate the CoDialogue Space and provide a QR code so participants can explore the platform on their own devices. We structured the session to integrate a live demonstration with discussion of the related research project. Our goal is not simply to highlight the technological features but to illustrate how participants have used the platform, how it is contextually grounded, and how it enhances learning experiences—aligning closely with the conference theme. Proposed Session Structure is

- 0–5 minutes: Overview of the project and introduction of the team
- 5–10 minutes: Interactive demonstration of the tool
- 10–15 minutes: Presentation of preliminary results from the pilot study
- 15–20 minutes: Discussion of planned tool improvements and future studies, including opportunities for audience collaboration
- 20–30 minutes: Question-and-answer session

Implications and conclusions

In this paper, we showcased CoDialogue Space, an AI-supported collaborative learning platform. Our design features demonstrate that the platform is contextually grounded, adaptable to various learning environments. We also presented the system architecture, offering a replicable model for researchers and technologists interested in creating or implementing similar tools. This project has implications for theory, methodology, and practice across the learning sciences and learning technologies.

The first implication lies in its theoretical grounding. Our platform is built on a clear AI-supported collaborative learning framework and enables the study of AI-mediated facilitation in authentic group settings. By implementing varied discursive moves, it allows researchers to examine when, how, and for whom specific facilitation strategies are effective. This supports investigations into mechanisms of productive talk—such as uptake, co-elaboration (Damşa & Ludvigsen, 2016), and epistemic agency (Scardamalia, 2002). The platform also permits controlled manipulation of AI behaviors (e.g., facilitation level, timing, frequency of interventions), creating conditions for causal inference in AI-assisted collaborative learning research. The system’s multi-user structure enables investigation of group-level constructs, such as collective regulation, shared metacognition, and distributed reasoning, which are difficult to capture in one-to-one AI systems (Law et al., 2021).

Second, our study generates a practical tool, readily usable in various learning environments because instructors can set learning topics modulate AI’s intervention while students can seamlessly engage in instructor-generated discussion chats. This platform design also creates a replicable research infrastructure: an open, instructor-configurable environment that can be deployed across varied disciplines, task types, and learner populations. This flexibility supports cross-context comparisons and the accumulation of generalizable evidence about AI-assisted CL.

Third, CoDialogue Space offers significant methodological contributions to research in the learning sciences and learning analytics. Our dashboard enables fine-grained, real-time analysis of collaborative learning processes. Unlike traditional CSCL and discourse analytics approaches that rely heavily on post-hoc coding of transcripts, our platform captures time-stamped interaction data, including conversational turns, AI interventions, participation patterns, and discourse trajectories. This allows researchers to study dynamic mechanisms such as breakdown repair, idea integration, and conceptual drift.

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Lowering Barriers for Designing AI-Embodied Robots for Learning

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Abstract: This paper presents RoboHearts AI, a web-based platform that enables educators, students, and researchers to easily design and deploy large-language-model (LLM)–powered characters on embodied robots such as Pepper, NAO, and LEGO-based extensions. By combining LLM-driven conversational capabilities with robots’ multimodal features, including gesture, animated movement, gaze, and visual display, RoboHearts AI supports naturalistic, adaptive, and contextualized interactions that extend beyond traditional screen-based chatbots. The paper introduces practical applications across educational and non-educational settings, and proposes two pedagogical scenarios: *learning with AI*, where robots act as tutors or learning companions, and *learning about AI*, where students gain AI literacy and deepen their understanding of AI by designing and iteratively refining robot characters. This hands-on demonstration guides participants in character creation and multimodal behavior design. This work lowers barriers to embodied AI adoption and highlights new opportunities for research, instructional innovation, and human-centered learning with AI-embodied robots.

Introduction and background

The rapid development of artificial intelligence (AI), especially generative AI, has made the integration of AI into a variety of applications significantly easier than before. AI-embedded robots with conversational capabilities can provide learning opportunities for learners through communication whenever necessary, as well as intelligent pathways to broader educational and societal needs. These systems combine the multimodal and social presence of robots with the conversational capabilities powered with AI, creating new opportunities for personalized, engaging, and contextualized learning experiences in classrooms and informal settings (Cardona et al., 2023; Følstad et al., 2021; Łukasik & Gut, 2025; Martin, 2025).

Recent advances in natural language processing and large language models (LLMs), which power today’s conversational agents, have enabled AI chatbots to engage in increasingly natural, contextualized, multi-turn dialogue. These capabilities have accelerated their adoption across diverse domains, including education, health, and customer service. In education specifically, a growing number of educators recognize that conversational AI can provide tutoring, on-demand and personalized feedback, and opportunities for learner autonomy at scale. These emerging possibilities, in turn, underscore the need to investigate responsible, human-centered approaches for integrating AI into teaching and learning ecosystems (Cardona et al., 2023; Følstad et al., 2021; Łukasik & Gut, 2025; Martin, 2025).

Embedding AI-driven conversational capabilities into physical robots has transformed them from purely task-oriented tools into interactive social partners capable of listening, responding, and adapting to learners’ needs. Unlike chatbots on computers or mobile devices, AI-enabled robots can draw on embodiment, gesture, gaze, and other nonverbal behaviors to enhance social presence and rapport, which can shape learners’ affective responses and perceptions of the interaction. These embodied interactions have the potential to reduce learner anxiety, increase motivation, and sustain engagement, particularly in domains such as language learning, communication skills, and other activities where relational interactions play a critical role (Park & Neumann, 2024).

AI-embodied robots can act as interfaces to curriculum-aligned content, analytics dashboards, and adaptive learning services, creating collaborative learning opportunities for teachers and students. Such platforms can enable individualized support, including just-in-time explanations and formative feedback, while also giving educators new ways to orchestrate classroom activities and monitor student progress (Cardona et al., 2023; Martin, 2025).

Chatbots in education have evolved from simple rule-based systems to advanced conversational agents capable of supporting complex teaching and learning processes across K-12, higher education, and lifelong learning contexts. The literature suggests potential benefits for learning support, motivation, and scalability; however, it also highlights methodological, ethical, and pedagogical challenges that remain as a concern (Aguilar, 2024; Eteng-Uket & Ezeoguine, 2025; Okonkwo & Ade-Ibijola, 2021; Thi Thuy Mai et al., 2024; Wollny et al., 2021). Research generally reports positive or mixed effects of chatbots on learning outcomes, engagement, and motivation, though effect sizes and robustness vary. Studies show that AI chatbots can improve comprehension, retention, and application of knowledge, while highlighting the need for grounded pedagogical principles to support learning (Aguilar, 2024; Eteng-Uket & Ezeoguine, 2025; Körner et al., 2025, 2025; Wollny et al., 2021).

The literature indicates that chatbots in educational settings commonly assume roles such as tutor, teaching assistant, coach, or peer. Pedagogical conversational agents, in particular, are often designed to deliver step-by-step instructions, formative feedback, and/or socio-emotional support. Research highlights the importance of empathetic, adaptive, and feedback-rich interactions, such as metacognitive prompts or affective feedback, in enhancing learner retention, metacognitive sensitivity, and overall engagement in chatbot-based learning environments (Aguilar, 2024; Körner et al., 2025; Wollny et al., 2021; Yin et al., 2025). However, reviews also highlight several critical challenges, including the need to align chatbot evaluation with pedagogical objectives, design for mentoring and sustained use, ensure data privacy and transparency, and mitigate risks related to bias and misinformation. Research on LLM-based tools, such as ChatGPT, identifies notable strengths, including personalization, immediate feedback, and support for idea generation, while also documenting significant limitations. These include generating inaccurate information, fabricated citations, hallucinated content, and persistent uncertainty regarding their impact on learners' motivation, overreliance, self-regulation, and long-term learning behaviors (Eteng-Uket & Ezeoguine, 2025; Thi Thuy Mai et al., 2024; Wollny et al., 2021).

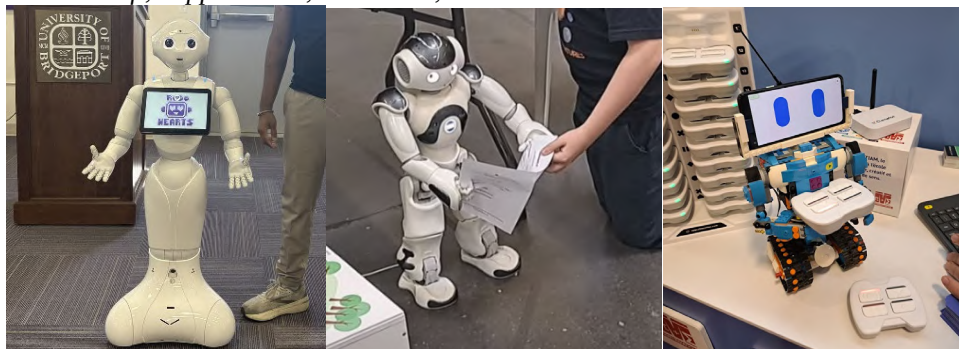
The literature underscores the ongoing need for research on the integration of chatbots in both education and broader societal contexts. At the same time, scholarship on AI-embodied partner or companion robots remains in its early stages, with many questions still unexplored. In response to these gaps, this paper introduces an AI-powered tool that enables educators, students, and other users to easily create AI-embodied robots tailored to support diverse learning goals and practical needs. This work aims to lower barriers to adoption while expanding opportunities for human-centered, embodied AI applications in educational settings.

RoboHearts AI – designing robot personality for AI-embodied robots

RoboHearts AI is an online system initially designed for Pepper and NAO humanoid robots (see Figure 1), originally developed by Aldebaran, providing users with an accessible interface for creating and deploying LLM-driven characters within embodied robotic platforms. Since the system is a web-based interface, there is no installation requirement, which could make it difficult to implement in school environments where the school IT department's permission is required to install additional applications. AI-embodied robots created with RoboHearts AI are multimodal, extending far beyond the capabilities of traditional chatbots by integrating visual displays with images and/or text, animated movements, and expressive facial behaviors. Each character can speak a specific language or switch between multiple languages. RoboHearts AI is currently in development to expand its capabilities to other robotic platforms.

Figure 1

From the left, Pepper Robot, Nao Robot, and LEGO extension with Android Phone



Functionality – How it works

With RoboHearts AI, a multilingual conversational platform equipped with a user-friendly Character Editor, users can create and refine multiple AI characters (see Figure 2). The Character Editor enables users to define each character through customizable instructions that specify personality, role, and scenario. Within the interface, users can adjust a range of parameters, including prompts, which can be typed and edited in the textbox on the platform, voice settings, and capabilities, such as animated movements, movements, and display behaviors. Once developed, characters can be deployed to robots by installing the corresponding application on each device.

RoboHearts AI can orchestrate those multimodal robot actions, including animated movements and gestures (wave, point, dance, and express emotions), tablet display (only with Pepper, show images, text, floor plans, or conversation prompts), and interactive suggestions (offer touch-ready responses on Pepper's tablet to

help conversations flow). The RoboHearts AI decides when to use these functions based on context and the character's instructions.

Using the Character Editor, users can also specify the animated movements the robot will perform. These movements can be defined through special commands or by embedding emojis within the text editor of the instruction window, where users create the prompt that defines a robot's behavior (see Figure 3). This allows users to seamlessly integrate expressive, context-appropriate animated movements into the robot's conversational responses.

Figure 2
RobotHearts AI Characters

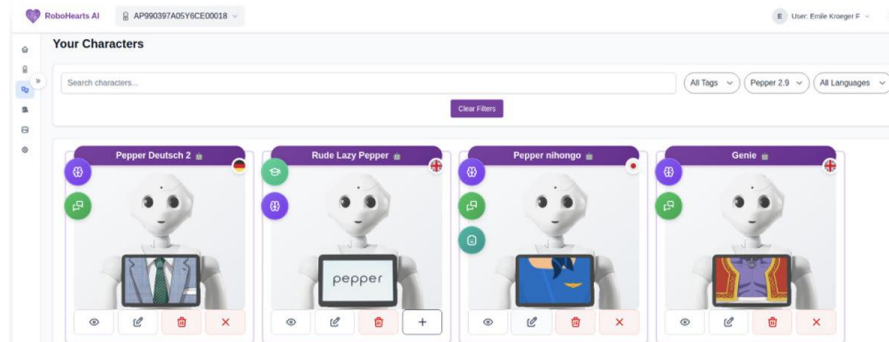
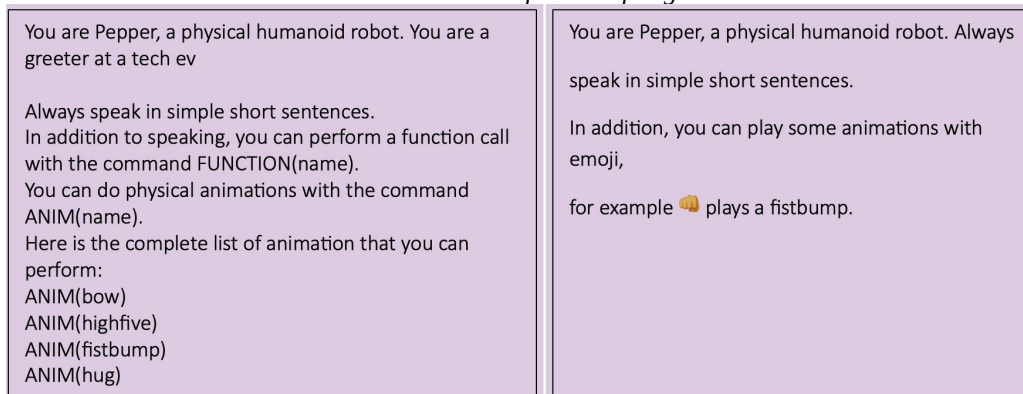


Figure 3
RobotHearts AI Characters Instruction Window for Prompting



Practical applications

RoboHearts AI offers a wide range of applications across educational and non-educational contexts. Current use cases demonstrate its versatility. It has been used in human-robot interaction research, deployed as a conversational agent for customer support in settings such as museums and retail environments, and integrated into educational contexts, including vocational training with nursing students (see Figure 4). These examples illustrate the system's capacity to support research, enhance user engagement, and facilitate skill development across diverse domains.

In educational contexts, we propose two primary scenarios: *learning with AI* and *learning about AI*. In the *learning with AI* scenario, teachers can design AI characters that align with a lesson's learning goals. AI-embodied robots can serve as learning companions for small groups in the classroom or as individualized tutors outside of class, providing rich learning experiences supported by on-demand responses and adaptive interactions, promoting sustained engagement. In the *learning about AI* scenario, students become the creators of robot characters using the RoboHeart AI platform. Students may already be familiar with generative AI tools, such as ChatGPT, on their personal devices. Although these tools are interactive, traditional generative AI chatbots do not inherently support students in *learning about AI*, as they offer limited visibility into how AI systems work, make decisions, or process information. Becoming a creator of robot characters provides students with valuable learning opportunities, allowing them to experiment with prompting and immediately observe the results through their interactions with AI-embodied robots. They may need to refine their prompts or adjust the robot's animated movements to make the human-robot interaction feel more natural. This iterative process not only deepens their

understanding of AI but also enriches their overall learning experience. Through the process of creating robot characters, students will learn 1) how prompting contributes to desired outcomes; 2) how LLMs control not only speech but also physical behaviors, including robot actions and movements; and 3) how to explore and test different use cases by deploying their robot characters with users and iteratively refining their designs. To explore student learning experience, the pre-/post-survey or interviews, and recording of student dialogue during the development and test process will shed light on how students deepen their understanding and curiosity towards AI.

Figure 4

AI-Embodied Robots in a Museum (left) and nursing training session (right)



Demonstration - Test and explore the potential of AI-embodied robots

This demonstration will offer a hands-on exploration of the RoboHearts AI platform. During the demonstration, participants will explore how prompting influences the AI-embodied robot's outcome - speech and physical behaviors by co-designing robot characters and deploying it in an AI-embodied robot.

First, we will demonstrate how RoboHearts AI works by showcasing a variety of pre-created characters aligned with the use cases described in the previous section, helping participants understand the platform's key functionalities. This demonstration will include advanced features such as speech customization, animated movements and gestures, and tablet display options (for Pepper robots only). Next, we will engage participants in a co-design activity, collaboratively creating a robot character using their input to illustrate how the platform's functionalities can be edited, customized, and deployed on the robot in real time. In addition, we will demonstrate the prescribed conversation option using QiChat, the scripting language for creating structured, interactive dialogues on NAOqi-enabled robots, such as NAO and Pepper. This feature allows users to design controlled and purpose-specific conversational flows, ensuring that the robot stays within predefined dialogue boundaries while still enabling meaningful interactions.

We will also introduce additional robot options, such as a LEGO Spike robot paired with an Android phone to integrate a RoboHearts AI character. This alternative is especially important for broadening access, as many classrooms and schools already have LEGO Spike kits and can adopt RoboHearts AI without purchasing new hardware.

Finally, we will invite participants to engage in discussions about potential use cases and learning activities beyond those we have identified, as well as explore possible research agendas that can advance our understanding of how AI-embodied robots can be integrated into teaching and learning environments and enhance student learning.

Conclusion

The rapid development of AI, particularly generative AI powered by LLMs, has made it possible for platforms, like RoboHearts AI, to support robots that interact and communicate more naturally with human users while leveraging the rich multimodal functionalities these robots already possess. RoboHearts AI broadens access to such powerful learning tools by offering affordable, easy-to-use features designed for non-technical users, including educators and students. This significantly expands opportunities for hands-on, experiential learning in both formal and informal educational settings. As AI-embodied robots become increasingly feasible and accessible, we can expect to see broader adoption in classrooms in the near future. This demonstration provides an important opportunity to experience and explore effective use cases, identify ways to enhance students' learning experiences, and ensure that AI-embodied robots have a positive impact on student learning, daily life, and society more broadly.

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TeamLearn: A Multimodal, AI-enhanced Dashboard for Reflection During Collaborative Problem-Solving

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Abstract: This demo presents TeamLearn, a multimodal collaboration analytics dashboard designed to facilitate reflection and provide feedback on developing group artifacts during long-term collaborative problem-solving projects. By integrating meetings, documents, and collaboration experience analytics, the dashboard offers visualisations that help groups monitor participation, task progress, and collaboration dynamics. Developed through a design-based research process, the TeamLearn dashboard provides structured opportunities for reflective dialogue. The demo will showcase the dashboard's core features and illustrate how it can be integrated into collaborative project activities.

Background

Collaborative problem solving (CPS) has become a central element of university teaching, particularly in programmes where students work on extended, open-ended projects that demand sustained coordination and joint knowledge construction. Despite its pedagogical value, collaboration of this kind is often difficult for students to manage. Prior research shows that groups frequently encounter problems such as uneven contributions, fragmented communication, and uncertainty about how their work aligns with shared goals (Damşa, 2014). These difficulties persist partly because CPS depends on several intertwined processes—understanding and advancing the task, monitoring strategies and progress, and maintaining effective social interaction—which students cannot easily observe or evaluate on their own (Graesser et al., 2020).

For teachers, supporting these processes is equally challenging. Much of the interaction that shapes collaboration occurs outside formal teaching events, leaving teachers with limited access to the day-to-day dynamics of group work. As a result, they often lack the information needed to intervene at timely moments, identify emerging issues, or help students adopt productive regulatory strategies (Schneider et al., 2021). In long-term, project-based settings, this lack of visibility means that misunderstandings, disengagement, or coordination breakdowns may only become apparent at the end of the project, when opportunities for formative guidance are largely lost (Muukkonen et al., 2020).

Tools that visualise traces of interaction such as learning analytics dashboards (LADs)—have been explored to support both students and teachers in understanding and reflecting on collaborative work. These tools can help make aspects of group activity more transparent, encouraging reflection on participation, task progress, or communication patterns (Kaliisa et al., 2024). However, existing dashboards typically present only basic or high-level indicators (e.g., the number of logins), limiting their usefulness when learners need deeper support in interpreting what the displayed data mean for their ongoing collaboration. Although advances in multimodal analytics and AI have broadened the types of data that can be analysed—from speech and documents to self-reports and behavioural cues—relatively few systems translate these capabilities into pedagogically grounded support for use in authentic higher education settings (Martinez-Maldonado et al., 2019). Recent reviews highlight that analytics tools often remain technically sophisticated but pedagogically under-integrated, leaving learners without adequate scaffolding to make sense of their collaborative processes (Esterhazy et al., 2025, Kaliisa et al., 2024).

To address these challenges, this demo introduces ‘TeamLearn’, a multimodal collaboration analytics dashboard explicitly designed to support reflection in small-group CPS. TeamLearn was developed through a participatory, design-based process involving teachers and students to create pedagogically meaningful visualisations. The dashboard integrates data from meetings, documents, and collaboration experience surveys to

offer groups shared, interpretable views of how their work is progressing. In this demo paper, we outline the pedagogical motivations and design principles behind TeamLearn, present its key features and underlying technical components, and discuss how it can be incorporated into instructional designs that embed structured, data-supported reflection throughout collaborative project work.

Introducing TeamLearn and its design process

TeamLearn is a multimodal collaboration analytics dashboard designed to help students navigate the complexities of CPS in project-based tasks. The system brings together data from group meetings, shared documents, and students' reported collaboration experiences, offering integrated visualisations that make central aspects of group work easier to interpret. The overarching goal of TeamLearn is to provide groups with a shared frame of reference that supports ongoing reflection, planning, and coordination during extended collaborative activities. The pedagogical foundations of TeamLearn draw on well-established perspectives from the learning sciences. CPS is widely understood as an activity that unfolds through the interplay of epistemic, metacognitive, and social processes, all of which must be monitored and negotiated for groups to be effective (Graesser et al., 2020, Järvelä et al., 2016). These processes, however, are typically implicit and distributed across interactions, tools, and time, making them difficult for students to observe, evaluate and act upon. TeamLearn is designed to make these dimensions more transparent by externalising participation patterns, task-related progress, and relational dynamics. In this way, the dashboard operates as a mediating artefact (Säljö, 2002), supporting groups in developing a more coordinated understanding of how their collaboration is unfolding. Its interface design follows human-centred and participatory principles (Giacomin, 2014), prioritising clarity, low cognitive load, and ease of interpretation.

The development of TeamLearn followed an iterative design-based research (DBR) process conducted over three course iterations. This process combined needs analysis, collaborative design sessions with students and teachers, and successive cycles of prototyping and refinement. Early stages focused on identifying the challenges students and instructors faced in monitoring long-term group work. At the same time, later iterations concentrated on translating those needs into multimodal analytics that were both technically feasible and pedagogically meaningful. Co-design activities played an important role in shaping which data were prioritised, how analytics should be visualised, and how the dashboard could best support reflection at different stages of collaborative work. While this demo paper provides only a high-level overview of the design trajectory, the complete methodological account—including empirical insights from each DBR stage is presented in Damşa et al. (2024) and other work in progress. The outcome of this multi-year process is a dashboard that integrates multimodal data streams and presents them through three interrelated components: meeting analytics, document analytics, and collaboration experience analytics. The following section describes these components in more detail, along with the technical mechanisms that support their generation.

The TeamLearn dashboard: Features and technical setup

TeamLearn brings together multimodal data from group meetings, shared documents, and students' self-reported collaboration experiences to create a unified environment for reflecting on collaborative problem solving (CPS) (See Figure 1). The system combines automated data processing pipelines with institutionally hosted AI services, such as transcription services to generate analytics, and large language models (GPT LLMs) to summarise documents and meeting transcripts in a way that is both pedagogically meaningful and aligned with European data-protection requirements (GDPR). The dashboard organises these analytics into three components: 1) Meeting Analytics, 2) Document Analytics, and 3) Collaboration Experience Analytics—each produced through dedicated pipelines that transform raw data into interpretable feedback.

Meeting Analytics is based on recordings of group meetings captured in Microsoft Teams, whether these occur entirely online or in hybrid/physical formats. Once a meeting is recorded, the audio is transferred to an automated processing pipeline. Speaker diarisation and extraction of speaking times and turn patterns are performed by combining Teams' transcription service (Azure AI Speech) and autotekst.uio.no (NB-Whisper), both secure, university-provided speech services that ensure accurate segmentation, which are then calculated and visualised using programming code. The content of the meeting is analysed by an internally hosted AI model (GPT LLM) guided by customised system prompts. These prompts specify how the discussion should be summarised, including identifying key topics and extracting up to 10 actionable items. This feature helps students recall the content of group meetings, monitor participation patterns, and track progress across meetings.

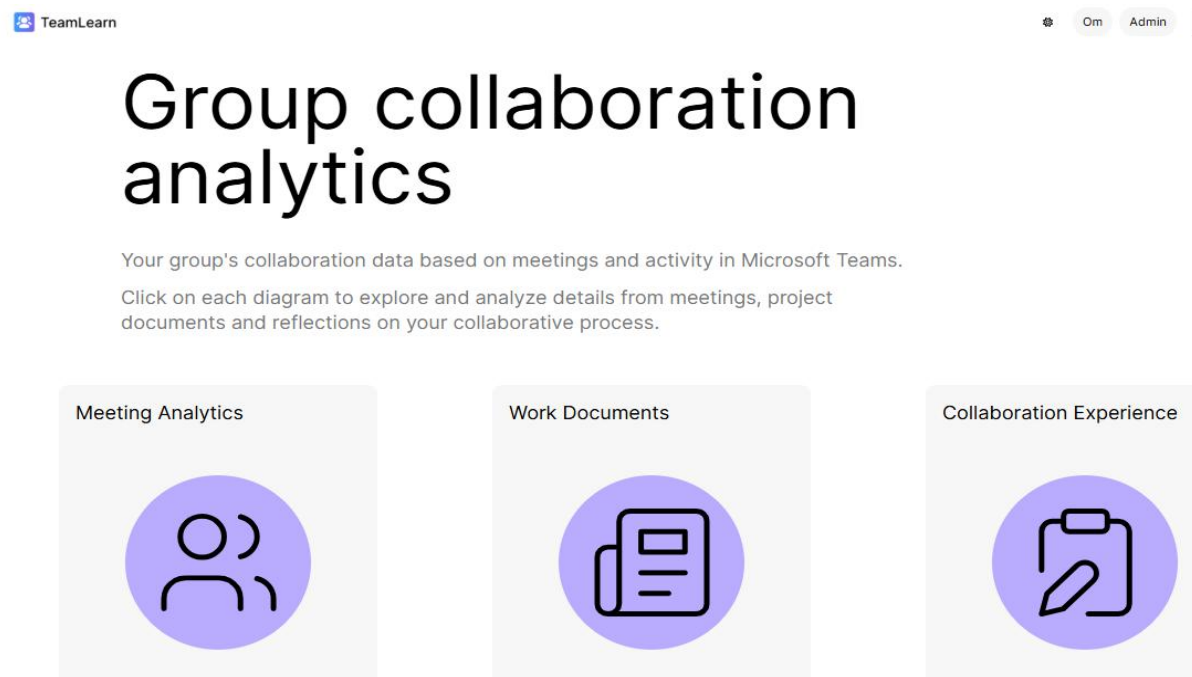
Document Analytics draws on the documents that groups produce and store in their Microsoft Teams (SharePoint) folders. TeamLearn retrieves these drafts through institutional integrations and applies AI-assisted analysis using a GPT LLM to generate summaries and formative feedback. This feedback is structured through

system prompts that encode the task requirements and expected elements of the group report. The resulting analytics highlight central ideas addressed in the draft and indicate aspects that may require further attention. The intention is to support epistemic monitoring by helping students assess how well their developing work aligns with the project's academic expectations.

Collaboration Experience Analytics visualises students' perceptions of their group processes based on short surveys administered at key stages of the project. These surveys are completed through a secure university survey system, and the dashboard automatically retrieves responses. The resulting visualisations show aggregated perspectives on dimensions such as communication, contribution, confidence in progress, and role clarity. By revealing areas of alignment or divergence in group members' perceptions, this component supports socially shared reflection on the relational and organisational dynamics of collaboration.

Figure 1

TeamLearn landing page



The TeamLearn landing page showing the three categories of collaboration analytics: meeting analytics, document analytics, and collaboration experience analytics.

Instructional use cases

TeamLearn is intended to be integrated into courses where students work on extended collaborative projects that require structured opportunities to reflect on their progress and collaboration dynamics. The dashboard can be embedded at predetermined points in the course—such as after major meetings, milestone submissions, or before supervision sessions—to help groups take stock of their participation patterns, task alignment, and shared experiences. Instructors can use the Meeting Analytics to support discussions about coordination and role distribution in supervision meetings. At the same time, Document Analytics can help students evaluate their evolving output against project requirements. The Collaboration Experience Analytics can serve as a prompt for conversations about communication quality, group climate, or emerging relational issues that might otherwise remain unspoken. Because each module offers a different perspective on collaboration, the dashboard can be used flexibly to support different pedagogical goals. For example, it may support reflective debriefs after major group activities, serve as a shared reference point during teacher–student consultations, or be incorporated into structured reflection tasks that encourage students to plan next steps. Although TeamLearn does not replace teacher guidance, its analytics can help students arrive at supervision sessions with a clearer understanding of their current situation, enabling more focused and productive conversations about how to move forward.

Planned demo session

The demonstration will be delivered as a presenter-led walkthrough of the TeamLearn dashboard. Because the system currently operates entirely within institutionally managed and protected environments, external users cannot access the live interface. All data including meeting recordings, documents, and survey responses—are stored and processed within secure university services, and analytics are visible only to members of each student group. These data-handling requirements ensure strong privacy and confidentiality, but also mean that the demo will rely on example data displayed by the presenter rather than participant interaction with the tool. During the session, the presenter will guide attendees through the dashboard's three components: Meeting Analytics, Document Analytics, and Collaboration Experience Analytics, illustrating how each module processes and visualises different forms of collaboration data. The walkthrough will highlight how the tool is used in practice within a course context, how analytics support structured reflection at various stages of group work, and how the system aligns with broader pedagogical aims related to CPS. The demonstration will conclude with a brief discussion inviting questions about potential pedagogical applications, future development directions, and how similar analytics infrastructures might be adopted in other instructional contexts.

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Introducing Scratch-Quantum: A Block-Based Interface for Quantum Computing Education

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Abstract: Quantum computing is increasingly recognized as a critical technological domain; however, its mathematical formalisms and Python-based programming interfaces pose substantial barriers for K–12 learners. To address these challenges, this study introduces Scratch-Quantum, a block-based environment that integrates quantum computing capabilities into Scratch. Leveraging IBM Qiskit as a backend, Scratch-Quantum enables learners to construct executable quantum circuits using visual blocks, allowing them to explore qubit initialization, gate operations, and measurement without writing code. This paper presents the design principles and system architecture of Scratch-Quantum and illustrates how learners can explore core quantum phenomena—such as superposition and entanglement—by constructing and running circuits within the tool. We propose this tool as an entry point for introducing quantum concepts into elementary and middle school classrooms by lowering syntactic difficulties and broadening access to quantum computing education.

Introduction

As global interest in quantum technologies accelerates, fostering quantum-literate future citizens has become a major educational priority (Bova et al., 2021; Nita et al., 2023). The Q-12 Education Partnership's K–12 Quantum Information Science and Engineering (QISE) Framework identifies foundational concepts—such as qubits, states, measurement, quantum gates, and entanglement—as essential components of early quantum learning (Q-12 Education Partnership, 2025). Several countries have recently begun incorporating quantum computing into public education, increasing demand for accessible tools that non-specialist teachers can use in classrooms (Seskir et al., 2022). Yet most existing instructional tools rely on text-based programming. These environments require learners to simultaneously acquire programming syntax and abstract mathematical representations, which generates extraneous cognitive load that diverts attention from quantum concepts themselves (Sweller, 1988; Yusuf et al., 2025).

To address these challenges, we developed Scratch-Quantum, a visual programming environment that embeds quantum computing functionality into the widely accessible Scratch platform. Scratch-Quantum allows learners to build quantum circuits through a familiar block-based interface and observe measurement outcomes, helping them focus on conceptual understanding. While the conceptual difficulty of quantum phenomena remains an inherent challenge, the block-based interface reduces syntactic barriers so that learners can devote more cognitive resources to grasping fundamental quantum concepts through experimentation and observation. Scratch-Quantum primarily targets elementary and middle school students (grades 5–9) who have basic familiarity with block-based programming and introductory probability concepts. The tool aims to broaden access to quantum computing education and provide a foundation for integrating quantum concepts across the STEM curriculum.

Theoretical framework

Scratch-Quantum is theoretically grounded in Constructionism, which holds that learners actively form knowledge by building and manipulating external artifacts (Papert, 1980; Papert & Harel, 1991). Scratch embodies this philosophy, enabling learners to actively construct computational concepts by creating meaningful projects such as animations and games (Resnick et al., 2009).

Two challenges characterize quantum computing education. The first is the syntactic barrier, stemming from the complexity of text-based programming languages. According to Sweller's (1988) cognitive load theory, recalling and typing code syntax generates extraneous cognitive load unrelated to the learning objective. Scratch-Quantum replaces code writing with block assembly, reducing this extraneous load and allowing learners to focus their working memory on understanding quantum concepts. The physical shapes of blocks provide visual cues about grammatical validity, shifting learners' attention from low-level syntax to higher-order reasoning (Bau et al., 2017).

The second is the conceptual barrier, arising from the counterintuitive nature of quantum phenomena such as superposition, entanglement, and probabilistic measurement (Seegerer et al., 2021). This barrier exists

regardless of the programming medium and cannot be resolved by a block-based interface alone. Scratch-Quantum addresses it in two ways. First, measurement outcomes are rendered as sprite behaviors, transforming abstract quantum states into observable phenomena. Second, histogram visualizations of repeated circuit executions and classroom dashboard aggregation allow learners to experience the relationship between the indeterminacy of individual measurements and collective probability distributions.

Existing web-based circuit simulators support learners in assembling and running quantum circuits but limit learners' interactions to circuit construction alone. QScratch (Escanez-Exposito et al., 2025) offers metaphorical experiences of quantum concepts but does not provide an environment for building and executing actual quantum circuits. Scratch-Quantum goes beyond these limitations by directly integrating executable quantum operations into a block-based programming environment.

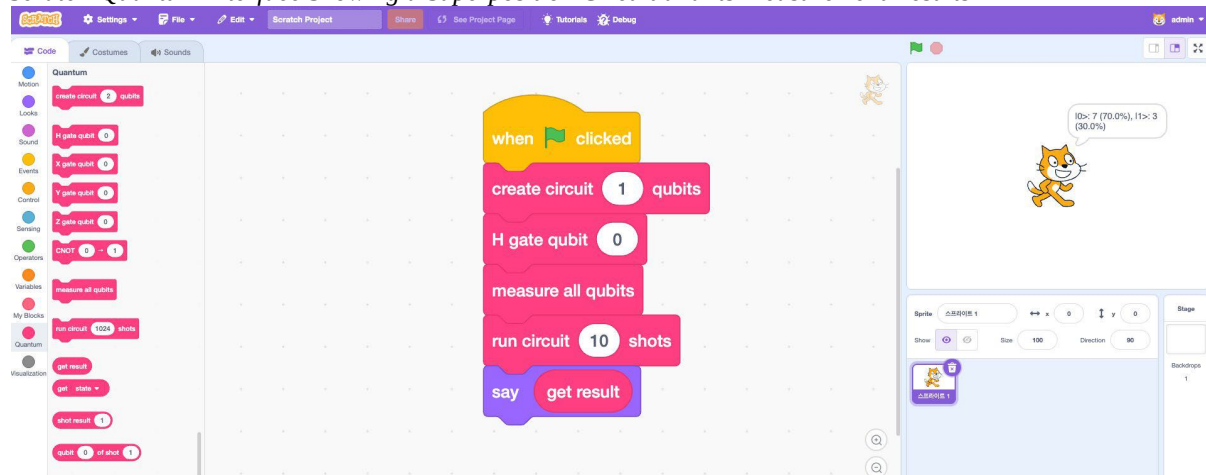
Quantum concepts are difficult to understand through explanation alone, and passive learning can increase the risk of misconception formation (Hu et al., 2024). In Scratch-Quantum, quantum blocks serve as what Papert (1980) called "objects to think with": learners construct circuits, predict outcomes, and explore the probabilistic nature of quantum measurement through repeated execution. For example, students can create a project in which a sprite's appearance changes based on quantum measurement results—applying an H gate places a qubit in superposition, so each execution produces a different sprite appearance depending on the collapsed state. Adding a CNOT gate creates entanglement, causing two sprites to always change to the same appearance, making the correlation between entangled qubits directly observable. This perfect correlation is possible only in a quantum context, giving learners a tangible experience of the difference between classical and quantum phenomena. Furthermore, by supporting hybrid programming that combines classical control structures (loops, conditionals) with quantum gates, the tool establishes a conceptual bridge toward text-based quantum frameworks such as Qiskit (Papert & Harel, 1991).

Design and features

Scratch-Quantum is a browser-based application hosted on a dedicated web platform built upon the open-source Scratch codebase, requiring no installation. The environment is designed to let students build and simulate quantum circuits. This section describes the core features of Scratch-Quantum and the learner experience it supports.

The central component is the Quantum Blocks library, which represents key Qiskit operations through an intuitive drag-and-drop interface. Single-qubit gates, multi-qubit gates, and measurement operations are each represented as distinct blocks. By assembling these blocks, learners can construct and execute quantum circuits without writing code (see Figure 1).

Figure 1
Scratch-Quantum Interface Showing a Superposition Circuit and its Measurement Results



Scratch-Quantum integrates quantum operations with classical Scratch logic, allowing learners to embed measurement results in conditional statements or run repeated executions through loops. Each block-based circuit is automatically translated into Qiskit code in the background, creating a pipeline between visual programming and standard quantum frameworks. This builds conceptual continuity and provides a steppingstone toward text-based quantum programming for learners who advance to higher levels.

To support classroom use, Scratch-Quantum includes a cloud-based Classroom Orchestration system that allows teachers to monitor learner progress in real time (see Figure 2). Students can share their screens using the Send Screen feature, and the Student Gallery dashboard enables the class to browse one another's projects, compare circuit designs, and reflect on outcomes. These features facilitate timely teacher feedback and create opportunities for peer learning. Together, these features position Scratch-Quantum as both an accessible introductory environment and a pathway toward more formal quantum programming. By situating authentic quantum computing within the familiar creative context of Scratch, the tool supports inquiry-based learning while maintaining conceptual fidelity to the underlying physics.

Figure 2
Student Gallery Dashboard Displaying Superposition Measurement Results Across Learners



System architecture and implementation

The architecture of Scratch-Quantum connects a familiar block-based programming environment with a high-performance quantum simulation engine, allowing learners to engage with quantum computing without installation or configuration barriers. The front-end is built on the Scratch GUI and Blocks framework, augmented with a custom quantum block library. Using blocks such as H gate, X gate, and measure qubit, learners can assemble visual representations of quantum circuits entirely within the Scratch canvas.

A middleware layer translates each block configuration into Qiskit-compatible Python code. When a learner runs a program, the translated script is sent to a server-side quantum simulation environment, where the circuit is executed and measurement results are returned to the Scratch interface. These results update sprite behavior or trigger conditional logic within the learner's project, enabling quantum circuit simulation within a visual programming paradigm.

To support real-world classroom use, Scratch-Quantum incorporates cloud-based real-time synchronization. Leveraging Supabase, the system manages project data, execution traces, and screen-sharing events. When learners use the Send Screen feature, their current state—including circuit configurations and experimental outputs—is uploaded to the cloud database. Teachers access this information via the Student Gallery dashboard, which visualizes the class's collective progress in real time. This orchestration infrastructure transforms Scratch-Quantum from an individual learning tool into a collaborative classroom platform, supporting monitoring, formative feedback, and whole-class reflection around quantum experimentation.

Interactive demo session

In the interactive session, participants will experience Scratch-Quantum from a learner's perspective. No prior knowledge of quantum computing is assumed; the session begins with a brief, jargon-free introduction to superposition and entanglement. For example, superposition is described as "a coin spinning in the air before it lands." Participants then explore the quantum block palette integrated into the familiar Scratch interface.

Participants first assemble an H gate followed by a Measurement block to implement simple quantum superposition, observing superposition and state collapse through sprite responses and histogram displays. Those who complete this activity then use a CNOT gate to create entanglement between two qubits. Finally, participants transmit their experimental results to the cloud using the Send Screen feature. These results are displayed in the Student Gallery, where the group can observe how repeated measurements collectively approximate a probability distribution. The session concludes by revealing the Qiskit code automatically generated from the block-based circuit, demonstrating the gradual transition from block-based to text-based quantum programming.

Implications and conclusion

Scratch-Quantum provides an environment in which elementary and middle school learners can approach quantum computing concepts through block-based programming. By replacing complex programming processes with manipulable visual structures, it reduces syntactic difficulties and allows learners to experience abstract quantum principles through concrete experimentation. While the block-based interface does not eliminate the inherent conceptual difficulty of quantum phenomena, it allows learners to focus cognitive resources on understanding superposition, entanglement and probabilistic measurement through direct experimentation.

Scratch-Quantum also holds promise as a research tool. Its integrated cloud infrastructure provides systematic records of learners' circuit designs, execution patterns, and measurement histories. These data offer researchers new opportunities to investigate how students construct quantum circuits and how misconceptions arise.

Scratch-Quantum currently supports only basic quantum gates (H, X, Z, CNOT) and measurement and has not yet been empirically validated in classroom settings. Future development plans include adding quantum protocols suitable for more advanced learners—such as quantum key distribution and simple search algorithms—and conducting empirical studies to verify learning effects in practice. Through these expansions, Scratch-Quantum aims to develop into a sustainable platform for both teaching and research in the emerging field of quantum education.

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Piano Lab Pro: An Exposure Based Mixed Reality Application to Mitigate Music Performance Anxiety

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Abstract: Piano Lab Pro (PLP) is an immersive experience for the Apple Vision Pro headset, designed to help piano students practice coping mechanisms for dealing with musical performance anxiety. To accomplish this, PLP allows students to perform a musical piece using a real piano, while having the Mixed Reality (MR) headset simulate a large performance space both visually and acoustically. During their performance, they experience dimmer house light, brighter stage light, and simulated audience presenting distracting behavior. PLP is designed in collaboration with faculty members from the school of music and addresses a practical need to help music students identify their anxiety triggers and practice effective coping mechanisms.

Introduction

Music performance anxiety (MPA) is a pervasive and well-documented challenge that musicians face both in professional and in instructional settings. While prevalence estimates vary, research shows that a substantial proportion of student musicians experience significant anxiety before or during performances (Barros et al., 2022). MPA can manifest cognitively as intrusive thoughts and self-criticism, physiologically as heightened arousal or muscle tension, and behaviorally as avoidance or diminished expressiveness (Matei & Ginsborg, 2017). For many musicians, these experiences are not exceptions, but a part of artistic practice and development.

We are developing the Piano Lab Pro (PLP) pedagogical toolkit that includes a Mixed Reality (MR) environment where music students practice playing the piano while experiencing MPA triggers. Integral to this innovation are student-facing and facilitator-facing resources that provide strategies to identify and to manage MPA, and to encourage reflecting on the experience to plan for improved anxiety management. Here, we discuss the design process of the PLP MR app, explain the interactive demo session structure, and outline our next steps.

Mixed reality to mitigate MPA

MPA triggers are multifaceted. Evaluative pressure from juries and audience members, unpredictable distractions such as environmental noise or technical malfunctions, and heightened self-awareness during public performance can intensify anxiety (Twitchell et al., 2025). Despite the prevalence of MPA, opportunities for students to practice managing MPA triggers in authentic settings remain limited. Access to recital spaces is constrained by scheduling, staffing, and cost, while live performances involve high psychological stakes that discourage experimentation and reflection. Music educators face persistent challenges in providing low-risk, repeatable contexts where students can safely confront performance stress and develop coping strategies through experience (Blair et al., 2022).

Extended Reality (XR) technologies, including Mixed Reality (MR), offer promising solutions to these challenges by recreating realistic performance environments in a controlled and supportive space. Immersive rehearsal enables repeated exposure to stress-inducing stimuli, such as audience presence, lighting, or subtle distraction, without the social or professional risks associated with live evaluation. Empirical research demonstrates that XR exposure training can reduce music performance anxiety compared to traditional relaxation methods (Bellinger et al., 2023). Studies in educational XR contexts show that exposure-based virtual experiences can prime students to manage anxiety more effectively and allocate greater cognitive resources to task performance (Hawes & Arya, 2021). These findings suggest that simulated practice environments, such as MR, can foster emotional regulation and cognitive readiness, supporting the goals of performance pedagogy.

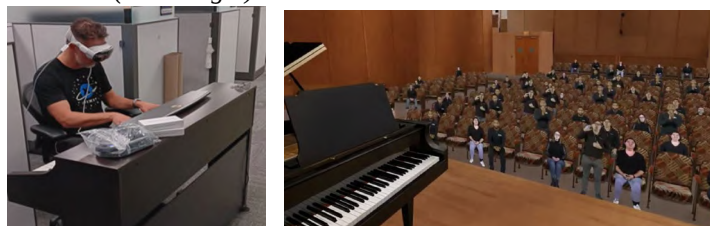
The Piano Lab Pro (PLP) pedagogical toolkit

The PLP Pedagogical Toolkit development is a collaboration between the Center for Academic Innovation and two faculty members from the School of Music, Theatre and Dance at the University of Michigan. The PLP Toolkit consists of the three components that together form a pedagogically coherent toolkit:

- **Student Preparation Guide:** This website is a self-directed resource that explains a range of MPA triggers, includes a self-diagnostic questionnaire for students to identify their triggers, and curates annotated resources to address MPA triggers.
- **Facilitator Guide:** The PLP MR app is a facilitated activity with a facilitator present to set up the PLP app on Apple Vision Pro headset, guide the students through their practice session, and debrief after the activity is completed. The facilitator guide prepares the facilitator to lead an effective PLP MR session.
- **PLP MR Application:** PLP uses Apple Vision Pro to immerse piano students in high-fidelity simulations of two music performance venues at the University of Michigan, offering realistic practice in high-pressure performance settings (Figure 1). PLP provides exposure to performance stress, paired with external scaffolding to support the development of stress-management strategies, helping students build confidence, reduce anxiety, strengthen their stage presence, and support more effective learning.

Figure 1

A Student Uses the PLP MR App to Play the Piano (on the left) in Front of a Virtual Audience (on the right).



PLP MR App Design Process

We adopted a design thinking approach (Dam, 2025) to develop and test the PLP.

Empathizes and define

A combination of semi-structured interviews with Piano students, site observations, and faculty consultations informed our design. Piano students represented diverse proficiency levels and technical skill which helped us glean a variety of preparation routines, internal experiences during performance, environmental triggers, and coping strategies in depth. For example, we noticed the importance of physical setup routines, the role of environmental distractions, and the need for pre-performance grounding strategies. We developed two personas: (1) students majoring in Piano who experience performance evaluation as core curriculum requirements; and (2) intermediate level musicians seeking confidence and stability despite inconsistent practice habits or moderate anxiety. To develop learner journeys, we finalized the personas' goals in a Design Jam.

Ideate

We synthesized Design Jam outputs into a series of design and development discussions to develop a learner journey and ideate the structure of the activity flow. The outcome of the ideation phase was the information architecture that outlined students' activity flow as they launch the PLP MR app and go through the performance:

- **Onboarding Steps:** after launching the PLP App, the student goes through
 - Welcome and introduction to the app
 - Setup the piano, sheet music, test audio, and confirms that set up is done correctly
- **Venue Selection and Setup:** using the self-diagnostic questionnaire, the student selects a venue
- **Default Play:** with no audience, no distractions, and no change in the lighting, the student can
 - Practice freely and become familiar with the venue.
 - Start a formal session via Start Performance
- **Performance:** the student can access Performance to
 - Return to Default Play using a Stop button
 - Cue Start Performance: house lights dim, stage lights increase, audience claps to signal Start
 - Play End-to-End with random audio and visual distractions
 - End Performance by selecting Stop and trigger audience applause

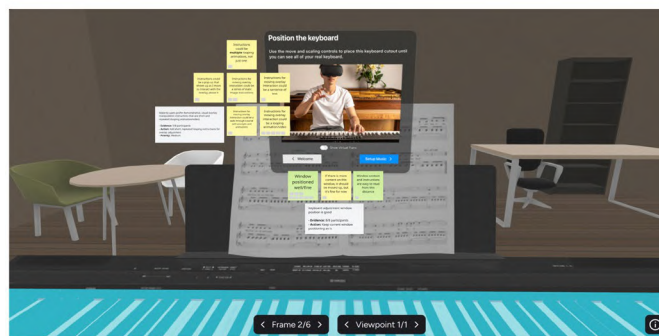
- Summary: the student reaches a Summary Page that shows the duration of the practice, distractions experienced, and an encouraging message. The student can then return to Default Play.

Prototyping tools and processes

We created minimal prototypes to test core interactions, gather feedback, and revise learner journey. In an iterative Build-Measure-Learn approach, we identified tasks that required user testing and task that could be simplified. Integral to prototyping were interaction design and MR environment construction processes.

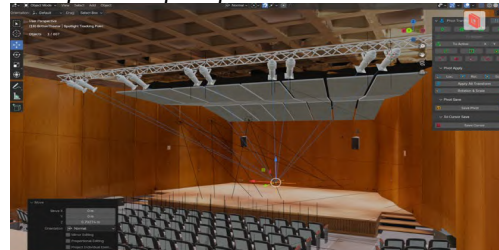
Interaction and experience design. While using the PLP app, students need to align the physical piano and the virtual piano. We translated the prototype into spatial layouts using ShapesXR and conducted an early assessment of the spatial clarity and onboarding usability of the PLP MR environment. Rapid spatial prototyping with ShapesXR enabled us to determine the placement of windows and instructions as students experience them inside a headset. For example, in Figure 2 a virtual box is overlaid on the physical piano keys and a virtual cutout is overlaid on the physical sheet music holder. The student receives information on how to align the virtual box and cutout with the physical piano. ShapesXR functioned different headsets, including the Meta Quest 3, giving us a broader pool of usability testers as it reduced constraints for testers who wore glasses.

Figure 2
A special Layout of a PLP Viewpoint in ShapesXR



MR Environment Modeling Development occurred in parallel with interaction design. Media experts and an environment artist documented two university music venues using photogrammetry, enabling the team to digitally reconstruct each space and preserve its atmosphere for immersive use in the experience. This reconstruction relied on aligning thousands of overlapping photos and performing spatial calibration to generate a high-resolution 3D mesh from the captures. The resulting high-poly scan was then optimized into a low-poly model, with high-resolution mesh detail and lighting baked into textures to preserve essential visual fidelity while enabling efficient real-time rendering on the Apple Vision Pro (Figure 3). Throughout this workflow, the environment artist and XR developer collaborated to pinpoint optimization opportunities and to calibrate the tradeoffs between visual richness and computational cost, ensuring smooth, stable performance.

Figure 3
The optimized file of the Britton Hall reconstruction.



Usability testing and the revised activity flow

We conducted two rounds of usability testing: (1) We focused on testing the ShapesXR prototype to understand window and instruction placement; (2) We examined the impact of environmental realism with the Britton Recital Hall 3D modeled environment, lighting control, and audience members. We revised the activity flow as follows.

Onboarding and venue selection

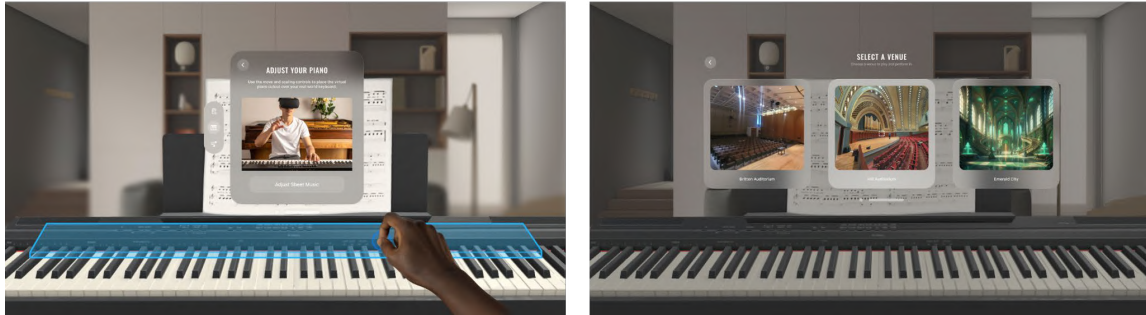
We re-ordered setup steps to first focus on high-level system setup. In this process, the student:

1. Sets up the “world position” and aligns the headset spatial orientation with their current position, especially the distance to the floor.
2. Connects the audio component routing the music from the piano to headset’s speakers.
3. Starts mixing and aligning the keys and sheet music of the virtual piano with the physical piano.
4. Selects a venue.

Figure 4 shows one of the prototypes, based on the evolving information architecture.

Figure 4

The PLP Prototype in Figma, Aligning the Physical and the Virtual Pianos (right) and Choosing a Venue (left)



Default play and performance

We merged “Default Play” and “Performance Mode” into a singular flow.

1. The student automatically enters a pre-performance, or Default Play, state with the house lights up and the crowd chatting. The student becomes acquainted with the space and mentally prepares to practice.
2. When ready, the student triggers the Performance Mode directly from the Default Play.
3. The house lights dim, the crowd quiets, and the stage lights feel brighter to isolate the student in a sudden “all eyes on me” moment. Based on our interviews, research, and usability testing, being the spotlight is a significant performance anxiety trigger. This is a new feature in our latest prototype.

Summary

At the time of writing this paper, the Summary step of the activity flow is in development.

1. The student indicates that they are done with a performance.
2. The audience clap and cheer.
3. The student leaves the venue. A “Summary” page displays the duration of performance, distractions encountered, and an encouraging statement. Students receive a copy of the summary to reflect on their performance over multiple sessions and to evaluate the impact of the experience to reduce MPA.

Next steps

The current version of the prototype provides more scaffolded instructions to foster students’ agency to progress through the steps. We are now refining the written instructions, editing looping videos to demonstrate actions in the headset, and building in user interface feedback.

Structure of the session

We will structure our interactive demo session in two parts:

- PLP app showcase and demo: A facilitator and a student will showcase a ten-minute practice session with the PLP pedagogical toolkit. Participants will observe student’s headset view. We will highlight unique design features and innovative solutions that we developed during the project.
- Interactive discussion: We will lead a discussion with participants to examine their experiences from practice or research regarding the longevity of pedagogically useful immersive applications, such as the PLP, as MR technology constantly evolves. We will also share design features that we scaled back due to implementation limitations.

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SEED: A Teachable-Agent System that Supports Elementary Students' Self-Explanation by Teaching an AI Chatbot

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Abstract: Self-explanation is a powerful strategy for conceptual understanding, yet elementary students struggle to produce coherent explanations without individualized scaffolding. LLM-based teachable agents offer a promising alternative—positioning the learner as teacher and the AI as student, leveraging the protégé effect to deepen understanding—yet most existing work targets undergraduate learners, leaving elementary students largely unaddressed. This demo introduces SEED (Self-Explanation Eliciting Dialogue), a prototype teachable-agent system that positions the child as the teacher and the AI as a novice learner. SEED features a role-reversal prompt engine, three-stage scaffolding flow, growth visualization, and auto-generated learning notebooks—all designed to sustain the protégé effect. This work demonstrates that LLM-based teachable agents, when designed around developmental principles, can meaningfully support self-explanation among elementary learners.

Introduction

Self-explanation is a well-established learning strategy that supports conceptual understanding and knowledge integration (Chi et al., 1994). However, learners often struggle to spontaneously generate high-quality self-explanations on their own (Renkl, 1997). This challenge is particularly pronounced among elementary students, who are still developing the metacognitive skills needed to monitor and regulate their own understanding (Flavell, 1979). Although prompting students to self-explain can substantially deepen their understanding (Chi et al., 1994), providing individualized scaffolding to every learner in a classroom of nearly thirty students remains a persistent challenge for a single teacher.

Recent generative AI systems are often designed to respond to users' questions immediately and deliver information with ease, which risks inducing cognitive offloading—the outsourcing of cognitive processing to AI—among students (Risko & Gilbert, 2016; Gerlich, 2025). A more productive alternative is the teachable agent model, in which the learner assumes the role of teacher and the AI assumes the role of student. This framework leverages the protégé effect—the tendency to invest greater effort and achieve deeper understanding when one believes their explanations are helping another learn (Chase et al., 2009).

Early teachable agent systems such as Betty's Brain demonstrated these effects (Leelawong & Biswas, 2008), yet relied on rule-based architectures that imposed limits on flexibility and scalability. More recently, researchers have begun exploring LLM-based teachable agents capable of open-ended dialogue (Jin et al., 2024; Kucharavy et al., 2025; Rogers et al., 2025). However, most of this work targets undergraduate learners, leaving the needs of elementary students largely unaddressed.

This demo introduces SEED (Self-Explanation Eliciting Dialogue), a teachable-agent prototype that positions the student as the "teacher" and the AI as the "learning student." Unlike recent LLM-based teachable agent systems, SEED is grounded in developmental and motivational design principles tailored for elementary learners—including psychologically safe, non-evaluative feedback and tangible reinforcement of the protégé effect through growth visualization and auto-generated learning notebooks. Our aim is not to present a polished product, but to share a working prototype and early classroom experiences that expose both the promise and the limitations of LLM-driven teachable agents.

System overview

Design principles

SEED was developed through a design-based research process involving literature review, needs analysis, and expert validation. Six design principles guided the prototype:

1. Developmentally Appropriate Scaffolding and Language Use
2. Inquiry-Driven Prompts for Explanation Construction
3. Role-Reversal interaction in a Teachable-agent Frame
4. Psychological Safety Through Non-Evaluative Responses

5. Teacher-Centered Monitoring Through a Simple Dashboard Interface
6. Character-Based Growth Visualization to Strengthen the Protégé Role

These principles were operationalized into nine detailed design guidelines through expert validation and were embedded into the system's prompt design, interface, and teacher dashboard.

Architecture

SEED follows a four-layer architecture:

- Interface layer: React-based app with (1) home screen, (2) student chat interface with messenger-style dialogue, (3) teacher dashboard
- Application layer: A lightweight application layer that handles HTTP requests from the client, logs conversations, tracks basic dialogue state (e.g., current phase in the three-stage flow), and formats API calls using structured system prompts.
- LLM layer: An LLM-based dialogue engine that generates child-directed responses and learning-notebook summaries. The system prompt encodes SEED's persona, design principles, and scaffolding strategies. Because SEED relies on LLM-based generation, these guidelines are instantiated in context-dependent ways consistent with the generative model's behavior, rather than as rigid rule-based operations.
- Data layer: A simple tabular data store that maintains student accounts, curriculum key concepts, growth levels, and conversation logs. Key concepts are pre-loaded per lesson and injected into the system prompt; the LLM draws on both these provided concepts and its pre-trained knowledge to interpret student explanations.

A three-stage scaffolding strategy supports explanation development:

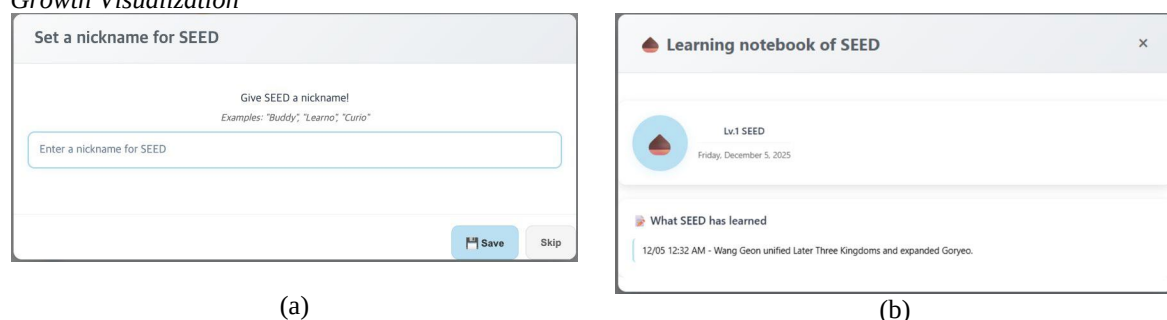
1. Elicitation: SEED adopts a curious student persona, inviting the learner to 'teach' the topic through free-form explanation.
2. Adaptive feedback: Guided by structured prompts, the LLM tends to follow one of four pathways—prompting error reconsideration, filling gaps, help-seeking support, or extending reasoning—using conceptual, procedural, metacognitive, and strategic supports.
3. Consolidation: SEED requests integrated re-explanation and generates a "My Learning Notebook" summary of what it has "learned". The notebook is generated for the student's reflection and does not update the LLM's parameters or persistent context.

Key features

SEED's role-reversal prompt engine positions the chatbot as a novice learner who consistently addresses the child as "teacher" and requests explanations ("I don't know—could you teach me?"). To deepen relational engagement, students are first invited to give SEED a personal nickname (Figure 1a), fostering a sense of ownership and responsibility toward their protégé from the outset. This ownership is further reinforced through auto-generated Learning Notebooks (Figure 1b) that document what SEED has "learned" from each teaching session, creating the impression that the student's explanations are genuinely building SEED's understanding—a key mechanism for sustaining the protégé effect. SEED also provides growth visualization that progresses from seed → sprout → leaf → flower → fruit as explanations accumulate, offering a tangible representation of the protégé's learning progress.

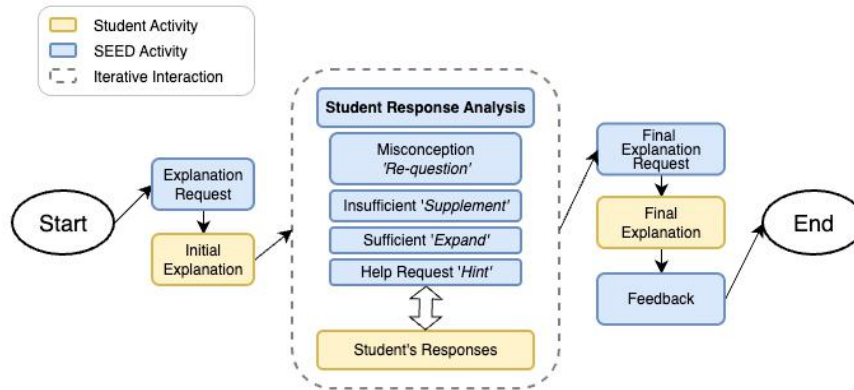
Figure 1

Learner-Facing Features of SEED: (a) Nickname Setup and (b) Auto-Generated Learning Notebook with Growth Visualization



Interaction scenario

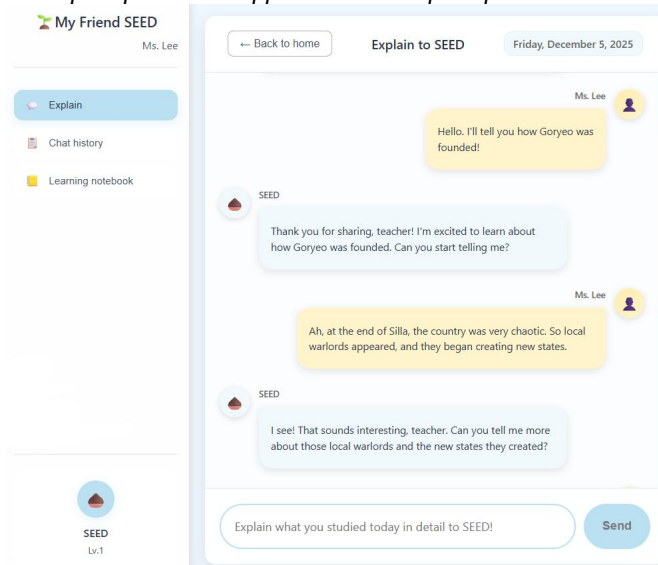
Figure 2
Interaction Flow



A typical SEED interaction begins with SEED's opening prompt inviting the student to explain a concept they have studied (Figure 2) (e.g., "Can you teach me what you learned today?"). The student then provides an initial explanation (e.g., "At the end of Silla, local warlords appeared and began creating new states"), after which SEED applies one of four scaffolded response modes. When a misconception is detected, SEED responds as a confused learner whose prior understanding conflicts with the student's account (e.g., "Hmm, that's a bit different from what I learned before—could you help me understand which part is right?"), prompting conceptual revision without signaling a simple error. When an explanation is incomplete, SEED asks targeted gap-filling questions to elicit missing elements, as illustrated in Figure 3, where SEED prompts the student to elaborate further on the local warlords and new states mentioned in their initial explanation. When a student requests help, SEED provides scaffolded hints that reduce cognitive load without revealing the answer (e.g., "That's okay—can you start by telling me just one thing you remember?"). When an explanation is sufficient, SEED extends reasoning through hypothetical and critical questions that encourage deeper thinking (e.g., "If you were the ruler at that time, what would you have done?"). This iterative exchange continues until SEED requests a final explanation, prompting the student to synthesize the full interaction into a consolidated response.

This protégé framing motivates learners to refine their explanations more actively. As students provide higher-quality explanations, growth visualization appears and a learning notebook is generated to reinforce metacognitive reflection.

Figure 3
Example of SEED's supplement-based prompt interaction



Implementation

SEED was implemented as a web-based prototype with a React front-end deployed on Vercel, a Google Apps Script back-end that handles API calls and logs conversations, Google Sheets as a lightweight data store, and an OpenAI GPT-4-based dialogue engine driven by structured system prompts. The prototype was iteratively refined through fifth-grade classroom deployment, teacher feedback, and conversation log analysis.

Preliminary findings

In an initial classroom deployment with fifth-grade students, many described SEED as "a younger student I want to help," suggesting that the protégé framing fostered psychological safety and a sense of responsibility. While some students initially questioned whether SEED truly did not know the material, this was mitigated by SEED's design as a curious and confused learner rather than a knowledge-withholding agent—a framing that positions SEED's responses as genuine expressions of uncertainty rather than implicit error signals. Engagement was further reinforced through the learning notebook and growth visualization, which supported the impression of SEED's ongoing learning over time.

Interactive demo session

This interactive demo will let participants:

- Experience SEED as a “teacher”. Participants will interact with SEED directly and observe how it responds adaptively to their explanations. For the conference demo, a topic accessible to international attendees will be used in place of the Korean history content shown in the examples above.
- Explore learner-facing features: growth visualization and the learning notebook. The demo will show how SEED represents its “learning progress” through growth levels and automatically generates a learning notebook summarizing what SEED has understood, reinforcing the protégé framing.
- Inspect the teacher dashboard. Participants can switch to the teacher view to review conversation logs, monitor students’ SEED levels, access auto-generated summaries, and observe how the dashboard supports lightweight classroom management.
- Examine SEED’s underlying prompt and data structures. The demo will showcase how SEED’s three-stage scaffolding strategy is implemented using prompt templates and structured data stored in Google Sheets, allowing participants to see how the system generates adaptive responses through LLM-based prompting rather than hard-coded rules.

Implications and conclusion

Taken together, SEED demonstrates that inverting the conventional AI-as-tutor framing—positioning the LLM as a novice learner rather than an expert explainer—can create conditions for deeper self-explanation and protégé-like engagement in elementary classrooms. For designers and teachers, our work highlights that how an AI is framed may matter as much as what it does. Limitations include a single subject domain and the probabilistic nature of LLM-based scaffolding. Future work will explore how teachable-agent framing scales across subjects, and investigate multimodal extensions—such as voice-based interaction—that may better support young learners who struggle with text input.

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Analyzing Embodied Engagement in Co-located Collaboration through Kinesthetic Modality

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Abstract: This demo aims to explore the potential of pattern mining in the kinesthetic modality. Multimodal collaboration analytics (MMCA) is an emerging and expanding field in CSCL and LA, emphasizing its capability to capture the digital trace of embodied engagement. However, few studies have explored the kinesthetic modality, leaving theoretical and methodological frontiers of how to handle accelerometer data. To narrow the gap, we developed a multimodal sensor badge and proposed an analysis pipeline by adapting pattern mining to accelerometer data to examine embodied engagement in collaboration. Eight groups of 3 undergraduates participated in collaborative problem solving, and one high-performing and one low-performing group were analyzed as a case study. The results suggest that pattern mining reveals nuanced but meaningful differences in engagement that could not be identified using conventional methods. Our analysis pipeline suggests the possible direction for deriving meaningful insights from kinesthetic modality data and extending theories of collaboration.

Introduction

Collaboration is multimodal in nature, and members' engagement is embodied in multimodal interactions. Therefore, although verbal interaction has remained a primary focus, nonverbal interactions should be considered in analyzing collaboration (Enyedy & Stevens, 2022; Ochoa, 2022). Building on this background, Multimodal collaboration analytics (MMCA) has been attracting more attention in Learning Analytics and CSCL (Esterhazy et al., 2025; Schneider, 2023), with advancements in pedagogical/practical, technological, methodological, and theoretical aspects (Esterhazy et al., 2025). Esterhazy et al. (2025) categorized multimodality into five types: namely, auditory, visual, tactile, physiological, and kinesthetic. According to Esterhazy et al. (2025), auditory and visual modalities have often been analyzed; however, the kinesthetic modality has received relatively less attention. Kinesthetic modality represents body movements like gesturing or interacting with material artifacts. It potentially sheds light on embodied engagement in collaboration, fostering deeper understandings of classroom orchestration, social regulation, and problem-solving (Prieto et al., 2016; Li et al., 2024; Zhao et al., 2023).

Previous studies have measured the amount of kinesthetic activity (body movements) to capture the embodied engagement in co-located collaboration. Several studies utilized accelerometer sensors to capture physical engagement with material artifacts in nursing simulation by calculating the intensity of body movement; in other words, the amount of activity (Buckingham Shum et al., 2019; Echeverria et al., 2019; Martinez-Maldonado et al., 2020). CPR activity performed on a manikin was detected from acceleration data collected by a wristband. Studies suggest that it is possible to estimate the embodied engagement based on acceleration data.

However, previous studies have mostly focused on nursing simulation settings, and theoretical and methodological challenges remain. Within co-located collaboration, members may interact with various material artifacts, such as computers and whiteboards. Theoretically, accelerometer data are potentially capable of capturing diverse forms of embodied engagement in collaboration, not only in nursing simulation. Moreover, from a methodological perspective, the pattern mining technique might enable us to gain further understanding of embodied engagement in collaboration. Previous studies have shown that pattern mining of the accelerometer data is useful for human activity recognition (Bijalwan et al., 2022; Matsubara et al., 2014). Examining not only the amount of activity but also the patterns of activity may reveal nuanced differences in embodied engagement.

In this demo, we explore the potential of pattern mining of accelerometer data in co-located collaborative problem-solving. In the following section, we first describe the multimodal sensor badge that we developed. We then present our analysis pipeline for examining the patterns of kinesthetic activity. Finally, through a case study, we demonstrate that the proposed approach can capture nuanced differences in embodied engagement.

Embodied engagement analysis with kinesthetic modality data

We developed a multimodal sensor badge, named Sensor-based Regulation Profiler (SRP) badge (Yamaguchi et al., 2022), to capture embodied process in co-located collaboration (Figure 1). The SRP badge is designed to

capture the embodied engagement, equipped with a triaxial accelerometer (ACC) that collects ACC data in 12 bits at 100 Hz. It also incorporates an infrared LED to detect face-to-face conditions and a sound pressure sensor to capture speech activity, but they are not explained in this demo. The SRP badge has a wireless time synchronization function in $\pm 30 \mu s$ accuracy, allowing millisecond time granularity for analysis.

Figure 1

Multimodal Sensor Badge (SRP Badge). (a) Appearance Attached to the Chest and Arms, (b) Sample Implementation in a Co-Located Collaboration



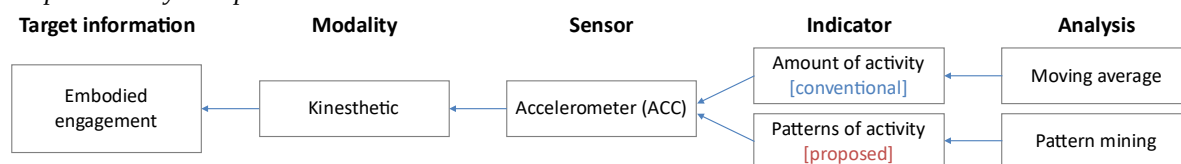
Figure 2 illustrates our analysis pipeline. Based on the frameworks of Esterhazy et al. (2025) and Ochoa (2022), we explain the analysis targets, modalities, sensors, and analytical procedures. To capture the digital traces of embodied engagement, SRP badges are attached to participants' chests and arms, and kinesthetic modality data are collected through the ACC sensors on the badges. From the ACC data, we extract two indicators: the amount and the patterns of kinesthetic activity. The amount of activity is analyzed using moving-average smoothing, and the patterns of activity are analyzed using a pattern-mining algorithm. The following sections describe each of these in detail.

The amount of activity is analyzed using moving-average smoothing, which is a commonly used conventional approach. The ACC collects triaxial (x-, y-, and z-axis) data (i.e., 3D vectors) at 100 Hz, and the instant amount of activity is calculated by the L2 norm (i.e., magnitude of the vector $\sqrt{x^2 + y^2 + z^2}$). Here, applying a moving average to the ACC data can reduce instantaneous noise and enable more appropriate capture of embodied engagement (Echeverria et al., 2019). In this demo, we calculated a one-second moving average (i.e., over 100 samples) and used it as the indicator of the amount of activity.

The patterns of activity are analyzed using the AutoPlait pattern-mining algorithm (Matsubara et al., 2014). AutoPlait discovers temporal patterns through unsupervised learning, enabling pattern detection and segmentation within large-scale time-series data (e.g., ACC-based motion-capture data). While the L2 norm (i.e., the amount of activity) captures only the magnitude of the ACC data, pattern mining considers both the magnitude and the direction of the ACC data. This potentially enables the detection of nuanced differences in embodied engagement by capturing subtle changes in body posture and movement.

Figure 2

Proposed Analysis Pipeline



Case study: Amount and patterns of kinesthetic activity in co-located CPS

Contexts and participants

Eight groups of 3 undergraduates (9 female, 15 male) participated in a 60-minute collaborative problem-solving (CPS) activity. Participants were asked to design a creative system or application to improve the user experience of university students. During the discussion, scaffolding for productive CPS was provided as needed. Participants were provided a computer and a whiteboard (Figure 3). Participants were free to access any generative AI tools.

Figure 3
Experimental Setting and Embodied Engagement (snapshots from the high-performing group)



Method

CPS sessions of eight groups were recorded and transcribed. CPS performance was assessed with the creativity framework (Runco & Jaeger, 2012). The SRP badges were attached to participants' chests and arms to capture kinesthetic modality data (i.e., ACC data). After preprocessing the collected ACC data to remove missing values and outliers, we extracted the amount and the patterns of activity: the amount of activity was calculated through moving-average smoothing, and the patterns of activity through pattern mining. In this demo, we present two cases from high- and low-performing groups, focusing on the interpretation of the indicators and video analysis.

Results

The results suggest that in the high-performing group, the relatively calm member engaged in small but important collective activities, whereas the inactive member in the low-performing group barely engaged in the activities.

The results of the moving-average smoothing analysis suggest that, in both the high- and low-performing groups, one member had a stable amount of activity, while the others had a large variance. Figure 4 illustrates the amount of activity based on the ACC data of participants' chest, right arm, and left arm. Figure 4(a) suggests that member A was more stable than the others. Video analysis tells us that member A mostly engaged in computer operations, while members B and C were more actively writing on the whiteboard and moving their arms for gestures. Thus, the amount of activity suggests that Member A engaged less in dynamic kinesthetic activities such as writing or gesturing. As well as member A in the high-performing group, member D in the low-performing group also demonstrated stable kinesthetic activity, indicating less dynamic movement during the CPS session. The video recording suggests that member D mostly remained standing and did not engage in activities such as writing, gesturing, or computer operations. However, the amount of activity, one of the conventional indicators, cannot distinguish the different types of engagement exhibited by Members A and D by itself.

Figure 4
Heatmaps of the amount of activity in the high- and low-performing groups. (a) High-performing group (members A, B, C), (b) Low-performing group (members D, E, F)

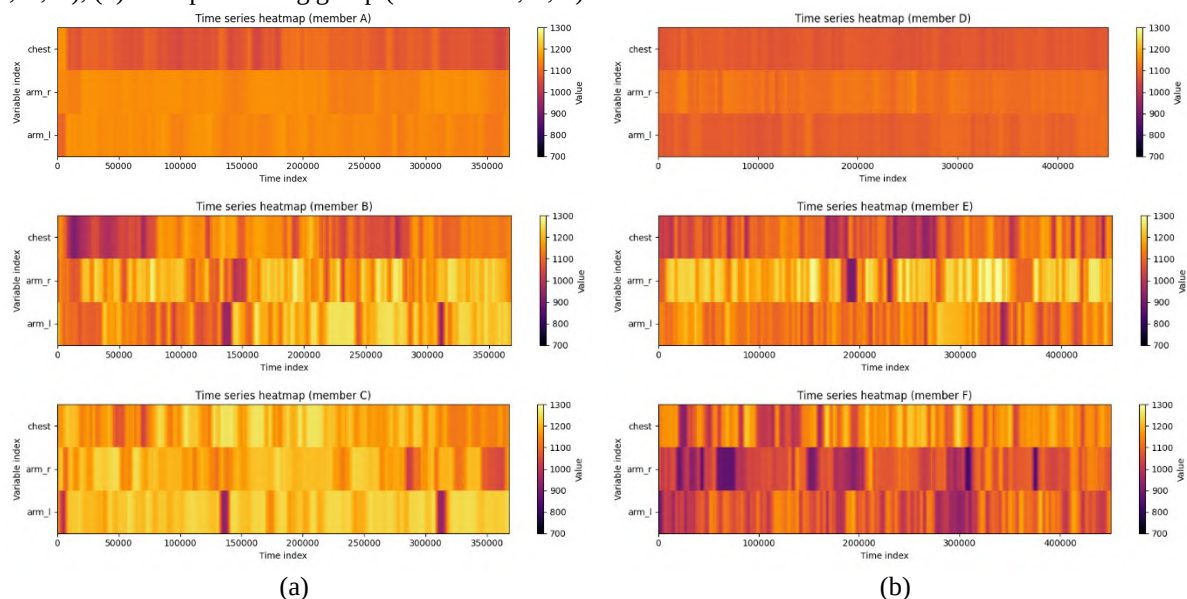
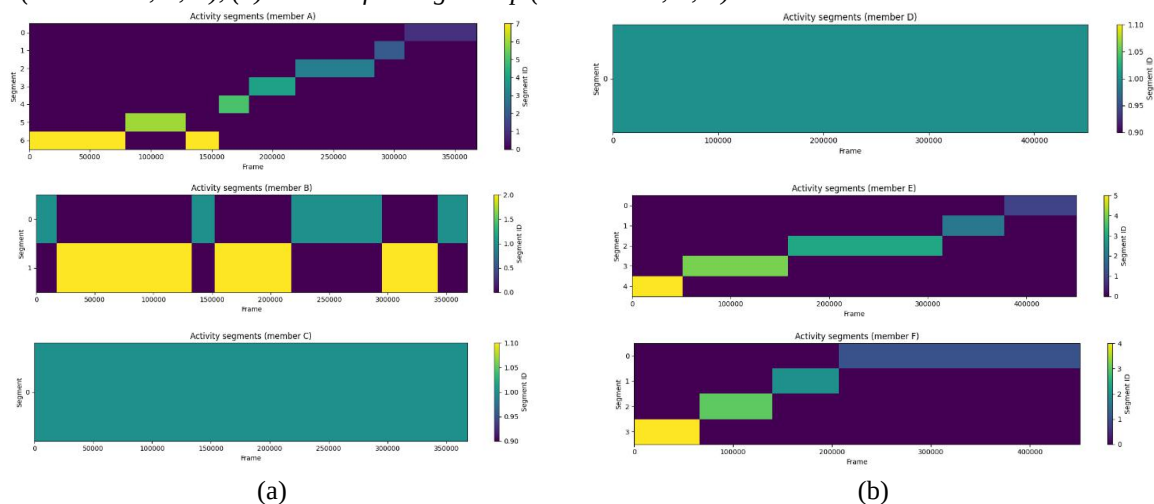


Figure 4(a) suggests that member A was more stable than the others, as well as member D in Figure 4(b).

The results of the pattern mining analysis revealed the different types of embodied engagement. Figure 5 shows the results of pattern mining by AutoPlait. Here, each colored bar indicates segments that were identified as the same pattern for each member. Figure 5(a) shows that member A exhibited multiple patterns of kinesthetic activity. The video recording indicates that, in addition to computer operation, different activities were observed, such as paying attention to the whiteboard. Combined with the information provided by the amount of activity, these results suggest that member A engaged in less dynamic but important activities. These activities are small enough to be overlooked, yet pattern mining can detect these nuanced differences. Member D, on the other hand, had only one segment of kinesthetic activity. This result aligns with the video analysis that member D was simply standing and watching the whiteboard or the screen, which implies less engagement by member D. Although member C also had only one segment, the amount of activity and video recording suggests consistent and dynamic activity, such as pointing and gesturing. In this way, combining the information of the patterns of activity with the conventional indicator, the amount of activity, enables the extraction of nuanced but meaningful insights that cannot be identified solely from activity intensity.

Figure 5

Results of the Patterns of Activity in High- and Low-Performing groups. (a) High-Performing Group (members A, B, C), (b) Low-Performing Group (members D, E, F)



Discussion

Through the case study, the combination of two indicators, the amount and patterns of kinesthetic activity, revealed nuanced differences in embodied engagement. Although both members A and D showed low amounts of activity, member A engaged in calm but important activities, whereas the inactive engagement of member D.

Our analysis pipeline provides both theoretical and methodological contributions to MMCA research. From a theoretical perspective, it extends the understanding of embodied engagement beyond nursing simulation. Further analysis of different kinesthetic patterns will deepen the insights of cognitive/socio-emotional interactions, potentially opening the discussion on actionable analytics for orchestration (Amarasinghe et al., 2024) and social regulation (Li et al., 2024). From a methodological perspective, it offers an interpretable MMCA approach based on pattern mining that can capture nuanced differences in kinesthetic activity. We argue that the proposed approach can indicate cues of learners' embodied engagement and can support further in-depth qualitative analysis.

A limitation remains. Despite recent attention to the integration of multimodal datasets (e.g., Buckingham Shum, 2019), our analysis focused solely on the kinesthetic modality. Further research is required to combine the kinesthetic modality with other modalities, such as visual, auditory, and tactile, in collaboration.

Interactive tools and demos

In this interactive demo session, we demonstrate the entire analysis pipeline, from data collection to data analysis. Participants will first experience data collection wearing the multimodal sensor badge, the SRP badge. We will then provide the case-study datasets and explain how to interpret the plots of the amount and patterns of kinesthetic activity, including video analysis. Through this session, participants will learn how to extract meaningful insights from kinesthetic modality data using wearable sensors, by iterative analysis of quantitative plots and qualitative video recording; differences in embodied engagement within and between members and groups will be explored.

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OctoStudio: Mobilizing Creative Learning in Community Contexts

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Abstract: OctoStudio is a mobile coding app co-designed with a global community of educators to provide equitable access to creative learning experiences for young people from all backgrounds. OctoStudio aims to enable learners to transform their phones and tablets into construction materials for creating interactive multimedia projects using block-based programming, device sensors, and culturally-inclusive visual assets. The app has been integrated across informal and formal learning environments globally, supporting cross-curricular learning, and helping to spread creative learning ideas and values. This interactive demo invites participants to create their own OctoStudio projects that combine physical and digital materials, explore applications for research and practice across diverse contexts, and engage in discussion about inclusive creative learning experiences that offer multiple pathways into computing for learners with varied backgrounds and interests.

Introduction

Supporting creative learning experiences for young people of all backgrounds requires equitable access to tools, resources, and learning environments that honor learners' unique identities and community contexts. In partnership with community-based educators around the world, OctoStudio was developed as a creative learning mobile app to support young people in creating what they imagine, transforming their phones and tablets into construction materials. Guided by the shared knowledge and expertise built through a community-of-practice, OctoStudio was designed to foster inclusive access to creating with technology (Rusk et al., 2024). This interactive demo aims to share how OctoStudio supports young people in engaging in creative learning experiences across diverse communities and learning contexts.

Background

Educational theorists and practitioners have long advocated for hands-on, project-based learning experiences. Constructionism was introduced as an alternative approach to traditional classroom instruction, theorizing that learning (in both formal and informal learning environments) happens most effectively when the learner is actively engaged in creating a meaningful artifact (Papert & Harel, 2002). Creative learning builds on this idea, describing a process where young people learn through making projects connected to their unique interests and passions, while collaborating with peers and embracing a playful spirit of tinkering and iteration (Resnick, 2017). Creative learning aims to support young people of all backgrounds in developing as creative, curious, collaborative, and caring learners (Resnick, 2025).

Constructionism and creative learning advocate designing for “low floors” for learners to get started easily, “wide walls” to create in connection with learners' diverse backgrounds and interests, and “high ceilings” to allow for increasingly complex creations (Resnick & Silverman, 2005). The creative learning philosophy has inspired the design of tools and communities such as the Computer Clubhouse (Resnick et al., 1998), the Brazilian Creative Learning Network (Valente & Burd, 2019), and Scratch, an online community and programming environment (Resnick et al., 2009).

During the COVID-19 global pandemic lockdowns of 2020 and the ensuing years, usage of online learning platforms like Scratch increased as young people sought to connect, create, and adapt for meaningful learning outside of school (Resnick, 2023; Scratch Foundation, 2020, 2021). However, the global pandemic also exacerbated existing disparities for learners around the world who did not have access to computers or a stable internet connection at home. Thus, educators saw a need for new tools that supported creative learning experiences in contexts where young people's primary access to technology was a mobile device (Rusk et al., 2024). Applying the affordances of mobile devices to constructionist learning theory and creative learning principles, OctoStudio has been developed to address this gap.

OctoStudio overview

OctoStudio is an app-based creative learning tool leveraging mobile devices as construction materials for young people to create with code, anytime, anywhere. Designed in a community-of-practice including educators around the world, OctoStudio aims to expand access to creative learning experiences with digital technologies, supporting young people who have often been excluded from such opportunities (Rusk et al., 2024). The app is available on phones, tablets, and Chromebooks and can be used offline after download. This feature is central to OctoStudio, addressing the realities of communities with no or intermittent internet access while broadening the environments in which computationally-rich learning can occur, from parks and trails to buses and trains.

The block-based coding environment allows young people to get started without prior coding experience and combines colorful blocks with icons and text, translated to over 31 languages. To create a project in OctoStudio, young people can work independently or collaboratively to first select a background or sprite. These can include emojis, photos, drawings, or images. The app leverages a standard emoji library and extends the default options with custom representations for a broader diversity of body types, hair types/styles, headwear, and cultural symbols (see Figure 1).

Figure 1

Example Standard Emojis, Paired with OctoStudio Adaptations



Sprites can be animated through programming blocks to add movement, sound, visual effects, and physical interaction. The coding environment builds on principles of designing for tinkability (Resnick & Rosenbaum, 2013) allowing learners to develop their understanding of computation while iteratively creating complex and meaningful interactive projects. OctoStudio enables new ways of creating and sharing on mobile devices by leveraging the portable form factor and sensors (e.g., accelerometer and gyroscope) embedded in mobile devices to support creation that combines physical and digital materials (see Figure 2). Young people can export their project as an animated GIF or video and share it with others using existing communication channels (e.g. WhatsApp, SMS, email, Bluetooth). Project files can also be shared between devices, supporting iteration, collaboration, and remixing.

Figure 2

A Diorama of an Animal City, Combining OctoStudio Projects with Cardboard



Contexts and communities

OctoStudio's design is informed by ongoing dialogue with partners who understand their communities' realities. Centering the perspectives and cultural practices of educators working in communities globally has enabled creative learning to spread in varied contexts around the world. This section will provide an overview of examples demonstrating OctoStudio as a part of learning experiences. The examples are situated in both informal and formal

learning environments and depict how OctoStudio has been used in research and practice to explore connections between computer science and other disciplines.

Informal

Designed to embrace the affordances of mobile devices, OctoStudio has been leveraged to support facilitated and drop-in style learning experiences in informal learning contexts. These include out-of-school time (OST) spaces, museums, libraries and community-based organizations.

In Massachusetts, USA, OctoStudio has been explored in public libraries through workshop themes like “bring a book to life”, where learners created projects that remixed book covers, animated literary characters, and extended book plots. Workshop facilitators collaborated with local librarians to select books that were popular in their community. The books were used alongside craft materials, inspiring a variety of projects (see Figure 3). OctoStudio’s multi-modal design supported children’s creative engagement with literacy beyond traditional written responses. Children could draw, animate, record their voices, add music, and program interactivity, choosing the modes of expression that best conveyed their understanding of and connection to the stories they read.

Figure 3

Children Using OctoStudio with Physical Craft Materials and Books.



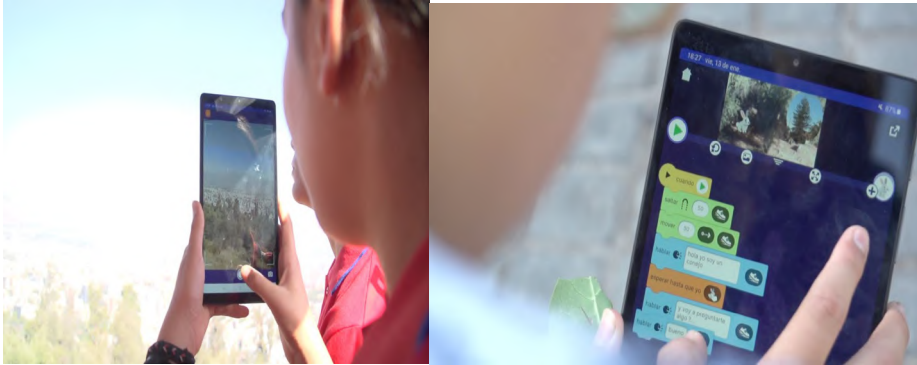
Formal

OctoStudio has been used in schools for cross-curricular learning experiences to integrate computing into subjects like English language arts (ELA) and history.

Seeing their students’ interest in OctoStudio, 4th-grade educators in Massachusetts, USA connected OctoStudio with their existing poetry curriculum. After writing odes and limericks, students created OctoStudio projects to perform and visualize their poems. They included art and photos to make their poems more personally meaningful. Additionally, OctoStudio features like voice recording and speech synthesis enabled learners to engage with their writing in new ways. Educators noted that animating their poetry in OctoStudio encouraged students to revisit their writing and iterate on their poems.

In another instance, educators in Chile have leveraged OctoStudio for place-based learning experiences outside of the classroom. On a field trip, middle school students explored a local historical park with OctoStudio. They photographed plants, insects, textures, and objects that caught their attention, and used these images as sprites and backgrounds in their projects (see Figure 4). OctoStudio’s offline and mobile design allowed the physical environment to become a part of students’ digital creations. Their coding projects served as expressions of observations and curiosity about the natural world around them.

Figure 4
Students Using OctoStudio During Their Field Trip.



Left: A student uses OctoStudio to capture the panoramic view of the park.

Right: A student's OctoStudio project showcasing curiosity about a local tree species.

Interactive demo

In this interactive demo, ISLS attendees will have the opportunity to participate in a learning experience with OctoStudio. We will begin by sharing an overview of the app and its features, followed by a brief group demo where the participants collaboratively make a project with the presenters. Participants will then be invited to explore OctoStudio individually and with others nearby to create their own projects that combine physical and digital materials. After this exploration, participants will share and reflect on their experiences, connecting to how they might envision using OctoStudio with the young people they support in research and/or practice. We will conclude the session by sharing learning resources and examples of OctoStudio in use across various contexts and discussing creative learning experiences that support multiple pathways into computing for learners across diverse backgrounds, interests, and needs.

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Putting the Data in MoDa: Integrating Agent-Based Modeling, Quantitative Data Analysis, and Teacher Responsivity to Investigate Complex Phenomena

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Abstract: This demo introduces key enhancements to MoDa: A free, open-source web based system designed to support students in making sense of complex systems. MoDa is a domain-specific, block-based computational modeling and analysis tool that allows students to build models side-by-side to real-world data. Our latest enhancements include integrating quantitative data in the system using the Common Online Data Analysis Platform and Activity Player to support structured data-rich investigations, allowing students to use a wider variety of real-world data sources to refine their model toward reproducing important features of system behavior both *qualitatively*, by reproducing important visual and dynamic behaviors, and *quantitatively* by allowing closer comparison of patterns, relationships, and variability.

Introduction and motivation

A central goal of science education is for students to interpret complex patterns in the world and reason about their potential causes. Sense-making of scientific phenomena and proposing solutions to problems often begins with finding patterns in data and attending to variability in those patterns. Students' lives are filled with such patterns (e.g., The water seemed warmer at my beach this summer - is it getting warmer everywhere?) and sources of variability (e.g., My cousins, who live across the country, aren't noticing higher water levels - why might that be?). Once initial trends emerge, students can propose causal mechanisms to explain the observed patterns.

Typically, modeling activities at the K-12 level engage students in making observations of a real-world phenomenon, constructing models that seek to explain or predict system behavior, and then validating and refining those models based on comparisons to the real-world phenomenon. Most often, such comparisons are *qualitative*—the simulation is compared to visual and behavioral characteristics of the real-world phenomenon (Schwarz et al., 2009), with less attention to its mathematical and probabilistic aspects. However, there is growing evidence that attending to *quantitative* characteristics can deepen student engagement. Analysis of statistical features of real-world data can introduce students to variability and error that can then be incorporated into models (Lehrer & Schauble, 2004). Similarly, prompting students to reason about quantitative details of simulation output, such as random noise, supports them in connecting micro and macro level behaviors of a system (Wilkerson & Wilensky, 2015). We extend this work to explore comparing simulated data to real-world data collected *in situ*, particularly large-scale public data streams.

MoDa is a tool to support integrating computational modeling and multiple forms of data analysis, in ways that are responsive to teachers' and students' needs. As a domain-specific, block-based modeling toolkit with embedded support to compare students' constructed models to real-world observations side-by-side (Wagh et al., 2022; Fuhrmann et al., under review) our latest design of MoDa adds support for quantitative and statistical analysis, through integrating students' model output and real-world datasets. In particular, it integrates with the Common Online Data Analysis Platform (CODAP), a web-based data analysis tool designed for middle and high school students, and widely used in schools (Finzer & Damelin, 2014). Because integrating this variety of modalities and evidence sources requires careful coordination, we are designing MoDa with a variety of highly adaptable features so that units can be adjusted in response to students' and teachers' immediate classroom needs. Below, we describe three aspects of this new design iteration: (1) A user interface for creating domain-specific agent-based blocks; (2) Exporting data from student models into CODAP for comparison with real-world data; and, (3) Leveraging Activity Player to support modular development and respond to teacher and student needs.

Quantitative analysis of models

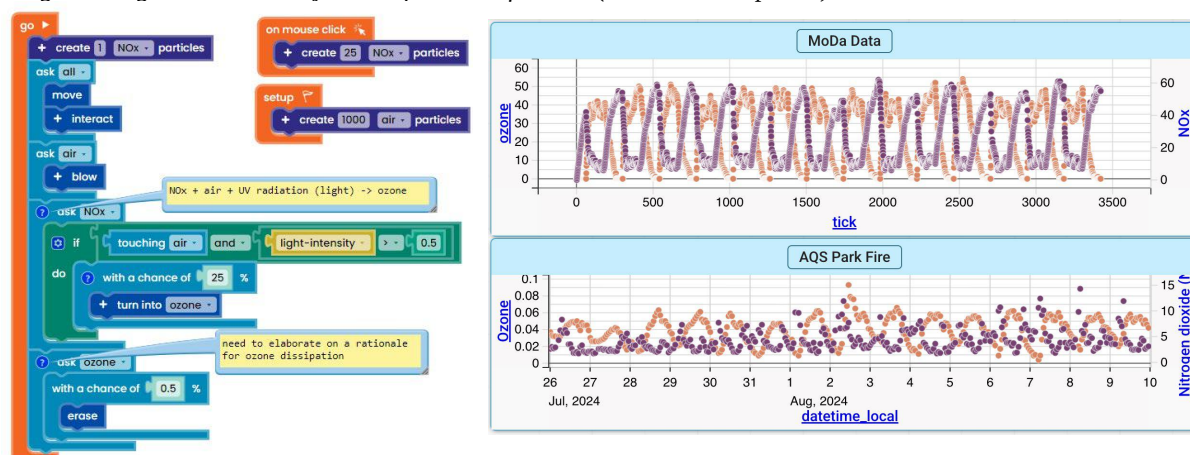
Data explorations support student reasoning about scientific relationships, patterns, and variability (Rivet & Ingber, 2017). Students engage in such explorations when they construct, analyze and use data. Because "data" can take many forms including simulation output, student-generated measurements, and public data (Lee &

Wilkerson, 2018), it is crucial for students to understand how different “data-texts” can work together to explore scientific phenomena and the different functions they serve toward building models (Duschl et al., 2021). It is also critical for students to attend to variability in the data so they can reason about patterns and associated levels of certainty in a phenomenon (Lehrer & Romberg, 1996; Lehrer & Schauble, 2004). Figure 1 features a block library focused on modeling air quality, along with the resulting simulated data and real air quality data.

MoDa is inspired by work on agent-based modeling (ABM) in education, and NetLogo in particular. ABM supports reasoning about causal mechanisms and probabilistic factors at the individual level, and simulates the behavior of elements in a complex system at different levels of analysis using simple probabilistic rules (de Jong et al., 2013; Wilensky & Resnick, 1999). By creating and studying these models, students can examine how population-level patterns emerge from individual-level processes (Wilensky & Reisman, 2006), supporting reasoning about mechanisms (Dickes et al., 2016; Hsiao et al., 2019) and probability (Abrahamson et al., 2006).

Figure 1

Programming and Data Analysis components of MoDa (Under Development)



The new design iteration of MoDa integrates domain-specific programming (left) with CODAP to allow model output (top right) to be analyzed and compared with real-world data (bottom right).

Data explorations and ABM are complementary practices. Both data explorations and ABM are rich scientific activities that involve multiple practices highlighted in science education reforms (e.g., NGSS Lead States, 2013). Uniting these practices through a coherent technology and curriculum has the potential to support students in reasoning about patterns in complex phenomena and their underlying causes in new ways. Emphasizing the probabilistic nature of systems, and the inexact science of extracting meaningful patterns from “messy” data, also aligns with important emerging ideas in computing education (what Tedre et al., 2021 call “CT 2.0”). Though our focus is on developing scientific models from data rather than machine learning, the emphasis on deriving meaning from data, parallel processing, and assessing the “correctness” of a model as evaluating its “accuracy at modeling some selected features of a phenomenon” are foregrounded in an approach that brings together quantitative data analysis and agent-based modeling. However, integrating these perspectives and levels of analysis is not straightforward for students and requires careful design and scaffolding (Wilkerson & Wilensky, 2015). That is what we aim to do through this work.

Responsivity to teachers’ pedagogies and student ideas

Introducing multiple sources of data across multiple scientific practices can be pedagogically challenging. To support teachers and students to meaningfully engage in this complex work, we aim to be attentive and adaptive to teachers’ needs and students’ ideas as they engage in this ambitious work. To do this, we draw on the notion of *responsivity in technology design*: Being responsive (Robertson et al., 2015) to students’ ideas through the design of a computational environment.

In prior work, we have considered how responsivity could be built into computational environments by designing for relatively rapid changes to the environment to reflect students’ ideas (Wilkerson et al., 2015). We identified multiple forms of responsivity ranging from renaming, adding, or removing existing blocks to better align with student ideas to extending out an existing blocks to enable students to pursue new investigations beyond the intended design to modifying the definition of a particular block, allowing for “finer tuning” of students’ models. We believe that attending to responsivity in this current project is especially important. When working

with real-world data, these types of responsivity can allow students to tap into the everyday language they may use to describe patterns, explore the quantitative dynamics of systems, and model more subtle differences between otherwise qualitatively similar patterns. Moreover, because integrating quantitative data analysis and ABM is pedagogically challenging work, enabling teachers to modify the computational environment to tailor to their students' ideas can help them better support their students.

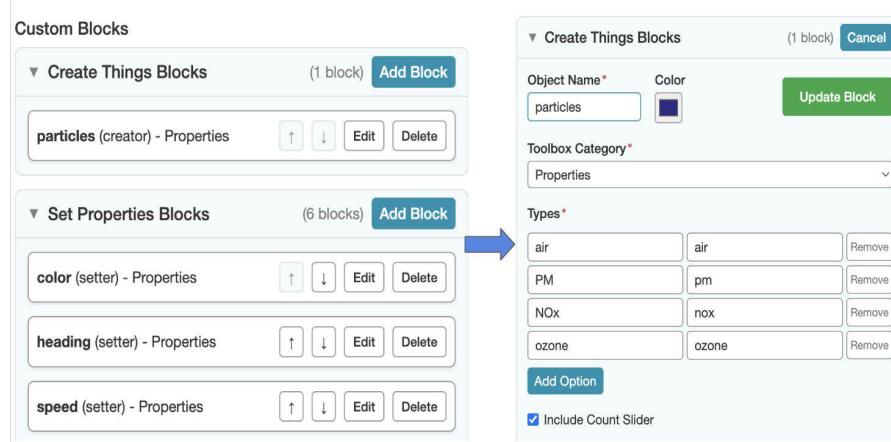
Goals and technology design

To combine the scientific practices of quantitative data exploration and ABM into a flexible platform, we are integrating data explorations in CODAP (Common Online Data Analysis Platform) with ABM in MoDa into a curriculum wrapper called Activity Player (The Concord Consortium, N.D.). The latest design of MoDa runs on a lightweight agent-based simulation engine called Atomic Agents (McNeill, 2022/2025). It can export simulated data directly into CODAP using Activity Player provides a platform to house MoDa and CODAP components, enable data transfer between the two so that simulated data can be directly compared with real-world sources, and provides curricular support infrastructure surrounding these technology components.

User interface for educators to create domain-specific agent-based blocks

An underlying vision for the latest iteration of MoDa is to democratize access to creating and modifying block libraries to support agent-based modeling investigations in classrooms. Our conjecture is that this can enable educators and educational designers (including teachers) to support modeling investigations that are responsive to students' ideas in ways that go beyond the research team's intent. To support this, we have been developing a visual interface to create and adapt domain-specific blocks (Figure 2) to design a block library for a particular unit. The interface is intended to support multiple levels of responsivity so that a user without much prior coding experience can modify names of entities, variables, and behaviors in the library, or remove blocks as needed, while users with more coding expertise could also modify the code for an existing block or generate a new block.

Figure 2
The Interface for Block Creation (Under Development)



Besides supporting block creation, we are envisioning additional ways to make MoDa responsive to teacher and student needs. For instance, earlier implementations of MoDa enacted *block unpacking*: Students being able to inspect, and modify the definition of a particular domain-specific block (Eloy et al., 2025). We envision making it possible for educators to select which blocks to make “unpackable” based on which mechanisms they would like their students to explore. For instance, depending on whether a teacher wants to focus on mechanisms for ozone production and/or dissipation, they can select which specific blocks to unpack so students can do a deeper dive into those specific scientific ideas (See Figure 1).

Simulating and Contextualizing quantitative data using ABMs and CODAP

To enable students to compare quantitative data from their ABMs with real world data, we are building an interface to export simulated quantitative data generated by students' ABM directly to CODAP for further exploration. This function builds on work that has embedded and exported data from simulations directly into CODAP (Finzer, 2014), further coupling such exports with real-world comparative datasets. To allow for flexible

use of these functions across contexts, students will be able to export (as relevant) one or more of the following kinds of data from their ABM: (1) Population data such as the number of a type of entity (e.g., “particles” in Figure 1); (2) Data about the environment in the form of “patch” variables; and (3) Sensors embedded in the model (e.g., a probe collecting data) or global reporter variables relevant to an activity.

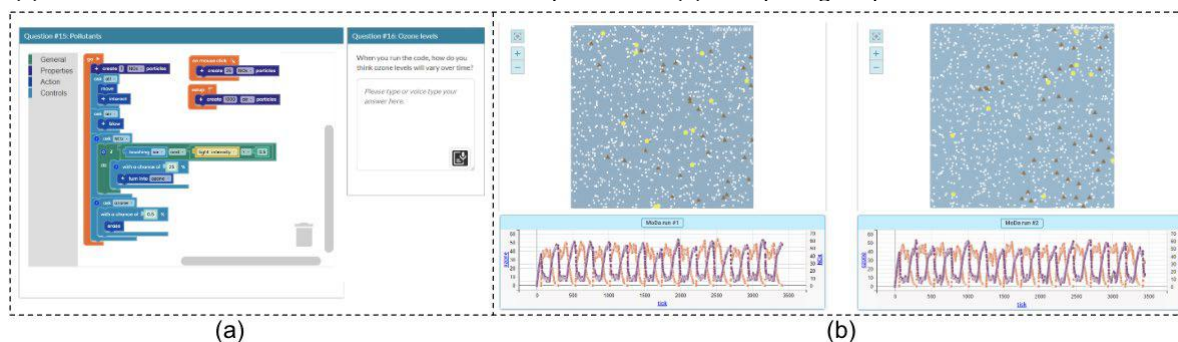
A central goal of the project is to support students to compare patterns in quantitative data from the real world and their ABMs for sense-making of mechanisms resulting in those patterns. In contrast to simpler systems, doing this in complex systems often involves connecting aggregate quantitative patterns to microscopic mechanisms that are described at a different unit of analysis - that of the agent. Hence, it is vital that each aggregate dataset exported from an ABM to CODAP be associated with the corresponding agent-level mechanisms that produced it. Our hope is that this can help students validate which agent-level mechanisms resulted in relevant aggregate trends that best match the real-world data. We are working towards providing this agent-level context to a dataset about the aggregate system in CODAP in two ways: (1) Enabling students to record and export visual data in the form of a video-recording of an agent-based simulation run; and, (2) Enabling students to “tag” their model code to the dataset in some way. Collectively, these can provide a context for students to compare quantitative trends in data to examine underlying agent-level mechanisms.

Modular design/integration of technology components

To enable responsivity to teachers’ pedagogical needs and students’ ideas, we will leverage the built-in modularity of Activity Player through both the design of the technology and curriculum. The technology described here around integrating ABM and quantitative data consists of four modular components: (1) Blocks-based ABM code; (2) The simulation; (3) Quantitative data from an ABM exported to CODAP; and (4) Real-world quantitative data represented in CODAP. Using Activity Player, we are currently exploring how these components can be combined in different ways to support a range of pedagogical and curricular goals. For instance, to support comparison of trends and variability in data from ABM and the real world, (3) and (4) can be adjacent (See Figure 1). On the other hand, to support code reflections (Wagh et al., under review), (1) can be used independently to ask students to predict what trends a model would result in (Figure 3a), or multiple model outputs can be combined to have students predict what output a program would result in (Figure 3b).

Figure 3a & 3b

(a) A code reflection for students to make a code prediction; (b) Comparing outputs from two model runs



Conclusion

This tech demo presents the most recent design iteration of MoDa, a tool to support integrating computational modeling and quantitative data analysis in ways that are adaptive and responsive to teachers’ and students’ needs. At this tech demo, participants will be able to interact with the Activity Player-based MoDa system. They will have a choice of exploring data and building models related to key air quality or water quality dynamics. They will also be able to preview accompanying curricular documents we are developing for the environment.

Authorship statement and Acknowledgements

AW initiated and conceptualized the paper. AW and MW wrote the initial draft. AE contributed to initial brainstorming and contributed data, analysis, and screenshots. All authors edited and approve the paper. Leslie Bondaryk and her team at Concord Consortium led technical development. Lucas Coletti developed and tested design prototypes and supported data cleaning. This work is supported by NSF Grant No. 2445609.

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***ProductiveMath*: A Research-Driven, AI-Powered Platform for Teacher-Generated Productive Failure Problems**

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Abstract: *ProductiveMath* is a generative-AI-powered platform designed to help teachers implement Productive Failure by making it easier to create open-ended, context-rich math problems that balance challenge and accessibility. This demo highlights how the platform supports teachers in the preparation phase while preserving teacher agency through review, editing, and regeneration features. The session is organized as two coordinated stations. At Table 1, attendees will see a concise slide walkthrough of *ProductiveMath*. At Table 2, participants will use *ProductiveMath* on laptops to select a standard, generate a problem, refine it with built-in guidance, and export a classroom-ready worksheet. Attendees may choose a quick or deeper exploration path. The demo concludes with a brief Q&A and a QR-code feedback form.

Introduction

Classrooms across the United States are facing a growing challenge as recent national assessments show substantial declines in middle-school mathematics performance (NAEP, 2024). These learning gaps are especially visible in algebra—an essential gateway subject that is often taught in ways students experience as abstract and disconnected from meaningful contexts (Dougherty et al., 2015; Common Core State Standards Initiative, n.d.; Gates, 2021). When instruction lacks relevance and opportunities for exploration, students have fewer chances to develop the deep reasoning skills needed for future learning.

Productive Failure (PF) addresses this challenge by engaging students in solving complex problems before formal instruction, prompting them to draw on prior knowledge, test ideas, and learn through mistakes (Kapur, 2008, 2012, 2015). Research shows this process can strengthen higher-order thinking, resilience, and conceptual understanding (Schwartz & Martin, 2004; Sinha & Kapur, 2021; Young et al., 2024). However, PF is difficult to implement because designing high-quality, open-ended, context-rich problems that balance challenge and accessibility requires substantial time, expertise, and creativity (Holmes et al., 2019; Darabi et al., 2018).

We developed *ProductiveMath* to address this issue. As a generative-AI-powered application, it supports teachers in PF's preparation phase by helping generate algebra problems aligned with productive struggle principles (Holmes, 2019; Zawacki-Richter et al., 2019; Seo et al., 2024). Importantly, *ProductiveMath* is designed to extend—not replace—teachers' capacity by preserving teacher agency to review, edit, and adapt tasks for diverse learners, making PF more practical for everyday classroom use.

Development overview

The development of *ProductiveMath* unfolded across five studies in a design-based research approach. Each study contributes a distinct layer of insight into how generative AI can support teachers in creating high-quality PF math problems. These studies, guided by design-based research principles, gradually refined both the pedagogical framework and the practical design of the platform.

Study 1 established what counts as a high-quality PF problem. A systematic review identified five core dimensions—prior knowledge activation, sweet-spot calibration, affective engagement, multiple representations/solution paths, and open-endedness—which were formalized into a 1–3 scoring rubric for each dimension (Kapur, 2008, 2012; Sinha & Kapur, 2021). The team then used the rubric to guide prompt engineering for GPT-4o and to rate 30 AI-generated problems; human raters found high quality overall ($M = 13.6$, $SD = 1.94$), strongest in prior knowledge activation and affective engagement.

Study 2 tested generalizability across grade levels and standards. Using the same prompt structure, GPT-4o generated 60 problems across seven Common Core domains. Human ratings remained high ($M = 13.59$, $SD = 1.28$), again strongest in prior knowledge activation and affective engagement, with moderate variation in open-endedness and multiple representations.

Study 3 examined teacher perceptions of problems and the early *ProductiveMath* prototype via surveys and hour-long interviews with seven middle-school teachers. AI-generated tasks averaged 17.19 ($SD = 2.98$) out

of 25, comparable to human-generated tasks (17.43, $SD = 3.55$) and higher than simple problems (13.29, $SD = 4.15$). Teachers praised engagement and productive challenge, but noted long text, occasional vagueness, demanding reading levels, and some standards misalignment; participatory design feedback led to requests for clearer formatting, simpler language, better visuals, and scaffolding, informing prompt and UI revisions grounded in HCI heuristics (Nielsen, 1994; Norman, 2013)--more details in the next section.

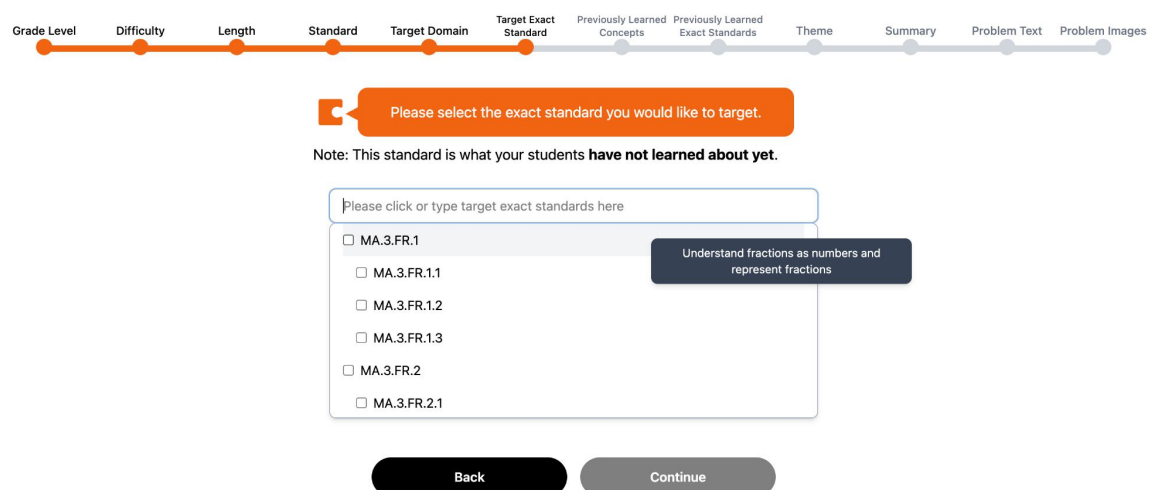
Study 4 evaluated the revised platform in five classrooms during full PF cycles (introduction, problem-solving, consolidation). Observations showed strong student engagement, collaboration, and resilience; student surveys (modified Intrinsic Motivation Inventory; Ryan, 1982) were generally positive, especially for effort, enjoyment, and perceived value. Teachers' PSSUQ ratings (Gonçalves da Silva e Souza & Canedo, 2018) indicated high usability: system quality ($M = 5.07$), information quality ($M = 4.83$), interface quality ($M = 4.93$), though consolidation facilitation varied.

Study 5 (ongoing) expands to 20 classrooms from 2nd to 12th grade to assess scalability, learning outcomes across settings, and large-scale logs of teacher and student interactions, supporting continued iterative improvement.

Design specifications

ProductiveMath (Rahimi et al., 2025) is designed around HCI (Human-computer interaction) principles (Park & Seda, 2018) and usability heuristics (Nielsen, 1994) to facilitate an intuitive and practical experience (Kheder, 2023). Nielsen's 10 usability heuristics includes: (1) Visibility and system status emphasizes the importance of keeping users informed about what the system is doing; (2) The match between the system and the real world relies on using familiar language and concepts; (3) User control and freedom supports the user's ability to reverse or exit unintended actions; (4) Consistency and standards emphasise familiar conventions across the interface to reduce confusion; (5) Error prevention encourages designers to anticipate and eliminate error-prone conditions before they occur; (6) Recognition rather than recall focuses on minimizing the user's cognitive load by keeping key information visible; (7) Flexibility and efficiency of use supports both novice and expert users by providing guided steps for beginners and offering shortcuts, customization, or advanced features for experienced users; (8) Aesthetic and minimalist design focuses on avoiding unnecessary information or decorative elements and prioritizing essential elements; (9) Help users recognize, diagnose, and recover from errors highlights presenting error messages that clearly identify the problem and suggest constructive solutions; and (10) Help and documentation points to providing support when users need guidance. Each design decision we made for ProductiveMath represents one or more of these heuristics. Figure 1 illustrates ProductiveMath's problem-generation process.

Figure 1
ProductiveMath Problem Generation



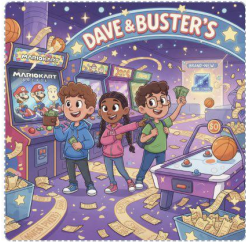
The horizontal progress bar provides visibility of system status. It shows the progress across key parameters and highlights the current stage. Instructions and teacher-facing language (e.g., "students have not learned about yet") support a match to real-world classroom concepts. User control and freedom are supported through persistent Back/Continue navigation and an editing option at the end of the problem generation.

Consistency is maintained through repeated step formatting and stable navigation. Error prevention and recognition, rather than recall, are supported by selecting from an existing standards list. Users can view the full explanations in plain language of the standard by hovering over it. Efficiency is supported through both browsing (novice-friendly) and search/type-to-filter (expert-friendly). Finally, the clean page layout reflects an aesthetic and minimalist design approach.

Figure 2
Example of an AI-Generated PF Problem Generated Using ProductiveMath.

Epic Arcade Race Challenge

StudentName
 GroupName
 Date



Alex, Jordan, and Sam were thrilled about their weekend trip to Dave & Buster's, a place full of exciting arcade games like Mario Kart, air hockey, and basketball hoops. Each wanted to win the most prize tickets, especially since they heard a rumor that the person with the most tickets could take home a brand-new game console. As they entered, gleaming machines buzzed and beeped, enticing them to start their fun-filled challenge. Each of them brought precisely \$20 to be spent wisely across the games.

Air hockey cost \$3 per game, Mario Kart was \$4 a race, and the basketball hoops charged \$2 for three rounds. They decided on a strategy: Alex loved Mario Kart and planned to play a minimum of 3 races; Jordan was an air hockey enthusiast, dedicating at least 4 games to this passion. Meanwhile, Sam felt the adrenaline rush in basketball, aiming for no less than 5 turns. Excitement filled the air as they did mental calculations, determining how their choices of games and spending could maximize ticket earnings.

To add a twist, they realized the tickets they earned from each game varied depending on skill. A typical round of basketball could yield 10 tickets if you score well, but mastering Mario Kart math could offer a bonus of up to 15 tickets per win. Innovative strategies were forming in their minds as they considered which game offered the best return on their limited budget and time. Air hockey proved trickier, with the potential to earn anywhere from 5 to 20 tickets based on performance.

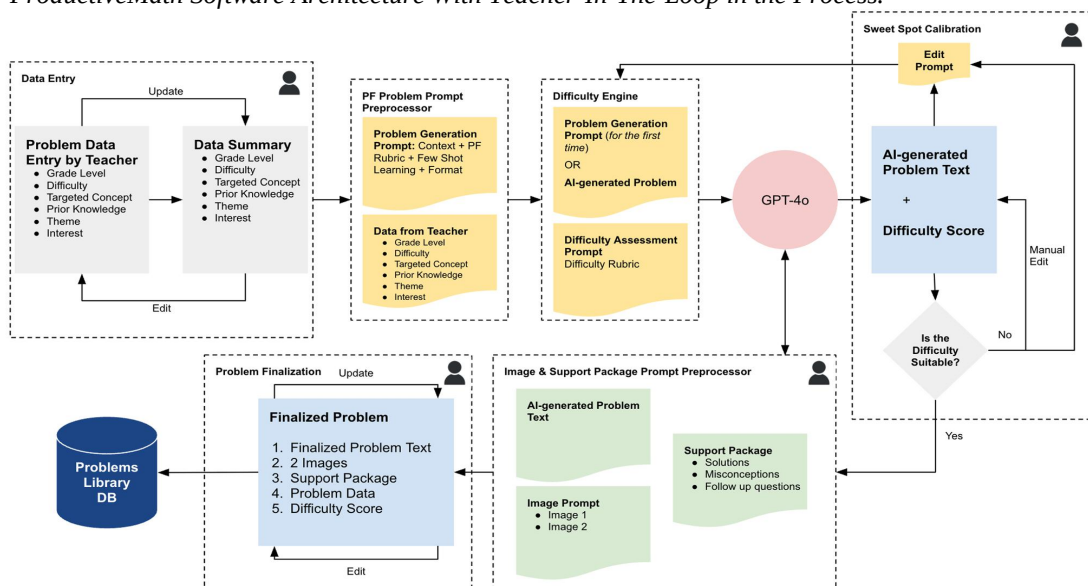


The challenge was beyond just playing; they needed keen observation and game mastery to win big. As they dove into their gaming spree, the task loomed: determine an optimal combination to achieve the most tickets, reasoning through variable spending and potential earnings. Should each stick strictly to their favorite games, or balance a mix for maximum prize potential? Craft different strategies, contemplate each allegiance, and propose how diverse game choices might lead to the highest score. What game allocations will bring home the coveted game console?

Software architecture

The workflow for *ProductiveMath* adapts a human-in-the-loop approach (Figure 3).

Figure 3
ProductiveMath Software Architecture With Teacher-In-The-Loop in the Process.



The frontend provides the user-facing interface where teachers enter problem data including grade level, difficulty, targeted concept, prior knowledge, theme, and interests, review AI-generated problems, adjust prompts for sweet-spot calibration, and finalize the problem by editing texts and choosing AI generated images. The backend contains the workflow logic that connects each stage of PF problem creation: it receives data from the frontend, formats it into prompts, routes them to GPT-4o, runs the difficulty engine, manages the iterative editing loop, and integrates results from the preprocessor including problem generation, difficulty assessment, image prompt, and support packages. The database stores all finalized outputs, including problem text, images, support packages, problem data, and difficulty scores, so the system can retrieve, update, and save PF problems in the library. The platform provides the infrastructure that hosts the system and enables coordination among components: the frontend collects teacher input, the backend processes it through prompt-preprocessors and GPT-4o, the difficulty engine ensures appropriate difficulty level for the problem, and the database preserves finalized problems, allowing the entire workflow to operate as an integrated PF problem creation system. As indicated in Figure 3, the teacher is in the loop throughout the process to make sure the PF problem is suitable for their students.

Intended Audience

ProductiveMath is designed primarily for grades 1 through 12 mathematics teachers who want to implement PF in their classrooms. The platform can support teachers who are already familiar with PF as well as those who are trying it for the first time and need more support in designing PF problems. *ProductiveMath* addresses everyday classroom demands—limited planning hours, varied students’ readiness levels, pressures to meet standards, and the desire to make math meaningful rather than procedural. Rather than expecting teachers to write PF tasks entirely on their own, *ProductiveMath* functions as a guided design partner. Designing a PF problem usually requires teachers to juggle many things at once: difficulty level, prior knowledge, standards, story context, and likely solution paths. *ProductiveMath* turns this workflow into a clear sequence of guided choices. Teachers decide on grade level, domain, concepts, and themes, then the AI generates a draft PF problem, and teachers read and adjust it so that the concepts, numbers, pictures, and language are correct for their students. This workflow mirrors the mental steps teachers should take when designing a PF problem but breaks that process into guided, manageable choices instead of forcing them to construct everything at once. Every step maintains teacher agency. Content can still be edited, shortened, reworded, or regenerated with one click, so teachers keep creative control without the time burden starting from scratch. *ProductiveMath* supports teachers beyond problem creation by attaching a support package to each task, including likely student solution paths, common misconceptions, and facilitation tips. Teachers can also export problems as classroom-ready PDF worksheets, aligning with existing routines. In addition, the Problems Library offers a shared repository of PF-aligned tasks filterable by grade, domain, and difficulty. Teachers can browse their own or colleagues’ problems, react to them, and adapt them rather than starting from scratch—reducing planning time and building a reusable bank of story-based PF tasks. Overall, these features are designed to lower cognitive load while preserving teacher agency: teachers retain instructional ownership while gaining practical resources to support productive struggle with clarity, engagement, and relevance.

Demo plan

We plan to have two side-by-side tables. Table 1 (Overview + Evidence): This station will feature a monitor or laptop running a short slide deck that introduces *ProductiveMath*—its purpose, how the platform was developed, and how teachers generate PF-aligned math problems through the system. The slides will highlight key findings from our five studies and walk through the workflow step by step. We will also include a small set of pre-loaded examples (including at least one “strong” PF task and one that needs revision) to illustrate how teacher review and editing are central to the process. Table 2 (Hands-on Try-It): This station will include three to four laptops for attendees to try *ProductiveMath* in real time. Each laptop will have a one-page “Quick Start” guide (and QR code) so participants can complete a short guided activity: select a standard, generate a problem, review/edit it using the PF rubric prompts, and export or save the task. Attendees can choose either a 2-minute “quick demo” path or an 8–10 minute deeper exploration path, depending on time and interest. To close the experience, we will invite participants to complete a brief QR-code feedback form (30–60 seconds) and then hold a short Q&A focused on usability, classroom fit, and implementation considerations across different instructional contexts.

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TopoLA Dashboard: Making Topological Learning Analytics Accessible for Exploring Learner Behavioral Structure

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Abstract: Topological Data Analysis (TDA) offers powerful methods for understanding structural patterns in learner populations, yet its application in learning analytics remains an emerging field with limited interpretive precedent. This interactive demo presents **TopoLA** Dashboard, a tool designed to make topological learning analytics accessible to educators and researchers without requiring topology expertise. By applying Zigzag Persistent Homology to behavioral data from the Open University Learning Analytics Dataset, the dashboard tracks β_0 —the number of connected behavioral components—across the course timeline, alongside composite indicators TEWI and TSI. We connect these constructs explicitly to CSCL concepts including social cohesion and co-regulated learning, while foregrounding a critical interpretive distinction: apparent β_0 convergence may reflect learner attrition rather than genuine behavioral alignment. Rather than providing definitive interpretations, the dashboard employs a three-layer framework—objective metrics, research-grounded pattern descriptions, and reflective prompts—to support educator sensemaking. We illustrate this framework with an authentic OULAD course example, demonstrating how rising β_0 and TEWI values in the weeks preceding an assessment deadline can prompt targeted instructional reflection. Participants will explore authentic OULAD data, experiment with topological parameters, and discuss design implications for emerging structural analytics.

Introduction

Learning analytics dashboards have become essential tools for understanding student engagement, with recent research documenting a shift from analytics-driven to learning-focused designs (Paulsen & Lindsay, 2024). However, a critical systematic review found no evidence that these dashboards improve academic achievement, with most studies reporting negligible or small effects (Kaliisa et al., 2024). This limitation is particularly significant in asynchronous learning contexts, where collective behavioral patterns at the group or cohort level remain largely invisible to educators.

Traditional learning analytics approaches struggle to capture the temporal evolution of behavioral structure within learner populations. Standard clustering methods provide static snapshots but cannot track how clusters persist, merge, or fragment across weeks. More importantly, even when structural patterns are detected, translating these mathematical abstractions into actionable educational insights remains challenging. Educators need tools that not only reveal structural dynamics but also support interpretation without requiring expertise in computational methods.

Topological Data Analysis (TDA), and specifically Zigzag Persistent Homology, offers a mathematical framework capable of tracking structural changes in non-monotonic temporal sequences (Carlsson & de Silva, 2010)—precisely the kind of data generated in flexible learning environments where learners enter and exit at different times. While TDA has been introduced to the learning analytics community as a promising approach (Munch, 2017), its practical application in educational dashboards remains emerging. More critically, there exists no established framework for interpreting topological indicators (such as β_0 , the number of connected components) in educational contexts.

From a learning sciences perspective, β_0 holds interpretive potential as a structural correlate of constructs central to computer-supported collaborative learning (CSCL): periods of low β_0 may reflect behavioral convergence associated with social cohesion, while sustained high β_0 may indicate fragmentation patterns linked to disengagement or divergent individual trajectories (Stahl et al., 2006). Crucially, however, apparent β_0 convergence may also result substantially from learner attrition rather than genuine behavioral alignment—a distinction that carries significant implications for interpreting co-regulation and collaborative learning dynamics (Järvelä & Hadwin, 2013). The dashboard's interpretive framework is designed precisely to surface such distinctions and support educators in responding appropriately.

This interactive demo presents *TopoLA* Dashboard, a tool designed to bridge the gap between methodological innovation in TDA and practical educational decision-making. Rather than presenting topological metrics as definitive answers, the dashboard provides layered interpretation support: a three-layer interpretation

approach that combines (1) objective metrics, (2) pattern descriptions grounded in learning analytics research, and (3) reflective prompts that invite educators to apply their contextual knowledge. This design philosophy acknowledges that TDA in education is an emerging field where interpretation frameworks are still developing.

The remainder of this paper describes the dashboard's architecture and key features, and outlines our interactive demo plan.

Tool overview

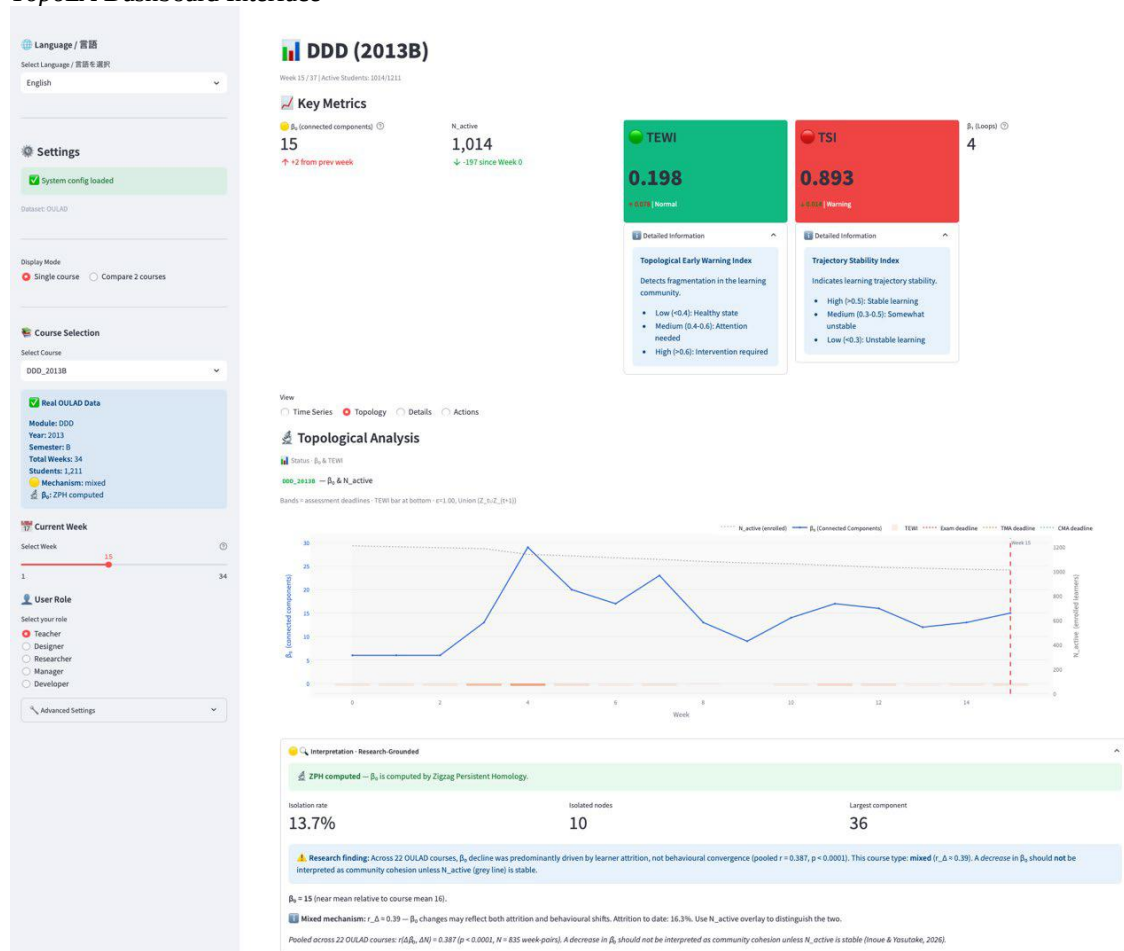
System Architecture

As shown in Figure 1, *TopoLA* Dashboard is a web-based interactive application designed for exploring behavioral structure dynamics in online learning environments. The system processes learning management system data through three stages:

- (1) Feature extraction from VLE activity logs—computing weekly click frequency vectors across activity types per learner, globally z-standardized to preserve temporal metric consistency
- (2) Topological analysis using Zigzag Persistent Homology (Vietoris-Rips complex, distance threshold $\varepsilon = 1.50$)
- (3) Interactive visualization with interpretive support.

Currently implemented as a proof of concept using the Open University Learning Analytics Dataset (OULAD; Kuzilek et al., 2017), the architecture is designed for future extension to other data sources through standardized data interfaces.

Figure 1
TopoLA Dashboard Interface



The interface displays alerts, key metrics (TEWI, TSI, β_0 , β_1) with detailed information panels, temporal visualization of topological features, and interactive controls for course selection, week navigation, and user role. Course DDD_2013B at Week 15 is shown.

Core visualization: β_0 temporal dynamics

The primary visualization displays β_0 (the number of connected components, or behavioral clusters) across the course timeline. Lower β_0 values indicate behavioral convergence—learners exhibiting similar patterns, a structural indicator potentially related to social cohesion and co-regulated learning—while higher values indicate fragmentation into diverse behavioral groups, which may reflect disengagement or divergent self-directed trajectories. Assessment deadlines are overlaid on the timeline, enabling educators to explore relationships between external constraints and structural changes.

Three-layer interpretation support

Rather than providing definitive conclusions, the dashboard employs a three-layer approach to support educator sensemaking.

Layer 1: Objective metrics

The dashboard displays quantitative measures without interpretation: current β_0 value, percentage change from previous week, and trend indicators. For example: " $\beta_0 = 15$ (increased 15% from Week 14)."

Layer 2: Pattern descriptions

Drawing on learning analytics research, the dashboard provides contextual descriptions of observed patterns. These descriptions use hedged language to avoid overclaiming: "Increasing β_0 values have been associated with behavioral diversification in prior studies of online learning environments."

Layer 3: Reflective prompts

The dashboard poses questions that invite educators to apply their contextual knowledge: "Consider: What instructional changes occurred this week? Are there assessment deadlines approaching? How does this pattern compare to previous course offerings?"

This design philosophy acknowledges that topological indicators require contextual interpretation that only educators with course-specific knowledge can provide.

Additional features

Course selection

Users can select from available courses to compare structural dynamics across different contexts, supporting cross-course pattern exploration.

Temporal navigation

Users can focus on specific time periods, zoom into deadline weeks, or view the entire course trajectory, supporting both detailed investigation and holistic pattern recognition.

Composite indicators: TEWI and TSI

Beyond raw β_0 values, the dashboard surfaces two composite indicators designed for educator accessibility (Inoue & Yasutake, 2026a). The Topological Early Warning Index (TEWI) quantifies structural fragmentation as the normalized rate of change in β_0 relative to a sliding window of preceding weeks (default $w = 3$), enabling timely detection of sudden fragmentation patterns that may warrant instructional attention. The Trajectory Stability Index (TSI) operates at the individual level, measuring each learner's behavioral consistency across weeks via the coefficient of variation of their activity vectors; higher TSI indicates a more stable, predictable engagement trajectory. Both indicators are normalized within each course, supporting temporal comparison while acknowledging cross-course comparison limitations. Together with β_0 and β_1 (the number of topological loops in the behavioral space), these four metrics populate the dashboard's key metrics panel visible in Figure 1.

Design rationale

The dashboard's conservative interpretive stance reflects the emerging state of topological data analysis in education. Unlike established metrics (e.g., click counts, completion rates), topological indicators lack

accumulated interpretive frameworks. By separating objective measurement from contextual interpretation, the dashboard positions itself as a tool for exploration rather than prescription—supporting educator inquiry while acknowledging the limits of current knowledge. A critical interpretive concern, surfaced explicitly by the reflective prompts, is that apparent β_0 convergence late in a course can substantially reflect learner attrition rather than genuine behavioral alignment. Distinguishing structural convergence from attrition-driven convergence is essential for making calibrated orchestration and scaffolding decisions.

Structure of the session

Participants will experience hands-on exploration of the dashboard, engaging directly with β_0 dynamics across authentic OULAD course data. Facilitators will demonstrate how to read the three-layer interpretation support and guide discussion on design implications for diverse learning contexts. No prior knowledge of topology is required. Participants can access the hosted dashboard on their own devices.

To illustrate, consider the course DDD_2013B at Week 15 shown in Figure 1. Layer 1 reports an elevated TEWI value alongside a rising β_0 , indicating a rapid increase in behavioral fragmentation relative to the preceding three weeks. Layer 2 contextualizes this pattern: increasing β_0 has been associated with behavioral diversification in prior studies of online learning environments, potentially reflecting divergent pre-deadline preparation strategies. Layer 3 then poses reflective questions: “What instructional events occurred this week? Is an assessment deadline approaching? Does this fragmentation pattern resemble what you observed in previous course offerings?” An educator with course-specific knowledge can cross-reference these prompts with their own records—recognizing, for instance, that Week 15 coincides with a major assessment release and that the fragmentation likely reflects legitimate strategic divergence rather than disengagement. This layered workflow exemplifies how the dashboard supports educator sensemaking without substituting for it.

Discussion and conclusion

TopoLA Dashboard represents an initial step toward making topological learning analytics accessible to educators and researchers. By providing layered interpretation support rather than definitive interpretations, the tool acknowledges the emerging state of this methodological approach while still offering practical utility for understanding behavioral structure dynamics. Preliminary empirical evidence from parallel OULAD analyses demonstrates that β_0 -based period segmentation outperforms fixed time-based approaches in 95% of OULAD courses (median variance ratio 4.10×; Inoue & Yasutake, 2026b), providing initial validation that topological indicators capture educationally meaningful temporal structure beyond arbitrary periodic boundaries.

Several limitations warrant acknowledgment. The current implementation uses OULAD as a proof of concept; generalization to other learning management systems requires validation. The three-layer interpretive framework, while designed to support educator sensemaking, has not yet been empirically evaluated for its effectiveness. Additionally, the relationship between topological indicators and learning outcomes remains an open research question— β_0 patterns may reveal structural dynamics without necessarily predicting academic success.

Future development will pursue three directions. First, we plan to extend data integration capabilities to support additional platforms through standardized interfaces. Second, we aim to integrate complementary analytical methods—including transition network analysis, higher-order network analysis, and persistent cohomology—expanding the range of structural and interactional patterns the dashboard can reveal and enabling cross-validation with established CSCL analytics frameworks. Third, and most critically, we will conduct user studies to evaluate how educators interpret and act upon topological indicators, producing empirically grounded interpretive guidelines and informing refinements to the interpretive support design.

We welcome feedback from the learning sciences community on how topological approaches might address analytical needs in diverse educational contexts. The source code will be made available to support replication and adaptation.

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Experimenting with Complex Phenomena in a Collaborative Mixed Reality Agent-Based Participatory Simulation

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Abstract: WIT is a collaborative platform that leverages agent-based participatory simulations and headset-based mixed reality to immerse learners in exploring complex phenomena. This session will demonstrate several key features of WIT through an example weather simulation, including its modeling of complex phenomena, modes of interaction, and data visualization tools. Preliminary findings that illustrate the platform's promise in encouraging learner inquiry of abstract phenomena, such as air pressure within complex systems, will also be discussed.

Introduction

Real-world phenomena often consist of complex systems. Understanding the processes and dynamics of these systems has been key to solving many real-world problems in the 21st century (Thurner et al., 2018) and benefits society-wide perspectives of our world (Serman, 1994). However, because these complex phenomena occur at scales that are not immediately visible or perceptible, they can be challenging to learn about (Wilensky & Resnick, 1999). Agent-based participatory simulations (p-sims) can support understanding complex phenomena as learners experience first-hand how local interactions create often unpredictable results across the system (e.g. Colella, 2013; Rates et al., 2016; Wilensky & Stroup, 1999). Mixed reality (MR) that overlays virtual objects on top of the real world has several affordances to support exploration of these phenomena. By representing otherwise invisible elements underlying complex phenomena as tangible virtual objects, MR supports direct exploration of these objects and their interactions (e.g., Roy & Radu, 2021). Moreover, MR can also support multiple learners to collaborate face-to-face in a shared physical space (Billinghurst & Kato, 1999).

The WIT platform

This paper reports on the design and development of a platform called We're in This Together (WIT). WIT leverages MR to create co-located, collaborative participatory simulations (p-sims) in which learners can explore complex phenomena in science together in a shared physical space (Roy et al., 2024). WIT combines the affordances of p-sims and agent-based models by immersing learners in an interactive 3D simulation of a complex phenomenon, enabling them to add or manipulate agents and the represented environment. The platform can synchronize up to 10 people using Meta Quest headsets, anchoring a virtual p-sim in a shared physical space such as a classroom. Our long-term goal is for the platform to serve as the foundation both for the development of new mixed reality p-sims and further research of the genre.

To investigate this genre of learning experiences in WIT, we are iteratively developing a functional prototype p-sim for learners to explore, interact with, and make sense of weather dynamics. We co-designed this p-sim with two middle school teachers, who suggested weather as a topic for its curricular relevance to middle school students and conceptual dependence on three-dimensional processes that are not directly visible and explorable. In the weather p-sim, air parcels float across the room, conditionally creating clouds that rain onto varied terrain biomes that are spread across the tabletops. Learners can interact with the simulation by heating or cooling the air parcels, creating pressure systems, or raising and lowering the terrain to create mountains and valleys. Learners can work in groups to collaboratively investigate how these modes of interaction affect weather and explore how collective actions can create long-term weather patterns such as increased rain frequency.

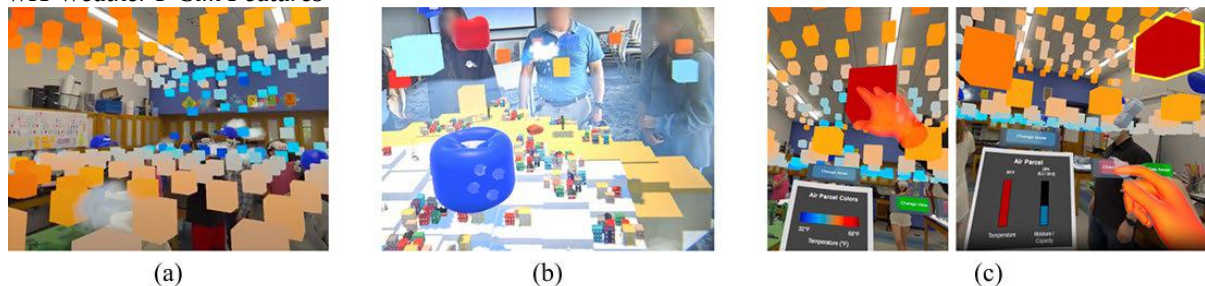
WIT design

Modeling complex phenomena

In WIT, our goal is to model the complexity of natural phenomena using agent-based modeling in a way that is physically immersive, explorable, and understandable for learners. Our initial prototypes used simple event-based models (EBM) with programmed discrete responses to high-level events (Baldwin et al., 2015). Although EBM supported rapid iteration of the overall experience, it lacked the depth for learners to inquire beyond high-level outcomes. Agent-based models (ABM) simulate the behaviors and interactions of individual agents that give rise

to global, emergent outcomes. In agent-based p-sims, learners control and/or take on the role of agents within a simulation to examine how their actions impact global outcomes in a system. We re-designed the weather p-sim's air and terrain systems from prescriptive EBM to emergent ABM, enabling learners to explore agent dynamics and discover the underlying rules of complex phenomena (Jacobson & Wilensky, 2006). While we developed the weather p-sim as a prototype, our platform modeling features are designed to be flexible and applicable to future WIT p-sims across different phenomena.

Figure 1
WIT Weather P-Sim Features



- (a) Air parcels circulate between upper and lower atmosphere layers, generating rain clouds based on humidity.
- (b) Terrain tiles track their individual altitude, temperature, and moisture values. Learners can raise or lower terrain to create mountains and use the object mode to create pressure systems (e.g. the dark blue object).
- (c) In the embodied mode, learners can change the temperature of their hand to heat or cool air parcels. Data can be visualized via objects' colors or data panel graphs.

Modeling fluids using smoothed-particle hydrodynamics

In the initial weather p-sim prototype, air was modeled as large, oblong “air masses” that floated across the room. Although the simplified model helped focus learners' attention on a few key tangible objects, it lacked the causal processes and emergent dynamics of weather that we aimed for learners to explore. For example, rain was programmed to form whenever air masses intersected, rather than emerging from underlying processes such as how cooling air decreases its capacity to hold moisture, limiting the depth of inquiry available to learners. Additionally, our consultations with meteorologists and climate scientists emphasized understanding 3-dimensional atmospheric dynamics. Although pressure systems are commonly understood as regions that push or pull air near the surface, they are more accurately modeled as vertical columns that circulate air between the upper and lower atmospheres.

To model the atmosphere's complexity, we redesigned the air masses to be represented as multiple small air parcels that dynamically influence neighboring parcels based on their temperature, moisture, and pressure (see Figure 1a). This is achieved using a modified version of smoothed particle hydrodynamics (SPH; Monaghan, 2005; Ni, 2025). This SPH representation of air enables learners to experiment with local interactions and investigate how their micro-level actions give rise to macro-level weather patterns.

Although implementing SPH achieved our goal of increasing the weather p-sim's capacity for representing key processes underlying weather in the atmosphere, it also increased the visual clutter of virtual objects in the p-sim. Our initial goal was to simulate a full 3D atmosphere, scattering air parcels throughout the entire room. However, this meant that many parcels floated along learners' eye level, obscuring the view of the rest of the simulation. The increased clutter interfered with the effects of learners' actions on the system: learners often could not tell whether air parcels were rising and falling by default or because of the influence of a pressure system object. To address this issue, we simplified the simulation to primarily represent the upper and lower atmospheres. Most air parcels move around on one of two layers: the lower atmosphere near the terrain or the upper atmosphere near the ceiling. Air parcels may fall or rise between the two layers at specific areas of high- or low-pressure respectively. This iteration maintains the complexity and spatiality of air while drawing learners' attention to key causal relationships, like how pressure systems influence the vertical movement of air.

Modeling environmental dynamics using a tile-based system

In the weather p-sim, terrain biomes like deserts and rainforests are projected onto tables throughout the physical room. We have developed a dynamic system that splits each table into a grid of terrain tiles (see Figure 1b). Each tile tracks its own temperature and moisture. A terrain tile's temperature value depends on its latitude and altitude. A tile's latitude is determined by its position in the room; one side of the room represents the cold North Pole (90° latitude), while the other side represents the hot equator (0° latitude). A tile's temperature decreases as it is raised

to create mountains. When a tile is rained on, its moisture value increases. Each tile's biome type changes depending on its moisture and temperature levels. For example, learners can transform a desert into grasslands by increasing the tile's moisture with persistent rain over time. This enables learners to experiment with how simulation agents and the agents' environment interact with each other.

Modes of interaction in WIT p-sims

To enable learners to explore and examine the dynamics of the phenomenon, we have designed different modes of interaction within the platform. The weather p-sim has three modes of interaction:

- *Object mode to create tangible virtual objects:* Learners can use their hands to place or move tangible virtual objects that behave as agents. In the weather p-sim, learners can place high- or low-pressure system objects (see Figure 1b) to create columns that make air parcels rise or fall between the upper and lower atmosphere layers. Our intention with this mode is to enable learners to add, visualize, and manipulate otherwise invisible elements as tangible objects and focus their investigations on these objects to create persistent changes in the system.
- *Embodied mode to directly control an agent:* Learners can use part of their body, like their hand, to change agents' properties. Our intention with this mode is to encourage learners to use movement for rapid testing. For instance, in the weather p-sim, learners can change the temperature of their hand to heat or cool nearby air parcels (see Figure 1c).
- *Environment mode to modify the simulated setting:* Finally, learners can make changes to the simulated environment to examine how these changes affect agents. For instance, in the weather p-sim, learners can grab terrain tiles and raise or lower them to create mountains and valleys. Learners can then observe how air parcels cool as they rise over mountains, often forming rain clouds along the mountainsides, or warm and remain dry as they descend valleys and canyons.

Data visualization in mixed reality

To support learners to connect their explorations in the p-sim to the represented phenomena, we have developed specific ways to visualize and engage with the generated data.

- *Embedded visualizations:* We have built data visualizations directly into the tangible objects in the p-sim so learners can track their changing properties. For example, air parcels change color according to their temperature and can be toggled to display how much moisture they have. Similarly, terrain tiles change color according to their biome type (e.g. tan deserts or dark green rainforests) and grow more plants as their moisture levels increase. These embedded visualizations are intended to enable learners to interact with the simulation and reason about how their actions affect the visualized properties simultaneously (e.g., Willett et al., 2016). By embedding data visualizations throughout the entire physical space, we also intend for learners to identify spatial patterns across the global system, such as how all air parcels in the upper atmosphere are colder than air parcels in the lower atmosphere.
- *Data sensors:* Learners can use data sensors to collect data from individual agents or environment tiles. While embedded visualizations are intended for learners to broadly estimate patterns in values of a property *across* multiple agents, data sensors are intended for learners to collect data from *one* agent at a time. Using a data sensor, learners can view numeric values and graphs for multiple properties on a panel. Learners can also trace the movement of an agent across space and over time. When a learner selects an air parcel using the data sensor, the parcel's movement is visualized in the simulation space via a trail tracing its recent path. This feature provides spatial information for a single agent over time so learners can analyze its movement behaviors.

By leveraging embedded visualizations and the data sensor tool, learners can investigate the modeled phenomenon in both breadth and depth.

Observations from pilot tests

Over the last year, we pilot tested WIT to investigate how learners interact with the simulation. As WIT represents a new genre of mixed reality p-sims, our goal is to present early analysis of how students engage with the medium. Two pilot tests were conducted with a total of four groups of 5-6 8th grade students (n=22) at a local middle school. In the pilot tests, we first gave students a brief orientation to how air parcels and terrain were represented in the model, then students spent between 10-30 minutes exploring the weather p-sim using Meta Quest 3 headsets. Three researchers and up to three of the students' teachers circulated the room to observe and ask students

prompting questions about their sensemaking process. At the end, the students participated in a 5-minute focus group exit interview to share their takeaways about the phenomenon and feedback on the experience.

We are currently analyzing video data from four student groups. This has involved generating content logs (Jordan & Henderson, 1995) of the videos to document how students used the different interaction modes and engaged with available data visualizations to make sense of weather dynamics. These content logs were used to generate initial observations and hypotheses about how students used these available features to explore weather dynamics, like how temperature and pressure influence rain patterns in this p-sim. Here, we share preliminary findings from our ongoing analysis.

Our initial analysis is revealing what we see as a key difference between how students engage in these two modes. As previously stated, in the weather p-sim, the students' goal was to influence precipitation in a geographical system. In general, students spent more time experimenting with pressure systems in the object mode. However, students more often verbalized correct observations about the causal relationship between temperature changes and precipitation while using the embodied mode.

In the embodied mode, students tended to quickly discover the relationship between temperature changes and precipitation. For example, one student, Alex, articulated a misunderstanding that warmer temperatures will result in greater precipitation when he stated, "We need to make it warmer so that [the air will] expel more rain." His friend, Jamie, corrected him: "No, it's cold... See Alex, the [moisture] number's going up." Jamie demonstrated this by holding their hand over an air parcel, causing the air parcel to turn dark blue and create a rain cloud. After testing this himself, Alex agreed, "Oh, nice... I'm wrong. I forgot how clouds worked. But you're right." After discovering and confirming this effect with each other, the students used the embodied mode to discover additional first-order relationships, such as how persistent rain changes terrain moisture level and how temperature affects air parcels' capacity for moisture.

In contrast, students engaged with the object mode in more exploratory ways and experimented with combinations of multiple pressure system objects more frequently. For instance, Riley walked over to a table and picked up another student's pressure system object, then stacked both of their low-pressure system objects together, which amplified the resulting upward airflow pattern. When a teacher asked what they were thinking, Riley observed, "It makes the air kind of cold? Like, I guess... it goes up and then it's cold," correctly describing the relationship between low-pressure systems, altitude, and temperature.

In both modes, students often identified first-order effects while exploring by themselves or with peers. Notably, students also identified higher-order, emergent effects when scaffolded with prompts from the researchers and teachers. For example, one researcher prompted Jamie to consider how high- and low-pressure systems could create wind. Jamie tested this by putting one high- and one low-pressure system near each other, then stepped back to observe, "They're going in like a big circle." Excited, one teacher asked, "What do you call that?" Jamie promptly answered, "Oh, convection currents!" correctly connecting back to prior knowledge they had learned the year before.

This pilot study illustrates how students have used the "embodied" and "object" modes to discover first-order effects and explore more complex relationships. As we continue to analyze the data, we are exploring the implications of this trend in terms of design and how to support learners' experimentation and sensemaking in the WIT p-sims. Our initial findings support the platform's promise as an immersive space for learners to explore spatial complex phenomena. In future work, we are interested in exploring how this genre of learning technology can support students to gain deeper insight into emergent relationships in complex phenomena. Future work may include swapping the abilities associated with the "embodied" and "object" modes to further examine how students use the affordances of each mode and analyzing how teachers scaffold students' inquiry inside the simulation.

Tech demo session

During the session, we will demonstrate several innovations that support (1) WIT's capacity for modeling complex phenomena by simulating agent-based interactions and dynamic environmental influence; (2) how learners can engage with the simulation through the embodied or object modes of interaction; and (3) exploration of data, leveraging the affordances of spatiality in mixed reality.

We will bring several Quest 3 headsets to set up the WIT p-sim in a space where participants can move around comfortably during the tech demo session. Participants will test out the features detailed in this paper in groups of up to 3 and have the opportunity to discuss design trade-offs and preliminary findings with our team. If any participants prefer not to use the VR headsets, they can watch a computer monitor screen to observe other participants' interactions with the p-sim via a third-person view.

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Visualizing Curricula in Higher Education: A Semantic Similarity Approach

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Abstract: Many institutions give students some freedom to choose courses across disciplines. Yet, existing course registration technologies often present courses as a textual list by discipline, leading to extraneous cognitive load and a lack of metacognitive awareness of how course decisions relate to the broader curriculum. To support students in making well-informed course choices, we develop a tool to visualize the entire curriculum based solely on course content. We describe its methodology, features, theoretical underpinning from learning sciences research, and ways to mindfully embed the tool into advising sessions.

Problem and motivation

In many institutions of higher education, particularly liberal arts colleges, students have considerable freedom to select courses from a broad range of fields. This freedom is intended to allow students to shape their own learning goals proactively and choose courses accordingly (e.g., Amherst College, n.d.; Brown University, n.d.). In practice, however, the exercise of this freedom is mediated by administrative technologies for course browsing and registration, which are often used to guide advising sessions. By structuring the information available to students and faculty, and by guiding their conversations in advising sessions, these tools play a sociomaterial role in shaping course selection and learning outcomes.

Many institutions rely on commercial, off-the-shelf registration platforms that often organize courses in a long list by discipline. While these technologies streamline administrative processes, they raise two challenges. First, reading through a long list can impose *extraneous cognitive load* (de Jong, 2009) on users, hindering their ability to make fully informed course selection decisions. Second, disciplinary boundaries and a lack of interactivity limit students' ability to develop *metacognitive awareness* (Flavell, 1976) of how their learning relates to the wider curriculum and their overall learning goals. Moreover, these two points are in fact related: the extraneous cognitive load imposed by such course scheduling tools can divert students' cognitive resources away from big-picture, metacognitive thinking. Together, these constraints can limit the scope and richness of conversations between students and advisors, raising the risk of making uninformed or unsatisfactory choices. Especially in liberal arts programs, such constraints may cause students to inadvertently gravitate toward familiar disciplines, contributing to a mismatch between students' exercise of freedom in course selection and these institutions' missions of promoting interdisciplinary learning.

Against this backdrop, we created the *Visual Open Curriculum*, a supplemental web application for course exploration and academic advising that is especially conducive to liberal arts curricula. First, the app lessens students' cognitive load by replacing lengthy course lists with a visual and interactive curriculum map, a semantic search engine, and a recommender function. Second, it organizes courses based on content rather than disciplinary classifications and includes a reflectional note-taking tool, which together promote conscious selection of courses according to learning goals. Moreover, recognizing that effective use of technology depends on both the users' mindful engagement and surrounding social contexts (Salomon et al, 1991; Rook et al, 2024), we provide explicit suggestions on how students and faculty can best integrate this tool into the broader sociomaterial environment of their advising sessions. Overall, the app aims to provide students with increased agency in navigating extensive curricular choices, informed by theories of learning.

Methodology and related literature

Previous studies within the field of learning analytics have developed visualizations and algorithms to map, predict, and recommend courses. They draw on different sources of information to analyze the curriculum: prerequisites (e.g., Parameswaran et al, 2011), past students' course-taking patterns (e.g., Pardos and Jiang 2020), grades (e.g., Jiang et al, 2019), course evaluations (e.g., Farzan and Brusilovsky 2006; Bendakir and Aimeur 2006) among others. The choice of which information source to use hinges on the objectives of the tool. For instance, if the objective is to recommend a series of courses that prepares a student well for some target course *X*, the tool should model prerequisite structures and how past students who have taken *X* fared in their courses prior to *X* (Jiang, Pardos, and Wei 2019).

In contrast to these studies, our objective is *not* to predict students' course selection or recommend the most viable path, but rather to facilitate discussion and exploration by identifying interesting courses for the student without being bound by disciplinary boundaries, grades, or prerequisites. We thus focus on content alone, drawing on course descriptions. In liberal arts institutions with more freedom in course-taking, or in humanities departments with fewer prerequisites, raw content similarity may very well dominate other factors in course choice. Approaches used in prior work fall short in this aspect, as none capture course content directly. For example, while students' course histories are correlated with content similarity (courses often taken successively are likely to be related), it may bias the algorithm toward courses in the same discipline and leave out courses that are similar but infrequently taken together. While traditional text analytics (e.g., bag-of-word models) struggle to capture semantic similarity, modern language models, which attend to the broader context of textual descriptions, are especially well suited for these purposes.

We employ publicly available course data from Amherst College, a selective private liberal arts college in Massachusetts, USA. Importantly, however, our methodology is general and can be used in any institution with textual course descriptions. Our procedure begins by extracting the descriptions of every course from the course catalog and cleaning these descriptions using OpenAI's GPT-4o, such that it retains only information about course content. We then calibrate a custom language model to convert these text descriptions into embedding vectors. An entirely zero-shot approach, e.g., using Open AI's GPT embedding model directly off the shelf, can be suboptimal precisely because they are trained on enormous corpora of internet text (Dell, 2025) and are prone to missing the subtle nuances in course content that students and faculty may care greatly about. We address this issue by conducting contrastive learning to tailor a smaller, open-source embedding model (Sentence BERT) to our particular dataset. We use cosine similarity to measure the similarity between two courses, and implement a combination of supervised contrastive learning (Khosla et al, 2020), alignment loss and uniformity loss (Wang and Isola, 2020) to make the model more attuned to the variation in course content. Finally, we define distance between a pair of courses simply as one minus the cosine similarity.

Since distances are defined pairwise, we have close to one million dimensions from the 13 thousand courses with valid descriptions offered in our 17-year sample period. To visualize this in 2D space, we use t-distributed stochastic neighbor embedding (t-SNE; Hinton and Roweis, 2002; Maaten and Hinton, 2008). While all dimensionality reduction techniques are inherently distortionary and involve trading off preservation of local vis-a-vis global structures, our experimentation reveals that t-SNE with a perplexity parameter of 50 offers good balance. Figure 1 shows an intuitive global spectrum of academic fields from the arts to the natural sciences, with similar courses and majors clustered near each other. There is also significant overlap of fields (as shown by the manually drawn rectangles in Figure 1) and mixing of departments, confirming that our map focuses solely on course content in a discipline-agnostic way. Further technical details and validation exercises are omitted for this short paper but are available in our longer documentation online.

Tool description

Visual map of the curriculum

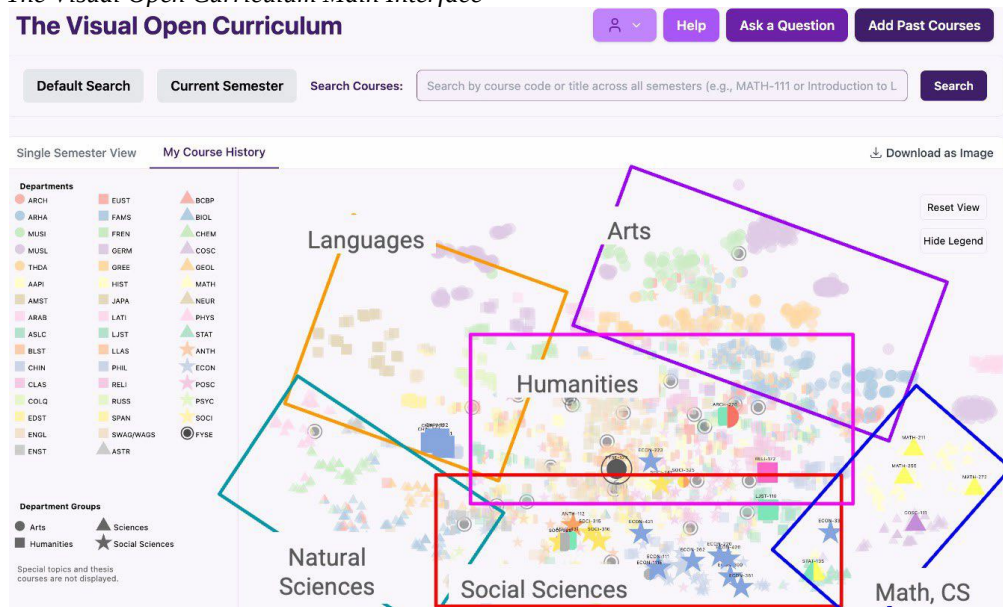
Many course scheduling tools do not allow students to compositely see all of their course options. Instead, they are presented with hundreds of courses in list form, alphabetically organized by disciplinary category. This places extraneous cognitive load on students, especially those attempting to explore multiple areas of their curriculum. It is well documented that extraneous cognitive load leads to disengagement in learning (de Jong, 2009; National Academies et al, 2018). Anecdotally, this sort of disengagement in the course exploration process can manifest as students 'settling' for courses within disciplines they are already familiar with.

In contrast to existing course registration tools such as Workday, the *Visual Open Curriculum* app was designed to decrease students' cognitive load by providing them with an interactive, visual map of the curriculum. The courses the student selects are emphasized visually, and for any mapped course, users may select the corresponding marker to view associated metadata: faculty, teaching schedules, course descriptions and course codes (Figure 2, left).

We also explicitly provide students with the "My Course History" view, where students and advisors can see in a single glance how the student's previous courses fit into the broader scope of the college's courses (Figure 1). This relational view supports their awareness of how they are constructing their path through higher education. Students can use the "My Course History" view to engage mindfully with the app and visualize their individual learning journeys holistically, supporting their decision to select courses that will build on previous courses or that will allow them to explore a new area. This sort of metacognitive awareness can support students in becoming conscious of their autonomous ability to set learning goals and select courses accordingly (National Academies et al, 2018). Furthermore, when students recognize that they have made an informed, autonomous decision, it

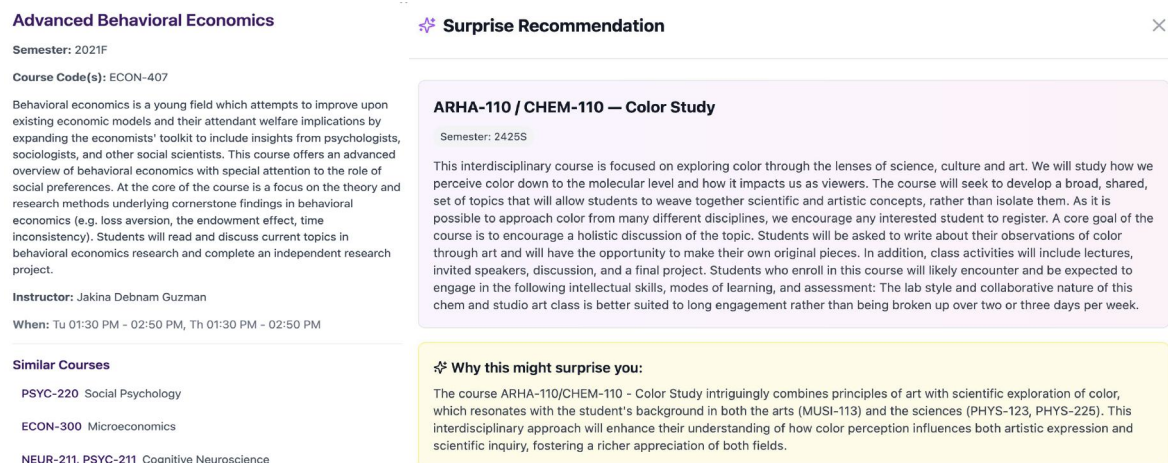
supports their motivation and commitment to these learning goals, even when they may encounter obstacles (Murayama et al, 2015).

Figure 1
The Visual Open Curriculum Main Interface
The Visual Open Curriculum



“My Course History View” with annotations overlaid indicating the academic field corresponding to each area of the map. Courses the student has taken are emphasized with larger markers and clearer colors, while the other courses offered during the same period are faded. As indicated in the legend, marker colors correspond to departments and shapes to broader field classifications.

Figure 2
Semantic Search Features



[Left side] An example course popup for ECON-407 “Advanced Behavioral Economics” with course description, semester code, teaching schedule and similar courses displayed.

[Right side] An example course recommendation for a student with previous exposure to Music & Physics: the “Color Study” course combines these interests, with detailed reasoning included in the yellow text block.

Semantic search

Existing course scheduling platforms are often characterized by restricted or underdeveloped searching functionality. Users can only look up courses for which they already possess the relevant course codes or words contained in the titles. To facilitate user exploration, we build three additional features that help redirect cognitive energy away from navigating the interface to actually engaging in course exploration.

First, we augment traditional keyword search by integrating a *semantic search engine*, applying the same methodology to the user's query to return the courses with the most similar descriptions. For example, a user might search "working with optics & lenses" and receive metadata on both an Advanced Astronomy elective (in the physics department) and an introductory course in photography (in the art department).

Second, in the popup for each course, we display the *most closely related courses* according to semantic similarity. For instance, a student, having particularly appreciated a course on Behavioral Economics in a past semester, can search for that course and immediately identify similar courses across all departments (Figure 2, left). This feature is not only immediately useful, but also addresses an important limitation of the map. Because dimensionality reduction distorts pairwise distances, the courses shown near a given course on the map are not always its true closest neighbors in the embedding space.

Lastly, we build a *recommender function* that combines the personalized insights of an academic advisor with an encyclopedic knowledge of all available courses. Specifically, we embed a chat-based LLM to return courses outside of the user's most frequently explored disciplines while maintaining the common thread that unites their areas of interest. For example, a physics or art student who might not usually consider taking a chemistry class could potentially be interested in "CHEM-110: Color Study" (Figure 2, right), but would likely never discover that class when perusing the traditional, opaque course registry.

Reflection and note-taking

Goal setting, often considered a key component of metacognitive thinking (Stanton, 2021), can improve learning outcomes both in terms of overall learning trajectories (Schippers et al, 2020) and specific learning tasks (Paulson and Bauer, 2011; Stanton, 2021). To this end, we provide students with an in-app space for reflection and note-taking, where they can bookmark courses of interest and add notes regarding their goals and motivation for taking each course. By connecting students' learning goals to a tangible map, we scaffold students' abilities literally "map out" different pathways they could take through the curriculum to reach these goals. Students will also have the option to share a snapshot of their mapped courses and reflections with their academic advisors. In this way, the app can not only provide material support through presenting information, but also aid in enhancing social relationships between students and their advisors.

Example useCase

Prior to the advising session, students will

- Conduct initial course exploration using the *Visual Open Curriculum*, e.g., zoom in on their past courses and look at neighboring courses to identify common themes, use the semantic search function to compile a list of courses that relate the student's key interests to other fields.
- Respond to prompts asking the student's motivations for their proposed course selection. Share that document with their advisors, improving knowledge sharing ahead of the actual advising meetings.

During advising meetings, the app functions as a visual support to enable better communication.

- Advisors will review the students' course choices to date in the "My Course History" tab. The map makes it clear when their studies are concentrated in one corner of the curriculum. Areas lacking highlighted markers stand out as potential areas of exploration.
- Advisors can propose alternative classes with common themes identified using the visual map and semantic search functions to avoid over-concentrating in specific departments.

Interactive demo session

We will project our web application on a screen. After briefly reviewing the methods, we will ask the participant to either play the role of an academic advisor or that of a hypothetical student. The author will play the other role. Together, we will review the app functionalities and conduct a mock advising session.

Conclusion

Traditional platforms often lack an overarching view of curricular offerings, hindering students' capabilities to align their learning journeys with personal goals as well as institutional ones. The *Visual Open Curriculum* app attempts to provide students with an accessible metacognitive scaffold, allowing them to chart their own paths through the map of possibilities that their education offers. We are currently preparing for a pilot intervention using this app at Amherst College in the context of first-year advising.

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Extending Immersive Learning with Smell: An Olfactory-Enhanced Virtual Reality System

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Abstract: We present VR-O, a virtual reality (VR) system that integrates olfaction into immersive learning environments. The project explores how adding scent to VR experiences can deepen engagement, support learning, and make simulations more authentic, particularly in domains where olfactory perception is central to expertise (e.g., water treatment, wine making). During this Interactive Tools and Demos session, participants will experience the olfactory VR setup firsthand and discuss pedagogical and technical implications for multisensory learning environments.

Introduction and rationale

Virtual reality (VR) has become a powerful medium for experiential learning, allowing learners to explore complex systems, perform procedural tasks, and engage with abstract concepts through embodied interaction (Asad et al., 2021). Yet most VR learning environments remain limited to visual and auditory modalities, omitting senses such as smell that are integral to how humans perceive and remember experiences (Melo et al., 2020).

Olfaction plays a particularly important role in many professional and everyday learning contexts. Domains such as wine making (Tempere et al., 2019), cheese production (Delahunty & Drake, 2004), and water treatment (Fox & Abbaszadegan, 2023; Spackman, 2024) depend on the ability to discriminate, describe, and identify odors. Teaching these skills is challenging: in some cases, physical materials may be costly, perishable, or difficult to standardize, while in other cases the opportunity to experience certain olfactory cues on the jobsite may happen only sporadically. These difficulties present an opportunity to create VR environments that integrate olfaction for use in educational settings. Specifically, integrating olfactory cues into VR learning environments can address physical, material, and temporal challenges by enabling: (1) lifelike practice in safe, reproducible conditions; (2) deeper engagement and motivation, as olfactory stimuli evoke strong emotional and mnemonic associations; and (3) richer multimodal learning, connecting smell with visual and linguistic representations. However, adding smell to digital environments presents technical hurdles such as accurate scent control, synchronization, and user comfort.

To explore these opportunities and challenges with a relevant test case, we designed our system to mimic a municipal water treatment facility's laboratory, wherein water quality is tested not only through chemical means, but also through quick sensory assessment. This design choice aligns with coursework currently offered at our university, in which students study water quality through the lenses of environmental chemistry and public health. Incorporating olfaction into this context is especially valuable because smell is a crucial indicator of water quality: odors can signal the presence of organic contaminants, disinfection byproducts, or algal blooms, as well as issues with filtration and distribution infrastructure (Fox & Abbaszadegan, 2023). In professional practice, operators rely on visual inspection and sensory evaluation to detect anomalies that may not be apparent through measurement instruments alone. By recreating a water laboratory in VR, our system provides opportunities for learners to practice identifying and interpreting odor cues in a controlled, repeatable environment, preparing them for real-world scenarios that demand both technical understanding and olfactory acuity.

System overview

The VR-O system comprises two tightly coupled subsystems: the wearable hardware that delivers odors to the user and the software stack that controls odor timing and intensity.

Hardware

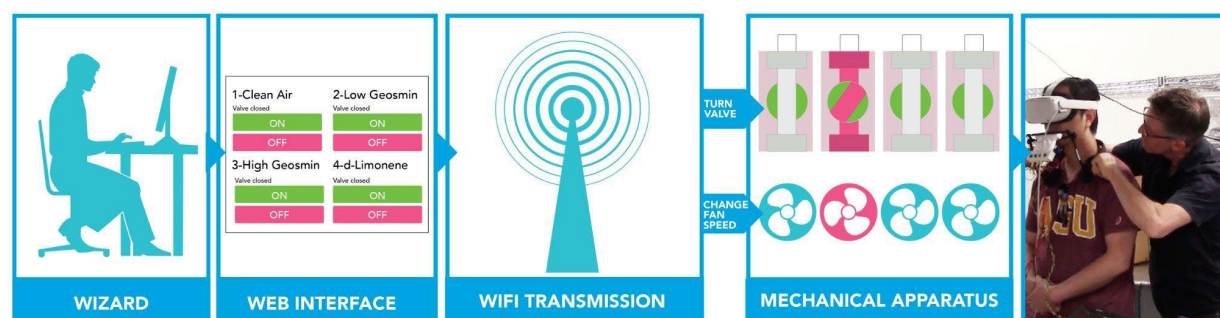
The wearable apparatus is a neck/shoulder rig modeled after a harmonica holder; the odor output is positioned directly beneath the user's nose. The device houses four channels: clean air, low-geosmin, high-geosmin, and

high-d-limonene. Each channel contains an odor chamber, a piezoelectric microblower, and a centrally mounted valve. Piezoelectric microblowers were chosen for their directional airflow and ability to act as a unidirectional seal, minimizing cross-contamination between channels. Odorants (liquid geosmin and d-limonene diluted in distilled water) are loaded into cut-to-size sponges positioned below the blower outlet to create headspace volatilization. Sponges rest atop self-healing injection ports so that odorant liquid can be added directly without opening the chamber or risking damage to the microblowers. For study sessions, we injected 15 mL of the prepared solution into each chamber prior to use; sponge size was chosen to provide headspace for steady volatilization during the session while limiting leakage.

Software and communications

Peripherals (microblowers, valves/servos, and digital potentiometers) are controlled by an ESP32 development board. The ESP32 connects wirelessly to a local hotspot hosted on the experiment laptop, permitting serial commands to be transmitted over Wi-Fi. To facilitate the Wizard of Oz control flow we used in our user study, we implemented a simple web page that exposes an “On/Off” control per channel. When the wizard clicks “On,” the web page issues serial commands that actuate the servo to open the valve and modulate the microblower speed; clicking “Off” closes the valve and stops the blower. A network diagram of this architecture (wizard interface → laptop hotspot → ESP32 → hardware) is shown in Figure 1.

Figure 1
Network Diagram of the Architecture



Odorants and intensity levels

Two odorants were selected for use in the system due to their relevance and distinct olfactory profiles: geosmin (an earthy/musty scent associated with algal and bacterial activity in surface water) and d-limonene (a citrus-like scent commonly present in industrial/byproduct contamination, but also a potential indicator of cyanobacterial dominance during high algal growth (Pozzer et al., 2022; Zhao et al., 2016)). The scent delivery device included four independent channels. To avoid participants developing an association between the sound of the motor and the delivery of an odorant, one channel was reserved for clean, filtered air with no added odorants. The remaining three channels were used to deliver commonly encountered odors in water treatment contexts. Geosmin was included because a residual earthy odor is often encountered in treated drinking water in arid regions and is safe at the dilutions used in this study, while d-limonene was selected as a contrasting odor representing atypical, potentially anthropogenic contamination. To further evaluate the sensitivity and controllability of the system, we incorporated two different concentrations of geosmin, prepared by diluting the liquid odorants in distilled water: d-limonene high = 1:10; geosmin high = 1:10; geosmin low = 1:100.

Wizard of Oz study

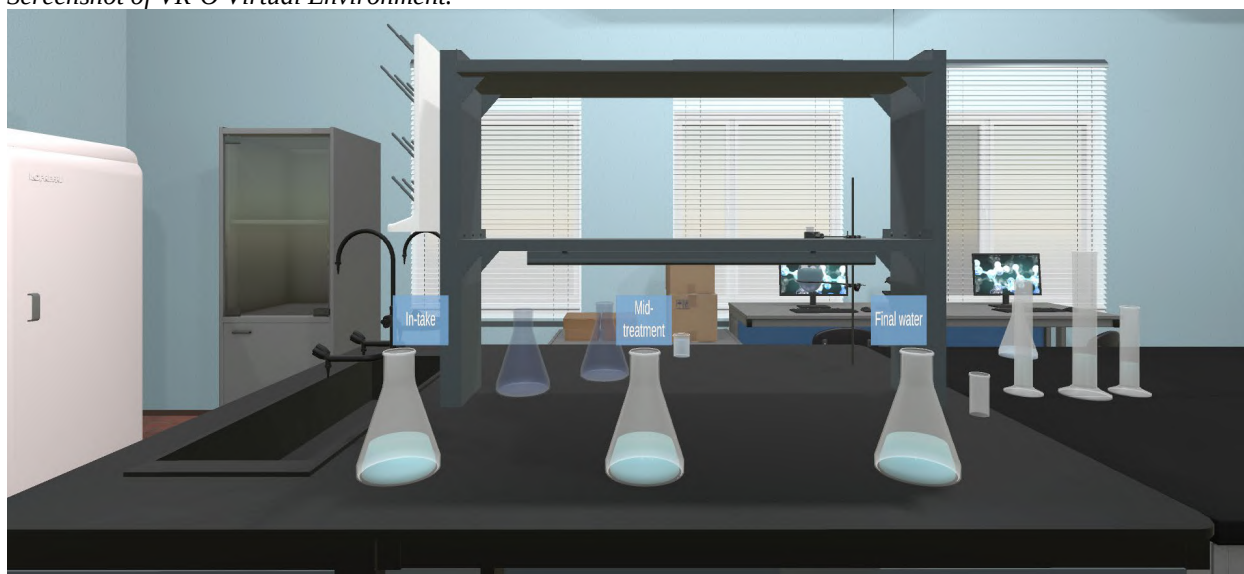
The VR-O system and study design were informed by authentic water treatment workflows. We developed the virtual environment in collaboration with Gary Burlingame, former director of the Philadelphia Water Bureau of Laboratory Services, whose professional experience in water-quality management shaped the scenarios and decision points used in the learning tasks. In municipal treatment facilities, water typically passes through three principal stages: (1) intake (raw water drawn from a natural source); (2) mid-treatment (filtration, sedimentation, and chemical treatment); and (3) final treatment (post-treatment processing prior to distribution). Odors commonly vary across these stages and can serve as useful cues for contamination, treatment efficacy, or process failure. By anchoring the VR-O experience in these three stages, the study simulated decision-making situations that reflect operational judgments relevant to water-quality assessment.

Method

We implemented a Wizard of Oz methodology (Martin & Hanington, 2012) to evaluate whether the VR-O system could support learners in detecting, identifying, and making treatment decisions based on odors encountered in a simulated water-quality laboratory. In this approach, participants interacted with the system as if it were fully autonomous, but a researcher managed key functions. Unlike in a traditional Wizard of Oz study where participants are not aware of the human in the loop, in our study, participants knew that a human researcher was operating the system and guiding the interaction. Specifically, during the study, the researcher narrated the scenarios and manually controlled scent delivery and timing. This method allowed us to assess the feasibility of the VR-O concept early in the design process without investing the time and resources required to build a fully autonomous system, and it enabled rapid iteration on non-technical elements such as scenario structure and the design of water-treatment tasks (figure 2).

Figure 2

Screenshot of VR-O Virtual Environment.



Participants saw a virtual workspace with three water samples, labeled “In-take,” “Mid-treatment,” and “Final water.”

Scenarios and odor delivery

Participants engaged in three scripted VR scenarios, each including three virtual water samples (intake, mid-treatment, and final water). Each scenario featured a unique combination of odor events and was based on information from expert interviews and real-world water-quality practices. Since one of the goals of the evaluation was to determine if participants could detect the presence of odors, some samples intentionally contained no odorant. In these cases, the system released clean, filtered air rather than remaining silent. As mentioned above, this was a critical design choice because it prevented participants from forming an unintended association between the sound of the device’s fan and the presence of an odor, thereby preserving the ecological validity of the task.

During each trial, participants were asked to smell the sample, state whether an odor was present or not, and if applicable, describe and then label the odor. After all three samples had been assessed, participants were asked to indicate what actions they would take as water-treatment technicians (e.g., whether additional treatment steps were required). Behind the scenes, the researcher/wizard triggered odor cartridges or clean-air release in real time to align with VR interactions.

Participants

A total of 23 students from Arizona State University, a large public university in the southwestern United States were recruited through mailing lists, online postings, and campus flyers. All participants were 18 years of age or older and completed a pre-screening questionnaire. Individuals with known olfactory impairments (e.g., allergies, nasal illness, recent head trauma) or susceptibility to motion sickness were excluded. Due to minor changes in the testing instrument after the first session, data from 22 participants were included in the final analyses.

Among the 22 usable participants (17 male, 5 female), the mean age was 23.5 years ($SD = 2.8$; range = 19–29). All were currently enrolled students, except for one who had recently completed a master’s degree. Participants had varied prior experience with VR: 22.73% reported no prior use, 27.27% had used VR once, 27.27% had used VR 2–3 times, 13.64% were occasional users, 4.55% were daily users, and 4.55% identified as power users.

To assess physical comfort and possible simulator-sickness effects, participants completed a 13-item symptom checklist before and after using the VR-O system. Items (e.g., nausea, dizziness, blurred vision) were rated on a 10-point Likert scale. A comparison of pre-exposure ($M = 15.61$, $SD = 2.20$) and post-exposure ($M = 14.72$, $SD = 1.49$) scores showed no significant change in symptoms, indicating that the system did not negatively affect participants’ wellness.

Results

A full paper containing detailed analyses is currently in preparation; here we report a brief summary of the core findings. To evaluate changes in participants’ ability to detect, identify, and respond to odors, we administered an 8-item pretest followed by an isomorphic 8-item posttest. Items assessed recognition of common water-treatment odorants and appropriate decision-making actions. A paired-samples t-test revealed a statistically significant improvement from pre-test ($M = 1.23$, $SD = 1.74$) to post-test ($M = 5.68$, $SD = 1.39$), $t(21) = -9.32$, $p < .001$, $d = -1.99$, indicating that participants performed substantially better after the VR-O intervention.

Interactive demo at ISLS

During the ISLS Interactive Tools and Demos session, attendees will experience a short olfactory-enhanced VR scenario and explore the system’s scent-delivery mechanisms. They will have the opportunity to examine the newest prototype of the VR-O device, which consolidates key design ideas from earlier iterations into a compact, portable unit. In this third-generation version, odorants are stored in subdivisions of a cylindrical chamber which is rotated to precise positions by a stepper motor; when a chamber is turned to the “open” position, its ports align to allow air to flow from an internal fan, over odorous sponges, and out through a TPU tube to deliver scent precisely to the user’s nose. An “open” position can deliver scent or a steady stream of clean, filtered air, which facilitates purging prior odorants and breaking the sensory association between airflow and scent delivery. In a “closed” position, the system seals off odorant chambers to prevent passive leakage and cross-contamination. Mounted via a backpack with types wrapping around the shoulder, the device routes the delivery tube along its straps to sit just beneath the user’s nose, enabling controlled and well-timed olfactory cues during the VR experience. Participants will be invited to discuss design considerations, technical choices, and pedagogical opportunities for using multisensory VR in instruction and research. This demo directly reflects the conference theme, Partnering with Purpose, by demonstrating how educational technologies can be developed through close collaboration with domain experts to address authentic learning needs. The VR-O system was created in partnership with water-quality professionals and aligns with concepts taught in current courses—particularly content that is difficult to convey in traditional classroom settings. By integrating expert knowledge with intentional technological design, the system exemplifies how research–practice partnerships can produce tools that enable meaningful, practice-grounded learning experiences.

Conclusion

This work demonstrates the potential of integrating olfaction into virtual reality to support more authentic, embodied, and practical learning experiences. By combining multisensory interaction with a portable, mechanically controlled scent-delivery device, the VR-O system offers a new avenue for teaching odor-based reasoning in domains such as water quality. The project addresses persistent classroom challenges (e.g., the difficulty of providing consistent, safe, and reusable odor samples) by enabling standardized, repeatable exposure to scents that are otherwise hard to access. At the same time, it tackles key engineering hurdles, including precise odor control, synchronization with VR events, and user comfort in portable systems. As VR continues to evolve, incorporating underutilized sensory modalities like smell can broaden the scope of experiential learning and create new opportunities for instruction, assessment, and research in complex, sensory-rich domains.

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Author contributions

Wylie led the writing of this submission. Lahey led the hardware design, with support from Dobos and Austin, and feedback from Likamwa. Likamwa led the software design, with additional support from Austin. Spackman and Wylie co-led the educational experimental design, with Islam leading the experimentation and Mannigel providing support. All authors read and approved the final version of the paper.

STEPS: An AI Tutor for Scaffolded Problem Solving in Introductory STEM Courses

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Abstract: Introductory STEM courses often reveal a disconnect between taught content and students' ability to solve complex, multistep problems. We present STEPS—an LLM-based tutor drawing on principles of deliberate practice—that scaffolds planning, step-by-step reasoning, and verification without giving away answers. Through iterative prompt engineering aligned with a validated problem-solving framework, STEPS was embedded into weekly assignments, quizzes, reflections, and lectures in large-enrollment chemistry and physics courses. We examined plan quality, solution correctness, student engagement, perceived helpfulness and confidence, and performance on targeted concepts. Results show high confidence, strong homework performance on STEPS-assisted problems, and increasing perceived helpfulness, with students shifting from answer-seeking to process-oriented inquiry and demonstrating growing AI literacy. This session will provide a brief overview of the framework, hands-on use of STEPS, and a guided workshop to craft authentically complex problems and translate them into STEPS-style prompts across STEM disciplines. We conclude with actionable guidance for pedagogically sound, learning-aligned AI integration in STEM courses.

Introduction

Chemistry and physics are among the most challenging introductory courses at the college level, and one of the frequently cited challenges in these courses is students' perceived disconnect between content covered in the course versus their ability to solve complex problems on homework and exams (Schwartz Poehlmann et al., 2024). Traditional supports (worked examples, office hours, etc.) provide little opportunity for scaffolded, deliberate practice of problem-solving process. On the other hand, while generative AI tools are widely available, their unstructured use for coursework can lead to answer-seeking, over-reliance, or inaccuracies that undermine learning and trust (Wang et al., 2025). Such unstructured use is especially problematic in the context of extensive research showing that productive struggles deepen students' learning (Kapur, 2016). In this work, we are addressing these two challenges by developing an AI tutor that explicitly scaffolds planning, step-by-step reasoning, and verification aligned with course goals, while embedded into coursework and assessments. Drawing on principles of deliberate practice, we developed and implemented an LLM-powered chatbot named STEPS, which stands for STEM Tutor for Effective Problem Solving, in order to scaffold students' problem-solving process and help them develop problem-solving expertise in introductory chemistry courses.

We developed STEPS by prompt-engineering an LLM using role definitions, behavior guidelines, and question sequences aligned with an established problem-solving framework in chemistry (Schwartz Poehlmann et al., 2024) and physics (Burkholder et al., 2020) to deliberately scaffold student problem-solving processes. We further piloted STEPS by embedding it into weekly assignments and complemented by quizzes with an individual planning phase followed by small-group discussion, homework reflections on exam performance where students identified a parallel problem and document problem-solving steps, and lectures that explicitly taught problem-solving processes. We measured how STEPS affects: (i) plan quality and adherence to a problem-solving framework, (ii) solution correctness, (iii) student question-asking behavior and engagement, (iv) students' perceived helpfulness and confidence, and (v) course performance on targeted concepts.

Theoretical framework

The deliberate practice approach emphasizes structured, goal-oriented learning that focuses on developing expertise through targeted practice and immediate feedback (Ericsson et al., 1993). Unlike rote memorization or passive study, deliberate practice requires learners to engage with challenges that stretch their current understanding, receive timely feedback, and refine their skills through repeated cycles of reflection and adjustment. In the context of college-level STEM education, this approach encourages students to move beyond surface-level comprehension toward deeper conceptual mastery, fostering habits of metacognition, persistence, and strategic problem-solving that are essential for long-term success in scientific disciplines. STEPS draws on the principles of deliberate practice to teach students problem solving, an essential skill in chemistry courses (Tsapalis, 2021), by: (i) dividing the skill into discrete subskills (problem solving → identifying relevant concepts, connecting relevant concepts to the given problem, identifying knowns and unknowns, etc.); (ii) designing learning activities suitable for learners (the instructor creates a problem based on recently covered content); (iii) providing specific, individualized, and timely feedback (students have access to STEPS when the problem set is released); and (iv) providing an opportunity to incorporate learned subskills (students are directed to solve the problem using the plan they wrote and to finish the rest of the problem set on their own). Learning activities with STEPS are embedded biweekly throughout the term, giving students regular opportunities to learn and practice problem solving.

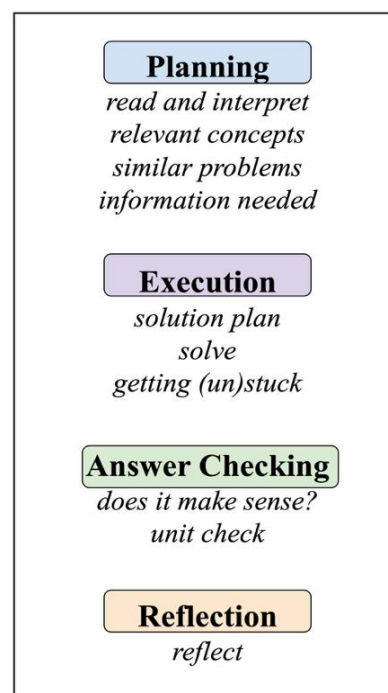
Design

In contrast to general-purpose AI tutors, STEPS is designed not to provide answers but rather to guide students to develop a step-by-step plan based on a validated problem-solving practices framework (Burkholder et al., 2020; Schwartz Poehlmann et al., 2024). This framework scaffolds the problem-solving process for physics (Burkholder et al., 2020) and chemistry (Schwartz Poehlmann et al., 2024) students using a template based on empirical studies of the problem-solving practices of expert scientists and engineers. Figure 1 outlines the problem-solving template in a chemistry context.

In order to get STEPS to guide students along the problem-solving process, we used best practices in prompt engineering along with key elements of the first two phases—planning and execution—of the problem-solving template. The prompt defines STEPS as a problem-solving coach that guides students through a structured question sequence: identifying relevant concepts, explaining their relevance, specifying needed details and equations, and creating a step-by-step plan. It also incorporates tutoring appropriate behavioral guidelines that prevent direct solution provision, maintain engagement, and redirect students back to the planning process when they seek complete answers or stray off task.

We partnered with teaching teams for introductory chemistry and physics courses to draft and refine the prompts for STEPS. Through weekly meetings, the teaching team and researchers: (i) created a real-world, complex problem requiring content and skills covered in the previous week's lectures; (ii) drafted, tested, and refined a prompt to support students through the problem-solving steps outlined in the problem-solving template (Figure 1); and (iii) discussed post-interaction survey results in aggregate to refine the prompt for the following implementation. Before deployment, our team conducted extensive testing by adopting different student personas, including answer-seeking students, students providing minimal responses, and engaged students offering detailed explanations, to evaluate the chatbot's behavior across varied interaction styles. Based on these tests, we refined further the behavioral guidelines and response patterns to handle different student approaches effectively. Figure 2 summarizes the iterative process used to develop prompts for STEPS. This collaboration between researchers and teaching teams ensured use of AI in the course aligned with learning goals and helped—rather than hindered (Wang et al., 2025)—student learning of not only course content but also AI literacy.

Figure 1
Outline of Problem-Solving Template.

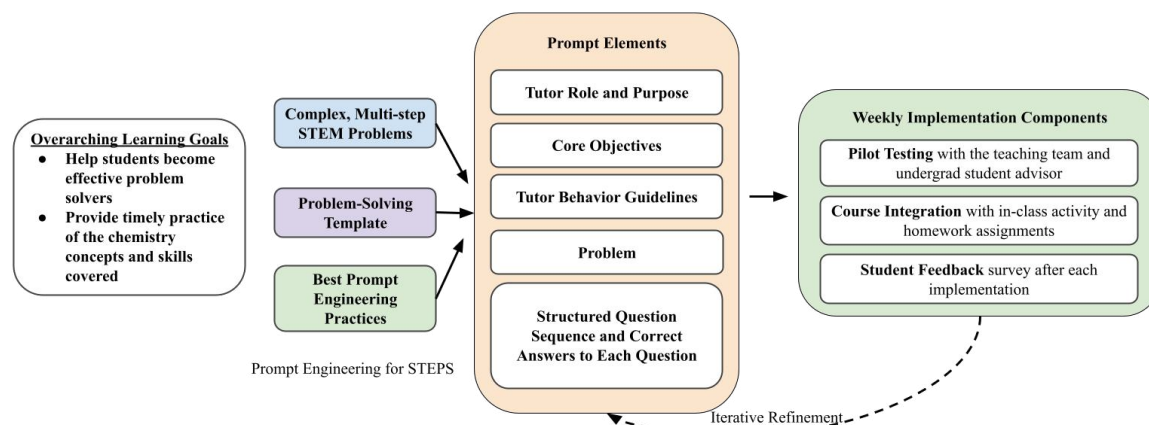


Implementation

While we deployed STEPS in several introductory chemistry and physics courses, in this present work, we focus on our initial deployment in an introductory chemistry course that taught foundational skills and concepts. Table 1 provides relevant details for this course. Students in the course interacted with STEPS on four separate occasions to solve a problem in a weekly problem set. Students were told to work with STEPS to develop a problem-solving plan for the assigned problem as a required part of the problem set. After developing a detailed plan to solve the problem, students solved the problem on their own (without STEPS) and submitted their work for assessment by the teaching team. After each interaction with STEPS, students took a post-interaction survey in which they uploaded the step-by-step plan they developed with STEPS, answered questions on perceived helpfulness of STEPS (Likert), confidence in their problem-solving plan (Likert), and general likes and dislikes (open response). Although interacting with STEPS was a required component of this course, students could opt in to have their data used for research purposes, and the study was approved by the Institutional Review Board (IRB#70070). For the four post-interaction surveys collected in this study, between 24 and 34 students consented per survey (out of 48 students total).

Figure 2

Iterative Process to Develop Prompts for STEPS.



Results

In the four post-interaction surveys collected during this study, students reported high confidence in their STEPS-supported problem-solving plans and achieved strong performance on three graded homework problems (see Table 2). Perceived helpfulness of STEPS increased across implementations, approaching “very helpful,” with students rating the tutor’s questions and feedback comparably. When asked what aspects of the interactions with STEPS were helpful in supporting their learning, students identified strength areas that aligned with STEPS’ pedagogical design: (i) providing guidance on how to solve the problem without directly giving solutions, (ii) asking questions to promote independent problem solving, (iii) providing helpful feedback and explanations, (iv) maintaining an encouraging and supportive tone, and (v) custom-designed for the course and chemistry problem solving. Open-ended responses were coded into thematic categories, and percentages reflect the proportion of consented students whose responses included each theme. Over the course of the term, students increasingly highlighted in their responses the value of ‘feedback/explanations’ (rising from 17% to 37% of consented students) and less often emphasized ‘guiding without solving’ (dropping from 42% to 15% of consented students), suggesting that the iterative prompt-engineering process produced improved prompts and that students were becoming more familiar with STEPS and its intended function.

When asked what aspects of the interactions with STEPS could be improved to support their problem solving, students most frequently requested clearer instructions/explanations (30%–41%) and fewer repetitive questions (7%–18%). Although prompt refinements reduced complaints about repetitive questioning by STEPS, these refinements resulted in an increase in students noting answer leakage (from 4% to 15%). Other improvement needs included more resources and examples, help extracting and organizing given values, and concern that an overly encouraging tone could mask errors. Post-course reflections from consented students indicate growing AI literacy: students assumed more agency, asked more and better-targeted process-oriented questions, and shifted from seeking solutions to planning their approach (e.g., identifying relevant concepts first). Their advice to future

users emphasized active engagement—brainstorming before asking, using STEPS to diagnose missteps, and asking follow-up ‘why’ questions—to deepen understanding rather than skip directly to answers.

Table 1
Course Details for STEPS Implementation.

Course	Introductory chemistry
Number of students	48
Number of weeks in term	10
Number of activities with STEPS	4
Number of conversations with STEPS	207
Average messages per chat (STD)	26.8 (17.9)

Table 2
Student Confidence and Solution Scores on STEPS-Assisted Problems.

Implementation number	Mean confidence (Likert)	Average score out of 5.00	N
1	3.92	-	24
2	4.29	4.71	34
3	4.39	4.80	29
4	4.26	4.59	27

Interactive session

This interactive session will begin with an overview of the problem-solving framework and how STEPS scaffolds problem-solving planning. Participants will then use STEPS on their own devices to work through a short, multistep problem and see the tutoring behaviors in action. Building on that experience, we will shift into a guided problem-creation workshop—participants will identify a target learning objective, articulate the core concepts and skills to assess, and craft an authentically complex real-world problem. The session will conclude with shared, actionable guidance for pedagogically sound AI integration in STEM courses.

Conclusions

Through this project, we demonstrate that STEPS, a prompt-engineered AI chatbot, grounded in a validated problem-solving practices framework and thoughtfully integrated into coursework, can help scaffold student problem-solving in introductory STEM courses. The successful design and implementation of the tool required interdisciplinary collaboration and iterative refinement and highlighted the importance of pedagogical expertise in LLM applications in educational contexts. In future implementations of STEPS, we will introduce best practices for maximizing its learning value based on this year’s data. We will also have instructors demonstrate effective ways of using the tool in class through expert modeling before having students use it independently. We are currently analyzing conversation histories to understand the types of interaction patterns that emerge among students and identify which interaction patterns lead to optimal learning outcomes. This analysis will help refine both the tool design and instructional approach for future implementations. Additionally, longitudinal studies are needed to assess whether the problem-solving skills and AI literacy students developed in these introductory STEM courses transfer across courses and contexts.

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“How Are You Solving That?”: Improving Learning in Virtual Reality With a Reflective AI Agent

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Abstract: Virtual reality (VR) environments provide rich contexts for learning, but learners often struggle to process deeper learning while using them. Reflective activities have been effective at increasing learning from VR, but few studies have engaged learners in reflection while in the learning environment. In this demonstration participants will use our bespoke VR escape room environment to demonstrate an AI-powered reflective agent who is modeled on escape room game masters. While solving logic-based puzzles, the reflective agent acts as a character in the escape room that encourages learners to discuss their problem-solving strategies. Participants in this demo will use the VR system and interact with the AI agent.

Introduction

Virtual reality (VR) is a technology that surrounds the user in a digital environment, immersing them with stereoscopic 3D images and sound that heighten their sense of “being there” in the virtual environment despite knowing where their physical body is (Slater, 2009). This increased sense of presence, along with heightening learners’ agency via full-body interactivity and a first-person perspective that allows for more “hands on” engagement, present affordances for learning that are distinct from other media (Dalgarno & Lee, 2010; Makransky & Petersen, 2021). However, studies of whether VR increases learning relative to less immersive media like video and text have found mixed results, with studies varying between negative, neutral, and positive effects that average out to a small positive advantage of VR (Wu et al., 2020). Recently, more research has looked beyond such a focus on the hardware to better understand the pedagogical designs within VR that support learning (Mayer et al., 2022). Further, there is a need for research to move beyond assessing only learning outcomes in terms of content retention and instead focus on higher-order skills and the process of learning in immersive digital environments.

We demonstrate a project that tackles the question of whether VR can be designed to support learning of higher-order skills by integrating metacognitive reflection as learners solve problems in the immersive environment. We built a VR escape room in which learners must solve logic-based puzzles to uncover the secrets of a prior society themed around gardening and healthy eating. As the learner solves the hands-on logic puzzles they are prompted by a character in the game, a spirit who holds the secrets to longevity and used to live in this house, to engage in metacognitive reflection during and after solving puzzles. This AI-powered system aims to improve upon prior examples of reflective activities that increased learning with immersive technologies but required learners to reflect outside of the VR environment. For example, studies have shown that segmenting a VR experience and asking learners to summarize what they have learned between segments (Parong & Mayer, 2018) or having them discuss what they learned with a peer after (Klingenberg et al., 2020) increased learning outcomes. Recently, some studies have found that discussions within the VR environment with peers is more effective than an individual reflection (Petersen et al., 2023). Our project looks to the potential of generative AI and large language models (LLMs) for encouraging such reflection without breaking immersion of the VR experience.

Related work

Reflection in virtual reality

Reflection in learning is an established activity that encourages learners to engage with learning of deeper structures beyond the surface features of a task. Several studies show that reflection improves learning outcomes in a variety of contexts, including in extended reality. Specifically in VR studies, reflection often takes place after the completion of the VR experience, and outside VR. For example, Parong & Mayer (2018) asked learners to remove their headset and write summaries after each segment, which increased learning and intrinsic motivation (Parong & Mayer, 2018). Klingenberg et al (2020) used post-VR processing activities like having learners teach what they just learned to another student, which increased learning retention, transfer, and self-efficacy. McGivney (2024) held reflective discussions with peers after using a VR experience to promote deeper learning.

As effective as these approaches are, they often require highly structured moments that occur outside of the VR experience, rather than promoting reflection while immersed in VR and engaged in the task, termed in-action reflection (Kleinman et al., 2025). However, recent research has highlighted the promise of encouraging people to reflect while using VR, as engaging with the immersive environment can be a highly generative activity (Wagener et al., 2023). One recent study found that learner reflection during a hands-on task in VR promoted higher-order thinking and engagement (Huang et al., 2024). Our AI-powered tool builds on this work to better embed reflective learning into the VR experience.

Current work: Design of a Reflective AI Agent for improving learning

Virtual reality learning environment

We developed an educational virtual reality escape room that utilizes puzzles to improve and measure logical thinking and problem-solving. The adoption of escape room as our learning environment draws from prior work by Kleinman et.al (2025), which recognized that escape rooms promote and require critical thinking and problem-solving skills comparable to traditional learning environments, and also have ‘game masters’ who facilitate events in the room but also provide hints or nudges to the players when they ask or become stuck in their progress. For this project, we created an AI agent based on the role and best practices of such game masters (Kleinman et al., 2025).

The single-player VR escape room was developed using Unity, the Meta XR CORE SDK 77.0.0, and the Open XR Plugin 1.14.0. Participants explore an abandoned house where they can access a kitchen and a greenhouse with five logic puzzles related to cooking and gardening that the participant must solve following a hub-and-spoke style: They solve four simpler (spoke) puzzles with simple logical problem-solving insights and unlock clues to the culminating “hub” puzzle that requiring more advanced logical insights. The spoke puzzles use hands-on interaction with game objects (watering plants, pouring sauce onto pasta), while the hub puzzle uses a more traditional grid interaction. Interaction is via hand tracking, not controllers, including gestures for teleporting beyond moving in the physical space allowed. Figure 1 depicts the layout of the escape room.

Reflection design

This reflection design addresses the challenges of reflection in VR by embedding it within the tasks at hand and allowing players to reflect during the problem-solving process as well as after. Players are prompted to reflect both in-action, while engaging in solving a puzzle. Figure 2 depicts an example of an AI-reflective prompt in the escape room.

The current version of the AI agent runs in the VR escape room via OpenAI’s API. It is not a hint-giving system, but a reflective dialogue agent designed to prompt metacognitive reasoning while being diegetic or embedded into the story. The agent adopts the persona of an elderly woman and maintains an emotionally neutral, grounded tone. It never breaks character, never references the system or environment, and never references itself. The interaction is a prompt system developed based on established practices by game masters (Kleinman et al., 2025). The prompts encourage the player to verbalize reasoning, evaluate their strategy, and articulate their assumptions about the puzzle (e.g., “What do you know that can help you solve the problem?”).

Demonstration

In this demonstration, participants will be able to try the VR escape room environment and interact with the AI agent. Typically, the entire escape room takes 20-30 minutes to complete, but for this demo we will offer participants a 5-10 minute experience. We will have 2-3 headsets available, and space permitting multiple people can use the demo at the same time. Typically, learners have freedom throughout the experience and must explore the space to discover the puzzles and objects needed to solve them. For this demonstration they will be directed to a puzzle, so they are able to begin solving it immediately and interact with the AI. Figure 3 illustrates the four “spoke” puzzles that participants can solve that utilize various interactions. Figure 4 illustrates a participant solving the “hub” puzzle being solved by a participant using a more traditional logic grid interaction. The AI is designed to prompt reflection based on the learners moves, such as idling, making irrelevant moves, or expressing frustration, and participants will be prompted based on these behaviors. Participants verbally respond to discuss their problem-solving process, and the AI will prompt them to elaborate when needed. Before and after they use the VR environment, participants can discuss the experience with the research team to learn more about the project and give feedback on the system.

Figure 1
A Map of the Escape room VR Environment.

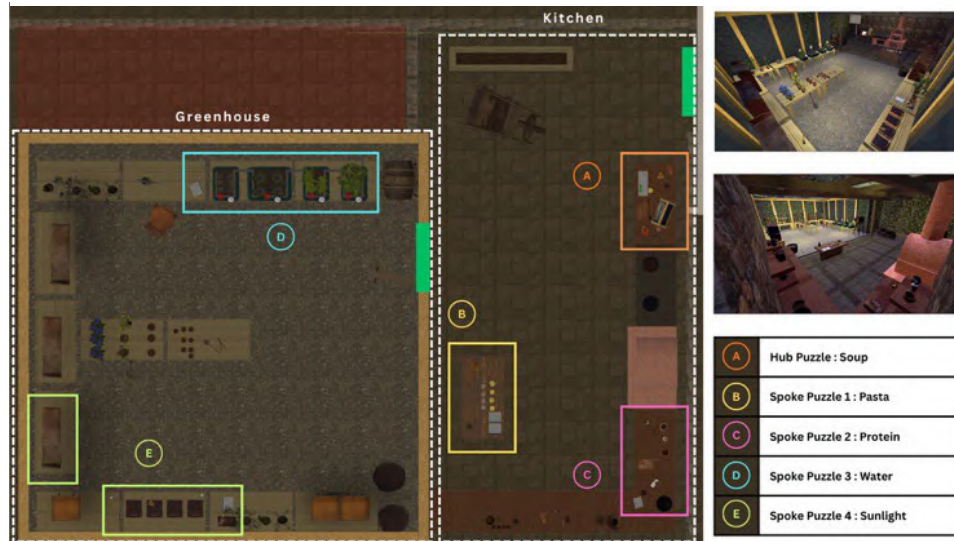
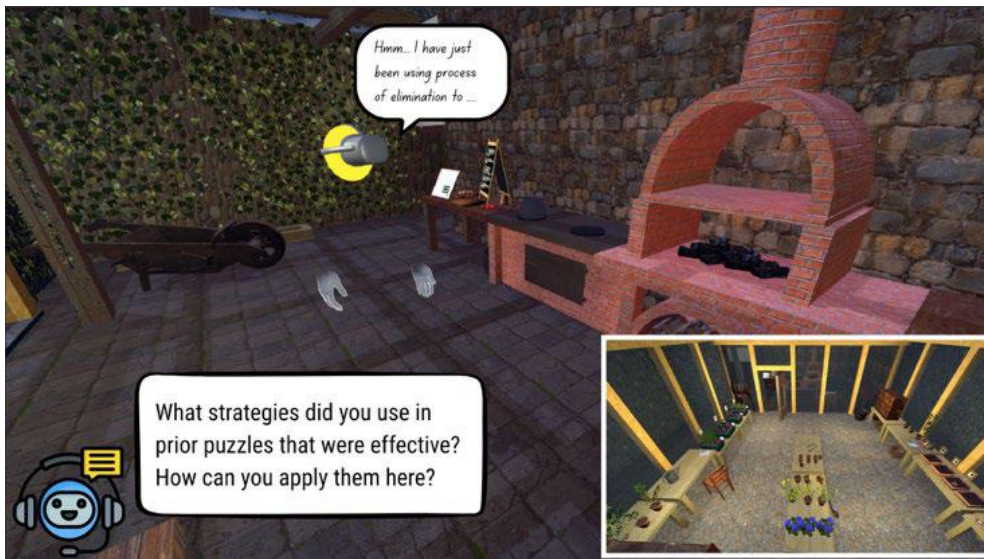


Figure 2
Example of Reflective Prompting in the VR Environment.

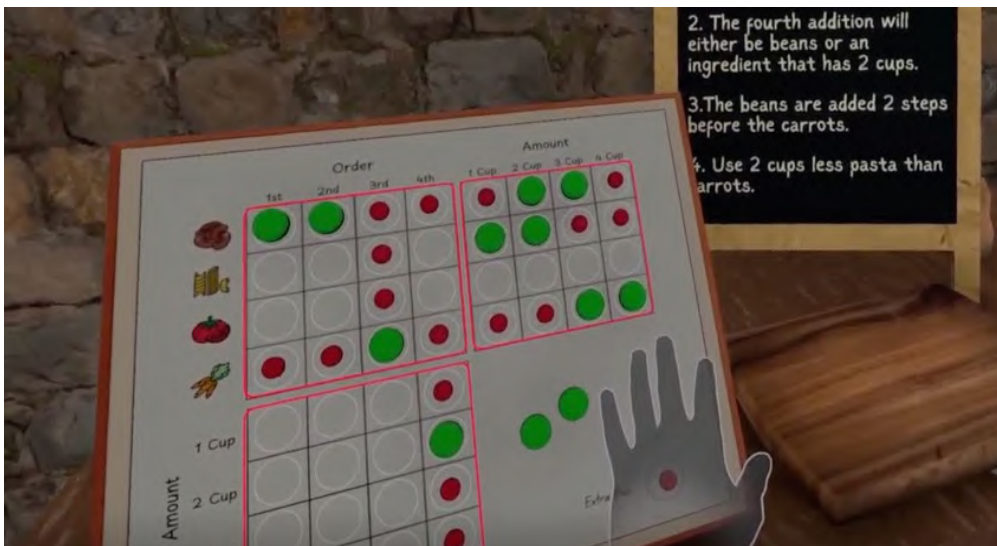


2

Figure 4
Four Spoke Puzzles Being Solved by a Participant.



Figure 3
Final Hub Puzzle in the Escape Room Being Solved by a Participant.



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Creating Learning Scaffolds for Engineering Design Using Concept Catalyst

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Abstract: K-12 teachers employ Engineering Design Challenges to help students learn about the Engineering Design Process hands-on. They use techniques like hard scaffolding questions to guide the students as they think through the different stages of the engineering design process. While useful, the creation of these questions adds to the teacher's preparation time for their classes. Concept Catalyst uses Large Language Models to assist teachers with the rapid creation of scaffold questions for engineering design challenges. Unlike open-ended chat, Concept Catalyst uses LLMs to summarize and decompose an engineering design challenge into the concepts that students will engage with, allow the teacher to visually manipulate and link related concepts, and to propose scaffolding questions for the teacher to modify or accept.

Introduction

K-12 engineering teachers help students learn about the Engineering Design Process (EDP) through design challenges, such as constructing a bridge or catapult. Engineering design challenges require students to break down a problem, think critically about it, develop an approach to address it, then iteratively implement, evaluate and redesign their approach. Writing documentation is a key practice used to teach the EDP—tables, graphs, diagrams, and other written references help students reflect on discussions, trade-offs between designs, and prototype failures (Mangold & Robinson, 2013).

To help students learn how to use documentation to iterate on their project solutions, K-12 engineering teachers use *scaffolding questions* (Douglas & Chiu, 2012). Soft scaffolding involves teachers dynamically offering support as they monitor students, while hard scaffolding involves anticipating and addressing students' needs ahead of time (Saye & Brush, 2002).

To create learning scaffolds, teachers draw on various teaching and professional experiences (Nadelson et al., 2014) to create and contextualize new curriculum and projects (Chiu et al., 2021). Reflective practices—in which teachers critically examine their teaching experiences and knowledge—are central to K-12 engineering teachers' iteration on classroom pedagogy. Reflective practices can improve teachers' understanding of engineering content (Radhakrishnan et al., 2021), discover strengths and weaknesses of their teaching (Mesutoglu & Baran, 2021), and revise lesson plans (Mesutoglu & Baran, 2021). However, creating scaffolds is both time intensive and challenging. Teachers must anticipate potential difficulties that students may face and tailor their curriculum to students' needs (Saye & Brush, 2002), adding to an intensive workload (Creagh et al., 2025).

Generative AI technologies hold the potential to support teacher activities, including lesson planning and assessment creation, thus decreasing teacher workload (Kasneci et al., 2023; Ravi et al., 2025). Teachers have expressed openness to using LLM-based tools designed to support project-based learning tasks (Ravi et al., 2025). While LLMs may not be reliable enough in a fully automated capacity (Schneider et al., 2025), they can be effectively used with teachers' oversight (Nagy et al., 2023), hence creating a collaborative system of teachers with AI that supports teacher agency and benefits both students and the teachers (Li et al., 2025; Yan et al., 2025).

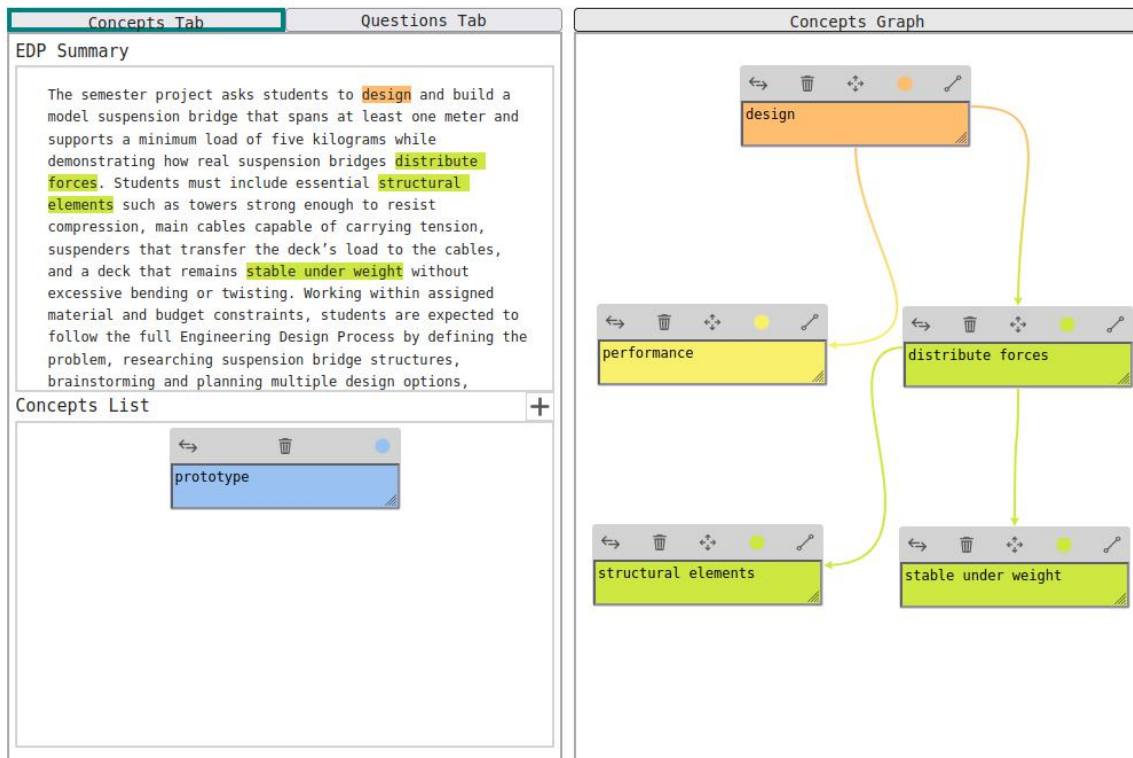
In this paper, we introduce an AI-based tool, Concept Catalyst, which helps teachers think through engineering design challenges using AI summarization, visualizations, and the synthesis of scaffolding questions once they have decided on which aspects of the problem to focus. Concept Catalyst is focused on helping teachers, not replacing them. The tool's design is based on prior work which introduced the framework to K-12 Engineering teachers in a Wizard-of-Oz study in which teachers reported positive feedback on the efficiency, quality, and diversity of content they wrote, explaining how it helped them more deeply engage with the EDP.

Concept catalyst

Concept Catalyst is a web-based tool that assists teachers in analyzing a design challenge and creating scaffolding questions for students. Concept Catalyst prompts teachers to highlight important concepts in paragraph summary of a design challenge, and then visually organize the concepts based on how they connect. Finally, teachers work with Concept Catalyst to synthesize scaffolding questions for their students.

Figure 1

A screenshot of the CONCEPTUALIZE step



The goal of Concept Catalyst is to reduce the amount of time taken in generating scaffolding questions. Concept Catalyst uses visualizations to support teachers' reflective practices with their curriculum while enabling them to easily leverage LLMs to generate scaffolding questions without the open-endedness of chat interfaces. Concept Catalyst avoids the issue of opacity of LLM-based decision making by keeping the decision making task entirely with the teacher, and employing the LLM as a generator of suggestions. However, we acknowledge that usage of this tool might still face structural issues, like the access of AI-supported tools being disproportionate between different socio-economic classes (Yan et al., 2025).

Concept Catalyst focuses teachers' attention on the Engineering Design Process (EDP) while keeping the LLM hidden behind an interface that guides teachers to create a concept map. Concept maps, knowledge graphs, and tree diagrams are all structured knowledge representations that can support the integration of new information with prior knowledge (Moreno & Mayer, 2007), a key aspect of reflection shown to help improve teacher practices (Ardito, 2022). Concept Catalyst uses teachers' concept maps to suggest scaffolding questions, which teachers can edit. Concept Catalyst is entirely teacher-facing and only outputs scaffold questions that can be printed and shared with students, who do not interact with the system.

Critically, Concept Catalyst places teachers in control of the scaffolds' content—helping teachers leverage the benefits of generative AI without removing their agency. Below, we detail how teachers work with Concept Catalyst through three stages: (1) Summarize, (2) Conceptualize and (3) Synthesize.

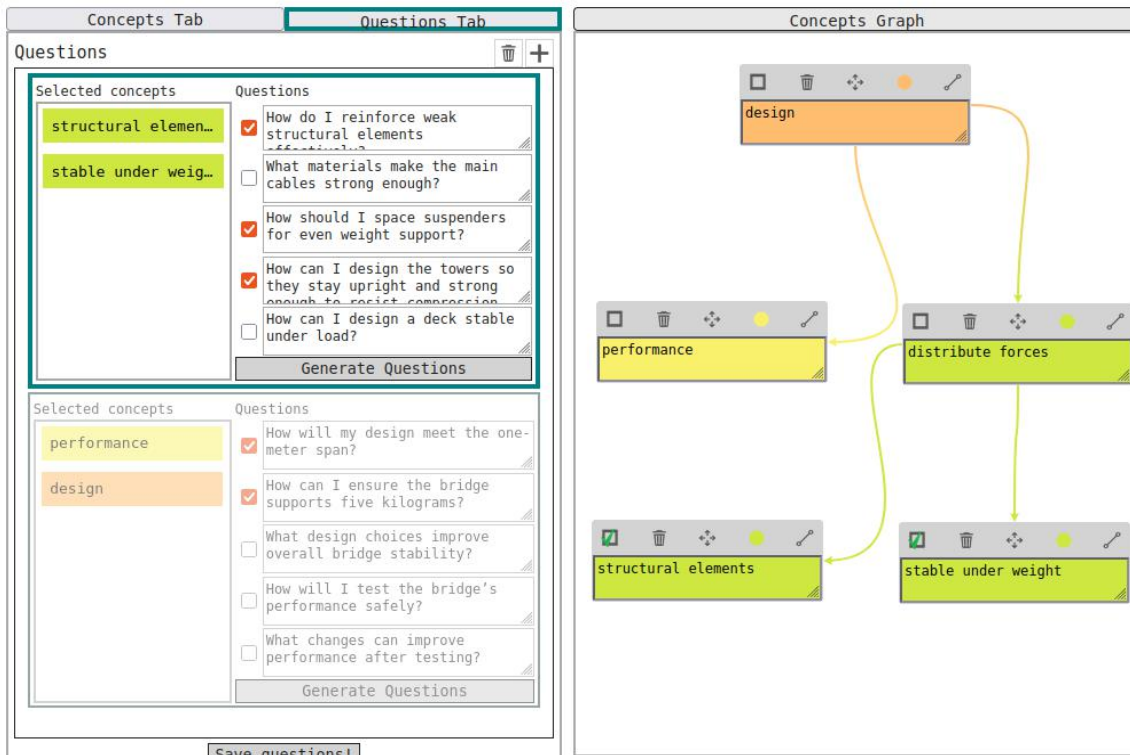
Summarize

In the SUMMARIZE step the teacher is prompted to input a paragraph summary of the design challenge or upload a unit map for the design challenge. If the teacher uploads a unit map, the tool processes it and displays a 200-word summary. Regardless, they can choose to modify it. Once the teacher is satisfied, they can move ahead by clicking on a 'Continue' button.

Conceptualize

In the CONCEPTUALIZE step, teachers are prompted to highlight key concepts that students should address in their design challenge and then visually connect the concepts that are related. The left half of the screen, titled

Figure 2
A screenshot of the *SYNTHESIZE* step



‘Concepts Tab’, holds two sections. The section on the top displays the summary that the teacher just approved. The bottom section, empty in the beginning, is a waiting area for the newly created concepts. The right half of the screen, titled ‘Concepts Graph’, is dedicated to a single section where teachers can move the concepts that they want to explore and connect them with lines.

The teacher begins by using their mouse to highlight any words or phrases in the summary that depict pertinent concepts. Every such highlight generates a concept-button in the waiting area. The teacher can also choose to create custom concepts by just clicking on the ‘+’ button. The teacher then moves the concept-button from the waiting area (lower left) to the graph area. The graph area provides two functionalities to the teacher: freely moving the concept-buttons around, and connecting them via lines. The teacher can explore the relationships between different concepts by moving the respective buttons closer or farther, upward, or downward, or by connecting them in a graph.

This visual exploration is aimed at providing the teacher with an avenue to apply their creativity and finding interesting concepts that inform the problem and the ways that they interplay.

The teacher can switch between these sections as many times as they need. Once they feel comfortable with the concept-graph, they can choose to switch to the Questions tab. See Figure 1 for a screenshot depicting the *CONCEPTUALIZE* step.

Synthesize

The *SYNTHESIZE* step is where the scaffolding questions are generated. This is the third screen of the tool, which shares its right half with the previous screen where the ‘Concepts Graph’ stays as it was left in the previous screen, while the left half of the screen, titled ‘Questions tab’, hosts question groups. The teacher attaches any number of concepts with a given question group by just clicking them. Each associated concept is previewed in the question group. Once the teacher deems the concept sufficient for a question group, they click the respective ‘Generate questions’ button. The tool generates five questions addressing the concepts that a student should consider while working on their project.

The teacher can generate any number of question groups with any combination of concepts they need. The teacher has complete freedom to accept, reject or modify the generated questions. The accepted questions are

added to a question bank. Once the teacher considers their work complete, they can end the process by clicking the 'Save questions' button which will give them a final preview of the approved questions and then print them to paper or a pdf document. See Figure 2 for a screenshot of this step.

Conclusions

Concept Catalyst is designed to reduce teachers' cognitive load while generating scaffolding questions for design challenges. At every single point of this interaction, the teacher can choose to accept, reject, or modify any artifacts created by the tool. The three stages—Summarize, Conceptualize and Synthesize—are not linear, allowing for an iterative flow that supports teachers' reflection of the problem as visualization of the concepts and their relationships gradually add clarity and suggest further possibilities.

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Show Your Work: A Gentle Introduction to Jupyter Notebooks for K-12 Educators

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Abstract: We are developing a suite of free, web-based, asynchronous introductory Jupyter notebooks designed for K-12 educators with little to no prior experience with programming or with computational notebooks. In contrast to typical “intro notebooks” which focus on technical features, the Show Your Work project builds on theory and findings from the learning sciences and science education to introduce computational notebooks as epistemic tools within specific school subject domains, guiding educators to explore the questions: What role can notebooks play in my specific classroom? What does it feel like to conduct an investigation using a Jupyter notebook? What epistemic and communicative practices are foregrounded and backgrounded by computational notebooks? These materials are likely of interest to ISLS members who work in computing education, computing-integrated STEM education, and/or teacher education, as well as to members who wish to learn more about computational narrative as a new area of focus within the learning sciences.

Motivation

Jupyter notebooks are a form of *computational notebook*, an artifact that allows multiple blocks of editable, executable code and its output to be interspersed with rich text in a single document in the spirit of Knuth’s “literate programming” (Knuth, 1984). The name, “Jupyter,” evokes three computer languages commonly associated especially with data analysis – Julia, Python, and R (Perez & Granger, 2015). Since its original release about a decade ago, Jupyter notebooks now support a wide variety of computational languages (Granger & Perez, 2021). While it is not the first example of computational notebooks, Jupyter has been widely adopted due to its availability as a free and open source tool, and the Jupyter infrastructure now forms the foundation of a variety of popular interactive computing environments such as Google Colab and Deepnote.

Although they were originally developed as a scientific research and communication tool, there is growing interest in using Jupyter notebooks at the K-12 level. Several research projects have studied the integration of Jupyter notebooks into secondary and intro college instructional materials (Fleischer et al., 2022; Son et al., 2021; Van Dusen, 2020). They are also becoming adopted as a curricular infrastructure including as part of Berkeley’s popular Data 8 “Introduction to Data Science” course activities (data6.org; data8.org), UCLA’s CourseKata high school mathematics curriculum (coursekata.org), and ProDaBi, a data/AI curricular package for secondary students (www.prodabi.de/en/). Most recently Jupyter Everywhere, a free, lightweight, simplified Jupyter interpreter interface that makes it easy for teachers and students to launch, edit, and share Jupyter notebooks without installing special software or services (jupytereverywhere.org).

Jupyter notebooks, as a popular example of a broader genre of computational notebooks, are likely to be of particular interest to the Learning Sciences community for theoretical reasons. In a recent paper, Odden and Zwickl (2025) described Jupyter notebooks as an example of diSessa’s notion of computational literacy: highlighting their integration of the *material* dimension of computational literacy through its support for explicit executable code, the *cognitive* (epistemic) dimension through users’s ability to test, revise, and keep note of the discoveries they make as they examine the output of their code, and the *social* (communicative) dimension through the ability to share this blend of code with text, images, and other explanatory media with others.

However, there is growing evidence that additional support is needed for Jupyter notebooks to be productively leveraged in K-12 classrooms. Most existing work describes a variety of technical and conceptual difficulties that teachers and students encounter when working with computational notebooks (e.g., Hanč et al., 2020; Johnson, 2020). While the potential for deep epistemic engagement through computational exploration and communication is high (Hüsing et al., 2024), without careful design or preparation notebooks are at risk of surface-level engagements in which students simply execute embedded code and read the output as if it were a printed text.

The need for teacher professional development

Beyond the need to address the technical challenges for teachers that are already well-documented in the literature, it is also important to offer pedagogical and epistemological support when introducing new epistemic tools in curricular materials. Ample work has demonstrated that introducing even “authentic” tools in secondary classrooms is only valuable to the extent that they are understood by teachers and students to support sensemaking (Stroupe et al., 2019). For example, students who are asked to use agent-based models to explore the particulate nature of matter are only successful insofar as they identify some compatibility between their representation of the phenomenon and the affordances of the tool (Wilkerson et al., 2018). Inspired from Russ’ (2014) distinction between epistemology “of” science versus epistemology “for” science, here we might similarly articulate computing not only as a practice “of,” but also “for” scientific reasoning.

Teachers play a critical role in helping students recognize the potential benefits of a given computational tool for advancing their understanding (Wagh et al., 2025). Thus to maintain disciplinary depth and support all students, teachers need to clearly understand the learning goals that a given activity or tool is meant to support, and be able to notice and respond to student thinking in service of those goals (Robertson et al., 2015). A first step to supporting such understandings is to examine how teachers, as learners themselves, might experience computational notebooks as a sensemaking tool.

Even if teachers do recognize the ways in which computational notebooks might function as an epistemic tool, there are still possible ambiguities regarding exactly what aspects of the analysis of complex datasets they can support. This is because data investigations are naturally interdisciplinary, and therefore their role within the curriculum can be ambiguous (Finzer, 2013). There are a number of paths an investigation can take. For example, in our own prior work, we found that teachers might emphasize dramatically different aspects of a data visualization, for instance, focusing on scientific, social, or mathematical learning goals (Wilkerson et al., 2025); a practice consistent with broader findings suggesting that teachers significantly shape how materials are enacted in their classroom (Remillard & Heck, 2014). Thus demonstrating how Jupyter notebooks or other computational narrative tools might integrate into a variety of disciplines, with a variety of potential (e.g. computational, statistical, or content-oriented) goals is consistent with what we know about teachers’ data integration practices.

Design principles

To address the need for epistemologically and pedagogically oriented teacher professional development experiences, we are developing (with a cohort of approximately 30 pre-service and in-service teachers, using a judicious informant design approach; Scaife et al., 1997) a suite of computational notebooks intended to form the basis of an interactive, self-paced introduction to Python notebooks for K-12 classroom educators. Our rationale, building on the preliminary findings from research described above, is that many practicing teachers are being introduced to notebooks through their integration into curriculum, without time to familiarize themselves with not only the technical but also and more importantly with the epistemological and pedagogical dimensions of these tools.

In a preliminary analysis of several existing “Introduction to Jupyter” notebooks developed both for practitioners and for educators, we identified an almost exclusive focus on their technical aspects. This included how a notebook can be run, edited, and what are some of its basic and advanced features. Several also included features that support pedagogical goals such as encouraging a tinkering disposition (“It is easy to break the code this way, but that is a normal part of coding!” from the YouCubed Jupyter introduction) or supporting the inquiry cycle (as demonstrated in CourseKata’s Jupyter introductory materials). Some existing introductory notebooks briefly addressed the epistemic role of notebooks in science, typically through appeals to authority.

Building on these initial findings, our sequence of notebooks intentionally foreground the pedagogical potential of notebooks as an epistemic tool, as grounded within use cases across a variety of school subjects. Each notebook is designed to explicitly communicate about eight principles we identified as desirable in our review of existing introductory notebooks including the *purpose* of leveraging a computational notebook for an activity; providing a *comparison* to other tools that might be used in such an activity; modeling and encouraging *documentation* both of code and of users’ thinking; describing the *planned use* of specific or unique features of the notebook to aid in analysis; supporting specific *code* learning goals; supporting specific *data* learning goals; supporting specific *content* learning goals, and encouraging productive *dispositions* toward notebook use.

Guided by these principles, our suite of notebooks provides teachers with repeated experiences in which they are expected to conduct inquiry using specific combinations of code, data, and content, and to encounter useful or unexpected findings in the process (see next section). As we make clear in our materials, our goal is not necessarily to advocate for the wholesale integration of notebooks into teachers’ work. Rather, as we state explicitly in our introductory notebook, we seek to “introduce [teachers] to what notebooks are, explore how they might be useful, and help [teachers] decide if they are right for [their] classroom.” (Intro Notebook).

The “Show Your Work” suite of notebooks

The “Show Your Work” sequence of activities is under active development, with five notebooks ready for sharing at the time of writing and an anticipated 10 ready by the time of the ISLS conference. When completed, the materials will include notebooks organized into four phases. Phase 1 provides a basic introduction and early data exploration experience using Jupyter notebooks. Phase 2 provides a variety of notebooks designed to introduce key code and data concepts that are most likely to be relevant to integrating Jupyter data explorations into particular school subjects such as chemistry, physics, statistics, and so on. For these first two phases, most materials are already well-developed and we are in the process of testing and refining them in collaboration with approximately 30 pre-service and in-service teachers.

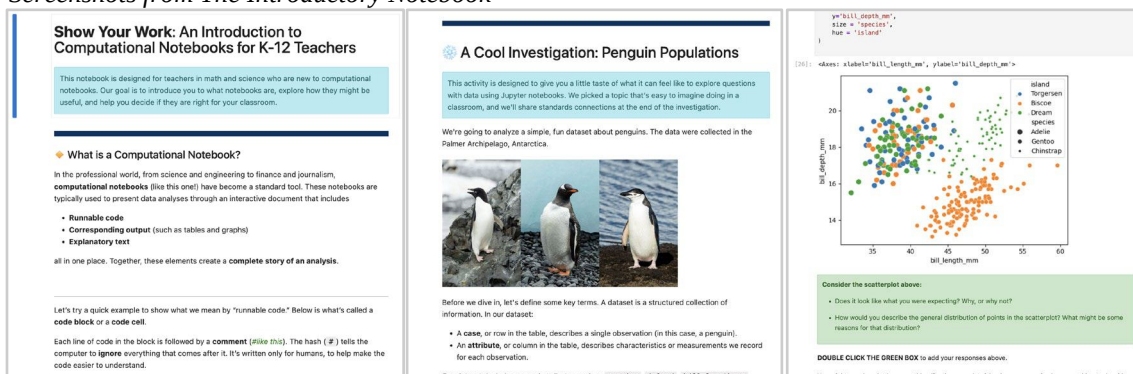
Phase 3 will include deeper dives into key computing and data concepts along with student case studies to encourage teacher noticing as mediated by Jupyter notebook interactions (e.g., Walkoe et al., 2017). Phase 4 will include materials to support teachers in adapting and creating their own notebooks for classroom use. This includes scaffolding their use of more traditional Jupyter documentation materials, and offering guidance on judicious use of generative AI tools, particularly those embedded in notebook development tools such as VSCode, whilst attending to the pedagogical and investigative principles introduced throughout the materials. These third and fourth phases are planned, though we anticipate that draft materials will be available for sharing by the time of the conference. Below, we elaborate on materials associated with Phase 1 and Phase 2 activities.

Phase I: Introducing notebooks as computational narratives

Our introductory notebook introduces teachers to the idea of a computational notebook as an “interactive lab notebook” in which they can keep, experiment with data and code, and communicate findings. The notebook guides teachers through the basic aspects of navigating, executing, and editing computational notebooks; diagnosing common errors and how to address them; and compares them to other epistemic tools used for data analysis in K-12 classrooms. Teachers then complete a data investigation activity using the well-known Palmer Penguins dataset (Horst et al., 2020), in which they encounter an unexpected non-linear relationship between bill depth and bill length in a population of penguins, and use visualization to explore what other factors (e.g. sex, species, mass, age, etc.) may help explain the irregular distribution (Figure 1).

Figure 1

Screenshots from The Introductory Notebook

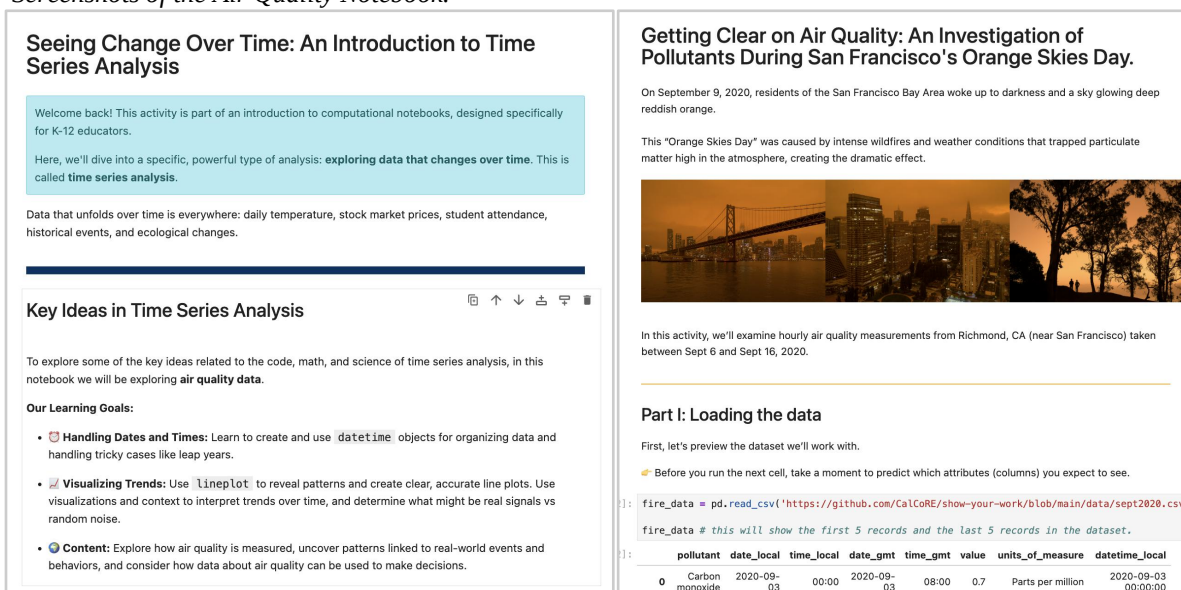


Including an introduction to notebooks (left), an inquiry activity (middle), and scatterplot code that users edit and supplement to address the inquiry question (right).

Phase II: Targeting key code and data concepts through “Topic Bundles”

The following set of notebooks focus on different disciplinary “bundles.” These are combinations of ideas from computing, data, and science that support specific kinds of subject matter explorations (such as GIS mapping to explore geological phenomena; exploring and visualizing basic descriptive statistics about distributions). Figure 2 features a notebook that introduces time series analysis through an exploration of air quality data as one such disciplinary “bundle.” The notebook introduces computational datetime as a specialised data structure, scaffolds users to generate and annotate lineplots, and to interpret time-series line plots with attention to scale, potential causes, and distinguishing between signal and noise. The notebook uses air quality data collected by local and national governmental organizations, and introduced related scientific concepts required to interpret patterns in the data (e.g. the Air Quality Index or AQI as a composite measure of air quality and PM2.5, Ozone, and Carbon Monoxide as specific air pollutants). through two interactive tasks.

Figure 2
Screenshots of the Air Quality Notebook.



The beginning segment introduces time series data and associated data structures and visualizations (left), and the major activity involves analyzing the different concentration patterns of three pollutants during a major fire event (right).

We are developing similar notebooks that connect to core topics in statistics, physics, engineering, and more (Table 1). For each, we work to identify one major phenomenon of interest for which patterns revealed through data and computing grant powerful insight into scientific mechanisms; 3-5 core code concepts that are required for the type of analysis that is introduced; and 3-5 data concepts related to the analysis.

Table 1
Completed, in Progress (), and Planned (^) Topic Bundles.*

Discipline	Phenomenon	Code Concepts	Data Concepts
Chemistry	Exploring patterns in air quality (AQI, Ozone, PM2.5, CO)	datetime lineplot	Time Series Data Scale Distinguishing Signal from Noise
Statistics	Typicality across contexts of test scores, housing prices	lists and dataframes histogram and boxplot	Measures of center, spread, and typicality
Physics*	Exploring force and motion	functions scatterplot with fit line	Linear regression Residuals
Engineering*	Optimizing the design of a cell phone	data filters	Optimization Filtering
English Language Arts^	Sentiment analysis of focal texts	strings frequency plots	Feature Extraction Bag of Words

Interactive demo plan and accessing “Show Your Work” notebooks

We look forward to possibly presenting this work at ISLS, which will allow us to learn from the community what might be other areas of need and to invite collaborators to contribute. By the time of the conference, we hope to have unified the notebooks into a single interactive digital interface that can be easily navigated, and we plan to have several laptops available for visitors to interact with the materials. Show Your Work project materials are being gradually released under a Creative Commons Attribution-NonCommercial 4.0 International (CC BY-NC 4.0) License, and some can be accessed at github.com/CalCoRE/show-your-work.

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TRAIL: A Human–AI Teaming Research Platform

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Abstract: TRAIL (Team Research and AI Integration Lab) is a research platform for studying human–AI teamwork and collaborative learning in postsecondary contexts. The system enables multi-condition experiments where human participants work together on a collaborative task with or without an AI teammate. AI teammate behaviors are configured through a personality interface grounded in the Big Five framework. TRAIL provides real-time team chat, automated team formation, integrated survey workflows, experiment chaining for longitudinal designs, and multi-level analytics. Its dual-memory architecture maintains coherent AI behavior across extended interactions, while rich datasets capture communication, trust, and team dynamics. This demo showcases how TRAIL functions as both a classroom and research tool, enabling controlled studies of collaboration, persona alignment, and group processes.

Introduction

TRAIL (Team Research and AI Integration Lab) is an experimental platform designed to support randomized controlled trials and structured classroom studies of human–AI collaboration built upon the framework proposed in Samadi et al. (2024). The platform enables researchers to manipulate key aspects of an AI teammate’s persona, including personality traits, backstory, communication style, domain expertise, and education level, allowing investigators to study how different configurations of human-like attributes influence collaborative processes. TRAIL provides a controlled environment where small teams work together on a collaborative task with or without an AI teammate, making it possible to examine how persona design shapes trust, coordination, and communication. A detailed account of results from our pilot study conducted on TRAIL, including analyses of socio-cognitive dynamics and comparative discourse patterns between human-only and human–AI teams, is presented in Choi et al. (2025).

AI technologies in education have traditionally been positioned as tools that automate feedback, provide explanations, or help students generate and refine ideas. However, recent research suggests that large language models are beginning to operate not only as tools but also as collaborators who participate in and shape group learning processes. Atchley et al. argue that tools like ChatGPT can serve as secondary collaborators by offering feedback, generating alternative perspectives, and supporting more productive student interactions during teamwork (Atchley et al., 2024). They further contend that learners must be prepared to collaborate with non-human agents whose reasoning and communication patterns differ from their own, noting that higher education will increasingly require collaboration skills that extend beyond human partners. This shift from AI as a tool to AI as a collaborator highlights the growing need for platforms that allow researchers to systematically study how AI partners influence trust, communication, and team dynamics.

The importance of persona in LLM-based agents has been underscored in recent work across human–AI interaction and agent modeling. Research shows that many AI systems still struggle to express human-like individuality, often producing generic or uniform patterns of communication that limit their appeal and effectiveness in social contexts. For example, Kruijssen and Emmons argue that AI agents frequently display bland and generic expressiveness and that personality expression is a key prerequisite for creating agents that feel more natural and engaging in human-facing applications. They further show that LLMs can express deterministic and consistent personalities when grounded in structured psychological frameworks such as the Big Five, supporting the development of more individualized and trustworthy AI teammates (Kruijssen & Emmons, 2025).

The Big Five model, also known as the Five Factor Model, characterizes personality along five dimensions: openness, conscientiousness, extraversion, agreeableness, and neuroticism (John & Srivastava, 1999). TRAIL uses this framework to parameterize AI teammate personas, allowing researchers to systematically vary personality traits and study their effects on team interaction. In related work, Samadi and Nixon (2026) examine how well LLM-based agents maintain assigned Big Five personality profiles during collaborative tasks, finding that alignment varies across traits and interaction contexts.

Across related research, persona grounding has been shown to influence the fidelity and stability of LLM-based agents. Park et al. demonstrate that generative agents behave more coherently when their underlying persona is constructed from rich, structured information rather than minimal demographic cues (Park et al., 2024). Their findings show that agents built from detailed interview data significantly outperform those built from demographic

or short persona descriptions in predicting human attitudes and behaviors. These results highlight that shallow persona conditioning can lead to weaker or stereotyped behavior, while richer persona scaffolds yield more accurate and stable identity modeling.

Together, these findings motivate the need for platforms that allow precise manipulation of persona elements in experimental settings. TRAIL addresses this gap by integrating personality-driven AI teammates into real-time collaborative environments, with interfaces that allow researchers to configure personality traits, backstory, linguistic style, domain expertise, and educational background. By pairing these configurations with controlled assignment across experimental conditions, TRAIL enables systematic investigation of how persona alignment, personality expression, and communication style shape human–AI collaboration. The platform has been deployed in both laboratory and undergraduate classroom settings, supporting research on trust formation, interpersonal dynamics, participation equity, and team performance.

Design goals and instructional needs

Learning sciences motivations

TRAIL addresses several challenges in studying hybrid collaboration. It allows researchers to compare human-only and hybrid teams across controlled experimental conditions, manipulate personality-driven AI behaviors, and capture structured data linking interaction patterns to perceived trust, inclusion, and effectiveness. The platform also supports instructors who want to embed human–AI teamwork into their courses.

Learning engineering perspective

TRAIL is grounded in learning engineering principles that emphasize the use of multi-source, authentic data for research and continuous improvement. Each session generates timestamped chat logs, participant identifiers, and pre and post survey responses that capture communication quality, trust, belonging, and demographic information. The AI’s long-term memory summaries provide interpretable traces of how the AI interprets team dynamics.

TRAIL computes behavioral and linguistic analytics using frameworks such as the Team Communication Toolkit (Hu, 2025) and Group Communication Analysis (Dowell et al., 2019). These tools identify sociocognitive roles, conversational dynamics, and participation patterns that reveal how teams coordinate and how AI personas shape team behavior. The platform supports both randomized experiments and longitudinal studies.

System architecture

TRAIL is built as a lightweight, web-based platform that supports real-time human–AI collaboration, randomized assignment, and multi-condition experimental design. The system consists of three integrated components: (1) an AI persona configuration interface, (2) a real-time collaboration environment, and (3) an analytics and data export module. These components are connected through a message-processing pipeline that batches incoming messages, integrates short-term and long-term memory, and generates personality-aligned AI responses.

At a high level, all participant messages sent through the chat interface pass through a batching and memory layer that integrates short-term conversational context with periodic long-term summaries. This design allows the AI teammate to maintain coherence over extended interactions without relying on full transcript replay. When appropriate, the system generates a personality-aligned response based on the configured persona, the current conversation state, and the team’s progress on the collaborative task. Notably, the AI does not respond to every message. Before generating a response, the system evaluates whether participation is appropriate based on contribution potential, conversational flow, and team dynamics, ensuring the AI does not disrupt natural human-to-human interaction. Responses are returned instantly through WebSocket communication, supporting fluid interaction across human-only and human–AI teams.

TRAIL’s backend stores complete chat histories, reflective AI memories, researcher-defined conditions, and survey data, enabling controlled studies with precise experimental assignment. Researchers can export raw or derived data for external analysis or use the built-in analytics dashboard to compare team participation, communication patterns, and condition-level outcomes. By combining a configurable AI agent, a real-time interaction environment, and structured data outputs, the architecture supports both classroom deployment and rigorous human–AI teaming research.

TRAIL also supports experiment chaining for longitudinal study designs. Researchers can link multiple experiments into ordered sequences where team compositions, group assignments, and participant nicknames persist across sessions. Each experiment in a chain can use different AI configurations, enabling within-subjects designs such as a baseline session without AI followed by sessions with varying AI personas. This capability

supports research on how sustained exposure to different AI teammates shapes trust, communication norms, and role distribution over time.

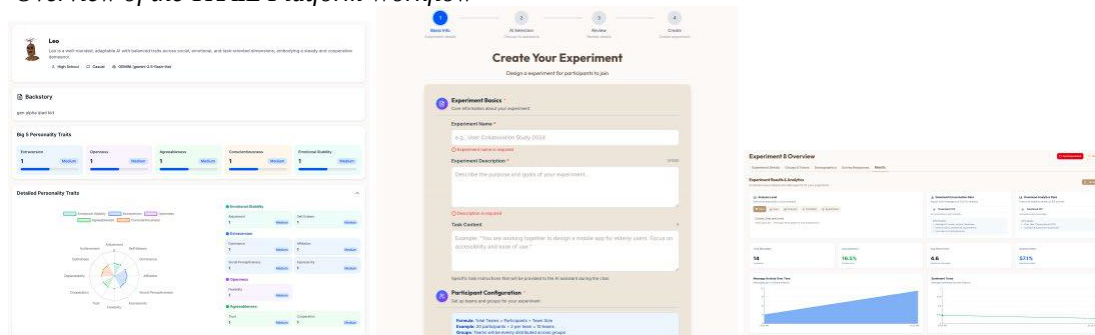
Use cases

Collaborative teamwork among postsecondary learners

TRAIL supports small teams working together on a collaborative task with or without an AI teammate. This setup enables investigation of human–AI collaboration dynamics in realistic team settings. Researchers can examine how students negotiate meaning, coordinate actions, and adapt communication strategies in the presence of different AI personalities. TRAIL’s analytics reveal patterns in participation, persona alignment, co-construction, and emergent sociocognitive roles across conditions (figure 1).

Figure 1

Overview of the TRAIL Platform Workflow



Overview of the TRAIL platform workflow, illustrating the three core stages of researcher interaction: (1) configuring AI teammate via the persona (2) configuring a collaborative task and experimental setup, and (3) accessing experiment-, team-, and user-level analytics for research and instructional use.

Studying trust and interpersonal communication in hybrid teams

TRAIL allows controlled manipulation of AI persona characteristics that are difficult to replicate with human confederates. Teams interact with AI partners that vary in warmth, assertiveness, initiative, or communication style. This supports research on trust formation, psychological safety, and adaptation to AI behaviors. The dual-memory system enhances interpretability by allowing the AI to reference earlier interactions and articulate reasoning processes.

Investigating personality-driven effects on coordination and role distribution

TRAIL’s persona and personality configuration tools allow researchers to examine how different AI teammate profiles influence coordination, leadership emergence, and role negotiation. A dominant persona may guide planning, while an affiliative persona may encourage consensus-building. TRAIL’s behavioral and linguistic data support analysis of how AI personalities influence team cohesion, communication flow, and sociocognitive role distribution.

Demo format and interaction plan

This interactive demo will guide attendees through the core components of TRAIL and allow them to experience the platform from both the researcher and participant perspectives. The session is structured around three short segments that mirror the lifecycle of a TRAIL study: configuring an AI teammate, participating in a live collaborative activity, and exploring automatically generated analytics.

Segment 1. Researcher view: Configuring an AI teammate

In the first segment, the demonstrator will introduce TRAIL’s researcher interface and show how AI teammates are created and customized. Attendees will see how personality traits, backstory, communication style, domain expertise, and education level can be configured to create expressive and human-like personas. The facilitator will walk through the process of setting up an experimental condition, including team size, timing, and random assignment to human-only or human–AI groups. This mirrors the setup process used in actual TRAIL research studies and highlights how the platform supports controlled experimentation.

Segment 2. Participant view: Live human–AI team interaction

Next, attendees will participate in or observe a short collaborative activity inside TRAIL’s real-time chat environment. Two parallel teams will be shown: one human-only and one working with a configured AI teammate. This side-by-side view allows attendees to compare how AI personas influence team interaction, initiative, and communication flow. The facilitator will briefly expose TRAIL’s dual-memory system by displaying how the AI generates and updates its reflective long-term memories, offering an interpretable look into how the AI tracks the team’s progress and dynamics.

Segment 3. Researcher view: Analytics and data capabilities

The final segment focuses on how TRAIL captures and organizes data for research and instruction. Using a pre-run dataset, the demonstrator will showcase TRAIL’s multi-level analytics: experiment-level summaries, condition comparisons, team-level measures, and individual participation indicators. Attendees will see how researchers can export chat logs, survey responses, and derived metrics for analysis with external frameworks such as the Team Communication Toolkit or Group Communication Analysis. This segment highlights how TRAIL functions as a research infrastructure for studying trust, communication, and socio-cognitive dynamics in human–AI teams.

Closing and discussion

The demo will conclude with an open discussion of use cases in classroom settings and research contexts. Attendees will be invited to explore potential collaborations, share ideas for integrating TRAIL into their own studies, and ask questions about implementation or experimental design.

Technical architecture

TRAIL uses a microservices architecture with a Django backend, Flask–SocketIO chat server, PostgreSQL, Redis caching, and a Next.js frontend. Components are containerized with Docker for scalability and cloud deployment. More information about the platform is available at <https://the-language-and-learning-analytics-lab.github.io/TRAIL-landing-page/>.

Discussion and future directions

TRAIL offers a flexible and extensible environment for studying human–AI collaboration, providing capabilities that can directly benefit the learning sciences community. First, TRAIL supports controlled experimentation with AI teammates at a level of precision that is difficult to achieve in classroom settings or commercial platforms. By allowing researchers to manipulate persona attributes, team configurations, and experimental conditions, TRAIL creates opportunities to investigate questions about trust, communication, and socio-cognitive dynamics in mixed human–AI teams. These affordances can help the community generate new empirical insights into how AI systems shape collaborative learning processes and how design choices influence learner experience.

Second, TRAIL lowers the barrier for conducting large-scale, multi-condition studies by automating random assignment, data collection, and workflow management. This makes it feasible for researchers to run replicable studies across institutions, contributing to the growing need for robust, cross-context evidence in human–AI interaction research. The platform’s structured data outputs, including chat logs, AI reflective memories, and derived communication metrics, also provide rich resources for methodological innovation in discourse analysis, multimodal learning analytics, and team interaction modeling.

Third, TRAIL has the potential to support instructional innovation. As AI becomes more embedded in collaborative work, educators need tools to help students practice communication, coordination, and critical thinking in settings that include AI partners. TRAIL provides a safe, configurable environment for giving students these experiences while allowing instructors to study how learners engage with AI personas and how team dynamics evolve when non-human agents are present.

Looking ahead, future development will focus on expanding TRAIL’s analytics capabilities, integrating additional modalities such as audio or physiological signals, and enabling more advanced forms of adaptive AI behavior. We also aim to cultivate a community of researchers and instructors who can share experimental templates, persona configurations, and datasets. By supporting collaborative development and data sharing, TRAIL can serve as a research infrastructure that advances collective understanding of human–AI teaming and contributes to new methodological and pedagogical directions in the learning sciences.

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Learning Black-GPT: Demonstrating the Creative Workflows of Black Artists Using Generative AI Tools

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Abstract: This demonstration presents an investigation of the creative workflows Black artists use when producing multimodal art with generative AI systems. Through screen-captured think-aloud protocols with three artists, we highlight routines that reveal culturally grounded practices rooted in hip-hop, Black Arts Movement, and broader African-American artistic lineages. The project argues for pluralistic AI design—an approach that recognizes multiple cultural repertoires and aesthetic traditions as foundational to AI system development and use. The session features edited workflow recordings and a live audience-guided demo that together illustrate how Black cultural knowledge, sociotechnical workflows, and ethical considerations shape AI-supported creative production and offer valuable insights for AI literacy learning environments that are culturally sustaining.

Background and problem space

AI tools are not neutral; they are designed through particular ethical and epistemological frameworks. As OpenAI CEO Sam Altman (2025) noted in an interview with Tucker Carlson, OpenAI's goal is not to make ChatGPT act like a human, but rather to reflect "the collective of all humanity." He argued that it is important that AI thinks pluralistically, and does not just represent one set of parochial values. At the same time, the ChatGPT model spec is a publicly shared document outlining the rules for behavior for Open AI models, and it is far from the value neutral goal Altman is hoping for. The rules and values in the model spec reflect the priorities and biases of its authors. This critique has longstanding precedent in the work of researchers such as Ruha Benjamin (2019, 2024), who point out algorithmic biases within AI systems.

The field has understandably focused on algorithmic bias, but the most promising new directions in learning sciences are not only critical—they are also imaginative. Tanksley (2024) advocates for "critical race algorithmic literacies" as tools to reimagine algorithms for liberatory purposes. Lizáraga (2023) likewise documents how communities of color enact "cyborg sociopolitical reconfigurations" through creative engagement with emerging technologies. We also draw on Saidiyah Hartman's (2008) concept of critical fabulation, which invites us to imaginatively fill in the silences and erasures in archives. Generative AI becomes a tool to produce visual, sonic, and textual counterhistories that "unforget" the past and imagine the yet unborn.

This shows the importance of investigating how pluralism is or is not encoded into AI systems and how various diverse communities engage with those systems in a multiplicity of ways. Users prefer different functionalities based on their cultural contexts. Specifically, AI artists engage with functionality based on the cultural histories of artmaking they emerge from. For example, African-American creatives might desire algorithms and routines informed by the aesthetics and ethos of African-American art history, including the Harlem Renaissance, the Black Arts Movement, and hip-hop. By studying AI creativity within Black expressive traditions, this project contributes to a pluralistic understanding of AI—one that recognizes how cultural contexts shape the ways people think, learn, and innovate with intelligent technologies. By examining the workflows of Black AI creatives, we deepen understanding of how communities innovate within, adapt, and reshape sociotechnical infrastructures.

Conceptual framework

Generative AI has particular potential to revolutionize *arts and creative education* (Zhou & Lee, 2024). Artists who use these systems engage in forms of problem-solving, pattern recognition, and reflective abstraction that parallel higher-order thinking skills central to human cognition. These artists thus embody *habits of mind* that are of keen interest to cognitive scientists and educators. However, previous studies of AI artistry have largely overlooked how cultural traditions—with their unique ways of making meaning, improvising, and reinterpreting—shape these human-AI interactions.

This study addresses that gap by focusing on Black AI creatives, who draw from African and African American lineages of expressive innovation. Prominent educational researchers focused on minoritized learners have long looked to Black activists as exemplars of the intellectual dispositions schools should nurture. Carol Lee's (2004, 2008) theory of cultural modeling draws from Toni Morrison's literary genius to design pedagogy that leverages the deep interpretive practices embedded in Black linguistic and expressive traditions. Similarly, scholars of hip-hop pedagogy (e.g., Alim, 2011; Love, 2019; Nzinga, 2023) have turned to hip-hop artists and

their cultural practices as models for culturally sustaining curricula. These approaches share the belief that Black cultural production is not only artistically rich, but intellectually generative—that the cognitive habits required to make resonant, socially conscious art are themselves pedagogical gold. Black AI creatives represent a new generation of artists whose habits of mind act as informative mental models for what should be nurtured in educational settings. This study therefore draws upon methods within the learning sciences of studying expertise (Chi et al., 2014) in which researchers try to develop a cognitive map of experts within a certain domain, in order to inform how this particular skill should be cultivated in novices.

The work of Black AI creatives exemplifies what comedian Karlous Miller (2025) called *Black-GPT*—a movement of artists and technologists using generative AI to extend the archive of the Black radical imagination. The participants in this study will be prominent artists within the Black-GPT milieu, including a few that our team has previously collaborated with:

- **Jidenna**, a Grammy award winning music performer who has produced *Oga City*, a generative AI film series drawing from his Nigerian-American heritage, using tools such as Kling, Veo, RunwayAI, and Higgsfield.
- **Jelie**, a multi-genre music producer integrating AI tools like Playbeat 4 as a generative drum sequencer, exploring how machine improvisation intersects with hip-hop production.
- **Kalonji Nzinga**, the project’s principal investigator, who has used Midjourney, OpenArt, and Veo to produce video works reimagining African and African-American mythologies (such as the resurrection of Osiris) as symbolic music videos for his hip-hop compositions.

These artists illustrate how generative AI can be used to remix cultural archives, visualize ancestral narratives, and prototype new aesthetic grammars. Their creative processes reveal cognitive patterns of improvisation, call-and-response prompting, and mythopoetic reasoning—habits that are simultaneously artistic, intellectual, and educational.

Methods: cognitive think-alouds and sociotechnical workflow mapping

We conducted screen-captured cognitive think-aloud sessions with three Black artists as they produced multimodal art using text, image, video, and audio-based generative AI systems. Artists narrated their decisions while referencing historical works, iterating prompts, adjusting parameters, and combining tools into multimodal production pipelines. Our methodological focus was on the workflow itself as a sociotechnical unit of analysis—capturing how cultural knowledge, technical skill, and improvisation interact across tools and modalities. Analysis of these cognitive think alouds surface tacit strategies often overlooked in AI and creativity research.

Figure 1
Generative AI Art Produced by Study Participants



(left) Screenshot of AI creative Jelie Jones explaining how she uses Playbeat 4 as an AI tool to generate drum sequences for House music. (right) Screenshot from Grammy award winning artist Jidenna’s film series *Oga City*, for which he used several AI image and video generation tools towards speculative fiction storylines.

Demo description

The demonstration will feature a curated set of short screen-recorded workflow clips from think-aloud sessions with two Black AI creatives: Jelie Jones and Jidenna. These clips reveal how artists integrate cultural repertoires, genre-specific aesthetic criteria, and improvisational decision-making into their use of generative AI systems. In Jelie’s workflows, we show how she uses Playbeat 4, an AI-powered rhythm generator, to produce House and Afro-House drum sequences. Participants will see how she employs Playbeat 4’s probabilistic pattern engine,

rhythmic mutation tools, and micro-timing features to explore alternative grooves, refine density and swing, and integrate patterns into Ableton. Her narration illuminates how Black dance-music traditions guide her evaluation of AI-generated rhythms and how she uses the tool's affordances to support improvisation and remix.

Additional clips highlight Jidenna's creative process for his speculative fiction project *Oga City*, in which he uses Midjourney and Runway to develop visual storyworlds. The recordings illustrate how he grounds prompts in diasporic aesthetic references—such as Nigerian photography, Afro-surrealism, and mythic symbolism—and how he chooses between Midjourney's high-fidelity still-image generation and Runway's motion tools for animating scenes, creating atmospheric transitions, and refining cinematic compositions. These clips demonstrate how the affordances of image and video models (e.g., stylization, temporal coherence, motion generation) interact with culturally grounded worldbuilding, and how artists negotiate representational bias, genre tropes, and alignment with Black speculative visual languages.

Following the recorded workflows, the session will include a brief live demonstration with audience participation, enabling attendees to experience these routines as learnable practices. Participants will propose cultural themes, aesthetic constraints, or musical/visual references, and the presenters will model the creative sequencing observed in the recordings—constructing prompts, grounding outputs in cultural knowledge, iterating across tools, and assembling multimodal artifacts. This live component illustrates how Black AI creative workflows can be adapted into participatory, culturally sustaining learning environments, while making visible the pluralistic design principles needed to support diverse artistic repertoires.

Findings: workflows and habits of mind of Black AI creatives

Analysis of the workflows revealed several recurring routines: (1) Cultural Referencing—artists consult Black artistic canons, including Harlem Renaissance photography, hip-hop production, and diasporic visual motifs; (2) Iterative Prompt Construction—artists refine prompts based on both aesthetic intuition and cultural grounding; (3) Modal Switching—artists move fluidly across image, video, and music generation to build layered artifacts; (4) Aesthetic Calibration—artists use genre-specific criteria rooted in Black expressive traditions; (5) Ethical Awareness—artists critique representational biases and negotiate problematic outputs; and (6) Remix Logic—artists treat AI outputs as raw material for creative transformation. These routines will be demonstrated in the exhibition through interactive audiovisual displays of focal artists' creative process.

Implications for learning sciences

This work contributes to understanding culturally sustaining AI literacy. By foregrounding Black creative workflows, we illustrate how AI systems can become generative spaces for metacognition, creative inquiry, and cultural modeling. The findings demonstrate the importance of grounding AI learning environments in pluralistic design principles and community cultural wealth. These findings elucidate design principles that inform the building of 1) generative AI tools 2) the creative workflows that incorporate these tools and 3) the learning environments that cultivate AI literacy. The workflows and habits of mind of diverse AI artists can be informative in mapping learning outcomes for culturally sustaining AI literacy.

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LumiBuilder: A Chatbot-Building Tool for Hallucination-Aware AI Literacy

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Abstract: Hallucinations, or confident but incorrect responses generated by large language models, present a critical challenge for young learners who increasingly turn to AI for schoolwork and everyday tasks. LumiBuilder is a web-based chatbot building tool designed to help children understand these limitations by constructing their own chatbots and testing how design choices influence model behavior. The tool integrates hallucination awareness scaffolds, including confidence displays, fact checking, model comparison, document verification, and repeated questioning, to support inquiry-based exploration of AI reliability. This interactive demonstration introduces LumiBuilder's core features and the design principles that shaped its development. Participants will explore the tool, observe how hallucinations emerge through varied prompts, and discuss implications for AI literacy, youth and AI interaction, and the design of learning environments that promote verification, reflection, and calibrated trust in AI systems.

Introduction

The rapid integration of generative AI into educational practice underscores a pressing need for young learners to develop robust mental models of how large language models (LLMs) operate, including their limitations, uncertainty, and error modes (Long & Magerko, 2020; Lee et al., 2021). *Hallucinations*, outputs that appear fluent and authoritative yet are factually incorrect, pose particular challenges because they can distort reasoning, undermine trust, and often go undetected by novice users (Ji et al., 2023). Most chatbot interfaces obscure the underlying mechanics of generative models, offering few opportunities for learners to investigate why errors occur or how to verify AI-generated information (Ng et al., 2021).

LumiBuilder was developed to address these challenges by shifting learners from passive AI use to active AI construction. Inspired by constructionist traditions that emphasize learning through building and debugging (Papert, 1980; Resnick, 2017), LumiBuilder engages learners in designing their own AI agents, specifying constraints, testing outputs, and iteratively refining their systems. Through this process, hallucinations become objects of inquiry rather than sources of confusion. The LumiBuilder sits within a larger AI literacy effort aimed at helping learners develop verification strategies, cultivate epistemic vigilance, and understand when and how to trust AI systems (Ng et al., 2021; Long & Magerko, 2020). By making AI behavior transparent and manipulable, the tool supports learners in building calibrated trust. This includes knowing when AI outputs are reliable and when additional verification is needed. This interactive demonstration showcases how design-based approaches can foster critical AI literacy skills that are essential for navigating an AI-mediated world. Moreover, the verification strategies LumiBuilder cultivates, such as source evaluation, cross-referencing, and evidence-based reasoning, may extend beyond AI-specific contexts. As learners increasingly encounter misinformation across social media, search engines, and AI-generated content, these practices could serve as transferable epistemic skills for navigating a broader information landscape. Future research should explore how hallucination-awareness scaffolds translate to other digital literacy contexts.

Related work

Efforts to define AI literacy have emphasized the need for learners to understand both what AI systems can do and what they cannot (Lee et al., 2021). Key competencies include recognizing uncertainty, probing system behavior, and understanding error sources. Research consistently shows that students often misinterpret AI fluency as correctness and lack systematic strategies to validate output (Leiser et al., 2024). Constructionism provides a framework for fostering deep understanding through the creation and debugging of computational artifacts (Papert, 1980; Kafai et al., 2020). Building on these foundations, the LumiBuilder enables students to author their own chatbots by defining purpose, personality, rules, and knowledge boundaries. Students experience firsthand how design choices shape AI behavior.

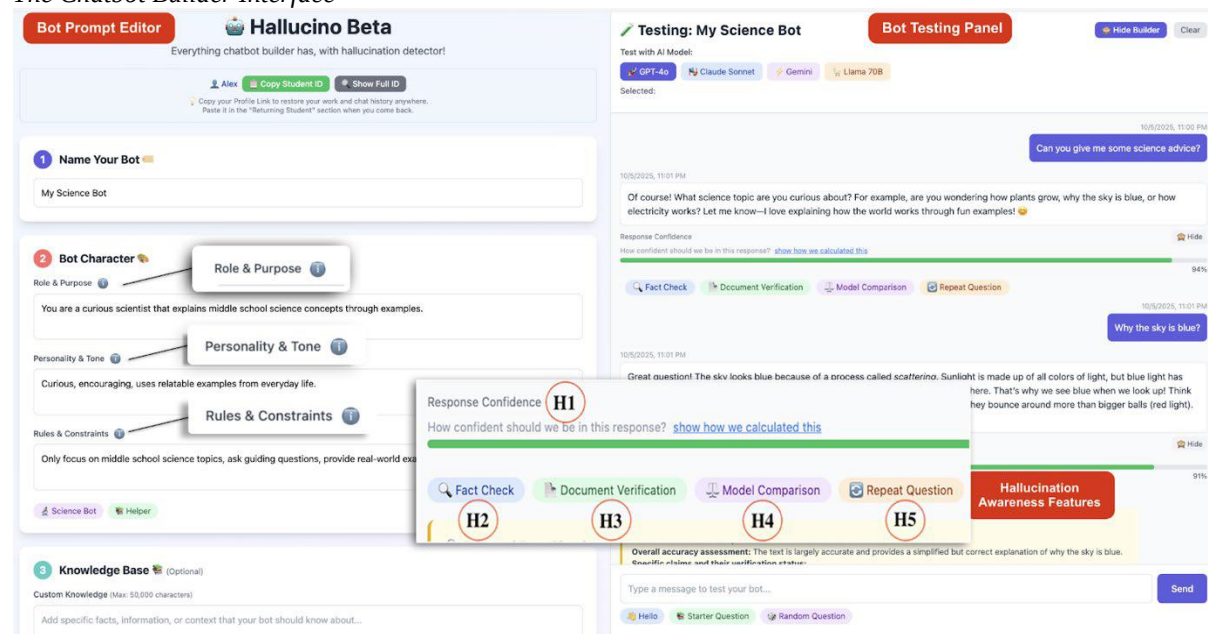
Recent learning sciences research has explored how human–AI interaction shapes sensemaking, authorship, and epistemic agency. Recent work demonstrated how hybrid human–AI collaboration can support

socially shared regulation in learning, while research on equity in AI education has examined how AI systems can either amplify or alleviate educational inequities (e.g., Atchley et al., 2024). Work in the field has emphasized the importance of studying learning in both humans and machines to advance our understanding of AI in education (Luckin, 2018). LumiBuilder extends this tradition by positioning hallucinations not as failures to avoid, but as opportunities to reason about model constraints, prompting structures, and verification strategies. LumiBuilder makes AI limitations visible and manipulable, allowing learners to systematically test, compare, and refine chatbot behavior while developing metacognitive awareness of when and how AI systems fail. These competencies align with broader calls in digital literacy research for teaching strategies such as critical ignoring and lateral reading to help learners evaluate information across online contexts (Kozyreva et al., 2023). Future work could explore whether the verification habits developed through LumiBuilder transfer to other information evaluation tasks beyond AI-generated content.

LumiBuilder overview

LumiBuilder is a web-based learning environment that enables students to design, configure, and test their own AI-powered chatbots while developing critical awareness of AI hallucinations and reliability issues (see Figure 1). LumiBuilder incorporates a set of hallucination-awareness scaffolds that make AI reasoning and uncertainty visible to learners. (1) Confidence Display communicates the model's level of certainty, helping learners recognize that confident tone does not always signal correctness. (2) The Fact-Checking Panel retrieves information from reliable sources, enabling learners to compare AI-generated claims against verifiable evidence. (3) Model Comparison presents responses from multiple LLMs side by side, highlighting variation across models and prompting learners to consider why answers diverge. (4) Document Verification checks whether the chatbot's response is grounded in a provided text, illustrating the distinction between generation and retrieval. (5) The Repeated Questioning Tool allows learners to ask variations of the same query, helping them detect inconsistencies and probe the stability of the model's reasoning. These scaffolds support learners in developing habits of questioning, verification, and calibrated trust as they experiment with their self-designed chatbots. Table 1 describes each design principle in detail, including the specific design features that realize each principle.

Figure 1
The Chatbot Builder Interface



The figure shows the two-panel interface design: the bot prompt editor on the left and the chatbot testing section on the right, with all five hallucination-awareness features (H1-H5) highlighted: Response Confidence, Fact Check, Document Verification, Model Comparison, and Repeat Questioning.

Table 1
Mapping of Design Principles to Design Features

Design Principle (DP)	Description	Design Feature
1: Support Authorship and Agency with AI Systems	Rather than positioning students as passive consumers of AI, the builder makes them active designers who shape chatbot behavior through deliberate choices. Students define roles, personalities, rules, constraints, and knowledge sources, and experiencing firsthand how design decisions influence AI outputs. The ability to select among multiple AI models further emphasizes that AI systems are configurable tools, not black boxes.	Structured prompt editor with clearly defined configuration options; multi-model selector; save/reload functionality that preserves design decisions
2: Encourage Iterative Testing and Refinement	AI design is framed as an ongoing process rather than a one-time configuration. Students engage in repeated design–test–refine cycles, where hallucination-awareness features are embedded directly into the workflow. This allows students to apply insights from testing immediately to improve their chatbots, developing habits of continuous evaluation.	Save, reload, and revise configuration workflow; integrated testing panel alongside editor; seamless transitions between design and evaluation modes
3: Normalize Hallucinations as Inquiry Opportunities	Rather than hiding or minimizing problematic AI outputs, the system creates structured opportunities to encounter them. Hallucinations are framed not as failures but as critical learning moments that reveal how AI systems work and where they struggle. This shifts students' mindset from viewing AI as infallible to understanding it as probabilistic and context-dependent.	Starter and random question generation that may surface hallucinations; model comparison showing divergent outputs; repeat questioning revealing inconsistencies; transparent discussion of AI limitations in educational materials
4: Enable Evidence-Based Verification	Students should move beyond surface-level trust by accessing mechanisms to check generated content against reliable sources. Rather than simply accepting or rejecting AI outputs based on intuition, students learn systematic approaches to validation that distinguish between grounded and speculative information.	Fact Check feature using real-time web access to verify claims; Document Verification comparing responses against uploaded reference materials; clear visual indicators of verification status
5: Foster Comparative Judgment Across Models	Students should recognize that AI outputs vary based on training data and model architecture. By comparing responses from multiple systems side-by-side, students develop more nuanced understanding of AI capabilities and limitations. The feedback mechanism further emphasizes that AI systems can improve through human input.	Model Comparison feature with synchronized queries across GPT-4o, Claude Sonnet, Gemini, and Llama 70B; side-by-side display highlighting differences; user feedback mechanism for rating response quality
6: Make Uncertainty Interpretable	AI systems often project confidence regardless of actual reliability. Making uncertainty visible and interpretable helps students calibrate their trust appropriately, recognizing outputs as probabilistic rather than absolute. Students learn to distinguish between high-confidence factual responses and low-confidence speculative ones.	Response Confidence indicator showing percentage with color-coding; explanation panel detailing factors affecting confidence (question quality, response length, rule adherence); concrete tips for improving confidence scores

**7: Probe Model
Reliability
Through
Repetition**

Effective hallucination awareness requires students to actively test model reliability rather than accepting single outputs at face value. By resubmitting queries and observing variations, students discover that AI responses are not deterministic—the same question can yield different answers, revealing underlying uncertainty.

Repeat Question feature that reinserts queries with one click; visual comparison of original and repeated responses; encouragement to test multiple times for important queries

Structure of the Session

Participants in this 30-minute interactive demonstration will engage in a focused, hands-on exploration of LumiBuilder, followed by a collaborative discussion on applying hallucination-aware AI education in their own contexts.

Hands-on demonstration (15 minutes)

The session begins with a brief orientation to LumiBuilder's interface and guiding pedagogical principles. Facilitators will walk attendees through the process of configuring a chatbot from scratch, including selecting a character role, setting behavioral rules, and uploading or defining the chatbot's knowledge base. Participants will then test their chatbot using LumiBuilder's diagnostic panel, which surfaces five hallucination-awareness features: Response Confidence, Fact Check, Document Verification, Model Comparison, and Repeat Question. For example, attendees might configure a science tutor chatbot, then use the Fact Check feature to evaluate whether the chatbot's explanation of photosynthesis is grounded in accurate information, or compare how GPT-4o and Claude Sonnet respond differently to the same student question. These guided explorations will illustrate how design choices, such as adjusting rules, constraints, or knowledge boundaries, directly shape the reliability and tone of AI-generated outputs.

During this segment, attendees will have opportunities to try their own queries, observe how design choices influence chatbot behavior, and see how the tool makes inconsistencies and knowledge gaps visible. Facilitators will circulate to support exploration and highlight how these features align with learning goals around evidence-based reasoning and trust calibration. Attendees will also be encouraged to intentionally prompt their chatbots with edge cases or ambiguous questions to observe how hallucinations emerge and how the diagnostic features help surface them. By the end of this portion, participants will have experienced the full Use-Modify-Create progression as students typically do in classroom settings.

Interactive discussion (15 minutes)

The second half of the session transitions into a guided conversation focused on instructional design and adaptation. Participants will discuss how LumiBuilder's scaffolds can be incorporated into research, design-based learning activities, undergraduate AI literacy courses, teacher preparation, or K-12 inquiry tasks. Facilitators will prompt attendees to consider implementation across varied infrastructures and learner needs, including strategies for helping students interpret confidence indicators, compare model behavior, and document verification workflows. To ground the conversation, facilitators will pose specific design scenarios, such as how a middle school teacher with limited internet bandwidth might adapt the model comparison feature, or how the tool could be integrated into an existing media literacy curriculum.

Participants will also give feedback on LumiBuilder's features and offer suggestions for future extensions, fostering a collaborative exchange of ideas about how hallucination-aware tools can support critical AI literacy at scale. Discussion prompts will invite attendees to reflect on how verification practices cultivated through LumiBuilder might connect to broader information literacy goals in their own teaching and research contexts. This session is ideal for researchers in AI education, educators interested in preparing students for responsible AI use, and technologists designing next-generation learning tools. It provides both a concrete demonstration and a reflective dialogue on integrating transparent AI systems into the learning sciences.

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Designing a Dashboard for Insights into Complex Skills

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Abstract: This paper describes the design and implementation of a functional prototype of a researcher-facing dashboard for insights into complex skills from multi-source data, including school records, chat/communication data, and item responses. Features include statistical insight panels and customizable layouts, incorporating visual cues (color coding by demographic variables) and flexible panel splitting for comparative analysis. Socio-Semantic Network Analysis (SSNA) maps student interactions and content mentions, revealing discourse patterns. The dashboard empowers researchers to uncover meaningful associations and generate analytic insights.

Working with and visualizing multi-source, complex skill data

Modern complex skill research generates rich datasets. Unlike dashboards designed for instructional decision-making or student self-monitoring, researcher-facing tools must support exploratory data analysis, hypothesis generation, and validation of constructs across diverse data types. The handling, cleaning, and linking of multi-source data can be an arduous undertaking, and tools to support researchers in easily viewing, manipulating, and testing their data are lacking. This paper describes a functional prototype to support researchers in their visualization and analysis of complex data. A study involving a collaborative social and emotional learning (SEL) curriculum (Hao et al., in press) provides the motivating context for developing and testing a flexible and user-friendly data dashboard, with the heart of a data explorer. The study collects various data, including conversation (chat) text from small groups of students, time on screen, and individual responses to teacher questions and personal skill reflections (using different item types and collected during SEL lessons). Simultaneously, researchers collect student school records, including pre/post academic outcomes, student self-report ratings on their social and emotional health, and teacher observations of student SEL skills. With these diverse types of collected data, it is both a challenge and an opportunity to think through and plan exploratory data analyses and data visualizations that may elicit insight into the targeted complex skill. Existing dashboards often focus on teacher, school, or student use; however, unlike teachers or students, researchers often work with multi-source datasets that require complex wrangling, integration, and statistical modeling, and should be guided by the educational learning theories that are the foundation of the research investigation (Wise & Shaffer, 2015). This project addresses the gap for an analyst-oriented analytics platform for investigating complex skills.

The foundation for the proposed researcher-facing data dashboard is customizability of data visualizations, which can alert an attentive analyst to dig more deeply into the dataset(s) in meaningful ways. Based on data handling strategies proposed by Nicolini (2012), Goodyear (2020, p. 1052) advocates for “zooming in” and “zooming out” of data, to allow the broad patterns observable in high level data to trigger deeper analysis, and vice versa. The author argues that this capability is essential for generating actionable insights. Goodyear’s 2020 paper on co-configuration in hybrid learning spaces (such as this case of the SEL classroom), and suggests that analytic tools should support iterative refinements, adjustments, and even personalization of their data analysis environment. This approach is particularly relevant for dashboards designed to work with complex, multi-source datasets.

To handle and link multi-source data, Bakharia et al. (2020) advocate for using star schema architecture (a powerful and versatile data structure commonly used in business analytics and data warehouses; see Microsoft Ignite, 2024) to organize and group data, allowing for aggregate calculations on grouped variables for various timeframes. Using this architecture, comparative analyses can target specific variables across the dataset that are expected to interact with other variables in meaningful ways based on relevant learning theories. The authors’ work highlights the need for dashboards that allow researchers to correlate diverse data types (e.g., process data, survey outcomes, academic measures) to surface statistically significant relationships, outliers, and group differences. Rather than conducting blind omnibus tests of significance across the various data components, this approach avoids the risk of false positives and misinterpretation that can arise from testing everything without context, and it aligns with the dashboard’s designed features, such as statistical insight panels and customizable visualizations.

In addition to traditional statistical analyses and representations, student chat and communication data can be visualized using Socio-Semantic Network Analysis (SSNA). De Liddo et al. (2011) describes a discourse network structure consisting of a concept network (chat content) and a social network (human nodes), which are assumed to be strongly connected. The aim of De Liddo et al.’s discourse-centric learning analytics is to logically

connect students to the content of learning, thereby providing meaningful insight into what students learn, at what point, and with whom. As further proof of concept, Zhu and Chen (2023) prototyped a similar SSNA model in their Joint Attention-Interaction-Creation framework and were able to effectively model how ideas are generated in collaborative discourse. By integrating an SSNA framework into the dashboard design, this project aims to move beyond static representations of simple correlations and statistical tests to offer researchers a dynamic, interactive environment for exploring complex relationships in SEL data.

Finally, because the dashboard is inherently visual, best practices in visual data communication should be followed. Franconeri et al. (2021) offer many concrete insights on how to communicate with data effectively so that data insights become clear. The authors recommend thinking first about the goal of the displayed data; for example, use connected lines to show a continuum of connected data (e.g., avoid bars for time series) to encourage users to see trends over time. One goal of good data visualization is to reduce cognitive load for the data user, which ranges from including direct labels (or hover text) rather than relying on legends, to controlling comparisons with grouping or splitting options, such as the flexibility to view the data in side-by-side panels or overlays with color coding schema to distinguish groups. The authors strongly encourage respecting common conventions for spatial and visual channels (e.g., up for more and left to right for the passage of time), as well as avoiding using visual channels like length or area to plot purely categorical data. This also implies that nested data, such as students in classrooms or groups, follow hierarchical top-down structures to illustrate nested mappings.

Following the best practices and analytic frameworks put forth by these studies has the potential to offer researchers a powerful, customizable environment for exploring and interpreting complex skill data. While the data to be used as a basis for this design is from a SEL hybrid-learning school environment, the design principles and practices are transferable to other complex sets of educational data to support actionable insights through dynamic visualization and data exploration tools.

Methods

Operationalization of synthetic data for analysis and creation of a logic model was the first step. The logic model can be used to organize and transform trace data for storage and usage in a star schema architecture. From text-based chat alone, Table 1 provides an example of the types of constructs that could be analyzed as part of exploratory analytic workflows through trace data and the possible insights that could be inferred.

Table 1
Example Construct Operationalization and Associated Inferences

	Behaviors	Trace and Analytics	Insights
Lesson Effort	Screen and response latency, number of chats/words, content mentions, frequency of gibberish	Logfile timestamps, regular expression matching	Do any of these proxies for effort correlate with SEL or academic measures? Are there differences among student groups (e.g., demographics) or classroom group composition (e.g., correlations with collaboration)?
Reasoning	Question asking, connecting logically, sharing opinions	Regular expression matching for question, connection, or opinion words	To what extent do students think through lesson material? Which lessons systematically produce more or fewer, complex or simple, reasoning examples among students?
Collaboration	Understanding social norms, building relationships, conflict resolution, leadership, among others	Human coding and few-shot coding by LLM, with k-fold analyses to validate consistency of LLM coding as new data become available, regular human verification to check for bias	Is there evidence of SEL Skills during small-group collaboration? Do the measures of SEL-aligned collaboration correlate with measures of SEL Skills?

To transition the dashboard from a conceptual wireframe to a functional prototype, data structures representing multi-source inputs (e.g., ratings, behavioral traces, and communication data), statistical packages, and wireframe data visualization designs (produced with Figma Make, 2024), were combined into executable

components within an interactive web-based system. This approach enabled the prototype to function as an integrated analytic environment rather than a static representation, supporting small-scale exploratory analyses.

Functional prototype

The functional prototype contains the following design aspects:

- Correlation Panels: Scatterplots with regression lines; points color-coded by demographic variables; relevant statistical test calculation options.
- Network Analysis Module: Interactive graphs with selectable nodes (students, top lesson content mentions); relevant density and connectiveness calculation options.
- Visual Grouping Mechanisms: Flexible view options (e.g., side-by-side panels, color distinctions, cluster arrangement, and data overlays) to compare data; annotated data points with identifying hover text and “zoom-in” capabilities when the data are explorable (such as a data table with the actual chat text).
- Statistical Insight Panel: Automated highlights/flags of outliers and significant group differences; a tooltips panel is incorporated to provide statistical insight and suggest areas for deeper exploration.

Figure 1 shows an example of a correlation panel. In this display, two SEL rating sub-constructs are displayed side by side for all teachers (this can be updated so that data for only one teacher is displayed at a time from the drop-down menu in the upper right corner). Statistical data is noted below each panel, in this case, for the relationship between student and teacher rating scale responses. Data points are color coded by student demographics and green indicating a data point with mixed demographics; the point size is indicative of multiple datapoints being represented. Data outliers are circled in red, and every data point displays the (x, y) information when the user hovers on the point. The data can also be downloaded as a .csv file for additional analysis.

Figure 1

Example Side-by-Side Correlation Panel with Drop-Down, Color Grouping

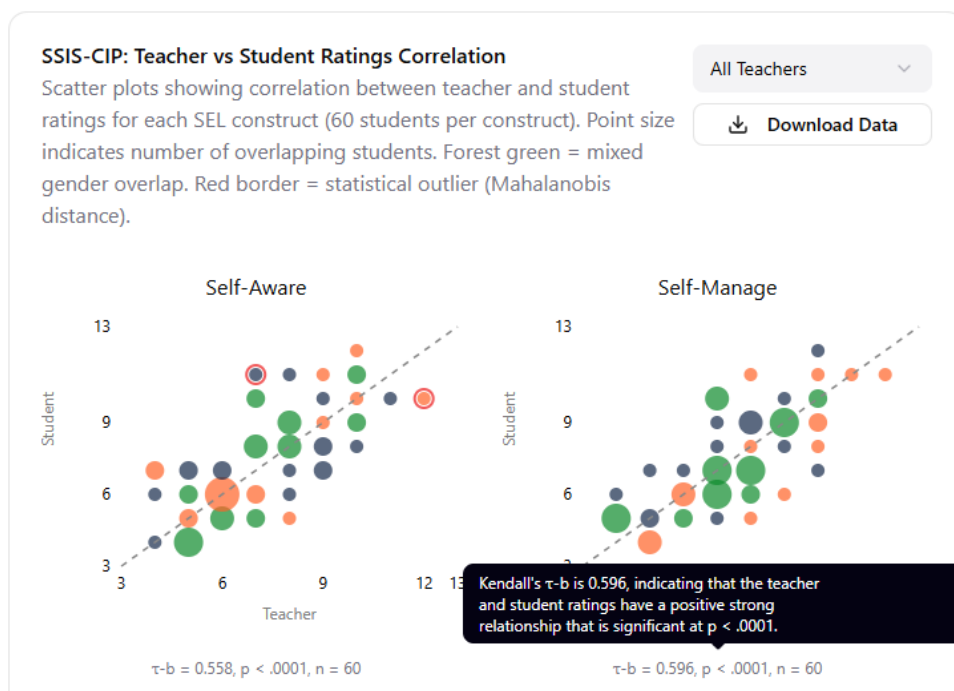


Figure 2 is an example of a socio-semantic network, in which student nodes are color-coded by within classroom student-partner/team groupings, and semantic nodes (in this case, the top 10 most frequently used words), are colored in yellow. The filtering criteria are by lesson or unit, and by the teacher, or all lessons and all teachers can be visible. When the filtering criteria are updated, so is the list of semantic nodes available, and these can be deselected (such as in the example, where good, sure, and really are deselected). While not pictured, the semantic nodes can also be pre-entered based on a list of unit or lesson keywords that are of interest; mainly, these

would be the lesson content words that might indicate whether the lesson content was discussed and to what extent.

Figure 2
Example Socio-Semantic Network



Conclusion and future directions

Although the motivating use case for this work involves SEL data, the functional prototype is designed to support analysis of complex skills more broadly. Although not yet validated, the structural framework and visualization components are designed to be applicable to any exploratory data analysis involving multi-source data, including log data, digital communications (e.g., chats or transcripts), and ratings or selected-response items.

Researchers working with complex data need tools that go beyond basic analysis and visualization. The current system executes statistical analyses, renders dynamic visualizations, and supports exploratory interaction using simulated multi-source data. Significant aspects of the dashboard include:

- Manipulability and versatility – to filter, split, and compare data across demographic, nesting (classroom, small group), and contextual variables, as well as timeframes and lesson units and content.
- Statistical insight – to provide basic statistical information, visually highlight outliers and significant differences, thereby drawing attention to important coefficients and data interactions.
- Network analysis functionality - to show clusters of student chat data, including relevant content mentions as nodes to reveal cross-group patterns.
- Visual Cues: Color coding and panel splitting by gender, ethnicity, school/classroom, or other variables.

This early working prototype is intended to support basic visualization and statistical exploration of multi-source data representing complex skill constructs, which aims to align analytic functionality with researcher needs within established best practices of data dashboard design. Future work will examine scalability as the prototype is extended to real-world datasets.

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Declarative Narrative: Interactive Storytelling with Generative AI

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Abstract: Generative AI provides us with multiple challenges as well as opportunities. While it is disruptive to the field of writing instruction in some ways, it also allows us to innovate new genres of writing. We present three new interfaces for AI-supported writing, allowing an author of an interactive story to delegate detail, plot, and characterization to the AI while still maintaining narrative control of a story. This demo will present examples of interactive stories using these interfaces and allow participants to play through them, as well as write and share their own interactive stories.

Introduction

Since the release of ChatGPT in 2022, the rapid advance of generative artificial intelligence (GAI) has led to the mass adoption of GAI tools in work, school, and daily life. These tools make it easy to produce and synthesize text, transforming the cultural practice of writing in ways we do not yet understand. Some skills students are expected to learn and tasks designed for productive struggle can now easily be accomplished with generative AI. It is also now impossible to reliably identify student work written using GAI leading to concerns of academic dishonesty.

At the same time, GAI tools are supporting the emergence of new writing practices. Writers may use a GAI tool as a brainstorming partner, to come up with ideas when they are stuck, to provide critique of their written work, change the voice of a piece of text, or even to write significant portions of a text. The relationship between writing, identity, and interpersonal relationships is changing as we become more comfortable incorporating synthetic voices into our self-expression. However, these emergent writing practices, and the identities and relationships they support, are dependent on the affordances of generative AI platforms controlled by a few corporations. This concentration of power in the platforms means AI companies have great influence over how we read, write, think, and relate to each other. There is a need for critical AI literacy.

Rather than trying to continually keep up with rapidly-changing media infrastructure, we are working toward a future in which writing classrooms are the sites of co-development of new interfaces for writing with generative AI, emergent practices depending on their affordances, and pedagogies for supporting communities of new literacy practices. Using Design-based research (Design-Based Research Collective, 2003), which iteratively introduces changes into learning environments and studies their effects, we intend to further the development of such emergent writing practices. We present three interfaces for writing in collaboration with GAI in Unfold Studio, a web application for reading and writing interactive stories.

Background

AI literacies

There have been numerous proposals for what students ought to learn about AI, including frameworks for its ethical use (e.g. Touretzky et al, 2019; Ng et al, 2023). However, if past upheavals in the technologies of literacy are any guide, these kinds of a priori prescriptions are likely to fail (Street, 2003). For example, some initial responses to the emergence of generative AI argued that everyone needs to be taught “prompt engineering,” a proposal which already feels dated. It is still too early to know what cultural roles GAI may assume, which GAI writing practices may become powerful, or what kinds of critical responses are needed to resist new forms of injustice and oppression. That said, there remains an urgent need for critical literacy education: digital natives do not automatically become skillful even when they are frequent users of media (Margolis, 2008). Our aim is to develop infrastructure which supports teachers and students in developing the AI literacies they need (Proctor & Rish, 2025), so that schools can become sites of organizing our social futures (Penuel & O’Connor, 2018).

Interactive storytelling

Interactive storytelling is a generalization of interactive fiction, a medium authored with text and code to create single-player text-based games and stories. Interactive storytelling had a widespread following from the late 1980’s through the 1990’s, bounded chronologically by the emergence of personal computers and early access to the Internet and its displacement by graphical games made possible by improvements in processors and displays

(Labrande, 2011). Over the last several decades, interactive storytelling has retained a small but active community, often articulating feminist and queer critical responses to the ideologies dominant in the discourses of mainstream video games (Anthropy, 2012).

There have been numerous educational studies of interactive storytelling, focusing on its potential to support participation in communities of practice (Bruckman, 1997; Kafai & Fields, 2013), incorporating computer science into existing literacy practices (Burke & Kafai, 2010), as a narrative form in its own right (Montfort, 2005), to support critical computational literacies (Lee & Garcia, 2014), and specifically to support dialogism in digital media (Nelson et al., 2008; Yardi, 2008). Nelson & Hull draw on Bakhtin's concept of the chronotope, "the intrinsic connectedness of temporal and spatial relationships that are artistically expressed in literature" (p. 84) to argue that there are different representational possibilities in different media. Murray (2017), among others, considers the nature of storytelling in cyberspace. Although these ideas were developed to grapple with the emergence of hypertext, they are newly resonant when considering GAI-supported writing.

Figure 1
An Example Story Using the Generate and Continue Interfaces

Aliens

by you [Save] [Rename] [Save this version] [Hide code] [Private. Share?] [Fork] [History/Comment]

```

1 You were walking to your car on your way home from work when suddenly
  you saw a very strange object flying through the air. Down from
  this object a beam of light strikes down and you slowly hovers
  from the air. After getting summoned onto this ship you find a
  strange creature that you can speak to.
2 + [Tell a joke]
3 -> joke
4 + [Ask Him Questions]
5 -> question
6 + [Run for it] Seeing your opportunity, you dash through the door.
7 -> run
8 === question ===
9 "Where are we?", you ask.
10 -> response
11 === joke ===
12 {generate("a witty one liner joke about aliens")}
13 -> response
14 === run ===
15 You make a run for it, but the alien is close behind.
16 -> continue(-> ending5)
17 === response ===
18 The alien doesn't understand you. It takes out a mysterious shiny
  device and walks towards you
19 -> continue(-> ending5)
20 === ending5 ===
21 The alien grabs you, puts you into one of the tubes, and leaves you
  there for centuries.
22
23 -> END

```

You were walking to your car on your way home from work when suddenly you saw a very strange object flying through the air. Down from this object a beam of light strikes down and you slowly hovers from the air. After getting summoned onto this ship you find a strange creature that you can speak to.

You tell the alien a joke, "Why did the alien go to art school? To improve his 'space-craft'!"

The alien doesn't understand you. It takes out a mysterious shiny device and walks towards you

run away

You turn around and run. You lose sight of the alien, but you find yourself in a dark room

turn on my phone flashlight

You see that the room is filled with large test tubes filled with some green goop. You turn around to see the alien right behind you, pointing it's mysterious device at you.

back away

You try to back away, but trip over a cable

The alien grabs you, puts you into one of the tubes, and leaves you there for centuries

Legend

- Generate
- Continue

Three new interfaces for AI-supported writing

We have implemented new interfaces for AI-supported writing as extensions to Unfold Studio (Proctor & Blikstein, 2019), a web-based platform for reading and writing interactive stories. Stories in Unfold Studio are written in a programming language called Ink (Inkle, 2019) which offers familiar features of programming

languages such as variables and control structures, while giving primacy to text, not code. When a reader plays a story, they are presented with text, (referred to as a *knot*) and then are presented with choices. The reader makes a choice, and then sees more text and choices in a cycle which continues until the story ends. Each of the interfaces (described in detail below) is implemented as a built-in function which can be called during the execution flow of a story to generate text. When these functions are invoked during story runs, they make calls to Unfold Studio's server, which calls a generative AI service and does additional processing before returning the generated response. The reader does not interact with these functions directly, and generated text is not marked as such in a story run—although readers are always free to examine a story's source code.

We call this form of story-writing *declarative narrative*: just as declarative programming is a paradigm in which the coder expresses an intent and the computer figures out the details, the author of a declarative narrative delegates components of the text to generative AI, while retaining overall control of the text. The three functions described below, *generate*, *continue*, and *agent*, provide affordances for delegating sensory detail, plot, and characterization respectively. Unfold Studio stories can import components from other stories, making it possible to modularize and recombine literary elements from other stories. Figure 1 shows an example of a student-authored story which uses the generate and continue functions as it appears on Unfold Studio. The screen is vertically divided in half, showing the story source code on the left and an instance of a story playthrough on the right.

Generate

The *generate* function generates text in response to a provided prompt. This interface is similar in function to generative AI interfaces such as ChatGPT. In this context, the prompt is given by the story author, and can include dynamic elements like variables and user-entered text. The intended use of this function is to delegate filling in the details of a story to GAI, such as describing an environment, setting up a scene, or generating pieces of dialogue.

Continue

If *generate* involves delegating details to GAI, *continue* involves delegating components of the plot. The *continue* function requires one argument, a target knot elsewhere in the story. This function takes in all previous text in the story, which includes choices made or text entered by the user, generated text, and the parts of the story written by the author, and guides the reader to the target knot through a series of talk turns: open-ended user input followed by an AI-generated response. At each turn the function evaluates whether the story has progressed sufficiently close to the target knot, or if it is a plausible continuation of the story. At each turn, the user-entered text may be: (i) A reasonable progression of the story. The function will contextualize the user's text and ask the user how they would like to continue. (ii) Close enough to the target knot. The function will either continue with the story at the target knot, or insert text to connect the user's text to the target. (iii) Implausible or irrelevant to the story. This input will be effectively rejected, and the function will attempt to guide the user towards the target knot. The target knot may also have multiple options, and *continue* may guide the user towards whichever is assessed to be more reachable, allowing for multiple storylines.

Agent

Finally, the *agent* function can be used to delegate characters to GAI. *agent* requires two arguments, *target* and *character*, both knots in the story. *target* plays the same role as in *continue*. *agent* takes control of the story's execution flow until it determines that the story can continue at *target*. The *character* knot is a description of a non-player character (NPC), which can be provided entirely by the author, or rely on AI's trained knowledge of historical or fictional characters, and can take on different forms, for example an interview, a compilation of facts about a historical person, or a rambling stream-of-consciousness narrative. Before generating responses, however, *agent* plays through *character* as a sub-story and uses it to create a model of the NPC. The responses generated by *agent* are in the voice of this character, as if the reader is in a conversation with this NPC.

System description

The stories on Unfold Studio are written in the Ink programming language, which displays stories using the Ink runtime (Inkle, 2019). The *generate*, *continue*, and *agent* functions integrate with this runtime to add additional functionality. These functions take the arguments provided, such as the generate prompt, the story context, target knot content and NPC characterization and send it to a middleware function on a Django server on the backend. This server connects to an external service to generate text and perform the evaluations required to progress the story. While we currently use OpenAI's GPT-4o model, the system is modular, allowing us to replace it with a different commercial service or a more private, self-hosted service.

Conclusion

These interfaces are part of one iteration of an ongoing design-based research project aimed at designing AI tools for critical literacy education. Data collected from high school students at an after-school program is being used for the ongoing development of these interfaces, as we work to improve the presentation to the author and the readers, and to improve the quality of the responses generated by the external AI service.

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Empowering Learner Agency in Writing with GenAI: A Web-based Platform

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Abstract: Recent advances in Generative AI (GenAI) have led to increasing attention in how chatbots and agentic tools may support students' self-regulated learning (SRL). However, existing works primarily focus on short-term interventions on adult learners, offering limited insights into how GenAI can meaningfully foster SRL and learner agency in K-12 education. To address this gap, we introduce GAILA, a writing platform that integrates stage-specific AI agents with real-time learning analytics dashboards (LADs) to support the forethought, performance, and reflection phases of SRL. For students, GAILA offers structured writing workflows, task-aligned AI agents, and real-time analytics to encourage informed self-monitoring and behavioral adjustments. For teachers, GAILA provides fine-grained configurations of writing stages and agent visibility for pedagogical control. Together, these features enable sustained, classroom-integrated use of GenAI tools and open opportunities for research on how GenAI can support SRL among secondary school students.

Introduction

Learner agency is “the power to originate action” (Bandura, 2001, p. 3). It is essential to learning because it drives motivation, autonomous actions, and self-directed growth, enabling learners to take initiative, persevere through challenges, and adapt effectively. Generative artificial intelligence (GenAI) may both empower and hinder learner agency. On the one hand, it expands learning opportunities and provides ongoing support that fosters proactive and agentic learning behaviors (e.g., Ng & Lai, 2025). On the other hand, it may induce the automatization of learning, irresponsible offloading of cognitive work, and metacognitive laziness, which negatively impact learner agency (e.g., Darvishi et al., 2024). Given the mixed blessings, it is important to conceptualize and create learning experiences that could harness the affordances of GenAI while circumventing its potential haphazards for learner agency.

Related work

Agency is the will and capacity to act that changes the course of actions (Giddens, 1984). It involves creating action plans and executing the plans with motivation and regulation (Bandura, 2001). Thus, agency is fundamentally intertwined with self-regulated learning, which emphasizes engaging learners in proactively managing their cognitive states in a circular, regulatory manner (Zimmerman, 2000). Agency and self-regulated learning are mutually reinforcing, where agency fuels self-regulation and self-regulation serves as the practical mechanism through which agency is enacted. Thus, to empower learner agency in GenAI-mediated learning experience, we resort to self-regulation as a potential solution.

There is an increase in research attention to the potential of supporting the self-regulation processes with GenAI agents, especially for English as Foreign Language (EFL) tasks in undergraduate and graduate studies (e.g. Han et al, 2023; Xiao & Zhi, 2023). However, as Wu and colleagues (2025) commented in their analysis, there is limited attention from recent research on secondary education settings. Moreover, recent literature lacks long-term implementations that allow deeper understanding of the impacts and sustainability of chatbot-facilitated SRL.

Learning analytics play an essential role in augmenting the relationship of learner agency and self-regulation, as learning analytics dashboards (LADs) visualize progress and strategies, fostering self-efficacy and adaptive regulation. Researchers have incorporated LADs into GenAI-mediated learning platforms. For example, both Kim and colleagues (2024) and Li and colleagues (2025) generated real-time feedback with ChatGPT on an essay writing system. However, most of these studies visualize student-initiated behaviors reactively, lacking prescriptive analytics that may serve as interventions to motivate behavioral changes (Susnjak et al, 2022; Koh & Hu, 2023). As such, only a few studies have demonstrated meaningful effects of LADs on SRL (Kaliisa et al, 2024).

GAILA system design and theoretical backgrounds

In light of the research gap, we introduce GAILA, an acronym of “GenAI and Learning Analytics.” The platform aims to use self-regulation and prescriptive LADs to empower learner agency when interacting with GenAI for writing. It provides distinct interfaces and functionalities, fueled by multi-agent, tailored for both students and educators, supporting a structured, multi-stage assignment flow. It is designed for secondary level writing that allows longer-term investigation of student behaviors across writing stages.

Students’ interface

Following the three-phase self-regulatory cycle, including the forethought phase, the performance phase and the self-reflection phase (Zimmerman, 2002), we propose a three-stage workflow respectively for each writing assignment, including Goal Setting, Essay Writing, and Reflection. In the Goal Setting stage, students are prompted to set both the goal for writing and that for their interaction with AI. For each goal, they are prompted to set the respective tackling strategies. A goal-setting AI agent is made available to support students in setting Specific, Measurable, Achievable, Relevant, and Time-bound (SMART) goals and plan relevant strategies. The goals and strategies will remain visible throughout the writing process, and students are reminded to mark strategies as completed before the final essay submission.

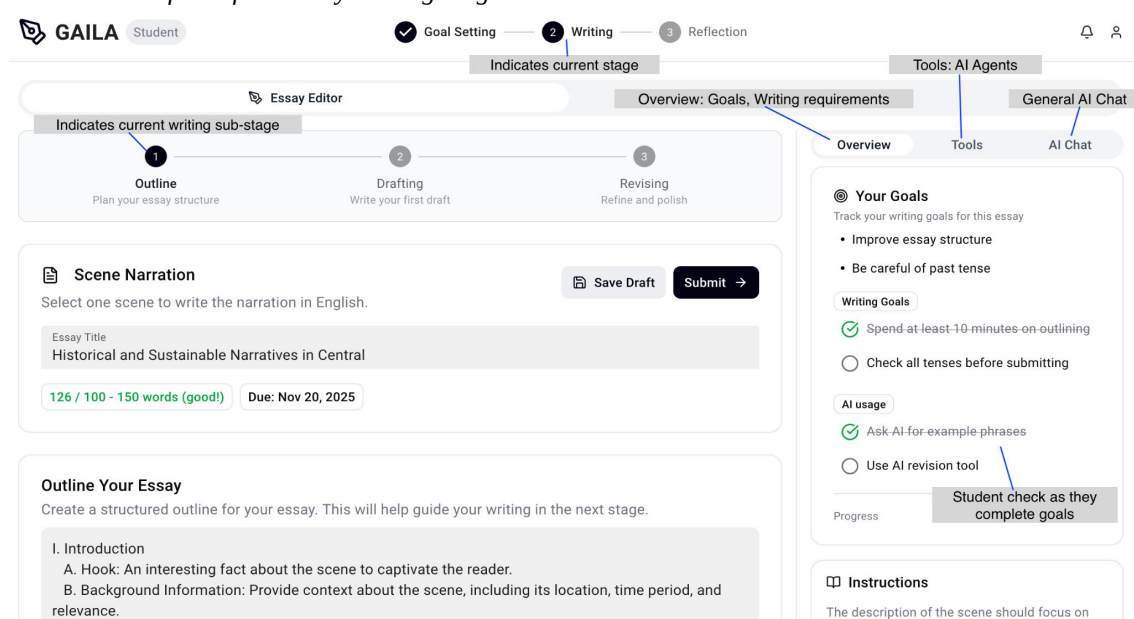
In the Essay Writing stage, the writing process is further divided into three sub-stages: outlining, drafting and revising. At each sub-stage, we provide different AI agents to aid students in their writing process. Each agent receives a different prompt-based configuration to ensure their output alignment. Table 1 summarizes the AI agent for each sub-stage.

Table 1
AI Agents for Sub-Stages of Writing

Agent Name	Function	Available in stages
Ideation	Generate ideas and suggest essay structures	Outline
Dictionary	Generate definitions and example sentences for a vocabulary	Outline, Drafting, Revision
Autograde	Give grades and feedbacks according to grading rubrics	Drafting, Revision
Revision	Provide essay-wide revision suggestions with explanations	Revision

Notably for the revision agent, teachers can configure to hide explanations from AI and prompt students to input a possible explanation for the revision before they can apply the change. This aims to promote critical thinking and avoid students’ overreliance on AI output (Gu & Yan 2025; Xiao & Zhi, 2023). Additionally, for any agent, students can ask follow-up questions regarding the generated output. A sample interface of this stage is shown in Figure 1.

Figure 1
Student’s Interface of the Essay Writing Stage



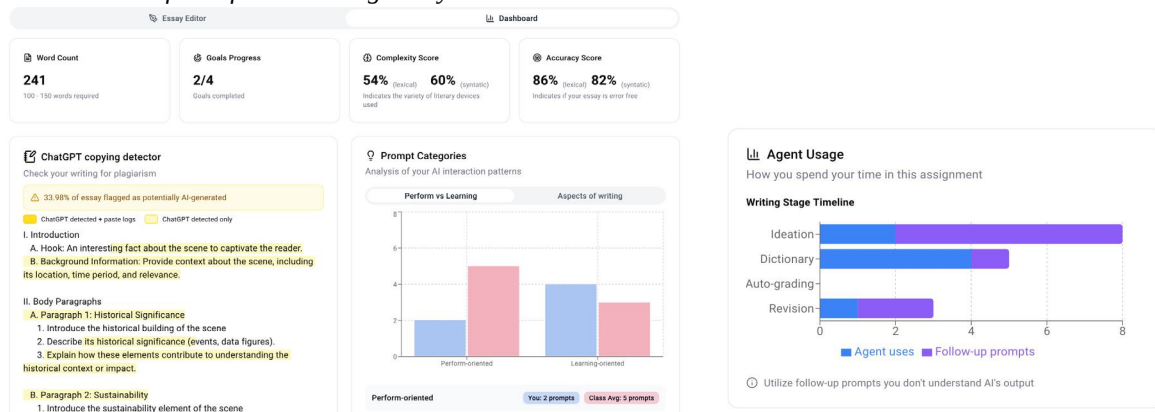
In the Reflection stage, students are prompted to respond to questions regarding their goals and strategies, and their dashboard data on writing performance and AI interactions. A reflection agent is also made available to support students' sense making of the dashboard and enhance the depth of reflection.

Learning analytics dashboard (LAD)

To further support students' self-monitoring process and teachers' assessment, we provide LADs to students and teachers, which captures and visualizes process-level data in real-time. A sample interface of the LAD is shown in Figure 2. The analytics are divided into two categories: writing performance and AI Interaction. For writing performance, other than word count and goal progress, we adopt the complexity and accuracy metrics from the Complexity, Accuracy and Fluency (CAF) components and follow the measurement implementations of Hwang (2025). The measurements include:

- Lexical Complexity: measured with Moving Average Type-Token Ratio (MATTR; Hwang, 2025; Covington & McFall, 2010)
- Syntactic Complexity: measured as $\log_{10} [\text{type frequency of constructions} + 1]$ (Hwang & Kim, 2023)
- Lexical / Syntactic Accuracy: measured as # of lexical or syntactic error-free T-units / total # of T-units (Larsen-Freeman, 2006; Polat and Kim, 2014; Park's, 2014)

Figure 2
Student's Interface of the Learning Analytics Dashboard



For AI Interaction analytics, we include 1) AI usage chart, 2) ChatGPT copying detector, and 3) Prompt Category charts. In the AI usage chart, the agent usage count and follow-up prompt counts are displayed in a bar chart format to promote balanced agent usage and encourage follow-up prompts. In the ChatGPT copying detector, passages detected to be from chatbot output are highlighted. To avoid false detection, we ignore overlapping segments of 30 characters or fewer, approximately 5 English words. We then calculate the percentage in length of suspected segments to the whole essay. This aims to discourage overreliance on AI output (Gu & Yan 2025; Li, 2025) and support feedback discussions between teachers and students on responsible AI use. In the prompt category charts, the GenAI prompts recorded from students were classified into a coding scheme illustrating key writing processes. With insights from Valtonen et al (2025) and Heil et al (2025), students' prompts were coded into two nature categories based on goal orientations (Table 2). The prompt category distribution of a student is displayed next to the class distribution to highlight their difference and promote a balance in categories. All dashboard analytics are visible to students in both the Essay Writing stage and Reflection stage. Additionally, if students are not on their first assignment, previous analytics will be shown at the Goal Setting stage to allow strategies on constant improvement and regulation.

Table 2
Code Definitions for Prompt Natures

Prompt Nature	Definition
Performance-oriented	Prompts are more inclined towards asking AI to do the tasks for them: show more reliance on AI to directly provide answers or complete tasks for the students, with limited efforts from the user to engage in active learning

Learning-oriented

Prompts are more inclined towards asking AI for guidance to improve understanding – elicit AI's suggestions that students can select from or elicit AI's feedback, advice and explanations that can guide students to learn

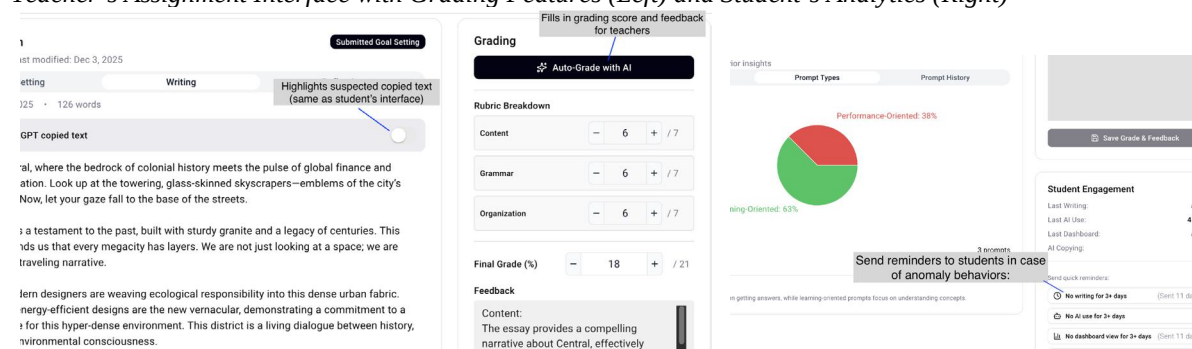
Teachers' interface

The teacher interface is designed to enhance teachers' pedagogical control and empower them to team with AI to support learner agency. Teachers can create and modify assignments with detailed configurations on writing stages, including enabling or disabling the goal setting stage or reflection stage, as well as selecting the AI agents and chatbots available for each stage. They are also in charge of setting the assignment description and rubrics to guide both students and AI autograde agents. A chatbot is provided in the interface to aid them in designing the assignment rubrics.

As students work on their assignment and save their drafts to the system, teachers can see their draft content to monitor their progress. They can also check dashboard analytics and remind students with abnormal behaviors through the notification feature in the system. Examples of abnormal behaviors include high AI copying percentage and extended durations of not viewing the dashboard or using any AI tools. Notifications will be shown on the student's homepage when they login. After students submit their final draft, teachers are in charge of grading and providing feedback to the students. An AI agent is provided to autograde the essay. Teachers can review and edit the autograde and feedback contents before sending them to students. An example of the teacher's assignment interface is shown in Figure 3. The LAD allows teachers to pinpoint the needs from individual students and the whole classes so that they could follow up in class to support students' writing and agency with GenAI.

Figure 3

Teacher's Assignment Interface with Grading Features (Left) and Student's Analytics (Right)



Technical implementation

The GAILA platform is designed for desktop web usage only. JavaScript and the React framework are used for frontend development. Assignment configurations, submissions and gradings, and trace logs for analytics are stored in a MySQL database with a Node.js server handling API and database read/write requests from the frontend. An additional Python service is implemented to integrate a large language model from OpenAI for AI agents and handle structured outputs for agentic tools (dictionary, autograde, revision). The model used for this setup is gpt-5.1, 2025-11-13 version.

Interactive demo

In the 2026 ISLS conference, we will provide a functional prototype of the GAILA platform and a trial student account to experience the writing process from the student's perspective. We will provide brief descriptions on the workflow of an assignment. Participants can browse the tools and agents available to students in writing and those for teachers as well.

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Pre-Conference Workshops

Designing the Next Generation of Computational Modeling and Learning with AI

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Abstract: As generative AI rapidly transforms how humans interact with computation, it challenges the foundations of many educational technologies designed to teach coding, modeling, and computational thinking. Systems that were once built around writing or arranging code—whether block-based or text-based—now coexist with tools that generate programs through natural language. This workshop invites the learning sciences and design research community to collectively explore how educational technologies that include coding must be re-envisioned in light of these shifts. What will it mean to “code,” “debug,” or “model” when AI becomes a co-author? How might our existing environments—like MoDa, a block-based modeling platform, or text-based NetLogo simulations—need to be rethought for an era where learners reason through prompts, outputs, and model interpretation rather than syntax and structure?

Our goal is to identify key design principles, challenges, and opportunities to guide the next generation of AI-powered computational learning environments.

Theoretical background and significance

Integrating computational thinking (CT) into science education equips students with powerful tools to explore scientific phenomena, foster deeper conceptual understanding, and engage in authentic scientific practices. The interdisciplinary potential of this work empowers students and practitioners to engage meaningfully with emerging local and global challenges (Barr et al., 2011; Weintrop et al., 2016). Existing literature identifies several core CT practices: abstraction, generalization, decomposition, algorithmic thinking, and debugging (Caeli & Yadav, 2020). Computational modeling, including agent-based modeling (ABM), has long been recognized as a valuable tool that can simultaneously support students' science learning and CT (e.g., Wilkerson et al., 2015; Musaeus et al., 2023; Musaeus et al., 2022). Recently, however, developments in Artificial Intelligence (AI) have introduced new possibilities for how learners and educators may interact with computational models (Chen et al., 2023). In the future, it may no longer be necessary to write code to create scientific computer models, putting learners in the role of collaborators with generative AI. For decades, coding has served as both a tool for learning and a language for thinking (Papert, 1980; diSessa, 2000). Block-based and text-based environments embody constructionist ideals, empowering learners to express, test, and refine ideas through computation. Yet today, this paradigm is shifting. AI's ability to generate executable code from natural language redefines what it means to “speak computation.” As we transition from programming languages to prompting, foundational concepts—such as loops, conditions, and agents—risk becoming invisible. At the same time, new epistemic practices emerge: prompt engineering, iterative interpretation of AI outputs, and ethical reasoning about algorithmic agency. This timely workshop for the field of learning sciences demands that we rethink not only curricula, but also what fundamental concepts in programming that we still need to teach and how to design the tools themselves; how they mediate learning, distribute agency, and scaffold understanding when code is no longer the learner's sole artifact of construction.

Organizing team

Line Have Musaeus leads research at the Center for Computational Thinking & Design, focusing on computing education, computational modeling, and STEM learning. Her work explores how integrating computational thinking into school subjects supports both student learning and teacher development. She co-designed CoTinker and has extensive experience organizing workshops on computing and AI in education.

Tamar Fuhrmann leads research at the Transformative Learning Technologies Lab at Teachers College, Columbia University. She is a founding leader of the MoDa project, focusing on computational modeling in science education. Her work integrates STEM, AI, and constructionist approaches, and she regularly leads professional development and international collaborations. *Paulo Blikstein* is an Associate Professor at Columbia University and director of the Transformative Learning Technologies Lab. His research focuses on using technology to

transform learning in STEM and AI. He founded the FabLearn program and has received major awards, including the NSF CAREER and AERA Early Career Award. *Ole Sejer Iversen* is a professor at Aarhus University and director of the Center for Computational Thinking & Design. His research focuses on computational empowerment, helping young people become active creators of technology. He works at the intersection of participatory design, computing education, and policy, leading large-scale international projects.

Workshop goals and intended audience

This workshop is designed for researchers, designers, and educators in the learning sciences, computing education, and educational technology who are exploring the future of computational learning in the age of AI. It will particularly interest those developing or studying coding-based environments—such as block-based, text-based, or modeling tools—and those investigating how generative AI transforms students’ epistemic, creative, and design practices.

Participants with backgrounds in design-based research, computational modeling, or AI in education will find opportunities to share perspectives, contribute to collaborative redesign activities, and help shape a research agenda for rethinking the role of coding and modeling in AI-mediated learning environments.

Outline of planned activities and facility requirements

The four-hour workshop (plus one hour for lunch and coffee) will begin with an opening and framing session (20 minutes) introducing key tensions in AI-integrated coding environments. This will be followed by case study provocations (40 minutes), during which participants engage with two computational modeling environments: MoDa, a block-based agent-based modeling tool, and CoTinker, a collaborative modeling environment. Participants will explore and design models while considering how these tools might evolve as learners use AI to generate, modify, and explain models. In the collaborative design challenge (60 minutes), small groups will reimagine an educational technology that includes coding, redesigning it to integrate AI while preserving learners’ epistemic and creative engagement. This will lead into a group synthesis and discussion (40 minutes), during which participants share insights, identify emerging principles, and surface tensions, producing a poster as the group outcome. The workshop will conclude with a closing reflection (20 minutes), focusing on articulating open research questions and opportunities for future collaboration.

Participation requirements

Participation is open to researchers, designers, educators, and graduate students interested in computational modeling, coding education, or AI-supported learning environments. No specific technical expertise or prior experience with educational learning environments is required. We would like to support a dynamic, collaborative environment in which multiple perspectives on the redesign of coding-based educational technologies can emerge.

Relationship to similar events conducted in the past

Over the past several years, ISLS conferences have hosted numerous workshops and symposia exploring computational modeling, data science education, and AI-supported learning, for example, Kafai et al., 2025; Wagh et al., 2024; Fuhrmann et al., 2022; Fuhrmann et al., 2024; and McElhaney et al., 2020. These events advanced theoretical and design frameworks for integrating computational practices into science and engineering education. However, few have directly addressed the implications of generative AI for redesigning the educational technologies that include coding, particularly how block-based and text-based modeling environments must evolve when learners can construct programs through natural language.

This workshop builds on that lineage but reframes the central question: not how to teach coding, but how coding itself—and the tools built around it—should be reimaged for an AI-mediated future. By convening researchers working at the intersection of computational modeling, design-based research, and AI in education, the session extends ongoing ISLS dialogues toward a new design agenda: understanding how AI reshapes the epistemic and pedagogical foundations of computational learning tools.

Expected outcome and outputs

The workshop will strengthen a community of practice focused on designing tools that preserve human sense-making and agency alongside AI assistance. It will generate open-ended questions that help rethink computational educational technologies in an AI-mediated context, and build a shared vocabulary around emerging hybrid practices where coding, prompting, and modeling intersect. Rather than defining what the future of coding should be, the session aims to open a dialogue on how our tools—and our field—must evolve to meet it.

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Stories of Practice for Upscaling Educational Innovation: Within and Between Research-Practice Partnerships

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Abstract: This full-day workshop invites Learning Scientists engaged in Research–Practice Partnerships (RPPs) to explore how stories of practice, linked with pedagogical design principles through AI-supported tools, can advance the upscaling of educational innovation. Prior to the conference, participants will use an ethically approved chatbot to elicit reflective narratives from teachers, highlighting their experiences in implementing and adapting innovations. During the workshop, participants will collaboratively analyze these stories and use a second chatbot to connect them to design principles within an existing repository, discussing implications for scaling across contexts. Through this hands-on and reflective process, participants will co-develop insights about fostering teachers' agency and ownership within RPPs. The workshop will conclude with a collective reflection on lessons learned and possible next steps for building a sustained community of practice around story–principle methodologies and ethical, narrative-based approaches to scaling educational innovation.

Overview

Grounded in the conference theme of "partnering with purpose", this full-day workshop responds to a growing need among Learning Scientists engaged in Research–Practice Partnerships (RPPs) to better understand how to capture and scale the local, practice-based knowledge that drives meaningful innovation. Many RPP teams at ICLS and beyond have been grappling with how to document, analyze, and share such knowledge in ways that recognize teachers' voices and support collective learning. This workshop invites participants to engage in a hands-on exploration of how stories of practice can serve as a bridge between local knowledge and scalable, cross-partnership learning. Prior to the conference, participants will be invited to collect identity-oriented stories of practice from teachers in their partnerships, using a dedicated, ethically approved AI-based chatbot that we will provide. This chatbot, previously validated as an effective elicitation tool, guides teachers in crafting reflective narratives that surface their experiences, challenges, and insights about implementing or scaling innovations. A short pre-conference online session will be offered to introduce the tool and demonstrate its use.

During the workshop, about 10 to 30 participants will join the organizing team to analyze these stories through collaborative reflection supported by another ethically approved AI-based chatbot. This second chatbot is designed to assist in linking stories of practice with pedagogical design principles within *INSIGHTS*, a digital environment that supports bottom-up knowledge building and serves as a shared repository for reification artifacts emerging across partnerships (Kali et al., 2024). Working in thematic groups led by organizing team-members, participants will first examine how such stories reveal, refine, or extend design principles and how this process deepens understanding within their own partnerships. A collective discussion will then address how linking stories and design principles, through AI-supported tools, can promote upscaling of educational innovation across and within RPPs. The discussion will also address the curation of the design principle repository within *INSIGHTS*, considering issues of selection, representation, and ethical authorship. Discussants will offer integrative commentary connecting workshop insights to the broader legacy of design-based research within the Learning Sciences. Our ultimate goal is to create a scholarly community around this practice - one that may later develop a

symposium and/or a special journal issue on how stories of practice that are connected with research-based design principles can foster inclusive learning within and across RPPs aiming to deepen and scale educational innovation.

Importantly, the AI tools used in this process, particularly the chatbot for story elicitation, were developed and implemented under approved ethical protocols. These tools ensure full privacy and transparency: no conversational data are stored, shared, or used for training large language models. Teachers retain full ownership of their narratives and explicitly approve which stories may be shared. Stories may be written in participants' native languages and, when needed, translated through a secure, locally managed process that ensures data privacy and fidelity of meaning. The stories themselves are not anonymized, allowing teachers' professional contributions to be acknowledged, while any pre-approval exchanges are permanently deleted and remain inaccessible to researchers or third parties. This design reflects our ethical commitment to participant agency, informed consent, and responsible data stewardship within RPP-based research.

Theoretical background and significance

Efforts to upscale educational innovation—to move beyond isolated projects and generate sustainable impact—require more than replicating successful interventions. Research identifies five interrelated dimensions of upscaling: spread, depth, sustainability, shift in ownership, and evolution (Coburn, 2003; Clarke & Dede, 2009; Law et al., 2015; Law et al., 2021). Spread refers to expanding the use of an innovation across classrooms or institutions; depth involves changes in pedagogical beliefs and practices; sustainability concerns maintaining the innovation over time; shift in ownership emphasizes practitioners' leadership; and evolution highlights adaptation and refinement across contexts. These dimensions underscore that upscaling is not only about growth in quantity but also about the quality and nature of educational change. Together, these dimensions show that educational change depends not only on transferring tools or practices but on building shared understanding, agency, and capacity among diverse actors. Within the Learning Sciences, such dynamics increasingly take shape through Research–Practice Partnerships (RPPs) that foster long-term collaboration between researchers, practitioners, and community members (Coburn & Penuel, 2016; Fishman et al., 2013; McKenney, 2016).

Yet even as RPPs aim to democratize innovation, many RPPs still face challenges of knowledge asymmetry and unequal voice—particularly when teachers' experiential knowledge remains implicit or underrepresented in the collective learning process (Datnow, 2020). To achieve meaningful upscaling, local insights must not only be surfaced but also translated into shared knowledge artifacts that circulate across partnerships and contexts. One promising approach lies in connecting stories of practice—teachers' reflective narratives of implementing innovations—with design principles—conceptual formulations that abstract and communicate lessons from practice (Bell et al. 2004; Kali, 2006; Kali & Linn, 2008).

We suggest that stories of practice (Sfard & Prusak, 2005; Luehmann, 2008) may help address this challenge, as they encapsulate tacit, situated knowledge about pedagogical enactment and identity. When elicited and shared within RPPs, such stories may function as “boundary objects” that invite multiple interpretations and foster dialogue among teachers, researchers, and other stakeholders (Akkerman & Bakker, 2011). Our previous work demonstrated that AI-based chatbots can scaffold the elicitation of identity stories, supporting teachers in articulating meaningful experiences that surface professional insights and shifts in identity. When collectively examined and linked to design principles, these stories become reified knowledge elements within a broader digital network, connecting diverse projects, partners, and perspectives (Aridor et al., 2022).

Building on this framework, this workshop proposes story–principle linking as a novel methodology for networked learning and reflection within and across RPPs. By engaging researchers in hands-on work with stories of practice from their own partnerships, elicited through AI-supported tools, we aim to collaboratively explore how such narrative-design interactions can: (1) Empower teachers' voices as co-constructors of knowledge; (2) Provide RPP leaders with a structured, research-based methods and tools for capturing and organizing local insights; and (3) Offer a mechanism for scaling innovations through conceptual connectivity and cross-partnership learning rather than replication (Kali et al., 2025). By positioning stories of practice as both reflective and epistemic artifacts, we propose a pathway toward upscaling educational innovation through relational, narrative, and principled learning.

Organizing team

To ensure an international, diverse, and representative dialogue that reflects the richness of the Learning Sciences community, the organizers convened an international group of scholars who represent a range of Research–Practice Partnerships (RPPs) dedicated to advancing educational innovation across varied domains, regions, and contexts. Each team member brings a complementary perspective on educational innovation. Rikke Magnussen will share stories of practice from the development of the Danish national *GreenEdTech* initiative and the *GetGreen* digital learning space, which hosts student–industry partnerships and supports collaboration on

addressing societal sustainability challenges. Camillia Matuk will share examples from various projects based out of NYU's RIDDLE lab, which partners researchers and schools in co-designing digital tools that support student-driven inquiry and data storytelling. Paulo Blikstein and Tamar Fuhrmann will share stories from the *Transformative Learning Technologies Lab (TLTL)*, where teachers collaborate with researchers to design and implement makerspace-based learning experiences across diverse school contexts worldwide, including Brazil and Hong Kong. Keren Aridor and Osnat Atias will draw on the Taking Citizen Science to School (TCSS) center, highlighting collaborations in which schools and scientists co-create citizen science projects that advance both scientific understanding and student learning.

Workshop goals and intended audience

The overarching goal of this workshop is to explore how stories of practice, elicited and analyzed through design principles and supported with AI tools, can advance the upscaling of educational innovation within and across Research–Practice Partnerships (RPPs). Participants will engage in hands-on collaboration linking stories of practice to pedagogical design principles, transforming local insights into shareable conceptual knowledge. The workshop will encourage critical reflection on the design principles within *INSIGHTS*—its curation, authorship, and evolving nature—and foster dialogue on how this process strengthens teachers' voices, enhances ownership of innovation, and supports sustainable change across partnerships. Beyond the session itself, the workshop aims to lay the groundwork for continued collaboration, through a joint symposium and future publication on inclusive, narrative-based approaches to scaling. By the end of the day, participants will have experienced and theorized a concrete methodology for connecting stories, design principles, and upscaling processes, gaining conceptual and practical tools adaptable to their own RPPs.

This workshop is designed for 10–30 participants, including educational researchers and practitioners who are actively involved in, or lead RPPs seeking to scale educational innovations. Participants are encouraged to draw on experiences from their own partnerships and, when possible, to bring stories of practice elicited in advance using the provided chatbot tool. Sharing stories is optional; others may participate as commentators and collaborators. The session will be most relevant for those interested in making teachers' tacit knowledge visible and actionable, experimenting with AI tools for reflection and knowledge building, discussing the curation of shared design principle repositories, and co-developing frameworks for scaling innovation across diverse educational networks. We expect a diverse group of participants representing multiple RPP contexts, from STEM and digital learning to community-based innovations, committed to advancing equitable, knowledge-rich researcher–practitioner collaboration. While most participants will come from the Learning Sciences community, we also welcome educational practitioners engaged in RPPs who wish to share their experiences and perspectives.

Outline of planned activities and facility requirements

This full-day workshop will combine pre-conference preparation, structured collaborative activities, and collective reflection. The design balances hands-on engagement with theoretical discussion, enabling participants to experience, analyze, and co-develop methods for connecting stories of practice and pedagogical design principles as a means of upscaling educational innovation. The workshop requires a flexible space that supports both plenary and group work, equipped with a projector, reliable internet access, and movable tables. Shared online workspaces and digital whiteboards will be used throughout the day to support collaborative visualization and documentation.

Prior to the workshop, registered participants will be invited to elicit identity-oriented stories of practice from teachers within their own RPPs using a dedicated AI-based chatbot that we will provide. The chatbot scaffolds story creation through reflective prompts that help teachers articulate their experiences and insights about implementing or scaling educational innovations. The chatbot does not replace human interaction; rather, it serves as a structured prompt system facilitated by an RPP team member who mediates the reflective dialogue with the teacher and records the evolving story. To ensure comfort and accessibility, an optional online orientation session will be offered in advance of the conference, introducing the tool and providing practical guidance for its effective use.

The full-day workshop will follow the schedule described in Table 1.

Table 1
Workshop Schedule

Time	Activity	Description
09:00–09:30	Opening and	Introductions, overview of workshop goals, and short theoretical

	framing	framing linking <i>stories of practice, design principles, and upscaling</i> .
09:30–10:30	Story sharing and reflection I	Participants who collected stories using the chatbot, share experiences from the elicitation process—how teachers engaged, what themes emerged, and how this shaped understanding of innovation.
10:30–11:00	Coffee break	
11:00–12:30	Story sharing and reflection II	Continuation of story presentations and collective reflection in plenary. Participants identify emerging themes, tensions, and knowledge types revealed by the narratives.
12:30–13:30	Lunch break	
13:30–15:00	Collaborative linking to design principles	Small thematic groups use the second AI-based chatbot and the design principle repository within <i>INSIGHTS</i> to connect stories to relevant principles, identify new or evolving ones, and discuss implications for upscaling within their own RPPs.
15:00–15:30	Coffee break	
15:30–16:30	Comments by discussants and summary	Discussants share integrative insights and guide a whole-group dialogue on the ethical issues such as: curation, evolution, and representation of design principles in collective knowledge building.
16:30–17:00	Next steps and closing discussion	Synthesis of workshop insights, planning for continued collaboration (e.g., joint symposium, future publications), and collective reflection on key takeaways.

Relationship to similar events conducted in the past

Our workshop continues a long-standing ISLS trajectory of developing shared infrastructures for design knowledge. Early workshops on design principles and knowledge infrastructures (Kali et al., 2002; Hoadley, Kali, & Linn, 2004) led to the creation of the Design Principles Database, a digital platform that, although no longer active, laid the conceptual foundation for the current design-principle repository within *INSIGHTS*. This trajectory evolved through later ISLS workshops focusing in directions such as emphasizing teachers' roles as designers of technology-enhanced learning, and the ethics of co-designing learning environments (Lui et al., 2025). The current workshop builds on this accumulated knowledge and extends it into new frontiers made possible by AI-supported tools, enabling teachers' practical wisdom and lived experience to play a more central, credited, and equitable role in the collective knowledge building of RPPs.

Expected outcome and outputs

By the end of the workshop, participants will have engaged in a structured, AI-supported process that transforms teachers' stories of practice into shared conceptual knowledge within their RPPs. They will gain enhanced methodological capacity through hands-on experience with the chatbots and continued access to the tools. The collaborative work will generate shared conceptual insights, represented as new entries within *INSIGHTS* that connect stories of practice to pedagogical principles and illustrate diverse pathways for scaling innovation. Beyond these tangible outcomes, the workshop will strengthen a growing community of Learning Scientists committed to narrative-based approaches to scaling educational innovation. It will pave the way for future scholarly dissemination through a joint symposium and a special journal issue. Finally, participants will engage in reflexive and ethical dialogue on curation, authorship, and inclusion in educational innovation.

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Imagining Possibilities for Bridging Research and Practice to Foster Mutual Learning and Lasting Impact

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Abstract: This four-hour interactive workshop invites participants to explore how expansive learning bridges the research–practice gap through equitable, sustained collaboration. Drawing on our *Hearing Each Other's Voices* framework for bidirectional partnerships and knowledge mobilization, the session engages researchers, practitioners, and educational leaders as co-learners in imagining new solutions and co-designing activities that foster mutual learning and transformation (Engeström & Sannino, 2017). Participants will experience an authentic researcher–practitioner exchange, reflect on challenges and opportunities from their own contexts, and contribute to a growing community committed to building shared language and co-created practices. Attendees will leave with practical tools, guiding principles for sustainable collaboration, and concrete steps for advancing partnerships that enhance the usability and impact of learning sciences research.

Workshop theme and goals

The workshop centers around the need to bridge the persistent gap between research and practice, which strongly relates to the 2026 ISLS theme of Partnering with Purpose. In our session, we will consider collaborative approaches that cross the boundaries between research and practice. We present a vision for bidirectional professional learning in which both researchers and practitioners learn from each other, challenge traditional hierarchies, and foster a dynamic exchange of knowledge to bring the worlds of research and practice together. Through this exchange, we aim to support participants to produce new research that is usable and actionable and fosters mutual learning.

This workshop provides a unique opportunity to gain both theoretical and practical insights to create new possibilities for research and practice. While frameworks such as design-based research (DBR), improvement science, design-based implementation research, and Research-Practice Partnerships (RPPs) have strengthened collaboration and provided models for working together, the field continues to struggle with impact and sustainability after formal studies conclude (Brown, 1992; Coburn et al., 2013; Levin, 2013; Penuel & Gallagher, 2017). There are few places where researchers and practitioners have opportunities to interact in sustained ways (Dragnić-Cindrić & Fusco, 2025). The purpose of this workshop is grounded in theoretical perspectives that advocate for explaining how researchers and practitioners engage in sustained collaboration.

Theoretical framing and background

Our workshop is grounded in expansive learning (Engeström & Sannino, 2017), which emphasizes collective activity, reflection, and the creation of new knowledge across boundaries. In this case, collaboration between researchers and practitioners becomes a site for mutual learning and innovation. The workshop interleaves three complementary perspectives that ground the HEOV project: 1) the relational and affective dimensions of collaboration, 2) design principles for equitable partnerships, and 3) knowledge mobilization and boundary crossing. Together, these frameworks guide how we conceptualize and facilitate learning across research and practice.

Relational and affective dimensions of collaboration

We build on scholarship highlighting the affective, social, and relational dimensions of sustaining partnerships (Potvin et al., 2024; Schneider & van der Linden, 2023; Walsh et al., 2021). This work also emphasizes the importance of capacity building and mutual ownership (Foster-Fishman et al., 2001; Franco, 2019), as well as

research on sustained partnerships between teachers, STEM coaches, and researchers that foster shared learning and long-term impact (Arastoopour Irgens et al., 2023; Borowczak et al., 2025; Datnow et al., 2025; Farrell et al., 2021; Giamellaro & Siegel, 2023; Susin et al., 2023). Within co-design processes, researchers and practitioners do more than just design curriculum or tools. Their relationships support deeper understanding and more meaningful impact, especially when the practitioner's context is centered in the collaboration. These relational and contextual elements are foundational for equitable collaboration and help teams intentionally co-construct conditions for collaboration (Victor et al., 2025). We also emphasize the importance of beginning with practitioners' needs, questions, and ideas when shaping collaborative work (Victor, 2026).

Design principles for equitable partnerships

Our model includes three conjectures that we present as design principles for equitable collaboration. First, ensuring every voice is heard requires intentionally disrupting power structures. Second, research articles serve as boundary objects that support dialogue between researchers and practitioners. Third, sustained spaces and activities for conversation help partnerships adapt ideas productively across contexts (McClain et al., 2024; Miller et al., 2025).

Knowledge mobilization and boundary crossing

The workshop will also discuss the important perspective of Knowledge Mobilization, the process of moving research knowledge into practical use. We examine how this process can be supported between the systems of research and practice. In this work, both researchers and practitioners act as brokers who help translate ideas to bridge research and classroom practice (Levin, 2013; Rycroft-Smith, 2022). Expansive learning provides a framework for understanding how this boundary crossing can generate shared understanding and new possibilities for collaboration between the two fields (Engeström & Sannino, 2017).

Workshop structure and activities

This four-hour interactive workshop is structured to share and extend the Hearing Each Other's Voices (HEOV) approach to equitable researcher-practitioner collaboration. Taking an expansive learning approach, we focus on a reciprocal process that enriches both domains, inviting participants to share, learn, and create together toward new theory and practice. Participants will engage in a progression of activities that move from reflection to enactment to co-design, mirroring the principles of expansive learning that guide the HEOV project. The design ensures that participants first ground their learning in their own "problems of practice," then experience an authentic researcher-practitioner exchange through role play and finally collaborate to develop strategies and tools for use in their own contexts. This structure balances clarity and flexibility, supporting active engagement while allowing participants to contribute new insights and adaptations that strengthen the HEOV model.

Learning goals and outcomes

Participants will:

- Identify key challenges in bridging the gap between research and practice in learning sciences and gain both theoretical and practical insights to create new possibilities for research-practice collaborations.
- Learn about the workshop team's model for collaboration and discuss key elements.
- Explore ways to support knowledge mobilization and the role of "brokers" in translating research.
- Connect with others in the ISLS community who are working to bridge research and practice.

These goals invite participants to move beyond reflection toward transformation by examining assumptions, practices, and conceptual frameworks in research and practice. We will explore questions such as:

- What should be preserved, and what must be transformed to create a more collaborative culture?
- How can we build shared language and joint inquiry that led to co-created solutions?

The workshop fosters boundary crossing and collective learning that support equitable and sustainable RPPs.

Time	Focus	Description
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0:00–0:30	Welcome, framing, and introductions	Overview of workshop goals, meet the participants and team, introduce the <i>Hearing Each Other's Voices</i> (HEOV) project, and set community norms for equitable participation.
0:30–1:00	Context mapping: Surfacing problems of practice	In small groups, participants identify a current “problem of practice” or “possibility for new practice” from their own contexts. Participants' ideas are captured on shared templates (digital or physical).
1:00–2:00	Role play: Experiencing and discussing the HEOV model	Using an authentic HEOV case, practitioner and researcher facilitators model showing how a partnership identifies needs, negotiates focus, and co-creates a Research Practice Brief (RPB). Participants then embody their roles (researcher/practitioner) to simulate a dialogue around their own issues. The conjectures of the HEOV model will be discussed.
2:00–2:15		Break
2:15–3:15	Co-design and adaptation	Building on insights from the role play, participants discuss the design of potential “translational products” (a RBP or alternatives e.g., partnership principles, reflection tools, alternative briefs) relevant to their own contexts. Facilitators circulate to support design thinking and highlight expansive learning moments.
3:15–3:45	Sharing and synthesis	Groups share key takeaways: insights, prototypes, and conditions for equitable collaboration. Facilitators synthesize across groups to highlight common principles for sustainable partnerships.
3:45–4:00	Closing reflections and next steps	Collective reflection on what participants will take forward and how they might continue connecting post-conference (e.g., through shared digital space or follow-up session).

Audience and relevance to ISLS

This session invites participation from:

- Learning scientists seeking to translate findings that can have durable impact
- Teachers, instructional coaches, and school leaders eager to work with researchers and share the needs they have for evidence-based practices in their school contexts
- Early-career researchers building RPPs or DBR projects
- Policy and design professionals focused on deeper impacts from research.

Aligned with the ISLS themes of design, collaboration, crossing boundaries, expansive thinking, and equity, the workshop will advance collective understanding of how closing the research practice gap can be theorized, operationalized, and practiced. Grounded in the theory of expansive learning, the session invites participants to experience and reflect upon the processes of boundary crossing and knowledge mobilization that occurs when researchers and practitioners co-create new knowledge.

Call for participation

We invite learning scientists, practitioners, and educators—principals, instructional coaches, district staff and administrators, teacher leaders, and coaches—to join us for this interactive, four-hour workshop on bridging research and practice. Grounded in expansive learning, this session engages participants as co-learners in a dynamic process of reflection, enactment, and co-design. Participants are encouraged to bring a piece of work from their own context, such as student data, a problem of practice, or a research artifact they wish to make more usable, to inform collaborative activities during the session. Together, we will explore what it takes to build equitable, sustainable partnerships that transform both research and practice. Participants will experience components of the *Hearing Each Other's Voices* project's approach firsthand, working alongside researchers and practitioners to identify problems of practice, experiment with collaborative tools such as Research Practice Briefs, and imagine new ways of mobilizing knowledge across contexts. Through these activities, we aim to co-create principles and practices that extend beyond the session to strengthen the broader ISLS community's capacity for collaborative, boundary-crossing inquiry.

About the organizers and project

The organizers of this workshop come from the *Hearing Each Other's Voices* project team, an NSF-funded initiative that explores how bidirectional partnerships between researchers and practitioners can strengthen the use and impact of research. The team brings expertise in the learning sciences, collaborative learning, teaching,

science education, and emerging technologies. Led by Cindy Hmelo-Silver and Judi Fusco, the team includes Joshua Danish, Haesol Bae, Krista Glazewski, Wendy Walter, Jessica McClain, Linette Victor, Jerika Miller, and Danielle Murphy.

Cindy Hmelo-Silver's work on problem-based and collaborative learning has shaped core theories of how learners and teachers co-construct knowledge, including her recent co-editing of a *Cognition and Instruction* special issue on co-design in teacher–researcher partnerships. Judi Fusco has led and studied research–practice partnerships since the 1990s, contributing to the field's understanding of networked learning communities and educator and researcher collaboration. Joshua Danish is well known for his research into the role of representations including drawings, simulations, embodiment, and play in supporting learning. He previously led the Representations for Teachers as Learners (RepTal) project, exploring how to help in-service teachers support learners in using representations in elementary science classrooms. Haesol Bae is a former classroom teacher and has been researching to understand how teachers can play a central role in supporting collaborative student learning while utilizing emerging educational technology, such as games and artificial intelligence. Krista Glazewski is recognized for her contributions in the areas of problem-based learning (PBL), collaborative learning, and computer-supported collaborative learning. Her research focuses on understanding how people learn in complex areas and how to design instructional systems that support effective learning processes supported by both foundational and emergent technologies. Wendy Walter is a former ELA, social studies, and multi-age teacher with over 25 years in teacher education. Her research focus is on Universal Design for Learning and curriculum design. Jessica McClain is a former middle and high school science teacher whose research centers on strengthening researcher–practitioner relationships and advancing asset-based pedagogies in P–20 educational spaces. Her work focuses on how educators and researchers can collaborate to design learning environments that recognize and build upon the diverse strengths students bring to the classroom. Linette Victor is a former chemistry professor and 6–12 STEM educator whose research focuses on connecting theory and practice through collaborative design. Her research explores how the design and implementation of professional learning experiences shape, support, and constrain the translation of evidence-based ideas into practical tools and strategies that can be readily adopted in real-world settings. Jerika Miller was a secondary English Language Arts teacher and has research interests in teacher learning and practitioner expertise, helping the team think critically about the practitioner perspective. Danielle Murphy is a former elementary teacher and an early-career learning scientist, who studies ways to foster teachers' epistemic agency in science education. Together, the team's expertise and scholarship provide the foundation for this workshop's emphasis on equitable, sustained collaboration between researchers and practitioners.

The HEOV research team, consisting of seven former practitioners, has been working together to develop a sustained community approach that provides a foundation for building trusting research-practitioner partnerships. For almost three years, the project members have matched researchers and practitioners in the United States to study the conditions necessary for developing equitable bidirectional partnerships. Through these partnerships, the team hopes to solve local problems, bring in new possibilities for learning, and highlight the importance of connecting practice to research (McClain et al., 2024; Victor et al., 2025). Over the life of the project, the team has worked with and learned from 21 practitioner-researcher teams. Five practitioners from the project researcher-practitioner pairs will be present to facilitate the workshop. The breadth of experience and expertise among the workshop facilitators demonstrates the relational infrastructure we advocate for in closing the gap between research and practice.

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Integrating the Learning Sciences into Technical Infrastructure for the Future of AI in Education

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Abstract: As artificial intelligence becomes increasingly influential in education, attention must extend beyond AI applications to the technical infrastructure on which those applications depend. Datasets, models, benchmarks, and governance mechanisms are already shaping the future of educational AI, yet these foundational resources are often developed without sufficient engagement with the learning sciences. This workshop explores how learning scientists can contribute to the design of such infrastructural public goods in ways that better reflect what is known about learning, while also addressing concerns related to safety, privacy, ethics, and equity. By bringing together researchers working at the intersection of AI and learning, the workshop seeks to identify promising lines of work, articulate key tensions, and stimulate future collaboration on learning-science-informed AI infrastructure.

Overview

The learning sciences offer strong conceptual foundations for the design of educational AI, drawing on longstanding research on scaffolding, feedback, self-regulation, and collaboration (Quintana et al., 2004; Hattie & Timperley, 2007; Järvelä et al., 2016; Sawyer, 2022). Yet these principles still only lightly shape the datasets, benchmarks, and model documentation on which AI systems depend. Training data rarely reflect what is known about productive learning interactions; benchmarks tend to measure task completion or factual accuracy rather than the quality of scaffolding or the depth of learner understanding; and model documentation seldom describes how a system's design relates to evidence on how people learn. Recent work by major AI developers signals growing interest in grounding educational AI in learning sciences principles (Jurenka et al., 2024; LearnLM Team, Google, 2025), but without more intentional integration into underlying infrastructure, educational AI risks prioritizing efficiency or engagement over meaningful support for learning. As past waves of "personalized learning" have shown, technology that optimizes for engagement without attending to how learning works can reproduce rather than address longstanding inequities in educational opportunity.

This challenge is particularly pressing as new efforts seek to build openly shared educational AI resources as public goods. The K–12 AI Infrastructure Initiative, a five-year, \$26 million effort led by Digital Promise and a coalition of partners, is funding the development of tutoring interaction corpora, learning-aligned benchmarks, and model documentation frameworks that embed pedagogical principles by design (Digital Promise, 2025; Roschelle et al., 2025). For such resources to genuinely advance learning, learning scientists must be actively involved in their co-design alongside AI engineers, education practitioners, and policymakers — contributing not only domain knowledge but also the critical perspectives needed to navigate questions of safety, privacy, ethics, and equity.

This workshop therefore explores how the learning sciences can be systematically integrated into these infrastructural public goods. It brings together learning scientists, AI developers, education practitioners, and funders to examine where and how learning sciences expertise is most needed in the development of technical infrastructure for educational AI. Its goals are to inform participants about emerging infrastructure efforts and the role learning sciences can play in shaping them, create space for participants to share current work and articulate concerns, identify promising directions for future research and investment, and stimulate the formation of cross-sector collaborations. Building on recent ISLS programming on AI and education (International Society of the Learning Sciences, 2022, 2024), the workshop positions the learning sciences not only as a field that evaluates AI applications after they are built, but as one that can help shape the foundations on which educational AI is constructed.

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GoGoData Participatory Sensing: Students, Educators and Researchers Building Collaborative Data Networks

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Abstract: In this half-day workshop, participants will engage with GoGoData, an open learning plugin that connects the GoGo Board microcontroller, sensors, Internet of Things (IoT) communication, and the Common Online Data Analysis Platform (CODAP) to support participatory sensing and collaborative exploration of real-time data. Beyond experimenting with the plugin's features for data collection, transmission, and interpretation, participants will analyze GoGoData as a design case that integrates constructionist principles with learning sciences perspectives on data literacy. Participants will, in turn, extend these theoretical and design lenses to examine how collaborative data networks can foster critical engagement with data in their own educational and research contexts.

Proposers/Organizers

Charles Pimentel

Charles Pimentel is a Research Fellow at the Transformative Learning Technologies Lab, supporting the documentation of the GoGo Board and the development of data-focused learning activities. He holds a Master's degree in Informatics from UFRJ and works in K-12 education with Programming, Artificial Intelligence, and Educational Robotics. Charles is a doctoral student at the Institute of Computing at the Federal University of Rio de Janeiro, researching Data Literacy, and a member of the FabLearn Fellows Cohort 3. He also teaches Design & Tech at Graded in São Paulo, developing tech-for-good projects connected to sustainability.

Vithoria Cabreira

Vithoria Cabreira is an undergraduate student in Computer Engineering at the Federal Technological University of Paraná (UTFPR). She supports the development of digital tools and web technologies for educational and research projects. Her interests include software development, web platforms, and the creation of digital solutions that improve user experience. Vithoria is currently strengthening her technical skills through practical projects and experimentation with new technologies.

Akio Goya

Akio is an Educational Technology Designer for the Transformative Learning Technologies Lab, coordinating the design, implementation, and documentation of GoGo Boards in Brazil, as well as the implementation of makerspaces in public schools. Akio holds a Master's degree in Instructional Technology and Media from Teachers College, Columbia University (USA) and a Master's and a Bachelor's degree in Computer Engineering from University of São Paulo (Brazil), and a Teacher Licensure in Mathematics (Middle and High School) from FIAR (Brazil).

Paulo Blikstein

Paulo Blikstein is an Associate Professor at Teachers College and an Affiliate Associate Professor of Computer Science at Columbia University, where he directs the Transformative Learning Technologies Lab and the Lemann Center for Brazilian Studies. His research focuses on how new technologies can transform the learning of science, engineering and computation. Paulo is the co-inventor of the GoGo Board, co-leading its research and development since 2001. Paulo has a Ph.D. from Northwestern University, a MSc. from the MIT Media Lab and an M.Eng. and B.Eng. from the University of São Paulo, Brazil.

Renato Russo

Renato is a doctoral candidate at the Technology, Media, and Learning program at Teachers College, Columbia University and a researcher at Transformative Learning Technologies Lab. In his research, Renato explores how learning technologies can promote critical and emancipatory knowledge construction practices. Renato earned a MA degree in Learning, Design and Technology from Stanford University Graduate School of Education.

Arnan (Roger) Sipitakiat

Arnan (Roger) directs the Teaching and Learning Innovation Center (TLIC) at Chiang Mai University in Thailand. As a faculty and researcher at the Computer Engineering Department, he also directs the Learning Inventions Laboratory (LIL). Arnan is the co-inventor of the GoGo Board, co-leading research and development on the toolkit since 2001. Arnan focuses on designing technologically-rich learning tools, such as the GoGo Board, that have allowed students to explore areas including robotics, interactive art, games, agriculture, and community development. Arnan earned his MSc. and Ph.D. from the Massachusetts Institute of Technology and B.Eng. from Chiang Mai University.

Intended audience

This workshop is intended for graduate students, educators, and researchers interested in exploring participatory sensing, critical data literacy, and learning technology design through hands-on experimentation. It particularly welcomes those studying or developing tools that connect physical computing, IoT, and real-time data exploration to promote collaborative inquiry and learning. More broadly, the session is open to all participants interested in integrating constructionist and learning sciences perspectives into the design of educational technologies. (Note that GoGoData, the GoGo Board, and CODAP are open-source platforms.)

Duration of the event

Half-day (3-4 hours, flexible depending on conference scheduling demands).

Description of the event

The GoGo Board was invented in 2001 as a low-cost and learner-friendly robotics and scientific sensing platform. In the years since, the project has undergone several design iterations, each affording new possibilities for students across diverse ages and backgrounds to explore powerful ideas in computing and STEM. In this workshop, participants will also engage with GoGoData (2025), a new open-source plugin that connects the GoGo Board to the Common Online Data Analysis Platform (CODAP) for real-time data communication and visualization. CODAP, developed by the Concord Consortium, has a long-standing history as an open-source environment for data analysis and modeling that supports inquiry-based learning. Together, GoGoData, the GoGo Board Kit version 7, and CODAP create a participatory sensing ecosystem that enables students to collect, share, and interpret live data while examining the learning sciences principles underlying the design of these tools [1-3].

Workshop Format

The workshop will be organized into five main activities.

I - Historical framing. Participants will begin with a brief discussion on the evolution of tangible interfaces within the Learning Sciences and basic education contexts. This discussion will be situated alongside the rise of big data practices and participatory or crowd sensing, showing how learner-generated, real-time data from distributed sensors can reshape inquiry, evidence making, and classroom discourse. The framing will surface design lessons on openness, transparency, and agency, linking data practices to issues of provenance, privacy, and ethics, and highlighting how collaborative data networks can support critical engagement and community action.

II - Exploring the GoGo Board 7 / GoGoData / CODAP. Participants will explore the core features of the GoGo Board version 7 and the GoGoData plugin, examining how each component emerged from research data and

iterative design decisions. Using onboard controls, they will visualize sensor values, directly before connecting to a computer to interact with GoGoData and its integration with the Common Online Data Analysis Platform (CODAP). Through this connection, participants will experience how real-time sensor data can be collected, transmitted, analyzed, and explored within CODAP's interactive environment. Facilitators will highlight how each feature embodies specific design principles that link tangible experimentation with data-driven inquiry.

III - Participatory sensing exploration. Participants will collaboratively select a variable to measure, such as temperature, luminosity, pollution, or sound level, using a GoGo Board 7 kit sensor. Working in small groups, they will connect the GoGo Board to CODAP through the GoGoData plugin and deploy sensors in different internet-connected areas across the UCI campus. Each group will collect and share real-time data through the open-source plugin integrated with the data analysis platform, contributing to a collaborative network of distributed measurements. As data accumulate, participants will analyze emerging patterns across locations, drawing inferences about environmental or contextual factors. The activity will conclude with a collective discussion where groups interpret their findings, reflect on methodological decisions, and connect their insights to broader perspectives on participatory sensing, open data collaboration, and critical data literacy in education.

IV - Conclusion and gallery walk. Participants will share and discuss the outcomes of their activities, reflecting on the lessons learned and insights gained throughout the workshop. The session will conclude with a collective show-and-tell, allowing groups to present their findings, highlight key observations from the participatory sensing exercise, and connect these experiences to broader questions about learning, design, and data literacy.

V - Post-workshop interaction. Optional follow-up opportunities will be provided for participants who wish to continue collaborating after the session. These interactions may include online discussions, exchange of design ideas, and updates on future developments of the GoGoData platform. Participants will be invited to share reflections, resources, and potential research connections that extend the learning and collaboration initiated during the workshop.

Throughout the workshop, facilitators will guide discussions connecting the design of GoGoData, the GoGo Board, and CODAP to learning sciences and constructionist frameworks, emphasizing how participatory sensing and data exploration are enhanced by open-source ecosystems. The session will highlight how openness, transparency, and accessibility in these tools empower learners and researchers to collaboratively collect, analyze, and interpret real-time data. Rather than serving as a technical training, the workshop offers a hands-on exploration of how open, theory-informed technologies developed through over twenty years of design-based research can foster agency, collaboration, and critical engagement with data.

Participation requirements

No requirements. We invite to join all participants interested in tangible interface design, educational robotics, and data science, irrespective of skill or experience levels.

Relationship to similar events conducted in the past

The author team conducted an NSF-funded project at the New York Hall of Science in collaboration with Teachers College, Columbia University, using the GoGo Board in data science camps for middle school students under the AISL program. We also did a physical computing workshop at ISLS 2024.

Facilities and equipment required

Facilities:

- Projector / screen for workshop facilitators
- Wifi connectivity

Equipment:

- Gogo Boards Kits (board and sensors - we will provide them)
- Participants will be asked to bring their laptops with an available USB-C port.

Minimal and maximal number of participants expected

8 - 15 participants

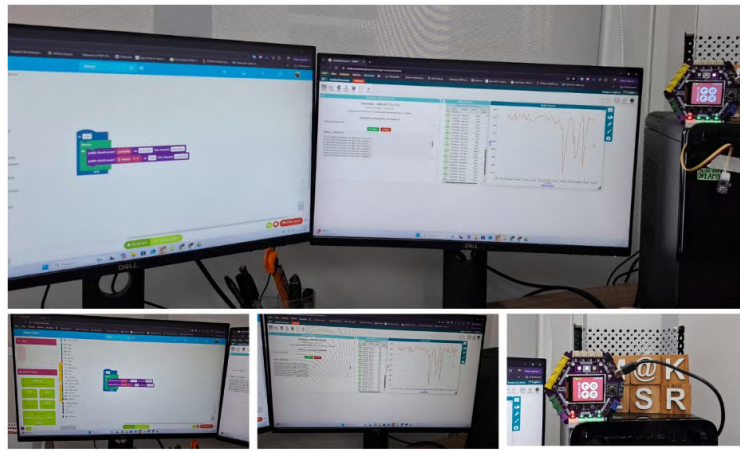
Draft call for participation

Calling all researchers, designers, and educators interested in exploring the power of participatory sensing to connect learning, data, and design.

In this half-day workshop, you will engage with GoGoData (2025), a new open-source plugin that links the GoGo Board microcontroller, sensors, and Internet of Things (IoT) communication to the Common Online Data Analysis Platform (CODAP) for real-time data exploration. Together, these tools create a participatory sensing ecosystem where learners and researchers can collaboratively collect, share, and interpret live data (Figer 1). Beyond experimenting with the platform's features, you will analyze GoGoData as a design case informed by constructionist and learning sciences perspectives, building on over two decades of iterative design research with the GoGo Board. This workshop offers a unique opportunity to experience how open-source technologies can empower communities to transform local data into meaningful insights and to apply these theoretical and design perspectives to your own educational research and development projects.

Figure 1

Participatory Sensing Ecosystem : GoGo Code - GoGo Data on CODAP - GoGo Board



GoGo Code and GoGoData CODAP plugin displaying sensor data from a GoGo Board.

Workshop format

The workshop is planned for in-person delivery, with the option of hybrid participation if necessary. The program will include interactive, hands-on activities supported by remote contributions from members of the international workshop team, ensuring collaboration and engagement across locations.

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Modelling Complex Learning Interactions—A Hands-on Workshop on Heterogeneous Interaction Network Analysis (HINA)

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Abstract: Learning is inherently a complex, multi-faceted process situated across learners, tools, and environments. This workshop introduces HINA (Heterogeneous Interaction Network Analysis), a methodological framework for modeling and analyzing complex learning interactions through networked relationships across diverse entities such as students, behaviors, and artifacts. By representing learning process data as heterogeneous interaction networks, HINA provides researchers with a means to uncover latent patterns, measure theoretical constructs, and derive actionable insights into learning processes. This half-day workshop features interactive discussions and a hands-on tutorial with the HINA webtool (hina-network.com), enabling participants to analyze their own data, followed by featured case studies and collaborative discussion on future applications in learning sciences. This workshop will contribute to the learning sciences community by disseminating a novel open-source tool for analyzing complex learning interactions, enhancing the methodological capacity for theoretical development.

Keywords: Learning Interactions, Network Analysis, Methodological Innovation, Learning Analytics

Organizer's background

Shihui Feng is an assistant professor in the Faculty of Education at the University of Hong Kong. Specializing in complex network analysis and learning analytics, her research focuses on developing new analytical methods and advancing theoretical frameworks to investigate collaborative learning dynamics and social interactions in digital environments. Her work has culminated in the development of HINA, a novel methodological framework that is implemented as an open-source tool for modeling, analyzing, and visualizing heterogeneous interactions in learning processes.

Kester Yew Chong Wong is a PhD student and part of the UCLAIT team at the UCL Knowledge Lab, University College London, UK. His research focuses on AI-based support for Collaborative Problem Solving (CPS). Kester is also a Teaching Assistant at the Nanyang Technological University, National Institute of Education (NTU/NIE) in Singapore under the Learning Sciences and Assessment Academic Group (LSA AG).

Tongguang Li is a Research Fellow at Monash University, specializing in the use of learning analytics to enhance adaptive instructional design and support self-regulated learning (SRL) processes. His current work involves the design of data-driven, GenAI-integrated intelligent scaffolding systems that promote SRL in computer-based learning contexts. As part of his fellowship, he also explores evidence-centred assessment in the context of learning with generative AI. His research investigates how GenAI tools can support writing development while driving innovation in assessment design.

Mutlu Cukurova is a Professor of Learning and Artificial Intelligence at University College London. He directs the (UCL Analytics & AI for Learning) UCLAIT team and engages in policy-making activities as an external expert (including UNESCO, IAEA, and the EC). He was the programme co-chair of the International Conference of AI in Education in 2020, is the Editor of the British Journal of Educational Technology, and an Associate Editor of the International Journal of Child-Computer Interaction.

Dragan Gašević is Distinguished Professor of Learning Analytics in the Faculty of Information Technology and the Director of the Centre for Learning Analytics at Monash University. Dragan's research interests center around data analytic, AI, and design methods that can advance understanding of self-regulated and collaborative learning. He is a founder and served as the President (2015-2017) of the Society for Learning Analytics Research. He is a

founding editor-in-chief of the *Journal of Learning Analytics* (2012-217) and *Computers and Education: Artificial Intelligence*.

Heterogenous Interaction Network Analysis (HINA)

Constructivist learning theories posit that knowledge is actively built by learners through interactions with their environment (Vygotsky, 1978; Piaget, 2013). At its core, learning is an interactive process emerging from a student's engagements with peers, instructors, and pedagogical tools (Bandura & Walters, 1977; Chi, 2009). These interactions are inherently heterogeneous, forming a network of relationships among diverse entities—such as students with learning artifacts. These heterogeneous interactions constitute the very fabric of the learning process and provide critical insights into student engagement and the impact of learning environments. To unlock these insights, there is a critical need for analytical methods capable of modeling these complex interactions to drive theoretical development in learning sciences.

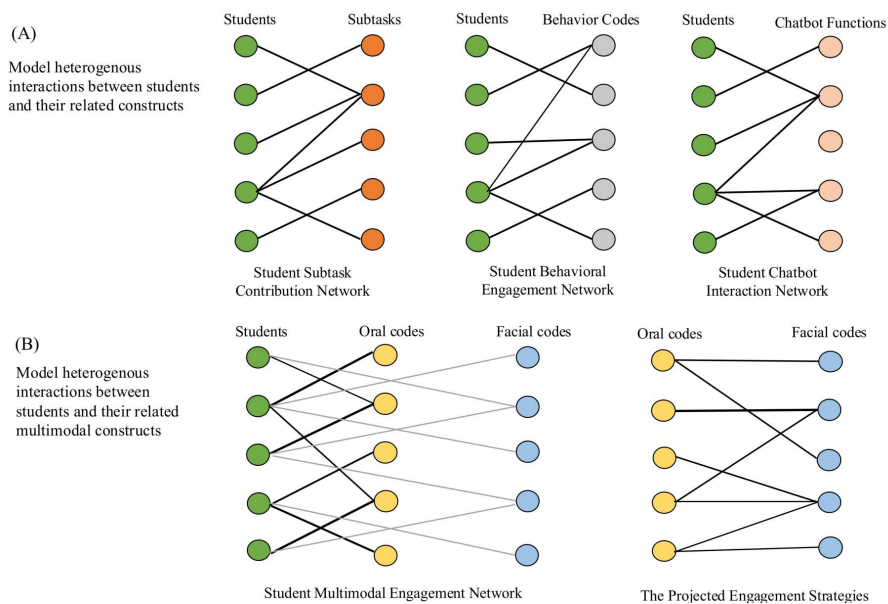
HINA (Heterogeneous Interaction Network Analysis) is a novel learning analytics framework designed for modeling, analyzing, and visualizing heterogeneous interactions in learning processes (Feng et al., 2025a). It formalizes multi-entity interactions in learning processes as Heterogeneous Interaction Networks (HINs), where nodes represent distinct entities—such as students, artifacts, or coded behaviors—and edges represent the relationships between them, such as engagement or association (see examples in Figure 1). Existing network-based learning analytical methods such as Epistemic Network Analysis (ENA) provide effective tools to analyze the relationships among a single type of nodes, for instance, modelling the co-occurrence of theoretical constructs within each pre-defined time window using dialogue data (Shaffer et al., 2016). HINA is fundamentally different in its focus on interactions between diverse entity types and in its suite of original analytical methods, which are tailored to measure and identify the underlying characteristics and patterns of the studied heterogeneous learning interactions.

HINA includes three-level original analytical methods, including 1) node-level analysis for measuring the quantity and diversity of interactions; 2) dyadic-level analysis for identifying statistically significant edges under a specified null model; 3) meso-level analysis for clustering nodes based on their heterogeneous interaction patterns with a novel non-parametric clustering method. In addition, HINA offers an interactive network visualization for presenting the analytical results and customizing network visualizations for feeding back the learning analytics results (see an example in Figure 2). Researchers can access these functions: by installing the HINA Python package (`pip install hina`) or by accessing the user-friendly HINA web tool at hina-network.com, which requires no coding experience.

As a peer-reviewed and empirically validated framework (Feng et al., 2025a; Feng et al., 2024; Feng et al., 2025b), HINA offers a flexible tool for modeling heterogeneous interactions across diverse learning contexts. Its applicability is especially pertinent to the growing field of human-AI collaboration. HINA provides an innovative method for analyzing student-AI interactions, enabling researchers to quantify contributions, identify engagement patterns, and detect key relationships. These capabilities offer new perspectives for understanding human learning with AI and for studying the dynamics of human-AI hybrid intelligence (Cukurova, 2025), positioning HINA as a valuable methodological tool for advancing the theoretical understanding of complex learning interactions in AI-assisted learning environments.

Figure 1

Illustrations of Heterogenous Interactions in Learning Processes Modelled by HINA (Feng et al., 2025b)

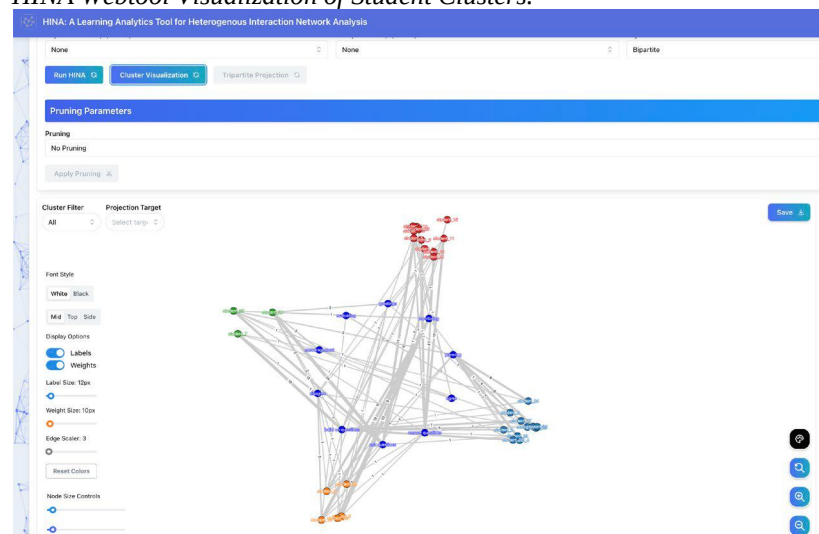


Workshop objectives

This workshop is highly interactive, aiming to provide researchers and practitioners with both the methodological details and practical skills to use HINA for analyzing learning process data. Participants will gain hands-on experience with the HINA web tool through guided analysis of real-world datasets, learning to identify interaction patterns, and generating actionable insights. We will explore HINA's applications through collaborative case studies in contexts like problem-solving and human-AI interaction, offering concrete models for participants' own research. The workshop will conclude with an interactive session dedicated to fostering collaborations, addressing participants' analytics needs, and charting future methodological directions. This dialogue aims to cultivate a sustained community of practice committed to collaboratively advancing method-enabled theoretical development in the learning sciences.

Figure 2

HINA Webtool Visualization of Student Clusters.



This snapshot shows the results of clustering students based on similar cognitive and metacognitive engagement patterns, identified using a novel Minimum Description Length (MDL) based community detection method.

Workshop details

Target Audience and Prerequisites

This workshop is designed for researchers and practitioners interested in learning process data, interaction analysis, and theory-driven modeling. It is particularly suited for those who have learning process datasets and are seeking new analytical approaches to make sense of them. The session is accessible to all; no prior network analysis or programming experience is necessary.

Workshop Agenda (half-day)

9:00-10:00 am	Welcome & Methodological Foundations of HINA
	Introduction to the core concepts and theoretical underpinnings of the Heterogeneous Interaction Network Analysis framework.
10:00-10:45 am	Interactive Session —Getting Started with HINA Web Tool
	covering dataset preparation, basic functionalities, and data visualization.
10:45-11:00 am	Break
11:00-12:00 pm	Featured presentations on HINA's applications. — Collaborative problem solving — Self-regulated learning — Human-AI interactions
12:00 -12:30 pm	Analyze Your Dataset & Interactive Discussions Analyzing participants' own dataset while co-designing future research/development

Logistics and requirements

No specialized equipment is required. Participants will need a laptop with a stable internet connection to access the HINA web tool.

Intended outcomes

Participants will gain practical experience by applying HINA to sample datasets in guided tutorials, building a foundation in network analysis and its application to learning data. They will leave with reusable analytical instructions and the skills to examine new research questions using HINA. Furthermore, this workshop is designed as a starting point for a sustained community effort; participant engagement and feedback will directly contribute to the tool's continued evolution, creating impact that extends far beyond the session itself.

Evaluation and follow-up

To evaluate and extend the workshop's impact, we will collect participant feedback on their takeaways and suggestions to guide HINA's iterative development. All materials—including tutorials, datasets, and scripts—will be made available on an open-access platform (e.g. Google drive) to facilitate immediate application. Furthermore, participants will be invited to a dedicated community forum to foster continued collaboration, troubleshooting, and co-development. This integrated approach ensures that the workshop contributes to sustained innovation and community-driven growth in learning sciences.

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Methodological Dialogues in the AI Era: Reconciling Tensions and Advancing Learning Sciences Research in STEM Education

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Abstract: This ISLS 2026 half-day workshop confronts the urgent methodological tensions in AI-enhanced STEM education research. We move beyond technical “how-to” to address deeper epistemological challenges: reconciling data-driven AI analytics with interpretative qualitative inquiry, neurocognitive with socio-cultural lenses, and automated analysis with human interpretation. Designed for learning scientists, STEM education researchers, and methodologists actively grappling with these paradigm tensions, the workshop employs keynote dialogues, deep-dive sessions, and collaborative synthesis. Participants will co-develop integrative frameworks, formulate practical guidelines for methodological evolution, and build a community dedicated to principled pluralism. Key outcomes include a position paper on methodological reconciliation, cross-paradigm research teams, and a special issue proposal. This session directly responds to the need for a refined, ethically grounded methodological toolkit to understand learning within increasingly complex, AI-mediated environments, fostering rigorous and impactful future research.

Organizers

- Sun Daner (Technology-Enhanced STEM Education, Emerging Technologies, Higher-Order Thinking)
- Chen Wenli (Computer-Supported Collaborative Learning, Techno-Pedagogical Design, Collaborative Learning Environments)
- Yang Yubin (Data-Driven Learning Assessment, Learning Analytics, STEM Education)
- Chen Gaowei (Learning Sciences; Artificial Intelligence in Education; Educational Dialogue)
- Morris, Siu Yung Jong (AI/VR/XR in Education, Gamified Learning, Teacher Professional Development)
- Pei Leisi (Educational Neuroscience, Mathematics Education, Handwriting Acquisition)
- Yang Yin (Cognitive Science in Learning, Digital Literacy, Emerging Technologies)
- Zhao Li (Classroom Teaching Behavior Research, Curriculum Resource Development, Media Literacy)
- Zhu Wangda (Interaction Design, AI in Education, UX Research Methods)
- Zhu Gaoxia (Student-Student Collaborative Inquiry Learning), Human-AI, Pedagogical Design)
- He Qiwei (Advancing Techniques in Behavioral Pattern Mining in Assessments for Education, Data Mining & Analysis, Education Data)

- Li Hongli (Educational Measurement, Quantitative Methods)
- Zhan Ying (Feedback in Higher Education, Online Formative Assessment, Interdisciplinary Curriculum Development and Implementation)

Intended audience

This workshop is designed for scholars and researchers at the forefront of methodological innovation and critical inquiry in technology-enhanced STEM education. It invites *Learning Scientists* reflecting on how AI redefines the study of cognition and collaboration; *STEM Education Researchers* integrating data-rich methods with theoretical rigor; *Educational Technology Scholars* critically examining AI's impact on pedagogy and research; and *Methodologists* navigating evolving paradigms of validity and ethics in data-intensive environments. We particularly welcome those actively grappling with core tensions between data-driven and theory-driven inquiry, scalable quantification and contextualized insight and who seek to collaboratively forge new pathways for rigorous, integrative, and ethically grounded research in the AI era.

Duration

Half-day (3.5 hours)

Description of the event

Theme and goals

This workshop directly confronts the methodological dualities and tensions reshaping learning sciences research in the era of AI-enhanced STEM education. Moving beyond mere innovation, our focus is on reconciling and productively advancing competing research paradigms, such as data-driven AI analytics versus interpretative qualitative inquiry, or neurocognitive evidence alongside socio-cultural perspectives. Our primary goals are to:

- Identify and critically examine the core methodological tensions introduced or amplified by AI in learning research.
- Co-develop integrative frameworks that honor paradigmatic strengths while enhancing overall methodological rigor.
- Formulate practical guidelines for the complementary evolution of both established and emerging research methods.
- Foster a sustainable community of scholars dedicated to methodological pluralism, dialogue, and principled advancement.

Ultimately, this effort is grounded in a deeper understanding of the essential, dynamic relationship between STEM education and learning sciences methodologies.

Theoretical background and relevance

The rapid integration of AI and data-intensive technologies in STEM education has generated complex methodological tensions, challenging the foundational approaches of the learning sciences. This workshop focuses explicitly on four critical dimensions of this challenge: the epistemological tension between predictive AI models and interpretative qualitative research; the methodological duality of integrating multimodal learning analytics with rich ethnographic classroom studies; the paradigmatic negotiation between cognitive-neuroscientific and socio-cultural frameworks for understanding STEM learning; and the practical challenge of maintaining methodological integrity while adopting emerging AI tools. Additionally, we confront the evolving nature of human–AI interaction in STEM learning environments, which disrupts traditional conceptions of collaboration, agency, and knowledge co-construction, requiring researchers to re-examine long-standing theories of social learning, metacognition, and distributed cognition in technologically mediated settings. This workshop thus responds to an urgent need to refine and expand our methodological toolkit in order to capture, interpret, and ethically engage with the nuanced and rapidly changing realities of learning in the AI era.

Outline of planned activities

1. Framing the Dialogues: Methodological Tensions in the AI Era (40 minutes)

- Keynote: “Beyond Either/Or: Navigating Methodological Pluralism in AI-Enhanced Learning Research”
- Panel discussion: Organizers present contrasting methodological perspectives on studying AI in STEM education

2. Methodological Dialogues: Deep-Dive Sessions (90 minutes)

- **Dialogue A: Data-Driven vs. Theory-Driven Approaches**
Focus: Balancing learning analytics with CSCL theoretical frameworks in STEM collaboration research
 - **Dialogue B: Neuro-Cognitive vs. Socio-Cultural Lenses**
Focus: Integrating educational neuroscience with classroom interaction analysis in mathematics education
 - **Dialogue C: Automated Analysis vs. Human Interpretation**
Focus: Combining AI-powered multimodal analytics in STEM problem-solving
 - **Dialogue D: Technology-Centric vs. Pedagogy-First Approaches**
Focus: Reconciling technological affordances with pedagogical principles in immersive STEM learning
- 3. Synthesis: Toward Methodological Reconciliation and Advancement (50 minutes)**
- Each dialogue group presents frameworks for methodological integration
 - Plenary discussion: “What does methodological advancement mean in the AI era?”
 - Development of principles for dual improvement of research methods
- 4. Practical Applications: Designing Methodologically Integrated Studies (30 minutes)**
- Participants apply workshop insights to their own research designs
 - Collaborative development of methodological guidelines for specific research contexts

Expected outcomes and contributions

1. A position paper: “Methodological Dialogues in AI-Era Learning Sciences: Principles for Reconciliation and Advancement”
2. Formation of 2-3 cross-paradigm research teams to pursue methodologically innovative projects
3. Proposal for a special issue on methodological innovation in learning sciences research

Participation requirements

We welcome both early-career and established scholars who are prepared to engage in reflective, constructive, and forward-looking dialogue on the future of research methods in the AI era. Prior experience with AI tools is not required, but a thoughtful stance on methodological challenges is essential.

Relationship to past events

While previous workshops have explored either specific methods or technological applications, this workshop uniquely focuses on the methodological dialogues and tensions emerging in AI-era learning sciences. It moves beyond technical “how-to” to address deeper epistemological questions about knowledge production in technology-enhanced STEM learning research.

Facilities and equipment required

- Projector and screen
- High-speed Wi-Fi
- Movable tables and chairs for group work
- Multiple whiteboards or digital displays for parallel group work
- Organizers will provide collaborative digital platforms
- Participants should bring laptops

Participant numbers

- Minimum: 8
- Maximum: 25

Recruitment strategy

1. ISLS Official Channels: Primary promotion through conference website, newsletter, and social media
2. Targeted Methodological Communities: Direct outreach to AERA SIGs on Advanced Studies of Methodology, Qualitative Research, and Learning Sciences; EARLI methodology-focused networks
3. Organizer Networks: Leveraging connections in learning sciences, STEM education, and educational technology communities

4. Special Interest Outreach: Engaging communities at the intersection of AI ethics, research methodology, and STEM education

Selection process

We will use a curated selection process to ensure:

- Balanced representation across methodological traditions
- Diversity of perspectives on key methodological debates
- Potential for constructive dialogue across paradigmatic boundaries
- Geographic and career-stage diversity

Selected participants will receive pre-workshop readings on methodological debates in learning sciences and be asked to prepare a brief reflection on their methodological positioning.

Beyond the Tutor: Co-Designing AI Support for Collaboration and Agency in Complex Learning Systems

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 Nathan Holbert, Teachers College, Columbia University, holbert@tc.columbia.edu
 Mike Tissenbaum, University of Illinois Urbana-Champaign, miketissenbaum@gmail.com
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 Ashita Bawankule, University of Illinois Urbana-Champaign, anb8@illinois.edu
 David A Hopping, University of Illinois Urbana-Champaign, davidah3@illinois.edu
 Santiago Ospina-Tabares, University of Illinois Urbana-Champaign, so27@illinois.edu

Abstract: This half-day workshop aims to address a critical question within the combined ISLS community: how might we create AI-supported learning environments that complement and enhance the human, social, and cultural dimensions of learning, specifically with regard to complex sociotechnical educational systems such as those found in highly learner-centered and/or constructionist environments?

Proposers/Organizers

Organizer	Background (including prior experience in conducting such events)
Blake Danzig, bpd2119@tc.columbia.edu, Teachers College, Columbia University	Blake Danzig is a doctoral student in the Technology, Media & Learning program at Teachers College, Columbia University, and a researcher in the Snow Day Learning Lab and Digital Futures Institute.
Nathan Holbert, holbert@tc.columbia.edu, Teachers College, Columbia University	Dr. Holbert is an expert in the design of play-based creative technologies. He has published extensively on constructionism and the Learning Sciences.
Mike Tissenbaum, miketissenbaum@gmail.com, University of Illinois Urbana-Champaign	Dr. Tissenbaum is an expert in design approaches for collaborative learning, and has run multiple workshops at ISLS and CSCL focusing on technology and AI supported learning and orchestration.
J.K. Voorhis, jkv2109@tc.columbia.edu, Teachers College, Columbia University	J Voorhis is a doctoral student in the Technology, Media & Learning program at Teachers College, Columbia University, and a researcher in the Snow Day Learning Lab and the Transformative Learning Technologies Lab.
Emmy Semprun, es4128@tc.columbia.edu, Teachers College, Columbia University	Emmy Semprun is a doctoral student in the Technology, Media and Learning program at Teachers College, Columbia University. She is a researcher in Snow Day Learning Lab and the Digital Futures Institute.
Ashita Bawankule, anb8@illinois.edu, University of Illinois Urbana-Champaign	Ashita Bawankule has facilitated training and learning modules and workshops with middle and high school students towards creating with emergent technologies towards identity and empowerment.
David A Hopping, davidah3@illinois.edu, University of Illinois Urbana-Champaign	Mr. Hopping has facilitated numerous workshops teaching technology enhanced makerspace concepts to middle and high school students, as well improvisation workshops with undergraduate students.
Santiago Ospina-Tabares, so27@illinois.edu, University of Illinois Urbana-Champaign	Santiago Ospina-Tabares is a doctoral student in Curriculum and Instruction at the University of Illinois at Urbana-Champaign. He has facilitated technology and makerspace oriented workshops with middle and high school students.

Themes and goals

Many commercial Large Language Model (LLM) based educational tools have been pitched as learning/teaching aids or tutors. Such tools are typically designed to be sources of knowledge—whether a teacher-replacing guide through content or a question and answer machine for students to turn to when stuck. Learning Scientists, however, recognize that learning isn't simply about the accumulation of information, but rather the social construction, connection, and application of domain-specific knowledge and community practices. This workshop addresses this critical gap: how might we create AI-supported learning environments that complement and enhance the human, social, and cultural dimensions of learning, specifically with regard to complex sociotechnical educational systems such as those found in highly learner-centered and/or constructionist environments?

The ISLS community has been actively engaged in the development of AI educational environments for decades and LS and CSCL researchers are currently creating a diverse range of LLM-powered tools that provide support in these environments. While the educational technology sector frequently claims their AI platforms are developed “grounded in learning science” (Google for Education, n.d.), foundational theories of learning and teaching are largely absent. And yet, emerging LLMs have been taken up by LS and CSCL researchers to create a diverse range of learning and teaching environments. By bringing together members of the ISLS connected community (ICLS and CSCL), the goal of this workshop will be to continue to expand the imaginary of AI-supported tools in complex sociotechnical systems. Participants of the workshop will have the opportunity to share and discuss existing and in-process LLM-based educational designs, and will work together to identify shared goals, values, and tensions around the design, use, and study of AI as a support for learning and teaching. The result of this workshop will be a community-facing report documenting these shared goals and tensions to open up new opportunities for expanding theory-backed AI powered social supports and to serve as a reference for ISLS researchers, designers, and partners.

Theoretical background and relevance to field and conference

The ongoing conversation in corporate spaces around the role of AI in education has largely centered around tool designs and implementations that (tacitly) enact instructionist models of learning—where the AI is the knowledgeable expert that learners should turn to for information, as well as the instructor who dictates the pace and order of instruction (Mishra et al., 2025; Resnick, 2024). Far less explored in these corporate spaces is the design of AI-powered tools that support the social and cultural aspects of learning which are extensively researched in both the Learning Sciences (LS) and Computer-Supported Collaborative Learning (CSCL) communities.

Recent work by LS and CSCL scholars has begun to identify best practices for designing learner-centered and collaborative AI tools. For example, Druga and Ko (2025) provide future design considerations for AI Copilots for youth. Kafai et al. (2025) convened CreateAI, a workshop that generated a report with important insights and questions, for how we might build (and evaluate) constructionist experiences for learning about AI. Anderson et al. (2025) brought together a group during last year's conference to share their explorations in designing AI as support for groups in the classroom as a step in this direction.

Constructionist and project-based learning environments center learner voices, interests, and experiences. In these complex sociotechnical systems, knowledge and expertise are distributed across people, materials, technologies, artifacts, and social interaction (Holbert et al., 2020). Constructionist design paradigms recognize that to engage with meaningful and powerful ideas and to feel a sense of self-efficacy, students need to have access to the proper scaffolding for testing their models of how they think the world works, not just be “spoon-fed” information (Papert, 1980). Learning happens in interactions with diverse supports and tools (Greeno, 2011) as learners draw from available resources to address problems they care about and make artifacts that are meaningful to them and their communities. These sociotechnical systems echo two notions that learning scientists generally agree upon: (1) knowledge is actively constructed by learners (Piaget, 1964) and (2) that the “best scaffolding provides this support in a way that helps children discover the answer on their own” (Sawyer, 2022, p. 11).

Moreover, CSCL research provides foundational principles and empirically tested design knowledge for the successful integration and implementation of AI tools that genuinely enhance human collaboration and knowledge building. From a CSCL perspective, the goal is to design technologies “to create artifacts, activities, and environments that enhance the practices of group meaning making” (Stahl et al., 2022, p. 416).

Recent systematic reviews reveal that while constructionist AI literacy education shows promise, significant gaps remain in how tools are embedded in practice and how sociotechnical complexities are addressed early in learning experiences (Lin et al., 2025). We see this workshop as in conversation with scholars throughout the ISLS communities that have begun to investigate how foundational LS and CSCL theories inform the use of AI in educational design (Kafai et al., 2025; Mishra et al., 2025; Resnick, 2024). Here we focus the community's attention on the sociocultural features of AI-supported learning environments and ask: In what ways can we design

and implement AI to contribute to the broad array of social, cultural, and human elements of complex sociotechnical learning systems?

Summary of activities

The session will open with an opportunity for each participant to briefly present their AI-integrated tools or implementation designs that focus on support in constructionist and learner-centered learning environments.

Using abstracts written by applicants during the application process, and leveraging Cultural-Historical Activity Theory (CHAT) (Engeström, 2001) as an organizing framework for analyzing sociotechnical learning systems, the workshop team will identify 3-5 areas of interest reflecting different dimensions of AI integration in constructionist learning activity systems related to participants' expertise. For example, categories might include one focused on relationships, community, and collaboration (e.g., access to expertise) and others attending to the division of labor (e.g., creating groups of learners). A key strength of CHAT is its emphasis on contradictions and tensions within activity systems—these are productive sites for understanding design trade-offs. As we expect new ideas will emerge in participants' presentations in the opening of the session, we will also create space for additional categories to emerge beyond those pre-planned.

Following presentations, participants will engage in small-group analysis sessions organized around these areas of interest. For example, one group might analyze how AI mediates learners' engagement with domain representations across multiple tools, while another examines tensions between AI-provided support and learner agency. This structure allows participants to develop deep insights into specific dimensions while building shared understanding across diverse implementations.

Groups will then reconvene for whole-group synthesis, sharing findings and collectively articulating design principles, values, and ongoing questions. Through facilitated discussion, participants will identify convergent patterns (e.g., common approaches to mediating collaboration while preserving exploration) and productive tensions (e.g., trade-offs between providing structure and maintaining open-endedness, or between AI efficiency and human relationship-building). The session will close with participants identifying priorities for future research and committing to collaborative documentation of insights. The culmination of the day's activity will result in a community facing interactive report.

Expected outcomes and contributions

In this specific community of people who are constructionist designers, there is a prevailing view that AI tools should not be the main collaborator, but should be an assistant in finding meaningful human collaborations. After the workshop concludes, outputs will be synthesized by organizers into a digital report documenting the group's articulated set of values and design directions - questions to start to articulate and solve through design work. This includes values articulated by workshop participants around the design, use and study of AI tools to support learning in learner-centered constructionist environments. The report will include tool examples shared at the workshop to showcase the diversity of the design space as well as central design goals, complex tradeoffs, and obstacles to successfully design and implement such tools. We will explore the potential for submitting as a special issue for the Journal of the Learning Sciences.

Relationship to similar events conducted in the past

The CreateAI project included a symposium at ICLS 2025 and a workshop held in Philadelphia. As exemplified in the project report (Kafai et al., 2025) CreateAI generated important insights and questions for how we might build (and evaluate) constructionist experiences for learning about AI. Inspired by this work, we extend this conversation and turn our attention to how AI and LLM-based technologies can support constructionist and project-based learning experiences that involve a wide range of topics. Where CreateAI asked "How do we teach about AI?", we ask "How might AI support collaboration, agency, and meaning-making in learning communities?" This workshop offers an opportunity to collectively discuss and document design considerations and methods leveraged by the ISLS community in LLM-powered learning designs.

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An Evaluation Framework for Pedagogical Large Language Models

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Abstract: Large language models (LLMs) are increasingly adopted as AI tutors, providing scalable, interactive assistance across a range of topics. Yet there remains a lack of a unified, comprehensive framework for evaluating the pedagogical behaviors these models exhibit across multiple tutoring sessions and varying learning scenarios. This workshop addresses this gap by introducing a two-layer evaluation framework that distinguishes between (1) foundational behaviors that LLMs must demonstrate to be considered pedagogically qualified, and (2) stylistic instructional behaviors that shape their effectiveness as tutors. Participants will refine and apply this framework through hands-on interactions with an LLM-powered tutoring system, engaging in a five-session sequence of introductory Python programming modules. Through this workshop, we aim to advance the application of generative AI tools in education settings in a safe and pedagogically effective manner.

Organizers' backgrounds

Eugene Park is a software engineer and researcher in the Personal Robots Group at the MIT Media Lab, where he leads research and system development on LLM-powered AI tutoring platforms. His work focuses on large language models and AI safety with applications in education, and his research agenda is centered on building pedagogically effective and reliable AI tutors.

Daniel Wendel is a lead software developer and research engineer in MIT's Scheller Teacher Education Program. Daniel has served as a core developer, curriculum designer, and research manager on projects spanning agent-based modeling, virtual reality, and AI-mediated learning, and his work focuses on designing scalable learning technologies that strengthen STEM understanding through interactive computer-based environments.

Dr. Grace C. Lin is a Learning Scientist and Assessment Designer at MIT whose work focuses on building children's and students' capacity to learn. As an educational researcher, Grace has led teams on educational intervention projects using innovative technology from cognitive training games to AI-powered conversational agents, and has contributed to building a practitioner-facing measures repository, particularly its rating systems, of early childhood measurements. More recently, her work has been focusing on child-centered AI, AI education for children and adults, the implications of AI technology, and its influences on human relationships.

Dr. Sharifa Alghowinem is a leading research scientist in the field of multimodal AI and Human-Robot Interaction (HRI). Currently stationed at MIT's Personal Robots Group, Dr. Alghowinem specializes in developing AI models to enhance human-robot interaction, with a particular focus on mood and behavior recognition using speech, gaze, and body movements to detect conditions such as engagement, depression and deception.

Workshop theoretical background

Large language models (LLMs) are increasingly deployed as AI tutors, offering personalized explanations, adaptive feedback, and conversational guidance across a wide range of learning contexts. Their capacity to generate interactive, context-aware instruction positions them as promising tools for expanding access to high-quality tutoring, particularly in domains such as programming, where LLMs have demonstrated considerable competence. However, despite their growing use in classrooms and learning platforms, the learning sciences community lacks a principled understanding of how these models teach and which pedagogical behaviors support or hinder learning.

Prior evaluations of AI tutors have typically focused on behavioral metrics that do not clearly distinguish between pedagogically harmful behaviors and those that are pedagogically effective (LearnLM Team et al., 2025; Macina et al., 2025; Maurya et al., 2025). While valuable, these metrics overlook a central insight from contemporary pedagogical theory: not all instructional traits are equally suitable across learning situations, nor do all teaching styles promote learning in the same way. Friesen & Su (2023) argue that pedagogy is not a fixed set of methods or universally "correct" practices, but rather a context-dependent negotiation between competing pedagogical tensions, such as constraint versus freedom, proximity versus distance, and protection versus exposure. Pedagogy, in this view, "is structured not so much by processes, methods, and outcomes, but by

irresolvable oppositions and the tensions between them”. This perspective highlights that effective teaching cannot be reduced to a single optimal style; instead, instructional behaviors must flexibly align with learners’ needs, contexts, and developmental stages. Bringing this lens to AI tutors underscores the need for an evaluation framework that differentiates *essential* pedagogical requirements from *stylistic* instructional variations, ensuring models are assessed not only for pedagogically correct behavior but also for their alignment with pedagogically effective behavior.

Moreover, due to increasing domain expertise of frontier LLMs, this can create an illusion of overestimation in their pedagogical performance (e.g. strong performance on programming tasks or reasoning benchmarks may not necessarily translate to an LLM’s ability to teach effectively). A lack of a concrete analysis of LLMs’ pedagogical behaviors risks the misuse of LLMs, and prevents researchers and practitioners from meaningfully developing the instructional quality and reliability of AI tutors.

To address this gap, we identified pedagogically harmful behaviors in LLM tutors, such as conceptual inconsistencies, topic drift, and failures to detect misconceptions, and formalize these issues as forms of *pedagogical misalignment*. Ensuring the absence of such behaviors represents a minimal requirement for safe and trustworthy AI tutoring. Building on this foundation, we introduce a two-layer evaluation framework for pedagogical alignment. The first layer, *pedagogical qualification*, captures whether an LLM satisfies baseline standards of instructional safety. The second layer, *pedagogical personality*, characterizes stylistic instructional traits, such as scaffolding depth, adaptivity, metacognitive prompting, and instructional tone, that shape the learning experience and likely modulate learning gains.

This workshop aims to refine and operationalize this framework, deepen our shared understanding of pedagogical traits in AI tutors, and collaboratively develop metrics for evaluating them. Through hands-on interaction with an LLM-powered tutoring system, participants will explore how pedagogical safety and pedagogical style intersect, and how these dimensions can inform the design of reliable, aligned, and learning-effective AI-based educational tools.

Intended audience and logistics

This workshop is designed for researchers and practitioners in educational technology, the learning sciences, AI/ML, and NLP who are interested in the reliable and effective use of generative AI for education. It is particularly relevant for scholars studying AI safety, AI in education, human-AI interaction, or instructional design, as well as developers building LLM-powered educational tools.

The workshop is planned as a full-day event to allow sufficient time for hands-on interaction with the LLM-based tutoring system across multiple sessions, group analysis and annotations of pedagogical behaviors, and structured discussions aimed at refining the proposed evaluation framework.

To support effective collaboration, we request a room equipped with round tables suitable for small-group work. If possible, we also request basic writing materials, such as sticky notes, paper, pens, and colored markers. Participants will be asked to bring their own laptops to interact with the AI tutor. The workshop is designed for a minimum of 8 and a maximum of 30 participants, ensuring meaningful interaction, hands-on guidance, and productive group collaboration.

Outline of planned activities

The workshop is organized as a full-day, highly interactive session combining hands-on experience with an AI tutoring system, individual analysis, and structured group discussions. Participants will engage in five short introductory Python programming sessions using an LLM-powered tutor, evaluate the tutor’s pedagogical behaviors, and collaboratively refine the proposed two-layer evaluation framework.

Schedule

- 09:00 – 09:30: Welcome and Introductions. (30 minutes)
- 09:30 – 10:40: Hands-on Interaction with AI Tutor: Modules 1–3. (70 minutes)
- 10:40 – 11:00: Break (20 minutes)
- 11:00 – 13:00: Hands-on Interaction with AI Tutor: Modules 4–5. Behavior Survey. (100 minutes)
- 13:00 – 14:00: Lunch Break (60 minutes)
- 14:00 – 15:30: Small-Group Annotation and Framework Refinement (90 minutes).
- 15:30 – 15:45: Break (15 minutes). Transition to whole-group synthesis.
- 15:45 – 17:00: Large-Group Discussion and Synthesis (75 minutes).

17:00 – 17:30: Closing Reflections and Next Steps (30 minutes).

Expected outcomes and contributions

This workshop will equip participants with a refined and practically grounded framework for evaluating the pedagogical behaviors of LLM-powered tutors. Through hands-on interaction with an AI tutoring system, individual and collaborative annotation exercises, and structured group discussions, participants will collaboratively sharpen the distinction between *pedagogical qualification*, the foundational behaviors required for safe and trustworthy tutoring, and *pedagogical personality*, the stylistic instructional traits that influence learning effectiveness. The workshop will produce a consolidated set of pedagogical metrics, including clarified definitions, improved categorizations, and a consensus-based ranking of behaviors along a harmful-to-stylistic spectrum. This refined framework and metric set will support future empirical research and systematic tool development practices.

Beyond methodological contributions, the workshop will foster a shared vocabulary and conceptual grounding for analyzing how LLM tutors behave across different learning scenarios. Participants will gain insight into the strengths and limitations of current AI tutoring systems, enabling them to better identify problematic behaviors, understand instructional variability, and connect model behaviors to established learning sciences theory. Thus, this workshop also aims to strengthen the research community at the intersection of AI education and AI safety by identifying open challenges and catalyzing collaborations across the learning sciences, AI/ML, and educational technology domains.

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A CSCL Taxonomy Building Workshop

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Abstract: This workshop aims at collaboratively developing a shared taxonomy for the Computer-Supported Collaborative Learning (CSCL) community that captures diverse epistemological, methodological and technological perspectives to help build a common ground for CSCL research. Through pre-workshop Delphi input, hands-on application, and iterative refinement, participants co-construct a framework to support clearer communication, cumulative knowledge building, and more transparent dissemination in CSCL proceedings and journals. The resulting taxonomy and follow-up activities aim to strengthen community identity, foster collaboration, and guide future CSCL research and publication practices.

Organizers

- **Martin Greisel:** Assistant professor/postdoc at the University of Augsburg, focusing on regulation in collaborative learning, peer feedback, and teacher training.
- **Burcu Şener:** Lecturer at Middle East Technical University, PhD candidate at Bogazici University, focusing on collaborative problem solving, collaborative debugging, computational thinking.
- **Sebastian Simon:** Postdoctoral fellow at the University of Le Mans with a PhD in Computer Science. During his PhD, he focused on enhancing mobile and co-located Computer-Supported Collaborative Learning with Augmented Reality (AR). In his current Postdoc position, he investigates generative AI agents as collaborative entities in co-located CSCL settings.
- **Conrad Borchers:** PhD candidate at Carnegie Mellon University, researching data science and HCI applications to improve educational processes.
- **Elham Tajik:** PhD student at the University at Albany, she focuses on the transformative role of generative AI in education and HCI.
- **Bahar Shahrokhian:** PhD candidate at Arizona State University, researching multimodal learning analytics and collaboration in co-located learning environments.

Intended audience and resources

This half-day workshop is intended for all researchers and educators interested and engaged in CSCL, as well as its related domains, including but not limited to: educational psychologists, computer scientists and learning scientists. We particularly welcome researchers from all domains to contribute their vision of computer-supported collaborative learning, with a minimum of 8 to a maximum of 40 participants. We can manage up to 30 in-person and 10 online (Zoom) participants. Regarding resources, we only need a large seminar room with WiFi and a projector that enables group work through flexible tables (i.e., no lecture hall with fixed rows). In-person participants will be asked to bring their personal laptops.

Description of the event

Theme and goals

In recent years, the field of Computer-Supported Collaborative Learning (CSCL) has continued to expand in scope, methods, and theoretical foundations. While this diversity reflects the richness and vitality of the field, it also creates challenges for integration and cumulative knowledge building. Researchers draw on a wide range of epistemological, methodological, and technological perspectives, and often work in parallel rather than in concert. This fragmentation can make it difficult to identify shared concerns, trace the development of ideas, or recognize complementarities among approaches. At the same time, the multiplicity of perspectives represents an invaluable resource. If systematically understood and connected, this multiplicity can foster deeper insights into learning processes and the design of collaborative technologies. Previous CSCL workshops have identified the need for a common language across the different domains that are present in the CSCL community. Common language

presents the first step to common ground, a concept that is key to the knowledge building in a scientific community, particularly in one that emphasizes the strength of co-existing paradigms, requiring awareness of semiotic and semantic differences (that may otherwise lead to issues like gatekeeping or siloed research, Kuhn, 1962). Particularly, the 2025 CSCL workshop participants have identified the need for community building in the CSCL field and the potential of a collaboratively created CSCL taxonomy as an artefact for building common ground.

Consequently, this workshop has two goals: (1) community building and (2) taxonomy development. To build community, the workshop brings together members of the CSCL community to overcome fragmentation and to harness the richness of the field's diversity. It will help strengthen a community identity by creating an awareness of what constitutes CSCL and foster a feeling of agency through a common activity centered around the community itself (Wise & Schwarz, 2017). The second goal is to develop a taxonomy for classifying future CSCL research, especially contributions to future conference proceedings and ijCSCL articles. This taxonomy aims to offer a shared framework that reflects the community's pluralistic character while enabling more coherent navigation, communication, comparison, and synthesis of research. As a tangible outcome and lasting artefact, it aims to promote collaboration across paradigmatic differences in the future. In addition, the process of developing such a taxonomy itself constitutes an effort of community building, thereby also serving the first goal of the workshop. In summary, by collaboratively developing and validating such a taxonomy, supported by a broad proportion of the CSCL community, the workshop seeks to lay the foundation for a more connected and integrated future for CSCL research.

Theoretical background and relevance to the field and conference

CSCL is inherently multidisciplinary, integrating perspectives from at least the learning sciences, educational sciences, psychology, and computer science. This diversity enriches the field but also introduces substantial variation in researchers' underlying epistemic stances.

According to the AIR model (Chinn et al., 2014), such stances encompass differences in 3 ways. First, researchers have different *aims*—they seek to achieve different things with their work. Second, they hold different *ideals*—they disagree on what counts as valid knowledge and good evidence and the criteria by which success is evaluated. Third, they use different *reliable processes*—they favor different methods for how data is created and interpreted, that is, the methods considered most appropriate to reach their aims.

While this pluralism fosters innovation and theoretical breadth, it also poses challenges for mutual understanding in our field. Researchers examining similar phenomena may struggle to find, understand, and interpret each other's research contributions because they pursue different aims, evaluate knowledge by different criteria, and employ different methods. This is reflected, for example, in the co-existence of different major theoretical traditions (e.g., constructivist and sociocultural, cognitive/information-processing), a focus on different mid-level theories or theory-driven pedagogical frameworks (e.g., knowledge building, cognitive load theory, theory of immersive collaborative learning) or interactional constructs (e.g., transactivity, socio-cognitive conflict) and competence frameworks (e.g., collaborative problem-solving skills), and different analytical and design methodologies (e.g., collaborative analytics or design-based research) (Brown, 1992; Dillenbourg, 1999; Graesser et al., 2018; Kirschner et al., 2018; Makransky & Petersen, 2023; Scardamalia & Bereiter, 2014; Vogel et al., 2023; Vygotsky, 1978; Wise et al., 2023). As a result, communication across perspectives can become fragmented, leading to disciplinary silos and limited knowledge integration within the CSCL community (Wise & Schwarz, 2017), thereby hindering the advancement of knowledge.

To remedy fragmentation, a workshop at the CSCL 2025 conference aimed at developing a shared paradigm (see also section "Relationship to similar events conducted in the past" below). However, one main outcome, also advocated by a panel of senior CSCL scholars, was that attempting to unify the CSCL community under a single theoretical or methodological framework would weaken the field's richness in diversity. Its multiperspectivity is not a weakness to be corrected, but a strength that enables deeper and more comprehensive inquiry. As a result, the goal should be to bridge epistemic diversity to enable constructive dialogue rather than to simply erase different perspectives and epistemologies.

A step toward bridging these differences is to establish a shared vocabulary that enables researchers to articulate their approaches in a comparable manner (Wise & Schwarz, 2017). When organized systematically, such a vocabulary can take the form of a taxonomy—a structured set of dimensions and descriptors that capture from which viewpoint researchers approach their investigation, which phenomenon they study, and how they study it. A CSCL taxonomy could thus provide a common reference point for describing research aims, methodologies, collaboration phenomena (e.g., participation, agency, different types of awareness, social and cognitive processes etc.), and the tools used to support them. In doing so, it would strengthen our ability to engage in cumulative dialogue while maintaining our diverse intellectual foundations.

After the development of such a taxonomy, one potential use case would be to implement the taxonomy into the submission process and make it a part of each contribution to the CSCL conference proceedings and to the ijCSCL. This would serve a threefold function: First, it would encourage authors to explicitly clarify their epistemic position and research orientation by selecting the appropriate descriptors. Sharpening authors' clarity about their epistemic aims and ideals helps raise awareness for what is considered rigorous research in the CSCL community, clarifying expectations for authors and easing review processes for journals, including ijCSCL. Second, it would guide readers in their expectations and understanding of CSCL papers to avoid misconceptions and mismatched expectations. Currently, readers may not immediately notice epistemological differences between them and the authors of a research contribution. If the contribution was clearly labeled based on a common taxonomy instead, this misconception could be avoided and readers would be informed of the epistemological prerequisites for reading a specific contribution. Third, the taxonomy would make it more likely that CSCL researchers find and read all the research contributions that are relevant to them. At the moment, researchers risk missing out on research contributions that may be informative to them as they are addressing the same phenomenon, just because these contributions do not use the same vocabulary they are familiar with. A shared taxonomy makes it possible to search for and filter research that addresses the same phenomenon but from different epistemic stances. Therefore, such a taxonomy increases the agency of readers to seek out different epistemic stances than their own (or the one they are looking for). Other potential use cases include newcomers' onboarding and providing the basis for review articles and field development.

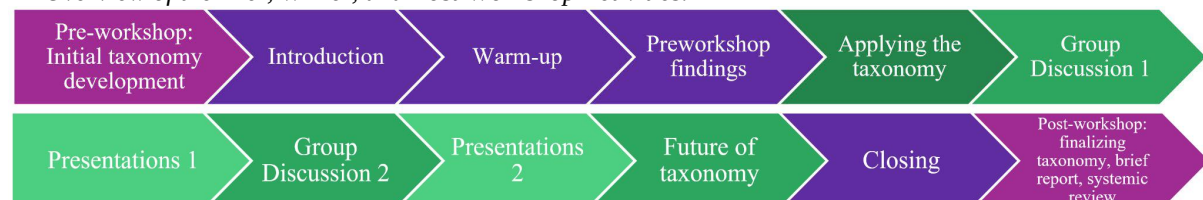
Outline of planned activities

Pre-workshop: Drafting a taxonomy

Before the workshop (see also Figure 1), we will collect the community members' comments and opinions on a draft of a potential taxonomy for CSCL. To this end, we will use an adapted version of the Delphi approach (Dalkey & Halmer, 1963). First, the organizing committee will prepare a first version of the taxonomy. Second, individual expert researchers whom we will select to represent the variety of perspectives within CSCL will be asked to enrich and refine the draft. Third, the revised version will be shared with the enrolled workshop participants online. The participants will be asked to give feedback on the taxonomy anonymously, and to state whether they would suggest any additions or deletions for the categories by justifying their answers. Using their input, workshop organizers will prepare the third version of the taxonomy prior to the workshop.

Figure 1

An Overview of the Pre-, While-, and Post-Workshop Activities.



While-workshop: Refining the taxonomy

After a short introduction (presenting context, aim and structure), a warm-up (sharing research interests and personal meaning of CSCL), and a short presentation of the latest taxonomy version, each participant applies the taxonomy to one's own paper (or a default paper as fallback). This activity familiarizes participants with the latest taxonomy draft and gives them firsthand experience using it for later evaluation.

Then, the participants (virtual and onsite) are put into random groups of three or four, and discuss their application experience, the aspects of CSCL they have found hard to categorize, the guiding principles or criteria they have used to evaluate the categories, and the parts of the taxonomy they think should change. The discussion is documented by adding comments on the related parts of the taxonomy shared on a Miro board. In each group, a member is asked to take the role of a moderator to ensure equal contribution. Then, groups share their discussions by highlighting the most problematic areas of the taxonomy or pressing concerns via pitch presentations. During the presentations, organizers note the to-dos for each part of the taxonomy.

Next, participants regroup by selecting which specific part of the taxonomy they would like to work on. These groups then work on the corresponding to-dos. Similar to the previous round, a member moderates the discussion. This is followed by a second round of pitch presentations, where one member from each group

presents ideas for improvement of the part of the taxonomy they worked on. Each pitch presentation is followed by a Q&A so that other participants can make suggestions as well.

Depending on the remaining time, participants are asked to share on the Miro board how they imagine the taxonomy to be used in CSCL research or design contexts, and how they envision the “success” of the taxonomy in terms of both academic and practical impact. The workshop finishes with participants sharing their contact information and stating whether they would like to be interested in post-workshop activities.

Post-workshop: Follow-up activities

If there are any unresolved areas regarding the taxonomy, we will invite the participants and the broader CSCL community to comment on the latest version of the taxonomy to finalize it and to collaborate on its implementation. Then, we will a) write a brief report, b) conduct a systematic review applying the taxonomy on a range of research works, and c) team up with future CSCL program committees and ijCSCL editors to integrate the taxonomy into the submission systems.

Expected outcomes and contributions

The primary workshop goal is to develop a CSCL taxonomy codebook, which will be published as a brief report for ijCSCL. Also, during the workshop, participants will critically examine, refine, and align their conceptual understandings of key dimensions of CSCL research and practice. In the long term, this shared, systematic framework can serve as a foundation for future empirical and theoretical work by informing the paper writing processes in the CSCL community and increasing the quality of the papers produced. We also envision that the taxonomy will be implemented in the submission processes of the ijCSCL and the CSCL conference.

Furthermore, a systemic review for CSCL research using the taxonomy will be conducted, presenting a significant contribution to the field to classify and evaluate previous CSCL research.

Moreover, collaboration among various CSCL researchers is expected to be strengthened following while- and post-workshop activities. The process of co-constructing the taxonomy might foster new connections and potential research partnerships, leading to a continued dialogue and joint publications around the use, validation, and further development of the taxonomy. The workshop thus aims not only to produce a tangible scholarly output but also to build a sustained, collaborative research network that advances the field of CSCL.

Relationship to similar events conducted in the past

The proposed workshop continues the work of last year’s CSCL workshop at ISLS 2025 in Helsinki to advance CSCL as a multi-paradigm field. To ensure continuity, this proposal has been shared with the previous organizers and support from the ijCSCL editors has been assured. Relatedly, it extends the 2024 *Reimagining and Advancing CSCL Research and Building CSCL Community* workshop held at the 2024 ISLS conference (Chan et al., 2024) by supporting community building and collective vision development across diverse scholars. It further draws on the 2018 senior-researcher panel on theory building during the symposium *Towards next steps for the CSCL Community: Advancing science and informing real world collaboration in Web 2.0* (Ludvigsen et al., 2016) and on the authors’ related research collaboration (Simon et al., 2025).

A description of how you will solicit participation including a draft call for participation that will be posted by organizers on the ISLS 2026 website if the event is accepted.

We will recruit participants through CSCL communication channels, including the conference website, mailing lists, and social media, as well as by directly inviting the experts from the first Delphi round (see “pre-workshop activities” above), contributors and panelists from prior CSCL workshops, the CSCL retreat 2025, the CSCL community committee, and recent ijCSCL authors. This combined open and targeted approach will ensure a diverse group representing different disciplines, career stages, and epistemic perspectives. In last year’s iteration of this workshop, a similar approach has led to a healthy participant turnout. If more than forty people apply, we randomly select participants to maintain a diverse group.

Draft call for ISLS website

We invite researchers, educators, and practitioners from across the CSCL community to join our half-day ISLS 2026 workshop, A CSCL Taxonomy Building Workshop. Participants will collaboratively refine a draft taxonomy that supports understanding across diverse theories, methods, and technologies in CSCL for later use in CSCL publications. The workshop builds on prior community efforts from CSCL 2024 and 2025 to strengthen cross-paradigm dialogue and shared understanding.

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Re-envisioning Assessment Challenges and Opportunities in the Generative AI Era through CSCL

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Abstract: Assessment in higher education faces unprecedented challenges, as generative AI now enables students to produce polished assignments in seconds. Many institutions have responded by reverting to pen-and-paper exams, a solution that mitigates some concerns but introduces other problems narrowing educational aims. This workshop provides a platform for collaborative exploration and dialogue of how CSCL can offer fresh perspectives and pathways for addressing these challenges and creating opportunities. Grounded in CSCL theory, design, and analytics, this workshop examines conceptualizing AI as a dialogic, collaborative partner; designing innovative assessments; assessing processes rather than products; and using collaborative analytics to support learning in AI-enriched environments. Key issues for inquiry include: purpose of assessment, design and fine-tuning of AI-supported assessment, AI-powered multi-modal analytics and feedback, equity and diversity, researcher–teacher collaboration, ethics-privacy, and public discourse. Together, we work towards re-envisioning assessment, synergizing research and practice, building collective knowledge and strengthening CSCL's impact and significance in the Gen AI era.

Organizing team

We are a group of CSCL researchers with shared interests in examining how CSCL can offer perspectives and directions for addressing assessment issues in higher education arising from the use of generative AI. Carol Chan conducts research on knowledge building, focusing on socio-cognitive-epistemic dynamics, reflective assessment, analytics, and teacher professional development. Xiao Hu's research is on learning analytics, automated assessment and feedback, and AI in education. Her research includes collaborative writing facilitated by real-time analytics dashboards, multimodal learning analytics in collaborative maker activity, and analytics-supported writing in AI environments. Baruch Schwarz's research in CSCL focuses on argumentation, dialogue, and reasoning, particularly examining the roles of technology in boosting collaborative argumentation. Stephen Fiore conducts research on Human-AI teaming, focusing on collaborative cognition, including knowledge building when collaborating with AI, and the role of artificial social intelligence in assessing collaborative learning. Jun Oshima examines how knowledge-building discourse among undergraduate /graduate students can be supported by Generative AI, and examines their interactions with AI from a network-science perspective. Wenli Chen conducts research on human-AI collaboration to augment human learning by fine-tuning generative AI informed by learning sciences theories, developing AI-powered multi-modal learning analytics to assess students' collaborative learning in classrooms. Manoli Pifarré examines and designs dialogic, technology-enhanced pedagogies that foster co-creation processes focusing on how interactive technologies—empowered by artificial intelligence and multimodal learning analytics—can assess, visualize, and support group-based creative problem-solving.

Intended audience

- CSCL researchers/ graduate students interested in collaboration, assessment and generative AI
- Researchers/graduate students from learning sciences working on generative AI related to assessment
- Higher-education instructors/practitioners navigating new assessment challenges in the AI era with interest in research-informed strategies and collaborative approaches.

Duration

A half-day workshop will be conducted as a HYBRID event, with onsite and remote participants connected by Zoom and online sharing tools. Participants may register to attend as onsite or virtual attendees.

Workshop description and goals

We are facing a global assessment crisis in higher education with the widespread use of AI tools: students can generate polished assignments in seconds, fueling surface learning, plagiarism, and metacognitive laziness, and leading to reliance on ChatGPT (Fan et al., 2025). Public discourse highlights the seriousness of this shift—for example, Bloomberg (May 27, 2025) asked, “Does college still have a purpose in the age of ChatGPT?” Many universities have reverted to in-person, pen-and-paper exams, which mitigate some risks but create others, curtailing the valuable educational goals of inquiry, problem-solving, and collaboration.

This workshop is premised on the idea that CSCL is well-positioned to offer directions for rethinking assessment in higher education in the AI age (Schwarz et al., in press). We build on decades of CSCL research on assessing collaboration (Cress et al., 2021), extending to CSCL theory, design, and analytics for assessment practice. Traditional assessments emphasize final products (e.g., essays) that AI can replicate easily. Conversely, CSCL foregrounds collaborative processes (e.g., co-construction, shared regulation) and CSCL designs such as argumentation, dialogic interaction and knowledge building, can offer new ways to scaffold and assess AI-human collaboration. Beyond polished output, CSCL analytics can make learning-collaboration processes visible for feedback and actionable improvement (Wise et al., 2021). This workshop provides a venue for sharing CSCL research and practices to address challenges and opportunities, reflecting on what has been accomplished and examining potential for further inquiry and impact. Our goals are to (a) identify and problematize assessment challenges brought by GenAI through a CSCL lens, (b) examine emerging CSCL theories, pedagogies, and analytics for AI-enriched assessment, and (c) reflect on assessment practice both as higher-education instructors and CSCL/learning sciences researchers. We seek to synergize efforts from workshop participants to build collective knowledge, create networks, to address these challenges and to strengthen CSCL’s impact.

Theoretical background and themes

Theoretical perspective. Since its emergence in the 1990s, CSCL has reconceptualized learning from individual knowledge acquisition to co-construction of knowledge (Ludvigsen & Steier, 2019). Current concerns about students copying AI-generated work reflect a continued reliance on surface-level, transmission-based conceptions of learning and assessment. Using a CSCL lens, Gen AI use in homework need not be prohibited; instead, it should be scaffolded to support agency, co-construction, and shared regulation (Järvelä, et al., 2023). The knowledge-building model posits collaboration as collective work on continually improvable ideas (Chan & van Aalst, 2018; Scardamalia & Bereiter, 2014). Similarly, AI-human collaboration can be conceptualized as iterative idea improvement rather than task completion. We need to shift from policing to scaffolding students to leverage AI as a dialogic partner, using AI-generated transcripts as evidence for learning and collaboration. CSCL knowledge-building perspectives could provide a theoretical foundation for unraveling the nature of AI-human collaboration while enriching CSCL theories in AI age.

Design-pedagogy-technology perspective. CSCL offers powerful approaches for aligning assessment, learning, and collaboration. Research in assessment has shown that assessment can be understood as assessment of learning (evaluating completed work), for learning (supporting improvement), and as learning (making thinking visible) (Wiliam, 2017). Beyond assessing completed work, CSCL approaches—argumentation, collaborative problem-solving, knowledge building, and maker work, etc.—provide rich opportunities for formative assessment that support collaboration and complement traditional exams/papers. Assessing group interaction and quality of talk, increasingly supported by AI (Hu et al., 2022), offers important indicators of learning. Reflective assessment enables the assessment of collaboration while also scaffolding interaction and idea development (Yang et al., 2024; van Aalst & Chan, 2007), and this could now be enriched with AI. In this way, AI becomes part of an assessment ecology that supports collaborative learning rather than merely reproducing assignment products.

Analytics perspective. Contemporary CSCL analytics research now makes it feasible to assess processes beyond easily replicable products. AI-enhanced collaborative analytics can trace participation, interactions, and revisions through log data, revealing whether students are engaging in meaningful knowledge construction or simply copying outputs (Ouyang et al., 2023). These process-level data can be fed back to teachers and learners through dashboards to support monitoring, reflection, and improvement. CSCL researchers are well-positioned to design AI-enriched analytics that make collaboration processes and idea development—previously invisible—now visible (Ba et al., 2024; Hu et al., 2022). This process-focused view emphasizes ongoing work tracing student engagement and activity rather than post-hoc grading through concurrent, embedded, and transformative assessment. Such analytics assess not only what students produce but how they collaborate—with AI and with one another—to co-construct and to build knowledge.

Issues for Discussion: This workshop provides a platform for participants to share ideas as we inquire into how CSCL could address challenges of assessment for new opportunities in the AI era. Grounded in theory, design, and analytics and ecological-ethical considerations, several areas of inquiry include:

- **Focus of assessment and expanding theories.** Assessment needs to shift from evaluating products to processes—examining how people learn and collaborate with AI. Emerging research on human–AI collaboration highlights cognitive, metacognitive, motivational, and socio-emotional processes, yet more work is needed—for example, on how students take agency and engage in dialogue with AI when tackling assessments to advance learning. Questions also arise about what knowledge, skills, domain-specific competencies, and collaboration processes should be assessed in AI-mediated assessment -- How can AI’s roles shift from merely providing content to supporting co-construction, regulation, and reflection? These shifts call for expanding CSCL theoretical perspectives and engaging with related fields on assessment. Continued inquiry is needed to extend CSCL frameworks to better capture the complexities of learning and collaboration in AI-mediated assessment.
- **Collaboration and AI-assessment designs.** Building on CSCL designs, researchers can work on productive assessment tasks to replace traditional ones easily emulated by AI and designing AI-enriched collaborative environments. Questions include what we are assessing, what the assessment activity involves, and what resources—peers, AI, and technology—are required. Reflective assessment can prompt students to reflect on their work with AI for ongoing improvement. As well, automatic assessment features can provide teachers with timely information to assess student engagement and to make “on-the-fly” decisions about what support is needed. Design experiments conducted on AI-assessment designs while examining the emerging human-AI collaboration process would help advance both research and practice..
- **Examining AI-supported multi-modal data analysis.** Collaborative analytics, combined with AI, can play a central role in assessing learning-collaboration processes via multi-modal data analyses. These could include automatic scoring and annotating student texts and conversations, applying expert-designed rubrics for evaluation, and identifying emerging patterns of behaviors. Information shown in dashboards can support teacher assessment and promote students’ reflection on ongoing work. These analyses open the door to assessing human–AI interactions and designing assessment tasks that capture these dynamics. How these tools and analytics can enhance assessment validity, scalable analysis, providing meaningful insights into collaborative learning processes are important questions.
- **Equity and diversity issues.** The advent of AI in assessment raises important questions about equity and diversity. AI use may widen existing gaps if students have unequal access to technology, differing levels of guidance, or varying digital literacy. AI technology itself can be biased due to an imbalance in data availability. At the same time, well-scaffolded AI use and mindfully-designed AI tools have the potential to narrow disparities by providing feedback and more accessible pathways for demonstrating student understanding. We need to consider how assessment design and practices can ensure fairness, how AI-based tools can be designed to reduce inequities, and what forms of support learners need to participate equitably in AI-mediated assessment environments.
- **Research–practice integration.** CSCL-AI assessment for impact involves research-practice synergy developing practical tools and strategies that teachers can use. This involves understanding teachers’ needs and communicating complex findings in accessible, actionable forms. Inquiry such as interviews with teachers about their successful experiences with AI-informed collaboration and assessment could enrich the integration. Supporting higher-education teachers in adopting new views of assessment, collaboration, and AI use—and aligning CSCL research with classroom realities—remain to be investigated.
- **Privacy, ethics, and limitations.** AI-supported assessment raises concerns about data privacy, ethical use, and system limitations. Collecting process data—such as interaction logs or conversational traces—requires careful attention to consent, transparency, and responsible data use. Ethical challenges include helping students use AI responsibly, preventing misuse of analytics, addressing bias in AI-generated judgments, and ensuring students are not harmed by algorithmic errors. Ongoing scrutiny of AI limitations is needed to ensure new CSCL-AI assessment practices remain trustworthy, fair, and beneficial.
- **Engagement with the public.** The rise of generative AI has sparked widespread public debate, offering CSCL researchers opportunities to engage broader audiences. Public engagement involves helping students, parents, policymakers, and other stakeholders understand the challenges and opportunities AI brings to assessment. By clarifying misconceptions, sharing evidence, success stories, and highlighting responsible uses, CSCL researchers can contribute to more informed public discourse.

Outline of planned activities (half-day workshop)

Before the workshop: Participants will complete a short survey (a) identifying their research interests in CSCL, assessment, and generative AI, (b) sharing examples of how they have used generative AI in classroom-assessment contexts; and c) proposing key questions for workshop discussion.

During the workshop: The workshop will be structured to maximize sharing, exchange & knowledge building.

- **Introduction.** Organizers introduce aims and expected outcomes, followed with participant introduction.
- **Invited panel discussion.** 3-4 researchers will be invited to give brief presentations that expand on key themes, also informed by participants' pre-workshop submissions.
- **Breakouts and group discussion**
 - Connecting and sharing. In rotating dyads, participants share the big idea or challenge they bring to the workshop, as well as what they hope to gain and contribute. Each person speaks and responds before moving to a new partner.
 - Knowledge sharing and building: In groups of four, participants rotate and discuss an issue or idea related to their work, focusing on opportunities to develop their research, pursue interdisciplinary collaboration, and take action related to key themes. AI tools could be used to stimulate and help summarize the discussion; each group shares one standout idea from their discussion (online participants will be supported with online collaboration tools)
- **Closing.** Synthesis of what we have learned and reflection on potential actions, including proposed activities on publication, projects, media writing, networking, and community building

After the workshop: Participants will form a Google group for communication connecting with CSCL committee /junior researchers and panel discussion planned coming year, opening to CSCL and learning sciences researchers.

Expected outcomes and contribution

The anticipated outcomes include workshop materials, publication of a paper, collaborative plans for projects and papers; and a Google group for communication for participants and interested ISLS members.

Participation requirements

Participation requires completing the survey on assessment, CSCL, and generative AI. The survey collects participants' experience with CSCL and assessment to be shared during the workshop to enrich dialogue.

Relationship to similar events conducted in the past (e.g., CSCL or ICLS)

This workshop is distinctive in addressing the assessment challenges facing higher education with the rise of generative AI. There could be similar events on CSCL multi-modal analytics and generative AI but few synergizing research and practice for broadening CSCL visions for impacts in assessment in Gen AI era.

Facilities and equipment required

We will use the conference's communication platform for remote participants (e.g., Zoom), possibly including other tools, such as Miro (<https://miro.com/login/>) or Padlet (<https://padlet.com/>) to support online collaboration.. For on-site participants, internet access, projectors, and a room large enough to accommodate breakout spaces for group discussions are required.

Minimal and maximum number of participants

We aim for 15 and 25 participants, with a balanced mix of in-person and remote participants, along with the organizing team and invited panelists.

Draft call for participation

About the Workshop: We invite researchers, graduate students, and practitioners to join this workshop to explore the roles of CSCL in advancing AI-supported assessment. With the rapid rise of generative AI, higher education faces urgent assessment challenges. We propose CSCL offers powerful lenses and tools for rethinking assessment challenges and how learning and collaboration can be assessed in AI-rich environments. This workshop provides a venue for sharing research, practices, and emerging ideas and future trends on AI-enriched assessment, including the role of AI as a collaborative partner for knowledge building, process-focused assessment, assessment design, multi-modal analytics, equity-ethics issues, teacher-researcher collaboration, and public engagement. Together, we will identify ways to address current challenges, examine theories and designs for CSCL-AI enriched approaches, build collective knowledge and reflect on implications for our roles as CSCL researchers and educators. Participants will engage in reflective inquiry, interactive discussions, and community building to reconceptualize assessment and to develop future collaborations on CSCL research for impact.

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Seeing in 360°: Practical and Theoretical Insights for Video Recording in Learning Sciences Research

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Abstract: This full-day workshop addresses the theoretical and practical challenges of using 360° video and multi-camera systems in qualitative research. Drawing on a multi-year research–practice partnership in secondary classrooms, we share insights for implementing a range of recording technologies. We will address theoretical issues, including how each configuration shapes what becomes visible, invisible, and analytically possible. We will also address practical challenges researchers encounter when working with complex video data, including: managing massive file sizes, synchronizing multiple videos, and establishing data storage and processing workflows. Through hands-on activities, attendees will collect, process, and analyze their own 360° footage and practice troubleshooting common pitfalls. We invite researchers at all levels to consider how 360° video and multi-camera set-ups can support responsible, rigorous, and theoretically informed inquiry.

Workshop overview

This full-day workshop is intended for researchers, including graduate students and emerging scholars, who are interested in using 360° video in their research projects. The workshop organizers have engaged in a multi-year research project using a diverse array of video data collection methodologies, including 360° cameras (e.g., GoPro MAX), standard cameras, and synchronized multi-camera setups. This extensive practical experience has yielded valuable methodological insights, which we are eager to share with the learning sciences community.

Workshop goals and participants

This workshop is designed for learning scientists who collect, analyze, or theorize with video data, including those who are interested in multi-modal analysis, emerging recording technologies, representations of practice, and ethical considerations surrounding visibility and video-based research. No prior experience with 360 video is required, and we welcome participants at any career stage. The goals of our session are twofold:

First, participants will consider the theoretical and analytic affordances of various video recording set-ups, including what is rendered visible and invisible with their methodological choices, and how those choices influence their analysis. It is tempting to think that collecting more data or richer data is always better; that is, that multi-camera set-ups are better than single-camera, or that more immersive 360° videos are preferable to wide-angle views. However, we argue that this is not necessarily so: even the best video recording renders some aspects of an event invisible and unrecoverable, and the technical difficulties of wrangling large data corpora may render some analyses impractical. As researchers, it is important to thoughtfully weigh these choices. In this session, we will engage learning sciences colleagues in deeper theorizing around these issues.

Second, we will share practical knowledge and insight into how to use complex video recording set-ups. Across two research–practice partnerships — Project SIGMa (Supporting Instructional Growth in Mathematics; Horn & Garner, 2022) and Project TAU (Teaching Amidst Uncertainty) — we have spent extensive time developing and troubleshooting various video recording set-ups. Using multi-camera set-ups, especially with 360° video, presents many nontrivial challenges: synchronizing multiple cameras, managing different files and file types, as well as processing and storing large files. Building on the theoretical conversation, participants will gain practical knowledge and skills to use 360° cameras and multi-camera set-ups — including knowledge of equipment (cameras, but also accessories such as SD cards and cables), software, data compression, and data storage requirements. We have refined and synthesized our data management processes to use with our large, multi-site research team, and we want to share this hard-won knowledge with others in the field.

Organizing team

We are a multi-generational research team involved in a multi-year research–practice partnership, through which we have captured rich, multi-perspective video records in teachers' classrooms. Elizabeth (senior research assistant) and Brette (co-PI) developed and implemented a variety of data collection protocols. Darryl (co-PI)

offered technical expertise to develop Mathematica programs to synchronize and process videos. Together, we offer rich experience in both the design and implementation of video data collection protocols.

Theoretical background

We resurrect Hall's (2000) question around video recording teaching and learning activities: "*How should we expect new technical media for recording human action, along with a growing suite of tools for their analysis, to shape the specific cultural practices of cognitive and educational researchers?*" (p. 649). As video recording technology has become increasingly sophisticated and complex in the quarter-century since Hall posed this question, it is vital to consider how such technology shapes our work as researchers and learning scientists. Moreover, Hall argues, video recording is not a neutral practice, and our choices for creating video records are theory-laden. In this workshop, we will consider a) how video both captures and (re)organizes activity and b) how video shapes analytic practices. We examine this through our use of 360° video and multi-camera set-ups.

How does multi-camera 360° video (re)organize activity?

Video recordings construct a particular *version* of an event, where decisions — around *what* to record and *from where* — frame the activity of interest in a particular way. Each camera has a fixed field of view, inevitably deleting some elements of the activity entirely; multiple cameras can offer *more* views, but cannot fully capture everything. Even within what is captured, some elements become foregrounded while anything at the periphery will be backgrounded. What remains in the video record then stands in for the "real" activity that can be studied, replayed, analyzed, and shared. The decisions behind what we are able to capture reflect and embed theoretical and anticipatory assumptions into the video record. These orientations then shape what becomes visible, emphasized, and relevant in any subsequent study of the interaction. When we make decisions about what to capture and what to foreground in the recording, we necessarily change what the event is.

This is true for any video record, but it's important to consider how these implications might be different for 360° video. Although 360° cameras *expand* the field of view, elements of activity will still be deleted, as they fall outside the audio range or line of sight. While background activity is generally visible, 360° cameras privilege what is physically closer to the camera. In other words, foregrounding is no longer a matter for framing, but of how the placement of the camera affords a particular, spatially organized perspective.

How does multi-camera 360° video shape viewing and analysis?

Video offers a durable record that can be replayed and edited. This allows us to rebuild fine-grained interactional detail of events, often to a level of detail that exceeds what can be captured using fieldnotes or audio recordings. While multi-camera and 360° recordings expand what is captured, they still present particular points of view. Across different cameras — or, by panning and zooming in a single 360 video — viewers follow different viewing paths can create different senses of "what happened," producing multiple possible versions of the event. Multi-camera and 360° video can restore some dimensions of the recorded event, affording different interpretive trajectories. Yet, these trajectories are still constrained within the original, recorded theoretical frame.

With any video record, what we select for analysis is theoretically and culturally structured (Derry et al., 2010). Multi-camera and 360° recordings change what counts as "selectable" because analysts choose spatial vantage points as well as temporal moments. For example, with 360° video, analysts can decide what becomes "in frame" *after* the fact, reshaping what becomes visible and analyzable. With a more complex record, the ways we navigate it — how we choose where to look or move throughout the record — require greater reflexivity and transparency about how we structure our analyses and the theoretical assumptions underlie those choices.

Ethical considerations

The use of video warrants explicit attention to ethics. Immersive and panoramic recordings expand what can be seen, potentially capturing learners who did not anticipate being recorded or activity not visible to teachers or researchers in real time. In an increasingly surveillance-oriented world, these expanded capacities raise questions about consent, circulation, data storage, and the right to opacity. Following Derry et al. (2010), we emphasize the need for transparency and care in decisions about what is captured, who views it, and how these records are used. In this workshop, we will explicitly address issues of visibility, vulnerability, and equity, and invite participants to consider how immersive media can support respectful and responsible research practices.

Practical issues for video data collection

In addition to discussing theoretical issues, we aim to equip participants with practical knowledge and skills to use complex video recording protocols in their research. As with any technology, there is a learning curve to using

new cameras and data collection protocols. However, we have found that using 360 cameras, especially in a multi-camera set-up, has presented substantial challenges.

Our development of these video methods stems from our own research-practice partnership projects (RPPs), in which we collaborate with secondary mathematics teachers to support their reflection and learning. In this work, we use video recordings to document classroom activity, using them as fodder for both teacher reflection and research analysis. These dual purposes have required us to find ways to make the videos usable and legible for multiple audiences. And while our data collection happens in classrooms, we argue that our processes are of interest to the broader learning sciences community — that is, the challenges we encountered and the solutions we have found are common across many learning environments.

Limitations of typical video technologies

A typical use of video for capturing classroom interaction includes a single video camera, with either a whole-class view or a focus on a particular group of students. In our RPP work, our teachers frequently use groupwork as a key part of their pedagogical practice. We conjectured that a typical single-camera set-up would be insufficient for our project, as we would need to choose between framing a whole class or one specific group.

In early iterations of our RPP work (Horn & Garner, 2022), we addressed this concern by using the Swivl platform. The Swivl captures one video stream with up to five separate audio tracks (one teacher and four student groups). This allowed us to select a single framing for the video (i.e., a whole-class, wide-angle view), but with more detailed audio data on student collaboration. However, the Swivl had serious limitations: Students were only recognizable if they spoke out loud. Students with similar voices were difficult to differentiate. And, of course, many aspects of interaction (e.g., gesture) were simply not available in the video record.

Affordances and challenges of multi-camera 360° video

To address these concerns, we sought more complex data collection methods that would allow for richer video data. However, with each technological innovation, we encountered new challenges. For the next iteration of our RPP work, we looked for platforms that would allow us to capture video recordings of multiple student groups. During our pilot work, we tried a multi-camera version of Swivl. Similar to the single-video, multiple-audio set-up, the multi-camera Swivl would enable us to easily synchronize multiple video streams. However, the platform requires strong and stable wifi; this is not always available in research settings, particularly in schools. We ultimately developed our own data collection protocol, consisting of: one whole-class camera (with audio from the teacher), a point-of-view camera worn by the teacher, and three 360 cameras at student groups.

This multi-camera set-up generally captures the rich record of classroom interaction that we were seeking. With 360 video, individual students are visible and audible — we can more readily tell who is speaking and how often. Students' body language and gestures are clear, allowing us to attend to nonverbal interactions. We can also better see what students have in front of them — their worksheets, whiteboards, and manipulatives. The point-of-view and whole-class cameras capture the teacher's movement and conversation throughout the lesson. Taken together, this approach creates rich video records that afford robust analyses of learning.

Yet the complexity of our data collection protocol presented additional challenges, requiring troubleshooting and innovation. In addition to common issues (e.g., battery life, file size), we needed to address:

1. *Video joining:* GoPro cameras record long sessions in smaller clips — about 15 minutes for a wide-angle view, and about 8 minutes for a 360 view. It is impractical to view a lesson in these small pieces. To address this, we developed a Mathematica program that joins the clips together.
2. *Video syncing:* With our multi-camera set-up, it was crucial to develop a way to synchronize the videos within a single lesson. To address this, we developed a second Mathematica program that allows us to synchronize and trim the videos automatically.
3. *Data transfer and uploading:* We needed ways to process, store and share our video data. The large files (up to 60 GB on a single camera) are nontrivial to move (e.g., to a hard drive or the cloud). We identified the equipment that impacts transfer and upload speeds to expedite the process.
4. *Data management:* Coordinating and managing many data files — especially with multi-camera set-ups and multiple data collection events across a large team of researchers — is challenging. We developed strategies for organizing, storing, and parsing our large data corpus.

In this workshop, we plan to share this practical knowledge to support other researchers in using similar equipment. Participants will receive documentation (e.g., workflows) as well as hands-on experience.

Outline of planned activities

In this full-day workshop, we plan to spend the morning session discussing the theoretical aspects of video data collection, including comparisons of the practical and analytic affordances of various methods. In the afternoon, we will address the practical considerations and share the Mathematica programs that we developed.

In the morning, we will begin with *introductions* — of both presenters and participants — and *framing of the workshop goals*. During this time, we will elicit the participants' contexts and experiences with video, allowing us to build on their knowledge and expertise. Then we will *introduce the theory* underlying the workshop, including key concepts from Hall (2000) and Derry et al. (2010); we will center this discussion on how 360 video reorganizes researchers' recording, representation, and viewing practices.

Next, we will have *invited speakers present* how they use 360 video in their own work — aiming for 3-4 brief presentations from different projects (including ours), illustrating the use of 360 video and multi-camera set-ups across classroom and out-of-school learning contexts. We will facilitate a discussion with speakers and participants to share thinking across diverse analytic goals and methodological choices.

To close out the morning portion of our workshop, we will offer a *comparative video demonstration*. Participants will view parallel video recordings of the same event using different media, including: whole-class video with targeted audio, standard wide-angle view, and a 360 view. After viewing the same event through different recording methods, we will facilitate a discussion of the affordances and constraints of each method.

In the afternoon session, we will begin by leading participants in *creating 360 video data* to work with. In small groups, participants will engage in a brief collaborative activity and record it. Before beginning the activity, we will facilitate groups' discussion of *how* to record the activity — each group can decide how to arrange the camera, themselves, and their materials (drawing on the theoretical discussion from the morning).

After recording their own engagement in the collaborative activity, groups will learn how to *process the video data they collected* — including joining and synchronizing the files using our Mathematica programs. Groups will *analyze their video data*, with an emphasis on understanding how their data collection choices impact what is available in the video record. Then we will facilitate a discussion to share insights across the groups. To close the workshop, we will prompt participants to *reflect on the take-aways* for their research.

Expected outcomes and contributions

Participants will leave this workshop with practical knowledge and tools for using 360° and multi-camera video in their research. The Mathematica programs for joining and synchronizing videos are a key contribution, as they solve persistent problems in video data collection. As researchers, it is crucial that our methodological approaches are theoretically sound and ethically grounded. To that end, participants will also grapple with the theoretical underpinnings and ethical implications of the data collection methods they choose to engage with.

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Community Workshops

Listening to the Land and Sea: Community Learning at Crystal Cove Conservancy

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Abstract: This community engagement workshop explores how environmental learning can be designed, enacted, and studied across interconnected land and sea systems. Centered at Crystal Cove State Park, participants engage in an immersive field-based experience that illustrates how public landscapes support connections among schools, community organizations, and professional science and land-management communities. Drawing on learning sciences perspectives, the workshop integrates guided exploration of inland and coastal ecosystems with structured reflection and collaborative sensemaking. Participants examine how learning designs incorporate scientific practices, local ecological phenomena, and community relevance, and how participation structures influence inquiry, equity, and civic reasoning. The workshop also considers how Crystal Cove's existing STEM programs provide a meaningful context for examining field-based learning in environmental and climate education. Participants collaboratively generate design principles and research questions to inform future educational efforts and study.

Overview and rationale for the community engagement workshop

Across the learning sciences, there is growing recognition that consequential learning is not confined to classrooms but emerges through participation in community life and sustained engagement with meaningful problems that matter to people, institutions, and shared environments. Community-based learning frameworks emphasize learning as a process of developing knowledge, identity, and agency through participation in culturally and socially organized practices that are embedded in everyday activity systems (Lave & Wenger, 1991; Rogoff, 2003). This perspective is particularly salient for climate and environmental education, where learning is inseparable from community histories, public decision-making, and collective stewardship of ecological systems that learners can directly observe, measure, and revisit over time (Bang et al., 2012; Fenwick et al., 2015).

Despite increased attention to community-based approaches, the field continues to grapple with how to design and study learning environments that meaningfully connect classrooms, communities, and public institutions at scale, while also supporting equity, civic participation, and rigorous disciplinary learning. Learning scientists face persistent questions about how learners move across settings, how participation structures shape belonging and legitimacy, and how public institutions and shared landscapes function as infrastructures for learning that bridge formal and informal education. Initiatives such as the Environmental and Climate Change Literacy Projects (ECCLPs), a multi-institutional partnership linking universities, school districts, informal learning organizations, and community partners, have emerged in response to these challenges by supporting coherence, relationship building, and mutual learning across educational sectors.

This community engagement workshop centers Crystal Cove State Park and the education programs of Crystal Cove Conservancy as an exemplary community-based learning ecology situated within this broader ECCLPs context that connects schools, informal learning environments, and professional science and land-management communities. Crystal Cove provides a distinctive opportunity to examine how learners engage in authentic scientific practices across interconnected land and sea systems, how they develop relationships to public natural resources, and how educators design and facilitate learning experiences that integrate disciplinary inquiry, civic reasoning, and cultural responsiveness. As a public landscape where scientific inquiry is deeply intertwined with conservation and coastal change, Crystal Cove offers a compelling site for examining learning at the intersection of science, community participation, and civic life (Ardoin & Heimlich, 2021); Sezen-Barrie, et al., 2023).

Theoretical background: Community-based learning

Learning as situated participation

Community-based learning builds from the premise that learning is not simply the accumulation of individual knowledge but a process of participation in social practices and the development of competence in activities that are valued within communities (Lave & Wenger, 1991). From this perspective, learning is inseparable from the

social relations, cultural norms, tools, and material conditions within which activity unfolds. Context is not a backdrop for learning; rather, it is constitutive of what is learned and who learners become through participation.

Complementary sociocultural perspectives emphasize that learning involves the co-construction of meaning through interaction, dialogue, and joint activity within historically developed practices (Rogoff, 2003). These commitments have shaped learning-sciences research on apprenticeship, collaborative inquiry, and learning across settings, and they provide a strong conceptual foundation for understanding community-based learning environments that span schools, informal programs, and public institutions.

Environmental and land-based contexts heighten the importance of this framing because ecological systems are material, dynamic, and consequential for communities. Learners encounter tangible evidence of change, such as shoreline shifts, erosion patterns, biodiversity indicators, and species distribution, and these observations are often connected to public concerns and local decision-making. These conditions create opportunities for learning that link scientific practice with lived experience and collective responsibility, enabling learners to interpret evidence in relation to community histories and future possibilities (Bang et al., 2012; Fenwick et al., 2015).

Community histories, relations, and learning infrastructures

From a community-based learning perspective, public landscapes and institutions are not neutral spaces but are shaped by layered social, political, and ecological histories that are unevenly experienced across communities. These histories matter for learning because they influence whose knowledge is recognized, how participation is organized, and what forms of engagement are legitimized. In learning-sciences research, shared environments can function as epistemic resources that support modeling, explanation, and argumentation, while also foregrounding questions of access, representation, and civic responsibility.

Research suggests that when learners engage with environmental issues that are embedded in community contexts they can revisit over time, they are more likely to sustain inquiry, develop systems-level reasoning, and coordinate evidence with explanation (Hmelo-Silver et al., 2007; Zimmerman & Weible, 2017). Repeated engagement across settings supports iterative cycles of data collection, interpretation, and revision, aligning with learning-sciences work on inquiry, modeling, and the development of scientific argumentation. At the same time, extended participation in community contexts can support affective and relational dimensions of learning, fostering motivation, care, and persistence with complex environmental problems.

Crystal Cove as a community-based learning ecology

Crystal Cove State Park is one of California's ecologically and culturally significant coastal landscapes. Established in 1979, the park encompasses approximately 3,936 acres along 3.2 miles of Pacific coastline and extends inland through coastal bluffs, chaparral canyons, and undeveloped backcountry within the South Coast Wilderness area (California State Parks, 2010). Located between Newport Beach and Laguna Beach in Orange County, Crystal Cove includes extensive beaches, tide pools, inland canyons, and a large marine conservation area and underwater park, creating continuity between terrestrial and marine ecosystems (California State Parks, 2011). Ecologically, Crystal Cove is a biodiversity-rich environment supporting diverse bird species, reptiles, amphibians, and rare coastal plants and animals found in Southern California (National Audubon Society, 2012; Californian Wildlife Region, 2013). The park supports a wide range of recreational and educational activities, including hiking, tide-pool exploration, and marine studies, which shape who encounters the park and how learning opportunities can be designed to be accessible and inclusive. In partnership with California State Parks, Crystal Cove Conservancy plays a critical role in expanding public learning opportunities through structured STEM programs. The Conservancy brings more than 7,000 low-income students to Crystal Cove each year, many for their first visit to the ocean, and provides programs at no cost to schools to reduce economic barriers. These programs exemplify community-based learning through sustained engagement, collaboration with scientists and resource managers, extended curricular sequences, and opportunities for youth to contribute data and interpretations that matter to real conservation and management questions.

More recently, Crystal Cove Conservancy has attracted interest from university-based researchers studying environmental and climate literacy. The Conservancy's STEM programs offer a valuable educational setting for examining how students engage with ecological phenomena, scientific practices, and conservation questions in field-based contexts. At this stage, engagement is exploratory and focused on understanding how existing programs contribute to student learning and teacher development. Any research activity is conducted in alignment with Crystal Cove's established educational mission and supports ongoing efforts to enhance public learning experiences within the park.

Figure 1

Crystal Cove State Park as Site for Learning about Fire Prone Habitats and Connection to Marine Life



Crystal Cove community engagement workshop: Activities, structure, and call for participants

Workshop activities and structure

This half-day community engagement workshop is designed for learning scientists, educators, and community partners interested in how community-based environmental learning is designed, enacted, and studied across interconnected land and sea systems. The workshop is intentionally structured as a focused, immersive experience rather than comprehensive professional development. Participants engage in a sequence of field-based and reflective activities that make visible how Crystal Cove functions as a community-based learning ecology. The workshop combines guided exploration of inland and coastal environments with structured opportunities for sensemaking, allowing participants to examine how learning designs integrate scientific practices, local phenomena, and community relevance. Across activities, facilitators foreground learning-sciences perspectives on participation, boundary crossing, systems thinking, and civic reasoning, while inviting participants to surface problems of practice and design questions from their own contexts. The workshop concludes with collaborative synthesis, where participants connect field experiences to broader implications for research, design, and partnership work in community-based learning. The timeline below outlines the flow of activities and focal goals for each segment.

Table 1
Timeline of Activities (3^{1/2} hours)

Time	Focus	Description
0:00–0:30	Welcome, framing, and orientation	Welcome participants and introduce facilitators and partners. Overview of workshop goals, agenda, and norms for equitable participation. Orientation to Crystal Cove as a community-based learning ecology.
0:30–1:00	Inland orientation and guided field walk	Guided walk through inland ecosystems, including coastal sage scrub, chaparral canyons, and restoration areas. Facilitators highlight how land-management challenges become sites for student inquiry and learning.
1:00–1:30	Context mapping and surfacing problems of practice	Small-group reflection grounded in field observations. Participants identify a problem of practice or possibility for new practice from their own contexts, using shared templates to connect observations to learning-sciences concepts.
1:30–1:45	Break	Short break and transition from inland systems to coastal environments.
1:45–2:15	Shoreline walk and cross-system connections	Guided beach walk connecting inland ecosystems to shoreline and marine processes. Focus on systems thinking, socio-scientific reasoning, and public spaces as learning infrastructures.
2:45–3:15	Collaborative sensemaking and design implications	Small-group discussion synthesizing insights from field experiences. Participants examine design principles, equity considerations, and research implications for community-based learning across contexts.
3:15–3:30	Sharing, synthesis, and next steps	Groups share key takeaways and questions. Facilitators synthesize across discussions and support reflection on future collaboration, research, and dialogue beyond the workshop.

Learning goals and outcomes

By the end of the workshop, participants will:

- Recognize Crystal Cove as a community-based learning ecology that supports learning across settings, institutions, and communities
- Identify design principles that support disciplinary learning, equity, and civic reasoning
- Generate research questions and methodological considerations relevant to studying learning across contexts and public landscapes
- Build connections with other participants interested in community-based learning, research–practice partnerships, and environmental education

Call for participation

We invite learning scientists, educators, and community partners to join this half-day community engagement workshop focused on community-based environmental learning at Crystal Cove State Park. This session is designed for participants interested in learning across settings, research–practice partnerships, equity-oriented design, and the role of public institutions as learning infrastructures (Akkerman & Bakker, 2011). Participants are encouraged to bring a question, problem of practice, or research interest related to community-based learning, environmental education, or learning across contexts. The workshop engages participants as co-learners through shared field experiences, structured reflection, and collaborative sensemaking, offering opportunities to connect theory with practice in a public landscape. Grounded in learning-sciences theory and situated within a living learning ecology, the Crystal Cove workshop offers participants an opportunity to examine how community-based learning is designed, experienced, and theorized, while building relationships that support future collaboration and inquiry within the ISLS community.

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Playful Learning in the Community: An interactive tour of community based early learning installations

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Abstract: Playful Learning Landscapes (PLL) transforms everyday community spaces like parks, bus-stops, and grocery stores into conduits for caregiver-child interactions and early learning. During this workshop, attendees will visit a series of PLL installations in Santa Ana, CA, which were co-designed by researchers from UC Irvine and Temple University, and families from the Santa Ana Early Learning Initiative, a local grassroots non-profit community organization. Attendees will engage with the PLL installations and learn about the design process and impact of the installations on community members and caregiver child interactions at PLL sites. The session will also demonstrate how this project was conducted in partnership with the City of Santa Ana's Parks and Recreation and Public Works departments who own and maintain the installations and have begun to replicate them using public funds in other community spaces. Challenges, lessons learned, plans for sustainability, and practical and theoretical insights will be discussed.

Playful learning in the community: An interactive tour of community based early learning installations

Informal learning environments like children's museums, zoos, and maker spaces are powerful contexts for supporting young children's science, technology, engineering, and math (STEM) learning and identity. Unfortunately, children from low-income families have less access to these spaces due to a variety of barriers like admission prices, transportation challenges, or even an awareness of their existence. In addition to these barriers, linguistically and ethnically minoritized families can feel alienated in these spaces which rarely center their languages, cultural values, and practices. Playful Learning Landscapes is an international initiative that brings informal learning installations to everyday spaces like parks, bus-stops, grocery stores, laundry mats, medical clinic waiting rooms, sidewalks, and walls (Hassinger-Das, 2018; Bustamante et al., 2019). Playful Learning Landscapes promote the caregiver-child conversations and interactions that decades of developmental and learning sciences research suggest lead to learning, such as conversational turns, STEM language, and hands-on engagement (Bustamante et al., 2020; Hassinger-Das et al., 2021; Pesch et al., 2025). In addition to designing PLLs based on the latest research in developmental and learning sciences, they are also designed in partnership with local families, so the installations build from families' values, routines, and cultural practices (Bermudez et al., 2023). Understanding learning as a cultural process situates family's participation in learning environments and design as dictated by values, routines, and cultural practices (Bang & Medin, 2010; Rogoff et al, 2007). Leveraging cultural assets, practices, and values can therefore increase relevance and significance for families, in turn increasing inclusivity, sustainability, and engagement.

Partners

The PLL work in Santa Ana, CA, is conducted in the context of a long-term research practice partnership between the Santa Ana Early Learning Initiative (SAELI), the University of California Irvine, and the City of Santa Ana. SAELI is a grassroots non-profit community organization bringing together local families, non-profit organizations, and leadership from the Santa Ana Unified School district to support families with children between 0-9 years of age. SAELI is led by director Wendy Gomez and uses a multigenerational approach through which *promotoras* (e.g., Chen et al., 2023) promote wellness through initiatives on child development, connecting families with resources, and local advocacy projects. On the university side, these projects are led by UC Irvine

School of Education Professors Andres Bustamante and June Ahn and Temple University Psychology Professor Kathy Hirsh-Pasek. This research team brings expertise in child development, early education, participatory design, and research practice partnerships. The City of Santa Ana supported the project through their Parks and Recreation Department and Public Works Department, with support from the City Manager's Office, Mayor's Office, and City Council. Other organizations and businesses included in this work were Northgate Gonzalez Market grocery chain, the Santa Ana Unified School District, First Five Orange County, and Orange County Health Care Agency.

In this RPP, we endeavored to center families' voices in all phases of the process from designing the installations, to deciding their locations, to the evaluation research, and dissemination of the findings. Importantly, trust and relationship-building with families were central to the design sessions. Personal stories and family values were shared through informal conversations over meals. These informal opportunities to connect allowed trust to be built over time, which uniquely positioned researchers and families as equals. We believe this allowed families to share openly and addressed traditional power dynamics between researchers and families. Decisions were taken jointly through parallel conversations with partners and interested parties, requiring transparency, trust-building, multiple channels of communication, rapid problem-solving, adaptation, and other forms of invisible relational labor. However, tensions emerged throughout the process particularly when navigating the intersection of large bureaucracies including federal funding, university systems, and city government. The project faced multiple challenges, including changing project timelines and material constraints, changes in organizational leadership, physical limitations of the public spaces designated for installations, and the complex social and political contexts of the community. Given existing power differentials between governmental partners and community stakeholders, our university team played a critical intermediary role, advocating on behalf of the community in certain cases. Whereas in other cases, our community partners stepped up and advocated for the project using their influence as voting constituents and activists in their community. Ultimately, the political capital of all stakeholders was necessary to bring this project to fruition.

Research design in the community

The Santa Ana PLLs were supported by an Advancing Informal STEM Learning (AISL) award from the National Science Foundation. The project began in 2020, and due to the pandemic, all design sessions in the first year of the project were held on Zoom. We held seven 90-minute co-design sessions where we learned about families histories and experiences with their community spaces, and used participatory design activities such as bags-of-stuff, photo-elicitation, story boarding, and storytelling to encourage families to share their goals and routines in their community spaces (for a detailed description of design process see Bermudez et al., 2023; Garcia et al., 2022; Chew et al., 2022).

Through an inductive thematic analysis of these co-design sessions, we surfaced community values, practices, and aspirations such as promoting intergenerational learning, and increasing family time. As part of these themes, we also surfaced ways in which families' own routines and practices already involved STEM concepts and practices. We iteratively combined these themes and existing practices with the science practices and crosscutting concepts from the Next Generation Science Standards (NGSS) and principles from developmental science for creating high-quality early learning environments that are active, engaging, iterative, social, meaningful, and joyful (Hirsh-Pasek et al., 2015). While a single design could not meet all the criteria, we sought balance across community values, developmental principles, and science learning goals (see Bermudez et al., 2023 for more detail). Further, we were intentional to align these practices with SAEI's three step design model that included engaging parents in *discovery & dreaming*, *designing*, and creating a *sustainability plan*. Using these design goals and guiding frameworks, we conducted prototype ideation sessions with SAEI families and then subsequent sessions where we created low-fidelity mock-ups of the prototypes for play testing and iteration during the second and third years of the project. Children and parents were interviewed to hear their feedback on the designs and learn about their experiences playing with the physical prototypes. These sessions were key to evaluate factors from usability to how and if families' interactions aligned with their values and practices. These sessions were held in person at their local Community Center, started with a shared meal, followed by a design activity, reflections and share outs. Through these sessions we iterated and refined the prototypes until they were ready to be shared with our fabrication partners.

Fabrication and launch

For fabrication, we contracted a Los Angeles based museum installation design shop. We worked closely with their engineers to adapt the co-designed prototypes for durability, usability, safety, cost efficiency, and regulation compliance. All designs were reviewed by City of Santa Ana Parks and Recreation or Public Works departments and UC Irvine's Facilities Management department. We worked collaboratively with the City of Santa Ana and

SAELI to decide where the installations would go, taking advantage of existing city renovation efforts to save costs and create efficiencies when possible. We also worked with legal representation from UCI and the City of Santa Ana to execute a Memorandum of Understanding that the PLL installations were being donated to the city by UCI and the City of Santa Ana would own and maintain them. This process took approximately two years from the beginning of the fabrication process to installation in public spaces. The installations were implemented between Fall 2024 and Spring 2025 with a grand opening ribbon cutting ceremony held on May 10th, 2025, as part of the NSF's 75th anniversary celebration and Mother's Day in many Latin American countries. The ceremony was hosted at Madison Park in Santa Ana and attended by the Mayor, Director of Parks and Recreation, several city council members, and the Dean of the UCI School of Education.

Data collection

Our data collection protocol included baseline observations of STEM language and caregiver child interactions at each of the five installation sites pre-installation as well as a matched comparison site (e.g., for a bus-stop installation we observed a comparison bus stop in the same neighborhood). We also trained SAELI parents to collect surveys at all PLL and control sites of caregiver's self-efficacy in teaching STEM to their children, perceptions of safety and learning opportunities in their community spaces, and attitudes around the connection between play and learning. We then performed the observational and survey data collection protocol post-installation to examine the impacts of the PLL's on caregiver child interactions and experiences in community spaces. While post-installation data collection is on-going we expect to present findings from these analyses at the ISLS workshop in June 2026. For a more detailed description of our analytical plans please refer to our project pre-registration on the Open Science Framework.

Workshop details

The tour will be hosted by UCI Associate Professor and project PI Andres Bustamante, UCI Professor and project Co-PI June Ahn, SAELI Director Wendy Gomez, and several SAELI parents who participated in the design and data collection processes. The hosts will offer insights into the project's co-design approach, the partnership with the city (challenges and opportunities), parents' roles as designers, data collectors, and project advocates, impact results, and future plans for policy agenda towards sustainability and scaling. The timetable of activities and sites to be visited are shown below (Table 1). We will also discuss several cross-site design considerations, including Playful Learning Principles, culturally responsive design, the use of public spaces as learning environments, multigenerational engagement, attention to accessibility, and low-burden participation that aligns with families' everyday routines.

Table 1
Timetable of Activities

Hour	Activity
8:00 am	Meet at the conference hotel at 8am PST on June 15 th , 2026. We will depart on a charter bus to tour the five PLL sites in Santa Ana.
8:30 am	Arrive at Santa Ana, touring the 5 different sites (see text below).
11:30 am	End of the tour. The charter bus will be taking people back to the conference hotel.

Site 1: Madison Park.

Madison Park features the largest of the Santa Ana PLL installation, Parkopolis, the life-sized board game for math and science learning. Parkopolis is a giant board game where children spin life sized dice and advance around the board completing math and science games like playing patterns on musical pipes, and measuring how far you can jump on a giant ruler. Through the co-design process, we learned about games like Basta, a category game that many SAELI families play, and integrated them into Parkopolis games. During the co-design sessions, many families expressed concerns about visiting their neighborhood parks and cited the bathroom facility in the back of Madison Park as a hotspot for illicit activities like drug use and prostitution. In response, the city demolished those bathrooms replacing them with new bathroom facilities in another part of the park, and we constructed Parkopolis where the old bathrooms were transforming a hot spot for criminal activity into a hub for early learning, a powerful example of community voice leading to impact in public infrastructure.

Site 2: Abacus bus stop.

The giant abacus at the bus stop on the corner of Main Street and McFadden Avenue in Santa Ana invites families to engage in conversations around numeracy and arithmetic. Many families in our design session reference using an abacus to learn math when they were children in Mexico and other countries in Latin America and wondered why their children's schools didn't use an abacus. This installation is a powerful symbol of scalability because it was not funded by our NSF grant. The city of Santa Ana was renovating bus stops in this area and due to the awareness of our projects, leveraged existing funds to execute the design.

Site 3: Northgate Gonzalez market.

Northgate is a family owned local grocery chain that many of the SAELI families shop at. During our co-design sessions, we used families stories about grocery shopping with their parents growing up or with their children now to develop signage that promote math and science interactions. Examples include a conversion chart between pounds and kilos at the deli and using your senses to pick out the best fruits and vegetables in the produce section. This installation is unique because it demonstrates how local business and the private sector can engage and support this work.

Site 4: Angels Park.

Angels Park features two PLL installations, *La Lotería* which is a giant version of the bingo style game that is very popular with local families, and *How Tall Am I?* which is a statue where children can measure their height in comparison to California animals. Both of these installations are a great example of intuitive design and leveraging existing knowledge to create engaging STEM learning experiences. *How Tall Am I?* invites immediate engagement through its visual design and integrated measurement markers, requiring only minimal instruction in the form of a single guiding question. Similarly, *Lotería* is among the most popular family games in this community, lengthy instructions are not necessary. Instead, we simply embed short STEM activities into the playing cards for families to integrate into the game play. In addition to supporting informal STEM learning, both of these installations were co-designed with the intention of supporting intergenerational learning which is a core value of Latine families.

Site 5: Washington elementary school.

The last stop on the tour will visit the *Yo Veo (I See)* Mural, an interactive art exhibit where children look for hidden objects that reflect local cultural and community activities. This is another example of a local institution adopting a project design. This mural was funded by the Santa Ana Unified School District, who were renovating this school and directed funds for murals. This school also features a different project, Fraction Ball, a redesigned basketball court for math learning.

Future directions

During the May 10th ribbon cutting event, the City of Santa Ana made an official proclamation that May will be Early Childhood Education month in the city in recognition of this innovative work. Building on this momentum, we are working with city officials on a resolution to integrate PLLs into the city's on-going process for new public amenities leveraging existing budget from local institutions. We plan to position SAELI as drivers of future PLL efforts where they engage local community members in a design process of new public amenities that center early learning and build from local families' funds of knowledge. This work represents a scalable and community centered vision for designing public infrastructure to promote child development and civic engagement.

Invitation to ICLS participants

This session examines the evolution of a project and partnership from initial creation through implementation, using this project as a live case study to surface challenges, lessons learned, and future directions from different points of view. In doing so, we demonstrate how *partnering with purpose* can generate meaningful community impact and offer useful insights for other scholars and initiatives seeking to engage in collaborative, community-centered work.

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Early Career Workshops

Engineering a Learning Environment Which Fosters Interaction and Thinking

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Abstract: My research investigates how Visible Thinking and dialogic teaching can be integrated to foster interaction, thinking, and a culture of dialogue in EFL higher education. Using design-based research, I have iteratively designed and refined thinking-routine-embedded instruction and am now extending this work along three strands: analyzing thinking and dialogic moves in student discourse, exploring the cultural and affective dimensions that sustain a thinking culture, and refining design principles through curricular-design alignment between conjecture mapping and the curricular spider web. This ongoing program aims to build a theoretically and contextually grounded framework explaining how language, interaction, and design co-mediate learning in East Asian EFL classrooms.

Introduction

My research seeks to understand and design learning environments that foster interaction, thinking, and a culture of dialogue in English-as-a-foreign-language (EFL) higher education. Grounded in a sociocultural perspective of interaction (Mercer, 2000), I examine how students' reasoning and meaning-making are mediated through talk and through the design of instruction.

During my doctoral study, I conducted a multi-iteration design-based research (DBR) project embedding Visible Thinking (Ritchhart et al., 2011) into university English courses. Through iterative refinement, I developed four instructional design principles that demonstrated how thinking-routine-based instruction can externalize students' reasoning, enhance collaborative dialogue, and nurture a supportive classroom ethos. After completing my PhD, my inquiry evolved from investigating whether thinking routines enhance learning to exploring how and why learning occurs through dialogic interaction and culturally situated classroom practices.

Across current and future projects, I aim to advance an integrated research program that connects three strands: (1) analyzing the intersection between thinking moves and dialogic moves; (2) examining the cultural dimension of thinking and classroom ethos, and (3) refining design principles through an aligned curricular-design framework.

Thinking and dialogic moves: Exploring the intersections and tensions in collective reasoning

The first strand focuses on unpacking how thinking moves and dialogic moves function together to mediate collaborative learning. My current analyses code students' discourse in two parallel layers: thinking moves—such as explaining, connecting, reasoning with evidence, and considering viewpoints (Visible Thinking)—and dialogic moves, adapted from the Scheme for Educational Dialogue Analysis (SEDA; Hennessy et al., 2016). Rather than adopting SEDA's full eight-cluster framework, I selected a subset of dialogic categories most relevant to my classroom context, including express, build on, reason, coordinate, and position. These categories capture the interactional mechanisms that keep the dialogue going—how students initiate ideas, sustain discussion, reason, and display agreement or disagreement. In many of my classroom tasks, these dialogic processes are interwoven with the joint creation of tangible objects, such as concept maps or character analysis diagrams. This task structure positions talk and object production as mutually shaping processes, echoing Damşa's (2014) view of collaborative learning as a multi-layered phenomenon where interaction, knowledge objects, and shared epistemic agency are tightly interconnected. Recognizing this interplay between discourse and object-making highlights the need to examine not only how thinking and dialogic moves unfold, but also how they may be embedded in broader collaborative practices. My goal is not to merge thinking and dialogic moves into a single system, but to examine their co-occurrences, sequences, synergy, and potential tensions.

This methodological stance reflects my theoretical grounding in Vygotsky's view of language as both a psychological and cultural tool (cited in Mercer, 2000), and Mercer's concept of dialogue as a "social mode of thinking" (Mercer & Littleton, 2007). Both perspectives highlight that thinking and dialogue are mutually constitutive: dialogue enriches thinking, while thinking gives substance to dialogue. However, analysis from my previous study (under review) shows that these two sets of moves can partially overlap or even conflict. Some dialogic moves (e.g., "build on") may serve interactional continuity without necessarily deepening reasoning. I therefore aim to explore the underlying mechanisms that may account for these overlaps and tensions in classroom

discourse, potentially revealing how local cultural contexts influence patterns of interaction and meaning-making. As Lefstein and Snell (2011) stress, classroom discourse cannot be easily generalized, as patterns of talk are shaped by distinct national and institutional cultures and may even shift within the same classroom depending on goals, topics, and activities. This insight resonates with my analysis of thinking and dialogic moves, which not only examines their micro-level co-occurrences but also considers how these moves are mediated by the objects, instructional design and by cultural expectations in EFL classrooms. Together, these perspectives foreground how dialogic interaction operates not in isolation but within broader learning ecologies, paving the way for examining how classroom culture shapes and is shaped by collaborative thinking practices.

Cultural dimension and classroom ethos: Building a culture of thinking

The second strand extends from micro-level discourse to the meso-level of classroom culture. In a recent cross-case exploratory study of two EFL university courses, I analyzed students' reflections and identified four interrelated elements shaping a culture of thinking (Ritchhart, 2015): teacher guidance and emotional support, peer interaction and supportive ethos, structured tasks that scaffold reasoning, and improved confidence and comfort in expression.

This line of work draws on sociocultural theory, Visible Thinking's eight cultural forces (Ritchhart, 2015), and dialogic teaching's notion of a dialogic culture (Alexander, 2020; Resnick et al., 2018). While these frameworks highlight different facets of classroom culture, they share a concern for the social and moral fabric sustaining intellectual engagement. Integrating them helps interpret my findings as manifestations of how cognitive, relational, and ethical dimensions converge into a shared ethos where thinking becomes a collective responsibility.

In East Asian EFL contexts, linguistic anxiety and hierarchical classroom norms often discourage students from taking intellectual risks (Yuan et al., 2022). Therefore, I aim to examine how teacher mediation and peer relationships foster mutual respect, safety, and belonging, and how such affective conditions enable more sustained and reasoned dialogue. Ethnographic observation and cross-case thematic analysis will help trace how these cultural elements emerge and evolve. Through this work, I seek to show how classroom ethos functions as both a precondition and a product of collective thinking.

Refining design principles through curricular-design alignment

The third strand advances my design-based research trajectory by refining the design principles that structure thinking- and dialogue-rich instruction. My earlier design principles emphasized general curricular alignment. Building on this foundation, I now aim to develop a more nuanced and context-sensitive framework that interweaves thinking, dialogic, and cultural dimensions. Firstly, thinking design defines the specific forms of reasoning instruction aims to cultivate and explores how these thinking actions can be observed and scaffolded. Secondly, dialogic design focuses on structuring talk opportunities and teacher prompts that move discussions beyond surface-level participation toward deeper reasoning. Thirdly, cultural design considers how contextual values, classroom norms, and relationships can be intentionally designed to sustain intellectual engagement.

Methodologically, this strand builds on my previous attempt to align the curricular spider web (van den Akker, 2013) with conjecture mapping (Sandoval, 2014). In that work, I sought to ensure curricular coherence by connecting two types of alignment: (1) aligning curricular objectives, thinking goals, learning tasks, environment design, and assessment; and (2) aligning the embodied instructional designs with their underlying theoretical principles. This alignment helped visualize how conjectures about learning mechanisms are realized within concrete curricular elements. Although that framework was produced as a conceptual outcome of my prior DBR, I now plan to implement it as a methodological model in my next phase. By integrating it with the three-phase DBR cycle (McKenney & Reeves, 2012; Gravemeijer & Cobb, 2006), I will iteratively design, enact, and analyze new instructional cycles that bridge theory, design, and practice. This model will allow me to observe more closely how designed tasks, classroom discourse, and cultural development interact, thereby generating empirically refined mid-level design principles relevant across EFL and dialogic learning contexts.

Future directions and community engagement

In the coming years, I plan to initiate a new cycle of DBR that integrates these strands into a unified program of inquiry. I also aim to collaborate with international researchers in dialogic teaching and learning sciences to explore cross-context adaptation of dialogic and thinking-based pedagogies. Through this program, I hope to contribute both a contextualized perspective from East Asian higher education and a theoretical framework that connects cognitive, social, and cultural dimensions of learning, advancing the shared goal of understanding how thinking and dialogue intertwine to transform classroom learning and culture.

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STEAM in the Learning Sciences: Foregrounding Heritage Algorithms in the Design of Learning Tools

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Abstract: My research examines how cultural practices (e.g., barbering, cosmetology, fashion design) and epistemologies (i.e., ways of knowing) can be learned and sustained while acquiring computer science (CS) skills. I draw on ethnocomputing and leverage heritage algorithms—hybrids of computing and culture—to create simulations of cultural practices using algorithms. These simulations can be used to teach cultural ways of knowing and making as well as CS skills. Overall, my research focuses on sustaining cultural making practices and cultivating the creative aspects of CS education through community-based participatory design research. In this statement, I share my previous work in CS education and ethnocomputing and situate future strands of my work in STEAM and the learning sciences.

Introduction

Earlier this year the editors of the *Journal of the Learning Sciences*, Erica Halverson and Leema Berland, asked for the field to expand “what counts as the learning sciences” (2025). Drawing on the field’s commitment to bring together diverse scholars, tools, and disciplines, Halverson and Berland point to this range as the source for continued growth and success in understanding learning and supporting all learners. While the field is fundamentally concerned with learning, what it means to learn, and how to design for learning, there is growing interest and participation in the maker movement and making in educational contexts that is coupled with the large role that the arts play in reimagining teaching and learning (Halverson & Sawyer, 2022; Lachney & Foster 2020). Given this shift to include arts, making, and creativity as central parts to learning, an integrated approach to science, technology, engineering, and mathematics (STEM) education has increasingly drawn on the mutual benefit of including the arts to improve STEM pedagogy and create learning environments that foster inclusivity and sociocultural knowledge production. This inclusion of the arts creates “STEAM”, or an integrated approach to science, technology, engineering, art, and mathematics.

Technologists and proponents of computing education have long argued that computing serves as a gateway or bridge for creativity in learning contexts by connecting creative expression and STEM learning (Papert, 1980; Kafai et al., 2014; Bennett et al., 2016; Bennett, 2021). In early work on ethnocomputing, Eglash and colleagues (2006) demonstrate the affordances of this relationship by creating a suite of tools called *Culturally Situated Design Tools*—digital tools that explore the computational and mathematical knowledge of textiles, architecture, stamps, and other practices from various Indigenous and vernacular cultures. My work lies at this intersection of culture, arts, learning, and technology and is situated in computer science (CS) education. Through the lens of ethnocomputing, I explore how we can sustain Black cultural ways of knowing, being, and doing while teaching and learning computational skills, highlighting the bidirectional relationship between computing and culturally situated making (Allen Kuyenga, Gladney, et al., 2023; Allen Kuyenga, Lachney, et al., 2023).

General Framework: Ethnocomputing in STEAM

Ethnocomputing recognizes the ways in which computing knowledge exists outside of Western assumptions of what counts as computing knowledge and challenges the dichotomies between cultural and technical knowledge and knowledge production. I explore the unique relationships between ethnocomputing and STEAM concepts like computational thinking (CT) through “contact zones” (Lachney et al., 2021). Contact zones are those epistemic encounters between different cultures where learning and creating are done in light of histories of violence and extraction (Allen Kuyenga, Lachney, et al., 2023). These zones allow for multidirectional translations between computing and cultural ways of knowing, being, and doing. These translations offer a buffer against CT and computing’s tendencies to subsume cultural lifeways and reproduce logics of imperialism, including assimilation and extraction (Lachney et al., 2021).

Ethnocomputing offers an alternative way of thinking about how we engage historically excluded groups of people and their knowledge in computing and computing education. One way I do this is by leveraging “heritage algorithms.” Bennett (2016) defines heritage algorithms as hybrids of computation and culture that circulate intergenerational value within communities. Indeed, heritage algorithms demonstrate how Indigenous and vernacular communities have developed and maintained technological creativity and ingenuity.

Examples of heritage algorithms can be found in cornrow braiding (Bennett, 2016), cosmetology (Lachney et al., 2019), barbering (Allen Kuyenga, Gladney, et al., 2023), and textiles (Bennett, 2016; Allen Kuyenga, 2025). These examples have offered a systematic way to analyze computational representations of creative expressions and cultural knowledge systems related to making and creativity.

Methods

In this research, I ask two main questions. First, I ask: *how can we leverage CS education to sustain the making, communal, and relational knowledge practices found within Black communities?* Second, I ask: *what computational knowledge, embedded within the making and communal practices of Black peoples, can be leveraged to teach CS skills?* These questions guide my collaborations with artists and learners, my design of learning environments and tools, and the analysis of teaching and learning that occur in these spaces and/or with these tools. To answer these questions, I employ Design-Based Research (DBR) as a primary methodological framework, because its iterative pattern of design and reflection is useful for understanding how, when, and why innovations work in complex contexts, like classrooms (The Design Based Research Collective, 2003; Kafai, 2005; Hoadley & Campos, 2022). DBR in conjunction with participatory methods support my commitment to design with, rather than for, communities, positioning students, teachers, and artisans as co-designers and co-researchers (Allen Kuyenga, 2025). These collaborative processes also allow me to investigate what happens when knowledge is exchanged within contact zones using ethnographic methods to attend to specific cultural and relational dimensions of learning and design. Lastly, I leverage narrative analysis to trace participant meaning making and creative expression while engaged in STEM education.

Future research

At the start of my career and as I formally enter the Learning Sciences community, I bring a research agenda that foregrounds computational learning as culturally grounded and relational in nature. I position my work in this community as part of a broader effort to “expand what counts as learning” and more specifically to expand what counts as CS learning and CS knowledge. To this end, I and my colleagues (2023) have previously presented *pathways* as an agentic alternative to the pipeline metaphor common in computing education. The concept of pathways constructs alternative understandings of *what counts as computing* that privilege student identities and desired paths—the expansive routes to engaging with computing through various entry points like barbering, cosmetology, and fashion design.

I will continue pursuing new pathways by developing other community-based participatory design research projects to create spaces and tools from the knowledge and perspective of community members, like artisans and designers. While it is still in the earliest stages, I am developing this work in community with others who share the priorities of bringing Black epistemologies and making practices to bear on how we teach, learn, and imagine CS education. I aim to co-design a series of virtual tools and curricular materials that integrate making practices, computational knowledge, and community perspectives on computing to provide robust educational materials for youth and adult learners in and around the U.S. state of Pennsylvania in cities like Pittsburgh, Harrisburg, and Philadelphia. These projects will work to meet the need of culturally sustaining CS education and provide opportunities for intergenerational learning in CS learning contexts. I hope that this work also brings together researchers, communities, and practitioners to have generative dialogue around what it means to do this work and commit to sustaining cultural making practices in a climate that increasingly devalues and erases these practices, their relevant knowledge systems, and their histories.

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Reimagining Literacy as Liberatory Learning: Centering Black Literacies in the Learning Sciences

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Abstract: This paper outlines a research program situated at the intersection of the Learning Sciences, Literacy Studies, and Black Studies. The work investigates how racialized histories of literacy shape contemporary learning environments and how Black learners and communities design countercurricula that affirm joy, resistance, and self-determination. Drawing from archival, ethnographic, and design-based methods, this research advances a theoretical framework of *bound literacy*—a concept describing how literacy has historically been constrained by antiblackness—and explores what learning might look like when literacy is “unbound.” The program contributes to the Learning Sciences by positioning Black literacies as sites of innovation, imagination, and design, offering frameworks and methodologies for understanding learning as a collective, historical, and liberatory act.

Overview

My research bridges the Learning Sciences, Literacy Studies, and Black Studies to examine how racialized histories of literacy shape learning and how Black communities design liberatory literacies in response. This paper outlines my research program on “bound” and “unbound” literacies and its contributions to theorizing learning as a site of freedom-making.

Theoretical framework: Learning as a liberatory and historical practice

My research begins from the premise that learning is not only cognitive or social but also historical and political. Building on sociocultural theories of learning (e.g., Vygotsky, 1978) and critical traditions within the Learning Sciences (e.g., Nasir et al., 2006), I conceptualize literacy as a designed social practice shaped by structures of race, power, and history. To theorize this relationship, I introduced the framework of *bound literacy* (Massengale, 2024), which traces how literacy in the U.S. popular imagination has been historically “bound” to chattel slavery and its afterlives. This reframes literacy not as a neutral skill set but as a cultural construct used to delimit who counts as a learner. By examining how literacy has been racialized—particularly through associations with Black deficiency—this framework exposes how antiblackness has shaped the epistemological foundations of schooling, assessment, and learning theory. At the same time, I view literacy as a site of resistance and design. Across historical and contemporary contexts, Black people have cultivated literacies that imagine freedom beyond dominant educational systems. Aligning with Learning Sciences scholarship that centers imagination, equity, and design (e.g., Gutiérrez & Jurow, 2018), I argue that Black literacies are world-building practices that create new ways of being, knowing, and learning.

Methodological approach: Bridging historical, ethnographic, and design-based inquiry

My research blends archival analysis, ethnographic fieldwork, and design-based research to trace how Black literacies emerge, persist, and transform across time. This multimodal approach positions literacy as a layered, intergenerational practice connecting past, present, and future learning ecologies. Through archival work, I analyze primary sources such as the Federal Writers’ Project (FWP) slave narratives to examine how formerly enslaved people described their learning and teaching practices. These narratives reveal literacy as both a tool of domination and a site of resistance, laying the groundwork for theorizing how racialized notions of literacy continue to shape educational discourse. Complementing this historical analysis, I conduct long-term ethnographic research with contemporary Black learners in community-based educational spaces. My dissertation centered on an informal enrichment group in the San Francisco Bay Area that fosters racial affirmation and creativity among Black children aged four to twelve. Over eighteen months of fieldwork—including participant observation, interviews, and multimodal documentation—I explored how children author their own literate identities and reimagine reading and writing as acts of joy and self-determination. Finally, I approach design-based research as a generative praxis rather than a prescriptive model. Building on insights from my ethnographic work, I developed and iteratively refined a series of Creative Literacies workshops that invite children to co-design literacy experiences through storytelling, art, and play. These workshops function as sites of learning,

design, and theory-building, illuminating how liberatory literacies can be cultivated and sustained in real-world contexts.

Research contributions to the learning sciences

This program of research contributes to the Learning Sciences in three key ways: First, my work *reframes literacy as a designed and historical construct*. By tracing the racialized histories that shape current learning practices, my research expands the field's understanding of how sociocultural contexts and power relations influence learning. I understand literacy not just as situated action (Lave, 1987) but as historically sedimented design—an inheritance that both constrains and enables future learning. Secondly, this work *centers Black literacies as sites of epistemic innovation*. My research foregrounds the creative and intellectual labor of Black learners and communities, showing how their everyday practices constitute theories of learning in their own right. This approach aligns with equity-oriented and justice-centered movements within the Learning Sciences (e.g., Esmonde & Booker, 2017), while also contributing new conceptual tools for studying learning as a mode of freedom-making. Finally, my research *integrates ethnographic and design-based methodologies*. The methodological synthesis I employ offers a model for how the Learning Sciences can bridge qualitative, historical, and design traditions. This allows me to weave archival and contemporary data into narrative analyses that honor the imaginative dimensions of learning. Design-based approaches then transform these insights into practical interventions that support liberatory pedagogies.

Future directions

Moving forward, I plan to extend this work in three interconnected directions. First, I will explore *intergenerational literacies* by examining how literacy learning unfolds across generations within Black families and communities. Preliminary findings from archival and ethnographic data suggest that literate identities are familial and relational—passed through stories, songs, and everyday acts of care. This analysis will inform a theory of intergenerational literacy learning that highlights continuity, inheritance, and imagination across time. Second, I am committed to *community-engaged design*. I plan to expand the Creative Literacies workshops into a pilot study that partners with educators, families, and children to co-create early literacy resources grounded in joy and self-determination. This work will translate theoretical insights into practice, producing materials that challenge deficit narratives and broaden the possibilities of literacy education. Finally, I will continue *theorizing unbound literacies* by extending my concept of *bound literacy*. I aim to explore what it means to unbind literacy from antiblackness, investigating how educational research and design can presuppose Black brilliance and creativity as foundational rather than exceptional. This work advances critical design traditions in the Learning Sciences by envisioning learning environments designed from and for liberation.

Conclusion

Across its theoretical, methodological, and practical dimensions, my research advances a vision of the Learning Sciences that takes seriously the historical and racialized conditions under which learning occurs. By centering Black literacies as both analytic and imaginative sites, my work contributes to ongoing efforts in the field to build more equitable, humanizing, and justice-oriented learning environments. Ultimately, I aim to cultivate a body of scholarship that helps the Learning Sciences reckon with its historical inheritances while designing toward freer, more expansive educational futures.

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Advancing a Learning Sciences Approach to Quantum Information Science in K-12

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Abstract: My research examines learning in complex, rapidly evolving technological domains, particularly quantum information science. Bridging multiple theoretical perspectives, I am currently pursuing two directions: designing learning environments that make ultra-complex STEM accessible to K-12 learners; and theorizing quantum computational thinking to identify concepts, practices, and perspectives that distinguish quantum from classical reasoning.

Research agenda and theoretical framing

In my research, I am committed to understanding how emerging technologies shape and support learning. I pursue this understanding both via conceptual inquiry and via the design of learning environments that help learners navigate complex, rapidly evolving domains. While some of my previous (e.g., Coelho & McCollum, 2025) and ongoing work involves empirical investigations using AI and NLP technologies to support teaching and learning across K-12 and post-secondary contexts (e.g., Coelho et al., 2025a), this submission focuses on my research in K-12 quantum information science (QIS).

My scholarship is primarily grounded in learning sciences traditions that view learning as situated and mediated by material and social interactions (e.g., Cole, 1998; Moll et al., 2006; Pea, 1985, 1993a, 1993b; Rogoff, 2003). I adopt an integrationist stance (Danish & Gresalfi, 2018), deliberately examining learning across one or more interrelated planes (Rogoff, 2008), while critically attending to the analytic affordances and limitations of this approach and maintaining a commitment to understanding the dynamic interplay among learners, social interactions, and context. I also draw on complementary perspectives that have traditionally emphasized the individual – such as constructivism (Kafai, 2012; Papert & Harel, 1991; Wilensky, 2020) – and cognitive sciences (e.g., Bandura, 2010) as generative for understanding and supporting meaning-making, agency, and motivation. I utilize these complementary perspectives to inform my conceptual and empirical contributions to K-12 quantum information science education as described in the subsequent sections. Thereafter, I discuss future directions centered on studying and designing for quantum computational thinking.

Current work

When I began reviewing the literature on K-12 quantum information science education (Coelho et al., 2025b), I was surprised to find that learning sciences researchers, who have spent decades studying how students understand complex scientific concepts, were almost entirely absent from the conversations. This absence seemed particularly striking given that learning quantum information science requires learners to grapple with ideas that defy classical intuition and mathematical formalisms that create barriers to accessibility even at the college level, let alone in K-12 settings. These are precisely the kinds of challenges that learning sciences researchers have developed expertise in addressing: making complex, abstract concepts accessible through appropriate representational tools, scaffolding, and cultural practices that support new ways of thinking. And just as this expertise is most needed, national initiatives are pushing for quantum education to begin in middle and high school (NSF, 2020). While computer science and physics education researchers (e.g., Franklin et al., 2020; Singh et al., 2022) have begun important work in this space, the conversations have yet to substantially engage with learning sciences frameworks and methods that could complement and extend these disciplinary efforts. Our review also showed that both theoretical grounding and design conjectures in K-12 QIS education research remain limited. Many initiatives simply ask, “Does it work?” in a Thomas Edison exploratory framework (Stokes, 2011), without interrogating *why* or *under what conditions* learning occurs, risking innovations that engage learners but do not reliably support deep understanding.

My position paper (Coelho et al., 2025d) argues for a bidirectional opportunity that neither field has fully recognized. QIS education could leverage learning sciences expertise in design-based research to iteratively develop quantum learning environments. This includes principled approaches for creating new representational infrastructures (Wilensky, 2020) that make quantum phenomena accessible while preserving mathematical integrity, as well as participation structures that allow learners to collectively make sense of quantum phenomena, negotiate meaning through shared representations, and transform their participation in disciplinary practices (Lemke, 2001; Pea, 1992; Rogoff et al., 1995). But equally important, and perhaps more provocative, is what

quantum information science offers the learning sciences: a domain that pushes us to reconsider our theories of conceptual change, representation, and learning at the boundaries of human intuition.

To put these conceptual arguments into practice, my recent empirical work (Coelho et al., 2025c) investigates how collaborative game making (Fields & Kafai, 2018) leveraging distributed expertise (Pea, 1993) can support students in developing initial mental models of quantum information science concepts and practice while cultivating curiosity, confidence, and identity – early pathways toward a “competent outsider” stance (Sandoval et al., 2025). We found that designing quantum-themed board games created accessible entry points supporting participation without requiring advanced formalism, though supplementary social and material support will also be incorporated in future iterations of the work. Throughout this process, groups mobilized diverse expertise and resources to materialize quantum ideas, illustrating how even quantum-adjacent activities can drive motivation, sensemaking, and participation in ultra-complex STEM domains.

Future work

Building on my prior work in K-12 quantum information science education, I have begun to articulate a tentative framework for quantum computational thinking (QCT), extending classical computational thinking to the distinct concepts, practices, and perspectives of quantum information science, while attending to the ethical, societal, and justice-related dimensions of access to quantum technologies (Coelho et al., 2025e). This exploration identifies key *concepts* (e.g., randomness, interference, superposition), *practices* (e.g., designing probabilistic algorithms, leveraging amplitude amplification), and *perspectives* (e.g., understanding computation as materially grounded, designing under indeterminacy) that distinguish QCT from classical computational reasoning, while also emphasizing equitable participation and the social and ethical stakes of access to quantum technologies. Conceptually, this work considers how engagement with QCT might prompt learners to develop non-classical intuitions, reason under uncertainty, and approach computation, abstraction, and scientific problem-solving in fundamentally new ways.

Looking forward, my K-12 research agenda focuses on two interrelated directions. First, inspired by the STEM novice/expert research literature (Bransford et al., 2000; Erickson et al., 2006) and Science-Technology-Studies investigations of laboratory life (e.g., Latour, 2005; Latour & Woolgar, 1986), I aim to ground a QCT framework in empirical studies of how quantum information scientists think and work, examining the concepts they use, the practices they enact, and the perspectives they bring to reasoning with quantum phenomena and technologies. Second, I plan to build on these insights by designing and iteratively refining K-12 learning environments that scaffold engagement with QCT, leveraging new representational infrastructures and collaborative practices that invite students into developmentally appropriate, quantum-adjacent forms of authentic inquiry. Such activities may include designing games, music, or visualizations that incorporate principles like amplitude amplification or probabilistic reasoning, enabling learners to make abstract quantum ideas tangible, expressive, and personally meaningful.

While practical quantum computers remain on the horizon, progress toward them is uncertain and uneven (Eisert & Preskill, 2025). They are not expected to replace classical systems; instead, future systems will likely integrate both paradigms (Preskill, 2018). The field is already shaped by substantial financial investment to address the enormous hardware challenges. Yet an even greater obstacle lies in identifying problems for which quantum approaches provide decisive advantages, beyond those canonical examples discovered decades ago. How we approach quantum reasoning and learning today will influence not only who gets to participate in shaping this emerging field, but also what kinds of problems we even recognize as solvable. The learning sciences contribution, then, is not auxiliary to quantum computing’s development; rather, it is central to its realization.

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Bodies, Participation, and Power: Learning as a Site of Inclusion, Exclusion, and Resistance

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Abstract: Youth bring diverse ways of knowing, communicating, and creating meaning to school, yet classrooms often privilege a narrow slice of these practices—treating difference as deficit and leaving many disabled, neurodivergent, queer, or trauma-impacted youth at the margins of participation. These exclusions affect belonging and access to civic, community, and disciplinary life. My research investigates how to design asset-oriented, participatory, and body-inclusive learning environments for young adolescents, where questions of identity and belonging are especially acute. Drawing on sociocultural and critical learning theories, participatory design-based research, and social models of ability and the body, I examine how multimodal repertoires can be recognized and leveraged as learning resources. This agenda advances three strands focused on designing multimodal learning environments that prioritize learner and educator dignity: (1) expanding conceptions of adolescent participation and literacy, (2) designing inclusive multimodal technologies, and (3) positioning youth as co-designers and co-theorists of their learning.

Strand 1: Expanding conceptions of youth participation

This strand is grounded in my experiences and those of current adolescents regarding secondary schools' narrow conceptions of participation. Secondary schooling has privileged print and verbal literacies while devaluing embodied, affective, and multimodal avenues for meaning-making. These framings disproportionately exclude multiply-marginalized adolescents and reinforce deficit orientations and hegemonic norms.

In my previous work, I employ a social model of disability to explore how schools 'disable' and 'enable' certain bodies, calling for researchers to shift responsibility for student performance away from the individual and onto the design of the system in which they operate (Vickery, 2023). In my dissertation, I introduce a five-dimensional inter-theoretical framework that conceptualizes the body as simultaneously thinking, feeling, observed, politicized, and autonomous. I also provide methodological guidance for conducting humanizing, dignity-affirming research with youth living in non-normative bodies (Vickery, 2025a) and employ critical resistance narrative analysis to examine how adolescents navigate and resist normative embodied expectations in schools (Vickery, 2025b).

Moving forward, I expand normative notions of participation in youth civic education. I extend this work into civic and technical domains—motivated by the current sociopolitical climate. Such work demands multidisciplinary and multimodal competencies that cannot be siloed within individual subject areas. To these ends, I hope to support adolescents' learning of media, data, digital, and scientific literacies through engagement with culturally relevant issues. Building from epistemic practices in science and engineering education (e.g., evidence evaluation, systems reasoning, modeling), I see opportunities to extend these into literacy/language education (rhetorical device usage, media & data literacy, interpretation of misinformation), social studies/history (complex sociopolitical systems, causal mechanisms), and computing education (engineering ethics, data privacy, transparency, ethical AI adoption). As political systems increasingly disenfranchise multiply-marginalized youth, these literacies become not abstract skills but conceptual and tangible tools for civic and social change. A society of care must acknowledge and design against these barriers.

Strand 2: Designing inclusive multimodal technologies

This strand is rooted in my experience as a UX designer and researcher, where access and inclusion were often addressed retroactively rather than prioritized from the start. In educational technologies, this reinforces deficit-oriented assumptions about learners whose practices do not align with normative expectations. This work centers on learners' histories, needs, and desires in the edtech design process, ensuring technologies are proactive, responsive, and adaptive to learners' needs.

In my recent work (Vickery & Danish, 2025), I share my rapid design and iterative refinement of a mixed-reality embodied learning environment for a reverse-inclusion summer camp for youth with moderate-to-severe disabilities, aimed at identifying and addressing barriers to expression and participation. In my 2024 work

on the GEM-STEP project (Vickery et al., 2024; NSF Grants #1908632 & #1908791), I led my team in integrating trauma-informed pedagogies into the facilitation and evaluation of mixed-reality embodied activities (where students use motion-tracking technologies to collaboratively model complex scientific phenomena), demonstrating how centering comfort, choice, and agency can amplify students' capacities for participation. In Zhou et al. (2025), we investigate how elementary learners' group-divergent embodied practices afford opportunities to learn about complex scientific phenomena. As a Visiting Research Scientist funded by the Engage AI Institute, I design and develop narrative-based, AI-scaffolded, gamified experiences for middle school students that are context-relevant and engage learners in critical thinking about food and environmental justice (Humburg et al., 2025).

I build on my dual background in software engineering and UX design to critically engage with how 'cutting-edge' technologies such as AI, XR, and asynchronous learning tools are implemented at scale. My prior work has often relied on retrofitting inclusion into technologies – moving forward, I hope to (1) establish trusting partnerships with schools and communities, (2) conduct on-the-ground needs assessments, (3) co-design contextually grounded solutions with these needs prioritized at the onset, and (4) iteratively evaluate implementation. Access and inclusion are not endpoints but baselines; innovation must be defined by—and accountable to—those most impacted.

Strand 3: Positioning youth as co-designers and co-theorists of their learning

This strand addresses a tension across education and research alike – that too often, youth are positioned as passive users of and participants in designs they had no role in shaping. Thus, these designs (which here include pedagogies, activities, tools, communities, and theories) fail to provide the necessary scaffolding for youth to thrive and learn within them. My work takes an asset-oriented stance, treating adolescents' perspectives and repertoires of practice as valuable resources for designing more sensitive, equitable, and emancipatory experiences for youth who would otherwise be forced to (at best) change themselves for the sake of compliance.

In prior work, I've offered autoethnographic accounts of how my traditional notions of research-practice partnerships were challenged and remade in a collaboration with local paraprofessionals, special educators, and youth with disabilities (Vickery & Peabody, Under Review). In my dissertation Article 2 (Vickery, 2025c), I drew on Vygotskian notions of mediation and affect to understand what affective, sensory, and lived experiences are activated for youth with non-normative embodied experiences in the presence of various instructional design elements (i.e. various activity/task framings, technologies, proximity of embodiment, sensory demands, forms of movement, & participation structures). Further, Article 3 positions youth as co-theorists of a framework for dignity-affirming, body-inclusive embodied learning environments (Vickery, 2025d).

This strand responds to a recurring theme across my projects—that "grown-ups just don't get it." This perceived ignorance strains opportunities for authentic educator–youth relationships and deepens youth distrust of educational systems. To address this, I aim to develop organized, protected "designer spaces"—digital and physical—for youth to co-design community-facing projects and professional development for educators and youth-service providers. Theoretically, this strand validates youth knowledge as legitimate civic and pedagogical insight; methodologically, it redefines youth's role in research beyond that of participants to that of educators and researchers with the means to help adults "do better."

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Overview of Research: BlackCrit and Social-Design Experiment

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Abstract: This research agenda examines how Black youth and teachers resist anti-Blackness through critical literacy, STEM learning, and counternarrative practices in urban schools. Grounded in Critical Race Theory and BlackCrit, the work positions race and anti-Blackness as central forces shaping educational experiences while centering Black joy, agency, and knowledge production. Using social design experiment, this research implements justice-oriented dual-credit courses and documents the pedagogical practices of Black teachers. Across iterative cycles of design and analysis, the studies aim to generate new theories of learning rooted in Black cultural wealth while materially improving educational opportunities, well-being, and academic pathways for Black youth.

Introduction

As a literacy and urban education researcher, my work focuses broadly on Blackness/anti-Blackness in education. Specifically, I seek to better understand the literate lives of Black youth in urban K-12 schools and the pedagogical practices of Black teachers. Amid sweeping federal and state reforms banning books and critical theories about race and racism (Alexander et al., 2023), Black teachers and Black youth stand on a precipice which we have seen many times before. The concentrated effort by school districts and politicians to recommit to colorblindness and the myth of meritocracy are only veiled attempts at maintaining the status quo: white supremacy (Crenshaw et al., 2019). We know these reforms, along with state sanctioned violence, cause significant harm to Black teachers and youth, so it begs the question(s):

- *RQ1:* What are the psychological and mental impacts of anti-Blackness on urban students' perception of K-12 education?
- *RQ2:* How can Black students' and teachers' resistance to anti-Blackness inform theories of learning?
- *RQ3:* How do Black urban youth leverage critical literacy practices and STEM learning to create counternarratives?

Theoretical framework

Critical race theory and blackCrit

This work is situated in Critical Race Theory (CRT) (Bell, 1980; Bell, 1992; Delgado, 1989; Delgado, 2001; Dixon & Lynn, 2013; Matsuda, 1993). CRT offers a framework for race and racism as central and unshakeable foundations of our society. The theory has several important tenets that are relevant to my work—like interest convergence and racial realism—but the most focal tenet in my research is the understanding of counternarratives (Delgado, 1989; Harris et al., 2023; Kinloch et al., 2020; Solórzano & Yosso, 2002). Counternarrative, or counterstorying, “represents the knowledges, perspectives, and experiences of Black people and others marginalized by racial oppression and white dominance” (Kinloch et al., 2020, p. 387). My work seeks to amplify the voices of Black students and teachers in a way that leverages counternarratives, specifically, in an educational context (Ladson-Billings & Tate, 2017). This can work in conjunction with elevating student voice in a dialogic setting (Segal et al., 2017). While CRT places an emphasis on the impact of race and racism more broadly, BlackCrit offers a parallel and connected framework aimed at better understanding anti-Black suffering, injustice, and educational practices (Dumas, 2016; Dumas & Ross, 2016; Givens, 2021a; Sobti and Welsh, 2023). These two interconnected frameworks directly inform the work I do with literacy and STEM in urban schools.

Methodology: Social design experiment and ethnography

My prospective research studies will utilize the tradition of social design experiment (Gutiérrez & Jurow, 2016). The main goal, informed by more traditional approaches to design research (Anderson & Shattuck, 2012; Barab et al., 2004; Collins et al., 2004; Tinoca et al., 2022), will be to implement educational interventions that materially improve the lives of Black youth in an urban school. Rather than focusing on transforming the social and material lives of the broader group coined “nondominant communities” (Gutiérrez, 2008), the BlackCrit theoretical framing works with this design experiment to specifically address the deleterious impacts of anti-Blackness in an educational setting.

Gutiérrez & Jurow (2016) keenly note that, “the relation between everyday knowledge and scientific knowledge is of particular theoretical interest to us as well as to the learning sciences” (p. 572). Through studies like this, I plan to elucidate the everyday knowledges of Black youth (counternarratives) by implementing a dual-credit college course at predominantly Black urban high schools in the Rust Belt Northeast. Additionally, as a teacher of this prospective course, I will build on my dissertation research to act as what Barab et al. (2004) refer to as a “change agent” (p. 255). The course will be a 100-level humanities course that focuses on the blending of science, technology, engineering, and math (STEM) and literacy. My work intersects with STEM through a shared goal of challenging the deficit narratives that marginalize minoritized youth in these disciplines often face from educators (van Es et al., 2025). Moreover, the course will aim to provide justice-oriented and inquiry-based practices (Damşa & Nerland, 2016; Luehmann et al., 2023) around STEM. This course will be based on similar work I have conducted with urban youth where a STEM and literacy design experiment were implemented (O’Neil-White & Hunter, in press). Following the delivery of courses like this, I will utilize design principles to assess the feasibility of Black counternarratives as a conceptual tool for learning. I have already attained approval through multiple school districts to conduct this work—with data collection already under way.

In addition to the social design experiment work, my work also extends into critical ethnography (Madison, 2012). Specifically, I have begun conducting research in a classroom in a first-ring suburban school to better understand the pedagogical practices of a Black male English teacher—a group that has been the victim of anti-Blackness and exclusion in the education field (Bell, 2017; Brockenbrough, 2015; King, 1993; Young & Young 2020). Through these ethnographic methods, I will disseminate findings that will provide an innovative Black male literacy teaching framework.

Lastly, my research has deep roots in my classroom teaching experience. As a longtime English teacher, the gap between the theoretical world of educational research and the challenges teachers are facing on the ground was often a difficult hurdle. I bring my experiences as a classroom teacher to much of the research I conduct, including a recent practitioner article in the *English Journal* that highlights my own efforts to decolonize my classroom to meet the needs of diverse students. While I’m no longer a classroom teacher, I’m keenly aware that teachers need research that can “meet them where they are at” in ways that are helpful, transformative, and refreshing.

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Tools and Methods to Support data Visualization Literacies through Programming

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Abstract: My research has focused on supporting the development of computational and data visualization literacies through the design of constructionist tools and the analysis of students' programming trajectories. During my PhD, I developed PlayData, a block-based programming environment that enables learners to construct data visualizations. To investigate how they engage with the tool, I created methods for capturing and analyzing process data from the environment, including a machine-learning approach for modeling programming trajectories and the use of Epistemic Network Analysis to characterize the connections among students' Data Visualization Programming moves over time. These approaches reveal how learners navigate open-ended programming tasks and how computational and data-related practices can mutually reinforce one another. Building on this foundation, my research agenda seeks to advance our understanding of how learners develop data and computational literacies and to inform the design of learning environments that foster meaningful engagement with computing.

Research background

Throughout my research trajectory, I have worked on two inter-related topics: (1) the *development of computational tools* rooted in constructionist learning perspectives, and (2) unveiling students' *learning trajectories* in open-ended tasks focused on programming and data visualization. My research is grounded in prior work on data visualization and computational literacies, both central to understanding and participating in the world today. Data visualization literacy refers to the ability to understand and create visual representations of data. It encompasses being able to read charts, but also to critique, modify, and design new representational forms for data - also referred to as meta-representational competencies (Azevedo, 2000; diSessa et al., 1991; diSessa & Sherin, 2000). Computational literacies (diSessa, 2001) refer to the idea of computing as a way of learning and participating in the world, related to "a set of practices situated in a sociocultural context which utilize external computational media to support cognition and communication" (Kafai & Proctor, 2022 p. 148). In my research, I am interested in understanding how these literacies can mutually support and enhance each other. Regarding the design of tools, I have developed PlayData, a block-based programming environment for constructing data visualizations, informed by a framework that draws upon taxonomies from the Information Visualization field (Fernandez, Freitas, et al., 2024). In order to understand how students engage with PlayData, I have worked on methods for collecting and analyzing on how learners create programs with the tool, described below.

Learning analytics and Epistemic Network Analysis

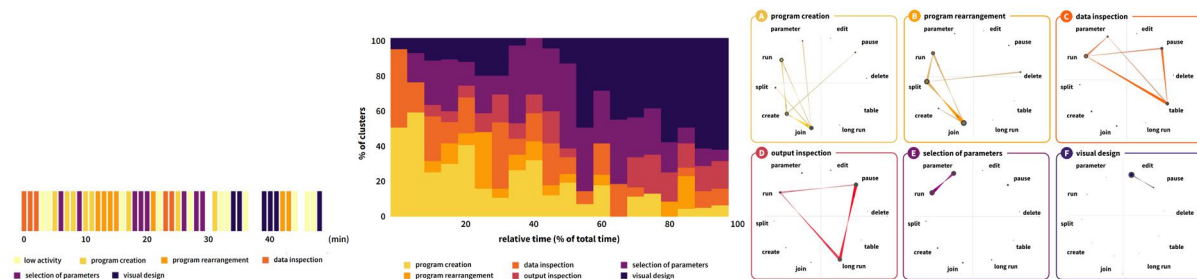
Significant efforts have been directed toward developing tools to support the assessment of learning in computing. However, many of these solutions still rely on analyzing final products, offering only a limited perspective on the practices in which students engage. The use of learning analytics techniques to analyze large volumes of data generated by log-capturing tools has expanded our understanding of learning processes in programming tasks, enabling more detailed descriptions of their pathways (Berland et al., 2014; Blikstein et al., 2014). Despite advances in this area, process-oriented metrics could still benefit from important improvements, including collecting information about the programming process as a continuous flow, rather than as isolated records captured in save or compile states; and considering the specific characteristics of different types of tasks, recognizing that trajectories vary according to context (for example, when developing a program to analyze data, the pathways followed will differ from those involved in creating code to solve a mathematical problem). Epistemic Network Analysis (ENA) can also contribute to examining how events in a programming session connect to one another and how these connections evolve over time. ENA is a quantitative ethnography technique that identifies connections among coded elements in data and represents them through a network model (Shaffer et al., 2016; Shaffer & Ruis, 2017). Although ENA has been widely used to analyze student discourse, log data in programming environments can also be explored using these techniques (Fernandez et al., 2025).

In my doctoral research, I developed an automated method using machine learning techniques to describe students' trajectories as they created data visualizations using the PlayData programming environment (Figure 1, left). Additionally, I used ENA to characterize, through connections between events, the main actions in which students engaged during these activities (Figure 1, right), which I termed Data Visualization Programming moves (DVP moves). By analyzing these connections temporally, I generated insights into the dynamics among these

actions and into how programming sessions evolve over time (further details can be found in Fernandez et al., 2025).

Figure 1

Left: Individual programming trajectory generated from clustering log data collected in a block-based programming environment. Center: Aggregated visualization over time. Right: ENA networks for each of the identified Data Visualization Programming moves (Fernandez et al., 2025).



Design-based research

The Design-Based Research (DBR) approach (Barab & Squire, 2004; DBRC, 2003) combines the systematic design of tools or intervention strategies implemented in real-world settings with reflective practices among those involved, considering the challenges and affordances encountered. This type of research is characterized by continuous cycles of design, enactment, analysis, and redesign, employing documentation methods that make it possible to connect theory with outcomes of interest. I have been working with DBR since my master's research (which involved the design of physical computing toolkit) and, more recently, in the development of PlayData (Fernandez, Blikstein, et al., 2024; Fernandez, Freitas, et al., 2024).

Moving forward

My broader goal as a researcher is to contribute to understanding how people learn computing and data science in contexts of authorship and design. Building on the background described in the previous section, the research agenda I plan to pursue unfolds across three interconnected strands encompassing methodological, pedagogical, and cognitive aspects.

Refinement of methods for analyzing learning trajectories: To describe, categorize, and visualize students' trajectories in programming tasks, I aim to keep developing a line of research centered on the application of machine-learning techniques combined with ENA and qualitative video analysis. The types of questions I am currently interested in understanding include: How do different techniques for segmenting data affect the results and patterns obtained from automated analyses of learning trajectories? What are the most effective ways of visualizing patterns related to learning trajectories to support feedback for teachers and learners?

Development of tools, resources, and pedagogical strategies for data visualization and computational literacies: Additionally, I am interested in contributing to the development of educational tools, resources and pedagogical strategies aimed at integrating computing and data science. This includes, for example, incorporating new features into the PlayData environment and documenting its affordances and challenges in introductory programming/data-science courses. I am also particularly interested in developing functionalities that support the analysis of learning trajectories in general-purpose environments such as Jupyter Notebook. Guiding questions for this line of research include: How do different pedagogical approaches, types of tasks, and forms of student organization (e.g., individual tasks or pair programming) influence trajectories in programming environments? How do the characteristics of the tools impact the trajectories of learners?

Understanding how interactions with AI tools influence programming learning: Generative-AI-based tools have radically transformed how people code and learn to code. Given the growing role that these tools play in CS learning (often without pedagogical intentionality), I have recently become interested in better understanding how interactions with them affect students' ways of thinking *about* and *with* their code. In this sense, I aim to investigate how learning trajectories unfold in both AI-supported and non-AI-supported contexts, examining how these interactions shape learners' strategies, conceptual understanding, and agency in the programming process.

Ultimately, I expect that my work may provide researchers and educators with new insights into how learners develop computational and data literacies, and to inspire the design of constructionist tools to engage them in productive explorations with computers.

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Developing Digital Humanities Pedagogical Content Knowledge

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Abstract: This research investigates the pedagogical affordances of digital humanities methods like text-mining and the resulting visualizations of text in support of disciplinary learning outcomes using co-design, classroom research, and qualitative interviewing. My work examines how visualizations of literary texts facilitate computational and literary thinking in English Language Arts classrooms and can cultivate data skepticism.

Introduction

While the field of digital humanities has developed pedagogical practices for training future digital humanists, less attention has been paid to how digital humanities methods, especially computational text analysis, can support learning outcomes in other disciplines. This research statement overviews my work in education at the intersection of digital humanities, English Language Arts, and data science, exploring how computational methods can support both literary literacy and data literacy.

Digital humanities pedagogy for humanists and data science

My research in this area centers on the pedagogical opportunities afforded by digital humanities methods, especially digital humanities methods related to text, and their implementation outside of digital humanities contexts. I study the ways in which digital humanities pedagogy can be leveraged in support of other disciplines' learning outcomes, including traditionally humanist learning outcomes and data science learning outcomes. My work is deeply indebted to constructionism (Papert, 1993) to understand critical making within digital humanities pedagogy, and it is committed to the principles of data feminism (D'Ignazio & Klein, 2020).

Since 2024, I have worked on *Building STEM Skills by Integrating Data Literacy and Text Analytics in English Language Arts* (National Science Foundation Grant No. 2241483). The project aims to develop and research the potential of integrated data literacy and text analytics instruction in middle school English Language Arts (ELA). Through co-design, classroom research, and participatory professional development with practicing English Language Arts teachers, we developed and tested text analytics tools, visualizations, and pedagogical techniques for use in ELA middle school classrooms. Within this project, I frequently find myself in the role of translator—working to bridge digital humanities, data science, and ELA concepts—and I have worked on pedagogical design, classroom research using video analysis, instrument design for qualitative interviews, and data analysis. We find that visualizations built on human-readable scales of text (e.g. a single novel, short story, or poem) afford productive computational and literary thinking, with students able to challenge, revise, and understand the computations using textual evidence and their own linguistic-experiential reservoirs (Nomura et al, 2025).

Developing this kind of pedagogical content knowledge also requires surfacing the existing meaning-making protocols students use when looking at visualizations of text—including their misconceptions, assumptions, and critical frameworks. My work, therefore, seeks to understand—through experimental conditions and through close reading of digital-humanities argument—what happens when an expert or when a student looks at a visualization of literary text. Existing work on graph comprehension highlights that domain knowledge interacts with graph literacy (Freedman & Shah, 2002, Shah & Hoeffner, 2002)—it is likely, therefore, that literary-critical and linguistic knowledge interact with graph comprehension for visualizations of text. We have adapted existing graph literacy instruments (e.g. Lai et al, 2016) to see if students interpret frequency differently when literature is involved, or if they understand ratios of lines in a play differently than ratios of seeds that sprouted. Our preliminary findings suggest that students do draw on literary funds of knowledge when they interact with text-based visualizations.

In addition to supporting more robust humanities pedagogy, this work also seeks to further develop and understand the mechanisms by which humanities data practices support broader data skepticism—the development of perspectives that consider the subjunctivities, contexts, and diverse meanings of data (D'Ignazio & Klein, 2020; Wild & Pfannkuch, 1999)—in data science and statistics education. How is working with text, especially literary text, different from working with other kinds of data in learning environments? What kinds of pedagogical content knowledge can we develop to support more robust data skepticism in both teachers and students?

In all of my classroom-based work, I explore the possibilities of minimal computing (Risam & Gil, 2022) for digital humanities pedagogy—recognizing that computing under conditions of constraint may be not

only necessary, but beneficial. My recent work in this vein includes an analysis of how erasure poetry can be a low-tech way to teach the relationship between text-mining and text-encoding (Messarra & Nomura, 2025). This approach, which blends the calls of data feminism to visceralize data (D'Ignazio & Bhargava, 2019) with the pedagogy of Tukey's *Exploratory Data Analysis* (1977) and Papert's constructionism (1993), has implications for humanities pedagogy, data science pedagogy, and digital humanities pedagogy at both the secondary and postsecondary level.

Future work

I plan to build on my work looking at the outputs of digital humanities research by diving into deeper questions of algorithmic and computational literacy, focusing on methods of operationalizing by hand or with a computer. In particular, I want to know how digital methods and tools can change, support, and develop metacognition in humanist rules of notice and signification (Rabinowitz, 1986) and literary-critical epistemic cognition (Lee et al, 2016) in specifically pedagogical contexts. Professional digital humanities practice has done significant work in the development of such rules to the ends of research-related utility, and some work in the vein of pedagogy designed to train future digital humanists, but less in the area of pedagogical utility for other learning outcomes. I hypothesize that the act of operationalizing a literary-critical concept, and the work of moving back and forth between computational outputs and text, can help students develop better humanist and algorithmic metacognition.

The work described in this paper has been on visualizations that are built on roughly-human readable scales—a single novel, a short story, a poem, a collection. Following calls to develop pedagogy that supports postdigital literary literacy, including corpus and database literacy (McLean Davies et al, 2020), I have piloted some work aiming to bridge these scales with those of professional digital humanities, including experimental protocols to surface the differences between single-text and corpus-level visualization for unfamiliar text. This work will help us better connect secondary and postsecondary digital humanities pedagogy, supporting more robust pipelines in humanities disciplines.

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Fostering Agentic and Actionable Collaborative Discourse in Analytics-Enhanced Learning Environments

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Abstract: My research explores how analytics-informed learning environments can foster agentic and actionable collaborative discourse in higher education. Drawing on sociocultural and dialogic theories, I conceptualize discourse as a core mechanism of knowledge co-construction and epistemic agency as learners' capacity to initiate inquiry, build on peer ideas, and shape shared understanding. Using a design-based, mixed-methods approach, I integrate computational discourse analysis with participatory design to trace epistemic engagement and develop analytics-based interventions that scaffold learners' discursive moves. Projects such as EntryPoint illustrate how analytics-based scaffolds and AI partners can support dynamic participation trajectories and amplify student voice. Ongoing work advances temporal modeling of discourse, expands human-centered analytics tools, and promotes inclusive design for AI-supported collaboration. This agenda contributes to computer-supported collaborative learning (CSCL) by advancing theory, methods, and tools that center learner agency and dialogic engagement in dynamic, data-rich learning environments.

Overview

My research lies at the intersection of CSCL, learning analytics, and human-AI interaction, with a central focus on how analytics-informed learning environments can foster agentic and actionable collaborative discourse. Grounded in sociocultural and dialogic theories of learning (John-Steiner & Mahn, 1996; Stahl et al., 2014), I view discourse not simply as evidence of learning but as the core mechanism through which learners articulate ideas, negotiate meaning, engage in knowledge co-creation, and construct shared understanding (Bereiter & Scardamalia, 2014; Van Aalst, 2009). A foundational commitment in my work is the conceptualization and scaffolding of epistemic agency, defined as learners' capacity to initiate inquiry, articulate ideas, take up peers' contributions, and shape the trajectory of joint intellectual work (Damşa et al., 2010). My research examines how epistemic agency is enacted and constrained in learning experiences mediated by analytics-based scaffolds and AI-powered agents, and how carefully designed interventions can support dialogically rich and agentic participation. To advance this agenda, I pursue two interrelated lines of inquiry, each reflecting a distinct orientation to learning analytics in CSCL contexts: (1) using learning analytics techniques as an analytical lens to model the structural, semantic, and temporal dynamics of discourse; and (2) designing and implementing analytics-informed interventions that serve as learning partners to scaffold dialogic participation and epistemic moves. Together, these lines of work position learners not only as recipients of information but as epistemic agents who actively shape knowledge co-construction in collaboration with both human and artificial partners. Through this framing, my research explores how analytics-informed interventions can be designed and implemented to support agentic and dialogically rich participation in authentic collaborative learning settings.

Theoretical approach

My research is grounded in sociocultural and dialogic theories of learning, which emphasize the social, situated, and mediated nature of meaning-making. Learning is conceptualized as a collaborative process of knowledge building in which understanding emerges through interaction with others, tools, and cultural artifacts (Bereiter & Scardamalia, 2014). In this view, discourse is not merely a vehicle for sharing ideas, but a generative space through which knowledge is co-constructed, contested, and transformed (Oshima et al., 2012; Van Aalst, 2009). Within this theoretical framing, collaborative discourse is positioned as the central mechanism of learning in CSCL contexts. Building on foundational work in collaborative knowledge building (Scardamalia & Bereiter, 2006; Van Aalst, 2009), my research examines how learners pose problems, elaborate ideas, and engage in sustained inquiry within discourse communities, as well as how dialogic exchanges foster shared understanding and collective cognitive responsibility. A central construct anchoring my research is epistemic agency, defined as learners' ability to take initiative, contribute substantively, and shape the direction of joint inquiry during discourse (Damşa et al., 2010; Nieminen & Ketonen, 2024). Epistemic agency is conceptualized as a distributed and emergent property, co-constructed through participation in sociomaterial activity systems. This perspective foregrounds how agency is enacted not only through individual cognition but also through relational, temporal, and material conditions within learning environments.

As collaborative learning increasingly unfolds in analytics-enhanced environments, my research extends these theoretical commitments to account for emerging sociotechnical actors. Specifically, my studies examine how AI agents (Jung, in preparation), analytics-based suggestion systems (Jung & Wise, 2025), and orchestration tools (Booth et al., Under Review) mediate learners' epistemic positioning, dialogic moves, and participation trajectories during knowledge co-construction. Within a design-based research framework, these perspectives guide the development of analytics-informed interventions that scaffold learners' discursive engagement (Jung et al., 2026). For example, tools such as EntryPoint (a discourse suggestion system for social annotation) are designed to prompt re-engagement, introduce epistemic contrast, and support temporal coordination, enhancing learners' capacity to participate meaningfully in discourse (Jung et al., 2026). These interventions are grounded in the belief that well-designed technologies can amplify dialogic space, invite student initiative, and sustain epistemic engagement over time. Ultimately, this theoretical approach bridges CSCL's foundational emphasis on discourse and knowledge co-construction with the emerging realities of AI-augmented learning environments such as MACIE (a real-time multi-agent orchestration engine for idea creation; Booth et al., Under Review). This provides a lens for critically examining how learners navigate human-AI collaboration, how tools can be designed to support dialogic learning, and how epistemic agency can be sustained within increasingly complex sociotechnical ecologies.

Methodological approach

To investigate collaborative learning in analytics-enhanced environments, my research adopts a design-based, mixed-methods approach that integrates computational learning analytics with qualitative and participatory design methods. My methodological stance is grounded in the premise that understanding collaborative discourse requires both fine-grained temporal analysis of interactional processes and context-sensitive insight into learners' perspectives, routines, and participation trajectories.

At the core of my methodological approach is computational discourse analysis, used to trace how learners co-construct knowledge over time. Techniques such as natural language processing (NLP) and social network analysis are employed to examine how ideas emerge, evolve, and dissipate in digital discourse spaces. These methods enable the identification of patterns of epistemic engagement (e.g., anchoring, elaboration, and integration) as they unfold in environments such as asynchronous social annotation and generative co-authoring tools (Jung & Lee, 2026). In addition, my research employs temporal analysis to unpack the sequential nature of discourse moves. For example, prior work has examined how learners respond to analytics-informed suggestions, take up peer ideas, and shift participation roles in relation to analytic scaffolds (Jung et al., 2026). These analyses support the development of temporal models of learning that move beyond static indicators to capture evolving epistemic trajectories across time and context.

Complementing these computational approaches, my research incorporates human-centered design methodologies to develop and evaluate analytics-informed interventions. Through co-design workshops, prototype testing, and participatory evaluation, my studies actively engage learners in shaping systems such as EntryPoint (Jung & Wise, under review). These activities surface students' needs, values, and concerns regarding analytics-mediated learning and ensure alignment between theoretical commitments and lived experience. Additional methods, including think-aloud protocols, reflective interviews, and interaction trace analysis, are also used to examine how learners interpret and appropriate analytics-based scaffolds in practice (Jung & Wise, 2025). Across projects, my research aims to develop methodological frameworks that are both rigorous and responsive, capable of capturing the complexities of human-analytics interaction, the situated nature of collaborative discourse, and the pedagogical considerations of analytics-mediated learning.

Future plans

My future research will advance analytics-informed interventions that support epistemic agency and actionable participation through three interrelated directions. First, my work will deepen temporal modeling of epistemic activity by leveraging NLP-based semantic and trajectory modeling to examine how learners co-develop ideas over time with peers and AI agents. This direction aims to move beyond static metrics toward responsive analytics that support real-time scaffolding of dialogic participation. Second, my research will expand the design and implementation of analytics-informed interventions that are personalized, context-aware, and aligned with learners' routines. Building on systems such as EntryPoint and MACIE, future work will develop and test adaptive scaffolds that respond to discourse dynamics and support re-engagement, epistemic contrast, and sustained inquiry in authentic learning contexts. Third, my research will extend into curricular innovation by integrating analytics-informed learning into educational data science and AI literacy initiatives, particularly for non-STEM learners. This work aims to support epistemic agency and critical engagement with data-intensive workplace scenarios.

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Supporting K-12 Science Teachers in Equitable Science Teaching through Co-design

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Abstract: My research explores how to support K-12 science teachers' professional development toward equitable and justice-centered teaching. Positioning teachers and students as those who decide what matters to them in their local contexts, I leverage design-based research to co-design, test, and co-refine pedagogical tools with practitioners. Through this process, I study teacher learning by drawing upon a professional identity development lens. In my most recent work, I study how to support middle school science teachers in customizing high-quality instructional materials and investigate how curriculum customization catalyzes teacher teaching. Looking forward, I hope to explore how community-based curriculum design and customization help science teachers create equitable learning experiences for their students.

Introduction

My research agenda focuses on K-12 science teachers' professional development (PD) as equity-centered practitioners. In my work, I use design-based research (DBR) (Cobb et al., 2003) to partner with teachers to design, test, and refine pedagogical tools to support them in developing equity- and justice-centered practices. Through this process, I draw upon learning as identity development as an analytic lens to study teacher learning (e.g., Holland et al., 1998; Wenger, 1998). Instead of lecturing teachers about the practices they should enact, my work starts with learning about teachers' and their students' needs in their local contexts, positioning them as the authoritative figures who decide what is important to them. From there, I collaborate with teachers to figure out how to achieve their pedagogical goals in the service of their students' needs. Equity is central to my work. In my most recent work, I mainly draw upon Philip and Azevedo's (2017) four equity discourses, whether when I work with teachers to unpack what equity means or when I operationalize it in data analysis. Instead of using this framework to evaluate the criticality of teachers' work from a deficit perspective, I focus on helping teachers move towards the discourses that are more critical, namely, expanding what counts as science and positioning science as a tool for social transformation. Methodologically, I adopt a variety of qualitative methods to develop rich accounts of teachers' lived experiences to inform future PD design.

Research trajectory

My scholarship on understanding how science teachers develop a professional identity as justice-centered practitioners began in graduate school. Throughout my tenure as a doctoral student, I worked extensively with pre-service science teachers (PSTs) in informal settings to support them in enacting justice-centered, ambitious science teaching (JuST) practices (Luehmann et al., 2024). My dissertation employed a phenomenological narrative design to examine how a group of six PSTs' engagement in designing and implementing culturally sustaining units anchored in a local rural community contributed to their professional identity development as JuST practitioners. In this work, I drew upon the figured world (Holland et al., 1998) to understand how the PSTs improvised their JuST identities by examining the cultural artifacts they created and their reflections on the design and implementation of the units. Meanwhile, I identified the significant encounters the PSTs recognized as instrumental to their learning across cases. My analysis reveals that centering the cultural relevance in their units allowed the PSTs to develop an expansive understanding of science and their roles as science teachers. It sheds light on the challenges the PSTs experienced and the support they required when they negotiated their JuST identities through their process of becoming.

During my post-doctoral training, my research has expanded to investigate how curriculum-based professional learning (CBPL) supports science teachers in making sense of equity and the pedagogical features embedded in high-quality instructional materials (HQIM). Theorizing curriculum customization as a professional learning site, my post-doctoral work has explored how middle school science teachers customize HQIM guided by their equity goals. Teachers' interaction with curricular materials and equity goals throughout the customization process potentially yields a more sophisticated teacher learning about both the curriculum and their approach to equity (Remillard, 2005). In the Customize project funded by the National Science Foundation, we employed the DBR approach to collaborate with science teachers from a local school district. I led the efforts at the local research site to engage the participating teachers in professional learning communities (PLCs) to customize an open-source

curriculum that employs a storyline instructional model to support students' equitable sensemaking. The major research question I pursue in this work is "How does curriculum customization catalyze teacher teaching?"

In a first-author manuscript from *Customize*, I foreground teachers' work of increasing curricular relevance to students. Bringing together a humanistic science lens (Aikenhead, 2005) and Philip and Azevedo's (2017) four equity discourses, the article investigates how the participating teachers grappled with the idea of "making the curriculum more relevant to students" by focusing on "who decides what is relevant for what equity purpose". Informed by Remillard's (2005) framework on the participatory relationship between teachers and the curriculum, I attended to the new or deeper knowledge teachers developed through their navigation of the customized lesson and the original lesson. Using thematic coding, I analyzed the video recordings of the PLC sessions, audio recordings of teachers' reflections on their enactment, and the teacher-generated artifacts. Findings highlight the complexity of "being relevant to students" within the equity discourses and reveal the participatory relationship the teachers navigated among themselves, their equity goals, and the curriculum (Remillard, 2005).

Alongside my post-doctoral project on teacher learning through curriculum customization, I continued collaborating with The JuST Lab to develop and refine the JuST framework and investigate how to support teachers' professional development as JuST practitioners. The JuST lab is a cross-site collaboration between the University of Rochester, the University of Connecticut, and Islandwood Environmental Center. This collaboration originated from our research team's efforts to co-design a cultural-setting unit with in-practice science teachers in the Greater Rochester area during the COVID-19 pandemic. In the unit, students investigated an anchoring phenomenon of how the pandemic disproportionately impacted people of color. They explored not only the science behind COVID-19 but also the social determinants of health that contributed to health inequities experienced by different population groups. The implementation of the initial unit allowed us to identify the design elements that the teachers recognized as helpful to create a safe space for students to share and discuss critical issues in science classrooms at the beginning of the school year. This initiative later developed into a DRK-12 project across the three research sites. Drawing upon DBR as the primary design, we sought to understand how teachers developed and revised JuST practices in networked PLCs (Thompson et al., 2019). Grounding our work in culturally sustaining pedagogy, we published the Justice-centering Ambitious Science Teaching framework of core practices in 2024 (Luehmann et al., 2024).

Future work

In the future, I will continue exploring how to support K-12 science teachers in equitable and justice-centered teaching by conducting DBR research. In particular, I will focus on engaging teachers, family members, and community members in the process of curriculum design, adaptation, and enactment, and study how this process catalyzes teachers' professional learning toward equity- and justice-centered science teaching. Ultimately, I aim to engage different stakeholders in participatory research design (Bang & Vosoughi, 2016) to investigate what new knowledge will be generated through community-based collaboration on curricular design and customization. For teacher learning, I am especially interested in 1) identifying design principles to support teachers in customizing HQIM to create more equitable learning experiences for their students through that type of collaboration, and 2) exploring how that collaboration will impact teachers' professional learning through a professional identity development lens.

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Teaching and Learning Mathematics with Artificial Intelligence

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Abstract: Novel technologies influence the curriculum, pedagogy, and learning in schools. My research explores the impacts of AI-based tools in mathematics education, attending to how these tools influence teaching practice and how students learn to think mathematically. My scholarly goal is to support mathematics and stochastics teachers in developing innovative and robust pedagogies that uphold cognitive demand in learning even as tools that automate humanistic thinking become endemic to schooling.

Overview

My research centers curriculum, pedagogy, cognition, and AI/LLM automation in mathematics. I am interested in how AI-based tools mediate students' mathematical thinking and teachers' development of expert teaching practices to support student thinking. My interests stem from my experience as a high school math teacher of seven years, where I learned that mathematics tools and curricular resources played a major role in lesson orchestration and what my students ultimately learned (Brown, 2011). I conceptualize teachers and learners as agentic, creative, and curious drivers of learning rather than recipients of societal changes influenced by developments in AI automation. I further envision teachers as lesson designers and orchestrators who, when faced with the latest technological disruptions, can design effective instruction based in part on multi-faceted forms of teacher knowledge, including knowledge of how students' interactions with tools impact their learning.

To support teacher learning with insights about AI-tools' impacts on students' mathematical thinking, I have recently begun to research students' AI-mediated problem solving processes (Delaney et al., 2025b). This work is in its second year and involves both well-defined and ill-defined mathematics tasks. I conduct video-recorded task-based interviews and observe the problem solving processes that surround students' decisions, evaluations, AI trust, and epistemic judgments of mathematics output produced by chatbots. I have initiated this work in both clinical and classroom settings with algebra, calculus, statistics, data science, and numerical reasoning tasks. The findings from these studies inform my own teaching practice and AI adaptations, and with this knowledge, I continuously work to help other teachers develop design skills for an AI-integrated future.

Establishing this program requires proficiency with the latest technologies. Under some conditions, technologies can create dynamic, inquiry-driven learning experiences that raise the cognitive demand required to solve nonroutine tasks, such as appraising the quality of a dataset or designing a facial recognition classifier (Delaney & Lee, 2024; Delaney & Selbach-Allen, 2025). In other circumstances, technology can derail students (or teachers) from their learning goals. My research program explores how teachers and students use novel technologies, the affordances and challenges that arise from their technology use, transformations on student and teacher thinking that are mediated by technology, and implications for learning.

Theoretical frameworks

I draw from theoretical frameworks that foreground the complexity of teaching and learning. I have several lines of research that build from different lineages of theory. When I conceptualize relationships between math curriculum and teaching, I draw from Brown's illustration of teaching as adaptative, improvisational design work, akin to orchestrating a jazz concert (Brown, 2011). His pedagogical design capacity framework formed the basis of my dissertation, where I studied how statistics teachers adapted and enacted a machine learning curriculum (Delaney, 2025). When I conceptualize teachers' mathematical professional learning with AI, I draw from teacher knowledge (Shulman, 1986) and teacher noticing (van Es & Sherin, 2008). What teachers attend to, interpret, and how they respond to students is directly related to their knowledge. Yet while teachers are still in the early stages of developing their knowledge, they may not yet have the multimodal forms of expertise to evaluate LLMs' output and/or claims about students' mathematical thinking. I am therefore developing theories about when and how mathematics teachers may prompt LLMs as cognitive supports, and what additional (human) supports they need to grow their pedagogical content knowledge and noticing abilities.

From the student perspective, I have recently explored knowledge building with AI-based tools (Scardamalia & Bereiter, 2020) and building expertise in mathematics with LLMs (Chi & Glaser 1985; Hiebert & Carpenter, 1992). Today's LLMs can imitate humanistic mathematical reasoning, and can complete most of K-12 mathematics tasks with minimal prompting specifications. It is therefore necessary for students and teachers to discern how to use LLMs to build knowledge and foster their own expertise rather than bypass the cognitive work involved in mathematics problem solving altogether. Recent work has begun to explore what human learning

and epistemic agency may look like when LLMs are involved in mathematics tasks, proposing that students' uses of LLMs are indicative of how much problem solving agency they retain versus offload to intelligent agents (Wu et al., 2025). I intend to extend this work and study how students' LLM uses mediate their moment-by-moment thinking and what agency they preserve as they develop problem solving skills.

Methods

I use a mixture of methods and research settings to enact my program. As a former high school teacher, I believe that observing teachers enact lessons with their own students can yield high-fidelity accounts of curricular and teaching interventions. At the same time, I use my own classroom teaching with pre-service mathematics teachers and aspiring undergraduate data scientists as a testbed for academic interventions. My methods build from design-based principles (Brown, 1992) that explore relationships between teaching, curricula, tools, and learning.

Core to my work is preserving teachers' design agency and decision-making authority in their classrooms. I use co-design methods (Delaney et al., 2025a; Severance et al., 2016; Xie et al., 2024) to position teachers as core decision-makers about the curriculum and methods that they use. I have sought co-design partnerships with industry partners to design mathematics and data science tasks with LLMs that are authentic to the work that practitioners do. I seek to position myself as a support for teachers during the curriculum design process who is respectful of the cognitive effort required to create high-quality materials (Potvin et al., 2024).

As Brown suggests, classroom environments are ecologically noisy and chaotic, and it is sometimes difficult to discern why teachers make decisions about teaching, curriculum, and technologies. To this end, I use clinical interviewing (diSessa, 2007) to understand student and teacher thinking processes. I have used clinical interviews to study reasoning about machine learning (Delaney et al., 2023), algorithmic bias (Lee et al., 2022), and teachers' selections of datasets for teaching statistics (Delaney & Lee, 2022; Lee & Delaney, 2021). I have learned to carefully design tasks and interview questions in these settings, and to consider my own positionality in relationship to the data produced. My analysis methods skew qualitative although I am trained as a data scientist and have recently begun to apply unsupervised clustering methods model patterns in students' thinking.

Research plans

I am building a research program to advance knowledge on human-AI interaction in mathematics. I plan to closely study students' thinking processes with LLMs during the next 1-3 years through a series of carefully-designed experiments so that I can develop concrete, nuanced recommendations for teachers and curriculum designers. Two of my current funded projects reflect these goals. In the first, I apply external conditions and observe whether they influence students' AI evaluations during mathematics problem solving with AI tools. I designed this variation because, in my exploratory round of AI and mathematics problem solving, I found that students' AI evaluations of mathematics were more superficial than I had expected, because there was no goal or incentive to obtain a mathematically-sound answer. In the second, a colleague and I have trained a deceptive chatbot to consistently give an incorrect answer. In task-based video-recorded interviews, we will measure whether students' epistemic trust and conference of mathematical authority is more robust toward the deceptive chatbot or a human tutor instructed to make the same mistake as the chatbot. Overall I am working to refine my study design and research practice so that I can recruit large enough student samples for randomized control trials. From these studies I aim to develop theories about human-AI mathematical interactions, including the degree to which any student-AI interaction truly qualifies as "collaboration" as the learning sciences conceive of it (e.g., Barron, 2003).

In addition, I am continuing to partner with school districts and study teacher co-design with AI-based lessons using insights from my research. I am leading a professional learning in data science for late elementary school, where my colleagues and I will focus on building teachers' co-designed lessons about algorithmic bias in school communities. With the same colleagues, we are working toward obtaining grant funding for AI literacy co-design that supports values and local ontologies in rural Borderlands California. Our goal is to propose models for acquiring *AI literacies* (Kafai et al., 2024) that do not undermine rural and transnational ways of knowing.

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Multimodal Interactions with AI: Human–AI Partnerships in Learning, Teaching, and Research

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Abstract: My research program examines how multimodal interactions with AI shape learning, teaching, and research. I investigate how learners build and express knowledge across verbal, embodied, and digital modalities, and how AI can meaningfully participate in these processes. My work spans three interconnected strands. First, I design AI-based systems that personalize learning by assessing cognitive processes through multimodal data triangulation. Second, I develop human–AI hybrid intelligence approaches for modeling complex, temporally layered learning processes, leveraging methods such as the interleaving approach, epistemic network analysis, and computer vision. Third, I design immersive XR learning environments in which AI agents scaffold embodied STEM learning through interactive multimodal guidance. Across these strands, my goal is to advance theoretically grounded and methodologically rigorous frameworks for understanding and designing human–AI partnerships that enhance multimodal sense-making and meaning-making in learning sciences and CSCL contexts.

Introduction

My work is driven by a simple but persistent question: *How do learners build and express knowledge when their learning unfolds across multiple modes, such as speech, gesture, digital tools, and now, interactions with AI?* In today's technology-rich learning environments, these multimodal interactions shape not only how students make sense of ideas but also how they share and demonstrate what they know. When we overlook these diverse ways of meaning-making and knowledge expression, we risk misrepresenting learners' abilities and missing opportunities to support them more effectively (Nathan, 2023).

Motivated by this concern, my research advocates for multimodal approaches to knowledge construction, expression, and assessment. I study how learning happens across verbal, embodied, and digital channels, and increasingly how these processes evolve when learners collaborate with AI systems. Much of my early work focused on human–human interaction—how people speak, gesture, and use digital tools while learning, collaborating, reasoning, and teaching. Recently, my work has expanded to examine human–AI interaction as a new and important layer of learning, teaching, and research, where AI can shape, scaffold, and even participate in meaning-making processes.

Across these commitments, my research spans three interconnected strands: (1) personalizing and supporting learning through AI systems, where I explore how AI can act as a reflective thinking-partner rather than an answer-giver; (2) partnering with AI as a research collaborator to model complex, temporally layered learning processes; and (3) designing immersive XR-based learning environments that provide AI-mediated multimodal interaction and scaffolding to support embodied STEM learning. Across these strands, my overarching goal is to advance our understanding of human–AI partnership in learning, teaching, and research—specifically, how humans and AI co-construct sense-making and meaning-making.

Personalizing and supporting learning through AI systems

The first strand of my work focuses on personalizing and supporting learning through AI systems, grounded in my expertise in assessing learners' cognitive skills and constructs through multimodal data triangulation. By integrating speech, gesture (or body movement), and digital traces collected throughout the learning process, I investigate how observable behaviors can serve as meaningful proxies for internal cognitive processes. For example, in Sung et al. (2024), I led the design and development of an intelligent tutoring system to support learning with visual representations in engineering education. Leveraging multimodal data triangulation, including gesture, log data, and formal and informal assessments, I demonstrated the feasibility of evaluating students' internal visualization skills. In another study (Sung et al., 2024), I modeled the temporal dynamics of self-regulated learning (SRL) in a technology-mediated biology course by integrating multimodal traces, including digital traces with think-aloud data. This work revealed distinct SRL processing patterns between achievement groups, demonstrating how multimodal traces can surface meaningful differences in learning processes.

Building on this foundation, I am extending this line of work to better understand how learners interact with technology, including AI, and how these interactions shape learning in AI-mediated contexts. Most recently,

in my graduate-level programming course (Fall 2025), I incorporated a constrained LLM-based AI assistant called *PingPong* (Beta version; Computational Policy Lab, 2025) as a virtual teaching assistant and observed how students and AI co-solved programming problems. At the University of Tennessee, Knoxville, I am now co-developing implementation guidelines and rubrics to help instructors adopt constrained AI models responsibly and effectively in their courses.

Human–AI collaboration in modeling multimodal learning processes

The second strand of my research focuses on human–AI hybrid intelligence in multimodal learning analytics (MMLA). Contemporary learning is deeply interactive and collaborative, requiring analytic methods that preserve the temporal and contextual richness of multimodal interactions to understand how meaning-making unfolds over time. In my prior work, I proposed an analytic approach called interleaving (Sung et al., 2021), which models “temporally entangled” multimodal interactions while retaining the contextual detail of each event. Using epistemic network analysis (ENA; Shaffer, 2017), this approach produces interpretable visualizations that support researchers, instructors, and learners in understanding the dynamics of learning. This work received the 2022 CSCL Naomi Miyake Best Student Paper Award, a recognition of its methodological contribution to the field.

Building on this foundation, I recently proposed a framework for human–AI hybrid intelligence in MMLA, in which humans and AI collaborate through interlinked feedforward and feedback processes (Sung, under review). These iterative cycles span data collection, analysis, and interpretation, positioning AI not just as a tool but as a partner in uncovering patterns and generating insights about learning. Empirically, I have employed computer vision and machine learning to automatically detect and track fine-grained learner behaviors. This work has enabled the interpretation of body movement as an indicator of engagement in collaborative learning (Sung & Nathan, 2024), the examination of verbal and nonverbal contributions to co-constructing embodied mathematical understanding (Sung & Nathan, 2025), and the detection of learners’ physical interactions with objects such as worksheets and keyboards (Lee et al., 2026). Together, these studies demonstrate how human–AI hybrid intelligence can deepen our understanding of complex, temporally layered multimodal learning processes through coordinated feedforward and feedback cycles.

Designing AI-mediated XR environments for embodied STEM learning

This strand of my work focuses on designing immersive XR learning environments, such as VR and AR, where AI acts as a collaborative learning partner. My goal is to help learners experience abstract STEM concepts through embodied and tangible interaction, making the learning process more intuitive, engaging, and conceptually grounded. In earlier project, I developed a movement-based simulation for pre-service mathematics teachers to experience and co-design embodied approaches to math learning (Sung & Nathan, 2024). More recently, I co-developed a motion-based AR geometry game for elementary students (Na & Sung, 2025). Using computer vision and augmented reality, this intervention was implemented in real classrooms and resulted in significant improvements in students’ geometry understanding. Building on this foundation, I am now co-leading a VR science learning project that features an AI agent capable of communicating through speech and gesture. This AI agent provides guidance and feedback in an interactive and embodied manner, enabling learners to engage with scientific ideas through multimodal dialogue and action.

Collectively, this line of research advances the design of immersive learning environments that enable embodied knowledge building with AI as an active collaborator. It deepens our understanding of how learners co-construct meaning with AI through multimodal interaction and informs the development of future educational interventions that leverage emerging immersive technologies.

Future directions

In future work, I will push each strand forward by design AI systems that serve as responsive and trustworthy thinking partners, creating hybrid intelligence approaches that make complex learning processes visible, and designing XR environments that bring embodied interaction and AI guidance into collaborative STEM learning. Through these efforts, I aim to contribute evidence-based frameworks for how human–AI partnerships can enhance learning, interaction, and sense-making in learning sciences and CSCL contexts.

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AI-Enhanced Sociocultural, Multimodal Approaches for Equitable Collaborative Learning: Learning, Power, and Multimodality

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Abstract: My research lies at the intersection of the learning sciences, learning analytics, and Science, Technology, Engineering and Mathematics education, and centers on making collaborative learning more equitable for K–12 learners, particularly multilingual, multicultural, and marginalized youth. Across three interconnected lines of inquiry, I position myself as a scholar who bridges the learning sciences and learning analytics, advancing theoretically grounded, situated, and equity-oriented approaches to illuminate the complexity of collaborative learning at multiple scales.

Introduction

I am a learning scientist committed to advancing equitable, multimodal learning. This commitment has shaped my work across diverse contexts—from developing nationally awarded K–12 collaborative physical computing curricula, to teaching in a rural Ugandan village and developing sustainable curricula that integrate local folk tales, music, and art into lessons, to training as a student teacher in a culturally diverse Australian public school, to co-founding the nonprofit Society of Technology for Education and Learning Analytics, and to my current research in learning sciences and artificial intelligence (AI). After completing my Ph.D. in Science, Technology, Engineering and Mathematics (STEM) Education at the University of Texas at Austin, I joined the University of Virginia as a tenure-track Assistant Professor in the School of Education and Human Development.

My work focuses on developing and deploying AI-enhanced multimodal learning analytics approaches across diverse global contexts. Drawing on sociocultural perspectives of learning (Engeström, 1987; Lave & Wenger, 1991; Vygotsky, 1978; Wertsch, 1998) and translanguaging theory (García & Kleifgen, 2019; García & Wei, 2014), I examine multimodal participation dynamics—“the interplay of verbal, nonverbal, physical, and/or digital contributions that learners make as they engage with peers, (shared) tools, and objects” (Lee et al., 2026b, p. 3)—and seek to make these dynamics more visible in authentic, co-located collaborative STEM learning settings. Traditional educational frameworks have long privileged narrow, monolingual, and Western-centric forms of participation, such as verbal or written contributions, marginalizing learners whose linguistic and cultural practices differ from dominant norms or institutionally sanctioned expectations (Annamma et al., 2013; Flores & Rosa, 2015; McDermott & Hood, 1982). My research is thus driven by the imperative to broaden, through more inclusive perspectives, the boundaries of what is recognized and valued as meaningful participation. I anchor my work in the understanding that STEM classrooms should invite students to observe, question, reason, create, and collaborate (Cohen et al., 1999), while recognizing diverse ways of knowing and learning. Equally central is the conception of collaborative learning as a space where access to learning is shared, fostering collective authority and responsibility rather than reproducing teacher-centered authority within group settings (Barron, 2003; Cohen et al., 1999; Langer-Osuna et al., 2020; Lee et al., 2026b). These perspectives shape how I design research.

Across projects, I engage participatory and justice-oriented approaches, including research–practice partnerships and participatory design research (Bang & Vossoughi, 2016). These approaches foreground lived experiences, sustained relationships, and attention to power so that tools and studies emerge from the needs and aspirations of those most impacted. Methodologically, video recordings of authentic, co-located group work serve as my primary data source, as they capture the complex embodied, relational, and material dimensions of participation. I integrate qualitative, quantitative, and computational approaches in unique ways to study participation and collaboration at multiple scales using video data, as well as to examine the affordances and limitations of AI-supported approaches in analyzing these data.

Research areas

My *first line of research* advances scalable, less intrusive, and ecologically valid multimodal learning analytics approaches for analyzing multimodal participation in co-located collaborative STEM learning environments using AI. Expanding my dissertation, I develop and deploy a theoretically grounded video analysis system that uses computer vision and neural networks to temporally track and visualize observable multimodal human activities (e.g., typing, talking, writing) to study collaborative group dynamics—an approach I refer to as *Activity Mapping*. This system and the associated visualization tool open new analytic possibilities by using AI to make nonverbal and tool-mediated interactions visible and to enrich subsequent quantitative and qualitative analysis. For example,

one of my recent studies (Lee et al., 2026b), which integrates the Activity Mapping approach with Epistemic Network Analysis (Shaffer, 2017) and qualitative video-based interaction analysis (Erickson, 2012; Jordan & Henderson, 1995), found that groups with dominant gender and language backgrounds exhibited distinct participation patterns, illuminating the varied ways in which roles, tasks, and ideas are negotiated and contributions are made. Importantly, my work is not oriented toward classifying, predicting, or labeling learners using AI. Rather, I design multimodal learning analytics approaches as augmentative tools that surface multimodal participation dynamics, enabling educators and researchers to notice the diverse, complex, and multimodal ways students participate. These approaches provide new analytical lenses that go beyond the limits of human observation or would otherwise require an impractical amount of time, supporting (not replacing) human critical interpretation. To learn more about this line of work, see Lee et al. (2026a, 2026b) and Lee (2025a). Ongoing work continues to expand the behavioral repertoire the system can detect, strengthen model performance, and prototype versions that support real-time teacher noticing in authentic classrooms (e.g., Lee et al., under review).

The *second area* examines how power and learning are co-constructed (Philip & Gupta, 2020) in the context of collaborative student groups, focusing on key mediational artifacts. I study power relations, including influence, access, and authority, such as who controls access, how behavior is shaped, and how these artifacts reflect and/or mediate broader societal structures and (in)equities. My recent research (Lee, 2025b, under review-b) investigates how access to a shared keyboard mediates power dynamics in a collaborative STEM learning group, focusing on how Latinx middle school students negotiate participation through social and intellectual authority (Langer-Osuna et al., 2020). Integrating the aforementioned AI-supported Activity Mapping approach with qualitative video-based interaction analysis (Erickson, 2012; Jordan & Henderson, 1995), quantitative analysis revealed shifting patterns of keyboard access, while qualitative findings exposed the nuanced ways these patterns reflected or obscured underlying authority dynamics (Lee, under review-b). Findings illustrate how the keyboard functioned not merely as a tool for input, but as a multifaceted mediational artifact serving various functions; how tool access does not inherently equate to authority; and how authority is not only asserted but must be socially ratified through peer uptake. These insights inform the design of more inclusive collaborative learning environments that honor sociocultural complexity. Having previously focused on Latinx multilingual learners, I am now expanding this work to more diverse multicultural and multilingual classrooms through new partnerships.

The *third area* centers on participatory design and sociocultural inquiry into learning analytics and AI-supported collaborative classrooms. Across partner hubs in Virginia, New Mexico, South Korea, and Brazil, educators and community leaders have voiced concerns that AI systems for education are often not designed with—or for—the communities most impacted. To challenge the Western, monolingual, and individualistic assumptions embedded in these systems, I plan to lead the Global Responsible AI in Education Collaborative (pending funding decision). This transnational, transdisciplinary initiative brings together researchers, practitioners, community partners, youth, AI developers, and policy experts from international hubs to co-develop globally grounded research agendas and responsible AI guidelines for teachers through cross-cultural collaboration, youth-led inquiry, and participatory design. The project ultimately seeks to reimagine what counts as knowledge, evidence, and participation in AI-supported education (Bang & Vossoughi, 2016), and to redefine “responsible AI” through equity-centered, multilingual, multimodal, and cross-cultural perspectives (Lee, under review-a). This work is currently underway as a pilot in one hub (funded by the University of Virginia IDEAs grant), where I lead a research–practice partnership with South Korean public schools to co-design learning analytics– and AI-supported classrooms that explore how such tools can be responsibly developed to advance equitable K–12 STEM learning across global contexts.

Contributions to the learning sciences

Across these interconnected lines of work, I contribute to the field by (1) advancing theoretically grounded, AI-supported multimodal learning analytics approaches that honor sociocultural complexity; (2) deepening understanding of learner power dynamics, participation, and tool mediation in K–12 collaborative learning; and (3) shaping globally responsive, equity-centered frameworks for responsible AI in education. As a researcher trained in the learning sciences and committed to global, multilingual, multimodal, and community-centered forms of innovation, my work seeks to make methodological, theoretical, and practical contributions that expand how the field conceptualizes and studies participation and collaboration, while also providing a technical pathway for studying these dynamics at scale in real-world classrooms. Grounding my studies in converging insights from sociocultural theories of learning, critical and decolonial education research, translanguaging, multimodal learning analytics, and participatory design research (Bang & Vossoughi, 2016; Engeström, 1987; García & Kleifgen, 2019; García & Wei, 2014; Lave & Wenger, 1991; Vygotsky, 1978; Worsley et al., 2021), I aim to continue focusing on the critical yet underexplored intersection of AI, multilingualism, multimodality, and equity (Lee, 2025a, under review-a; Lee et al., 2026b).

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Learning With and About AI: Building Capacity for Equitable Participation in an AI-Shaped Future

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Abstract: My research investigates how learners and educators understand, design with, and collaborate with AI systems, and how these sociotechnical interactions shape equity and participation. Using design-based research, participatory design, and qualitative and computational methods, I study AI education and disciplinary integration of AI in K–12 contexts. My work aims to design equitable learning environments, develop theoretical models of learner engagement with AI, and build cross-disciplinary approaches to teaching with and about AI.

Theoretical and methodological foundations that inform my work

My research sits at the intersection of the learning sciences, data science and artificial intelligence (AI) education. I am particularly interested in how educators and learners understand, design with, and collaborate with AI systems, and how these sociotechnical interactions shape equity and participation in society. In my work, I frame learning as a socially and materially mediated practice, drawing on sociocultural perspectives that conceptualize knowing as participation in activity systems rather than simply the acquisition of decontextualized concepts (Vygotsky, 1987; Wertsch, 1991). Within this view, AI tools and data infrastructures function as cultural tools (Cole, 1996; Saxe, 1991) that shape what becomes visible, what is considered authoritative, and whose perspectives are deemed valid as knowledge. I also draw on constructionist (Holbert et al., 2020; Kafai & Resnick, 1996; Papert & Harel, 1991) and design-oriented traditions that treat building computational artifacts as a powerful avenue for learning. When students build or critique AI applications and data stories, they are engaging in epistemic work that reveals their assumptions about what counts as evidence, how models relate to the world, and whose experiences are encoded in data (Vakil et al., 2022). I am particularly attentive to how young learners develop both conceptual and sociotechnical understandings of the affordances and limits of AI technologies, and how educators can support these competencies in ways that foster agency, critique, and creativity.

Methodologically, my work draws on design-based research (Hoadley & Campos, 2022), participatory design (Cumbo & Selwyn, 2022), and qualitative approaches commonly used within the Learning Sciences. I employ design-based research to iteratively co-design data science and AI resources with educators and community partners, while developing mid-level theoretical constructs regarding AI literacy and learner–AI collaboration. Participatory design approaches with youth and teachers position them as co-designers of curricula, games, and AI experiences, helping surface the tensions they experience when trying to integrate AI into their local contexts and disciplinary practice. In terms of analysis, I primarily employ qualitative methods, including thematic and discourse analysis of interviews and observational data, to examine how students and teachers make sense of AI tools and concepts in situ. I complement these with quantitative and computational approaches, including epistemic network analysis (ENA) (Marquart et al., 2018; Shaffer, 2017) to explore patterns in how learners and educators engage with, reason about, and respond to AI. I am also beginning to explore LLM-assisted coding and modeling workflows that can support, but not replace, human interpretive work with complex datasets (Arastoopour Irgens & Adisa, 2024). Through this combined theoretical and methodological orientation, I aim to contribute to the Learning Sciences by conceptualizing AI and data-driven technologies as part of the social and epistemic infrastructure of learning.

Past and current work

My research trajectory began with questions about how young learners come to understand, apply, and critique AI systems. In my doctoral work, I collaborated with community afterschool educators to design learning environments that situate machine learning concepts in sociopolitical contexts. I led studies examining how children interrogated datasets, reasoned about model outputs, and built their own AI applications. To achieve this, I collected and analyzed a wide range of data sources, including students' code snippets, design sketches, digital interaction logs, discourse transcripts, and survey data. I used both ENA and thematic analysis to map the connections students made across concepts as they interacted with AI systems, supportive adults, and peers. My analysis showed that when students are given opportunities to troubleshoot and design AI systems, their computational thinking practices improved and their confidence in questioning and proposing solutions to

algorithmic limitations increased. Alongside this work with youth, I participated in several projects focused on K–12 data science and computing education. Within research–practice partnerships in elementary schools, I collaborated with teachers to design data science units that integrated students’ interests (such as music) with data modeling and interpretation (Herro et al., 2022). These projects examined how teachers understood data science, how school structures influenced their capacity to adopt new curricula, and how students engaged with data storytelling and visualization (Adisa et al., 2024). Collectively, this work contributed to emerging conversations in the Learning Sciences about how data practices can be introduced in ways that are meaningful, culturally relevant, and equitable for younger learners. During my doctoral studies, I also became increasingly interested in how teachers and students perceived the role of emerging generative AI technologies in education. I conducted survey and interview studies examining high school and middle school students’ perspectives on tools such as ChatGPT, including their hopes, concerns, and ideas about appropriate educational uses. These projects documented a range of stances, from enthusiasm to ambivalence and resistance, and underscored the need for grounded AI literacy support for both students and educators (Adisa & Adefisayo, 2025; Famaye et al., 2024).

My current postdoctoral work at Stanford builds directly on these experiences and moves more centrally into the AI literacy and human–AI collaboration space within the Learning Sciences and computer-supported collaborative learning (CSCL) communities. Recognizing the pivotal role teachers play in shaping learners’ critical engagement with technology, I co-design AI literacy resources across subject areas with teachers through the CRAFT (Classroom-Ready Resources About AI For Teaching) project. This includes analyzing how teachers conceptualize AI literacy within their disciplines, how they design activities that connect AI concepts to core disciplinary practices, and how school and policy contexts shape what is possible. I am particularly drawn to cases where teachers design AI lessons that stretch conventional disciplinary boundaries—for example, exploring the intersection of AI and art, AI and environmental science, or AI and statistics in ways that invite critique and imagination. These efforts allow me to study teacher learning in PD contexts, including how educators make sense of rapidly evolving AI technologies, how they negotiate district expectations and ethical concerns, and how they connect AI literacy goals to their existing pedagogical commitments. In parallel, I examine learner–AI collaboration in problem-solving contexts, an area that has become increasingly important as generative AI systems participate more directly in students’ cognitive and creative work. I conceptualize human–AI collaboration as a form of joint activity that redistributes cognitive, social, and metacognitive work across human and machine actors. To investigate this, I have designed and implemented studies examining how students negotiate meaning, agency, and communication with AI when solving problems individually and in groups. Using epistemic network analysis alongside fine-grained interaction analysis of video and interview data, I trace how cognitive moves (such as strategy generation and evaluation), social moves (such as turn-taking and justification), and metacognitive moves (such as planning and monitoring) are distributed among students and AI systems. My preliminary findings suggest that productive human–AI collaboration is characterized by students’ active regulation of the AI’s role—including engaging in independent work, questioning AI outputs, and integrating AI-generated ideas with their own reasoning. Across these strands, my past and current work contribute a line of inquiry focused on how learners and educators come to understand AI systems, how they collaborate with AI in problem solving, and how AI literacy can be integrated into disciplinary learning in ways that foreground agency, critique, and creativity.

Future work

My future research agenda will focus on designing and studying learning environments that enable learners to collaborate productively with AI to learn, create, and engage in STEM problem-solving. Building on my current projects, I plan to partner with educators and students to co-design AI-powered digital learning environments that promote STEM learning while cultivating critical and reflective engagement with AI. My focus here is on understanding how AI agents and human contributions jointly shape cognitive, communicative, and metacognitive outcomes, and what kinds of support help learners maintain agency in the presence of increasingly capable tools. On the design side, I plan to develop lightweight tools that encourage students to engage in productive forms of collaboration, provide transparency into AI suggestions, and capture data that can be repurposed for timely feedback and teacher insight. Methodologically, I will advance the use of epistemic network analysis and natural language processing to model how students connect disciplinary content with AI tools and practices, and to characterize the evolving interaction patterns that emerge in hybrid human–AI teams. Relatedly, I intend to continue articulating and testing cross-disciplinary frameworks for integrating AI literacy into K–12 instruction. Drawing on my experience working with teachers across AI, data science, and various disciplinary contexts, I will collaborate with educators in multiple subjects to design and study AI-integrated units that align with disciplinary epistemologies while fostering a critical understanding of AI.

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Doctoral Consortium

What Makes Actions Matter in Virtual Reality? Examining the Effects of Conceptual Congruence and Sensorimotor Engagement on Geometric Reasoning

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Abstract: This dissertation examines how physical actions in virtual reality influence geometric reasoning through two complementary studies. Study 1 synthesizes nine experiments ($N=774$) using integrative data analysis to establish baseline mechanisms of embodied geometric reasoning. Study 2 employs a 2×2 experimental design (Target $N=200$), manipulating conceptual congruence and sensorimotor engagement of physical actions in VR environments to test how these action characteristics affect geometric reasoning. Participants engage with virtual geometric shapes via hand tracking, with multimodal data capturing verbal reasoning, gestures, and VR interactions. This work advances theoretical understanding of embodied learning and evidence-based design principles for immersive learning environments.

Motivation and research questions

Extended reality (XR) technologies hold promises for engaging and intuitive mathematics learning, yet significant research gaps remain. Current studies often lack a solid theoretical foundation, focus more on *whether* XR is effective rather than *why*, under-explore the unique potential of body movement, and devote less attention to mathematics learning than other STEM fields.

This dissertation, guided by embodied learning theories, aims to understand how body movement influences mathematics learning in XR. It specifically investigates the effects of physical actions on geometric reasoning through two research questions: (RQ1) How do physical actions and gestures impact geometric reasoning in non-VR settings? and (RQ2) How do interactions with geometric shapes, with different levels of conceptual congruence and sensorimotor engagement, influence geometric reasoning in a VR environment?

Theoretical background

The intersection of embodied learning theory and XR technology motivates this investigation, with geometry providing an ideal context where physical actions naturally support mathematical reasoning. The Grounded and Embodied Mathematical Cognition (GEMC) framework (Nathan & Walkington, 2017) demonstrates that task-relevant actions that are designed to highlight key mathematical properties facilitate geometric reasoning. In an experiment, Walkington et al. (2022) found that the cognitive relevance of actions matters more than movement itself: task-relevant actions significantly improved reasoning, particularly when accompanied by explanatory gestures, while irrelevant actions showed no benefit despite similar motor engagement.

Virtual reality presents unique opportunities and challenges for embodied mathematics learning. Johnson-Glenberg's (2018) embodied learning framework for VR identifies two critical dimensions that this dissertation directly examines: *conceptual congruence*, which is the alignment between body movement and learning content, and level of *sensorimotor engagement*, the degree of body involvement. This framework hypothesizes that higher congruency and greater body involvement lead to better learning outcomes. Skulmowski and Rey's (2018) taxonomy of immersive learning further suggests that physical manipulations that are more aligned and integrated with the task create stronger memory traces through the generation of motor codes, though they caution that excessive interactivity without conceptual relevance may increase extraneous cognitive load.

These theoretical perspectives converge on key predictions for VR geometry learning: (1) The conceptual congruence of virtual manipulations should matter more than general interactivity, and (2) active manipulation should outperform passive observation, but only when actions align with mathematical content.

Methods

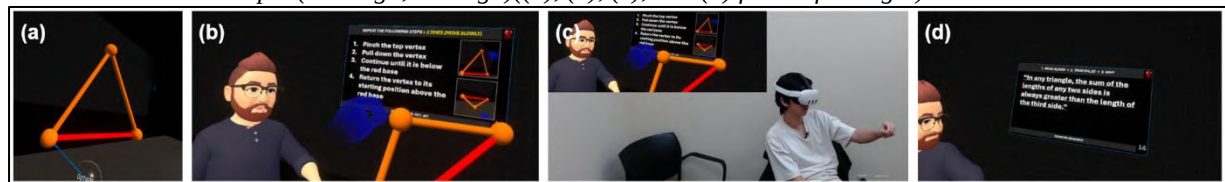
Study 1 synthesizes raw data from 9 independent randomized controlled experiments ($N=774$). Taking an Integrative Data Analysis approach (Curran & Hussong, 2009), it employs Item Response Theory to harmonize measures across heterogeneous datasets, then uses structural equation modeling within a multiple-indicator multiple-cause (MIMIC) framework to test pathways and estimate effects of task-relevant/irrelevant actions on geometric reasoning outcomes via latent embodied cognitive mechanisms.

Study 2 is a 2×2 between-subjects experiment in which participants (Target $N=200$) are randomly assigned to conditions, manipulating two key attributes of actions they are directed to perform in VR: Conceptual Congruence (CC; High vs. Low) and Sensorimotor Engagement (SE; High vs. Low). The study involves two phases. During the *baseline phase*, participants are asked to prove or disprove six geometric conjectures outside VR. In the *intervention phase*, participants evaluate the truth of six new conjectures. They don a Meta Quest 3 headset that, through an app called CurioXR, displays custom-designed virtual geometric shapes in a VR environment. Participants interact with each shape either by direct manipulation (SE High) or by watching a video (SE Low), with the interaction being either relevant (CC High) or irrelevant (CC Low) to the conjecture that immediately follows. After each interaction, the shape disappears, and participants evaluate the truth of a conjecture about the shape's concepts or properties.

For example, the triangle inequality states, “*In any triangle, the sum of the lengths of any two sides is always greater than the length of the third side.*” Participants are first presented with a virtual triangle (Figure 1a), unaware that the triangle inequality conjecture will follow. Then, depending on their condition assignment, they are asked to do one of the following: (1) Move the top vertex closer to the bottom side of the triangle (CC High, SE High; Figures 1b, 1c), (2) watch a video clip of someone else moving the top vertex closer to the bottom side (CC High, SE Low), (3) move the whole triangle downward (CC Low, SE High), or (4) watch a video clip of someone else moving the triangle downward (CC Low, SE Low). After completing the interactions, the triangle disappears, and participants read and either prove or disprove the conjecture (Figure 1d). The condition-specific intervention unit is repeated across six conjectures, with the order counterbalanced.

Figure 1

Intervention Unit Example (CC High, SE High)(a), (b), (c), and (d) from left to right)



Preliminary findings and data gathered

Study 1 preliminary analysis of five experiments ($N=425$) revealed that task-relevant actions show positive indirect effects on deductive geometric reasoning ($\beta^* = .062$, $p = .001$), while task-irrelevant actions impair geometric reasoning ($\beta^* = -.047$, $p = .001$), supporting the hypothesis that cognitive benefits of embodied interactions depend on *conceptual congruence* with mathematical content rather than general motor activation. For Study 2, multiple data streams will capture the embodied geometric reasoning process: video recordings of verbal geometric justifications, co-speech gestures, and screen recordings documenting VR hand interactions and shape manipulations (See Figure 1c). Geometric reasoning will be assessed using the qualitative coding rubric developed by Nathan and colleagues (2021) regarding generalizable, logical, and operational reasoning.

Expected contributions

This dissertation makes several contributions to the learning sciences and mathematics education. First, it systematically investigates what makes physical actions effective in VR by isolating conceptual congruence and sensorimotor engagement as design dimensions, providing evidence-based principles for VR learning environments rather than simply comparing different media. Second, it demonstrates how integrative data analysis can promote cumulative science in the learning sciences while advancing methods for studying embodied cognition through multimodal VR data collection. Finally, this research addresses the notable paucity of experimentally controlled, theoretically motivated studies on thinking and learning with immersive technology.

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Measuring and Supporting Active Listening Skills in Collaborative Problem Solving through Multimodal AI

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Abstract: Collaborative learning is central to education, yet it often involves unequal participation—some voices dominate while others remain unheard. Such imbalance constrains both individual learning and group outcomes. Prior research has primarily focused on regulating speaking while overlooking active listening, which remains difficult to assess and support in practice. This dissertation addresses this gap by positioning active listening as a measurable and learnable collaborative skill. I develop the Theory–Measurement–Evaluation (TME) framework, which operationalizes active listening constructs into multimodal behavioral indicators and builds interpretable multimodal learner models. Grounded in these explainable evaluations and pedagogical principles, I design a dialogic feedback system that translates analytic insights into actionable guidance to support reflection and skill development. A pre-post, between-subjects experiment will evaluate its effectiveness.

Background and goals

Collaborative learning is central to education, not only as a way to build understanding across perspectives, but also as preparation for civic life, where collaboration is essential. Yet participation in collaboration is rarely equal. Studies consistently show that some students dominate while others are sidelined, which limits both individual learning and group outcomes (Barron, 2003). Most interventions to date have focused on efforts to design equitable conditions—structuring tasks, scripting activities, or assigning group roles (Cress et al., 2021). Far less attention, however, has been given to students' own capacity to notice and regulate unequal participation within their groups. A promising but underdeveloped pathway for building such capacity is *active listening*. The National Communication Association defines active listening as “the process of receiving, constructing meaning from, and responding to spoken and/or nonverbal messages” (Worthington & Bodie, 2017, p. 214). This definition highlights core skills such as hearing, understanding, remembering, interpreting, evaluating, and responding (HURIER model, Brownell, 2023). In collaborative learning settings, active listening requires interpreting and responding to peers' contributions in ways that make space for diverse voices and foster shared understanding. Despite its importance, active listening is rarely taught in schools. Teachers often lack the time and resources to observe subtle interactional dynamics and provide feedback, and even research has largely emphasized regulating speaking (e.g., turn-taking) over listening. As a result, one of the most critical levers for improving equitable collaboration—students' ability to listen actively—has remained underdeveloped.

This dissertation addresses this gap by pursuing three interrelated research inquiries: *RQ1*. What are the theoretical and practical considerations for effective analytics-supported feedback systems that aim to develop collaboration skills, and what does the current state of research reveal about such interventions? *RQ2*. How can active listening be characterized and measured in small-group collaboration using multimodal learning analytics? *RQ3*. Based on these measurements, how can we design and evaluate pedagogically grounded, AI-generated dialogic feedback that helps students improve their active listening skills?

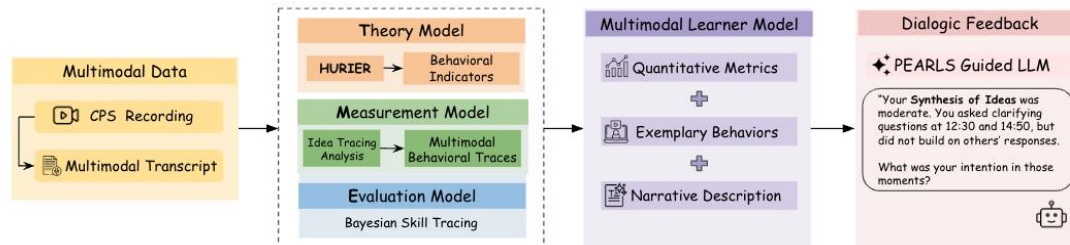
Methodology, data and preliminary findings

To answer *RQ1*, I proposed the Theory–Analytics–Pedagogical Feedback (TAP) theoretical framework (Huang & Ochoa, 2025), extending prior learning analytics models to integrate theory, measurement, and feedback for the design and evaluation of systems that foster collaboration skill development. Using TAP as an analytical lens, I conducted a semi-systematic review of existing collaboration feedback systems and identified four major gaps in current research and practice. These findings underscore the need for systems that move beyond raising awareness to actively supporting the development of collaboration skills.

To answer *RQ2*, I developed the Theory–Measurement–Evaluation (TME) framework (Huang et al., 2026b) (Figure 1). TME transforms multimodal data (audio and video) into interpretable learner models by linking a *theory model* that operationalizes active listening constructs into behavioral indicators, a *measurement model* that automatically extracts contextualized multimodal traces using Idea Tracing Analysis (ITA; Huang et al., 2026a), and an *evaluation model* based on uncertainty-aware probabilistic modeling. The framework was initially validated using data from an exploratory lab study with 30 college students collaborating in nine groups on an

open-ended design challenge. Multimodal interaction data were captured using a 360-degree conferencing system and smart pens, and audio was transcribed and speaker-diarized to support verbal data analysis.

Figure 1
Bridging Explainable Modeling and Actionable Feedback



To address RQ3, Building on the multimodal learner model produced by TME, feedback is delivered through a LLM-mediated conversational interface structured by the PEARLS framework and its advocacy-inquiry method (Eppich & Cheng, 2015) (Figure 1). The system presents construct estimates and exemplary moments, and provides personalized actionable guidance. In this design, the LLM operates within predefined pedagogical constraints rather than as an open-ended generative agent. A randomized between- and within-subjects, two-session experiment with 90 college students is planned. Feedback will be randomized at the group level, between two sessions. Outcomes will assess pre-post changes between conditions in self-awareness, motivation, and behavioral change measured through TME across sessions.

Expected contributions

This dissertation makes theoretical, methodological, and practical contributions to CSCL research. *Theoretically*, it expands our understanding of collaborative processes by reconceptualizing active listening as a visible, measurable, and learnable process that shapes inclusion and knowledge co-construction. *Methodologically*, the proposed TME framework enables theory-grounded multimodal measurement of collaboration skills. ITA advances collaborative discourse analysis by treating ideas not only as outcomes of interaction but also as an analytic lens for interpreting verbal and nonverbal behaviors across multimodal data. *Pedagogically*, this work demonstrates how explainable assessments can be translated into actionable, dialogic interventions that support self-regulated collaboration skill development, amplify diverse voices, and promote more equitable participation in CSCL environments.

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Developing Systems Thinking Related to Sustainability Among Urban Indian Middle School Students Using Cognitive Apprenticeship

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Abstract: The present study focuses on developing systems-thinking among middle school students (11-14 years) through a cognitive apprenticeship-based learning approach. The study aims to develop interventions to foster students' systems thinking, as an important competence for solving sustainability problems. The study will first examine experts' understanding of systems and processes as they solve system-related problems. Subsequently, the insights from it will be integrated with the cognitive apprenticeship learning approach to develop interventions for middle school students.

Research goals

This study aims to design an effective learning environment to develop systems-thinking skills among urban Indian middle school students, focusing on sustainability concepts and related problems. Specifically, my research focuses on understanding how systems thinking related to water sustainability can be developed among urban Indian middle school students, and my research questions are:

How do middle school students think about water as a system from a sustainability perspective?

How do experts think about systems from a sustainability perspective? How does a cognitive apprenticeship-based learning environment facilitate changes in systems thinking about water sustainability among middle school students?

What is the effect of cognitive apprenticeship-based intervention on students' systems thinking about water sustainability?

Background of the project

Sustainability problems are complex because they involve associations with different stakeholders, multiple perspectives, the ambiguous nature of the concepts surrounding them, and competing values (Morris & Martin, 2009). Given the complexity of such problems, systems thinking is an important skill for solving them (Tomkinson, 2009). Systems thinking is a higher-order thinking skill that focuses on analyzing a complex topic to identify its parts and put them together to gain a holistic view of the system (Kali et al., 2003). It helps to view the "Sustainable world" as a system and to analyze its essential components and their interconnections (Strachan, 2009). It also helps us understand the complex nature of the interconnected world and apply our diverse learning experiences (Strachan, 2009).

Research suggests that a practice-based environmental education intervention, embedded in a context that promotes community participation, helps develop people's systems-thinking (Dutta & Chandrasekharan, 2018). However, how systems-thinking evolves among learners has not been captured in this research. Also, the research does not map the specific features of the intervention with different kinds of changes in systems thinking. Past research also suggests that experience-based learning does not always help improve systems thinking skills (Sadira, 2021), as quoted by Ekselsa et al. (2023). Explicit emphasis must be placed on the interconnections between structures operating at different organizational levels (Raved & Yarden, 2014). Together, this literature suggests the need to explore other learning approaches to develop systems-thinking among middle school students.

Cognitive apprenticeship is a learning approach in which concepts and facts are situated in the context of their use (Collins et al., 1987). It is based on the principles of learning theories such as situated learning, communities of practice, communities of learners, scaffolding, articulation, and reflection (Collins & Kapur, 2014). This study aims to use the cognitive apprenticeship model to combine explicit scaffolding and practice-based interventions to develop key components of systems-thinking among middle school students. The aim is for students to develop these skills to address the sustainability problems they encounter. One such problem that we have chosen for our research is water sustainability, as water is a natural resource and an intrinsic part of the environment. Students would be able to connect with sustainability issues related to water.

Methodology

The research design for this research will be based on educational design research (EDR), a type of research design that aims to solve an educational problem through an iterative process and generate new knowledge through scientific inquiry (McKenney & Reeves, 2013). The present work will follow the EDR model, using qualitative, quantitative, and mixed-methods approaches to address the research questions. An expert study in the initial stage of the research will provide insights into how experts solve sustainability problems using a systems-thinking framework. These findings will be combined with the features of cognitive apprenticeship to design a learning environment for students to develop their systems-thinking. In the later stage, the study to understand the effectiveness of the designed learning environment will be conducted with the Indian middle school students (ages 11 to 14 years).

Data gathered and expected findings

To synthesize experts' understanding of systems and the processes and skills they use to solve sustainability problems, I have interviewed five experts. The experts identified for the study were academicians and practitioners working in the fields of hydrology, climate change, carbon sequestration, ecology, and environmental science. They were identified using purposive sampling. I used a think-aloud protocol, in which experts were given a primary problem related to a sustainability issue to solve, followed by a semi-structured interview to understand their analysis of systems and reasoning, either in support of their proposed solution for the system or in opposition to alternative solutions. I am in the process of analysing the data using thematic analysis, with a systems-thinking framework. I expect to gain deeper insights into how experts use systems thinking to solve sustainability problems. These findings, i.e., the systems thinking approaches used by experts to solve sustainability problems, will be externalized for students during the modeling phase of the cognitive apprenticeship-based learning environment. This will help students build their conceptual models of the processes used in systems thinking.

Expected contributions

This research aims to contribute to a cognitive apprenticeship-based learning environment to address the problem of water sustainability among urban Indian middle school students. It is conjectured that the learning environment will have features such as real-world problems, collaboration and dialogue among students, problematizing and structuring scaffolds, and reasoning. The expert study in this research will contribute to understanding the systems-thinking model used by experts to solve sustainability problems. Another contribution of this study will be to theorize how such a learning environment fosters systems-thinking among students.

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Co-designing and Implementing Reflective Tools and Routines with Children in Cooperative Inquiry

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Abstract: Reflection is a valuable component of learning and design processes, yet motivating children to reflect in informal learning contexts can be challenging. My dissertation examines this issue in a university-based Cooperative Inquiry co-design team where children and adults collaborate as equal design partners. Leveraging a DBR approach, I conduct two studies. The first investigates how children co-design reflection tools and processes that align with their values and mental models. The second explores how they use these systems over time and how this shapes their developing self-concept as designers. Through this research, I will contribute an adaptable infrastructure for authentically integrating reflection into child co-design, insights into how DBR can meaningfully include children as co-design stakeholders, and a deeper understanding of how children think, learn, and develop their identities through co-design.

Motivation and background

The learning sciences have long documented the importance of reflection for a range of learning processes (Dewey, 1986; Moon, 2004; Schön, 2017), with classrooms traditionally serving as important spaces for developing children's reflective learning. Though less structured and systematic, informal learning contexts can also offer rich environments for reflection and identity development, but it can be challenging to motivate children to pause and reflect in the absence of standard classroom norms and enforcing structures (Gardner et al., 2006).

As a doctoral student at the University of Maryland with an interest in how children learn and develop their identities through co-design, I have experienced some of these challenges firsthand through my participation in UMD KidsTeam—an intergenerational co-design team where university faculty and students collaborate with children between the ages of 7 and 12 to design new child-facing technologies. Guided by the Cooperative Inquiry (CI) co-design framework (Druin, 1999; Guha et al., 2013), which aims to involve children as equal collaborators with adults across the entire design cycle, our team employs a variety of established CI 'design techniques' aimed at making co-design accessible, engaging, equitable, and empowering for children. One such technique that has been mentioned in the literature, but which has received considerably less attention, is 'journaling', which has been described as a tool for "design partners to reflect on their experiences by writing or drawing" (Guha et al., 2013). Despite this directive, CI scholars and practitioners have observed child design partners' lack of intrinsic motivation to engage in reflective journaling and have reported challenges in communicating the goals and value of this technique in a manner that resonates with children's personal interests and values (Blinder et al., 2025). Issues with implementing reflective journaling in CI are further complicated by logistical challenges (e.g., time constraints, the need for adult writing support), ideological challenges (e.g., adult discomfort with the solitary school-like nature of written reflection) and adult and child design partners' conflicting conceptualizations of journaling (with children prioritizing documenting and remembering design ideas and adults prioritizing process- and identity-oriented reflection) (Blinder et al., 2025).

These challenges indicate an opportunity to explore additional approaches toward fostering purposeful sustained reflection within CI and other child co-design contexts. In my dissertation, I capitalize on this opportunity by engaging UMD KidsTeam children in contemplating the role reflection can play in their lives and identities as designers through co-designing new developmentally accessible and personally meaningful reflection tools and routines to be used in our co-design context. I then examine how children's engagement with these reflection approaches and their self-perceptions as designers evolve over time.

Methodology

I draw on Design-based Research (DBR) (Wang & Hannafin, 2005), CI co-design (Druin, 1999; Guha et al., 2013), and case study methodologies (Stake, 1995) to investigate the following research questions:

RQ1: *How can tools and routines for documented reflection be integrated into co-design in a manner that resonates with children's values and mental models?*

RQ2: *How do children interact with co-designed reflective tools and routines over time?*

RQ3: *How do children's perceptions of themselves as designers evolve as they interact with co-designed reflective tools and routines over time?*

I answer these questions through two phases of qualitative research. In *Study 1*, I use CI to investigate RQ1 by facilitating a series of co-design sessions with UMD KidsTeam to prototype reflection tools and routines to be implemented in our KidsTeam context. In *Study 2*, I use case study research to investigate RQ2 and RQ3 over a period of 4-6 months to understand how children interact with the reflection tools and routines developed during Study 1 and how their self-perceptions as designers concurrently evolve. The data I am collecting for these qualitative studies include co-design artifacts, children's documented reflections, my observational field notes and reflective memos, and audio and video recordings of co-design sessions, children's individual and group interviews, and adults' session debrief discussions.

I am analyzing the data from *Study 1* using reflexive thematic analysis (Braun & Clarke, 2019). Across *Study 2*, I will engage in continuous cycles of data collection and analysis (Stake, 1995) by iteratively coding my collected data to identify patterns and themes related to children's reflection content, modalities, and behaviors. Though I will primarily use inductive coding for these analyses, I may apply deductive codes derived from established leveled reflection frameworks (Fleck & Fitzpatrick, 2010; Moon, 2004) to answer RQ2. Finally, I will analyze relationships between how children reflect and how they see themselves as designers and use these findings to select two or three representative focal learners for in-depth case studies to communicate my findings.

Collected data and preliminary results

I have conducted four ideation and prototyping co-design sessions with KidsTeam, which resulted in two prototypes that we user-tested and refined during a 1-week KidsTeam summer camp. *Design 1* is an immersive environment called "Reflection Islands", through which children generate multimodal accounts of their design ideas, each of which is represented as a customizable 'island' on a virtual map. Children grow and develop their islands by selecting from a menu of prompts (or creating their own) to reflect about the decisions and processes that informed each of their designs. They can also visit and create pathways across their own and teammates' islands by reflecting on overarching patterns they observe across this broader landscape. *Design 2* is a non-competitive card game called "Memory String" that uses predetermined prompts to elicit participants' accounts of their design experiences and learning, while encouraging players to understand and discuss their shared and divergent perspectives as designers. The game's color-coded cards represent different categories of KidsTeam-relevant reflection, as determined by child design partners, including: "favorite memory of processes"; "sad/emotional", "funniest", and "lightbulb" moments; and "random [creative] challenges".

Expected contributions

Through this research, I aim to make the following design, methodological, and empirical contributions: (1) a refined and tested adaptable infrastructure (consisting of tools, routines, implementation guidelines, etc.) that can be used to promote a more authentic and developmentally appropriate integration of reflection into co-design with children in a manner demonstrated to support their growth and development as designers; (2) a process model for a how DBR—which often involves co-design with adult practitioners (e.g., teachers) and testing in authentic classroom contexts with K-12 students—can incorporate co-design with young learners as a primary stakeholder group by positioning them as full design partners rather than as mere testers, users, and/or informants; and (3) an enhanced understanding of how children think, learn, and develop their identities through co-design.

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Science at Work: Science and Learning among Coffee Roasters

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Abstract: For more than a century science education has been guided by a promise that schools will prepare students to use science in everyday life. Yet little research has examined how science is actually used in everyday life. My dissertation takes this gap seriously by studying science and learning as they occur outside of school. Drawing from extended ethnographic fieldwork, I analyze how professionals in the coffee industry use, teach, and learn science in their daily work.

Goals of the research

My dissertation has three goals: First, to document how science is used outside of laboratories, which have long defined what counts as science. Second, to analyze how learning unfolds through the embodied, material, and social practices that constitute such work. Lastly, to translate those findings into principles for designing science learning environments. Taken together, these goals address a long-standing gap in science education and learning sciences research: the disconnect between the things learned in schools and those used in everyday life. My dissertation argues that reimagining science education-and education more generally-requires renewed attention to the diverse settings in which science and learning occur.

Background of the project

Since the late 1800s science education has aimed to prepare students for life after school (Rudolph, 2023). The most recent of such reforms-the *Framework for K-12 Science Education* (NRC, 2012)--continues this tradition. My dissertation responds to two issues in this research and design paradigm.

First, the forms of science that people use in everyday life are not the forms of science practiced by professional scientists (cf. Aikenhead, 2006; Eisenhart & Finkel, 1998; Irwin, 1995; Wynne, 1995). Despite these differences, how people use science in everyday life has received very little attention, with the majority of "how science happens" being defined by professional scientists (Bang et al., 2012; Warren et al., 2020). The result is that designs for science learning environments are based upon professional scientists' activities because their everyday analogs are underrepresented. The second issue stems from the first: Because everyday settings have received less attention, how *learning* occurs in these settings is also underrepresented (Nasir et al., 2020).

Stated from another angle, one that is slightly more critical but hopefully generative: One would assume, if K-12 education aims to prepare young people for life after school, that research on *their lives after school* would be a priority. This has historically not been the case. Instead, the bulk of science education research examines learning that happens in schools, among young people, organized by professional teachers, and measures what these groups do against the activities of scientists.

My dissertation takes these issues seriously. It does so by studying people outside of schools and professions typically recognized as scientific. I chose to study people in the coffee industry as representatives of the teaching, learning, and science that happen in everyday life because my background as a barista prior to graduate school sensitized me to the forms of science that occur across the industry. My dissertation develops takes these forms seriously and uses them to expand images of science and learning.

Methods

My dissertation draws from video-based fieldwork that I conducted at multiple field sites (Becker & Geer, 1957; Emerson et al., 2011; Erikson, 2006). Since 2022 I've conducted over 600 hours of video-based fieldwork, primarily at two Chicago companies, focusing mostly on the practices of coffee roasters. Additional data come from national events, farms, and learning environments I've co-designed.

The corpus includes audiovisual recordings of roasting sessions, including five multi-year apprenticeships, quality control sessions (e.g., tasting coffee), and classes we've taught; fieldnotes and analytic memos; semi-structured interviews; and artifacts such as data logs, graphs, and instructional materials.

I use three analytic approaches to answer three research questions. Briefly before elaborating below: First, I use a deductive coding approach (Saldana, 2021) that borrows the Science and Engineering Practices (SEPs) and Disciplinary Core Ideas (DCIs) from the *Next Generation Science Standards* (NGSS Lead States, 2013) to identify how roasters use recognized scientific practices; second, I conduct

interaction analysis of audiovisual data (Hall & Stevens, 2015) to examine roasters' practices *on their own terms*, including the teaching and learning of those practices; lastly, I develop a novel approach to studying learning-as-becoming by using membership categorization analysis (Sacks, 1992; Fitzgerald & Housley, 2015) to trace how apprentices move from novice to recognized expert status through talk-in-interaction. Together these methods link fine-grained analyses of moment-to-moment activity with coarse-grained representations of developmental trajectories across years.

Data gathered and preliminary findings

RQ1 - How do coffee roasters use science? Across hundreds of video episodes, I detail how roasters' practices align closely with the SEPs, yet also extend beyond them. Roasters, for instance, form hypotheses, select tools, decide on variables, collect data, and revise procedures based on new evidence. Crucially, these investigations integrate the human body as an analytic instrument: roasters smell, taste, touch, listen to, and visually inspect every coffee they work with.

RQ2 - How do roasters teach and learn their craft? Learning to roast coffee is organized through *apprenticeship* arrangements in which novices first work shoulder-to-shoulder with experienced roasters before starting independent work (Davey & Stevens, 2024). Early in their apprenticeships, teachers scaffold the learning environment so apprentices coordinate an increasing number of information sources, including the body's senses and workplace instruments. Once apprentices can produce "good-enough" roasts-after only three or four days of training-they are expected to work independently, make mistakes, and reflect on those experiences with colleagues. This brief and intense period of learning complicates fundamental research in the learning sciences that depicts apprenticeships as unfolding over many years.

RQ3 - How can these findings inform learning design? To translate insights from ethnographic fieldwork into the design of learning environments, I co-designed and co-taught a series of workshops with coffee professionals at Northwestern University. Students designed and carried out experiments to produce the "perfect cup," balancing quantifiable variables (temperature, brew time, extraction) with sensory ones (flavor, aroma, mouthfeel). Preliminary analyses of video (44 hours) and survey data show that students developed alternative conceptions of experimentation and began to view science as something they could do with their own bodies, not just with lab instruments. These workshops demonstrate how designing disciplinary practices based on extended fieldwork in understudied sites of science can reach similar outcomes valued by reforms.

Expected contributions

My dissertation documents scientific practices in non-laboratory workplaces. By constructing continuities between everyday and professional practice, my work challenges long-standing hierarchies about who counts as a scientist and what counts as science.

Second, my findings invert an assumption in science and science education: that sensory information is subjective and unreliable (e.g., Varelas et al., 2021). Coffee roasters demonstrate that perception-seeing, tasting, smelling, hearing, and feeling-is not opposed to scientific reasoning but fundamental to it (e.g., Jaber & Hammer, 2016). This part of my dissertation has implications for how science educators value bodily experiences and for designing learning environments that integrate the body with more normatively recognized scientific instruments.

Third, few learning sciences studies trace development across years in naturally occurring settings. By weaving video data and extended fieldwork together, my dissertation advances an approach for studying learning as it unfolds across extended timescales and sociotechnical systems.

Fourth, translating findings from roasteries to designed environments advances design principles for the learning sciences by arguing for designs that: treat the body as a legitimate way of knowing; encourage investigation of the everyday; and connect canonical concepts to the practices through which people make sense of the world.

Lastly, by showing how science exists and flourishes in workplaces, my research offers educators ways to help students see themselves as participants in science and seeks to reconnect disciplinary education with the diverse, lived experiences of people's everyday lives.

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Reimagining Early AI Literacies: A Care-Centered Co-Design Study with Parents and Young Children

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Abstract: While AI literacy has gained increasing attention in education, existing frameworks often privilege technical competencies and workforce preparation, overlooking the developmental, relational, and ethical concerns central to early childhood education. Grounded in socio-constructionist theories and an ethics-of-care lens, my dissertation reimagines AI literacy as a care-centered, co-constructed practice within family ecologies. This three-paper dissertation includes: (1) a care-centered critical review of AI literacy frameworks that advances a new conceptualization of early AI literacies in ECE; (2) a techno-ethnographic study investigating parents' beliefs, practices, and contexts shaping children's AI learning at home; and (3) participatory co-design with parent-child dyads to iteratively develop a Family AI Literacy Starter Kit. These studies bridge conceptual, perceptual, and design perspectives by identifying gaps in current frameworks, mapping parental engagement, and co-creating developmentally appropriate resources toward justice-oriented early AI literacies in families.

Background and goals

Artificial intelligence (AI) now permeates children's daily lives through voice assistants, "smart" toys, and algorithmic recommendation systems, shaping a generation of "AI natives" who integrate intelligent systems into their cognitive, social, and imaginative worlds from an early age (Druga et al., 2017; Yang, 2025). While these technologies can support personalization and creative exploration, they also raise concerns about privacy, bias, overreliance, and the erosion of relational, human-centered learning (Berson et al., 2025). These tensions have fueled growing interest in AI literacy, broadly defined as the competencies needed to understand, evaluate, and engage with AI responsibly and ethically (Long & Magerko, 2020; Mills et al., 2024). However, much of the work targeting younger learners either adapts frameworks designed for older students (e.g., Su & Yang, 2023; Williams et al., 2019) or offers little actionable guidance for practice (e.g., Luo et al., 2024), often overemphasizing technical skills and workforce preparation while overlooking developmental appropriateness and the need for situated, meaningful sense-making in early childhood. Families, as a crucial "third place" for early AI learning, bring cultural knowledge, values, and relational practices that shape how children encounter and interpret AI (Druga et al., 2022). Yet, their lived expertise and everyday practices remain underexamined.

To address these gaps, this dissertation centers on the lived expertise of parents and children and adopts a co-design approach to surface meanings, practices, and needs that current top-down AI literacy frameworks often overlook. It begins with a critical review that introduces an ethics-of-care lens and proposes three interconnected framings of early AI literacies—functional, relational, and critical. Building on this foundation, Phase 2 adopts a techno-ethnographic approach to investigate how families negotiate and enact AI literacy in everyday routines and what factors shape these practices. Phase 3 then moves from perception to practice by engaging parent-child dyads as co-designers of a developmentally appropriate, low-burden Family AI Literacy Starter Kit for home use. Together, these phases address the overarching research question: *How are early AI literacies conceptualized, negotiated, and enacted within family contexts?*

Theoretical framework

This dissertation is grounded in an integrated theoretical framework that bridges sociocultural and constructionist perspectives with contemporary framings of AI literacies. Rooted in the view that knowledge is co-constructed through making, dialogue, and participation in culturally meaningful activity (Vygotsky, 1978; Papert, 1980), this study understands learning as both a creative and relational process. Drawing on Freirean literacy and multiliteracies (Freire, 1970; New London Group, 1996), the study views AI literacies as multimodal, socially situated, and emancipatory practices that empower families to "read and rewrite" the technological world. Extending Kafai et al.'s (2020) computational literacies framings, the dissertation adopts three complementary lenses—functional, relational, and critical—to examine how learners understand how AI works, engage with it in human-AI and human-human relationships, and reflect on its ethical and societal implications. By foregrounding attentiveness to needs, responsibility in action, competence in practice, and responsiveness to feedback, the ethics of care framework (Tronto, 2013; Porayska-Pomsta & Asare, 2025) repositions AI literacies as a socially situated

and relational practice. Together, these integrated perspectives ground the study's methodological, co-design, and analytic choices, enabling a care-centered approach.

Methodology

This dissertation employs a three-phase, mixed-methods design. Phase 1 is a critical review (Paré et al., 2015) of AI literacy frameworks and review papers from 2015 to 2025, synthesizing their theoretical and pedagogical implications for early childhood education. Phase 2 uses a techno-ethnographic mixed-methods approach (Pink et al., 2016; Kien, 2008), combining an online survey and home-based inquiry with 8-10 parent-child dyads to capture both the breadth and depth of parents' perspectives and mediation practices on family technology use. Phase 3 adopts a participatory co-design methodology (DiSalvo et al., 2017), engaging parent-child dyads in iterative, home-based activities that explore adaptable AI literacy themes. In this way, design functions as both a method for generating data and a means of refining practical supports.

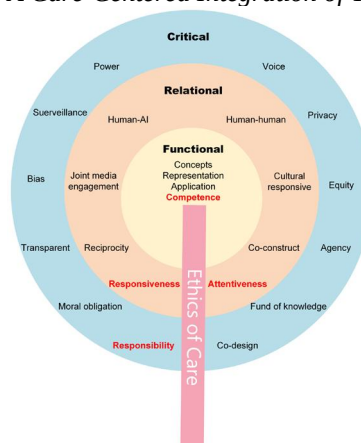
Data analysis follows an iterative, multi-layered strategy. In Phase 1, framework and thematic synthesis (Booth et al., 2016) map existing frameworks onto the functional, relational, and critical framings and interpret them through the ethics-of-care lens. In Phases 2, survey data are analyzed using descriptive and basic inferential statistics. Qualitative data, including open-ended survey responses, home visit notes, recordings, and interview transcripts, are analyzed using reflexive thematic analysis and in vivo coding (Braun & Clarke, 2006). In Phase 3, family-recorded interactions are analyzed through discourse and interactional analysis (Jordan & Henderson, 1995; Goodwin, 2000), triangulated with weekly reflections, interviews, fieldnotes, family-created artifacts, and researcher memos through artifact and content analysis. Across all three phases, the early AI literacies framework is continuously refined through movement between theory, empirical inquiry, and design.

Preliminary findings

Phase 1 resulted in an Early AI Literacies Framework as shown in Figure 1. The functional framing, at the core, defines why AI is taught and which concepts, tools, and representations are emphasized. Surrounding this core, relational practices make functional ideas learnable and create space for critical reflection. The critical framing forms the outer layer, connecting to broader sociocultural and ethical contexts. An ethics of care may serve as connective tissue guiding implementation: it grounds the functional in developmental sensitivity, deepens the relational through empathy and reciprocity, and sharpens the critical through moral accountability and justice.

Phase 2 and Phase 3 data collection will be conducted from Spring 2026 to Fall 2027, primarily in participants' homes. Participants will be recruited through a Chinese school, a Girl Scout troop, a public school, and local libraries to capture a wide range of family experiences and perspectives on AI.

Figure 1
A Care-Centered Integration of Three Framings in Early AI Literacies



Expected contributions

This dissertation bridges theory, perception, and practice to reimagine AI literacy as a care-centered, co-constructed process within family ecologies. It aims to contribute: (1) a theoretically grounded early AI literacies framework that provides conceptual and practical guidance for learning and design; (2) empirical insights into how parents conceptualize and negotiate AI in daily life; and (3) evidence of family co-learning through participatory co-design, resulting in practical, low-barrier tools such as the Family AI Literacy Starter Kit.

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Reimagining Mathematics through Children's Play with Video Data

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Abstract: This two-cycle participatory design study (PDR) develops a novel methodology for including young children in design processes for their own mathematical learning: “playing” with classroom video data. I partner with twelve K-2 students from one school, integrating student-teacher-researcher video analysis and co-design as a mechanism for reimagining participation possibilities in mathematics classrooms. I examine implications of play-based methodological spaces for children’s recognition—contributing new understandings of how young children, particularly with multiply marginalized identities, can inform theories of learning, and transform power-laden discourses of mathematical competence in schools.

Background

Power-laden discourses on *what counts as mathematical participation* remain deeply entrenched within K-12 mathematics, including accountability logics aimed at defining “failing” schools and students and replacing young children’s curiosities with short-term skills to optimize performance gains (Miller & Almon, 2009). Moreover, such ideologies intersect with systems of oppression to reify *who is recognized as mathematically competent* by stratifying racially minoritized and economically disadvantaged children into low ability groups, thereby limiting their access to opportunities to learn, including advanced mathematics coursework in future grades (Cobb & Russell, 2015). In contrast, humanizing mathematics advocates for a joint pedagogy *with* students—produced for the purpose of “naming” and “transforming” extant systems (Freire, 1970). Efforts to promote sociopolitical consciousness are well documented in secondary settings. Yet, methods for advancing K-2 students’ liberatory action remain undertheorized, as schools and researchers are forewarned that children’s “nonsensical,” countersocial thought patterns render them unreliable informants (Yoon & Templeton, 2019). Indeed, Wohlwend (2018) describes children’s play as an “untidy, just-fine-as-is literacy that young children choose and use to make sense of their worlds” (p. 301). However, such forms of sensemaking are rarely taken up as relevant to curricular priorities, and, as a result, children’s subversive provocations more often go unheard (Yoon & Templeton, 2019). The result is that children have been historically excluded from weighing in on decisions about their mathematical learning. To extend humanizing mathematics to K-2 students, the field needs methodological innovation to recognize children’s imaginative expertise and analytic modes.

Objectives and research questions

This PDR study develops one such novel methodology: “playing” with classroom video data alongside young children. I conjecture that by recognizing children’s playful video noticings, educators may come to recognize their students’ mathematical competencies in transformative ways. I examine the following research questions:

1. How do K-2 students and their teachers analyze video data of their mathematical learning?
2. How do K-2 students leverage video data to reimagine mathematical learning designs?
3. What are the implications of K-2 students’ reimagined learning environments for their participation possibilities and recognition as mathematically competent in mathematics classrooms?

Methodologies and data gathered

Informed by critical childhood studies (Yoon & Templeton, 2019) and a situative theory of learning (Lave & Wenger, 1991), my PDR study partners with three teachers and four K-2 students from each of their classrooms (twelve students total), whom I selected based on their positioning as “less mathematically competent.” To establish a starting point for co-design, I filmed each classroom’s “business-as-usual” mathematics instruction, from which I selected “critical events” (Jordan & Henderson, 1995) of each focal child’s normative participation possibilities. Each child and teacher discussed the identified clips in multiple settings: child-researcher, teacher-researcher, child-teacher-researcher, and peer-teacher-researcher. Using their video noticings as a design “springboard,” students then engaged in peer-teacher-researcher co-design sessions to reimagine existing design features of their school’s mathematics curriculum, including tools and participation structures (Sandoval, 2014), which their teachers and I then formalized as lesson plans. Finally, each teacher enacted these lessons in their classrooms, which I again filmed. I collected additional data sources, including pre- and post-interviews.

To answer Research Question 1, I will use emergent coding (Strauss & Corbin, 1990) to compare actors’ video noticings, and to consider what students and teachers can potentially learn from one another when attending

to different aspects of mathematical participation. To respond to Research Question 2, I will apply interaction analysis (Jordan & Henderson, 2019) to salient “ethnographic chunks” (p. 57) from video-recorded co-viewing and co-design sessions to trace the function of such methodological spaces for facilitating adult-child co-design of school mathematics. Lastly, to answer Research Question 3, I will apply pattern codes from RQ1 to video-recorded co-design sessions. I will identify relationships between children’s video noticings and design features (Sandoval, 2014). I will then identify salient classroom lesson episodes that illustrate these design features in context to examine the interactional conditions under which students are recognized as mathematically competent (or not) in classrooms. I will nuance findings with pre- and post-interviews for potential shifts in students’ and teachers’ conceptions of what it means to be mathematical.

Expected findings and contributions

I used conjecture mapping (Sandoval, 2014) to inform learning designs and anticipated outcomes. At a high level, I conjecture that learning designs guided by children’s video noticings will expand what it means to be mathematical. Namely, my pilot study found that as children analyzed and shared their “favorite parts” of classroom video, they created playful mathematical plotlines (Vescio, 2025), which challenged teachers’ and researchers’ noticings of the same data (Vescio & Gresalfi, under review). The present study extends this work by amplifying new subject-subject relations in design, in turn opening new subject-object relations (Bang & Vossoughi, 2016). By “seeing” mathematics with children, I posit that educators may also “see” their students and the discipline in new ways and, in so doing, redefine mathematical participation and competence.

Ultimately, I expect that children’s reimagined designs will advance child-informed theories for supporting mathematical learning. Thus, this study attends to calls within the learning sciences for participatory research that directly engages with theories of learning. For instance, this dissertation could offer revised processes of conjecture mapping (Sandoval, 2014) for relating processes of partnering to designs and outcomes for learning. Relatedly, and perhaps most obviously, I aim to refine video viewing with children as a mechanism for wandering outside of the typical research gaze and for understanding how young children conceptualize their own mathematical learning (Yoon & Templeton, 2019). Therefore, this PDR study will also contribute empirical narratives of humanizing mathematics with K-2 students (Freire, 1970), while disrupting deficit framings of historically marginalized learners—instead capturing their curiosity, creativity, and playfulness.

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Implementing and Advancing Computational Action in the Context of a Near-Peer Mentorship Model within a Maker Camp

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Abstract: This three-part dissertation investigates the emergent theory of computational action within the context of a multi-year summer maker camp. The first two papers define the first two pillars of computational action and investigate how they can be influenced by the design of informal maker camps. The final paper will investigate the implementation of a near-peer mentorship model within a camp, to present case studies of each of the middle-high school mentors' experiences. It will investigate the design decisions within the mentorship model that influenced mentors' computational action. This final paper will extend the theory of computational action and how it can be utilized to develop a sustainable community of practice where the ownership of computational learning shifts towards the learners' communities.

Background

Scholars and practitioners over the last few decades have been preparing learners to navigate the rapidly developing, technologically driven world (Guzdial, 2022). In this effort, Wing (2006) proposed computational thinking, which despite its significant influence, does not consider how students' computational learning should enable them to utilize technology meaningfully towards their lives and communities (Kafai & Proctor, 2022).

In response, computational action (Tissenbaum et al., 2019) focuses on learners using and creating with computing to develop their computational identities and digital empowerment. Following a constructionist approach (Papert, 1980), computational action suggests that students should create artifacts with their identities and communities in mind, and thus, engage with computing as producers in personally relevant ways.

My dissertation is embedded in the computational action-driven Connected Spaces (C/S) summer maker camps, where middle-high school participants created projects with computing. I investigate their experiences to understand the influence of camp design on their computational identities and digital empowerment.

Goals of the research

My dissertation will consist of three individual journal articles, exploring students' experiences of two pillars of computational action – computational identity and digital empowerment – and subsequently inform the theory through my findings. Table 1 outlines the first two articles, which will investigate C/S camp design elements from 2022, 2023, and 2024 camps. For the doctoral consortium, I will primarily discuss the third article, and its potential contribution to my dissertation, to computational action, and how it will inform my research trajectory.

Table 1

Outline of the First Two Articles in the Three-part Dissertation

Research Questions	Context & methods	Preliminary/anticipated findings
- How can computational identity be observed in maker education? - How can mind maps and brainstorming activities in maker curricula contribute to students' computational identities?	Qualitative analysis of 2022 participants' final interviews, maker artifacts, room video/audio recordings.	- C/S participation impacted students' computational identity factors – interest, usefulness, confidence, competence, recognition, perception of community, and reduced external resistance. - Mind mapping and brainstorming activities supported students' making towards prior interests, communities, and their future communities; the maker community impacted students' experiences.
- How can digital empowerment be observed within students' maker camp participation? - How can curriculum, activities, and environments be designed to enable makers' digital empowerment?	Design-based research through qualitative analysis of 2022-2024 curriculum, post-camp interviews, student artifacts.	- Themes of design alterations emerged across the years that contributed to the dimensions of digital empowerment – impact, meaningfulness, choice/agency, critical consciousness, community of support, and efficacy. - Students' digital empowerment experiences were influenced by design decisions for instance, conversation with familiar community leader led to critically conscious and impactful projects.

In the third paper, I will investigate the unique near-peer mentorship model implemented in the C/S 2025 camp. I will investigate the computational identity and digital empowerment experiences of the four middle-high school mentors as they assisted their middle-school peers towards making mobile apps, and as they worked with the research team to co-design and facilitate activities that enabled mentees to design meaningful apps. I will answer the research questions: 1) how did the experience of being near peer mentors and co-designers influence mentors' computational identities and digital empowerment? and 2) what were some effective methods of positioning students as near-peer mentors in a maker camp?

Methodology and expected findings

At the end of the 2025 camp, I conducted and recorded semi-structured interviews with the four near-peer mentors. I audio-recorded daily focus-group debrief sessions where the mentors and I discussed the activities of the day, what went well, and how we could prepare for the upcoming days. The mentors participated in a day-long training session before the camp, which was recorded; each camp day was recorded, and student and mentor artifacts were visually documented. Conducting holistic coding (Saldaña, 2021) of this data I will create a quintain case study (Stake, 2005), to present individual case studies of each mentor's experience. These individual cases will inform a collective case study of the mentorship model to investigate the curriculum design decisions that supported mentors' computational action experiences. The theories of computational identity and digital empowerment developed in the first two papers will inform the coding and themes within this final study.

Each of the four mentors demonstrated unique mentorship approaches. For instance, Jade took on the role of the lead mentor among the group, Jasmine was intent on establishing sense of belonging and fun within the camp, Aria was the youngest and had no previous experience with C/S, and Talia often took on the role of a student rather than a mentor. By presenting individual case studies of the mentors, I will investigate how these varied lived experiences and approaches influenced their computational identity and digital empowerment in the camp. Through these individual case studies, I anticipate finding themes about the mentorship model that influenced the mentors' experiences, such as working as a team of four, co-designing curriculum that enabled students to make meaningful apps, and participating in the mentor training. I will use these themes to present features of the mentorship model that impacted the mentors' computational action experiences.

Contributions

This dissertation's contributions will be two-fold: 1) developing and establishing two pillars of the computational action theory (computational identity and digital empowerment), and 2) investigating the multi-year implementation of the computational action driven design of the Connected Spaces summer camp. The third article in particular will investigate computational action within the context of a constructionist mentorship model. While the theory of computational action emphasizes students becoming producers of and with technology, the third article extends this approach, to understand how computational action can be implemented to establish a sense of belonging and sustainable community of practice among new and returning students. In this manner, this article seeks to shift the ownership of computational learning and creation towards learners through a computational action driven near-peer mentorship model.

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Designing AI-mediated Feedback for Knowledge Building: Comparative Inquiry and Design-Based Research

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Abstract: This dissertation examines how generative AI mediates feedback and shapes knowledge-building processes in higher education. Through three connected studies, it analyzes how students and expert educators engage with AI in academic writing and collaborative inquiry and how these interactions influence their epistemic framing. Grounded in dialogic feedback frameworks and Knowledge Building theory, the research identifies differences in the structure and processes of AI-supported knowledge-building dialogue and explores how generative AI can be integrated into project-based learning environments that support collaborative idea improvement. Together, the findings reconceptualize feedback in AI contexts and provide analytic and design insights for fostering epistemic cognition and sustained knowledge building.

Background

Recent work has reframed feedback from information transmission to a process-oriented practice grounded in interaction and meaning-making (Winstone et al., 2022). In this view, feedback emerges through iterative dialogue in which learners interpret, negotiate, and use information. This perspective aligns with dialogism, which conceptualizes knowledge as emerging through interaction among people and artifacts (Bakhtin, 1981). Generative AI introduces a new form of participation in this process by providing immediate and elaborated responses, yet prior research shows that while learners value its efficiency, they question its contextual and relational sensitivity (Otaki et al., 2026).

To examine how AI can support deeper inquiry, this study adopts Knowledge Building (KB) theory, which frames learning as the collaborative improvement of ideas (Bereiter & Scardamalia, 2014). From this perspective, productive inquiry is sustained through discourse practices such as proposing ideas, evaluating their promise, and engaging in reflective dialogue (Bereiter & Scardamalia, 2016). However, recent work suggests that AI-generated responses require critical scrutiny, as they may be incomplete or weakly reasoned (Chen & Zhu, 2023). This raises a central question: how does generative AI shape the interpretive work and discourse processes that sustain knowledge building?

Goals of the research

First, the dissertation examines how students and expert educators differ in their epistemic framing when engaging with GenAI, focusing on how expectations and interpretive practices shape dialogic feedback. Next, it analyzes how these groups differ in the structure and processes of AI-mediated, knowledge-building dialogue during complex problem-solving. Finally, it explores how GenAI can be integrated into project-based, KB-oriented courses and identifies design principles for learning environments that support sustained collaborative inquiry. Together, these goals position GenAI as a mediational resource for sustaining epistemic agency and collective idea improvement in knowledge-building environments.

Methodology

The dissertation employs complementary methods across three studies. Study 1 adopts a quantitative ethnography approach (Shaffer, 2017), combining focus groups with students and educators to examine differences in epistemic framing when engaging with AI feedback. Study 2 investigates AI-mediated collaborative problem-solving using discourse data from student pairs and expert educators, analyzing how ideas are constructed and developed through interaction. Study 3 is a design-based research intervention embedded in a semester-long undergraduate course, tracing how students iteratively engage with AI in sustained inquiry. Across studies, network-based approaches, including Epistemic Network Analysis and KBDeX (Oshima et al., 2012), are used to model the structure and evolution of discourse. This multi-method design enables a process-oriented understanding of how AI participates in knowledge-building activity.

Data and preliminary findings

Study 1 (Otaki et al., 2026) revealed contrasting epistemic framings of AI-generated feedback. Students emphasized efficiency, clarity, and emotional safety, positioning AI as a low-risk resource for initial improvements. In contrast, educators foregrounded contextual understanding, relational continuity, and ethical trust, highlighting AI's limitations in addressing nuanced and affective aspects of feedback. These differences indicate that AI is interpreted through distinct epistemic lenses, shaping how it is incorporated into feedback practices.

Study 2 extended this analysis to collaborative problem solving. Students frequently organized their reasoning around AI-generated ideas, producing coherent but externally guided arguments. Their discourse networks showed limited integration between evaluation, problem reframing, and meta-dialogue. In contrast, expert educators engaged in iterative cycles of defining, reframing, and evaluating problems, using AI selectively to test and refine ideas. Their discourse exhibited more integrated epistemic structures linking evidence, critique, and reflection. These findings suggest that differences in epistemic framing shape not only how AI is used, but also how inquiry is organized.

Study 3 examines how generative AI can be integrated into sustained inquiry within a KB-oriented course. Preliminary observations indicate that productive use of AI emerges when it is positioned as one resource among others, rather than as a primary driver of activity. This highlights the importance of designing learning environments that support the coordination of AI, peer interaction, and instructional scaffolding.

Expected contributions

This dissertation contributes to a deeper understanding of how GenAI mediates dialogic feedback and knowledge-building practices in higher education. By examining students and expert educators across writing, argumentation, and project-based inquiry, it clarifies the epistemic differences that shape human–AI interaction and demonstrates how these differences influence the structure and depth of knowledge-building dialogue. In the Doctoral Consortium, I hope to contribute to discussion from a theoretically grounded perspective on feedback and KB theory, methodological perspective in network analyses including ENA and KBDeX, and practical insights gained through facilitating project-based learning environments, in ways that help deepen our community's collective inquiry.

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Machine Epistemology as a Lens to AI Ethics Education: Forging Early Adolescents' Epistemic Insights through Constructionist Learning in Upper Primary and Junior Schools

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Abstract: K-12 AI ethics education often focuses on topics including bias or privacy but rarely addresses how AI systems generate decisions in ways that create ethical challenges. This gap limits students' ability to grasp the underlying sources of these ethical issues. Drawing on machine epistemology, which conceptualizes AI as an epistemic agent with distinctive ways of knowing and reasoning, this design-based research investigates how constructionist learning can foster students' epistemic and ethical insights about AI. The study was implemented during a two-week summer program with thirty-four upper primary and junior secondary students in Hong Kong, integrating hands-on AI activities with guided reflection. Data from classroom interactions, interviews, and student artifacts were analyzed using inductive thematic coding and a deductive framework based on levels of epistemic insight. Preliminary findings show a developmental progression from surface-level descriptions of AI behavior toward emerging conceptual and critical reasoning, alongside persistent misconceptions about AI's reasoning processes.

Goals of the research

This dissertation aims to design and investigate a constructionist learning environment that supports early adolescents' epistemic insights about AI. Specifically, it seeks to:

- Develop and refine a constructionist pedagogical framework that integrates hands-on AI activities with guided reflection to support students' epistemic and ethical explorations of AI.
- Examine how students' epistemic insights develop as they engage with an AI ethics curriculum.
- Identify recurring misconceptions and reasoning challenges that arise during the learning process.

Background of the project

The rise of AI brings innovation alongside ethical concerns related to bias, transparency, and autonomy (Adams et al., 2023). Addressing these issues requires understanding how AI *knows*—how systems generate, justify, and sometimes fail in producing knowledge. Yet, most K-12 AI ethics interventions focus on what AI does rather than on the epistemic roots of ethical challenges (Wiese et al., 2025). *Machine epistemology* offers a productive lens for deepening students' reasoning by treating AI systems as epistemic agents that generate knowledge through statistical inference, pattern recognition, and probabilistic reasoning. This perspective helps students understand why AI systems may be biased, brittle, or opaque (Barelli et al., 2024).

Drawing on *epistemic insight* (Billingsley et al., 2018) as a theoretical anchor, this study examines learners' abilities to reflect on the nature and limits of AI knowledge, and how these epistemic processes connect to real-world consequences. To address this gap in K-12 AI ethics education, we leverage constructionism (Papert, 1980). While constructionist approaches support engagement, scholars highlight that careful scaffolding is needed to foreground AI's epistemic mechanisms and connect technical processes to ethical reasoning (Lin et al., 2025). Against this backdrop, this study examines how early adolescents develop epistemic insight within a constructionist AI ethics curriculum and identifies the challenges they encounter.

Methodology

This study employed a design-based research (DBR) approach to iteratively design and investigate a constructionist learning environment that supports early adolescents' epistemic insights about AI (Barab & Squire, 2004). The eight-session summer camp involved 34 upper-primary and junior secondary students in Hong Kong and integrated constructionist activities—such as training machine learning (ML) models with Google Teachable Machine (GTM) and creating poems with generative AI—paired with guided reflection.

An embedded qualitative case study of 11 focal students captured diverse engagement and achievement profiles. Data sources included video-recorded lessons, fieldnotes, student artifacts, semi-structured interviews,

and teacher debriefings. Analysis combined *inductive thematic coding* with *deductive coding* of epistemic levels (descriptive, conceptual, critical/integrative) to trace emergent dimensions of students' epistemic reflections and assess the varying degrees of sophistication in their reasoning. All procedures followed ethical protocols with informed consent.

Preliminary findings

Student development in epistemic insight

Analysis revealed a three-level developmental trajectory of students' epistemic insights. At the *descriptive* level, students focused on surface features of AI behavior and often relied on literal or anthropomorphic explanations. At the *conceptual* level, they recognized AI's reliance on patterns, data quality, and human input, and began identifying structural limits in machine reasoning. At the *critical/integrative* level, some students coordinated multiple epistemic considerations—such as ML's probabilistic reasoning, paradigm differences, and ethical consequences—to evaluate how and why AI produces particular outcomes.

Across levels, students' reasoning clustered into four dimensions: the nature of AI knowledge, human involvement in AI knowing, limits of AI reasoning, and comparative reasoning across AI paradigms. For example, in a banana-classification task using GTM, descriptive reasoning centered on visible actions ("we give AI pictures so it can learn"), conceptual reasoning involved data-pattern explanations and recognition of probability, and critical/integrative reasoning connected misclassification to broader ethical issues such as fairness and bias.

Learning challenges and difficulties

Despite growth, students showed persistent challenges. Some exhibited *cross-level struggles*, recognizing the limits of simple data labelling yet defaulting to fixed positions without weighing context or trade-offs. Others demonstrated *dimensional struggles*, conflating ML and generative AI as generic "pattern-matching" systems or treating prompting as equivalent to training. These flattened distinctions constrained their ethical reasoning and signaled the need for explicit scaffolds that foreground paradigm differences and epistemic uncertainty.

Expected contributions

Theoretically, this study advances an epistemic perspective on AI ethics by positioning epistemic insight as a core lens for understanding how young learners interpret AI's knowledge sources, reasoning processes, and limitations. It also offers fine-grained evidence of how such insight develops across AI paradigms. Practically, it generates design principles for constructionist AI learning environments that foreground epistemic mechanisms—rather than technical procedures alone—by leveraging students' naïve ideas as productive resources and integrating hands-on making with structured reflection. These contributions support the design of equitable, human-centered AI literacy practices that strengthen learners' epistemic agency and ethical reasoning when engaging with issues such as opacity, authorship, and human-AI co-agency.

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Learning with the Seasons: Lands and Waters-Based Temporality in Learning and Mobility

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Abstract: The natural world is an active learning environment and provides a discursive context in which humans make meaning with one another. One way to experience collective meaning making in this learning environment is through group forest walks by walking, storying, and mapping lands and waters which Marin et al. (2020) refer to as ‘learning on-the-move.’ Water specifically creates a visual and sonic landscape where humans shift their attention and movement in response to it. This micro-ethnographic study uses video-based interaction analysis to understand the moment-to-moment unfolding of learning by paying attention to the semiotic layering of talk and movement while on forest walks during monsoon, winter, and summer seasons. The objective of this study is to explore how water shapes learning processes and learning environments as participants attend to relations with water through shifts in their movements, bodies, and conversation in relation to water while on forest walks. Broader implications of this study include using Indigenous ways of knowing to resist erasure and respond to increasing privatization of traditional lands and waters in the Global South.

Project background

Water mediates and organizes human relationships to the natural world, teaching us to attune to the sights, sounds, and seasonal rhythms of nature. It is also an active subject in organizing human attention and mobility, creating opportunities for humans to follow its sound in search for it, developing sensorimotor skills to swim, float, or traverse it. Knowing about water, however, is different from water-based ways of knowing. Indigenous communities have long relied on water-based ways of knowing about the world, developing kinship with waters and water life, and adapting to change as relationships to water change through the seasons. Alongside these longstanding practices are also issues of climate-change, environmental (in)justice, and increasing privatization of the lands and waters that Indigenous peoples call their home. I do not imagine the natural world to be a pristine, uncontested concept, but recognize that it is saturated with ongoing political struggle, colonial pasts and presents, and are sites where Indigenous futures are made.

This dissertation examines how water-based ways of knowing organize human attention, coordination, and mobility, and contribute to resilient Indigenous communities, helping communities adapt to the seasonal rhythms of nature, especially in the context of ongoing dispossession of traditional lands and privatization of forests for industrialization. Participants of this ethnographic study are multigenerational members of the Siddi and Halakki communities, two Indigenous communities that share space in the Uttara Kannada region of southern India where public forests are becoming increasingly privatized and local paper and cardboard processing mills are polluting rivers and waterways. Amidst this violence to their environment, this study asks how learning with water can be a liberatory pedagogical tool, helping communities re-member relationships with lands and waters that are attempted to be dismembered by neoliberal development.

Goals of the research

Mapping, storying, and walking with land and water are longstanding Indigenous activities, shaping understandings of water through sensory experience and the production of land- and water-based artifacts. Forest walks are a focal activity in this study during which Siddi and Halakki members walk through land, interact with the sights and sounds of water, and build meaning by conversing with one another, centering land in their re-storying. These forest walks bring together multigenerational participation in cultural and linguistic practices, connecting the often-separated worlds of adults and children (Rogoff, 2003). As such, a variety of ages are represented in Siddi and Halakki group forest walks.

Specific research questions during these activities include:

1. How does water organize attention, storying, and mobility for learning?
2. How do Siddi and Halakki members on forest walks learn to “read” water (Bruce et al., 2023) while on-the-move? What does water-based literacy and reading look like?
3. How do lands and waters guide human attention in multiple and simultaneous ways (Rogoff & Correa-Chavez, 2005) while on forest walks?

Methodology

The goal of my project is to methodologically unsettle normative time-space ontologies and human-nature divides (Bang & Marin, 2015) in the learning sciences. This requires attending to onto-epistemic heterogeneity (Warren et al., 2020) and recognizing Indigenous ways of knowing (Smith, 2012). Research methods are powerful when they can foreground ways of knowing and making-meaning about the world that are often rendered invisible. This dissertation uses participant-generated video-based ethnography using interaction analysis traditions (Jordan & Henderson, 1995; Derry et al., 2010), viewing humans and water as co-participants in an activity. Although any person may choose to document their forest walk, this study focuses on subjects age 7 or above. This is to ensure that each participant has adequate ability to walk, operate a camera, and converse with other subjects.

Walking through land presents opportunities for water to organize human attention and guide movement across multiple sensory dimensions. “Talk, gesture, and movement form a layering of semiotic resources that provide opportunities for the transformation of action and learning” (Goodwin, 2013; Melander, 2012). Forest walks offer an opportunity to observe how people relate to one another, make relations with the natural world, move through the world with their bodies, and organize attention while jointly participating in an activity (Marin & Bang, 2018; Marin, 2020). I do this over several seasons to understand how individuals change through time while practices endure through time (Esmonde, 2016). The seasons and seasonality are Indigenous, land-based forms of temporality, allowing us to understand human development along non-linear temporal scales. Together, forest walks as a research methodology offer us a glimpse into the way that relationality, place, and movement shape human development over several seasons.

Data corpus and preliminary findings

My fieldwork is taking place over a period of 9 months, split into three rounds of data collection and organized by the seasons of the Uttara Kannada region. During each round, I have collected video data of forest walks and have generated weekly fieldnotes from daily jottings, comprising one complete ‘set’ of data. As I have already begun my fieldwork, I am completing the forest round of data collection from Nov 2025 to Feb 2026, comprising the winter season with eight sets of data. My second round of data collection will be the summer season from April 2026 to June 2026 where I anticipate collecting eight sets of data. The final season will be the monsoon from July 2026 to September 2026 where I anticipate collecting eight more sets of data, totaling 24 sets of data for saturation.

Videos from forest walks are content-logged and transcribed three-dimensionally (Mondada, 2016; Rymes, 2015) to include human action, speech, and the sonic landscape with sights and sounds of the natural world such as water, birds, wind, and trees. Birds chirping, the wind blowing, leaves rustling, and water flowing are all ways that nature provides a discursive context for meaning making (Ingold, 2007, 2010; Ingold & Vergunst, 2008). Codes generated from transcript analysis were raised into categories and themes using a grounded theory approach (Charmaz, 2001).

A summary of important findings thus far includes: (1) Water plays a central role in organizing sense-making as participants used water-forms such as streams and lakes to make sense of other life such as deer frequenting a place. Furthermore, water was used to make sense of social structures such as caste and social history of place as participants inquired about the unequal access to water and water-infrastructure for activities such as farming, which is predominantly controlled by elite/upper-caste landowners. (2) Water organizes movement as participants make bodily movements in relation to the sound of water while on forest walks.

Expected contributions

Walking and storying the land on repeated forest walks offer new opportunities for encounter and making relations with land and water (Marin & Bang, 2018) and demonstrate how water shapes learning environments by directing the attention and movement of people through land. However, such opportunities for place and sense-making cannot exist in isolation from histories of oppression and violence such as private property, race, and in South Asian contexts, caste. Designing forest walks as a learning activity by attending to place stories creates “an opening to grappling with the ethical potentialities of plural more-than-human worldings where both the human and more-than-human shape and are shaped by political, economic, and cultural forces.” (Kirskey & Helmreich, 2010; p. 545; Nxumalo, 2016, p. 644). My work with the Siddi and Halakki communities not only includes questions about how the natural world organizes the attention and movement of learners on forest walks, but how knowledge about the history of the land and its people can make these walks a politically liberatory act. In this way, place is not a neutral landscape or a blank slate on which relationships are made but an event (Massey, 2005) that has many stories in the making.

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Designing for Radical Imagination: Teacher Speculation, Collective Futuring, and Social Praxis

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Abstract: This dissertation examines how learning environments for K–12 teachers can be designed to foster radical imagination through speculative activities. Data were collected across four mathematics teacher education courses at a large U.S. university, and a core group of participants joined a collective inquiry group to explore how to create elements of their imagined futures in their present classrooms. Anticipated findings include design principles that promote radical imagination among teachers, the forms of collectivity that emerged within the inquiry group, and the focal points of teachers’ radical reimaginings of mathematics teaching and schooling.

Goals and background

The aim of the present study is to design and analyze various learning environments where K-12 mathematics teachers engage with radical imagination and collective futuring. Radical imagination is the act of crafting new possibilities for the future that are not constrained by systemic structures or the status-quo of the “possible.” Neoliberal logics typically locate the source and the solution of (educational) problems in the individual, and thus leave unexamined the structures and collective action needed to challenge systemic forces (Barker & Scheele, 2016). Radical imagination by collectives of teachers, and fostering that imagination through teacher education, intentionally endeavors to rewrite the dominant power structures that maintain the status quo of capitalist, white supremacist, and settler-colonial systems in education. This is of particular importance in mathematics teacher education, as the discipline of mathematics is viewed as politically neutral (Harouni, 2015), which ultimately serves to maintain hierarchies of power and underscores patriarchal ways of knowing (Hesse-Biber et al., 2004). Specifically, this research explores: (1) How can learning environments in mathematics education be designed to contribute to the development of teacher radical imagination, and (2) How can collectives of teachers act on their imagined future in the present?

Study design and methods

This qualitative dissertation comprises of three studies with data collected in the mathematics teacher education department of a large U.S. university. Across the studies, 40 participants consented to include their data for analysis, and I collected over 170 speculative writing artifacts. In study 1, I incorporated identical speculative writing prompts into four education courses to explore the content of teachers’ imagined futures. The participants’ written responses to the speculative prompts will be analyzed through thematic narrative analysis following the methods from my pilot study (Claiborne & Cordero-Siy, 2024).

Study 2 is a design-based research study (DBR; The Design-Based Research Collective, 2003) surrounding the co-design of an elementary math methods course to foster radical imagination. Consistent with recent critical approaches in DBR (e.g. Bang & Vossoughi, 2016), I apply this theoretical and analytic approach to analyze how the learning environment enabled the expression of radical imagination by the prospective elementary teachers. I will analyze their speculative writing, course artifacts, classroom recordings, individual interviews, and co-design meetings with my faculty co-instructor attending to how their expressions enable or constrain the transformation of educational institutions.

For Study 3, I facilitated collective inquiry groups for participants from the courses to collectively act on their imagined futures. Aligned with participatory methods (Bang & Vossoughi, 2016), the teachers were co-designers and researchers, shaping the aims of each session and leading their own speculative experiments to take up their futures in their classrooms and communities. The analysis of the video recording and artifact data from the sessions is one of the foci I will address during the consortium.

Preliminary findings

Ongoing analysis of the speculative writing across the courses shows that teachers engaged directly with key issues they encounter in their daily work and imagined alternatives to constraining schooling practices. The teachers’ writing revised and expanded the meaning of school mathematics and challenged longstanding norms such as ability grouping, tracking, mathematics as apolitical, and the boundaries between school mathematics and youths’ lives and communities. In the design of the elementary math methods course, my co-designer and I have

grappled with design considerations together, memoing about the tensions the teachers face between imagining new practices and supporting youth within the current U.S. system. In the speculative workshops, the teachers have identified various areas of their daily practice in schools to implement their imagined futures, and the cross-grade, cross-subject collectivity emerging among them offers rich material for analysis and discussion.

Expected contributions

This dissertation will offer theoretical and practical contributions. Through the analysis of multiple learning environments where teachers engage in speculative writing, I will posit empirically-informed design principles to promote radical imagination among teachers. The aim is to inform practice in teacher education and contribute to the growing field of speculative pedagogies for and with teachers. Further, through the collective inquiry groups, my aim is to share considerations for participatory futuring by teachers towards enacting justice-driven practices in the present (within and/or beyond schools), and to highlight the collective nature of these futuring practices.

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Epistemic Engagement with Epistemically Elastic Content

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Abstract: This dissertation aims to investigate how non-experts engage with content that presents itself as apparently sound and structurally logical, but dismisses larger contextual factors that make it problematic. I describe how I take a mixed-methods approach to propose a study to be implemented with youth and young adults in Rio de Janeiro, Brazil. Additionally, I present preliminary findings that result from the first phase of this research project.

Project motivation and research goals

Imagine someone holds a controversial stance such as “raiding an official building is legitimate as long as this is done to rescue a country from collapse.” Although that may sound contradictory, to many people that is the result of a well-structured reasoning chain. In fact, research has sustained that a sort of public pedagogy is in place to “educate” people so that holding such views actually feels part of becoming a literate citizen (Greenhalgh et al., 2021). In my dissertation, I am interested in studying the intricacies of the processes through which people construct knowledge in ways that seem agentic but result in controversial stances.

I am especially intrigued by how processes of agentic knowledge construction find support in ostensibly reasonable argumentation. As Duncan and Chinn (2025) put it, those epistemically fragile arguments are “structurally sound”—that is, they pass many criteria of formal analysis of argumentation, such as relevance, (superficial) sufficiency of evidence, and nature of justification—but are built on weak epistemic processes and on suboptimal evaluative criteria. From that motivation, I arrive at the core guiding inquiry for this dissertation: non-experts construct meaning from content that is epistemically elastic but enveloped in structurally sound arguments; how does that construction of meaning take place in highly social settings that are information-heavy, emotionally-charged, and technology-infused? Examples of those settings are instant messaging apps and social media, spaces that enable a distorted form of learning—that is, learning that is effective but used as a springboard to advance harmful political agendas (Russo et al., 2024).

My motivation stated above drives the two components that I intend to address in the Doctoral Consortium: my Methods and Findings. I operationalize that motivation as one main research question and two subquestions that lay the groundwork to answer it:

- (RQ) How do non-experts engage epistemically with epistemically elastic content, that is, content that presents itself as apparently sound and structurally logical, but dismisses larger context that makes it problematic?
 - (RQ 1.1) What epistemic criteria and processes do non-experts rely on as they construct meaning from that type of content?
 - (RQ 1.2) How do pieces of intuitive knowledge and naturalized axioms (Philip, 2011) support those non-experts’ understanding of that content, scaffolding further knowledge construction?

In formulating these questions, I refer mainly to literature on *epistemic cognition* (Sandoval, 2017), particularly the AIR framework proposed by Chinn et al., (Chinn et al., 2020); supplementarily, I refer to literature related to Knowledge-in-pieces (KiP) by diSessa and colleagues (diSessa et al., 2015; Philip, 2011) to investigate how epistemic processes operate on intuitive knowledge and assumptions about the social world.

Methodology summary

I am running a mixed-methods research project to be implemented in phases, employing both in-depth clinical interviews and a cognitive task implemented online. By triangulating data from these two sources, I seek to identify how participants’ sociocultural repertoires and lived experience interact with new evidence. Qualitative data will include video recordings of task-based interviews which I will use to explore students’ criteria and processes in detail and to understand how they recruit pieces of intuitive knowledge as part of their interpretation of evidence. Once I finish analysis of qualitative data, I will use the insights to design the cognitive task.

Anticipated findings based on interview data

Here, I refer to interviews conducted with a convenience sample of 14 Brazilian participants (age 16-23, Mdn = 19; 11 undergraduate students) recruited in Rio de Janeiro in October, November, and December, 2025. Participants read a scenario in which a fictional council member asks his social media audience to help him take

a stance: should he vote for or against a bill to ban non-local informal beach vendors? Participants were then asked to mark their stance on a pro/against continuum. Next, they were presented 4 arguments, one at a time; after reading each argument out loud, they could either keep their stance or change, and were asked to explain why they chose so. Finally, they were asked to rank the arguments according to relevance and to reflect on possible effects of the bill. Although the sample and setting represent limitations for my study, the type of data I obtained revealed to be appropriate for my analytic approach. Below, I present snippets of preliminary analyses organized by my two supporting questions.

In answering RQ1.1, I draw on epistemic cognition literature (Chinn et al., 2014) to investigate the relevance participants assign to each type of argument. Preliminary analysis showed that testimony was the most relevant source for participants and that drawing on personal experience played a key role in their integration of the evidence presented, more so than other criteria like source trustworthiness.

In answering RQ1.2, I am interested in the knowledge elements that students “recruit” to support their interpretations. In my analysis, I draw on KiP to go back and forth between the data and an emerging theory I develop to help explain the role of those elements in students’ interpretations. In two examples in my pilot data, ML (Female, 22, Law major) and AN (Male, 18, recent high-school graduate), both participants recruited the knowledge element people migrate to search for opportunities. However, throughout the interview, each participant developed different stances as they progressed in their reasoning: whereas ML sustained her support for immigrants, AN considered more restrictive policies, contrasting that knowledge element against new evidence and arguments.

Expected contributions

With my dissertation, I hope to examine in depth the intuitive knowledge and epistemic criteria non-experts employ as they engage with ideas that have consequential societal implications. I anticipate that deeper examination can inform designs that prepare students to navigate “epistemically messy” settings (Duncan et al., 2018) where evidence of poor quality is presented and form the basis for structurally sound and consequential stance-taking. I also expect to add multidisciplinary to epistemic cognition scholarship that has already made significant contributions to the natural sciences. The topics I aim to address in my studies have deep moral and cultural dimensions, and I intend to incorporate dimensions that relate to other literacies (e.g., data, media). Additionally, my dissertation can expand the international reach of the learning sciences by working with Brazilian participants. This adds to our community’s search for epistemological pluralism and, importantly, helps illuminate how diverse populations navigate the challenges that motivate this work.

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Arts Gallery and Performance

Arts, Gallery and Performance

Track Introduction

The ISLS 2026 Arts, Gallery & Performance Submissions highlight creative, practice-based contributions that expand how the learning sciences conceptualize and communicate knowledge. Featuring visual art, installations, performances, and other embodied and multimodal forms, the track invites submissions that explore learning through aesthetic, cultural, and experiential lenses, complementing traditional research formats. By foregrounding participation, interpretation, and public engagement, the track creates a shared space for dialogue across disciplines and offers new pathways for understanding learning as situated, aesthetic, and socially embedded.

Micro- and Macrocossms in Accessible Reality: VR/AR Glasses as Partners in Learning

This work presents a visual narrative of immersive learning through affordable smartphone-based VR/AR technology. The photographs capture a student exploring two scientific domains: the vast scale of the solar system and the microscopic structure of molecules. Using simple VR glasses and a smartphone, the learner navigates planetary space and constructs molecular models atom by atom.

The images highlight moments of curiosity, concentration, and discovery, emphasizing the human experience within digital environments. By focusing on accessible tools rather than expensive laboratory equipment, the work invites viewers to reflect on how emerging technologies can democratize science education.

The series connects artistic documentation with learning sciences perspectives such as embodied learning and situated cognition, illustrating how physical interaction with digital worlds can deepen understanding of abstract scientific concepts.

Oleksii Voronkin

Links: Conversations of Sound and Body

Links: Conversations of Sound and Body is a live, improvisational dialogue between music and movement. Beginning with solo piano and expanding through spontaneous participation, the piece explores how learning emerges through relationships between sound, body, and attention. Grounded in constructionism and arts-based research, Links dissolves the boundary between artist and audience, inviting participants to co-create meaning and experience partnership as presence-fleeting, relational, and alive.

Roey Kafri (and You!)

Performance Score for Academic Conferences

1. In lieu of an abstract or paper, submit this score to an academic conference. The performance has begun.
2. If your submission is rejected, the performance ends. You may bow to your audience of reviewers if you wish.
3. If your submission is accepted, print the following materials:
 - a. A script or paper for a different submission you are presenting elsewhere at the conference.
 - b. An academic paper, article, or book chapter written by someone else you have enjoyed recently.
 - c. The call for papers from the conference.
 - d. Your CV.
4. Once printed, shuffle the pages of these materials together into a random order.
5. Download a Powerpoint template. Change the first slide to include the title of your submission, your name, and institutional affiliation. Change the last slide to say “thank you” and include your name and email. Do not change any of the other slides except-

Peter J Woods

How Do We Stay With It? Art for Climate Justice

This zine emerges from Art, Education and Climate Justice, a course where students use the arts to inquire into the climate crisis and develop a personal “stance” for engaging its challenges. Co-created by students and faculties, it gathers drawings, collages, photographs, craft, and mixed-media reflections from twenty-eight students. We conceptualize that the works collected here are not meant as representations or products of learning, but as living artifacts of inquiry. The pieces trace climate emotions, interdependence, inequities, and the historical systems shaping climate vulnerability. Each page is a part of a larger conversation: How can we stay with climate grief and still imagine otherwise? How might making art help us think with the world rather than about it? We see this zine as an archive of art that can mediate learning about climate justice, and how learning becomes a practice of attention, relation, and responsibility.

Jack Beebe, Emilie Bymers, Kaylee Caldier, Peighton Clapper, Alejandro Clausen, Addison Craig, Steph Do, Brandon Durland, Hailey Fernandez, Merve Haklidir, Agnes Huang, Archana Kalluru, Isabella Kaplun, Jonah Kowal, Connie Mina Lee, Giada Lesothebe, Sabrina Lopez, Jaymie Ni, Aileen Nichols, Nancy Price, Chuen Tan, Sammi Tomlinson, Hafsa Usman, Aliya Walmsley, Ella Ann White, Katie Rose Wilbur, Sherin Youssef, Elahe Zare

Looking Through the Lens: How Youth Practice Perspective-Shifting Through Filmmaking

Spy Hop’s “Voices of the West” program is a weeklong media production intensive offered throughout Utah. Youth participants engage in 30 hours of hands-on learning, working in small groups to produce three short films: a silent film featuring an emotion, character, and prop; a parody of a “YouTube tutorial”; and a documentary spotlighting their community. In tandem with a research study on “perspective-shifting” in youth film production, four films have been selected to demonstrate the complex, multimodal artistic decisions young filmmakers make as they consider different points of view. “Severo” is a reflective documentary about a local restaurant; “How to be a Superhero” uses the tropes of YouTube tutorials to subvert our expectations of the “superhero” and the “how-to” format; “The Seed” takes us into the perspective of a woman experiencing anxiety after reading an Instagram post; and “Huish Movie Theater” seeks to evoke nostalgia for moviegoing and community spaces.

Student Artists (pseudonyms: Wes, Sophia, Greta, Joy, Lucas, Alice), Researchers (Jill Beale, Kayla DesPortes) and Spy Hop Program Director (Adam Sherlock)

POV

I am (re)searching interactions. In the summer of 2025, I spent six days with 12 educators in and along Lake Erie, building connections with waters, lands, creatures, and each other. In my greed for data, I chose to do video-based interaction analysis, that slippery tool that makes us feel like we can make sense of others’ point-of-view. I asked educators to wear chest-mounted GoPro Max cameras to capture 360° video. The output lets us spin and spin, wresting point-of-view away from the educators and into researchers’ eyes and hands. This experimental video, backed by the rhythms of water and words, reconfigures images taken during a macroinvertebrate data collection activity educators conducted in a small tributary to the Lake. It is an invitation to wonder about the point-of-view of creatures we ask to teach us about water health, and to embrace the unresolved messiness of wading and listening without expecting answers.

Marijke Hecht

Nobody Lives Without Regrets

Nobody Lives Without Regrets is a work of interactive speculative fiction in which you, the reader, interview Carmen Torres, a formerly prominent researcher of AI-aided learning, in the year 2056. Depending on the questions you ask, you can guide Carmen's story to one of twenty-two unique conclusions, each conveying a different speculative future for the integration of AI, education, and society. Inspired by the framework of speculative education, this work instructs the reader to fix their gaze not on the present moment but on the various potential futures that this moment could produce. Through Carmen's story and your interaction with it, the piece asserts the centrality of the individual's decisions in shaping the future of our shared society at large. It says that through difficulty and fear, hope always exists for a better future if we work for it.

Hazel Marshall

#SlowLS

This video/performance explores time, movement, observation, and rest through the experience of a scholar disabled at the intersection of 'academia x chronic illness.' Multiple videos of the subject in the midst of daily campus foot traffic will playback at different speeds, along with several textual provocations, allowing viewers to consider the rushed pace of 'two-a-year' academia. Viewers are prompted to consider how the pace at which we engage with research may change the kinds of work that become possible. #SlowLS asks us to consider our pace through our pipeline: from grant to human relation, analysis, rest, and writing. #SlowLS asks us to consider whether we are holding space for humanizing journeys, and the complicated ways disability might help us push back on the oppressive rush of academia.

D Teo Keifert

What's Edgier than Edgy? Education, baby! Making Art with Children for Research + Dissemination

This entry to the Arts Gallery & Performance events is a collection of artifacts collaboratively made with young children as part of the Learning, Being, and Doing (LBD) in Families study. This submission features three research artifacts co-created by children, parents, and uni-based researchers. These artifacts include: (A) A construction of cardboard, paper, marker, and tape creation meant to illustrate "perspective" in conversations; (B) A cardboard and paper representation of a family at a campfire with the imagined additions of a uni-based researcher's two cats; (C) A video co-created by a family and uni-researcher that could be shared in research presentations to introduce our shared work. We will prepare accompanying story cards to elaborate on how each artifact represents a blending of child + parent + uni-based researchers' repertoires. The co-created artifacts represent innovative co-research methodology for dissemination to audiences that matter to collaborative research teams, especially children.

D Teo Keifert, Andria Ellis, Little Poof, Bunny, Carrot, Cheetah, Owl, Unicorn

Life is a Pure Flame: Thermochromic Kirigami Lotus

In the face of today's overlapping crises — environmental, social, existential — it is easy to become preoccupied with endings. This rising tide of anxiety and death-focus can leave individuals and communities searching for meaning and comfort amidst uncertainty. In response, we seek to revive ancient traditions and reintroduce acts of communal making and letting go.

Inspired by Eastern cultural traditions and Western psychological insights, and considering the current age in which origami itself is being reconceptualized as a powerful tool for scientific innovation, we envision a new kind of origami lotus—one that brings together ancient symbolism and modern technologies. By incorporating materials such as thermochromic paint, we can vividly visualize transformation and growth, turning each lotus into a living metaphor for change and renewal. For children, the project offers a cognitively engaging experience; for more mature participants, it serves as a ritual for personal reflection on life's cycles.

Jacob Puthipiroj, Supanat Deawrattanakun

The Look Again Quilt

For millennia, humans used pictures to tell stories. Narrative quilts are a traditional form of pictorial storytelling, dating back centuries. The Look Again quilt is a modern example of a narrative quilt used for science communication and engagement.

Since blankets are familiar and often evoke positive feelings, quilting as a science communication media may help avoid top-down cognitive processing that results in public disengagement with science. In addition, quilts can be gamified and create not only storytelling, but interactive opportunities.

The Look Again quilt features two realistic eyes, one open and the other closed at the top of the quilt, above an array of blocks, each representing different aspect of biology. The closed eye side is monochromatic and represents the angst and prevalence of stereotypes when many consider biology. For example, chemical equations represented as spinning blocks and straight-line stitching to represent rigid structures. In contrast, the open eye side is full color with blocks designed to evoke a sense of wonder and curiosity at the ways biology is part of one's life. For example, including blocks with wavy stitching and layered fabrics that represent the ecosystem at a beach.

Overall, the quilt encourages people to look at biology again and see the beauty of biology around them. Engagement with the quilt can come in the form of (A) Finding a block and using it to tell a biology story about the concept presented there (B) Playing co-designed I Spy (find a block that you think represents genetic engineering) (C) Asking people to reflect on what they notice about their perceptions and feelings of biology when they look again. How people engage with the quilt presents many opportunities for research on how people make sense of the relevance of biological information to their lives.

Melanie Peffer

Patchworking conversations

We created this piece with the goal to spark conversations about art, creativity, and connections. We invite everyone to participate in the making of an evolving artwork throughout the conference, embodying the theme "Partnering with purpose" by blurring boundaries between artists and observers, and acts as individual or collective. The work challenges the idea that artistic talent is innate and instead takes an exploratory approach to creation; a capability of expression set in motion through partnering with self, others, and materials. The collage we began by affixing pieces of fabric to paper grows when we ask all patrons to "find a material that matches your feeling and attach it onto the collage". In addition to becoming an exploration of material agency and a document of a collective journey, this work demonstrates that artmaking is not limited to self-identified artists, and that art-making can foster compassion through participation, reflection and connection.

Anastasia Y. Goodwin, Lexie Zhao, Naomi Field, Lana Ćosić

Reflexivity as Collaborative Poetic Process

Grounding in learning sciences scholarship on the embodied, affective, and relational nature of learning, we invite conference attendees to explore their own learning in and through their research through the co-creation of a collaborative poetic reflexivity installation. To begin, the installation will include a collection of poems written by members of our research team and others (with their permission) in response to prompts including: What does your/our research mean to you? What feelings, images, memories call you to your work? What in your work are you grappling with? We offer poetic processes of personal and collaborative inquiry as cognitive–affective tools for reflexive transformations that can strengthen our connections to our own complex humanity, to our research partners and their complex humanity, and to our aspirations for ourselves, our work, and our worlds.

Sofia Tancredi, Nicole Louie, Rosanne Luu, Siqi Huang, The Mattering Collective

Of Stars and Constellations: Culture and Identity within Collaborative Interactive Computational Art

Artistic computing education offers opportunities to explore disciplinary intersections that can support culturally sustaining experiences with technology. We present five computational art pieces created by 15 NYC high school students as part of CWP 2.0—an art & tech residency co-designed and facilitated by NYU researchers and Community Word Project Teaching Artists. Students combined visual and literary arts with electronics and coding as they reflected on oneself as a “star” in a constellation within their communities and society. Each piece includes sensors and output components the students programmed and embedded into sculptural and wearable artifacts. The artifacts are representative of various societies the learning community envisioned at the intersections of their cultures and identities. The pieces include: Threads of Our Community—a woven tapestry, An Ode to Identity—an interactive crown, An Ode to Peace—an interactive hat/pot, An ode to Unity—an interactive cape, and An Ode to Music-ing—an interactive music box & headphones.

Kayla DesPortes, Francisco Castro, Elizabeth Leonard, Cass Scheirer, Nichelle Ryan, Sabria Islam, Kathleen McDermott, and the 15 high school student cohort of CWP 2.0 2025

Marimbol (musical instrument)

We showcase two marimboles (bass lamellophone instruments) built by students of fandango music who participated in community-based instrument building workshops. These instruments were built collaboratively by over 20 participants under the guidance of a master luthier in a home that was temporarily repurposed into a grassroots workshop and cafeteria space in order to accommodate the collective, participatory building and learning process.

Leopoldo Novoa Matallana, Yared Portillo, Nathan Eddy, Miguel Mayagoytia, Cascabel de Dos Ríos

Intergenerational Arts-based Storytelling: Fostering Solidarity Learning Toward Climate Justice

Arts-based storytelling can counter epistemic injustice in climate education by centering lived experiences of minoritized communities. Drawing on critical participatory ethnography with Arab American immigrant youth, families, and teachers, we co-analyze collaborative climate change mapmaking as intergenerational knowledge work. Art fostered solidarity by legitimizing lived realities, incubating new onto-epistemologies, linking local and global experiences, and nurturing critical consciousness for climate justice advocacy. We share the artwork of our participants

Wisam Sedawi, Angela Calabrese Barton, Rachel Sherwin, Batoul Abdallah, FamJam Project

Dreams of Blindness

Dreams of Blindness (2025) is a VR documentary and participatory installation that explores the inner worlds of visually impaired people through their dreams. Based on interviews with 22 visually impaired participants in China, the work presents multiple dream narratives through participants’ authentic voices and dream-like visualizations. The project employs AI-assisted processes, from transcript analysis to visual generation, to support the creative translation of non-visual sensory experiences while minimizing authorial bias. By foregrounding emotional and sensory dimensions of dreaming, the film invites audiences to reconsider dominant assumptions about blindness. Alongside the VR experience, a participatory Dream Book encourages viewers to visualize and record their own dreams, transforming the installation into a shared reflective space. Through this arts-based research approach, the work seeks to foster empathy, dialogue, and critical reflection on disability, perception, and human imagination.

Li Jiang

We Keep Us Safe: Art for Expanding Abolitionist Imaginaries

This collection of works brings together both drafts and completed artworks co-created by the Abolitionist Futures Collective in Winnipeg, Manitoba. These pieces emerged from a series of workshops attended by artists and community members to imagine safety and care beyond traditional policing and carceral systems. The works included here are: lino prints, posters, and the drafts created across three different workshops. They reflect ongoing conversations the participants had about harm, care, conflict, and collective responsibility in a city shaped by increased policing and systems of incarceration.

These artworks are not simply expressions of abolitionist political ideas and imaginaries, they also demonstrate how those ideas are formed. Through making art together, the participants experimented with different artistic and political possibilities, built on each other's thinking, and expanded what can be imagined. The result is a living archive of collective sense-making.

Abolitionist Futures Collective

The Void: Using Arts-Based Autoethnography to Grapple with Genealogy and the Epistemicide of “Official Family Trees” in the Wake of the United States Census

I am a Black, nonbinary, queer person who was raised in Black girlhood in a city at the intersection of the Rust Belt and Appalachia. I use art to produce knowledge that is familiar, yet challenging. This visual arts piece utilizes autoethnography as a tool to reflexively engage with questions of the past, genealogy, historiography, fugitive spaces, and the interaction between anti-blackness and misogynoir as I react to a book created by a close relative. I am learning more about my genealogy while actively navigating how the dominant historic memory continuously serves our current white supremacist hegemony. Reflexivity is required in the process of learning oneself, allowing individuals to engage with excluded ontologies and epistemologies in a way that produces knowledge and makes meaning. Through this process of learning about my family and historiography, I discovered the role of authentic learning as an integrated process of both knowing and becoming.

Maya S. Malik

Pedagogical Zine Library

In the Learning Sciences community, there has been a history of engaging youth, educators, and other community members with research through curricula, programs, professional development, and more. We see potential in expanding engagement with creative forms such as zines, which are self-published and self-distributed work often in the form of printed text and images. In this art gallery work, we will share a Zine Library that features zines informal educators have created in our workshops over the last 3 years, featuring the “What Equity Means to Me,” and “Now What?” zines. In the spirit of making and sharing zines, we encourage other ISLS participants to share zines via social media or invitations to other researchers who have created research zines. During the conference, we will invite people to add to the zine library, either through engaging with our interactive zines or creating their own zines using the mini-zine format.

Mimi Shalf, Ricarose Roque

What's In A Game? Framing Undergraduate Improvisers Contributions as Legitimate Participation in Improv's Oral History

This work is a collection of improv forms (also known as games) developed by students in my undergraduate game-studies course, "Improv for Games." Students first learned games during class and by watching performances. Then, they modified the rules of those games. Finally, they all created games that they performed for each other.

My students, few of whom considered themselves improvisers before taking my course, created brilliant new forms that demonstrate a deep understanding, and they deserve to have that recognized. I have tried to motivate my pedagogical approach with the idea of meaningful making. I want my students to feel that they have contributed something towards furthering an art form and that they have become better designers in the process. These improv games (demonstrated by members of Spicy Clamato, the University of Illinois' longest running improv troupe) are "good enough" to be performed, enjoyed, and modified for years to come.

David A. Hopping, Noah Eliot Meyerhoff, Adam Vinn, Austin J. Reitz, Carly McConnell, Chayapol Corsello, Chinmay Ambasht, Daniel Blus, Eswyn Burd, Henry Bonzak, Julianne Liu, Louie Santos, Matthew Lugardo, Morgan Mueller, Natalia Restrepo, Rushi Kakkirala, Zanaiya Stephens

Reorienting Cartographies

This mixed-media mapping piece layers leaf-based pigments with tracing paper drawings of local infrastructures to challenge the assumed neutrality of cartographic tools. The plant pigments foreground land and the more-than-human world, while the infrastructural overlay visually "murks" these relations, mirroring how dominant maps overwrite ecological orientations. Drawing on sociocultural and decolonial theories of learning, the work reframes mapping as a relational practice that reorients viewers toward land-centered ways of knowing

Naomi G. Lau

A Live Magic Show to Introduce Magic a Pedagogy for Creativity

This interactive magic show will demonstrate how magic can be used to facilitate students' creativity and esign thinking. Through a series of live magic performances with narrative explanation, this session illustrates how watching, learning, and performing magic can help promote flexible thinking, challenge assumptions, and teach user-centered creative design principles.

Tong Li

"We Know Something You Don't Know": A Live Actual Play Performance

This presentation will be an improvisational "actual play" performance of the table-top roleplaying game "We Know Something You Don't Know." This game, designed by the authors to support educator learning, guides players in creating characters based on commonly marginalized troublemaker archetypes in classrooms. We invite listeners to consider ways the creation of "complex yet capable" troublemakers in co-constructed schools opens space for educators to challenge power hierarchies in figured worlds of schooling and cultivate critical civic empathy.

Karis Jones, Virginia Killian Lund, Zia Pochkhanawala, Peter Woods

Traces of Transformation: An Artist-in-School-Community Project

Traces of Transformation is a co-created artist-in-school-community project emerging from Fall of Artica (FoA), developed through a partnership between Artivation Projects (represented by artist Kathy H. Zhou) and the interdisciplinary Art and Design course (IDC) at University of Toronto Schools, led by teacher Robin Michel.

Building on the Culture Shoes Project, students designed shoes embodying their FoA avatars, using gameplay to surface ideological tensions — safety vs. freedom, now vs. future, utility vs. culture. Through research, discussion, and critical reflection, each student imprinted a personal message onto their shoe sole. These Adobe Illustrator designs were realized as laser-cut rubber stamps at the UTS innovation hub.

Echoing Allora and Calzadilla's politically engaged interventions, this critical-making work asks: What marks do we leave behind as learners, and how do these traces shape the paths available to those who follow?

Kathy H. Zhou, Robin Michel, Rubina Khan

Latina/x Voces: Comunidad, Creaciones, y Casa

“Latina/x Voces: Comunidad, Creaciones, y Casa” is a celebration of a multi-year Participatory Design Research project aimed at amplifying the presence of Latina/x youth in museums through co-creation of exhibitions. This exhibition includes selected artworks created by Latina/x youth during after-school programs and a summer camp between 2021-2025. These artworks were displayed previously at the Krannert Art Museum in Champaign, IL. Each year, the artists identified themes that represented their collective works. The “Comunidad” exhibition showcased languages, community spaces, and cultural foods. The “Creaciones” exhibition explored the intersection of Latina/x and youth cultures. Finally, the “Casa” exhibition examined what it means to gather at home for formal and informal occasions. In addition to increasing authentic representation of Latina/x youth in museums, this project has contributed to research about multilingual and multicultural literacy practices, participatory methodologies, and visitors’ appreciation for and sense of belonging in exhibitions designed by marginalized groups.

Project co-directors are Catherine Dornfeld Tissenbaum, Mónica González Ybarra, and Idalia Nuñez. “Core” artists include Darly Chen Man, Evolet Cruz Hernandez, Ashley Rebolledo Hernández, Kasey Rivas Lagos, and Mayra Valencia. Past artists include Gabriela Aguilar, Marvel Briones-Rebolledo, Maria Francisco, Leilani Hernandez, Rubi Lopez Felix, Yuridia Lopez Felix, Joyce Miralez.

Culture & Identity: Creating Meaningful Spaces Through Collaborative Art

This mural encompasses a variety of emotions, feelings, and perspectives from artists in their path to understanding culture and how it shapes their identity. I believe that creating a collaborative mural is a beautiful visual because it showcases how culture can connect and bring us closer together while also celebrating the differences in all of our cultures and identities and how that uniqueness makes us who we are. Additionally, I think the collaboration and reflection allowed for us to also express the nuances of culture, such as: how cultures can be mixed, how culture can be more than race, gender, sexuality but that communities also build culture, and that culture is not a one sized fits all experience but an ever evolving and growing feeling that constantly changes through time.

Sanjana Vinjamuri, Allison Starks

Contracted: Collaborative Learning through Musical Chaos

This project envisions a music making experience to explore productive friction and the value/tradeoffs of rapid feedback through spontaneous collaborative music making in community spaces. In this work, we will engage a group of people in co-creating “music” through independent manipulation of the elements that make that music. The goal of the project is to help people focus on the sources of information that working together with instant feedback (e.g., the music all can hear) enables people to both successfully collaborate and come into “light conflict.” This explores boundaries of productive versus unproductive friction while creating a meaningful musical experience both for the collaborative participants and others around them.

Ryan Lay, Matthew Berland, Peter Woods, Roey Kafri

Art Card Drawing Video

Ji Yoon Chung is an interdisciplinary artist, educator, and researcher pursuing a PhD in Integrated Studies in Education at McGill University and a Visiting Assistant Professor in the University of New Mexico Art Department. As a visual art educator in a children’s hospital, her research explores how arts education can be meaningfully integrated into hospital school settings through inclusive and adaptive approaches that respond to the changing needs of pediatric patients.

Her studio practice investigates how fleeting and immaterial experiences can be translated into tangible forms. Working across drawing, photography, sculpture, and digital fabrication, she often dissolves ink from printed photographs to evoke the erosion of memory and the emergence of new meanings.

Chung holds an MFA from the Rhode Island School of Design, an Ed.M. from Harvard Graduate School of Education, and a BFA from Seoul Women’s University. Her work has been exhibited in Providence, Boston, Reykjavik, and Seoul.

Ji Yoon Chung, Su Bin Kang

MOVE about it: getting groups thinking together through movement

We are a group of Learning Scientists, Dancer-Choreographers, Scientists, & friends who have breathed together, danced together, and learned a lot together over the past 4 years as the design team on the Choreographing Science Project. We designed for expansive embodied learning between middle schoolers and scientists, curious as to how dancing together might destabilize epistemic hierarchies. We began from a starting point that any movement can be dance and any dance can support collective learning. We value a continual uplifting of moving bodies, a dedication to creativity, a rigorous practice of empathy, and an openness for change. We produced a deck of movement prompt cards—MOVE about it—to capture our flavor of supporting groups to create improvisational choreography. If we need to move in order to know then the best way to share what we know is to move about it with others.

Lauren Vogelstein, Rebecca Steinberg, Dionne Champion, Curtis Thomas, Lara Appleby, Aditi Wagh

Intersect: A Mixed Media Collage Series & Working Session

I approach artwork as a collaboration with unruly matter: pigments bloom and resist, papers buckle and tear, materials misbehave. A central concern in my work is agency—not only as a human capacity, but the ways materials, tools, and environments participate in shaping what becomes possible. I work in active relationship with these affordances and resistances, following a posthumanist, distributed sense of agency in which decisions and outcomes are negotiated among bodies, ideas, and materials. As a disabled and chronically-ill scholar and artist, my practice is interleaved with rest, pauses that allow introspection. The processes I use reveal worth and desire in what could be labeled ‘flawed’ or ‘ruined’—metaphors for ways society is disabling. My recent work has focused on ways arts can support the development of critical consciousness, bolster commitment to justice, equity, and inclusion, and enhance visions of radical hope for engineering and science faculty.

Vanessa Svihla

Thresholds of Knowing

Quilting connects art, craft, design, storytelling, political resistance, and cultural preservation. Coupling the fiber arts with electronics can expand opportunities for storytelling and learning. In this gallery submission, a quilter and two art-computing education researchers reflect on their distributed process of integrating quilting with electronics and present a triptych of large-format electronics quilts, each centering an unsung woman in technology:

- Miriam E. Benjamin: the initiator, inventor of the initial airplane call button and hotel call button [signals]
- Angela Ruiz Robles: the illuminator, inventor of one of the first e-readers [mechanisms]
- Gladys Brown West: the locator, contributed to mathematically modeling earth supporting the development of GPS [mathematics]

We communicate how their signals, mechanisms, and mathematics have impacted the ways in which we codify our existence, our knowledge, and our presence. We will present a 5min video documentary of our asynchronous, distributed process highlighting the opportunities for cultural learning through narratives.

Kayla DesPortes, Mia Shaw, Earamichia Brown

Research for Humans: Social Practice Art as Scholarly Communication

Research for Humans is an interactive multimodal installation using social practice art to address a global epistemic crisis. Trust in reality wavers as traditional institutions lose their footing, while technology accelerates the need for reinvention. What is truth, how do we know it, and how might we design ways forward?

Inspired by centuries of automata, the work reflects our enduring fascination with what it means to be alive and the uneasy balance between human vitality and machine logic. Visitors approach a roving cart overseen by an animatronic peddler — part mystic, part minstrel — and converse by talking or typing. Responses drawn from published research are printed in real-time on cards: one to keep, one added to a growing repository.

Social practice art transforms research into communal inquiry. Here, academic knowledge becomes something felt, embodied, and carried home — a provocation toward braver thinking and social action.

Emily Reardon, Lucius Von Joo, OreOluwa Badaki, Chris Moffett

Mathematizing, Visualizing, and Power: A Collection of Community- Centered Data-Art by Appalachian Youth

This collage poster shares photographs and artist statements of artistic data visualizations created by East Tennessean youth in Mathematizing, Visualizing, and Power (MVP): Appalachian Youth Becoming Data Artists for Community Learning, a multi-year NSF-funded project. Across five sites over three years, approximately 150 Appalachian youth participated in this project to connect data science, art, and community life. They were invited to ask questions that mattered to their communities, collect and analyze data from their communities, and transform what they found into artistic data visualizations. The featured data-art works in this poster spotlight how youth used data to tell stories about community issues they care about and to communicate those stories in creative, personal ways. By bringing together data investigation and artistic expression, this project created a meaningful and community-centered pathway for youth to engage with both data science and the arts.

MVP Data Artists

Visualizing Justice: Arts-Based Inquiry in Community-Engaged Climate Education

As a PhD student and "artist-scientist," my work investigates the sociopolitical tensions within climate education. This project utilizes arts-based visual methodologies to analyze a graphic novel authored by two 6th-grade students in a systemically disenfranchised community. When linguistic analysis proved insufficient to capture the "multiplicity of meaning" within our partnership, I turned to visual inquiry to surface subjugated voices and document my own analytic process.

The resulting images function as both data analysis and emergent findings. They map the complex interplay of economic disparity, climate violence, and the political threats facing DEI initiatives. By integrating my own positionality, specifically the contradictions between my working-class roots and current position as a middle-class researcher, the artwork makes visible the ethical friction inherent in justice-oriented research. This practice moves beyond simple representation, using art as a moral and political tool to critique power dynamics and resist the reproduction of inequity in collaborative spaces.

Rachel Sherwin

Shaunk: An Embodied Computing Approach to Music Learning through Culturally Responsive Design

This design-based research study explores interdisciplinary interactions between the Arts and Computer Science in a Music classroom. We co-designed Shaunk, a robotic Dhol (a Punjabi percussion instrument), with music instructors and studied it as part of an after-school program for middle and high school students. We conducted a mixed-methods analysis of video data, pre- and post-assessments, and semi-structured interviews to understand how embodied interaction with a musical robot shapes learning engagement, fluency, collaborative practice, and students' perceptions of computing. Our analysis shows increases in practice time (150%), task completion (from 60% to 95%), and engagement duration. Qualitative analysis of student interviews suggests that students appreciated the connections between cultural practice and computing. We contribute design principles for culturally responsive embodied learning environments and discuss the implications of integrating computing into authentic cultural contexts

Harjot Singh, Preeti Raman

Unlearning

Unlearning is an interactive gallery that provocatively introduces and explores (un)learning as an active, political, communal process through collaborative collage-making. It invites the visitors to engage with 18 original collages, each exploring personal moments and journeys when default settings, rituals, practices, or learnings suddenly felt odd. It utilizes the dance of collaging between constructing and deconstructing to process another one between learning, unlearning, and relearning.

What, then, must be (un)learned? Our individual and collaborative pieces reveal how learning often means naturalizing harmful categories and systems, beliefs that become so deeply sedimented they feel like common sense. We learned that nationality is fixed, safety comes from carceral institutions, creativity is innate talent, voice must hide subjectivity, care should be withheld, bodies must be controlled. These weren't abstract concepts but embodied, ritualized performances that shaped who we understood ourselves to be and what possibilities we could imagine.

Abu Abdelbagi, Emmy Semprun, Shell Avenant, Maelin Aquino, Giles Bullen, Blake Danzig, Rhea Jafer, Andrea Kim, Sonia Kim, Dengjianing Li, Yafang Niu, Natasha Ryan, Madison Smith, Valeria Tedesco, Lucius Von Joo, Layla Yan, Lucy Zhang, Xinge Zhang

Learning Autotheoretical Trans* Body: Transforming What Was to What Is and Must Be Through Recreational Bodybuilding

My work explores trans* body politics capturing its complexities and subjectivities, locating our bodies as sites of inquiry. I draw majorly on Lugones's coloniality of gender to frame the colonial imposition upon our bodies, dictating how our bodies can(not) be, in which spaces we can(not) occupy. We're often deemed deviant and pathological because we're such a novelty in society when in fact we have existed for centuries, erased through the colonial imposition. I study my body through its praxis of recreational fitness and bodybuilding as a method of resistance and advocacy for the trans* body. My modes of inquiry are portraiture and autotheory, approaches that bridge arts, science, and reflexivity, offering potent analyses of intersectionality toward the work of social justice and change. Wherein, I delineate the nuances of the trans* body politics, positioning this sculpting and moving as a learning of the self and the other.

Breneil Malcolm

“On the fly”: A collaborative game for musical improvisation

“On the Fly” is a game designed to support musicians in practicing collaborative improvisation. This game provides generative constraints for musicians to enact, which pushes musicians to shape and structure their improvisations in real time without tying them to a particular genre or musical tradition. The primary game mechanic is a deck of cards that display simple line graphs or contours. Musicians pick a card and improvise a performance together that enacts the contour depicted. To increase the challenge of the game, musicians roll a die to determine what aspect of the music is described by the contour card. By providing a structure for musicians to explore and develop their improvising, “On the Fly” functions simultaneously as a format for musical performance, a pedagogical tool, and a research instrument. Through these overlapping functions, this game helps to advance our understanding of improvisation and theorize how it is learned and cultivated.

Ethan Rubin, Prawit Siriwat, Israel Flores

Crafting non-Euclidean geometry with yarn: The case of the hyperbolic plane

Aligning craft, mathematics, and socio-political activism, this installation art piece ventures into non-Euclidean geometry to crochet the hyperbolic plane in a range of colors and fabrics. While Euclidean geometry follows a well-known set of axioms such as a plane is a flat surface, non-Euclidean geometry directly subverts those. For centuries taught impossible to model physically, the hyperbolic plane, a surface which curves away from itself at every point, was successfully crocheted at last by a female mathematician, showing that craft can be a vehicle for even advanced mathematics. Thereafter and further gaining traction, crocheting hyperbolic planes became the focal point of the Crochet the Coral Reef project, drawing attention to climate change and showing that craft can also be a salient vehicle for socio-political activism. This living art piece welcomes visitor interactions and creative exploration toward collectively crafting a coral reef of our own.

Nickolina Yankova

Where I'm from Somos Familia: Community Prints that Inspire Interest-Driven Design

This screen-printed community story quilt is composed of individual prints representing family and community stories. Created at an art-making workshop in our maker studio, the making process and artwork produced serve as artifacts that inform AIMS Center's design-based research project. Interests and community assets revealed through the art-making at this and other workshops will inform AIMS efforts to design a makerspace where youth and families feel they belong, leading AIMS toward a community codesigned makerspace.

Lora Cawelti, Elin Anderson, Aileen Rizo, Paul Reimer, AIMS Center Staff

Interactive Mosaics: Youth Creative Computing with ArtAlgo

Interactive Mosaics presents a collection of algorithmic artworks created by rural middle school students using ArtAlgo, a creative computing platform that pairs p5.js coding with Aiko, a conversational AI design partner. Over six workshop sessions, students explored color as a computational medium, transforming HEX values and palette generation into deliberate artistic choices. They built interactive mosaics that respond to touch, sound, and motion through Makey Makey physical computing kits, bridging digital abstractions with tangible, embodied experience. Aiko supported students only when asked, preserving their creative agency while offering responsive scaffolding around color, mood, and structure. The resulting works illustrate how computational thinking and creativity develop together when young people use code as an expressive medium. This collection invites viewers to consider how purposeful human-AI partnership and community-centered design can make computational expression accessible and personally meaningful for youth in under-resourced settings.

STEM Club from West Middle School, North Carolina

OTHERLAND: The Cultural, Perceptual Knowledge ABOUT, FROM, and WITH the Others

OTHERLAND is an interactive installation that invites participants to step into a projected North American landscape, transforming static viewing into an embodied encounter. Within a 3m × 3m immersive environment, participants' gestures generate rippling waves of light that respond dynamically to movement. Based on a 2024 prototype by Glorie Gao's team, the installation draws on East Asian landscape aesthetics, in which water symbolizes philosophical depth and relational harmony. Real-time ambient sounds amplify this multisensory dialogue, challenging hierarchies privileging visual or cognitive knowledge over somatic and relational knowing (Abrahamson, 2014; Glenberg, 2010).

Glorie Q. Gao, Kathy H. Zhou

Acting School and the Art of Unlearning

Kailea Saplan is an actor, teaching artist, and scholar originally from the island of Hawai'i. Through interdisciplinary projects and partnerships, they explore connections between art (especially theatre), education, and the learning sciences, hoping to transform how we educate future generations. The motivation behind all their work is to support growing diversity and to foster curiosity and compassion for the other. In 2025, Kailea graduated with their Ph.D. in Curriculum and Instruction from the University of Wisconsin – Madison, and they are currently pursuing an M.F.A. in Acting from California Institute of the Arts.

Kailea Saplan, Ph.D., Tina Kronis, Heidi Scheller

Mapping Our Journeys: From Artist to Teacher

In this collaborative artist book, we engage in reflective narrative to explore our learning journeys including what drove us as students and artists to want to pursue teaching. We employ connected arts learning to conceptually ground our reflections and artwork in our culturally connected and creative educational opportunities. The context of our learning community is central to this project. We are a cohort of students and instructors at UCLA VAPAE, where we are participating in a sequence of three classes that span a full academic year. As a close cohort, we have developed community norms, made art together, and explored culturally sustaining and teaching, socioemotional learning, and supportive classroom environments. We've also shared our surprises, struggles, and successes as we were placed at school sites and taught for the first time. Through our artistic reflections, we hope to share a window to ourselves as a community of artists and teachers.

Lora Cawelti, Victoria Atilano, Megan Diaz, Natalie Elias, Koko Hara, Sam Jones, Luis Martinez, Kiki Ouyang, Elisabeth Park, Ale Rivas-Vargas, L Siswanto, Kate Stone

Learning in Balance: Cultural Jewellery as Knowledge Carrier

Learning in Balance explores how learning can be sensed through the body, materials, and cultural symbols. Inspired by John P. Miller's philosophy of holistic education, the project translates the Taoist concept of Yin and Yang into an embodied model of learning that balances intuition and analysis, body and mind, art and science. The jewellery series draws on Chinese traditional constellations from Taiji Yin-Yang to the Four Symbols, and the Twenty-Eight constellations. In this framework, Taiji represents wholeness, Yin-Yang reflects cycles of action and reflection, the Four Symbols suggest different modes of learning, and the constellations evoke interconnected knowledge networks.

Through wearable jewellery art forms, abstract educational knowledge becomes tactilely embedded experiences. NFC tags connect the pieces to digital reflections, inviting audiences to consider where Balance appears in their own learning and interact in the digital space. Jewellery becomes a cultural knowledge carrier where learning can be worn, felt, and remembered.

Yuna(Lei) Zhang

Reimagining Reading Through an Interactive Digital Book

The experience of reading is sometimes viewed as a passive occurrence. However, new possibilities for active—and even creative interactivity—open up when affordances of emerging technologies are leveraged for social and creative opportunities. This arts gallery and performance submission endeavors to ask participants to work in collaboration with artificial intelligence (AI) and augmented reality (AR) / virtual reality (VR) to create an engaging, socially-constructed and surprising digital book. Through the act of co-creation, reading morphs into a collaborative, experiential experience which aims to make the act itself more active, communal, and impactful.

Rana Al-Johani, Lourdes Keochgerien, Yuhao Qiu, Allison Starks

Papyrophilia vs. Mathophobia: Exploring paper invitation to constructionist mathematics learning

This installation explores how paper can serve as an aesthetic, artistic invitation into constructionist mathematics learning. We present a tiered, pyramid-like landscape of interactive paper and cardboard artifacts—pop-ups, tessellations, curved folds, and geometric solids—each designed as an object-to-think-with that foregrounds beauty, tactility, and mathematical structure. Building on Seymour Papert's vision of "soap-sculpture math" and his analysis of mathophobia, we ask what it means to treat mathematical work as gallery-worthy, proudly displayable and shareable art, rather than an anxiety-inducing exercise. Our design also responds to critiques of maker spaces that foreground high-tech tools and narrow participation: by beginning with familiar, low-cost materials and manual manipulation, we lower perceived technical barriers while still connecting to digital design and fabrication workflows and deeper mathematical thinking.

David M. Zikovitz, Agnese Del Zozzo, Francesca Fiore, Alberto Montresor

Major/Minor

Major/Minor is an interactive multimedia installation that invites audiences to transform contemporary political discourse into audio and video outputs. Through the mapping of forms of video-generated speech through algorithms that transpose majority and minority sentiments and other forms of political speech into major and minor musical scales, this piece considers and synthesizes the vast scale of news and information in contemporary media, generating audio and visual compositions that reshape daily media feeds into new and ambivalent expressions of our digital ecosystems, echo chambers and refracted realities.

Sam Mejias, Zuzana Bakova, Jackie Brady, Kevin Ingco, Negin Pazandeh, Shoro Roy

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